

Part I

Molecular Components of Cells



All life depends on water; all organisms are aqueous chemical systems. (Waves in Oahu, Hawaii, Brad Lewis/Liaison International)

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“ . . . everything that living things do can be understood in terms of the jiggings and wiggings of atoms.”

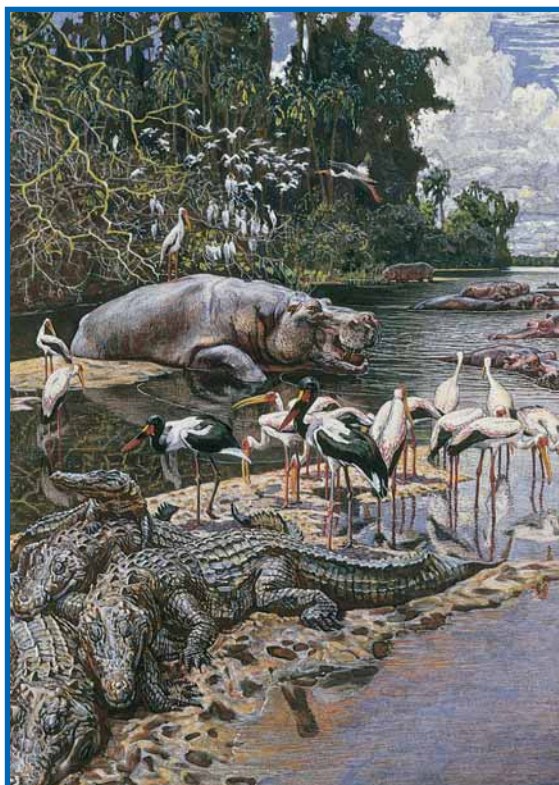
RICHARD P. FEYNMAN
Lectures on Physics
Addison-Wesley Publishing Company, 1963

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Chapter 1

Chemistry Is the Logic of Biological Phenomena



“Swamp Animals and Birds on the River Gambia,” c. 1912 by Harry Hamilton Johnston (1858–1927). (Royal Geographical Society, London/The Bridgeman Art Library.)

Molecules are lifeless. Yet, in appropriate complexity and number, molecules compose living things. These living systems are distinct from the inanimate world because they have certain extraordinary properties. They can grow, move, perform the incredible chemistry of metabolism, respond to stimuli from the environment, and, most significantly, replicate themselves with exceptional fidelity. The complex structure and behavior of living organisms veil the basic truth that their molecular constitution can be described and understood. The chemistry of the living cell resembles the chemistry of organic reactions.

Indeed, cellular constituents or **biomolecules** must conform to the chemical and physical principles that govern all matter. Despite the spectacular diversity of life, the intricacy of biological structures, and the complexity of vital mechanisms, life functions are ultimately interpretable in chemical terms. *Chemistry is the logic of biological phenomena.*

logic • a system of reasoning, using principles of valid inference

1.1 • Distinctive Properties of Living Systems

The most obvious quality of **living organisms** is that they are *complicated and highly organized* (Figure 1.1). For example, organisms large enough to be seen with the naked eye are composed of many **cells**, typically of many types. In turn, these cells possess subcellular structures or **organelles**, which are complex assemblies of very large polymeric molecules or **macromolecules**. These macromolecules themselves show an exquisite degree of organization in their intricate three-dimensional architecture, even though they are composed of simple sets of chemical building blocks, such as sugars and amino acids. Indeed, the complex three-dimensional structure of a macromolecule, known as its **conformation**, is a consequence of interactions between the monomeric units, according to their individual chemical properties.

Biological structures serve functional purposes. That is, biological structures have a role in terms of the organism’s existence. From parts of organisms, such as limbs and organs, down to the chemical agents of metabolism, such as enzymes and metabolic intermediates, a biological purpose can be given for each component. Indeed, it is this functional characteristic of biological structures that separates the science of biology from studies of the inanimate world such as chemistry, physics, and geology. In biology, it is always meaningful to seek the purpose of observed structures, organizations, or patterns, that is, to ask what functional role they serve within the organism.

Living systems are actively engaged in energy transformations. The maintenance of the highly organized structure and activity of living systems depends upon their ability to extract energy from the environment. The ultimate source of energy is the sun. Solar energy flows from *photosynthetic organisms* (those organisms able to capture light energy by the process of photosynthesis) through

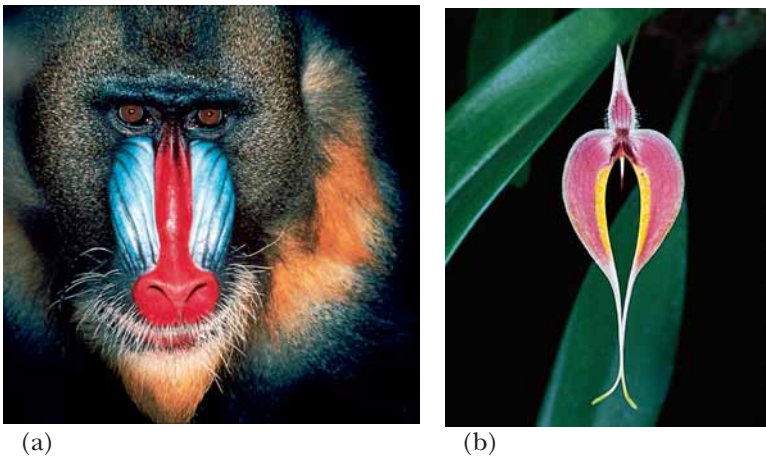


FIGURE 1.1 • (a) Mandrill (*Mandrillus sphinx*), a baboon native to West Africa. (b) Tropical orchid (*Bulbophyllum blumei*), New Guinea. (a, Tony Angermayer/Photo Researchers, Inc.; b, Thomas C. Boydon/Marie Selby Botanical Gardens)

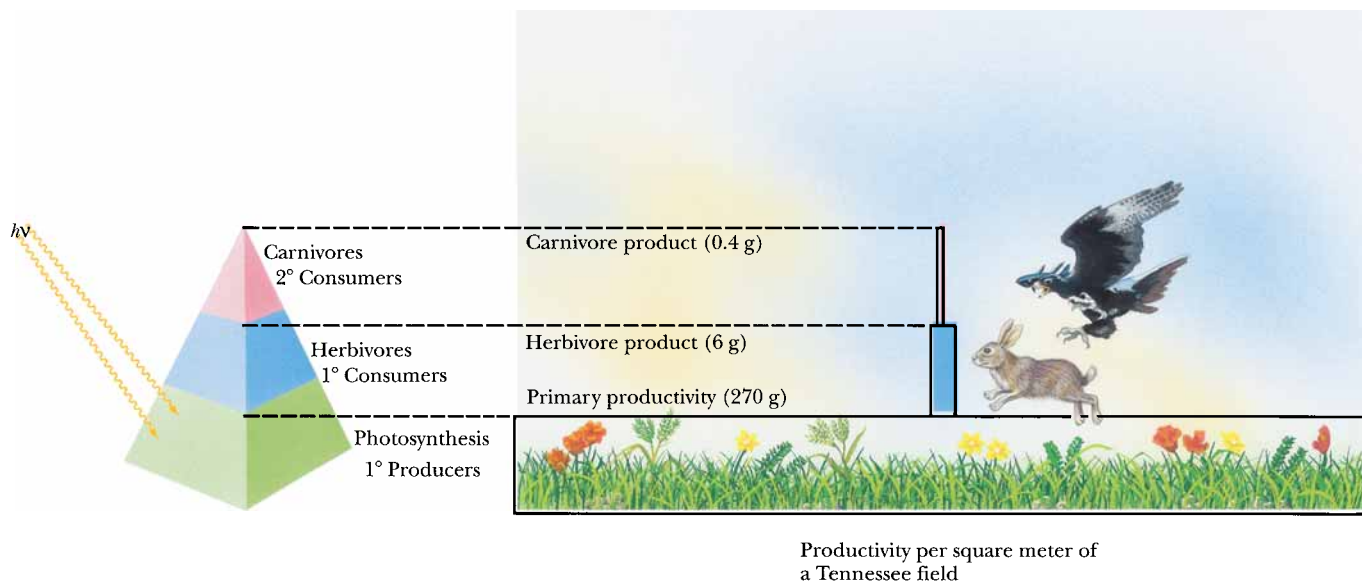


FIGURE 1.2 • The food pyramid. Photosynthetic organisms at the base capture light energy. Herbivores and carnivores derive their energy ultimately from these primary producers.

food chains to herbivores and ultimately to carnivorous predators at the apex of the food pyramid (Figure 1.2). The biosphere is thus a system through which energy flows. Organisms capture some of this energy, be it from photosynthesis or the metabolism of food, by forming special energized biomolecules, of which **ATP** and **NADPH** are the two most prominent examples (Figure 1.3). (Commonly used abbreviations such as ATP and NADPH are defined on the inside back cover of this book.) ATP and NADPH are energized biomolecules because they represent chemically useful forms of stored energy. We explore the chemical basis of this stored energy in subsequent chapters. For now, suffice it to say that when these molecules react with other molecules in the cell, the energy released can be used to drive unfavorable processes. That is, ATP, NADPH, and related compounds are the power sources that drive the energy-requiring activities of the cell, including biosynthesis, movement, osmotic work against concentration gradients, and, in special instances, light emission (bioluminescence). Only upon death does an organism reach equilibrium with its inanimate environment. *The living state is characterized by the flow of energy through the organism.* At the expense of this energy flow, the organism can maintain its

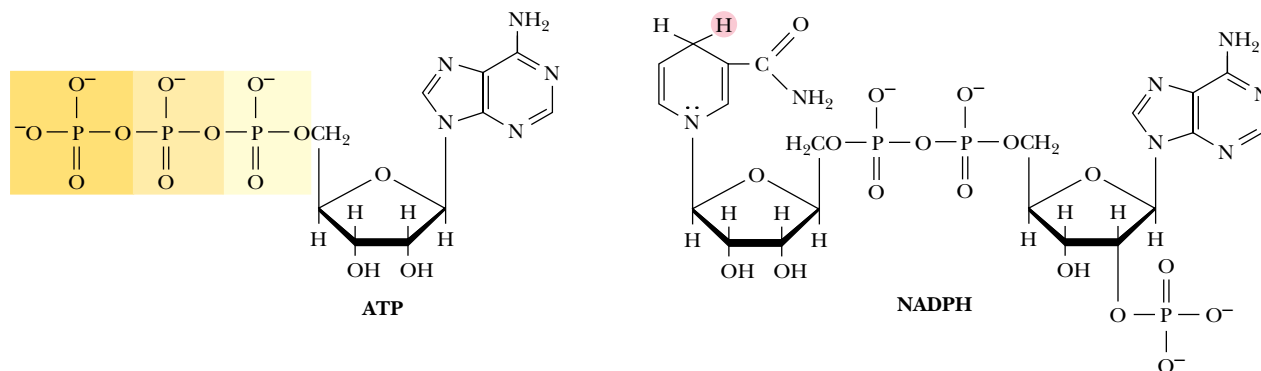


FIGURE 1.3 • ATP and NADPH, two biochemically important energy-rich compounds.

intricate order and activity far removed from equilibrium with its surroundings, yet exist in a state of apparent constancy over time. This state of apparent constancy, or so-called **steady-state**, is actually a very dynamic condition: energy and material are consumed by the organism and used to maintain its stability and order. In contrast, inanimate matter, as exemplified by the universe in totality, is moving to a condition of increasing disorder or, in thermodynamic terms, maximum entropy.

Living systems have a remarkable capacity for self-replication. Generation after generation, organisms reproduce virtually identical copies of themselves. This self-replication can proceed by a variety of mechanisms, ranging from simple division in bacteria to sexual reproduction in plants and animals, but in every case, it is characterized by an astounding degree of fidelity (Figure 1.4). Indeed, if the accuracy of self-replication were significantly greater, the evolution of organisms would be hampered. This is so because evolution depends upon natural selection operating on individual organisms that vary slightly in their fitness for the environment. The fidelity of self-replication resides ultimately in the chemical nature of the genetic material. This substance consists of polymeric chains of deoxyribonucleic acid, or **DNA**, which are structurally complementary to one another (Figure 1.5). These molecules can generate new copies of themselves in a rigorously executed polymerization process that ensures a faithful reproduction of the original DNA strands. In contrast, the



(a)



(b)



(c)

FIGURE 1.4 • Organisms resemble their parents. (a) Reg Garrett with sons Robert, Jeffrey, Randal, and grandson Jackson. (b) Orangutan with infant. (c) The Grishams: Andrew, Rosemary, Charles, Emily, and David. (a, William W. Garrett, II; b, Randal Harrison Garrett; c, Charles Y. Sipe)

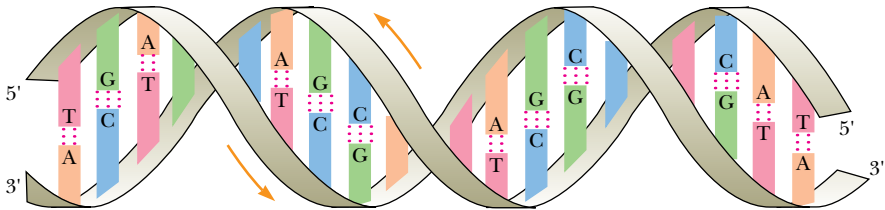


FIGURE 1.5 • The DNA double helix. Two complementary polynucleotide chains running in opposite directions can pair through hydrogen bonding between their nitrogenous bases. Their complementary nucleotide sequences give rise to structural complementarity.

complementary • completing, making whole or perfect by combining or filling a deficiency

molecules of the inanimate world lack this capacity to replicate. A crude mechanism of replication, or specification of unique chemical structure according to some blueprint, must have existed at life’s origin. This primordial system no doubt shared the property of **structural complementarity** (see later section) with the highly evolved patterns of replication prevailing today.

1.2 • Biomolecules: The Molecules of Life

The elemental composition of living matter differs markedly from the relative abundance of elements in the earth’s crust (Table 1.1). Hydrogen, oxygen, carbon, and nitrogen constitute more than 99% of the atoms in the human body, with most of the H and O occurring as H₂O. Oxygen, silicon, aluminum, and iron are the most abundant atoms in the earth’s crust, with hydrogen, carbon, and nitrogen being relatively rare (less than 0.2% each). Nitrogen as dinitrogen (N₂) is the predominant gas in the atmosphere, and carbon dioxide (CO₂) is present at a level of 0.05%, a small but critical amount. Oxygen is also abundant in the atmosphere and in the oceans. What property unites H, O, C, and

Table 1.1

Composition of the Earth’s Crust, Seawater, and the Human Body*					
Earth’s Crust		Seawater		Human Body†	
Element	%	Compound	mM	Element	%
O	47	Cl [−]	548	H	63
Si	28	Na ⁺	470	O	25.5
Al	7.9	Mg ²⁺	54	C	9.5
Fe	4.5	SO ₄ ^{2−}	28	N	1.4
Ca	3.5	Ca ²⁺	10	Ca	0.31
Na	2.5	K ⁺	10	P	0.22
K	2.5	HCO ₃ [−]	2.3	Cl	0.08
Mg	2.2	NO ₃ [−]	0.01	K	0.06
Ti	0.46	HPO ₄ ^{2−}	<0.001	S	0.05
H	0.22			Na	0.03
C	0.19			Mg	0.01

*Figures for the earth’s crust and the human body are presented as percentages of the total number of atoms; seawater data are millimoles per liter. Figures for the earth’s crust do *not* include water, whereas figures for the human body do.
†Trace elements found in the human body serving essential biological functions include Mn, Fe, Co, Cu, Zn, Mo, I, Ni, and Se.

N and renders these atoms so suitable to the chemistry of life? It is their ability to form covalent bonds by electron-pair sharing. Furthermore, H, C, N, and O are among the lightest elements of the periodic table capable of forming such bonds (Figure 1.6). Because the strength of covalent bonds is inversely proportional to the atomic weights of the atoms involved, H, C, N, and O form the strongest covalent bonds. Two other covalent bond-forming elements, phosphorus (as phosphate —OPO_3^{2-} derivatives) and sulfur, also play important roles in biomolecules.

Biomolecules Are Carbon Compounds

All biomolecules contain carbon. The prevalence of C is due to its unparalleled versatility in forming stable covalent bonds by electron-pair sharing. Carbon can form as many as four such bonds by sharing each of the four electrons in its outer shell with electrons contributed by other atoms. Atoms commonly found in covalent linkage to C are C itself, H, O, and N. Hydrogen can form one such bond by contributing its single electron to formation of an electron pair. Oxygen, with two unpaired electrons in its outer shell, can participate in two covalent bonds, and nitrogen, which has three unshared electrons, can form three such covalent bonds. Furthermore, C, N, and O can share two electron pairs to form double bonds with one another within biomolecules, a property that enhances their chemical versatility. Carbon and nitrogen can even share three electron pairs to form triple bonds.

Atoms	e [−] pairing	Covalent bond	Bond energy (kJ/mol)
H · + H ·	→ H : H	H — H	436
· C · + H ·	→ · C : H	$\begin{array}{c} \\ -\text{C}-\text{H} \\ \end{array}$	414
· C · + · C ·	→ · C : C ·	$\begin{array}{c} \quad \\ -\text{C}-\text{C}- \\ \quad \end{array}$	343
· C · + · N ·	→ · C : N ·	$\begin{array}{c} \\ -\text{C}-\text{N} \diagup \diagdown \end{array}$	292
· C · + · O ·	→ · C : O ·	$\begin{array}{c} \\ -\text{C}-\text{O}- \\ \end{array}$	351
· C · + · C ·	→ C :: C	$\diagdown \text{C} = \text{C} \diagup$	615
· C · + · N ·	→ C :: N ·	$\diagdown \text{C} = \text{N}-$	615
· C · + · O ·	→ C :: O	$\diagdown \text{C} = \text{O}$	686
· O · + · O ·	→ · O : O ·	- O — O -	142
· O · + · O ·	→ O :: O	O = O	402
· N · + · N ·	→ N :: N	N ≡ N	946
· N · + H ·	→ · N : H	$\diagdown \text{N}-\text{H}$	393
· O · + H ·	→ · O : H	- O — H	460

FIGURE 1.6 • Covalent bond formation by e[−] pair sharing.

Two properties of carbon covalent bonds merit particular attention. One is the ability of carbon to form covalent bonds with itself. The other is the tetrahedral nature of the four covalent bonds when carbon atoms form only single bonds. Together these properties hold the potential for an incredible variety of linear, branched, and cyclic compounds of C. This diversity is multiplied further by the possibilities for including N, O, and H atoms in these compounds (Figure 1.7). We can therefore envision the ability of C to generate complex structures in three dimensions. These structures, by virtue of appropriately included N, O, and H atoms, can display unique chemistries suitable to the living state. Thus, we may ask, is there any pattern or underlying organization that brings order to this astounding potentiality?

1.3 • A Biomolecular Hierarchy: Simple Molecules Are the Units for Building Complex Structures

Examination of the chemical composition of cells reveals a dazzling variety of organic compounds covering a wide range of molecular dimensions (Table 1.2). As this complexity is sorted out and biomolecules are classified according to the similarities in size and chemical properties, an organizational pat-

Table 1.2

Biomolecular Dimensions			
The dimensions of mass* and length for biomolecules are given typically in daltons and nanometers, [†] respectively. One dalton (D) is the mass of one hydrogen atom, 1.67 × 10 ⁻²⁴ g. One nanometer (nm) is 10 ⁻⁹ m, or 10 Å (angstroms).			
Biomolecule	Length (long dimension, nm)	Mass	
		Daltons	Picograms
Water	0.3	18	
Alanine	0.5	89	
Glucose	0.7	180	
Phospholipid	3.5	750	
Ribonuclease (a small protein)	4	12,600	
Immunoglobulin G (IgG)	14	150,000	
Myosin (a large muscle protein)	160	470,000	
Ribosome (bacteria)	18	2,520,000	
Bacteriophage ϕX174 (a very small bacterial virus)	25	4,700,000	
Pyruvate dehydrogenase complex (a multienzyme complex)	60	7,000,000	
Tobacco mosaic virus (a plant virus)	300	40,000,000	6.68 × 10 ⁻⁵
Mitochondrion (liver)	1,500		1.5
<i>Escherichia coli</i> cell	2,000		2
Chloroplast (spinach leaf)	8,000		60
Liver cell	20,000		8,000

*Molecular mass is expressed in units of daltons (D) or kilodaltons (kD) in this book; alternatively, the dimensionless term *molecular weight*, symbolized by Mr, and defined as the ratio of the mass of a molecule to 1 dalton of mass, is used.

[†]Prefixes used for powers of 10 are

10 ⁶	mega	M	10 ⁻³	milli	m
10 ³	kilo	k	10 ⁻⁶	micro	μ
10 ⁻¹	deci	d	10 ⁻⁹	nano	n
10 ⁻²	centi	c	10 ⁻¹²	pico	p
			10 ⁻¹⁵	femto	f

tern emerges. The molecular constituents of living matter do not reflect randomly the infinite possibilities for combining C, H, O, and N atoms. Instead, only a limited set of the many possibilities is found, and these collections share certain properties essential to the establishment and maintenance of the living state. The most prominent aspect of biomolecular organization is that macromolecular structures are constructed from simple molecules according to a hierarchy of increasing structural complexity. What properties do these biomolecules possess that make them so appropriate for the condition of life?

Metabolites and Macromolecules

The major precursors for the formation of biomolecules are water, carbon dioxide, and three inorganic nitrogen compounds—ammonium (NH_4^+), nitrate (NO_3^-), and dinitrogen (N_2). Metabolic processes assimilate and transform these inorganic precursors through ever more complex levels of biomolecular order (Figure 1.8). In the first step, precursors are converted to **metabolites**, simple organic compounds that are intermediates in cellular energy transformation and in the biosynthesis of various sets of **building blocks**: amino acids, sugars, nucleotides, fatty acids, and glycerol. By covalent linkage of these building blocks, the **macromolecules** are constructed: proteins, polysaccharides, polynucleotides (DNA and RNA), and lipids. (Strictly speaking, lipids contain relatively few building blocks and are therefore not really polymeric like other macromolecules; however, lipids are important contributors to higher levels of complexity.) Interactions among macromolecules lead to the next level of structural organization, **supramolecular complexes**. Here, various members of one or more of the classes of macromolecules come together to form specific assemblies serving important subcellular functions. Examples of these supramolecular assemblies are multifunctional enzyme complexes, ribosomes, chromosomes, and cytoskeletal elements. For example, a eukaryotic ribosome contains four different RNA molecules and at least 70 unique proteins. These supramolecular assemblies are an interesting contrast to their components because their structural integrity is maintained by noncovalent forces, not by covalent bonds. These noncovalent forces include hydrogen bonds, ionic attractions, van der Waals forces, and hydrophobic interactions between macromolecules. Such forces maintain these supramolecular assemblies in a highly ordered functional state. Although noncovalent forces are weak (less than 40 kJ/mol), they are numerous in these assemblies and thus can collectively maintain the essential architecture of the supramolecular complex under conditions of temperature, pH, and ionic strength that are consistent with cell life.

Organelles

The next higher rung in the hierarchical ladder is occupied by the **organelles**, entities of considerable dimensions compared to the cell itself. Organelles are found only in **eukaryotic cells**, that is, the cells of “higher” organisms (eukaryotic cells are described in Section 1.5). Several kinds, such as mitochondria and chloroplasts, evolved from bacteria that gained entry to the cytoplasm of early eukaryotic cells. Organelles share two attributes: they are cellular inclusions, usually membrane bounded, and are dedicated to important cellular tasks. Organelles include the nucleus, mitochondria, chloroplasts, endoplasmic reticulum, Golgi apparatus, and vacuoles as well as other relatively small cellular inclusions, such as peroxisomes, lysosomes, and chromoplasts. The **nucleus** is the repository of genetic information as contained within the linear sequences of nucleotides in the DNA of chromosomes. **Mitochondria** are the

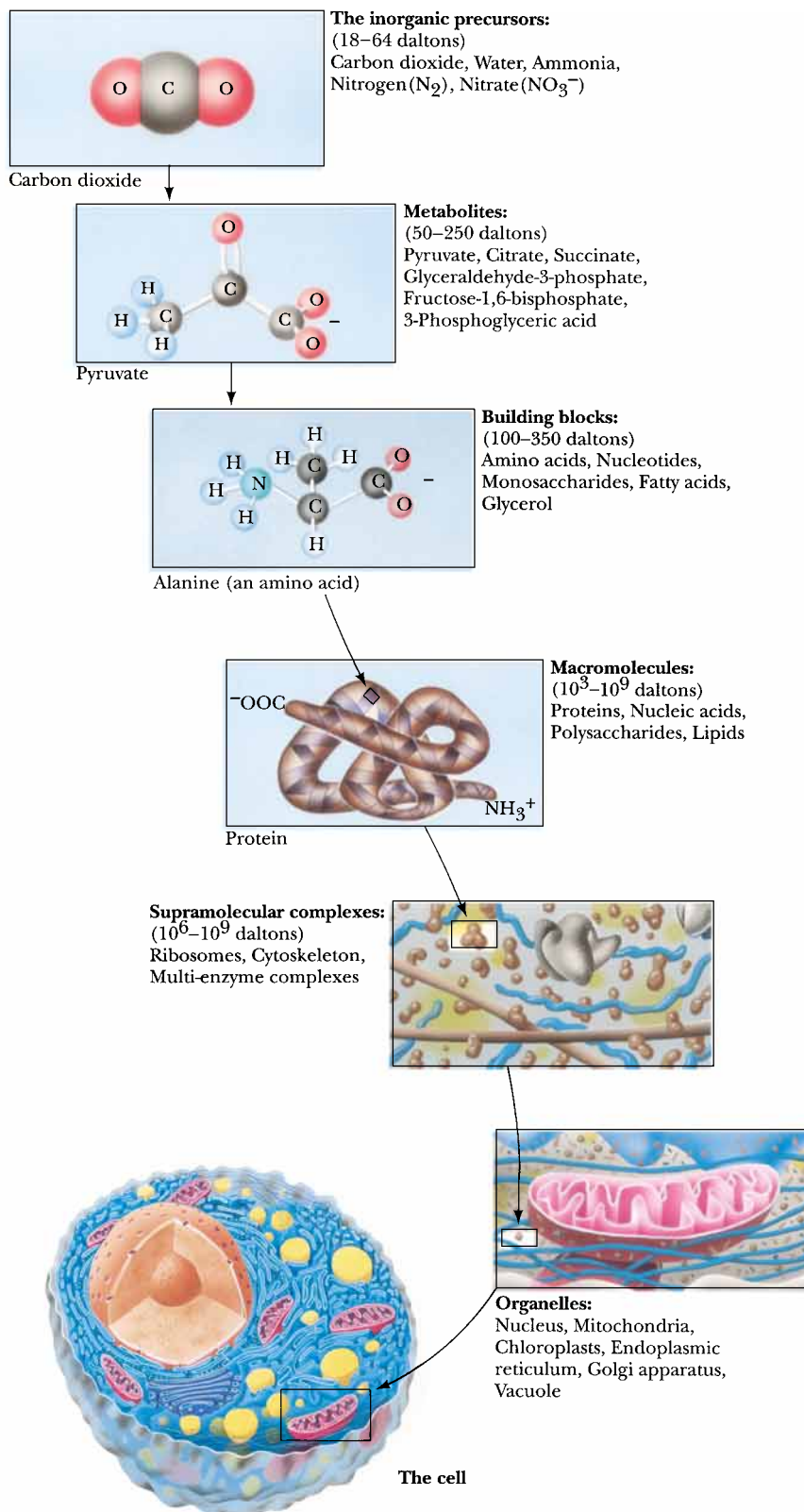


FIGURE 1.8 • Molecular organization in the cell is a hierarchy.

“power plants” of cells by virtue of their ability to carry out the energy-releasing aerobic metabolism of carbohydrates and fatty acids, capturing the energy in metabolically useful forms such as ATP. **Chloroplasts** endow cells with the ability to carry out photosynthesis. They are the biological agents for harvesting light energy and transforming it into metabolically useful chemical forms.

Membranes

Membranes define the boundaries of cells and organelles. As such, they are not easily classified as supramolecular assemblies or organelles, although they share the properties of both. Membranes resemble supramolecular complexes in their construction because they are complexes of proteins and lipids maintained by noncovalent forces. **Hydrophobic interactions** are particularly important in maintaining membrane structure. Hydrophobic interactions arise because water molecules prefer to interact with each other rather than with nonpolar substances. The presence of nonpolar molecules lessens the range of opportunities for water–water interaction by forcing the water molecules into ordered arrays around the nonpolar groups. Such ordering can be minimized if the individual nonpolar molecules redistribute from a dispersed state in the water into an aggregated organic phase surrounded by water. The spontaneous assembly of membranes in the aqueous environment where life arose and exists is the natural result of the hydrophobic (“water-fearing”) character of their lipids and proteins. Hydrophobic interactions are the creative means of membrane formation and the driving force that presumably established the boundary of the first cell. The membranes of organelles, such as nuclei, mitochondria, and chloroplasts, differ from one another, with each having a characteristic protein and lipid composition suited to the organelle’s function. Furthermore, the creation of discrete volumes or **compartments** within cells is not only an inevitable consequence of the presence of membranes but usually an essential condition for proper organellar function.

The Unit of Life Is the Cell

The **cell** is characterized as the unit of life, the smallest entity capable of displaying the attributes associated uniquely with the living state: growth, metabolism, stimulus response, and replication. In the previous discussions, we explicitly narrowed the infinity of chemical complexity potentially available to organic life, and we previewed an organizational arrangement, moving from simple to complex, that provides interesting insights into the functional and structural plan of the cell. Nevertheless, we find no obvious explanation within these features for the living characteristics of cells. Can we find other themes represented within biomolecules that are explicitly chemical yet anticipate or illuminate the living condition?

1.4 • Properties of Biomolecules Reflect Their Fitness to the Living Condition

If we consider what attributes of biomolecules render them so fit as components of growing, replicating systems, several biologically relevant themes of structure and organization emerge. Furthermore, as we study biochemistry, we will see that these themes serve as principles of biochemistry. Prominent among them is *the necessity for information and energy in the maintenance of the living state*. Some biomolecules must have the capacity to contain the information or “recipe” of life. Other biomolecules must have the capacity to translate this information so that the blueprint is transformed into the functional, organized structures essential to life. Interactions between these structures *are* the processes of life. An orderly mechanism for abstracting energy from the environment must also exist in order to obtain the energy needed to drive these processes. What properties of biomolecules endow them with the potential for such remarkable qualities?

Biological Macromolecules and Their Building Blocks
Have a “Sense” or Directionality

The macromolecules of cells are built of units—amino acids in proteins, nucleotides in nucleic acids, and carbohydrates in polysaccharides—that have **structural polarity**. That is, these molecules are not symmetrical, and so they can be thought of as having a “head” and a “tail.” Polymerization of these units to form macromolecules occurs by head-to-tail linear connections. Because of this, the polymer also has a head and a tail, and hence, the macromolecule has a “sense” or direction to its structure (Figure 1.9).

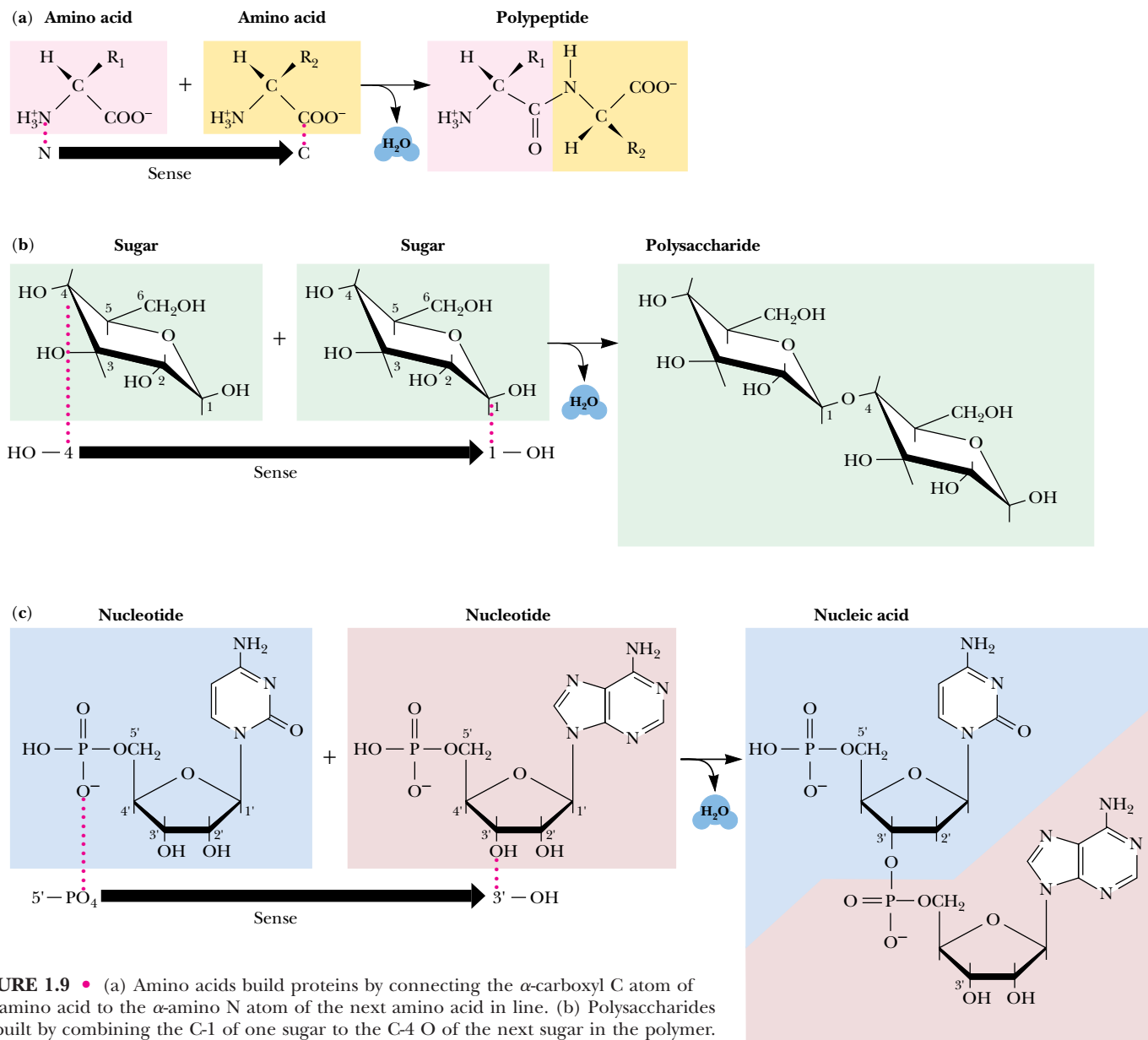
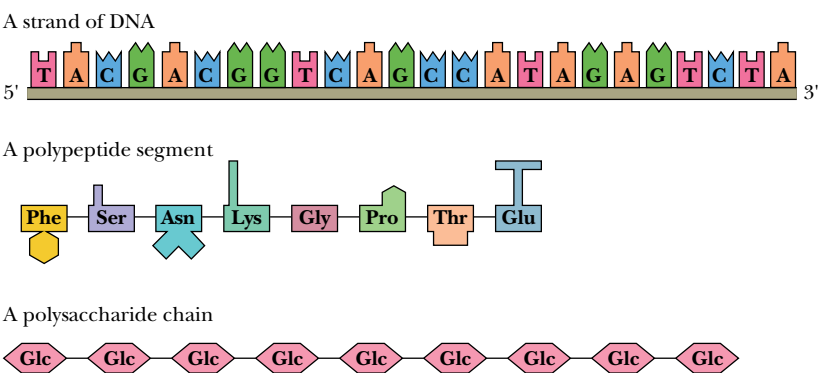


FIGURE 1.9 • (a) Amino acids build proteins by connecting the α -carboxyl C atom of one amino acid to the α -amino N atom of the next amino acid in line. (b) Polysaccharides are built by combining the C-1 of one sugar to the C-4 O of the next sugar in the polymer. (c) Nucleic acids are polymers of nucleotides linked by bonds between the 3'-OH of the ribose ring of one nucleotide to the 5'-PO₄ of its neighboring nucleotide. All three of these polymerization processes involve bond formations accompanied by the elimination of water (dehydration synthesis reactions).

FIGURE 1.10 • The sequence of monomeric units in a biological polymer has the potential to contain information if the diversity and order of the units are not overly simple or repetitive. Nucleic acids and proteins are information-rich molecules; polysaccharides are not.



Biological Macromolecules Are Informational

Because biological macromolecules have a sense to their structure, the sequential order of their component building blocks, when read along the length of the molecule, has the capacity to specify information in the same manner that the letters of the alphabet can form words when arranged in a linear sequence (Figure 1.10). Not all biological macromolecules are rich in information. Polysaccharides are often composed of the same sugar unit repeated over and over, as in cellulose or starch, which are homopolymers of many glucose units. On the other hand, proteins and polynucleotides are typically composed of building blocks arranged in no obvious repetitive way; that is, their sequences are unique, akin to the letters and punctuation that form this descriptive sentence. In these unique sequences lies meaning. To discern the meaning, however, requires some mechanism for recognition.

Biomolecules Have Characteristic Three-Dimensional Architecture

The structure of any molecule is a unique and specific aspect of its identity. Molecular structure reaches its pinnacle in the intricate complexity of biological macromolecules, particularly the proteins. Although proteins are linear sequences of covalently linked amino acids, the course of the protein chain can turn, fold, and coil in the three dimensions of space to establish a specific, highly ordered architecture that is an identifying characteristic of the given protein molecule (Figure 1.11).

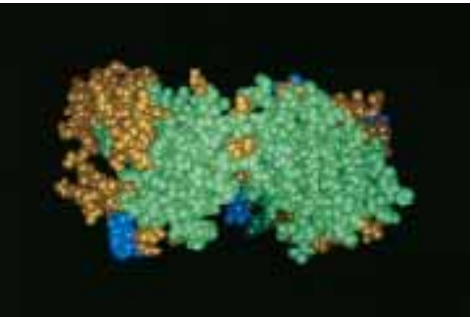


FIGURE 1.11 • Three-dimensional space-filling representation of part of a protein molecule, the antigen-binding domain of immunoglobulin G (IgG). Immunoglobulin G is a major type of circulating antibody. Each of the spheres represents an atom in the structure.

Weak Forces Maintain Biological Structure and Determine Biomolecular Interactions

Covalent bonds hold atoms together so that molecules are formed. In contrast, **weak chemical forces** or **noncovalent bonds**, (hydrogen bonds, van der Waals forces, ionic interactions, and hydrophobic interactions) are intramolecular or intermolecular attractions between atoms. None of these forces, which typically range from 4 to 30 kJ/mol, are strong enough to bind free atoms together (Table 1.3). The average kinetic energy of molecules at 25°C is 2.5 kJ/mol, so the energy of weak forces is only several times greater than the dissociating tendency due to thermal motion of molecules. Thus, these weak forces create interactions that are constantly forming and breaking at physiological temperature, unless by cumulative number they impart stability to the structures generated by their collective action. These weak forces merit further discussion because their attributes profoundly influence the nature of the biological structures they build.

Table 1.3

Weak Chemical Forces and Their Relative Strengths and Distances

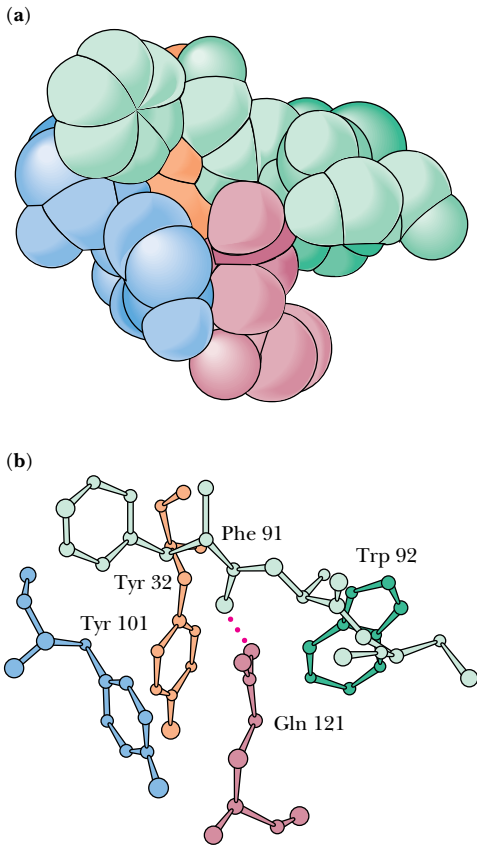
Force	Strength (kJ/mol)	Distance (nm)	Description
Van der Waals interactions	0.4–4.0	0.2	Strength depends on the relative size of the atoms or molecules and the distance between them. The size factor determines the area of contact between two molecules: The greater the area, the stronger the interaction.
Hydrogen bonds	12–30	0.3	Relative strength is proportional to the polarity of the H bond donor and H bond acceptor. More polar atoms form stronger H bonds.
Ionic interactions	20	0.25	Strength also depends on the relative polarity of the interacting charged species. Some ionic interactions are also H bonds: $\text{—NH}_3^+ \cdots \text{—OOC—}$
Hydrophobic interactions	<40	—	Force is a complex phenomenon determined by the degree to which the structure of water is disordered as discrete hydrophobic molecules or molecular regions coalesce.

Van der Waals Attractive Forces

Van der Waals forces are the result of induced electrical interactions between closely approaching atoms or molecules as their negatively-charged electron clouds fluctuate instantaneously in time. These fluctuations allow attractions to occur between the positively charged nuclei and the electrons of nearby atoms. Van der Waals interactions include dipole–dipole interactions, whose interaction energies decrease as $1/r^3$; dipole-induced dipole interactions, which fall off as $1/r^5$; and induced dipole-induced dipole interactions, often called **dispersion** or **London dispersion forces**, which diminish as $1/r^6$. Dispersion forces contribute to the attractive intermolecular forces between all molecules, even those without permanent dipoles, and are thus generally more important than dipole–dipole attractions. Van der Waals attractions operate only over a limited interatomic distance and are an effective bonding interaction at physiological temperatures only when a number of atoms in a molecule can interact with several atoms in a neighboring molecule. For this to occur, the atoms on interacting molecules must pack together neatly. That is, their molecular surfaces must possess a degree of structural complementarity (Figure 1.12).

At best, van der Waals interactions are weak and individually contribute 0.4 to 4.0 kJ/mol of stabilization energy. However, the sum of many such interactions within a macromolecule or between macromolecules can be substantial. For example, model studies of heats of sublimation show that each methylene group in a crystalline hydrocarbon accounts for 8 kJ, and each C—H group in a benzene crystal contributes 7 kJ of van der Waals energy per mole. Calculations indicate that the attractive van der Waals energy between the enzyme lysozyme and a sugar substrate that it binds is about 60 kJ/mol.

FIGURE 1.12 • Van der Waals packing is enhanced in molecules that are structurally complementary. Gln¹²¹ represents a surface protuberance on the protein lysozyme. This protuberance fits nicely within a pocket (formed by Tyr¹⁰¹, Tyr³², Phe⁹¹, and Trp⁹²) in the antigen-binding domain of an antibody raised against lysozyme. (See also Figure 1.16.) (a) A space-filling representation. (b) A ball-and-stick model. (From *Science* 233:751 (1986), figure 5.)



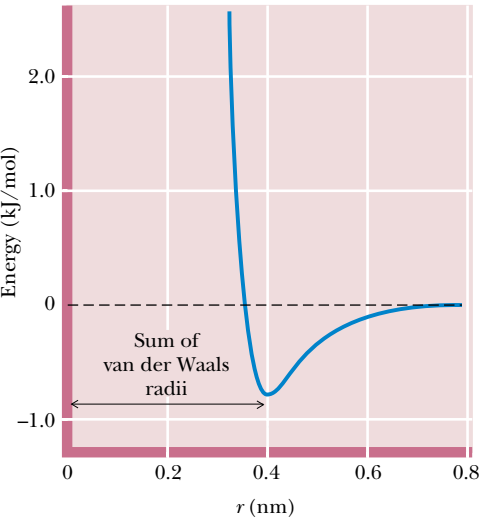


FIGURE 1.13 • The van der Waals interaction energy profile as a function of the distance, r , between the centers of two atoms. The energy was calculated using the empirical equation $U = B/r^{12} - A/r^6$. (Values for the parameters $B = 11.5 \times 10^{-6}$ kJnm¹²/mol and $A = 5.96 \times 10^{-3}$ kJnm⁶/mol for the interaction between two carbon atoms are from Levitt, M., 1974, *Journal of Molecular Biology* **82**:393–420.)

When two atoms approach each other so closely that their electron clouds interpenetrate, strong repulsion occurs. Such *repulsive* van der Waals forces follow an inverse 12th-power dependence on r ($1/r^{12}$), as shown in Figure 1.13. Between the repulsive and attractive domains lies a low point in the potential curve. This low point defines the distance known as the **van der Waals contact distance**, which is the interatomic distance that results if only van der Waals forces hold two atoms together. The limit of approach of two atoms is determined by the sum of their van der Waals radii (Table 1.4).

Hydrogen Bonds

Hydrogen bonds form between a hydrogen atom covalently bonded to an electronegative atom (such as oxygen or nitrogen) and a second electronegative atom that serves as the hydrogen bond acceptor. Several important biological examples are given in Figure 1.14. Hydrogen bonds, at a strength of 12 to 30 kJ/mol, are stronger than van der Waals forces and have an additional property: H bonds tend to be highly directional, forming straight bonds between donor, hydrogen, and acceptor atoms. Hydrogen bonds are also more specific than van der Waals interactions because they require the presence of complementary hydrogen donor and acceptor groups.

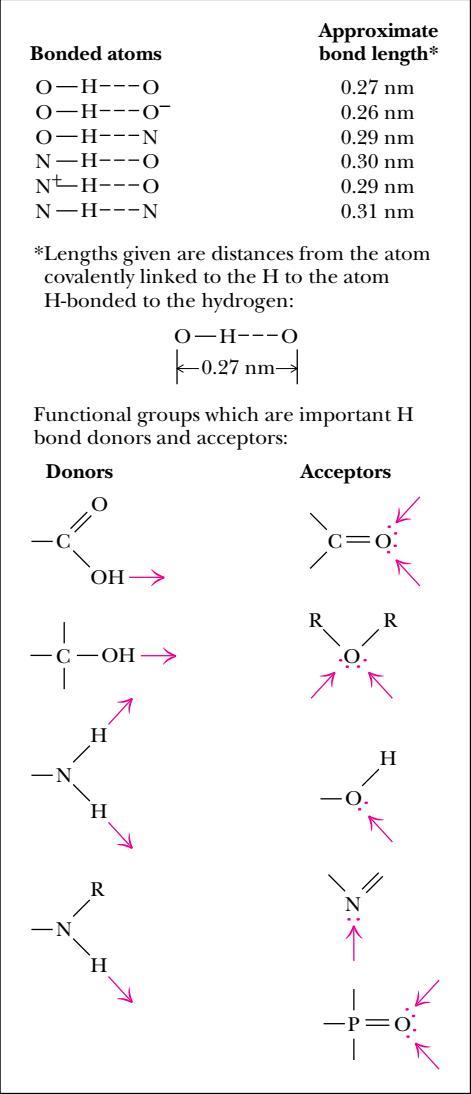
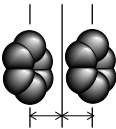


FIGURE 1.14 • Some of the biologically important H bonds and functional groups that serve as H bond donors and acceptors.

Table 1.4
Radii of the Common Atoms of Biomolecules

Atom	Van der Waals radius, nm	Covalent radius, nm	Atom represented to scale
H	0.1	0.037	
C	0.17	0.077	
N	0.15	0.070	
O	0.14	0.066	
P	0.19	0.096	
S	0.185	0.104	
Half-thickness of an aromatic ring	0.17	—	

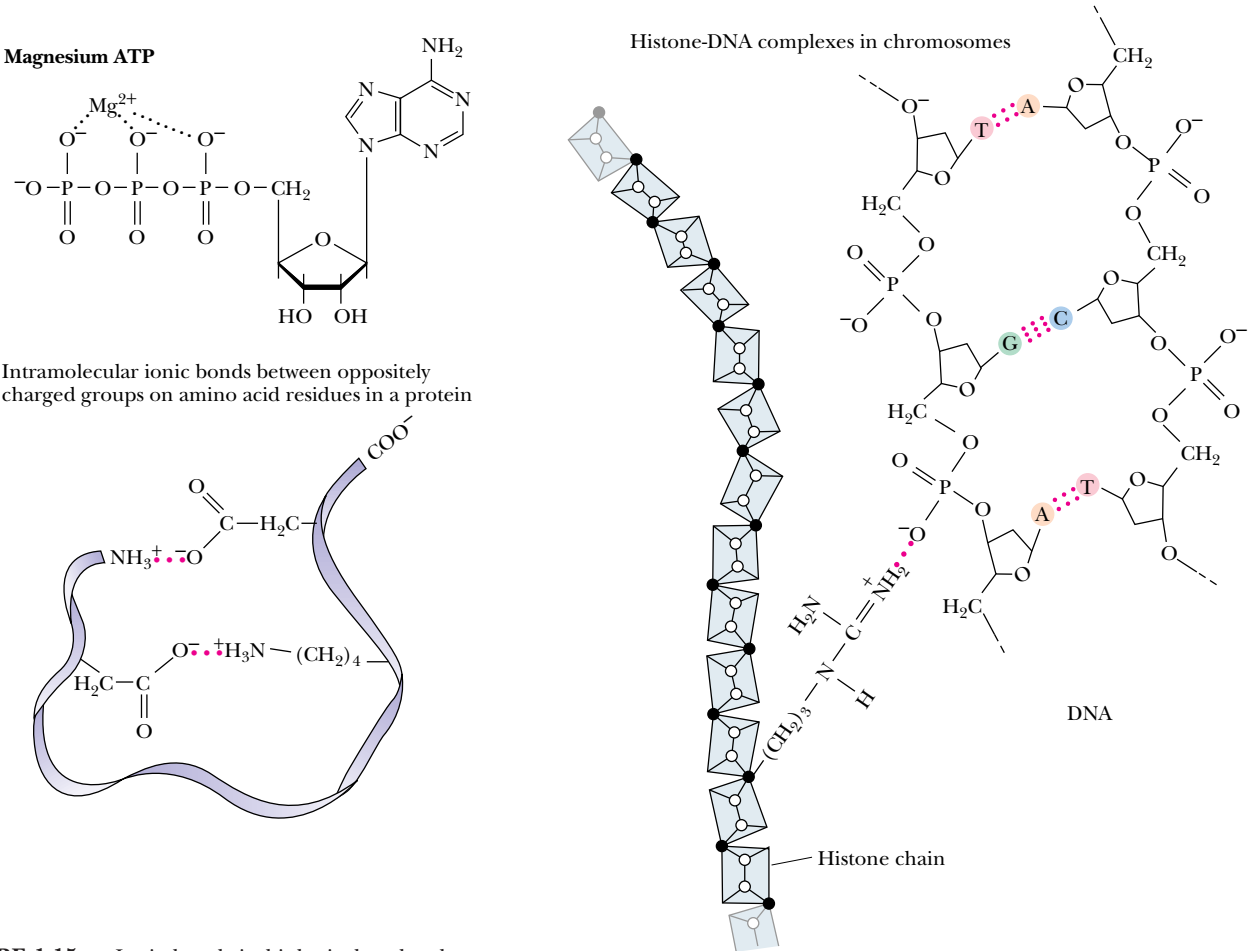


FIGURE 1.15 • Ionic bonds in biological molecules.

milieu • the environment or surroundings; from the French *mi* meaning “middle” and *lieu* meaning “place”

ligand • something that binds; a molecule that is bound to another molecule; from the Latin *ligare*, meaning “to bind”

Hydrophobic Interactions

Hydrophobic interactions are due to the strong tendency of water to exclude nonpolar groups or molecules (see Chapter 2). Hydrophobic interactions arise not so much because of any intrinsic affinity of nonpolar substances for one another (although van der Waals forces do promote the weak bonding of nonpolar substances), but because water molecules prefer the stronger interactions that they share with one another, compared to their interaction with nonpolar molecules. Hydrogen-bonding interactions between polar water molecules can be more varied and numerous if nonpolar molecules coalesce to form a distinct organic phase. This phase separation raises the entropy of water because fewer water molecules are arranged in orderly arrays around individual nonpolar molecules. It is these preferential interactions between water molecules that “exclude” hydrophobic substances from aqueous solution and drive the tendency of nonpolar molecules to cluster together. Thus, nonpolar regions of biological macromolecules are often buried in the molecule’s interior to exclude them from the aqueous milieu. The formation of oil droplets as hydrophobic nonpolar lipid molecules coalesce in the presence of water is an approximation of this phenomenon. These tendencies have important consequences in the creation and maintenance of the macromolecular structures and supramolecular assemblies of living cells.

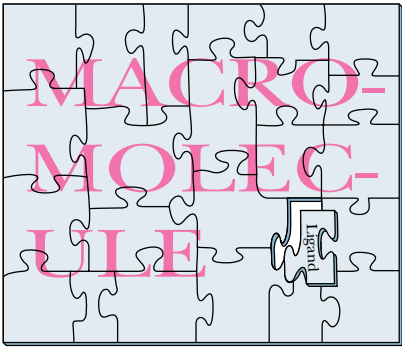
Structural Complementarity Determines Biomolecular Interactions

Structural complementarity is the means of recognition in biomolecular interactions. The complicated and highly organized patterns of life depend upon the ability of biomolecules to recognize and interact with one another in very specific ways. Such interactions are fundamental to metabolism, growth, replication, and other vital processes. The interaction of one molecule with another, a protein with a metabolite, for example, can be most precise if the structure of one is complementary to the structure of the other, as in two connecting pieces of a puzzle or, in the more popular analogy for macromolecules and their **ligands**, a lock and its key (Figure 1.16). This principle of **structural complementarity** is the very essence of biomolecular recognition. *Structural complementarity is the significant clue to understanding the functional properties of biological systems.* Biological systems from the macromolecular level to the cellular level operate via specific molecular recognition mechanisms based on structural complementarity: a protein recognizes its specific metabolite, a strand of DNA recognizes its complementary strand, sperm recognize an egg. All these interactions involve structural complementarity between molecules.

Biomolecular Recognition Is Mediated by Weak Chemical Forces

The biomolecular recognition events that occur through structural complementarity are mediated by the weak chemical forces previously discussed. It is important to realize that, because these interactions are sufficiently weak, they are readily reversible. Consequently, biomolecular interactions tend to be transient; rigid, static lattices of biomolecules that might paralyze cellular activities are not formed. Instead, a dynamic interplay occurs between metabolites and macromolecules, hormones and receptors, and all the other participants instrumental to life processes. This interplay is initiated upon specific recognition between complementary molecules and ultimately culminates in unique physiological activities. *Biological function is achieved through mechanisms based on structural complementarity and weak chemical interactions.*

Puzzle



Lock and key

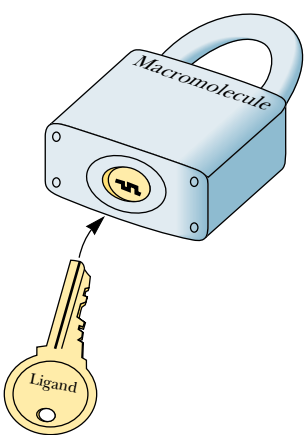
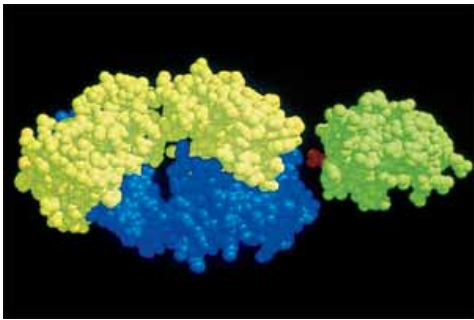
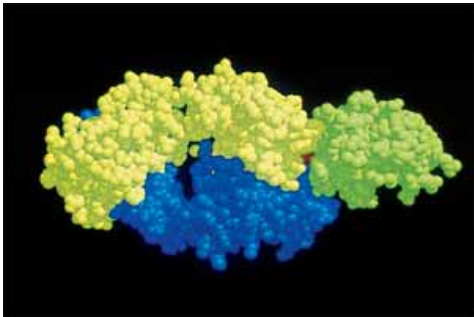


FIGURE 1.16 • Structural complementarity: the pieces of a puzzle, the lock and its key, a biological macromolecule and its ligand—an antigen–antibody complex. (a) The antigen on the right (green) is a small protein, lysozyme, from hen egg white. The part of the antibody molecule (IgG) shown on the left in blue and yellow includes the antigen-binding domain. (b) This domain has a pocket that is structurally complementary to a surface protuberance (Gln¹²¹, shown in red between antigen and antigen-binding domain) on the antigen. (See also Figure 1.12.) (photos, courtesy of Professor Simon E. V. Phillips)

(a)



(b)

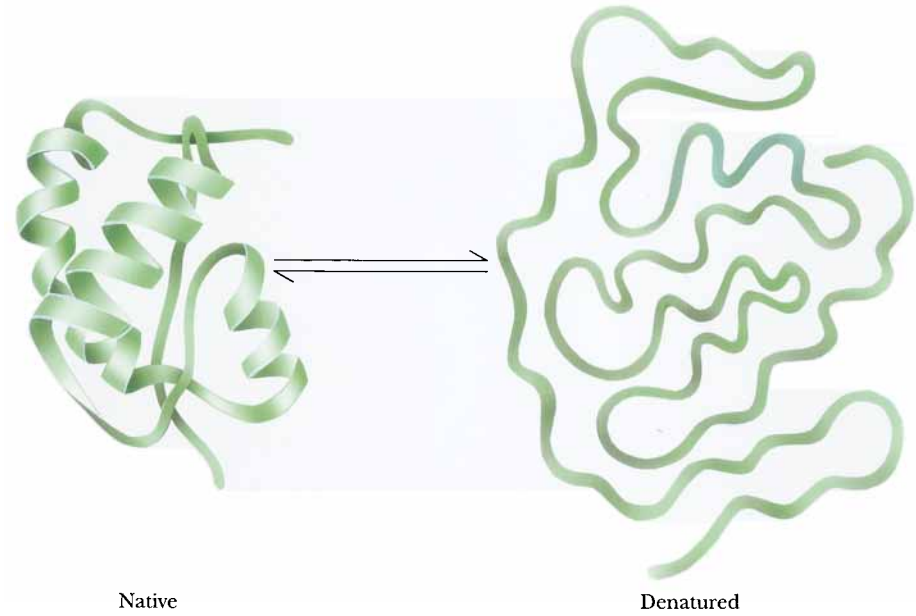


This principle of structural complementarity extends to higher interactions essential to the establishment of the living condition. For example, the formation of supramolecular complexes occurs because of recognition and interaction between their various macromolecular components, as governed by the weak forces formed between them. If a sufficient number of weak bonds can be formed, as in macromolecules complementary in structure to one another, larger structures assemble spontaneously. The tendency for nonpolar molecules and parts of molecules to come together through hydrophobic interactions also promotes the formation of supramolecular assemblies. Very complex subcellular structures are actually spontaneously formed in an assembly process that is driven by weak forces accumulated through structural complementarity.

Weak Forces Restrict Organisms to a Narrow Range of Environmental Conditions

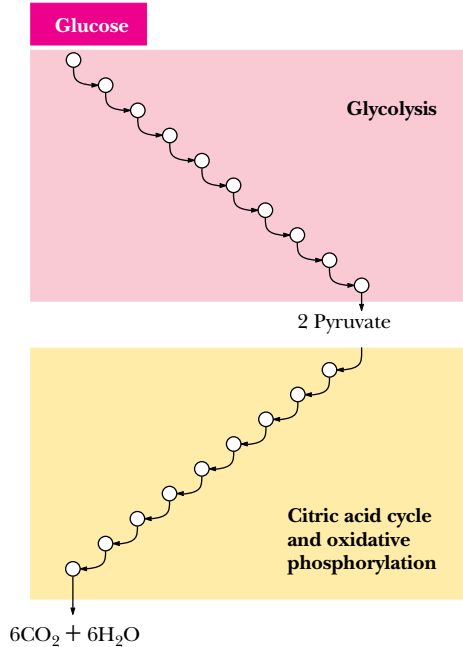
The central role of weak forces in biomolecular interactions restricts living systems to a narrow range of physical conditions. Biological macromolecules are functionally active only within a narrow range of environmental conditions, such as temperature, ionic strength, and relative acidity. Extremes of these conditions disrupt the weak forces essential to maintaining the intricate structure of macromolecules. The loss of structural order in these complex macromolecules, so-called **denaturation**, is accompanied by loss of function (Figure 1.17). As a consequence, cells cannot tolerate reactions in which large amounts of energy are released. Nor can they generate a large energy burst to drive energy-requiring processes. Instead, such transformations take place via sequential series of chemical reactions whose overall effect achieves dramatic energy changes, even though any given reaction in the series proceeds with only mod-

FIGURE 1.17 • Denaturation and renaturation of the intricate structure of a protein.



The combustion of glucose: $C_6H_{12}O_6 + 6O_2 \longrightarrow 6CO_2 + 6H_2O + 2,870 \text{ kJ energy}$

(a) In an aerobic cell



(b) In a bomb calorimeter

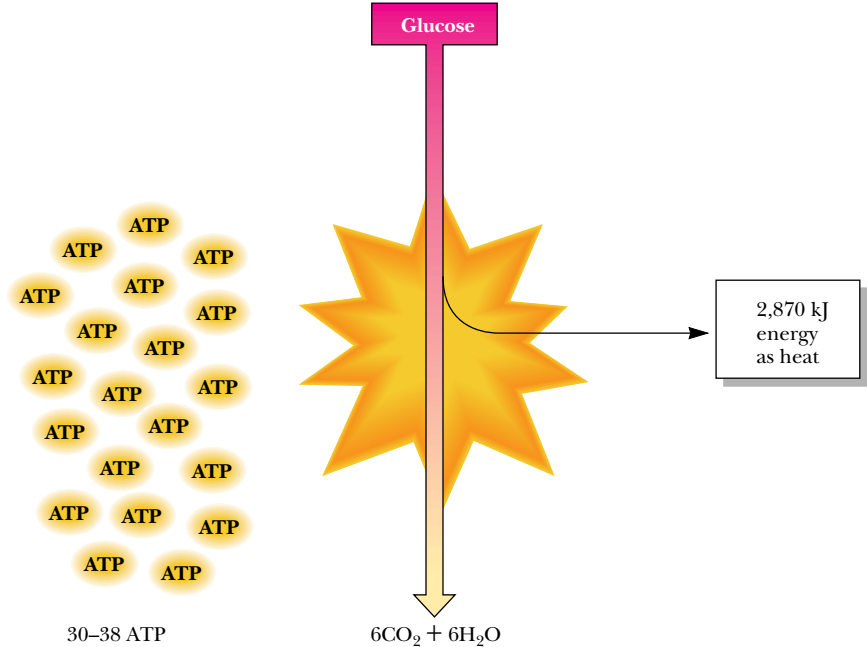


FIGURE 1.18 • Metabolism is the organized release or capture of small amounts of energy in processes whose overall change in energy is large. (a) For example, the combustion of glucose by cells is a major pathway of energy production, with the energy captured appearing as 30 to 38 equivalents of ATP, the principal energy-rich chemical of cells. The ten reactions of glycolysis, the nine reactions of the citric acid cycle, and the successive linked reactions of oxidative phosphorylation release the energy of glucose in a stepwise fashion and the small “packets” of energy appear in ATP. (b) Combustion of glucose in a bomb calorimeter results in an uncontrolled, explosive release of energy in its least useful form, heat.

est input or release of energy (Figure 1.18). These sequences of reactions are organized to provide for the release of useful energy to the cell from the breakdown of food or to take such energy and use it to drive the synthesis of biomolecules essential to the living state. Collectively, these reaction sequences constitute cellular **metabolism**—the ordered reaction pathways by which cellular chemistry proceeds and biological energy transformations are accomplished.

Enzymes

The sensitivity of cellular constituents to environmental extremes places another constraint on the reactions of metabolism. The rate at which cellular reactions proceed is a very important factor in maintenance of the living state. However, the common ways chemists accelerate reactions are not available to cells; the temperature cannot be raised, acid or base cannot be added, the pressure cannot be elevated, and concentrations cannot be dramatically increased. Instead, biomolecular catalysts mediate cellular reactions. These catalysts, called **enzymes**, accelerate the reaction rates many orders of magnitude and, by selecting the substances undergoing reaction, determine the specific reaction taking place. Virtually every metabolic reaction is served by an enzyme whose sole biological purpose is to catalyze its specific reaction (Figure 1.19).

Metabolic Regulation Is Achieved by Controlling the Activity of Enzymes

Thousands of reactions mediated by an equal number of enzymes are occurring at any given instant within the cell. Metabolism has many branch points, cycles, and interconnections, as a glance at a metabolic pathway map reveals

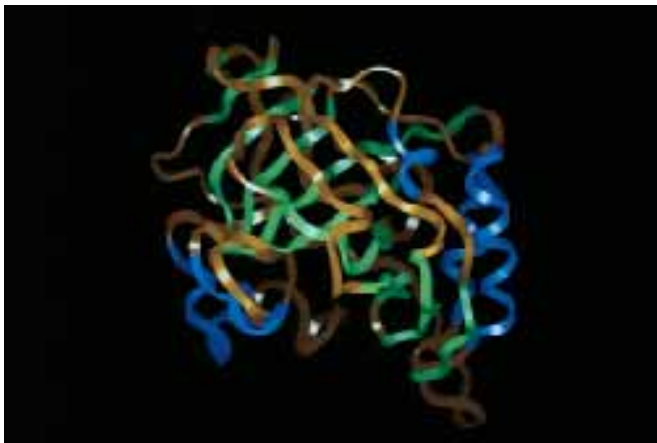
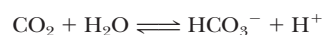
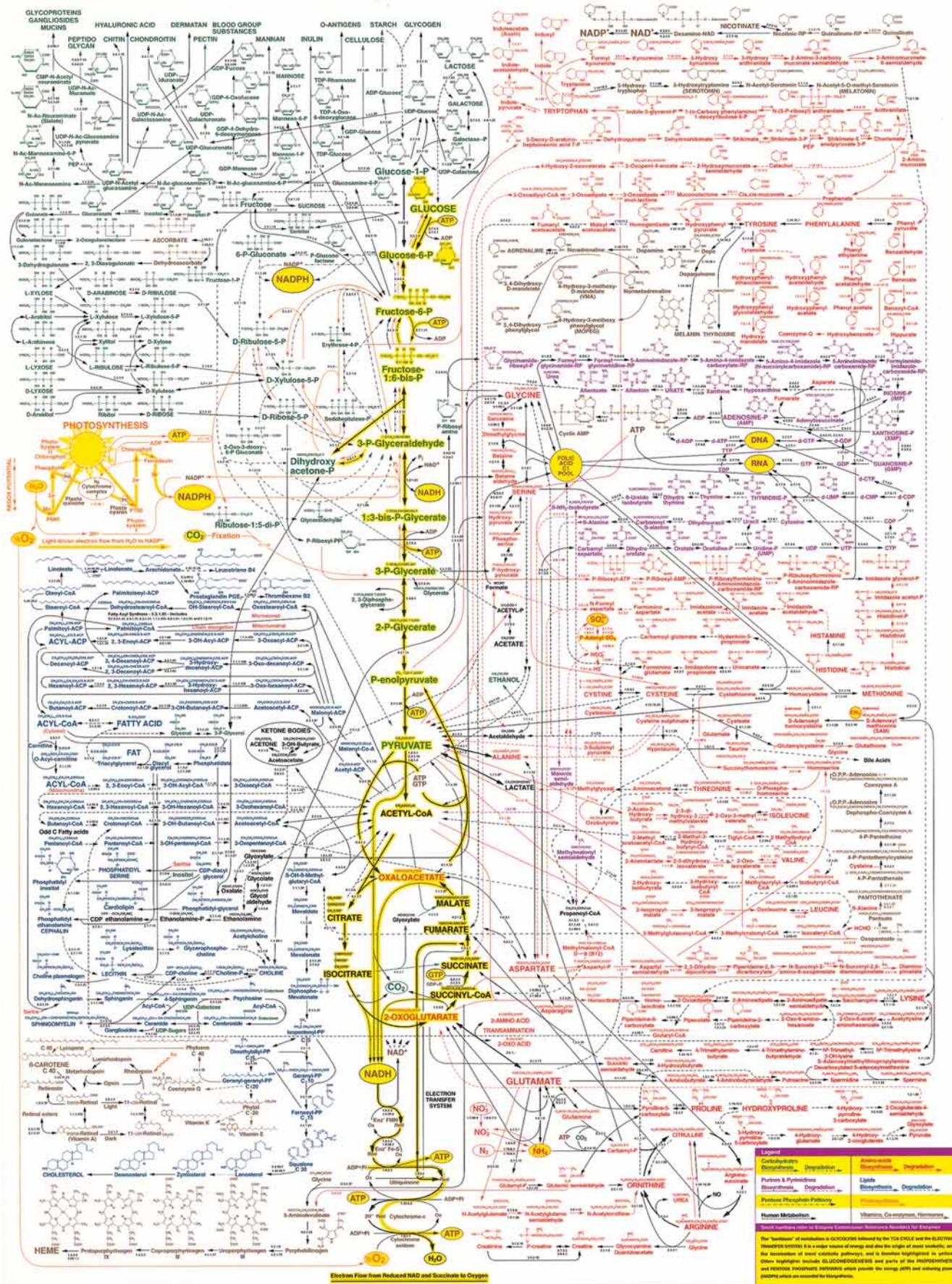


FIGURE 1.19 • Carbonic anhydrase, a representative enzyme, and the reaction that it catalyzes. Dissolved carbon dioxide is slowly hydrated by water to form bicarbonate ion and H^+ :



At 20°C, the rate constant for this uncatalyzed reaction, k_{uncat} , is 0.03/sec. In the presence of the enzyme carbonic anhydrase, the rate constant for this reaction, k_{cat} , is $10^6/\text{sec}$. Thus carbonic anhydrase accelerates the rate of this reaction 3.3×10^7 times. Carbonic anhydrase is a 29-kD protein.



◀ **FIGURE 1.20** • Reproduction of a metabolic map. (Courtesy of D. E. Nicholson, University of Leeds and Sigma Chemical Co., St. Louis, MO.)

(Figure 1.20). All of these reactions, many of which are at apparent cross-purposes in the cell, must be fine-tuned and integrated so that metabolism and life proceed harmoniously. The need for metabolic regulation is obvious. This metabolic regulation is achieved through controls on enzyme activity so that the rates of cellular reactions are appropriate to cellular requirements.

Despite the organized pattern of metabolism and the thousands of enzymes required, cellular reactions nevertheless conform to the same thermodynamic principles that govern any chemical reaction. Enzymes have no influence over energy changes (the thermodynamic component) in their reactions. Enzymes only influence reaction rates. Thus, cells are systems that take in food, release waste, and carry out complex degradative and biosynthetic reactions essential to their survival while operating under conditions of essentially constant temperature and pressure and maintaining a constant internal environment (**homeostasis**) with no outwardly apparent changes. *Cells are open thermodynamic systems exchanging matter and energy with their environment and functioning as highly regulated isothermal chemical engines.*

1.5 • Organization and Structure of Cells

All living cells fall into one of two broad categories—**prokaryotic** and **eukaryotic**. The distinction is based on whether or not the cell has a nucleus. Prokaryotes are single-celled organisms that lack nuclei and other organelles; the word is derived from *pro* meaning “prior to” and *karyote* meaning “nucleus.” In conventional biological classification schemes, prokaryotes are grouped together as members of the kingdom Monera, represented by bacteria and cyanobacteria (formerly called blue-green algae). The other four living kingdoms are all eukaryotes—the single-celled Protists, such as amoebae, and all multicellular life forms, including the Fungi, Plant, and Animal kingdoms. Eukaryotic cells have true nuclei and other organelles such as mitochondria, with the prefix *eu* meaning “true.”

Early Evolution of Cells

Until recently, most biologists accepted the idea that eukaryotes evolved from the simpler prokaryotes in some linear progression from simple to complex over the course of geological time. Contemporary evidence favors the view that present-day organisms are better grouped into three classes or lineages: eukaryotes and two prokaryotic groups, the **eubacteria** and the **archaea** (formerly designated as **archaeobacteria**). All are believed to have evolved approximately 3.5 billion years ago from a common ancestral form called the **progenote**. It is now understood that eukaryotic cells are, in reality, composite cells derived from various prokaryotic contributions. Thus, the dichotomy between prokaryotic cells and eukaryotic cells, although convenient, is an artificial distinction.

Despite the great diversity in form and function, cells and organisms share a common biochemistry. This commonality, although long established, has received further validation through **whole genome sequencing**, or the determination of the complete nucleotide sequence within the DNA of an organism. For example, the recently sequenced genome of the archaeon *Methanococcus*

jannaschii shows 44% similarity to known genes in eubacteria and eukaryotes, yet 56% of its genes are new to science. Whole genome sequencing is revolutionizing biochemistry as the protein-coding sequences of newly revealed genes outpace our understanding of what the proteins are and what they do.

Structural Organization of Prokaryotic Cells

Among prokaryotes (the simplest cells), most known species are eubacteria and they form a widely spread group. Certain of them are pathogenic to humans. The archaea are remarkable because they can be found in unusual environments where other cells cannot survive. Archaea include the **thermoacidophiles** (heat- and acid-loving bacteria) of hot springs, the **halophiles** (salt-loving bacteria) of salt lakes and ponds, and the **methanogens** (bacteria that generate methane from CO₂ and H₂). Prokaryotes are typically very small, on the order of several microns in length, and are usually surrounded by a rigid **cell wall** that protects the cell and gives it its shape. The characteristic structural organization of a prokaryotic cell is depicted in Figure 1.21.

Prokaryotic cells have only a single membrane, the **plasma membrane** or **cell membrane**. Because they have no other membranes, prokaryotic cells contain no nucleus or organelles. Nevertheless, they possess a distinct **nuclear area** where a single circular chromosome is localized, and some have an internal membranous structure called a **mesosome** that is derived from and continuous with the cell membrane. Reactions of cellular respiration are localized on these membranes. In photosynthetic prokaryotes such as the **cyanobacteria**,

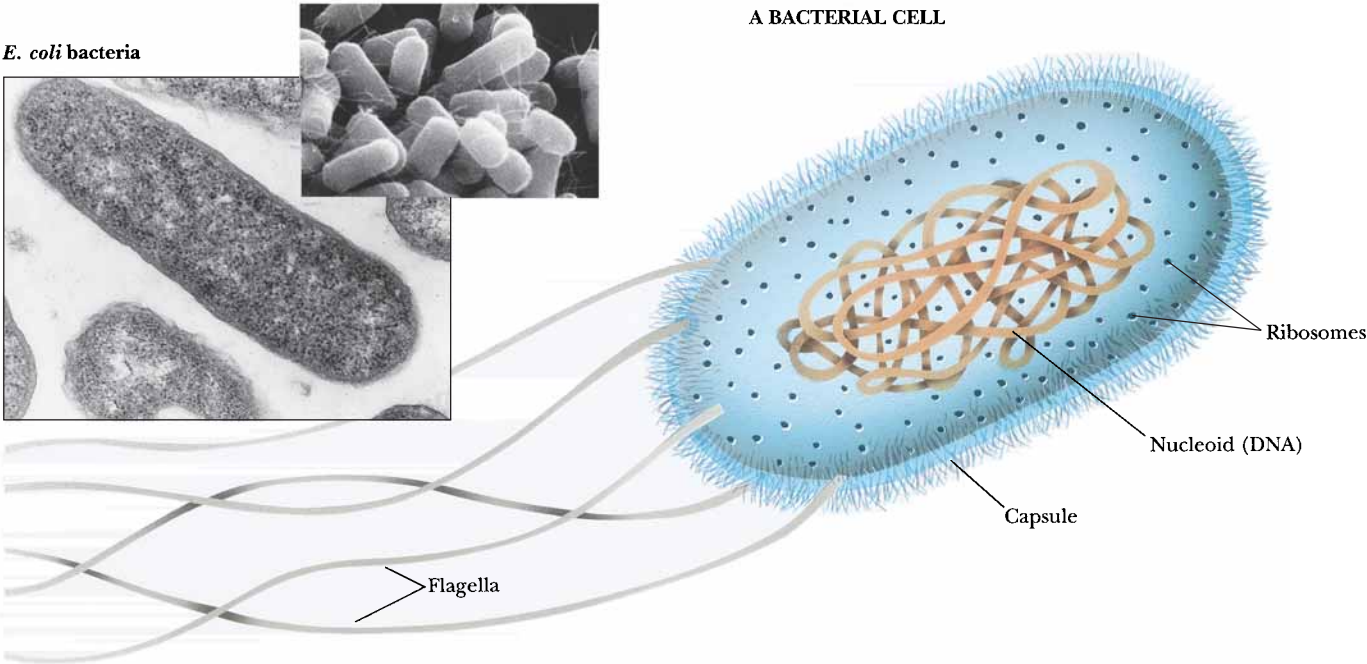


FIGURE 1.21 • This bacterium is *Escherichia coli*, a member of the coliform group of bacteria that colonize the intestinal tract of humans. *E. coli* organisms have rather simple nutritional requirements. They grow and multiply quite well if provided with a simple carbohydrate source of energy (such as glucose), ammonium ions as a source of nitrogen, and a few mineral salts. The simple nutrition of this “lower” organism means that its biosynthetic capacities must be quite advanced. When growing at 37°C on a rich organic medium, *E. coli* cells divide every 20 minutes. Subcellular features include the cell wall, plasma membrane, nuclear region, ribosomes, storage granules, and cytosol (Table 1.5). (photo, Martin Rotker/Phototake, Inc.; inset photo, David M. Phillips/The Population Council/Science Source/Photo Researchers, Inc.)

Table 1.5

Major Features of Prokaryotic Cells

Structure	Molecular Composition	Function
Cell wall	Peptidoglycan: a rigid framework of polysaccharide cross-linked by short peptide chains. Some bacteria possess a lipopolysaccharide- and protein-rich outer membrane.	Mechanical support, shape, and protection against swelling in hypotonic media. The cell wall is a porous nonselective barrier that allows most small molecules to pass.
Cell membrane	The cell membrane is composed of about 45% lipid and 55% protein. The lipids form a bilayer that is a continuous nonpolar hydrophobic phase in which the proteins are embedded.	The cell membrane is a highly selective permeability barrier that controls the entry of most substances into the cell. Important enzymes in the generation of cellular energy are located in the membrane.
Nuclear area or nucleoid	The genetic material is a single tightly coiled DNA molecule 2 nm in diameter but over 1 mm in length (molecular mass of <i>E. coli</i> DNA is 3×10^9 daltons; 4.64×10^6 nucleotide pairs).	DNA is the blueprint of the cell, the repository of the cell's genetic information. During cell division, each strand of the double-stranded DNA molecule is replicated to yield two double-helical daughter molecules. Messenger RNA (mRNA) is transcribed from DNA to direct the synthesis of cellular proteins.
Ribosomes	Bacterial cells contain about 15,000 ribosomes. Each is composed of a small (30S) subunit and a large (50S) subunit. The mass of a single ribosome is 2.3×10^6 daltons. It consists of 65% RNA and 35% protein.	Ribosomes are the sites of protein synthesis. The mRNA binds to ribosomes, and the mRNA nucleotide sequence specifies the protein that is synthesized.
Storage granules	Bacteria contain granules that represent storage forms of polymerized metabolites such as sugars or β -hydroxybutyric acid.	When needed as metabolic fuel, the monomeric units of the polymer are liberated and degraded by energy-yielding pathways in the cell.
Cytosol	Despite its amorphous appearance, the cytosol is now recognized to be an organized gelatinous compartment that is 20% protein by weight and rich in the organic molecules that are the intermediates in metabolism.	The cytosol is the site of intermediary metabolism, the interconnecting sets of chemical reactions by which cells generate energy and form the precursors necessary for biosynthesis of macromolecules essential to cell growth and function.

flat, sheetlike membranous structures called **lamellae** are formed from cell membrane infoldings. These lamellae are the sites of photosynthetic activity, but in prokaryotes, they are not contained within **plastids**, the organelles of photosynthesis found in higher plant cells. Prokaryotic cells also lack a cytoskeleton; the cell wall maintains their structure. Some bacteria have **flagella**, single, long filaments used for motility. Prokaryotes largely reproduce by asexual division, although sexual exchanges can occur. Table 1.5 lists the major features of prokaryotic cells.

Structural Organization of Eukaryotic Cells

In comparison to prokaryotic cells, eukaryotic cells are much greater in size, typically having cell volumes 10^3 to 10^4 times larger. Also, they are much more complex. These two features require that eukaryotic cells partition their diverse

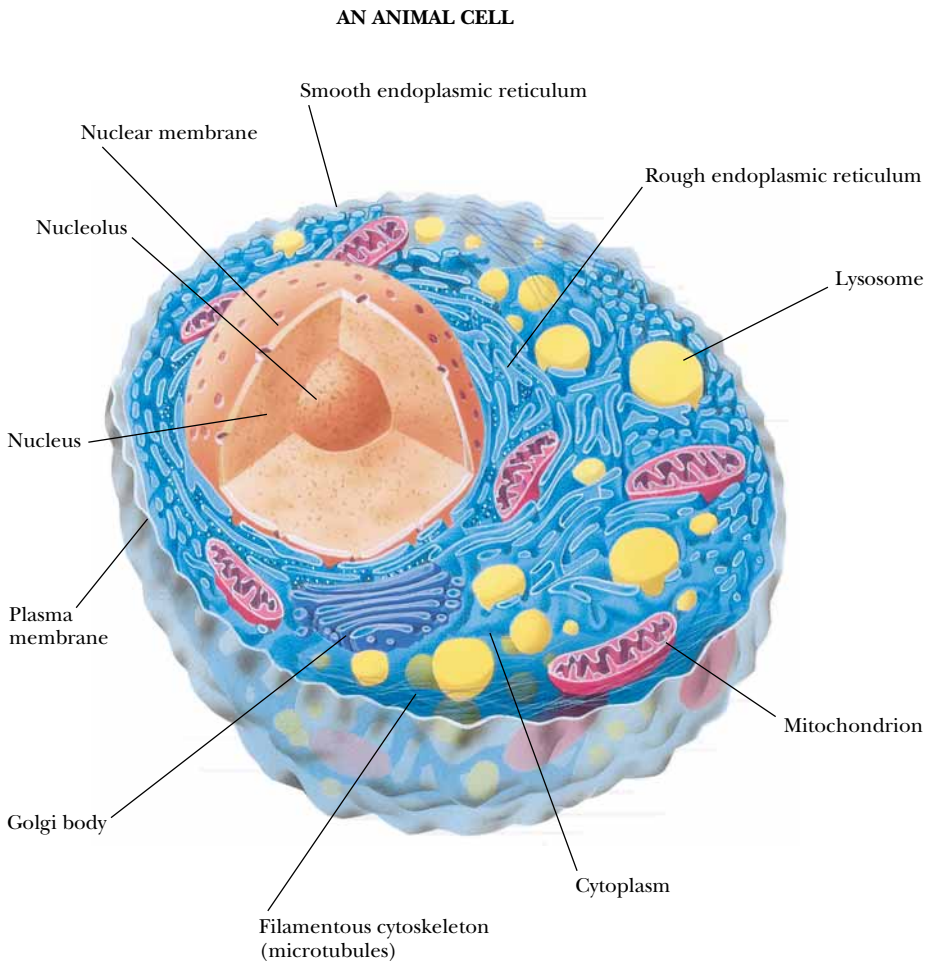
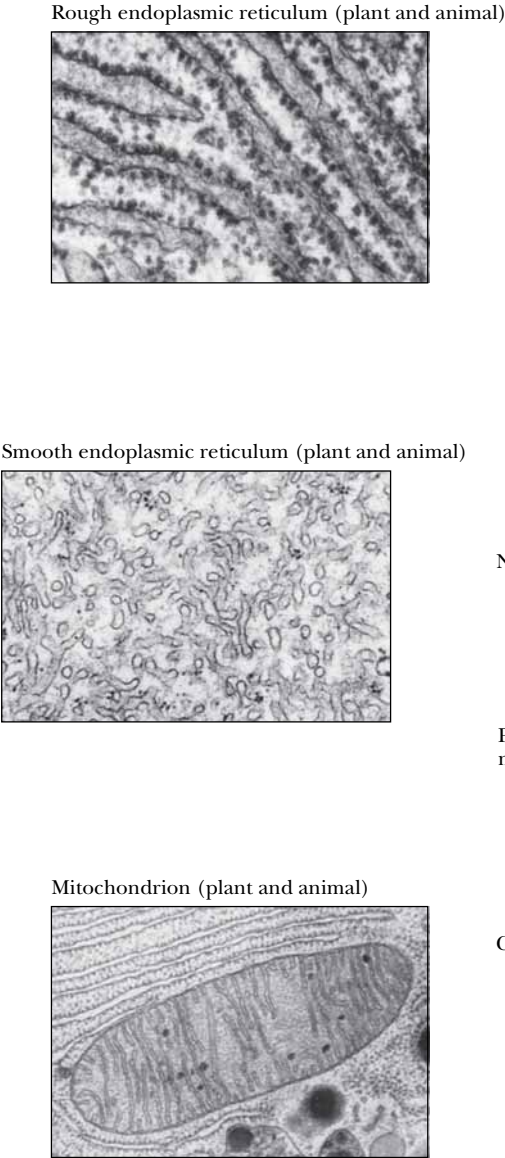


FIGURE 1.22 • This figure diagrams a rat liver cell, a typical higher animal cell in which the characteristic features of animal cells are evident, such as a nucleus, nucleolus, mitochondria, Golgi bodies, lysosomes, and endoplasmic reticulum (ER). Microtubules and the network of filaments constituting the cytoskeleton are also depicted. (photos, top, Dwight R. Kuhn/Visuals Unlimited; middle, D.W. Fawcett/Visuals Unlimited; bottom, Keith Porter/Photo Researchers, Inc.)

metabolic processes into organized compartments, with each compartment dedicated to a particular function. A system of internal membranes accomplishes this partitioning. A typical animal cell is shown in Figure 1.22; a typical plant cell in Figure 1.23. Tables 1.6 and 1.7 list the major features of a typical animal cell and a higher plant cell, respectively.

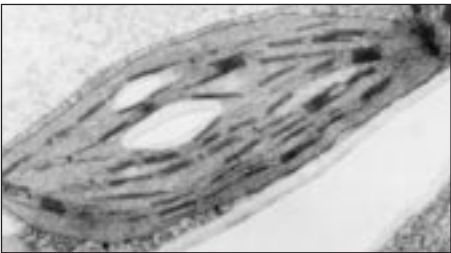
Eukaryotic cells possess a discrete, membrane-bounded **nucleus**, the repository of the cell’s genetic material, which is distributed among a few or many **chromosomes**. During cell division, equivalent copies of this genetic material must be passed to both daughter cells through duplication and orderly partitioning of the chromosomes by the process known as **mitosis**. Like prokaryotic

Table 1.6

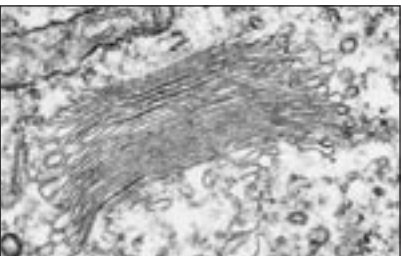
Major Features of a Typical Animal Cell

Structure	Molecular Composition	Function
Extracellular matrix	The surfaces of animal cells are covered with a flexible and sticky layer of complex carbohydrates, proteins, and lipids.	This complex coating is cell-specific, serves in cell – cell recognition and communication, creates cell adhesion, and provides a protective outer layer.
Cell membrane (plasma membrane)	Roughly 50:50 lipid:protein as a 5-nm-thick continuous sheet of lipid bilayer in which a variety of proteins are embedded.	The plasma membrane is a selectively permeable outer boundary of the cell, containing specific systems—pumps, channels, transporters—for the exchange of nutrients and other materials with the environment. Important enzymes are also located here.
Nucleus	The nucleus is separated from the cytosol by a double membrane, the nuclear envelope. The DNA is complexed with basic proteins (histones) to form chromatin fibers, the material from which chromosomes are made. A distinct RNA-rich region, the nucleolus, is the site of ribosome assembly.	The nucleus is the repository of genetic information encoded in DNA and organized into chromosomes. During mitosis, the chromosomes are replicated and transmitted to the daughter cells. The genetic information of DNA is transcribed into RNA in the nucleus and passes into the cytosol where it is translated into protein by ribosomes.
Mitochondria	Mitochondria are organelles surrounded by two membranes that differ markedly in their protein and lipid composition. The inner membrane and its interior volume, the matrix, contain many important enzymes of energy metabolism. Mitochondria are about the size of bacteria, ≈1 μm. Cells contain hundreds of mitochondria, which collectively occupy about one-fifth of the cell volume.	Mitochondria are the power plants of eukaryotic cells where carbohydrates, fats, and amino acids are oxidized to CO ₂ and H ₂ O. The energy released is trapped as high-energy phosphate bonds in ATP.
Golgi apparatus	A system of flattened membrane-bounded vesicles often stacked into a complex. Numerous small vesicles are found peripheral to the Golgi and contain secretory material packaged by the Golgi.	Involved in the packaging and processing of macromolecules for secretion and for delivery to other cellular compartments.
Endoplasmic reticulum (ER) and ribosomes	Flattened sacs, tubes, and sheets of internal membrane extending throughout the cytoplasm of the cell and enclosing a large interconnecting series of volumes called <i>cisternae</i> . The ER membrane is continuous with the outer membrane of the nuclear envelope. Portions of the sheetlike areas of the ER are studded with ribosomes, giving rise to <i>rough ER</i> . Eukaryotic ribosomes are larger than prokaryotic ribosomes.	The endoplasmic reticulum is a labyrinthine organelle where both membrane proteins and lipids are synthesized. Proteins made by the ribosomes of the rough ER pass through the outer ER membrane into the cisternae and can be transported via the Golgi to the periphery of the cell. Other ribosomes unassociated with the ER carry on protein synthesis in the cytosol.
Lysosomes	Lysosomes are vesicles 0.2–0.5 μm in diameter, bounded by a single membrane. They contain hydrolytic enzymes such as proteases and nucleases which, if set free, could degrade essential cell constituents. They are formed by budding from the Golgi apparatus.	Lysosomes function in intracellular digestion of materials entering the cell via phagocytosis or pinocytosis. They also function in the controlled degradation of cellular components.
Peroxisomes	Like lysosomes, peroxisomes are 0.2–0.5 μm single-membrane–bounded vesicles. They contain a variety of oxidative enzymes that use molecular oxygen and generate peroxides. They are formed by budding from the smooth ER.	Peroxisomes act to oxidize certain nutrients, such as amino acids. In doing so, they form potentially toxic hydrogen peroxide, H ₂ O ₂ , and then decompose it to H ₂ O and O ₂ by way of the peroxide-cleaving enzyme catalase.
Cytoskeleton	The cytoskeleton is composed of a network of protein filaments: actin filaments (or microfilaments), 7 nm in diameter; intermediate filaments, 8–10 nm; and microtubules, 25 nm. These filaments interact in establishing the structure and functions of the cytoskeleton. This interacting network of protein filaments gives structure and organization to the cytoplasm.	The cytoskeleton determines the shape of the cell and gives it its ability to move. It also mediates the internal movements that occur in the cytoplasm, such as the migration of organelles and mitotic movements of chromosomes. The propulsion instruments of cells—cilia and flagella—are constructed of microtubules.

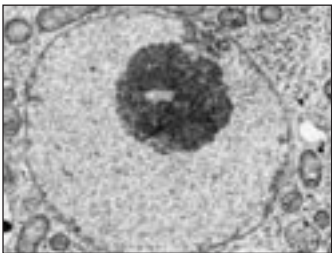
Chloroplast (plant cell only)



Golgi body (plant and animal)



Nucleus (plant and animal)



A PLANT CELL

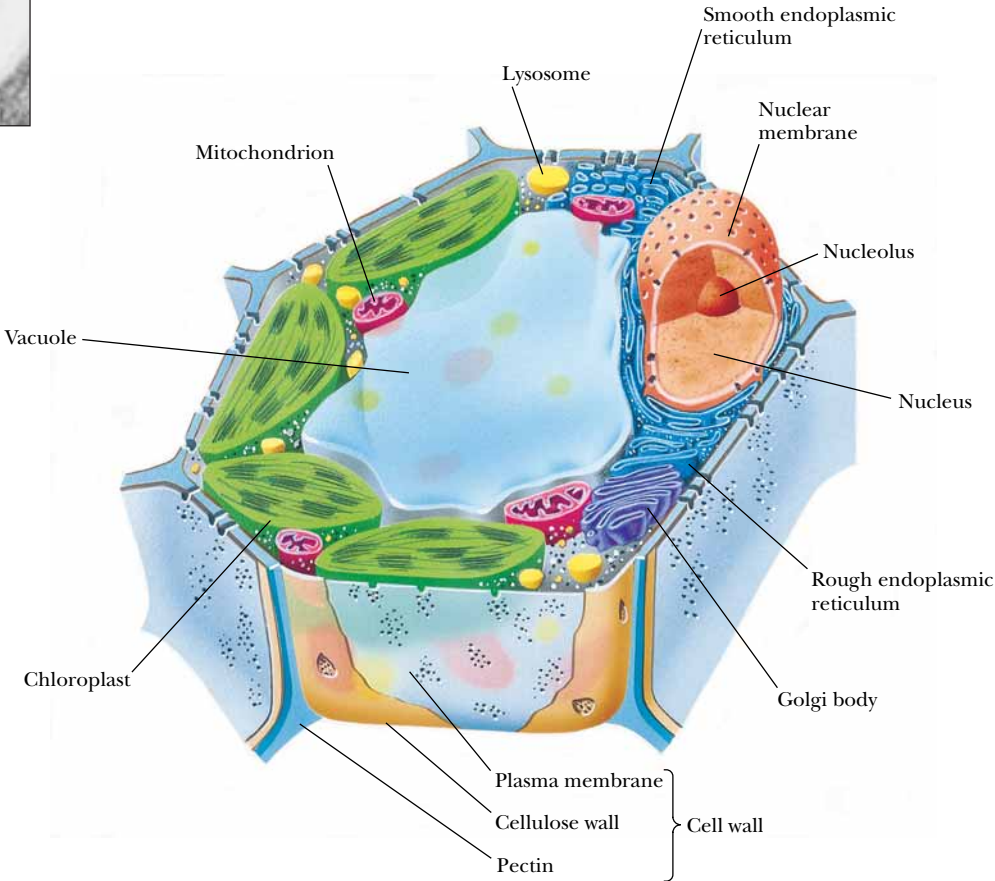


FIGURE 1.23 • This figure diagrams a cell in the leaf of a higher plant. The cell wall, membrane, nucleus, chloroplasts, mitochondria, vacuole, ER, and other characteristic features are shown. (photos, top, middle, Dr. Dennis Kunkel/Phototake, NYC; bottom, Biophoto Associates)

cells, eukaryotic cells are surrounded by a plasma membrane. Unlike prokaryotic cells, eukaryotic cells are rich in internal membranes that are differentiated into specialized structures such as the **endoplasmic reticulum (ER)** and the **Golgi apparatus**. Membranes also surround certain organelles (**mitochondria** and **chloroplasts**, for example) and various vesicles, including **vacuoles**, **lysosomes**, and **peroxisomes**. The common purpose of these membranous partitionings is the creation of cellular compartments that have specific, organized metabolic functions, such as the mitochondrion’s role as the principal site of cellular energy production. Eukaryotic cells also have a **cytoskeleton** composed of arrays of filaments that give the cell its shape and its capacity to move. Some eukaryotic cells also have long projections on their surface—cilia or flagella—which provide propulsion.

Table 1.7

Major Features of a Higher Plant Cell: A Photosynthetic Leaf Cell		
Structure	Molecular Composition	Function
Cell wall	Cellulose fibers embedded in a polysaccharide/protein matrix; it is thick ($>0.1\ \mu\text{m}$), rigid, and porous to small molecules.	Protection against osmotic or mechanical rupture. The walls of neighboring cells interact in cementing the cells together to form the plant. Channels for fluid circulation and for cell–cell communication pass through the walls. The structural material confers form and strength on plant tissue.
Cell membrane	Plant cell membranes are similar in overall structure and organization to animal cell membranes but differ in lipid and protein composition.	The plasma membrane of plant cells is selectively permeable, containing transport systems for the uptake of essential nutrients and inorganic ions. A number of important enzymes are localized here.
Nucleus	The nucleus, nucleolus, and nuclear envelope of plant cells are like those of animal cells.	Chromosomal organization, DNA replication, transcription, ribosome synthesis, and mitosis in plant cells are grossly similar to the analogous features in animals.
Chloroplasts	Plant cells contain a unique family of organelles, the plastids, of which the chloroplast is the prominent example. Chloroplasts have a double membrane envelope, an inner volume called the stroma , and an internal membrane system rich in thylakoid membranes, which enclose a third compartment, the thylakoid lumen . Chloroplasts are significantly larger than mitochondria. Other plastids are found in specialized structures such as fruits, flower petals, and roots and have specialized roles.	Chloroplasts are the site of photosynthesis, the reactions by which light energy is converted to metabolically useful chemical energy in the form of ATP. These reactions occur on the thylakoid membranes. The formation of carbohydrate from CO_2 takes place in the stroma. Oxygen is evolved during photosynthesis. Chloroplasts are the primary source of energy in the light.
Mitochondria	Plant cell mitochondria resemble the mitochondria of other eukaryotes in form and function.	Plant mitochondria are the main source of energy generation in photosynthetic cells in the dark and in nonphotosynthetic cells under all conditions.
Vacuole	The vacuole is usually the most obvious compartment in plant cells. It is a very large vesicle enclosed by a single membrane called the tonoplast . Vacuoles tend to be smaller in young cells, but in mature cells, they may occupy more than 50% of the cell’s volume. Vacuoles occupy the center of the cell, with the cytoplasm being located peripherally around it. They resemble the lysosomes of animal cells.	Vacuoles function in transport and storage of nutrients and cellular waste products. By accumulating water, the vacuole allows the plant cell to grow dramatically in size with no increase in cytoplasmic volume.
Golgi apparatus, endoplasmic reticulum, ribosomes, lysosomes, peroxisomes, and cytoskeleton	Plant cells also contain all of these characteristic eukaryotic organelles, essentially in the form described for animal cells.	These organelles serve the same purposes in plant cells that they do in animal cells.

1.6 • Viruses Are Supramolecular Assemblies Acting as Cell Parasites

Viruses are supramolecular complexes of nucleic acid, either DNA or RNA, encapsulated in a protein coat and, in some instances, surrounded by a membrane envelope (Figure 1.24). The bits of nucleic acid in viruses are, in reality, mobile elements of genetic information. The protein coat serves to protect the nucleic acid and allows it to gain entry to the cells that are its specific hosts. Viruses unique for all types of cells are known. Viruses infecting bacteria are called **bacteriophages** (“bacteria eaters”); different viruses infect animal cells and plant cells. Once the nucleic acid of a virus gains access to its specific host, it typically takes over the metabolic machinery of the host cell, diverting it to the production of virus particles. The host metabolic functions are subjugated to the synthesis of viral nucleic acid and proteins. Mature virus particles arise by encapsulating the nucleic acid within a protein coat called the **capsid**. Viruses are thus supramolecular assemblies that act as parasites of cells (Figure 1.25).

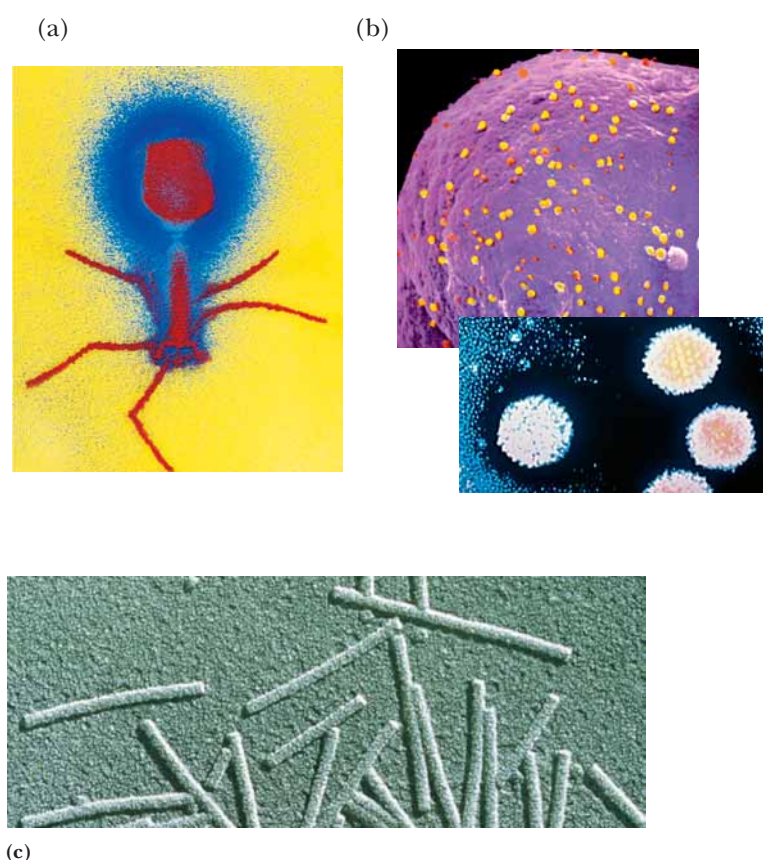


FIGURE 1.24 • Viruses are genetic elements enclosed in a protein coat. Viruses are not free-living and can only reproduce within cells. Viruses show an almost absolute specificity for their particular host cells, infecting and multiplying only within those cells. Viruses are known for virtually every kind of cell. Shown here are examples of (a) a bacterial virus, bacteriophage T₄; (b) an animal virus, adenovirus (inset at greater magnification); and (c) a plant virus, tobacco mosaic virus. (a, M. Wurtz/Biozentrum/University of Basel/SPL/Photo Researchers, Inc.; b, Dr. Thomas Broker/Phototake, NYC; inset, CNRI/SPL/Photo Researchers, Inc.; c, Biology Media/Photo Researchers, Inc.)

Often, viruses cause the lysis of the cells they infect. It is their cytolytic properties that are the basis of viral disease. In certain circumstances, the viral genetic elements may integrate into the host chromosome and become quiescent. Such a state is termed **lysogeny**. Typically, damage to the host cell activates the replicative capacities of the quiescent viral nucleic acid, leading to viral propagation and release. Some viruses are implicated in transforming cells into a cancerous state, that is, in converting their hosts to an unregulated state of cell division and proliferation. Because all viruses are heavily dependent on their host for the production of viral progeny, viruses must have arisen after cells were established in the course of evolution. Presumably, the first viruses were fragments of nucleic acid that developed the ability to replicate independently of the chromosome and then acquired the necessary genes enabling protection, autonomy, and transfer between cells.

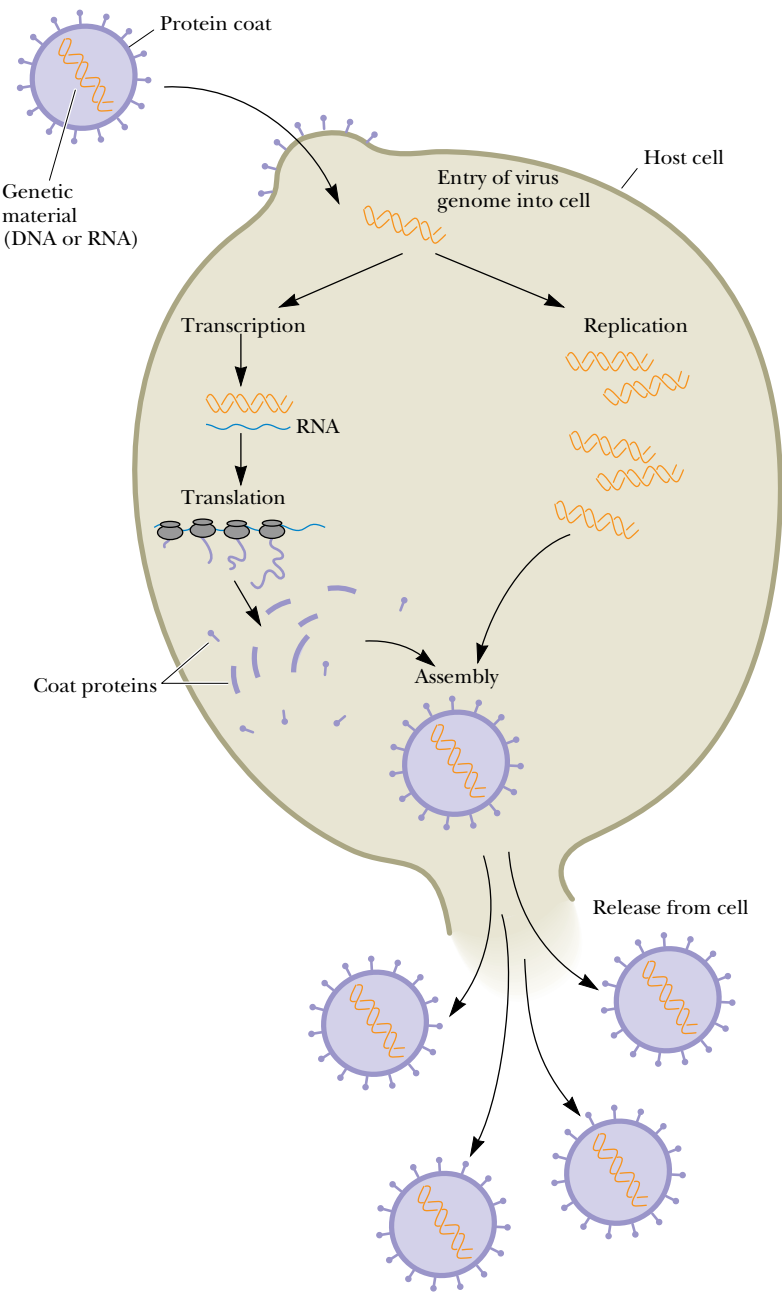


FIGURE 1.25 • The virus life cycle. Viruses are mobile bits of genetic information encapsulated in a protein coat. The genetic material can be either DNA or RNA. Once this genetic material gains entry to its host cell, it takes over the host machinery for macromolecular synthesis and subverts it to the synthesis of viral-specific nucleic acids and proteins. These virus components are then assembled into mature virus particles that are released from the cell. Often, this parasitic cycle of virus infection leads to cell death and disease.

PROBLEMS

1. The nutritional requirements of *Escherichia coli* cells are far simpler than those of humans, yet the macromolecules found in bacteria are about as complex as those of animals. Since bacteria can make all their essential biomolecules while subsisting on a simpler diet, do you think bacteria may have more biosynthetic capacity and hence more metabolic complexity than animals? Organize your thoughts on this question, pro and con, into a rational argument.

2. Without consulting chapter figures, sketch the characteristic prokaryotic and eukaryotic cell types and label their pertinent organelle and membrane systems.

3. *Escherichia coli* cells are about $2\text{ }\mu\text{m}$ (microns) long and $0.8\text{ }\mu\text{m}$ in diameter.

a. How many *E. coli* cells laid end to end would fit across the diameter of a pin head? (Assume a pinhead diameter of 0.5 mm .)

b. What is the volume of an *E. coli* cell? (Assume it is a cylinder, with the volume of a cylinder given by $V = \pi r^2 h$, where $\pi = 3.14$.)

c. What is the surface area of an *E. coli* cell? What is the surface-to-volume ratio of an *E. coli* cell?

d. Glucose, a major energy-yielding nutrient, is present in bacterial cells at a concentration of about 1 mM . How many glucose molecules are contained in a typical *E. coli* cell? (Recall that Avogadro's number $= 6.023 \times 10^{23}$.)

e. A number of regulatory proteins are present in *E. coli* at only one or two molecules per cell. If we assume that an *E. coli* cell contains just one molecule of a particular protein, what is the molar concentration of this protein in the cell?

f. An *E. coli* cell contains about 15,000 ribosomes, which carry out protein synthesis. Assuming ribosomes are spherical and have a diameter of 20 nm (nanometers), what fraction of the *E. coli* cell volume is occupied by ribosomes?

g. The *E. coli* chromosome is a single DNA molecule whose mass is about 3×10^9 daltons. This macromolecule is actually a linear array of nucleotide pairs. The average molecular weight of a nucleotide pair is 660, and each pair imparts 0.34 nm to the length of the DNA molecule. What is the total length of the *E. coli* chromosome? How does this length compare with the overall dimensions of an *E. coli* cell? How many nucleotide pairs does this DNA contain? The average *E. coli* protein is a linear chain of 360 amino acids. If three nucleotide pairs in a gene encode one amino acid in a protein, how many different proteins can the *E. coli* chromosome encode? (The answer to this question is a reasonable approximation of the maximum number of different kinds of proteins that can be expected in bacteria.)

4. Assume that mitochondria are cylinders $1.5\text{ }\mu\text{m}$ in length and $0.6\text{ }\mu\text{m}$ in diameter.

a. What is the volume of a single mitochondrion?

b. Oxaloacetate is an intermediate in the citric acid cycle, an important metabolic pathway localized in the mitochondria of eukaryotic cells. The concentration of oxaloacetate in mitochondria is about $0.03\text{ }\mu\text{M}$. How many molecules of oxaloacetate are in a single mitochondrion?

5. Assume that liver cells are cuboidal in shape, $20\text{ }\mu\text{m}$ on a side.

a. How many liver cells laid end to end would fit across the diameter of a pin head? (Assume a pinhead diameter of 0.5 mm .)

b. What is the volume of a liver cell? (Assume it is a cube.)

c. What is the surface area of a liver cell? What is the surface-to-volume ratio of a liver cell? How does this compare to the surface-to-volume ratio of an *E. coli* cell (compare this answer to that of problem 3c)? What problems must cells with low surface-to-volume ratios confront that do not occur in cells with high surface-to-volume ratios?

d. A human liver cell contains two sets of 23 chromosomes, each set being roughly equivalent in information content. The total mass of DNA contained in these 46 enormous DNA molecules is 4×10^{12} daltons. Since each nucleotide pair contributes 660 daltons to the mass of DNA and 0.34 nm to the length of DNA, what is the total number of nucleotide pairs and the complete length of the DNA in a liver cell? How does this length compare with the overall dimensions of a liver cell? The maximal information in each set of liver cell chromosomes should be related to the number of nucleotide pairs in the chromosome set's DNA. This number can be obtained by dividing the total number of nucleotide pairs calculated above by 2. What is this value? If this information is expressed in proteins that average 400 amino acids in length and three nucleotide pairs encode one amino acid in a protein, how many different kinds of proteins might a liver cell be able to produce? (In reality, liver cells express at most about 30,000 different proteins. Thus, a large discrepancy exists between the theoretical information content of DNA in liver cells and the amount of information actually expressed.)

6. Biomolecules interact with one another through molecular surfaces that are structurally complementary. How can various proteins interact with molecules as different as simple ions, hydrophobic lipids, polar but uncharged carbohydrates, and even nucleic acids?

7. What structural features allow biological polymers to be informational macromolecules? Is it possible for polysaccharides to be informational macromolecules?

8. Why is it important that weak forces, not strong forces, mediate biomolecular recognition?

9. Why does the central role of weak forces in biomolecular interactions restrict living systems to a narrow range of environmental conditions?

10. Describe what is meant by the phrase "cells are steady-state systems."

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“If there is magic on this planet, it is contained in water.”

LOREN EISLEY
Inscribed on the wall of the National
Aquarium in Baltimore, MD

OUTLINE

- 2.1 • Properties of Water
- 2.2 • pH
- 2.3 • Buffers
- 2.4 • Water’s Unique Role in the Fitness of the Environment

Chapter 2

Water, pH, and Ionic Equilibria



Some of the magic: Students and teacher view a coral crab in Graham’s Harbour, San Salvador Island, the Bahamas. (Lara Call)

Water is a major chemical component of the earth’s surface. It is indispensable to life. Indeed, it is the only liquid that most organisms ever encounter. We alternately take it for granted because of its ubiquity and bland nature or marvel at its many unusual and fascinating properties. At the center of this fascination is the role of water as the medium of life. Life originated, evolved, and thrives in the seas. Organisms invaded and occupied terrestrial and aerial niches, but none gained true independence from water. Typically, organisms are constituted of 70 to 90% water. Indeed, normal metabolic activity can occur only when cells are at least 65% H_2O . This dependency of life on water is not a simple matter, but it can be grasped through a consideration of the unusual chemical and physical properties of H_2O . Subsequent chapters establish that water and its ionization products, hydrogen ions and hydroxide ions, are crit-

ical determinants of the structure and function of proteins, nucleic acids, and membranes. In yet another essential role, water is an indirect participant—a difference in the concentration of hydrogen ions on opposite sides of a membrane represents an energized condition essential to biological mechanisms of energy transformation. First, let's review the remarkable properties of water.

2.1 • Properties of Water

Unusual Properties

In comparison with chemical compounds of similar atomic organization and molecular size, water displays unexpected properties. For example, compare water, the hydride of oxygen, with hydrides of oxygen's nearest neighbors in the periodic table, namely, ammonia (NH_3) and hydrogen fluoride (HF), or with the hydride of its nearest congener, sulfur (H_2S). Water has a substantially higher boiling point, melting point, heat of vaporization, and surface tension. Indeed, all of these physical properties are anomalously high for a substance of this molecular weight that is neither metallic nor ionic. These properties suggest that intermolecular forces of attraction between H_2O molecules are high. Thus, the internal cohesion of this substance is high. Furthermore, water has an unusually high dielectric constant, its maximum density is found in the liquid (not the solid) state, and it has a negative volume of melting (that is, the solid form, ice, occupies more space than does the liquid form, water). It is truly remarkable that so many eccentric properties should occur together in a single substance. As chemists, we expect to find an explanation for these apparent anomalies in the structure of water. The key to its intermolecular attractions must lie in its atomic constitution. Indeed, *the unrivaled ability to form hydrogen bonds is the crucial fact to understanding its properties.*

Structure of Water

The two hydrogen atoms of water are linked covalently to oxygen, each sharing an electron pair, to give a nonlinear arrangement (Figure 2.1). This “bent” structure of the H_2O molecule is of enormous significance to its properties. If H_2O were linear, it would be a nonpolar substance. In the bent configuration, however, the electronegative O atom and the two H atoms form a dipole that renders the molecule distinctly polar. Furthermore, this structure is ideally suited to H-bond formation. Water can serve as both an H donor and an H acceptor in H-bond formation. The potential to form four H bonds per water molecule is the source of the strong intermolecular attractions that endow this substance with its anomalously high boiling point, melting point, heat of vaporization, and surface tension. In ordinary ice, the common crystalline form of water, each H_2O molecule has four nearest neighbors to which it is hydrogen bonded: each H atom donates an H bond to the O of a neighbor, while the O atom serves as an H-bond acceptor from H atoms bound to two different water molecules (Figure 2.2). A local tetrahedral symmetry results.

Hydrogen bonding in water is cooperative. That is, an H-bonded water molecule serving as an acceptor is a better H-bond donor than an unbonded molecule (and an H_2O molecule serving as an H-bond donor becomes a better H-bond acceptor). Thus, participation in H bonding by H_2O molecules is a phenomenon of mutual reinforcement. The H bonds between neighboring molecules are weak (23 kJ/mol each) relative to the H—O covalent bonds (420 kJ/mol). As a consequence, the hydrogen atoms are situated asymmetrically

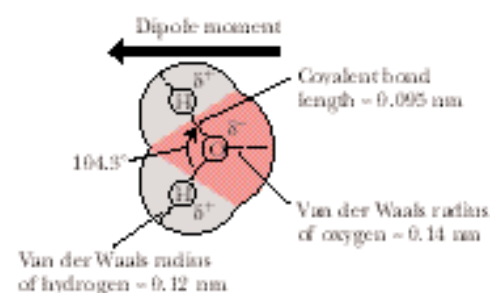


FIGURE 2.1 • The structure of water. Two lobes of negative charge formed by the lone-pair electrons of the oxygen atom lie above and below the plane of the diagram. This electron density contributes substantially to the large dipole moment and polarizability of the water molecule. The dipole moment of water corresponds to the O—H bonds having 33% ionic character. Note that the H—O—H angle is 104.3°, not 109°, the angular value found in molecules with tetrahedral symmetry, such as CH_4 . Many of the important properties of water derive from this angular value, such as the decreased density of its crystalline state, ice. (The dipole moment in this figure points in the direction from negative to positive, the convention used by physicists and physical chemists; organic chemists draw it pointing in the opposite direction.)

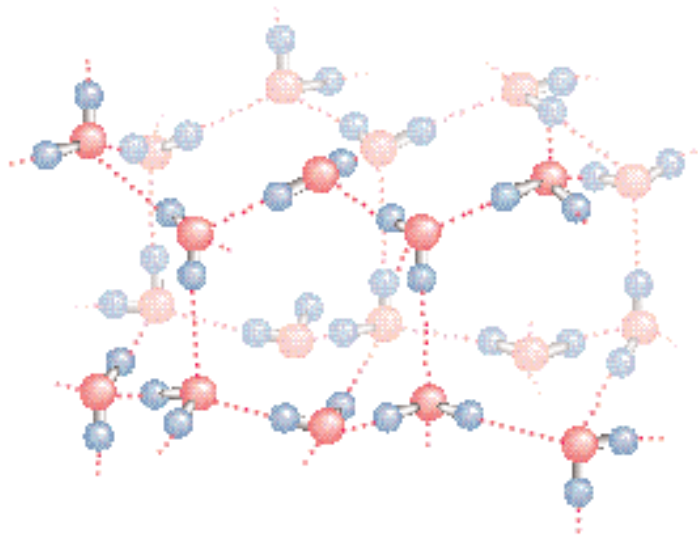


FIGURE 2.2 • The structure of normal ice. The hydrogen bonds in ice form a three-dimensional network. The smallest number of H_2O molecules in any closed circuit of H-bonded molecules is six, so that this structure bears the name *hexagonal ice*. Covalent bonds are represented as solid lines, whereas hydrogen bonds are shown as dashed lines. The directional preference of H bonds leads to a rather open lattice structure for crystalline water and, consequently, a low density for the solid state. The distance between neighboring oxygen atoms linked by a hydrogen bond is 0.274 nm. Because the covalent H—O bond is 0.095 nm, the H—O hydrogen bond length in ice is 0.18 nm.

between the two oxygen atoms along the O—O axis. There is never any ambiguity about which O atom the H atom is chemically bound to, nor to which O it is H-bonded.

Structure of Ice

In ice, the hydrogen bonds form a space-filling, three-dimensional network. These bonds are directional and straight; that is, the H atom lies on a direct line between the two O atoms. This linearity and directionality mean that the resultant H bonds are strong. In addition, the directional preference of the H bonds leads to an open lattice structure. For example, if the water molecules are approximated as rigid spheres centered at the positions of the O atoms in the lattice, then the observed density of ice is actually only 57% of that expected for a tightly packed arrangement of such spheres. The H bonds in ice hold the water molecules apart. Melting involves breaking some of the H bonds that maintain the crystal structure of ice so that the molecules of water (now liquid) can actually pack closer together. Thus, the density of ice is slightly less than the density of water. Ice floats, a property of great importance to aquatic organisms in cold climates.

In liquid water, the rigidity of ice is replaced by fluidity, and the crystalline periodicity of ice gives way to spatial homogeneity. The H_2O molecules in liquid water form a random, H-bonded network with each molecule having an average of 4.4 close neighbors situated within a center-to-center distance of 0.284 nm (2.84 Å). At least half of the hydrogen bonds have nonideal orientations (that is, they are not perfectly straight); consequently, liquid H_2O lacks the regular latticelike structure of ice. The space about an O atom is not defined by the presence of four hydrogens, but can be occupied by other water mole-

cles randomly oriented so that the local environment, over time, is essentially uniform. Nevertheless, the heat of melting for ice is but a small fraction (13%) of the heat of sublimation for ice (the energy needed to go from the solid to the vapor state). This fact indicates that the majority of H bonds between H₂O molecules survive the transition from solid to liquid. At 10°C, 2.3 H bonds per H₂O molecule remain, and the tetrahedral bond order persists even though substantial disorder is now present.

Molecular Interactions in Liquid Water

The present interpretation of water structure is that water molecules are connected by uninterrupted H bond paths running in every direction, spanning the whole sample. The participation of each water molecule in an average state of H bonding to its neighbors means that each molecule is connected to every other in a fluid network of H bonds. The average lifetime of an H-bonded connection between two H₂O molecules in water is 9.5 psec (picoseconds, where 1 psec = 10⁻¹² sec). Thus, about every 10 psec, the average H₂O molecule moves, reorients, and interacts with new neighbors, as illustrated in Figure 2.3.

In summary, pure liquid water consists of H₂O molecules held in a random, three-dimensional network that has a local preference for tetrahedral geometry but contains a large number of strained or broken hydrogen bonds. The presence of strain creates a kinetic situation in which H₂O molecules can switch H-bond allegiances; fluidity ensues.

Solvent Properties

Because of its highly polar nature, water is an excellent solvent for ionic substances such as salts; nonionic but polar substances such as sugars, simple alcohols, and amines; and carbonyl-containing molecules such as aldehydes and ketones. Although the electrostatic attractions between the positive and negative ions in the crystal lattice of a salt are very strong, water readily dissolves salts. For example, sodium chloride is dissolved because dipolar water molecules participate in strong electrostatic interactions with the Na⁺ and Cl⁻ ions, leading to the formation of **hydration shells** surrounding these ions (Figure 2.4). Although hydration shells are stable structures, they are also dynamic. Each water molecule in the inner hydration shell around a Na⁺ ion is replaced on average every 2 to 4 nsec (nanoseconds, where 1 nsec = 10⁻⁹ sec) by another H₂O. Consequently, a water molecule is trapped only several hundred times longer by the electrostatic force field of an ion than it is by the H-bonded network of water. (Recall that the average lifetime of H bonds between water molecules is about 10 psec.)

Water Has a High Dielectric Constant

The attractions between the water molecules interacting with, or **hydrating**, ions are much greater than the tendency of oppositely charged ions to attract one another. The ability of water to surround ions in dipole interactions and

FIGURE 2.3 • The fluid network of H bonds linking water molecules in the liquid state. It is revealing to note that, in 10 psec, a photon of light (which travels at 3 × 10⁸ m/sec) would move a distance of only 0.003 m.

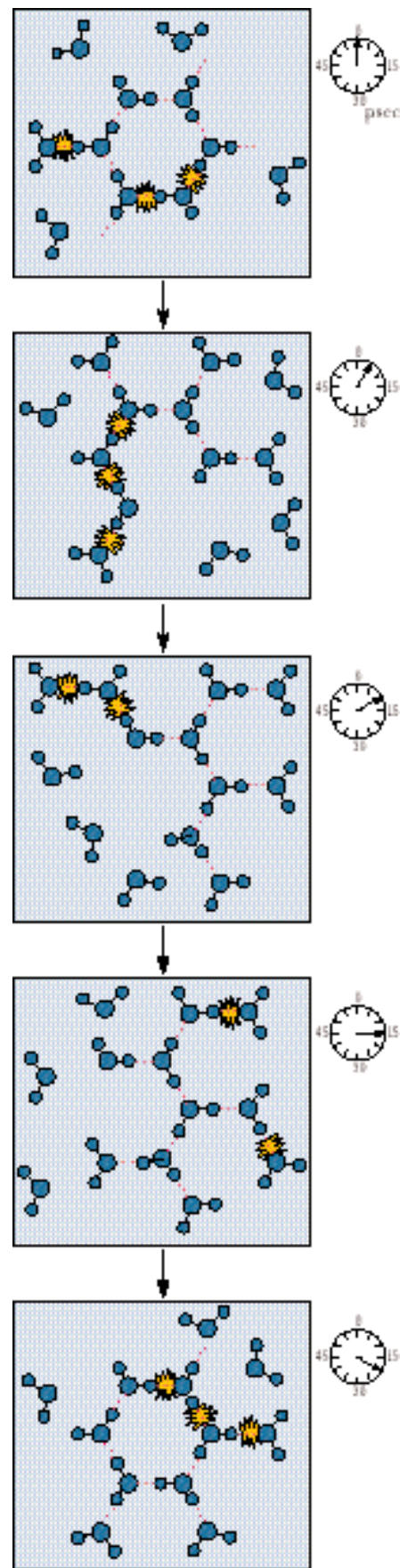
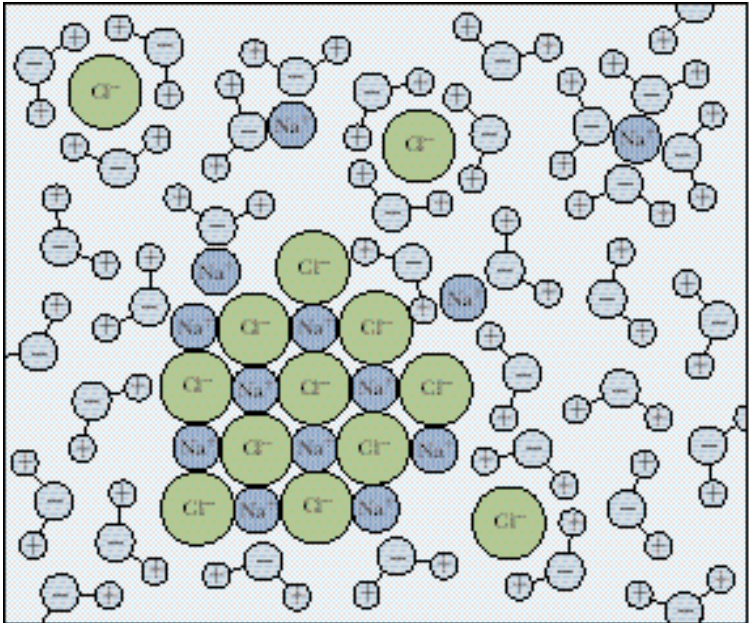


FIGURE 2.4 • Hydration shells surrounding ions in solution. Water molecules orient so that the electrical charge on the ion is sequestered by the water dipole. For positive ions (cations), the partially negative oxygen atom of H₂O is toward the ion in solution. Negatively charged ions (anions) attract the partially positive hydrogen atoms of water in creating their hydration shells.



diminish their attraction for one another is a measure of its **dielectric constant**, D . Indeed, ionization in solution depends on the dielectric constant of the solvent; otherwise the strongly attracted positive and negative ions would unite to form neutral molecules. The strength of the dielectric constant is related to the force, F , experienced between two ions of opposite charge separated by a distance, r , as given in the relationship

$$F = e_1 e_2 / Dr^2$$

where e_1 and e_2 are the charges on the two ions. Table 2.1 lists the dielectric constants of some common liquids. Note that the dielectric constant for water is more than twice that of methanol and more than 40 times that of hexane.

Water Forms H Bonds with Polar Solutes

In the case of nonionic but polar compounds such as sugars, the excellent solvent properties of water stem from its ability to readily form hydrogen bonds with the polar functional groups on these compounds, such as hydroxyls, amines, and carbonyls. These polar interactions between solvent and solute are stronger than the intermolecular attractions between solute molecules caused by van der Waals forces and weaker hydrogen bonding. Thus, the solute molecules readily dissolve in water.

Hydrophobic Interactions

The behavior of water toward nonpolar solutes is different from the interactions just discussed. Nonpolar solutes (or nonpolar functional groups on biological macromolecules) do not readily H bond to H₂O, and, as a result, such compounds tend to be only sparingly soluble in water. The process of dissolving such substances is accompanied by significant reorganization of the water surrounding the solute so that the response of the solvent water to such solutes can be equated to “structure making.” Because nonpolar solutes must occupy space, the random H-bond network of water must reorganize to accommodate them. At the same time, the water molecules participate in as many H-bonded

Table 2.1

Dielectric Constants* of Some Common Solvents at 25°C

Solvent	Dielectric Constant (D)
Water	78.5
Methyl alcohol	32.6
Ethyl alcohol	24.3
Acetone	20.7
Acetic acid	6.2
Chloroform	5.0
Benzene	2.3
Hexane	1.9

*The dielectric constant is also referred to as *relative permittivity* by physical chemists.

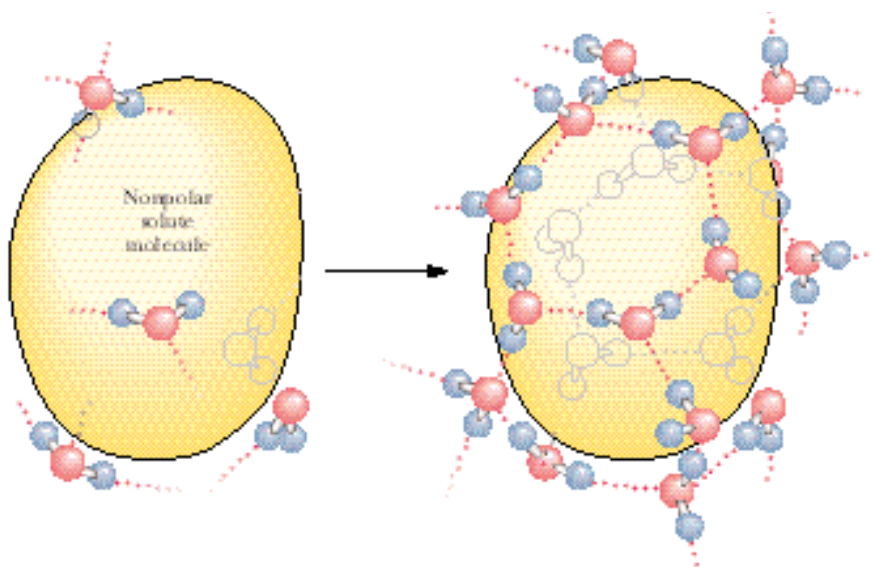


FIGURE 2.5 • Formation of a clathrate structure by water molecules surrounding a hydrophobic solute.

interactions with one another as the temperature permits. Consequently, the H-bonded water network rearranges toward formation of a local cagelike (**clathrate**) structure surrounding each solute molecule (Figure 2.5). This fixed orientation of water molecules around a hydrophobic “solute” molecule results in a hydration shell. A major consequence of this rearrangement is that the molecules of H_2O participating in the cage layer have markedly reduced orientational options. Water molecules tend to straddle the nonpolar solute such that two or three tetrahedral directions (H-bonding vectors) are tangential to the space occupied by the inert solute. This “straddling” means that no water H-bonding capacity is lost because no H-bond donor or acceptor of the H_2O is directed toward the caged solute. The water molecules forming these clathrates are involved in highly ordered structures. That is, clathrate formation is accompanied by significant ordering of structure or negative entropy.

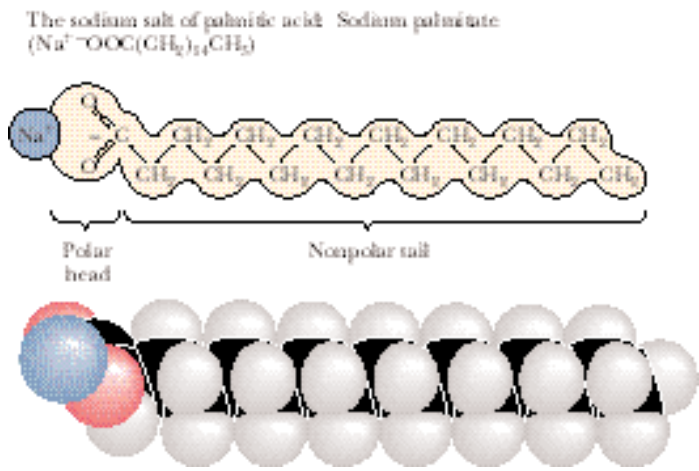
Under these conditions, nonpolar solute molecules experience a net attraction for one another that is called **hydrophobic interaction**. The basis of this interaction is that when two nonpolar molecules meet, their joint solvation cage involves less surface area and less overall ordering of the water molecules than in their separate cages. The “attraction” between nonpolar solutes is an entropy-driven process due to a net decrease in order among the H_2O molecules. To be specific, hydrophobic interactions between nonpolar molecules are maintained not so much by direct interactions between the inert solutes themselves as by the increase in entropy when the water cages coalesce and reorganize. Because interactions between nonpolar solute molecules and the water surrounding them are of uncertain stoichiometry and do not share the equality of atom-to-atom participation implicit in chemical bonding, the term *hydrophobic interaction* is more correct than the misleading expression *hydrophobic bond*.

Amphiphilic Molecules

Compounds containing both strongly polar and strongly nonpolar groups are called **amphiphilic molecules** (from the Greek *amphi* meaning “both,” and *philos* meaning “loving”), also referred to as **amphipathic molecules** (from the Greek *pathos* meaning “passion”). Salts of fatty acids are a typical example that

amphiphilic molecules, amphipathic molecules • compounds containing both strongly polar and strongly nonpolar groups

FIGURE 2.6 • An amphiphilic molecule: sodium palmitate. Amphiphilic molecules are frequently symbolized by a ball and zig-zag line structure, where the ball represents the hydrophilic polar head and the zig-zag represents the nonpolar hydrophobic hydrocarbon tail.



has biological relevance. They have a long nonpolar hydrocarbon tail and a strongly polar carboxyl head group, as in the sodium salt of palmitic acid (Figure 2.6). Their behavior in aqueous solution reflects the combination of the contrasting polar and nonpolar nature of these substances. The ionic carboxylate function hydrates readily, whereas the long hydrophobic tail is intrinsically insoluble. Nevertheless, sodium palmitate and other amphiphilic molecules readily disperse in water because the hydrocarbon tails of these substances are joined together in hydrophobic interactions as their polar carboxylate functions are hydrated in typical hydrophilic fashion. Such clusters of amphipathic molecules are termed **micelles**; Figure 2.7 depicts their structure. Of enormous biological significance is the contrasting solute behavior of the two ends of amphipathic molecules upon introduction into aqueous solutions. The polar ends express their hydrophilicity in ionic interactions with the solvent, whereas their nonpolar counterparts are excluded from the water into a hydrophobic domain constituted from the hydrocarbon tails of many like molecules. It is this behavior that accounts for the formation of membranes, the structures that define the limits and compartments of cells (see Chapter 9).

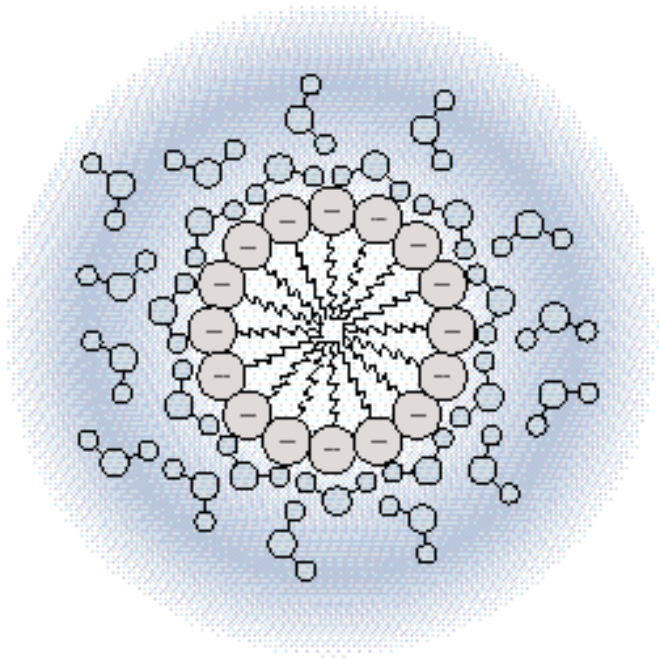


FIGURE 2.7 • Micelle formation by amphiphilic molecules in aqueous solution. Negatively charged carboxylate head groups orient to the micelle surface and interact with the polar H_2O molecules via H bonding. The nonpolar hydrocarbon tails cluster in the interior of the spherical micelle, driven by hydrophobic exclusion from the solvent and the formation of favorable van der Waals interactions. Because of their negatively charged surfaces, neighboring micelles repel one another and thereby maintain a relative stability in solution.

Influence of Solutes on Water Properties

The presence of dissolved substances disturbs the structure of liquid water so that its properties change. The dynamic hydrogen-bonding pattern of water must now accommodate the intruding substance. The net effect is that solutes, regardless of whether they are polar or nonpolar, fix nearby water molecules in a more ordered array. Ions, by the establishment of hydration shells through interactions with the water dipoles, create local order. Hydrophobic effects, for different reasons, make structures within water. To put it another way, by limiting the orientations that neighboring water molecules can assume, solutes give order to the solvent and diminish the dynamic interplay among H_2O molecules that occurs in pure water.

Colligative Properties

This influence of the solute on water is reflected in a set of characteristic changes in behavior that are termed **colligative properties**, or properties related by a common principle. These alterations in solvent properties are related in that they all depend only on the number of solute particles per unit volume of solvent and not on the chemical nature of the solute. These effects include freezing point depression, boiling point elevation, vapor pressure lowering, and osmotic pressure effects. For example, 1 mol of an ideal solute dissolved in 1000 g of water (a 1 *m*, or molal, solution) at 1 atm pressure depresses the freezing point by 1.86°C , raises the boiling point by 0.543°C , lowers the vapor pressure in a temperature-dependent manner, and yields a solution whose osmotic pressure relative to pure water is 22.4 atm. In effect, by imposing local order on the water molecules, solutes make it more difficult for water to assume its crystalline lattice (freeze) or escape into the atmosphere (boil or vaporize). Furthermore, when a solution (such as the 1 *m* solution discussed here) is separated from a volume of pure water by a semipermeable membrane, the solution draws water molecules across this barrier. The water molecules are moving from a region of higher effective concentration (pure H_2O) to a region of lower effective concentration (the solution). This movement of water into the solution dilutes the effects of the solute that is present. The osmotic force exerted by each mole of solute is so strong that it requires the imposition of 22.4 atm of pressure to be negated (Figure 2.8).

Osmotic pressure from high concentrations of dissolved solutes is a serious problem for cells. Bacterial and plant cells have strong, rigid cell walls to contain these pressures. In contrast, animal cells are bathed in extracellular fluids of comparable osmolarity, so no net osmotic gradient exists. Also, to minimize the osmotic pressure created by the contents of their cytosol, cells tend

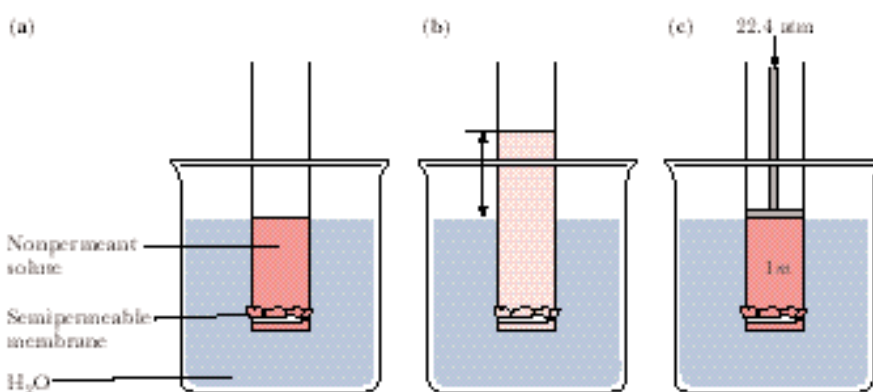


FIGURE 2.8 • The osmotic pressure of a 1 molal (*m*) solution is equal to 22.4 atmospheres of pressure. (a) If a nonpermeant solute is separated from pure water by a semipermeable membrane through which H_2O passes freely, (b) water molecules enter the solution (osmosis) and the height of the solution column in the tube rises. The pressure necessary to push water back through the membrane at a rate exactly equaled by the water influx is the osmotic pressure of the solution. (c) For a 1 *m* solution, this force is equal to 22.4 atm of pressure. Osmotic pressure is directly proportional to the concentration of the nonpermeant solute.

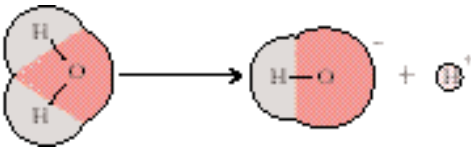
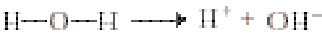


FIGURE 2.9 • The ionization of water.

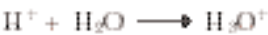
to store substances such as amino acids and sugars in polymeric form. For example, a molecule of glycogen or starch containing 1000 glucose units exerts only 1/1000 the osmotic pressure that 1000 free glucose molecules would.

Ionization of Water

Water shows a small but finite tendency to form ions. This tendency is demonstrated by the electrical conductivity of pure water, a property that clearly establishes the presence of charged species (ions). Water ionizes because the larger, strongly electronegative oxygen atom strips the electron from one of its hydrogen atoms, leaving the proton to dissociate (Figure 2.9):



Two ions are thus formed: protons or **hydrogen ions**, H^+ , and **hydroxyl ions**, OH^- . Free protons are immediately hydrated to form **hydronium ions**, H_3O^+ :



Indeed, because most hydrogen atoms in liquid water are hydrogen-bonded to a neighboring water molecule, this protonic hydration is an instantaneous process and the ion products of water are H_3O^+ and OH^- :



The amount of H_3O^+ or OH^- in 1 L (liter) of pure water at 25°C is 1×10^{-7} mol; the concentrations are equal because the dissociation is stoichiometric.

Although it is important to keep in mind that the hydronium ion, or hydrated hydrogen ion, represents the true state in solution, the convention is to speak of hydrogen ion concentrations in aqueous solution, even though “naked” protons are virtually nonexistent. Indeed, H_3O^+ itself attracts a hydration shell by H bonding to adjacent water molecules to form an H_9O_4^+ species (Figure 2.10) and even more highly hydrated forms. Similarly, the hydroxyl ion, like all other highly charged species, is also hydrated.

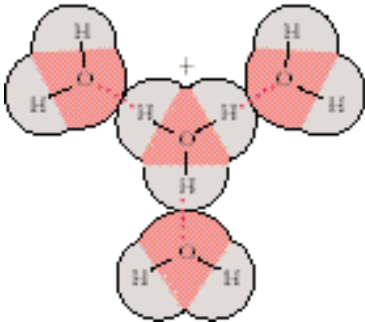


FIGURE 2.10 • The hydration of H_3O^+ . Solid lines denote covalent bonds; dashed lines represent the H bonds formed between the hydronium ion and its waters of hydration.

Proton Jumping

Because of the high degree of hydrogen bonding in water, H^+ ions show an apparent rate of migration in an electrical field that is vastly greater than other univalent cations in aqueous solution, such as Na^+ and K^+ . In effect, the net transfer of a proton from molecule to molecule throughout the H-bonded network accounts for this apparent rapidity of migration (Figure 2.11).

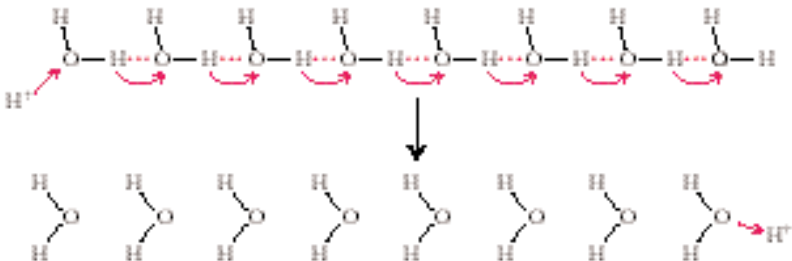
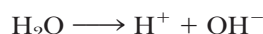


FIGURE 2.11 • Proton jumping via the hydrogen-bonded network of water molecules.

That is, the H-bonded network provides a natural route for rapid H^+ transport. This phenomenon of **proton jumping** thus occurs with little actual movement of the water molecules themselves. Ice has an electrical conductivity close to that of water because such proton jumps also readily occur even when the water molecules are fixed in a crystal lattice. Such conduction of protons via H-bonded networks has been offered as an explanation for a number of rapid proton transfers of biological significance.

K_w, the Ion Product of Water

The dissociation of water into hydrogen ions and hydroxyl ions occurs to the extent that 10^{-7} mol of H^+ and 10^{-7} mol of OH^- are present at equilibrium in 1 L of water at 25°C.



The equilibrium constant for this process is

$$K_{\text{eq}} = \frac{[\text{H}^+][\text{OH}^-]}{[\text{H}_2\text{O}]}$$

where brackets denote concentrations in moles per liter. Because the concentration of H_2O in 1 L of pure water is equal to the number of grams in a liter divided by the gram molecular weight of H_2O , or $1000/18$, the molar concentration of H_2O in pure water is 55.5 *M* (molar). The decrease in H_2O concentration as a result of ion formation ($[\text{H}^+], [\text{OH}^-] = 10^{-7}$ *M*) is negligible in comparison, and thus its influence on the overall concentration of H_2O can be ignored. Thus,

$$K_{\text{eq}} = \frac{(10^{-7})(10^{-7})}{55.5} = 1.8 \times 10^{-16}$$

Because the concentration of H_2O in pure water is essentially constant, a new constant, **K_w**, the **ion product of water**, can be written as

$$K_{\text{w}} = 55.5 K_{\text{eq}} = 10^{-14} = [\text{H}^+][\text{OH}^-]$$

The equation has the virtue of revealing the reciprocal relationship between H^+ and OH^- concentrations of aqueous solutions. If a solution is acidic, that is, of significant $[\text{H}^+]$, then the ion product of water dictates that the OH^- concentration is correspondingly less. For example, if $[\text{H}^+]$ is 10^{-2} *M*, $[\text{OH}^-]$ must be 10^{-12} *M* ($K_{\text{w}} = 10^{-14} = [10^{-2}][\text{OH}^-]$; $[\text{OH}^-] = 10^{-12}$ *M*). Similarly, in an alkaline, or basic, solution in which $[\text{OH}^-]$ is great, $[\text{H}^+]$ is low.

2.2 • pH

To avoid the cumbersome use of negative exponents to express concentrations that range over 14 orders of magnitude, Sørensen, a Danish biochemist, devised the **pH scale** by defining **pH** as *the negative logarithm of the hydrogen ion concentration*¹:

$$\text{pH} = -\log_{10} [\text{H}^+]$$

¹To be precise in physical chemical terms, the *activities* of the various components, *not* their molar concentrations, should be used in these equations. The activity (*a*) of a solute component is defined as the product of its molar concentration, *c*, and an *activity coefficient*, γ : $a = [c]\gamma$. Most biochemical work involves dilute solutions, and the use of activities instead of molar concentrations is usually neglected. However, the concentration of certain solutes may be very high in living cells.

Table 2.2

pH Scale				
The hydrogen ion and hydroxyl ion concentrations are given in moles per liter at 25°C.				
pH	[H ⁺]		[OH [−]]	
0	(10 ⁰)	1.0	0.00000000000001	(10 ^{−14})
1	(10 ^{−1})	0.1	0.0000000000001	(10 ^{−13})
2	(10 ^{−2})	0.01	0.000000000001	(10 ^{−12})
3	(10 ^{−3})	0.001	0.00000000001	(10 ^{−11})
4	(10 ^{−4})	0.0001	0.0000000001	(10 ^{−10})
5	(10 ^{−5})	0.00001	0.000000001	(10 ^{−9})
6	(10 ^{−6})	0.000001	0.00000001	(10 ^{−8})
7	(10 ^{−7})	0.0000001	0.0000001	(10 ^{−7})
8	(10 ^{−8})	0.00000001	0.000001	(10 ^{−6})
9	(10 ^{−9})	0.000000001	0.00001	(10 ^{−5})
10	(10 ^{−10})	0.0000000001	0.0001	(10 ^{−4})
11	(10 ^{−11})	0.00000000001	0.001	(10 ^{−3})
12	(10 ^{−12})	0.000000000001	0.01	(10 ^{−2})
13	(10 ^{−13})	0.0000000000001	0.1	(10 ^{−1})
14	(10 ^{−14})	0.00000000000001	1.0	(10 ⁰)

Table 2.3

The pH of Various Common Fluids

Fluid	pH
Household lye	13.6
Bleach	12.6
Household ammonia	11.4
Milk of magnesia	10.3
Baking soda	8.4
Seawater	8.0
Pancreatic fluid	7.8–8.0
Blood plasma	7.4
Intracellular fluids	
Liver	6.9
Muscle	6.1
Saliva	6.6
Urine	5–8
Boric acid	5.0
Beer	4.5
Orange juice	4.3
Grapefruit juice	3.2
Vinegar	2.9
Soft drinks	2.8
Lemon juice	2.3
Gastric juice	1.2–3.0
Battery acid	0.35

Table 2.2 gives the pH scale. Note again the reciprocal relationship between [H⁺] and [OH[−]]. Also, because the pH scale is based on negative logarithms, low pH values represent the highest H⁺ concentrations (and the lowest OH[−] concentrations, as *K_w* specifies). Note also that

p*K_w* = pH + pOH = 14

The pH scale is widely used in biological applications because hydrogen ion concentrations in biological fluids are very low, about 10^{−7} *M* or 0.0000001 *M*, a value more easily represented as pH 7. The pH of blood plasma, for example, is 7.4 or 0.00000004 *M* H⁺. Certain disease conditions may lower the plasma pH level to 6.8 or less, a situation that may result in death. At pH 6.8, the H⁺ concentration is 0.00000016 *M*, four times greater than at pH 7.4.

At pH 7, [H⁺] = [OH[−]]; that is, there is no excess acidity or basicity. The point of **neutrality** is at pH 7, and solutions having a pH of 7 are said to be at **neutral pH**. The pH values of various fluids of biological origin or relevance are given in Table 2.3. Because the pH scale is a logarithmic scale, two solutions whose pH values differ by one pH unit have a 10-fold difference in [H⁺]. For example, grapefruit juice at pH 3.2 contains more than 12 times as much H⁺ as orange juice at pH 4.3.

Dissociation of Strong Electrolytes

Substances that are almost completely dissociated to form ions in solution are called **strong electrolytes**. The term **electrolyte** describes substances capable of generating ions in solution and thereby causing an increase in the electrical conductivity of the solution. Many salts (such as NaCl and K₂SO₄) fit this category, as do strong acids (such as HCl) and strong bases (such as NaOH). Recall from general chemistry that acids are proton donors and bases are pro-

ton acceptors. In effect, the dissociation of a strong acid such as HCl in water can be treated as a proton transfer reaction between the acid HCl and the base H_2O to give the **conjugate acid** H_3O^+ and the **conjugate base** Cl^- :



The equilibrium constant for this reaction is

$$K = \frac{[\text{H}_3\text{O}^+][\text{Cl}^-]}{[\text{H}_2\text{O}][\text{HCl}]}$$

Customarily, because the term $[\text{H}_2\text{O}]$ is essentially constant in dilute aqueous solutions, it is incorporated into the equilibrium constant K to give a new term, K_a , the acid dissociation constant (where $K_a = K[\text{H}_2\text{O}]$). Also, the term $[\text{H}_3\text{O}^+]$ is often replaced by H^+ , such that

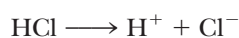
$$K_a = \frac{[\text{H}^+][\text{Cl}^-]}{[\text{HCl}]}$$

For HCl, the value of K_a is exceedingly large because the concentration of HCl in aqueous solution is vanishingly small. Because this is so, the pH of HCl solutions is readily calculated from the amount of HCl used to make the solution:

$$[\text{H}^+] \text{ in solution} = [\text{HCl}] \text{ added to solution}$$

Thus, a 1 M solution of HCl has a pH of 0; a 1 mM HCl solution has a pH of 3. Similarly, a 0.1 M NaOH solution has a pH of 13. (Because $[\text{OH}^-] = 0.1 \text{ M}$, $[\text{H}^+]$ must be 10^{-13} M .)

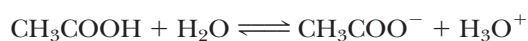
Viewing the dissociation of strong electrolytes another way, we see that the ions formed show little affinity for one another. For example, in HCl in water, Cl^- has very little affinity for H^+ :



and in NaOH solutions, Na^+ has little affinity for OH^- . The dissociation of these substances in water is effectively complete.

Dissociation of Weak Electrolytes

Substances with only a slight tendency to dissociate to form ions in solution are called **weak electrolytes**. Acetic acid, CH_3COOH , is a good example:



The acid dissociation constant K_a for acetic acid is 1.74×10^{-5} :

$$K_a = \frac{[\text{H}^+][\text{CH}_3\text{COO}^-]}{[\text{CH}_3\text{COOH}]} = 1.74 \times 10^{-5}$$

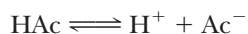
The term K_a is also called an **ionization constant** because it states the extent to which a substance forms ions in water. The relatively low value of K_a for acetic acid reveals that the un-ionized form, CH_3COOH , predominates over H^+ and CH_3COO^- in aqueous solutions of acetic acid. Viewed another way, CH_3COO^- , the acetate ion, has a high affinity for H^+ .

EXAMPLE

What is the pH of a 0.1 M solution of acetic acid? Or, to restate the question, what is the final pH when 0.1 mol of acetic acid (HAc) is added to water and the volume of the solution is adjusted to equal 1 L?

ANSWER

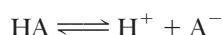
The dissociation of HAc in water can be written simply as



where Ac^- represents the acetate ion, CH_3COO^- . In solution, some amount x of HAc dissociates, generating x amount of Ac^- and an equal amount x of H^+ . Ionic equilibria characteristically are established very rapidly. At equilibrium, the concentration of HAc + Ac^- must equal 0.1 *M*. So, $[\text{HAc}]$ can be represented as $(0.1 - x)$ *M*, and $[\text{Ac}^-]$ and $[\text{H}^+]$ then both equal x molar. From $1.74 \times 10^{-5} = ([\text{H}^+][\text{Ac}^-])/[\text{HAc}]$, we get $1.74 \times 10^{-5} = x^2/[0.1 - x]$. The solution to quadratic equations of this form ($ax^2 + bx + c = 0$) is $x = (-b \pm \sqrt{b^2 - 4ac})/2a$. However, the calculation of x can be simplified by noting that, because K_a is quite small, $x \ll 0.1$ *M*. Therefore, K_a is essentially equal to $x^2/0.1$. This simplification yields $x^2 = 1.74 \times 10^{-6}$, or $x = 1.32 \times 10^{-3}$ *M* and pH = 2.88.

Henderson–Hasselbalch Equation

Consider the ionization of some weak acid, HA, occurring with an acid dissociation constant, K_a . Then,



and

$$K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]}$$

Rearranging this expression in terms of the parameter of interest, $[\text{H}^+]$, we have

$$[\text{H}^+] = \frac{K_a[\text{HA}]}{[\text{A}^-]}$$

Taking the logarithm of both sides gives

$$\log [\text{H}^+] = \log K_a + \log_{10} \frac{[\text{HA}]}{[\text{A}^-]}$$

If we change the signs and define $\text{p}K_a = -\log K_a$, we have

$$\text{pH} = \text{p}K_a - \log_{10} \frac{[\text{HA}]}{[\text{A}^-]}$$

or

$$\text{pH} = \text{p}K_a + \log_{10} \frac{[\text{A}^-]}{[\text{HA}]}$$

This relationship is known as the **Henderson–Hasselbalch equation**. Thus, the pH of a solution can be calculated, provided K_a and the concentrations of the weak acid HA and its conjugate base A^- are known. Note particularly that when $[\text{HA}] = [\text{A}^-]$, $\text{pH} = \text{p}K_a$. For example, if equal volumes of 0.1 *M* HAc and 0.1 *M* sodium acetate are mixed, then

$$\text{pH} = \text{p}K_a = 4.76$$

$$\text{p}K_a = -\log K_a = -\log_{10} (1.74 \times 10^{-5}) = 4.76$$

(Sodium acetate, the sodium salt of acetic acid, is a strong electrolyte and dissociates completely in water to yield Na^+ and Ac^- .)

Table 2.4

Acid Dissociation Constants and pK_a Values for Some Weak Electrolytes (at 25°C)

Acid	K_a (M)	pK_a
HCOOH (formic acid)	1.78×10^{-4}	3.75
CH ₃ COOH (acetic acid)	1.74×10^{-5}	4.76
CH ₃ CH ₂ COOH (propionic acid)	1.35×10^{-5}	4.87
CH ₃ CHOHCOOH (lactic acid)	1.38×10^{-4}	3.86
HOOCCH ₂ CH ₂ COOH (succinic acid) pK_1^*	6.16×10^{-5}	4.21
HOOCCH ₂ CH ₂ COO [−] (succinic acid) pK_2	2.34×10^{-6}	5.63
H ₃ PO ₄ (phosphoric acid) pK_1	7.08×10^{-3}	2.15
H ₂ PO ₄ [−] (phosphoric acid) pK_2	6.31×10^{-8}	7.20
HPO ₄ ^{2−} (phosphoric acid) pK_3	3.98×10^{-13}	12.40
C ₃ N ₂ H ₅ ⁺ (imidazole)	1.02×10^{-7}	6.99
C ₆ O ₂ N ₃ H ₁₁ ⁺ (histidine–imidazole group) $pK_R^†$	9.12×10^{-7}	6.04
H ₂ CO ₃ (carbonic acid) pK_1	1.70×10^{-4}	3.77
HCO ₃ [−] (bicarbonate) pK_2	5.75×10^{-11}	10.24
(HOCH ₂) ₃ CNH ₃ ⁺ (<i>tris</i> -hydroxymethyl aminomethane)	8.32×10^{-9}	8.07
NH ₄ ⁺ (ammonium)	5.62×10^{-10}	9.25
CH ₃ NH ₃ ⁺ (methylammonium)	2.46×10^{-11}	10.62

*These pK values listed as pK_1 , pK_2 , or pK_3 are in actuality pK_a values for the respective dissociations. This simplification in notation is used throughout this book.
† pK_R refers to the imidazole ionization of histidine.
Data from *CRC Handbook of Biochemistry*, The Chemical Rubber Co., 1968.

The Henderson–Hasselbalch equation provides a general solution to the quantitative treatment of acid–base equilibria in biological systems. Table 2.4 gives the acid dissociation constants and pK_a values for some weak electrolytes of biochemical interest.

EXAMPLE

What is the pH when 100 mL of 0.1 *N* NaOH is added to 150 mL of 0.2 *M* HAc if pK_a for acetic acid = 4.76?

ANSWER
100 mL 0.1 *N* NaOH = 0.01 mol OH[−], which neutralizes 0.01 mol of HAc, giving an equivalent amount of Ac[−]:



0.02 mol of the original 0.03 mol of HAc remains essentially undissociated. The final volume is 250 mL.

$$pH = pK_a + \log_{10} \frac{[Ac^-]}{[HAc]} = 4.76 + \log (0.01 \text{ mol}) / (0.02 \text{ mol})$$

$$pH = 4.76 - \log_{10} 2 = 4.46$$

If 150 mL of 0.2 *M* HAc had merely been diluted with 100 mL of water, this would leave 250 mL of a 0.12 *M* HAc solution. The pH would be given by:

$$K_a = \frac{[H^+][Ac^-]}{[HAc]} = \frac{x^2}{0.12\text{ M}} = 1.74 \times 10^{-5}$$

$$x = 1.44 \times 10^{-3} = [H^+]$$

$$\text{pH} = 2.84$$

Clearly, the presence of sodium hydroxide has mostly neutralized the acidity of the acetic acid through formation of acetate ion.

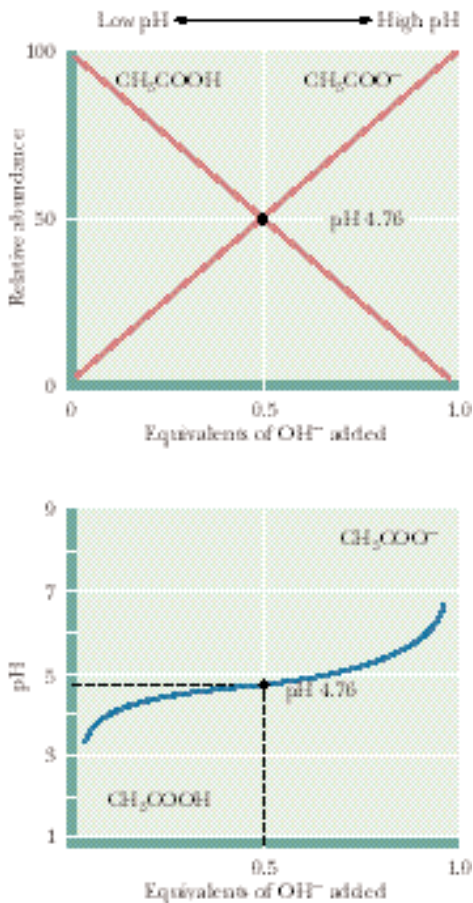


FIGURE 2.12 • The titration curve for acetic acid. Note that the titration curve is relatively flat at pH values near the pK_a ; in other words, the pH changes relatively little as OH^- is added in this region of the titration curve.

Titration Curves

Titration is the analytical method used to determine the amount of acid in a solution. A measured volume of the acid solution is titrated by slowly adding a solution of base, typically NaOH, of known concentration. As incremental amounts of NaOH are added, the pH of the solution is determined and a plot of the pH of the solution versus the amount of OH^- added yields a **titration curve**. The titration curve for acetic acid is shown in Figure 2.12. In considering the progress of this titration, keep in mind two important equilibria:

- $HAc \rightleftharpoons H^+ + Ac^- \quad K_a = 1.74 \times 10^{-5}$
- $H^+ + OH^- \longrightarrow H_2O \quad K = \frac{[H_2O]}{K_w} = 5.55 \times 10^{15}$

As the titration begins, mostly HAc is present, plus some H^+ and Ac^- in amounts that can be calculated (see the Example on page 45). Addition of a solution of NaOH allows hydroxide ions to neutralize any H^+ present. Note that reaction (2) as written is strongly favored; its apparent equilibrium constant is greater than 10^{15} ! As H^+ is neutralized, more HAc dissociates to H^+ and Ac^- . As further NaOH is added, the pH gradually increases as Ac^- accumulates at the expense of diminishing HAc and the neutralization of H^+ . At the point where half of the HAc has been neutralized, that is, where 0.5 equivalent of OH^- has been added, the concentrations of HAc and Ac^- are equal and $pH = pK_a$ for HAc. Thus, we have an experimental method for determining the pK_a values of weak electrolytes. These pK_a values lie at the midpoint of their respective titration curves. After all of the acid has been neutralized (that is, when one equivalent of base has been added), the pH rises exponentially.

The shapes of the titration curves of weak electrolytes are identical, as Figure 2.13 reveals. Note, however, that the midpoints of the different curves vary in a way that characterizes the particular electrolytes. The pK_a for acetic acid is 4.76, the pK_a for imidazole is 6.99, and that for ammonium is 9.25. These pK_a values are directly related to the dissociation constants of these substances, or, viewed the other way, to the relative affinities of the conjugate bases for protons. NH_3 has a high affinity for protons compared to Ac^- ; NH_4^+ is a poor acid compared to HAc.

Phosphoric Acid Has Three Dissociable H^+

Figure 2.14 shows the titration curve for phosphoric acid, H_3PO_4 . This substance is a *polyprotic acid*, meaning it has more than one dissociable proton. Indeed, it has three, and thus three equivalents of OH^- are required to neutralize it, as Figure 2.14 shows. Note that the three dissociable H^+ are lost in discrete steps, each dissociation showing a characteristic pK_a . Note that pK_1 occurs at $pH = 2.15$, and the concentrations of the acid H_3PO_4 and the conjugate base $H_2PO_4^-$ are equal. As the next dissociation is approached, $H_2PO_4^-$

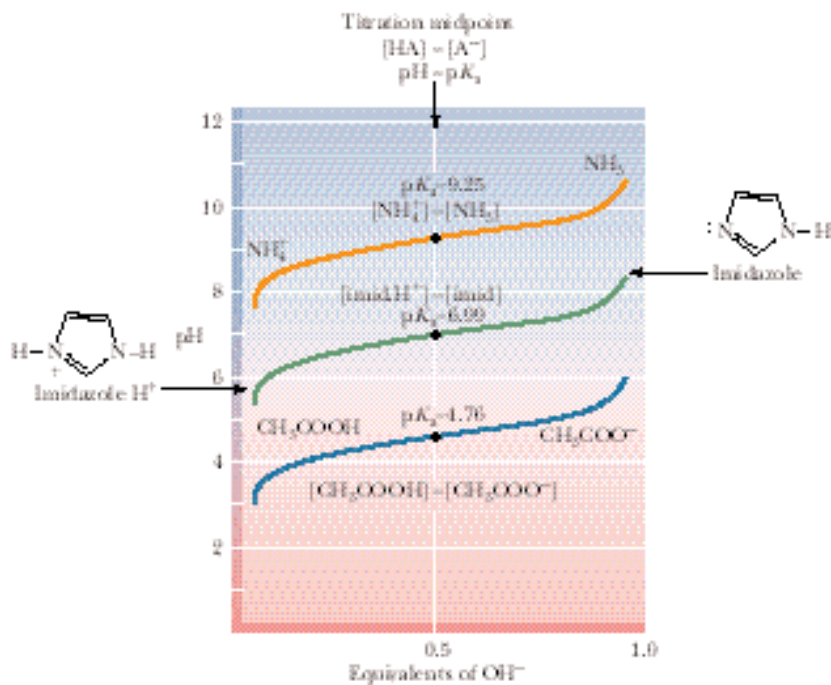


FIGURE 2.13 • The titration curves of several weak electrolytes: acetic acid, imidazole, and ammonium. Note that the shape of these different curves is identical. Only their position along the pH scale is displaced, in accordance with their respective affinities for H^+ ions, as reflected in their differing pK_a values.

is treated as the acid and HPO_4^{2-} is its conjugate base. Their concentrations are equal at pH 7.20, so $pK_2 = 7.20$. (Note that at this point, 1.5 equivalents of OH^- have been added.) As more OH^- is added, the last dissociable hydrogen is titrated, and pK_3 occurs at pH = 12.4, where $[HPO_4^{2-}] = [PO_4^{3-}]$.

A biologically important point is revealed by the basic shape of the titration curves of weak electrolytes: in the region of the pK_a , pH remains relatively unaffected as increments of OH^- (or H^+) are added. The weak acid and its conjugate base are acting as a buffer.

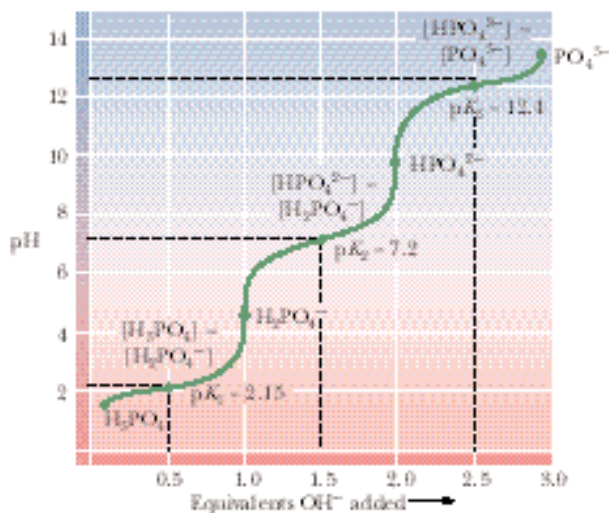


FIGURE 2.14 • The titration curve for phosphoric acid. The chemical formulas show the prevailing ionic species present at various pH values. Phosphoric acid (H_3PO_4) has three titratable hydrogens and therefore three midpoints are seen: at pH 2.15 (pK_1), pH 7.20 (pK_2), and pH 12.4 (pK_3).

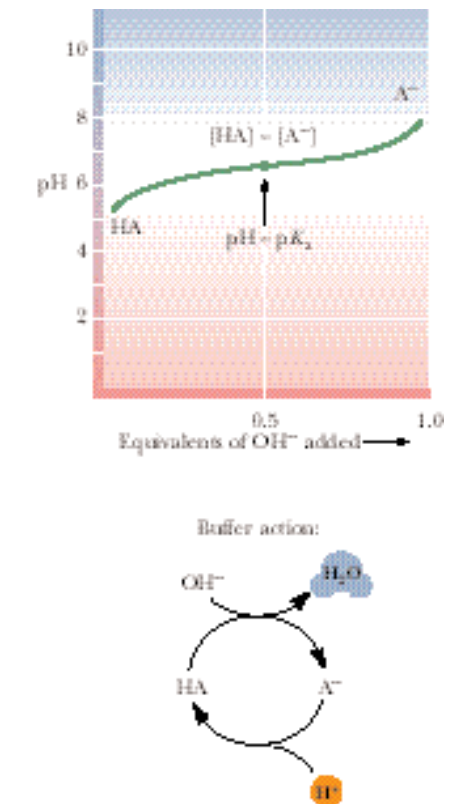
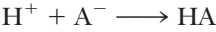


FIGURE 2.15 • A buffer system consists of a weak acid, HA, and its conjugate base, A⁻. The pH varies only slightly in the region of the titration curve where [HA] = [A⁻]. The unshaded box denotes this area of greatest buffering capacity. Buffer action: when HA and A⁻ are both available in sufficient concentration, the solution can absorb input of either H⁺ or OH⁻, and pH is maintained essentially constant.

2.3 • Buffers

Buffers are solutions that tend to resist changes in their pH as acid or base is added. Typically, a buffer system is composed of a weak acid *and* its conjugate base. A solution of a weak acid that has a pH nearly equal to its pK_a by definition contains an amount of the conjugate base nearly equivalent to the weak acid. Note that in this region, the titration curve is relatively flat (Figure 2.15). Addition of H⁺ then has little effect because it is absorbed by the following reaction:



Similarly, added OH⁻ is consumed by the process



The pH then remains relatively constant. The components of a buffer system are chosen such that the pK_a of the weak acid is close to the pH of interest. It is at the pK_a that the buffer system shows its greatest buffering capacity. At pH values more than one pH unit from the pK_a, buffer systems become ineffective because the concentration of one of the components is too low to absorb the influx of H⁺ or OH⁻. The molarity of a buffer is defined as the *sum* of the concentrations of the acid and conjugate base forms.

Maintenance of pH is vital to all cells. Cellular processes such as metabolism are dependent on the activities of enzymes, and in turn, enzyme activity is markedly influenced by pH, as the graphs in Figure 2.16 show. Consequently, changes in pH would be very disruptive to metabolism for reasons that become apparent in later chapters. Organisms have a variety of mechanisms to keep the pH of their intra- and extracellular fluids essentially constant, but the primary protection against harmful pH changes is provided by buffer systems. The buffer systems selected reflect both the need for a pK_a value near pH 7 and the compatibility of the buffer components with the metabolic machinery of cells. Two buffer systems act to maintain intracellular pH essentially constant—the phosphate (HPO₄²⁻/H₂PO₄⁻) system and the histidine system. The pH of the extracellular fluid that bathes the cells and tissues of animals is maintained by the bicarbonate/carbonic acid (HCO₃⁻/H₂CO₃) system.

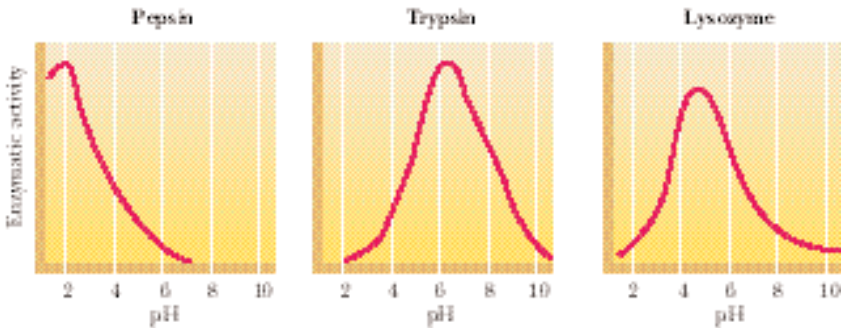


FIGURE 2.16 • pH versus enzymatic activity. The activity of enzymes is very sensitive to pH. The pH optimum of an enzyme is one of its most important characteristics. Pepsin is a protein-digesting enzyme active in the gastric fluid. Trypsin is also a proteolytic enzyme, but it acts in the more alkaline milieu of the small intestine. Lysozyme digests the cell walls of bacteria; it is found in tears.

Phosphate System

The **phosphate system** serves to buffer the intracellular fluid of cells at physiological pH because pK_2 lies near this pH value. The intracellular pH of most cells is maintained in the range between 6.9 and 7.4. Phosphate is an abundant anion in cells, both in inorganic form and as an important functional group on organic molecules that serve as metabolites or macromolecular precursors. In both organic and inorganic forms, its characteristic pK_2 means that the ionic species present at physiological pH are sufficient to donate or accept hydrogen ions to buffer any changes in pH, as the titration curve for H_3PO_4 in Figure 2.14 reveals. For example, if the total cellular concentration of phosphate is 20 mM (millimolar) and the pH is 7.4, the distribution of the major phosphate species is given by

$$pH = pK_2 + \log_{10} \frac{[HPO_4^{2-}]}{[H_2PO_4^-]}$$

$$7.4 = 7.20 + \log_{10} \frac{[HPO_4^{2-}]}{[H_2PO_4^-]}$$

$$\frac{[HPO_4^{2-}]}{[H_2PO_4^-]} = 1.58$$

Thus, if $[HPO_4^{2-}] + [H_2PO_4^-] = 20\text{ mM}$, then

$$[HPO_4^{2-}] = 12.25\text{ mM} \quad \text{and} \quad [H_2PO_4^-] = 7.75\text{ mM}$$

Histidine System

Histidine is one of the 20 naturally occurring amino acids commonly found in proteins (see Chapter 4). It possesses as part of its structure an imidazole group, a five-membered heterocyclic ring possessing two nitrogen atoms. The pK_a for dissociation of the imidazole hydrogen of histidine is 6.04.



In cells, histidine occurs as the free amino acid, as a constituent of proteins, and as part of dipeptides in combination with other amino acids. Because the concentration of free histidine is low and its imidazole pK_a is more than 1 pH unit removed from prevailing intracellular pH, its role in intracellular buffering is minor. However, protein-bound and dipeptide histidine may be the dominant buffering system in some cells. In combination with other amino acids, as in proteins or dipeptides, the imidazole pK_a may increase substantially. For example, the imidazole pK_a is 7.04 in **anserine**, a dipeptide containing β -alanine and histidine (Figure 2.17). Thus, this pK_a is near physiological pH, and some histidine peptides are well suited for buffering at physiological pH.

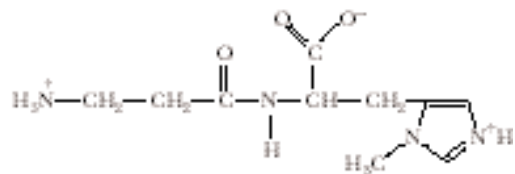
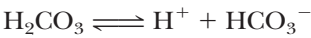


FIGURE 2.17 • Anserine (*N*- β -alanyl-3-methyl-L-histidine) is an important dipeptide buffer in the maintenance of intracellular pH in some tissues. The structure shown is the predominant ionic species at pH 7. pK_1 (COOH) = 2.64; pK_2 (imidazole- N^+H) = 7.04; pK_3 (NH_3^+) = 9.49.

The Bicarbonate Buffer System of Blood Plasma

The important buffer system of blood plasma is the bicarbonate/carbonic acid couple:



The relevant pK_a , pK_1 for carbonic acid, has a value far removed from the normal pH of blood plasma (pH 7.4). (The pK_1 for H_2CO_3 at 25°C is 3.77 (Table 2.4), but at 37°C, pK_1 is 3.57.) At pH 7.4, the concentration of H_2CO_3 is a minuscule fraction of the HCO_3^- concentration, and thus the plasma appears to be poorly protected against an influx of OH^- ions.

$$\text{pH} = 7.4 = 3.57 + \log_{10} \frac{[\text{HCO}_3^-]}{[\text{H}_2\text{CO}_3]}$$

$$\frac{[\text{HCO}_3^-]}{[\text{H}_2\text{CO}_3]} = 6761$$

For example, if $[\text{HCO}_3^-] = 24 \text{ mM}$, then $[\text{H}_2\text{CO}_3]$ is only $3.55 \text{ }\mu\text{M}$ ($3.55 \times 10^{-6} \text{ M}$), and an equivalent amount of OH^- (its usual concentration in plasma) would swamp the buffer system, causing a dangerous rise in the plasma pH. How, then, can this bicarbonate system function effectively? The bicarbonate buffer system works well because the critical concentration of H_2CO_3 is maintained relatively constant through equilibrium with dissolved CO_2 produced in the tissues and available as a gaseous CO_2 reservoir in the lungs.²

“Good” Buffers

Not many common substances have pK_a values in the range from 6 to 8. Consequently, biochemists conducting *in vitro* experiments were limited in their choice of buffers effective at or near physiological pH. In 1966, N.E. Good devised a set of synthetic buffers to remedy this problem, and over the years the list has expanded so that a “good” selection is available (Figure 2.18).

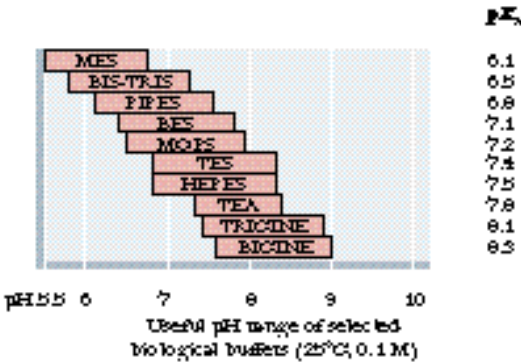


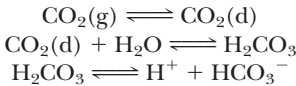
FIGURE 2.18 • The pK_a values and pH range of some “good” buffers.

²Well-fed human adults exhale about 1 kg of CO_2 daily. Imagine the excretory problem if CO_2 were not a volatile gas!

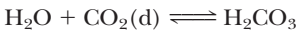
DEEPEERLOOK

How the Bicarbonate Buffer System Works

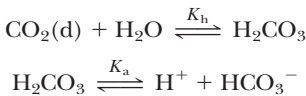
Gaseous carbon dioxide from the lungs and tissues is dissolved in the blood plasma, symbolized as CO₂(d), and hydrated to form H₂CO₃:



Thus, the concentration of H₂CO₃ is itself buffered by the available pools of CO₂. The hydration of CO₂ is actually mediated by an enzyme, *carbonic anhydrase*, which facilitates the equilibrium by rapidly catalyzing the reaction



Under the conditions of temperature and ionic strength prevailing in mammalian body fluids, the equilibrium for this reaction lies far to the left, such that about 500 CO₂ molecules are present in solution for every molecule of H₂CO₃. Because dissolved CO₂ and H₂CO₃ are in equilibrium, the proper expression for H₂CO₃ availability is [CO₂(d)] + [H₂CO₃], the so-called total carbonic acid pool, consisting primarily of CO₂(d). The overall equilibrium for the bicarbonate buffer system then is



An expression for the ionization of H₂CO₃ under such conditions (that is, in the presence of dissolved CO₂) can be obtained from K_h, the equilibrium constant for the hydration of CO₂, and from K_a, the first acid dissociation constant for H₂CO₃:

$$K_h = \frac{[\text{H}_2\text{CO}_3]}{[\text{CO}_2(\text{d})]}$$

Thus,

$$[\text{H}_2\text{CO}_3] = K_h[\text{CO}_2(\text{d})]$$

Putting this value for [H₂CO₃] into the expression for the first dissociation of H₂CO₃ gives

$$K_a = \frac{[\text{H}^+][\text{HCO}_3^-]}{[\text{H}_2\text{CO}_3]}$$

$$= \frac{[\text{H}^+][\text{HCO}_3^-]}{K_h[\text{CO}_2(\text{d})]}$$

Therefore, the overall equilibrium constant for the ionization of H₂CO₃ in equilibrium with CO₂(d) is given by

$$K_aK_h = \frac{[\text{H}^+][\text{HCO}_3^-]}{[\text{CO}_2(\text{d})]}$$

and K_aK_h, the product of two constants, can be defined as a new equilibrium constant, **K_{overall}**. The value of K_h is 0.003 at 37°C and K_a, the ionization constant for H₂CO₃, is 10^{−3.57} = 0.000269. Therefore,

$$\begin{aligned} K_{\text{overall}} &= (0.000269)(0.003) \\ &= 8.07 \times 10^{-7} \\ \text{p}K_{\text{overall}} &= 6.1 \end{aligned}$$

which yields the following Henderson–Hasselbalch relationship:

$$\text{pH} = \text{p}K_{\text{overall}} + \log_{10} \frac{[\text{HCO}_3^-]}{[\text{CO}_2(\text{d})]}$$

Although the prevailing blood pH of 7.4 is more than 1 pH unit away from pK_{overall}, the bicarbonate system is still an effective buffer. That is, at blood pH, the concentration of the acid component of the buffer is less than 10% of the conjugate base component. One might imagine that this buffer component could be overwhelmed by relatively small amounts of alkali, with consequent disastrous rises in blood pH. However, the acid component is the total carbonic acid pool, that is, [CO₂(d)] + [H₂CO₃], which is stabilized by its equilibrium with CO₂(g). The gaseous CO₂ buffers any losses from the total carbonic acid pool by entering solution as CO₂(d), and blood pH is effectively maintained. Thus, the bicarbonate buffer system is an *open system*. The natural presence of CO₂ gas at a partial pressure of 40 mm Hg in the alveoli of the lungs and the equilibrium



keep the concentration of CO₂(d) (the principal component of the total carbonic acid pool in blood plasma) in the neighborhood of 1.2 mM. Plasma [HCO₃[−]] is about 24 mM under such conditions.

2.4 • Water’s Unique Role in the Fitness of the Environment

The remarkable properties of water render it particularly suitable to its unique role in living processes and the environment, and its presence in abundance favors the existence of life. Let’s examine water’s physical and chemical properties to see the extent to which they provide conditions that are advantageous to organisms.

As a *solvent*, water is powerful yet innocuous. No other chemically inert solvent compares with water for the substances it can dissolve. Also, it is very impor-

Blood pH and Respiration

Hyperventilation, defined as a breathing rate more rapid than necessary for normal CO_2 elimination from the body, can result in an inappropriately low $[\text{CO}_2(\text{g})]$ in the blood. Central nervous system disorders such as meningitis, encephalitis, or cerebral hemorrhage, as well as a number of drug- or hormone-induced physiological changes, can lead to hyperventilation. As $[\text{CO}_2(\text{g})]$ drops due to excessive exhalation, $[\text{H}_2\text{CO}_3]$ in the blood plasma falls, followed by decline in $[\text{H}^+]$ and $[\text{HCO}_3^-]$ in the blood plasma. Blood pH rises within 20 sec of the onset of hyperventilation, becoming maximal within 15 min. $[\text{H}^+]$ can change from

its normal value of 40 nM (pH = 7.4) to 18 nM (pH = 7.74). This rise in plasma pH (increase in alkalinity) is termed **respiratory alkalosis**.

Hypoventilation is the opposite of hyperventilation and is characterized by an inability to excrete CO_2 rapidly enough to meet physiological needs. Hypoventilation can be caused by narcotics, sedatives, anesthetics, and depressant drugs; diseases of the lung also lead to hypoventilation. Hypoventilation results in **respiratory acidosis**, as $\text{CO}_2(\text{g})$ accumulates, giving rise to H_2CO_3 , which dissociates to form H^+ and HCO_3^- .

tant to life that water is a “poor” solvent for nonpolar substances. Thus, through hydrophobic interactions, lipids coalesce, membranes form, boundaries are created delimiting compartments, and the cellular nature of life is established. Because of its very high dielectric constant, water is a medium for ionization. Ions enrich the living environment in that they enhance the variety of chemical species and introduce an important class of chemical reactions. They provide electrical properties to solutions and therefore to organisms. Aqueous solutions are the prime source of ions.

The *thermal properties* of water are especially relevant to its environmental fitness. It has great power as a buffer resisting thermal (temperature) change. Its heat capacity, or specific heat (4.1840 J/g°C), is remarkably high; it is ten times greater than iron, five times greater than quartz or salt, and twice as great as hexane. Its heat of fusion is 335 J/g. Thus, at 0°C, it takes a loss of 335 J to change the state of 1 g of H_2O from liquid to solid. Its heat of vaporization, 2.24 kJ/g, is exceptionally high. These thermal properties mean that it takes substantial changes in heat content to alter the temperature and especially the *state* of water. Water’s thermal properties allow it to buffer the climate through such processes as condensation, evaporation, melting, and freezing. Furthermore, these properties allow effective temperature regulation in living organisms. For example, heat generated within an organism as a result of metabolism can be efficiently eliminated by evaporation or conduction. The thermal conductivity of water is very high in comparison with other liquids. The anomalous expansion of water as it cools to temperatures near its freezing point is a unique attribute of great significance to its natural fitness. As water cools, H bonding increases because the thermal motions of the molecules are lessened. Hydrogen bonding tends to separate the water molecules (Figure 2.2), and thus the density of water decreases. These changes in density mean that, at temperatures below 4°C, cool water rises and, most importantly, ice freezes on the surface of bodies of water, forming an insulating layer protecting the liquid water underneath.

Water has the highest *surface tension* (75 dyne/cm) of all common liquids (except mercury). Together, surface tension and density determine how high a liquid rises in a capillary system. Capillary movement of water plays a prominent role in the life of plants. Lastly, consider *osmosis*, the bulk movement of water in the direction from a dilute aqueous solution to a more concentrated one across a semipermeable boundary. Such bulk movements determine the shape and form of living things.

Water is truly a crucial determinant of the fitness of the environment. In a very real sense, organisms are aqueous systems in a watery world.

PROBLEMS

- Calculate the pH of the following.
 - $5 \times 10^{-4} \text{ M HCl}$
 - $7 \times 10^{-5} \text{ M NaOH}$
 - $2 \text{ }\mu\text{M HCl}$
 - $3 \times 10^{-2} \text{ M KOH}$
 - 0.04 mM HCl
 - $6 \times 10^{-9} \text{ M HCl}$
- Calculate the following from the pH values given in Table 2.3.
 - $[\text{H}^+]$ in vinegar
 - $[\text{H}^+]$ in saliva
 - $[\text{H}^+]$ in household ammonia
 - $[\text{OH}^-]$ in milk of magnesia
 - $[\text{OH}^-]$ in beer
 - $[\text{H}^+]$ inside a liver cell
- The pH of a 0.02 M solution of an acid was measured at 4.6.
 - What is the $[\text{H}^+]$ in this solution?
 - Calculate the acid dissociation constant K_a and $\text{p}K_a$ for this acid.
- The K_a for formic acid is $1.78 \times 10^{-4} \text{ M}$.
 - What is the pH of a 0.1 M solution of formic acid?
 - 150 mL of 0.1 M NaOH is added to 200 mL of 0.1 M formic acid, and water is added to give a final volume of 1 L. What is the pH of the final solution?
- Given 0.1 M solutions of acetic acid and sodium acetate, describe the preparation of 1 L of 0.1 M acetate buffer at a pH of 5.4.
- If the internal pH of a muscle cell is 6.8, what is the $[\text{HPO}_4^{2-}]/[\text{H}_2\text{PO}_4^-]$ ratio in this cell?
- Given 0.1 M solutions of Na_3PO_4 and H_3PO_4 , describe the preparation of 1 L of a phosphate buffer at a pH of 7.5. What are the molar concentrations of the ions in the final buffer solution, including Na^+ and H^+ ?

FURTHER READING

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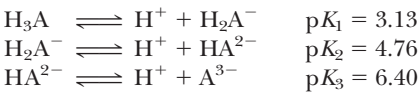
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- BICINE is a compound containing a tertiary amino group whose relevant $\text{p}K_a$ is 8.3 (Figure 2.18). Given 1 L of 0.05 M BICINE with its tertiary amino group in the unprotonated form, how much 0.1 N HCl must be added to have a BICINE buffer solution of pH 7.5? What is the molarity of BICINE in the final buffer?
- What are the approximate fractional concentrations of the following phosphate species at pH values of 0, 2, 4, 6, 8, 10, and 12?
 - H_3PO_4
 - H_2PO_4^-
 - HPO_4^{2-}
 - PO_4^{3-}
- Citric acid, a tricarboxylic acid important in intermediary metabolism, can be symbolized as H_3A . Its dissociation reactions are



If the *total* concentration of the acid *and* its anion forms is 0.02 M , what are the individual concentrations of H_3A , H_2A^- , HA^{2-} , and A^{3-} at pH 5.2?

11. a. If 50 mL of 0.01 M HCl is added to 100 mL of 0.05 M phosphate buffer at pH 7.2, what is the resultant pH? What are the concentrations of H_2PO_4^- and HPO_4^{2-} in the final solution?

b. If 50 mL of 0.01 M NaOH is added to 100 mL of 0.05 M phosphate buffer at pH 7.2, what is the resultant pH? What are the concentrations of H_2PO_4^- and HPO_4^{2-} in this final solution?

12. If the plasma pH is 7.4 and the plasma concentration of HCO_3^- is 15 mM , what is the plasma concentration of H_2CO_3 ? What is the plasma concentration of $\text{CO}_{2(\text{dissolved})}$? If metabolic activity changes the concentration of $\text{CO}_{2(\text{dissolved})}$ to 3 mM , and $[\text{HCO}_3^-]$ remains at 15 mM , what is the pH of the plasma?

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A theory is the more impressive the greater is the simplicity of its premises, the more different are the kinds of things it relates and the more extended is its range of applicability. Therefore, the deep impression which classical thermodynamics made upon me. It is the only physical theory of universal content which I am convinced, that within the framework of applicability of its basic concepts, will never be overthrown.

ALBERT EINSTEIN

OUTLINE

- 3.1 • Basic Thermodynamic Concepts
- 3.2 • The Physical Significance of Thermodynamic Properties
- 3.3 • The Effects of pH on Standard-State Free Energies
- 3.4 • The Important Effect of Concentration on Net Free Energy Changes
- 3.5 • The Importance of Coupled Processes in Living Things
- 3.6 • The High-Energy Biomolecules
- 3.7 • Complex Equilibria Involved in ATP Hydrolysis
- 3.8 • The Daily Human Requirement for ATP

Chapter 3

Thermodynamics of Biological Systems



Sun emblem of Louis XIV on a gate at Versailles. The sun is the prime source of energy for life, and thermodynamics is the gateway to understanding metabolism. (Giraudon/Art Research, New York)

The activities of living things require energy. Movement, growth, synthesis of biomolecules, and the transport of ions and molecules across membranes all demand energy input. All organisms must acquire energy from their surroundings and must utilize that energy efficiently to carry out life processes. To study such bioenergetic phenomena requires familiarity with **thermodynamics**, a collection of laws and principles describing the flows and interchanges of heat, energy, and matter in systems of interest. Thermodynamics also allows us to determine whether or not chemical processes and reactions occur spontaneously. The student should appreciate the power and practical

value of thermodynamic reasoning and realize that this is well worth the effort needed to understand it.

Even the most complicated aspects of thermodynamics are based ultimately on three rather simple and straightforward laws. These laws and their extensions sometimes run counter to our intuition. However, once truly understood, the basic principles of thermodynamics become powerful devices for sorting out complicated chemical and biochemical problems. At this milestone in our scientific development, thermodynamic thinking becomes an enjoyable and satisfying activity.

Several basic thermodynamic principles are presented in this chapter, including the analysis of heat flow, entropy production, and free energy functions and the relationship between entropy and information. In addition, some ancillary concepts are considered, including the concept of standard states, the effect of pH on standard-state free energies, the effect of concentration on the net free energy change of a reaction, and the importance of coupled processes in living things. The chapter concludes with a discussion of ATP and other energy-rich compounds.

3.1 • Basic Thermodynamic Concepts

In any consideration of thermodynamics, a distinction must be made between the system and the surroundings. The **system** is that portion of the *universe* with which we are concerned, whereas the **surroundings** include everything else in the universe (Figure 3.1). The nature of the system must also be specified. There are three basic systems: isolated, closed, and open. An **isolated system** cannot exchange matter or energy with its surroundings. A **closed system** may exchange energy, but not matter, with the surroundings. An **open system** may exchange matter, energy, or both with the surroundings. Living things are typically open systems that exchange matter (nutrients and waste products) and heat (from metabolism, for example) with their surroundings.

The First Law: Heat, Work, and Other Forms of Energy

It was realized early in the development of thermodynamics that heat could be converted into other forms of energy, and moreover that all forms of energy could ultimately be converted to some other form. The **first law of thermodynamics** states that *the total energy of an isolated system is conserved*. Thermodynamicists have formulated a mathematical function for keeping track of heat transfers and work expenditures in thermodynamic systems. This function is called the **internal energy**, commonly designated ***E*** or ***U***. The internal energy depends only on the present state of a system and hence is referred to as a **state function**. The internal energy does not depend on how the system got there and is thus **independent of path**. An extension of this thinking is that we can manipulate the system through any possible pathway of changes, and as long as the system returns to the original state, the internal energy, *E*, will not have been changed by these manipulations.

The internal energy, *E*, of any system can change only if energy flows in or out of the system in the form of heat or work. For any process that converts one state (state 1) into another (state 2), the change in internal energy, ΔE , is given as

$$\Delta E = E_2 - E_1 = q + w \tag{3.1}$$

where the quantity *q* is the *heat absorbed by the system from the surroundings*, and *w* is the *work done on the system by the surroundings*. **Mechanical work** is defined

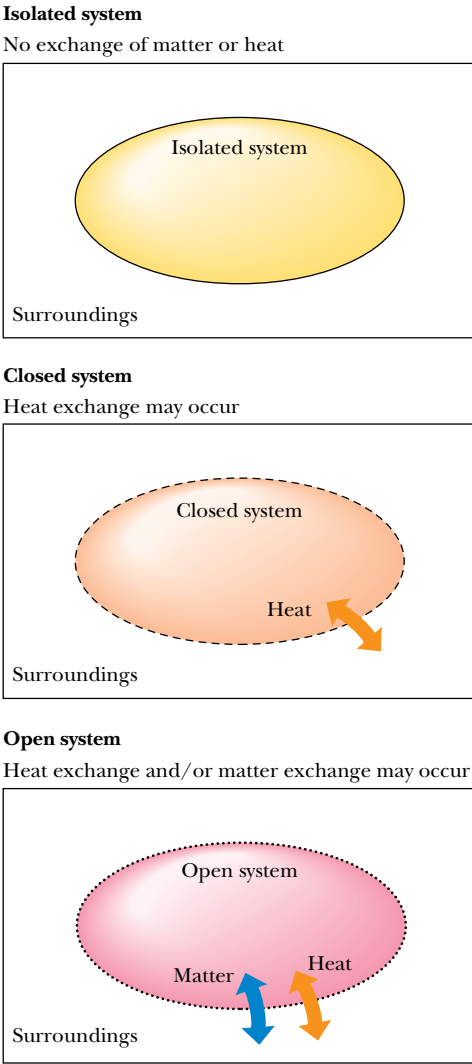


FIGURE 3.1 • The characteristics of isolated, closed, and open systems. Isolated systems exchange neither matter nor energy with their surroundings. Closed systems may exchange energy, but not matter, with their surroundings. Open systems may exchange either matter or energy with the surroundings.

as *movement through some distance caused by the application of a force*. Both of these must occur for work to have occurred. For example, if a person strains to lift a heavy weight but fails to move the weight at all, then, in the thermodynamic sense, no work has been done. (The energy expended in the muscles of the would-be weight lifter is given off in the form of heat.) In chemical and biochemical systems, work is often concerned with the pressure and volume of the system under study. The mechanical work done on the system is defined as $w = -P\Delta V$, where P is the pressure and ΔV is the volume change and is equal to $V_2 - V_1$. When work is defined in this way, the sign on the right side of Equation (3.1) is positive. (Sometimes w is defined as work done *by* the system; in this case, the equation is $\Delta E = q - w$.) Work may occur in many forms, such as mechanical, electrical, magnetic, and chemical. ΔE , q , and w must all have the same units. The **calorie**, abbreviated **cal**, and **kilocalorie (kcal)**, have been traditional choices of chemists and biochemists, but the SI unit, the **joule**, is now recommended.

Enthalpy: A More Useful Function for Biological Systems

If the definition of work is limited to mechanical work, an interesting simplification is possible. In this case, ΔE is merely the *heat exchanged at constant volume*. This is so because if the volume is constant, no mechanical work can be done on or by the system. Then $\Delta E = q$. Thus ΔE is a very useful quantity in constant volume processes. However, chemical and especially biochemical processes and reactions are much more likely to be carried out at constant pressure. In constant pressure processes, ΔE is not necessarily equal to the heat transferred. For this reason, chemists and biochemists have defined a function that is especially suitable for constant pressure processes. It is called the **enthalpy**, H , and it is defined as

$$H = E + PV \tag{3.2}$$

The clever nature of this definition is not immediately apparent. However, if the pressure is constant, then we have

$$\Delta H = \Delta E + P\Delta V = q + w + P\Delta V = q - P\Delta V + P\Delta V = q \tag{3.3}$$

Clearly, ΔH is equal to the heat transferred in a constant pressure process. Often, because biochemical reactions normally occur in liquids or solids rather than in gases, volume changes are small and *enthalpy and internal energy are often essentially equal*.

In order to compare the thermodynamic parameters of different reactions, it is convenient to define a *standard state*. For solutes in a solution, the standard state is normally unit activity (often simplified to 1 *M* concentration). Enthalpy, internal energy, and other thermodynamic quantities are often given or determined for standard-state conditions and are then denoted by a superscript degree sign (“°”), as in ΔH° , ΔE° , and so on.

Enthalpy changes for biochemical processes can be determined experimentally by measuring the heat absorbed (or given off) by the process in a *calorimeter* (Figure 3.2). Alternatively, for any process $A \rightleftharpoons B$ at equilibrium, the standard-state enthalpy change for the process can be determined from the temperature dependence of the equilibrium constant:

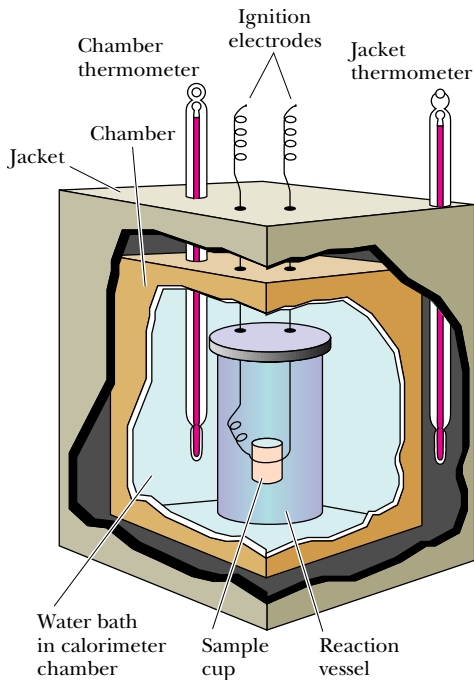


FIGURE 3.2 • Diagram of a calorimeter. The reaction vessel is completely submerged in a water bath. The heat evolved by a reaction is determined by measuring the rise in temperature of the water bath.

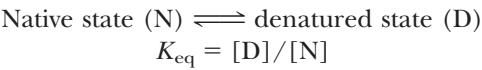
$$\Delta H^\circ = -R \frac{d(\ln K_{\text{eq}})}{d(1/T)}$$

(3.4)

Here R is the *gas constant*, defined as $R = 8.314 \text{ J/mol} \cdot \text{K}$. A plot of $R(\ln K_{\text{eq}})$ versus $1/T$ is called a **van't Hoff plot**.

EXAMPLE

In a study¹ of the temperature-induced reversible denaturation of the protein chymotrypsinogen,



John F. Brandts measured the equilibrium constants for the denaturation over a range of pH and temperatures. The data for pH 3:

$T(\text{K})$:	324.4	326.1	327.5	329.0	330.7	332.0	333.8
K_{eq} :	0.041	0.12	0.27	0.68	1.9	5.0	21

A plot of $R(\ln K_{\text{eq}})$ versus $1/T$ (a van't Hoff plot) is shown in Figure 3.3. ΔH° for the denaturation process at any temperature is the negative of the slope of the plot at that temperature. As shown, ΔH° at 54.5°C (327.5 K) is

$$\Delta H^\circ = -[-3.2 - (-17.6)] / [(3.04 - 3.067) \times 10^{-3}] = +533 \text{ kJ/mol}$$

What does this value of ΔH° mean for the unfolding of the protein? Positive values of ΔH° would be expected for the breaking of hydrogen bonds as well as for the exposure of hydrophobic groups from the interior of the native, folded protein during the unfolding process. Such events would raise the energy of the protein–water solution. The magnitude of this enthalpy change (533 kJ/mol) at 54.5°C is large, compared to similar values of ΔH° for other proteins and for this same protein at 25°C (Table 3.1). If we consider only this positive enthalpy change for the unfolding process, the native, folded state is strongly favored. As we shall see, however, other parameters must be taken into account.

Table 3.1

Thermodynamic Parameters for Protein Denaturation

Protein (and conditions)	ΔH° kJ/mol	ΔS° kJ/mol · K	ΔG° kJ/mol	ΔC_p kJ/mol · K
Chymotrypsinogen (pH 3, 25°C)	164	0.440	31	10.9
β -Lactoglobulin (5 M urea, pH 3, 25°C)	−88	−0.300	2.5	9.0
Myoglobin (pH 9, 25°C)	180	0.400	57	5.9
Ribonuclease (pH 2.5, 30°C)	240	0.780	3.8	8.4

Adapted from Cantor, C., and Schimmel, P., 1980. *Biophysical Chemistry*. San Francisco: W.H. Freeman, and Tanford, C., 1968. Protein denaturation. *Advances in Protein Chemistry* **23**:121–282.

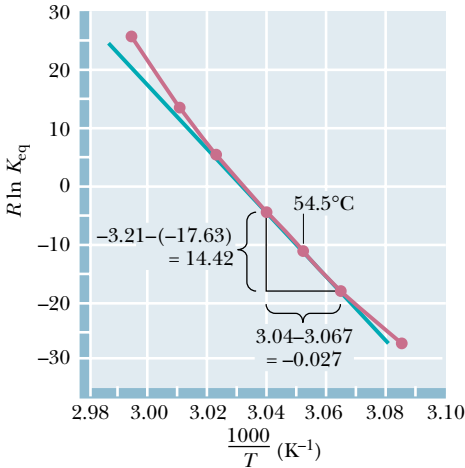


FIGURE 3.3 • The enthalpy change, ΔH° , for a reaction can be determined from the slope of a plot of $R \ln K_{\text{eq}}$ versus $1/T$. To illustrate the method, the values of the data points on either side of the 327.5 K (54.5°C) data point have been used to calculate ΔH° at 54.5°C. Regression analysis would normally be preferable. (Adapted from Brandts, J. F., 1964. *The thermodynamics of protein denaturation. I. The denaturation of chymotrypsinogen*. *Journal of the American Chemical Society* **86**:4291–4301.)

¹Brandts, J. F., 1964. The thermodynamics of protein denaturation. I. The denaturation of chymotrypsinogen. *Journal of the American Chemical Society* **86**:4291–4301.

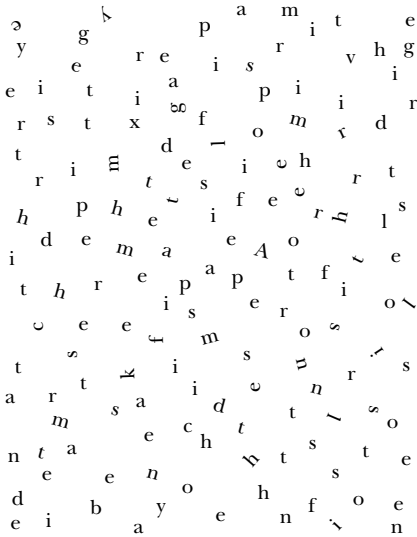
A D E E P E R L O O K

Entropy, Information, and the Importance of “Negentropy”

When a thermodynamic system undergoes an increase in entropy, it becomes more disordered. On the other hand, a decrease in entropy reflects an increase in order. A more ordered system is more highly organized and possesses a greater information content. To appreciate the implications of decreasing the entropy of a system, consider the random collection of letters in the figure. This disorganized array of letters possesses no inherent information content, and nothing can be learned by its perusal. On the other hand, this particular array of letters can be systematically arranged to construct the first sentence of the Einstein quotation that opened this chapter: “A theory is the more impressive the greater is the simplicity of its premises, the more different are the kinds of things it relates and the more extended is its range of applicability.”

Arranged in this way, this same collection of 151 letters possesses enormous information content—the profound words of a great scientist. Just as it would have required significant effort to rearrange these 151 letters in this way, so large amounts of energy are required to construct and maintain living organisms. Energy input is required to produce information-rich, organized structures such as proteins and nucleic acids. Information content can be thought of as *negative entropy*. In 1945 Erwin Schrödinger took time out from his studies of quantum mechanics to publish a delightful book entitled *What is Life?* In it, Schrödinger coined the term *negentropy* to describe the negative entropy changes that

confer organization and information content to living organisms. Schrödinger pointed out that organisms must “acquire negentropy” to sustain life.



The Second Law and Entropy: An Orderly Way of Thinking About Disorder

The **second law of thermodynamics** has been described and expressed in many different ways, including the following.

- 1. Systems tend to proceed from *ordered* (*low entropy* or *low probability*) states to *disordered* (*high entropy* or *high probability*) states.
- 2. The *entropy* of the system plus surroundings is unchanged by *reversible processes*; the entropy of the system plus surroundings increases for *irreversible processes*.
- 3. All naturally occurring processes proceed toward **equilibrium**, that is, to a state of minimum potential energy.

Several of these statements of the second law invoke the concept of **entropy**, which is a measure of disorder and randomness in the system (or the surroundings). An organized or ordered state is a low-entropy state, whereas a disordered state is a high-entropy state. All else being equal, reactions involving large, positive entropy changes, ΔS , are more likely to occur than reactions for which ΔS is not large and positive.

Entropy can be defined in several quantitative ways. If W is the number of ways to arrange the components of a system without changing the internal

energy or enthalpy (that is, the number of microscopic states at a given temperature, pressure, and amount of material), then the entropy is given by

$$S = k \ln W \quad (3.5)$$

where k is Boltzmann's constant ($k = 1.38 \times 10^{-23} \text{ J/K}$). This definition is useful for statistical calculations (it is in fact a foundation of *statistical thermodynamics*), but a more common form relates entropy to the heat transferred in a process:

$$dS_{\text{reversible}} = \frac{dq}{T} \quad (3.6)$$

where $dS_{\text{reversible}}$ is the entropy change of the system in a reversible² process, q is the heat transferred, and T is the temperature at which the heat transfer occurs.

The Third Law: Why Is “Absolute Zero” So Important?

The **third law of thermodynamics** states that the entropy of any crystalline, perfectly ordered substance must approach zero as the temperature approaches 0 K, and at $T = 0 \text{ K}$ *entropy is exactly zero*. Based on this, it is possible to establish a quantitative, absolute entropy scale for any substance as

$$S = \int_0^T C_P d \ln T \quad (3.7)$$

where C_P is the *heat capacity* at constant pressure. The heat capacity of any substance is the amount of heat one mole of it can store as the temperature of that substance is raised by one degree. For a constant pressure process, this is described mathematically as

$$C_P = \frac{dH}{dT} \quad (3.8)$$

If the heat capacity can be evaluated at all temperatures between 0 K and the temperature of interest, an absolute entropy can be calculated. For biological processes, *entropy changes* are more useful than absolute entropies. The entropy change for a process can be calculated if the enthalpy change and *free energy change* are known.

Free Energy: A Hypothetical but Useful Device

An important question for chemists, and particularly for biochemists, is, “Will the reaction proceed in the direction written?” J. Willard Gibbs, one of the founders of thermodynamics, realized that the answer to this question lay in a comparison of the enthalpy change and the entropy change for a reaction at a given temperature. The **Gibbs free energy**, G , is defined as

$$G = H - TS \quad (3.9)$$

For any process $A \rightleftharpoons B$ at constant pressure and temperature, the *free energy change* is given by

$$\Delta G = \Delta H - T\Delta S \quad (3.10)$$

²A reversible process is one that can be reversed by an infinitesimal modification of a variable.

If ΔG is equal to 0, the process is at *equilibrium*, and there is no net flow either in the forward or reverse direction. When $\Delta G = 0$, $\Delta S = \Delta H/T$, and the enthalpic and entropic changes are exactly balanced. Any process with a nonzero ΔG proceeds spontaneously to a final state of lower free energy. If ΔG is negative, the process proceeds spontaneously in the direction written. If ΔG is positive, the reaction or process proceeds spontaneously in the reverse direction. (The sign and value of ΔG do not allow us to determine *how fast* the process will go.) If the process has a negative ΔG , it is said to be **exergonic**, whereas processes with positive ΔG values are **endergonic**.

The Standard-State Free Energy Change

The free energy change, ΔG , for any reaction depends upon the nature of the reactants and products, but it is also affected by the conditions of the reaction, including temperature, pressure, pH, and the concentrations of the reactants and products. As explained earlier, it is useful to define a standard state for such processes. If the free energy change for a reaction is sensitive to solution conditions, what is the particular significance of the standard-state free energy change? To answer this question, consider a reaction between two reactants A and B to produce the products C and D.



The free energy change for non-standard-state concentrations is given by

$$\Delta G = \Delta G^\circ + RT \ln \frac{[C][D]}{[A][B]} \tag{3.12}$$

At equilibrium, $\Delta G = 0$ and $[C][D]/[A][B] = K_{eq}$. We then have

$$\Delta G^\circ = -RT \ln K_{eq} \tag{3.13}$$

or, in base 10 logarithms,

$$\Delta G^\circ = -2.3RT \log_{10} K_{eq} \tag{3.14}$$

This can be rearranged to

$$K_{eq} = 10^{-\Delta G^\circ/2.3RT} \tag{3.15}$$

In any of these forms, this relationship allows the standard-state free energy change for any process to be determined if the equilibrium constant is known. More importantly, it states that *the equilibrium established for a reaction in solution is a function of the standard-state free energy change for the process*. That is, ΔG° is another way of writing an equilibrium constant.

EXAMPLE

The equilibrium constants determined by Brandts at several temperatures for the denaturation of chymotrypsinogen (see previous Example) can be used to calculate the free energy changes for the denaturation process. For example, the equilibrium constant at 54.5°C is 0.27, so

$$\Delta G^\circ = -(8.314 \text{ J/mol} \cdot \text{K})(327.5 \text{ K}) \ln (0.27)$$

$$\Delta G^\circ = -(2.72 \text{ kJ/mol}) \ln (0.27)$$

$$\Delta G^\circ = 3.56 \text{ kJ/mol}$$

The positive sign of ΔG° means that the unfolding process is unfavorable; that is, the stable form of the protein at 54.5°C is the folded form. On the other hand, the relatively small magnitude of ΔG° means that the folded form is only slightly favored. Figure 3.4 shows the dependence of ΔG° on temperature for

the denaturation data at pH 3 (from the data given in the Example on page 59).

Having calculated both ΔH° and ΔG° for the denaturation of chymotrypsinogen, we can also calculate ΔS° , using Equation (3.10):

$$\Delta S^\circ = -\frac{(\Delta G^\circ - \Delta H^\circ)}{T} \tag{3.16}$$

At 54.5°C (327.5 K),

$$\Delta S^\circ = -(3560 - 533,000 \text{ J/mol})/327.5 \text{ K}$$

$$\Delta S^\circ = 1,620 \text{ J/mol} \cdot \text{K}$$

Figure 3.5 presents the dependence of ΔS° on temperature for chymotrypsinogen denaturation at pH 3. A positive ΔS° indicates that the protein solution has become more disordered as the protein unfolds. Comparison of the value of 1.62 kJ/mol·K with the values of ΔS° in Table 3.1 shows that the present value (for chymotrypsinogen at 54.5°C) is quite large. The physical significance of the thermodynamic parameters for the unfolding of chymotrypsinogen becomes clear in the next section.

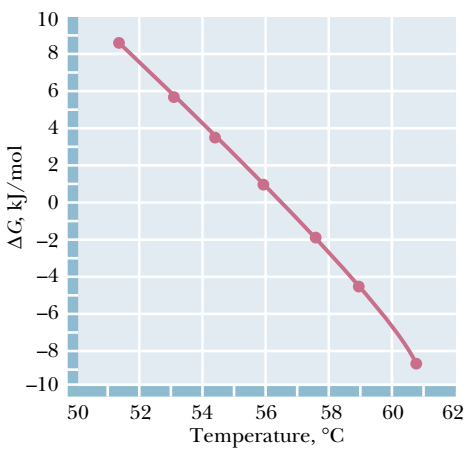


FIGURE 3.4 • The dependence of ΔG° on temperature for the denaturation of chymotrypsinogen. (Adapted from Brandts, J. F., 1964. *The thermodynamics of protein denaturation. I. The denaturation of chymotrypsinogen.* Journal of the American Chemical Society **86**:4291–4301.)

3.2 • The Physical Significance of Thermodynamic Properties

What can thermodynamic parameters tell us about biochemical events? The best answer to this question is that a single parameter (ΔH or ΔS , for example) is not very meaningful. A positive ΔH° for the unfolding of a protein might reflect *either* the breaking of hydrogen bonds within the protein or the exposure of hydrophobic groups to water (Figure 3.6). However, *comparison of several thermodynamic parameters can provide meaningful insights about a process.* For example, the transfer of Na^+ and Cl^- ions from the gas phase to aqueous solution involves a very large negative ΔH° (thus a very favorable stabilization of the ions) and a comparatively small ΔS° (Table 3.2). The negative entropy term reflects the ordering of water molecules in the hydration shells of the Na^+ and Cl^- ions. This unfavorable effect is more than offset by the large heat of hydration, which makes the hydration of ions a very favorable process overall. The negative entropy change for the dissociation of acetic acid in water also reflects the ordering of water molecules in the ion hydration shells. In this case, however, the enthalpy change is much smaller in magnitude. As a result, ΔG° for

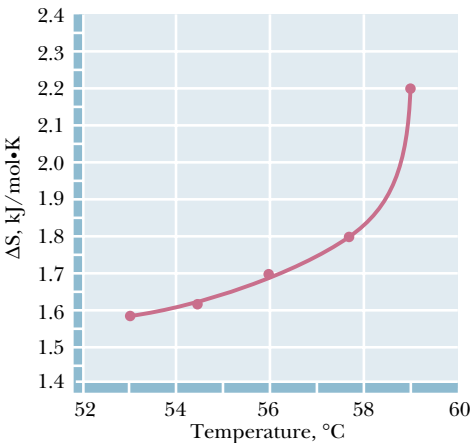


FIGURE 3.5 • The dependence of ΔS° on temperature for the denaturation of chymotrypsinogen. (Adapted from Brandts, J. F., 1964. *The thermodynamics of protein denaturation. I. The denaturation of chymotrypsinogen.* Journal of the American Chemical Society **86**:4291–4301.)

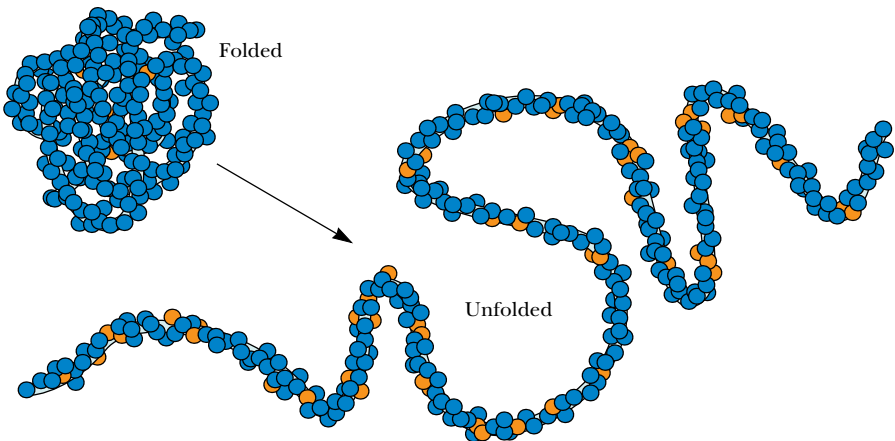


FIGURE 3.6 • Unfolding of a soluble protein exposes significant numbers of nonpolar groups to water, forcing order on the solvent and resulting in a negative ΔS° for the unfolding process. Yellow spheres represent nonpolar groups; blue spheres are polar and/or charged groups.

Table 3.2
Thermodynamic Parameters for Several Simple Processes*

Process	ΔH° kJ/mol	ΔS° kJ/mol · K	ΔG° kJ/mol	ΔC_P kJ/mol · K
Hydration of ions [†] $\text{Na}^+(\text{g}) + \text{Cl}^-(\text{g}) \longrightarrow \text{Na}^+(\text{aq}) + \text{Cl}^-(\text{aq})$	−760.0	−0.185	−705.0	
Dissociation of ions in solution [‡] $\text{H}_2\text{O} + \text{CH}_3\text{COOH} \longrightarrow \text{H}_3\text{O}^+ + \text{CH}_3\text{COO}^-$	−10.3	−0.126	27.26	−0.143
Transfer of hydrocarbon from pure liquid to water [‡] Toluene (in pure toluene) $\longrightarrow$ toluene (aqueous)	1.72	−0.071	22.7	0.265

*All data collected for 25°C.
[†]Berry, R. S., Rice, S. A., and Ross, J., 1980. *Physical Chemistry*. New York: John Wiley.
[‡]Tanford, C., 1980. *The Hydrophobic Effect*. New York: John Wiley.

dissociation of acetic acid in water is positive, and acetic acid is thus a weak (largely undissociated) acid.

The transfer of a nonpolar hydrocarbon molecule from its pure liquid to water is an appropriate model for the exposure of protein hydrophobic groups to solvent when a protein unfolds. The transfer of toluene from liquid toluene to water involves a negative ΔS° , a positive ΔG° , and a ΔH° that is small compared to ΔG° (a pattern similar to that observed for the dissociation of acetic acid). *What distinguishes these two very different processes is the change in heat capacity* (Table 3.2). A positive heat capacity change for a process indicates that the molecules have acquired new ways to move (and thus to store heat energy). A negative ΔC_P means that the process has resulted in less freedom of motion for the molecules involved. ΔC_P is negative for the dissociation of acetic acid and positive for the transfer of toluene to water. The explanation is that polar and nonpolar molecules *both* induce organization of nearby water molecules, *but in different ways*. The water molecules near a nonpolar solute are *organized but labile*. Hydrogen bonds formed by water molecules near nonpolar solutes rearrange more rapidly than the hydrogen bonds of pure water. On the other hand, the hydrogen bonds formed between water molecules near an ion are less labile (rearrange more slowly) than they would be in pure water. This means that ΔC_P should be negative for the dissociation of ions in solution, as observed for acetic acid (Table 3.2).

3.3 • The Effect of pH on Standard-State Free Energies

For biochemical reactions in which hydrogen ions (H^+) are consumed or produced, the usual definition of the standard state is awkward. Standard state for the H^+ ion is 1 M, which corresponds to pH 0. At this pH, nearly all enzymes would be denatured, and biological reactions could not occur. It makes more sense to use free energies and equilibrium constants determined at pH 7. Biochemists have thus adopted a modified standard state, designated with prime (') symbols, as in $\Delta G^{\circ'}$, K'_{eq} , $\Delta H^{\circ'}$, and so on. For values determined in this way, a standard state of 10^{-7} M H^+ and unit activity (1 M for solutions, 1 atm for gases and pure solids defined as unit activity) for all other components (in the ionic forms that exist at pH 7) is assumed. The two standard states can be related easily. For a reaction in which H^+ is produced,



the relation of the equilibrium constants for the two standard states is

$$K_{\text{eq}}' = K_{\text{eq}}[\text{H}^+] \quad (3.18)$$

and $\Delta G^{\circ'}$ is given by

$$\Delta G^{\circ'} = \Delta G^{\circ} + RT \ln [\text{H}^+] \quad (3.19)$$

For a reaction in which H^+ is consumed,



the equilibrium constants are related by

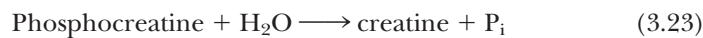
$$K_{\text{eq}}' = \frac{K_{\text{eq}}}{[\text{H}^+]} \quad (3.21)$$

and $\Delta G^{\circ'}$ is given by

$$\Delta G^{\circ'} = \Delta G^{\circ} + RT \ln \left(\frac{1}{[\text{H}^+]} \right) = \Delta G^{\circ} - RT \ln [\text{H}^+] \quad (3.22)$$

3.4 • The Important Effect of Concentration on Net Free Energy Changes

Equation (3.12) shows that the free energy change for a reaction can be very different from the standard-state value if the concentrations of reactants and products differ significantly from unit activity (1 *M* for solutions). The effects can often be dramatic. Consider the hydrolysis of phosphocreatine:



This reaction is strongly exergonic and ΔG° at 37°C is -42.8 kJ/mol . Physiological concentrations of phosphocreatine, creatine, and inorganic phosphate are normally between 1 *mM* and 10 *mM*. Assuming 1 *mM* concentrations and using Equation (3.12), the ΔG for the hydrolysis of phosphocreatine is

$$\Delta G = -42.8 \text{ kJ/mol} + (8.314 \text{ J/mol} \cdot \text{K}) (310 \text{ K}) \ln \left(\frac{[0.001][0.001]}{[0.001]} \right) \quad (3.24)$$

$$\Delta G = -60.5 \text{ kJ/mol} \quad (3.25)$$

At 37°C, the difference between standard-state and 1 *mM* concentrations for such a reaction is thus approximately -17.7 kJ/mol .

3.5 • The Importance of Coupled Processes in Living Things

Many of the reactions necessary to keep cells and organisms alive must run against their **thermodynamic potential**, that is, in the direction of positive ΔG . Among these are the synthesis of adenosine triphosphate and other high-energy molecules and the creation of ion gradients in all mammalian cells. These processes are driven in the thermodynamically unfavorable direction via *coupling* with highly favorable processes. Many such *coupled processes* are discussed later in this text. They are crucially important in intermediary metabolism, oxidative phosphorylation, and membrane transport, as we shall see.

We can predict whether pairs of coupled reactions will proceed spontaneously by simply summing the free energy changes for each reaction. For example, consider the reaction from glycolysis (discussed in Chapter 19)

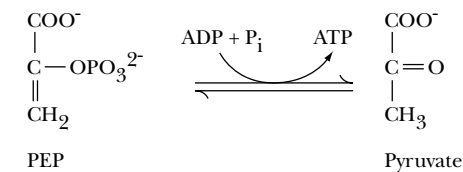


FIGURE 3.7 • The pyruvate kinase reaction.

involving the conversion of phospho(enol)pyruvate (PEP) to pyruvate (Figure 3.7). The hydrolysis of PEP is energetically very favorable, and it is used to drive phosphorylation of ADP to form ATP, a process that is energetically unfavorable. Using values of ΔG that would be typical for a human erythrocyte:

$$\text{PEP} + \text{H}_2\text{O} \longrightarrow \text{pyruvate} + \text{P}_i \qquad \Delta G = -78 \text{ kJ/mol} \qquad (3.26)$$

$$\text{ADP} + \text{P}_i \longrightarrow \text{ATP} + \text{H}_2\text{O} \qquad \Delta G = +55 \text{ kJ/mol} \qquad (3.27)$$

$$\text{PEP} + \text{ADP} \longrightarrow \text{pyruvate} + \text{ATP} \qquad \text{Total } \Delta G = -23 \text{ kJ/mol} \qquad (3.28)$$

The net reaction catalyzed by this enzyme depends upon coupling between the two reactions shown in Equations (3.26) and (3.27) to produce the net reaction shown in Equation (3.28) with a net negative $\Delta G^\circ'$. Many other examples of coupled reactions are considered in our discussions of intermediary metabolism (Part III). In addition, many of the complex biochemical systems discussed in the later chapters of this text involve reactions and processes with positive $\Delta G^\circ'$ values that are driven forward by coupling to reactions with a negative $\Delta G^\circ'$.

3.6 • The High-Energy Biomolecules

Virtually all life on earth depends on energy from the sun. Among life forms, there is a hierarchy of energetics: certain organisms capture solar energy directly, whereas others derive their energy from this group in subsequent processes. Organisms that absorb light energy directly are called **phototrophic organisms**. These organisms store solar energy in the form of various organic molecules. Organisms that feed on these latter molecules, releasing the stored energy in a series of oxidative reactions, are called **chemotrophic organisms**. Despite these differences, both types of organisms share common mechanisms for generating a useful form of chemical energy. Once captured in chemical form, energy can be released in controlled exergonic reactions to drive a variety of life processes (which require energy). A small family of universal biomolecules mediates the flow of energy from exergonic reactions to the energy-requiring processes of life. These molecules are the *reduced coenzymes* and the *high-energy phosphate compounds*. Phosphate compounds are considered high energy if they exhibit large negative free energies of hydrolysis (that is, if $\Delta G^\circ'$ is more negative than -25 kJ/mol).

Table 3.3 lists the most important members of the high-energy phosphate compounds. Such molecules include *phosphoric anhydrides* (ATP, ADP), an *enol phosphate* (PEP), an *acyl phosphate* (acetyl phosphate), and a *guanidino phosphate* (creatine phosphate). Also included are thioesters, such as acetyl-CoA, which do not contain phosphorus, but which have a high free energy of hydrolysis. As noted earlier in this chapter, the exact amount of chemical free energy available from the hydrolysis of such compounds depends on concentration, pH, temperature, and so on, but the $\Delta G^\circ'$ values for hydrolysis of these substances are substantially more negative than for most other metabolic species. Two important points: first, high-energy phosphate compounds are not long-term energy storage substances. They are transient forms of stored energy, meant to carry energy from point to point, from one enzyme system to another, in the minute-to-minute existence of the cell. (As we shall see in subsequent chapters, other molecules bear the responsibility for long-term storage of energy

Table 3.3

Free Energies of Hydrolysis of Some High-Energy Compounds*

Compound (and Hydrolysis Product)	$\Delta G^{\circ'}$ (kJ/mol)	Structure
Phosphoenolpyruvate (pyruvate + P _i)	−62.2	
3',5'-Cyclic adenosine monophosphate (5'-AMP)	−50.4	
1,3-Bisphosphoglycerate (3-phosphoglycerate + P _i)	−49.6	
Creatine phosphate (creatine + P _i)	−43.3	
Acetyl phosphate (acetate + P _i)	−43.3	
Adenosine-5'-triphosphate (ADP + P _i)	−35.7 [†]	
Adenosine-5'-triphosphate (ADP + P _i), excess Mg ²⁺	−30.5	
Adenosine-5'-diphosphate (AMP + P _i)	−35.7	

(continued)

Table 3.3

Continued

Compound (and Hydrolysis Product)	$\Delta G^{\circ'}$ (kJ/mol)	Structure
Pyrophosphate ($P_i + P_i$) in 5 mM Mg^{2+}	−33.6	
Adenosine-5'-triphosphate ($AMP + PP_i$), excess Mg^{2+}	−32.3	(See ATP structure on previous page)
Uridine diphosphoglucose (UDP + glucose)	−31.9	
Acetyl-coenzyme A (acetate + CoA)	−31.5	
S-adenosylmethionine (methionine + adenosine)	−25.6 [‡]	

supplies.) Second, the term *high-energy compound* should not be construed to imply that these molecules are unstable and hydrolyze or decompose unpredictably. ATP, for example, is quite a stable molecule. A substantial activation energy must be delivered to ATP to hydrolyze the terminal, or γ , phosphate group. In fact, as shown in Figure 3.8, the *activation energy* that must be absorbed by the molecule to break the $O-P_\gamma$ bond is normally 200 to 400 kJ/mol, which is substantially larger than the net 30.5 kJ/mol released in the hydrolysis reaction. Biochemists are much more concerned with the *net release* of 30.5 kJ/mol

Compound (and Hydrolysis Product)	ΔG° (kJ/mol)	Structure
Lower-Energy Phosphate Compounds		
Glucose-1-P (glucose + P _i)	−21.0	
Fructose-1-P (fructose + P _i)	−16.0	
Glucose-6-P (glucose + P _i)	−13.9	
sn-Glycerol-3-P (glycerol + P _i)	−9.2	
Adenosine-5'-monophosphate (adenosine + P _i)	−9.2	

*Adapted primarily from *Handbook of Biochemistry and Molecular Biology*, 1976, 3rd ed. In *Physical and Chemical Data*, G. Fasman, ed., Vol. 1, pp. 296–304. Boca Raton, FL: CRC Press.
†From Gwynn, R. W., and Veech, R. L., 1973. The equilibrium constants of the adenosine triphosphate hydrolysis and the adenosine triphosphate-citrate lyase reactions. *Journal of Biological Chemistry* **248**:6966–6972.
‡From Mudd, H., and Mann, J., 1963. Activation of methionine for transmethylation. *Journal of Biological Chemistry* **238**:2164–2170.

than with the activation energy for the reaction (because suitable enzymes cope with the latter). The net release of large quantities of free energy distinguishes the high-energy phosphoric anhydrides from their “low-energy” ester cousins, such as glycerol-3-phosphate (Table 3.3). The next section provides a quantitative framework for understanding these comparisons.

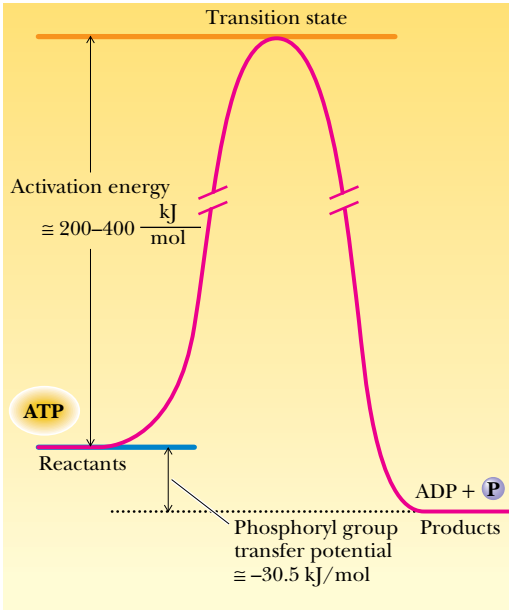


FIGURE 3.8 • The activation energies for phosphoryl group-transfer reactions (200 to 400 kJ/mol) are substantially larger than the free energy of hydrolysis of ATP (−30.5 kJ/mol).

ATP Is an Intermediate Energy-Shuttle Molecule

One last point about Table 3.3 deserves mention. Given the central importance of ATP as a high-energy phosphate in biology, students are sometimes surprised to find that ATP holds an intermediate place in the rank of high-energy phosphates. PEP, cyclic AMP, 1,3-BPG, phosphocreatine, acetyl phosphate, and pyrophosphate all exhibit higher values of $\Delta G^{\circ'}$. This is not a biological anomaly. ATP is uniquely situated between the very high energy phosphates synthesized in the breakdown of fuel molecules and the numerous lower-energy acceptor molecules that are phosphorylated in the course of further metabolic reactions. ADP can accept both phosphates and energy from the higher-energy phosphates, and the ATP thus formed can donate both phosphates and energy to the lower-energy molecules of metabolism. The ATP/ADP pair is an intermediately placed acceptor/donor system among high-energy phosphates. In this context, ATP functions as a very versatile but intermediate energy-shuttle device that interacts with many different energy-coupling enzymes of metabolism.

Group Transfer Potential

Many reactions in biochemistry involve the transfer of a functional group from a donor molecule to a specific receptor molecule or to water. The concept of **group transfer potential** explains the tendency for such reactions to occur. Biochemists define the group transfer potential as the free energy change that occurs upon hydrolysis, that is, upon transfer of the particular group to water. This concept and its terminology are preferable to the more qualitative notion of *high-energy bonds*.

The concept of group transfer potential is not particularly novel. Other kinds of transfer (of hydrogen ions and electrons, for example) are commonly

Table 3.4

Types of Transfer Potential			
	Proton Transfer Potential (Acidity)	Standard Reduction Potential (Electron Transfer Potential)	Group Transfer Potential (High-Energy Bond)
Simple equation	$AH \rightleftharpoons A^- + H^+$	$A \rightleftharpoons A^+ + e^-$	$A \sim P \rightleftharpoons A + P_i$
Equation including acceptor	$AH + H_2O \rightleftharpoons A^- + H_3O^+$	$A + H^+ \rightleftharpoons A^+ + \frac{1}{2} H_2$	$A \sim PO_4^{2-} + H_2O \rightleftharpoons A-OH + HPO_4^{2-}$
Measure of transfer potential	$pK_a = \frac{\Delta G^\circ}{2.303 RT}$	$\Delta \mathcal{E}_o = \frac{-\Delta G^\circ}{n\mathcal{F}}$	$\ln K_{eq} = \frac{-\Delta G^\circ}{RT}$
Free energy change of transfer is given by:	ΔG° per mole of H^+ transferred	ΔG° per mole of e^- transferred	ΔG° per mole of phosphate transferred

Adapted from: Klotz, I. M., 1986. *Introduction to Biomolecular Energetics*. New York: Academic Press.

characterized in terms of appropriate measures of transfer potential (pK_a and reduction potential, $\mathcal{E}_o$, respectively). As shown in Table 3.4, the notion of group transfer is fully analogous to those of ionization potential and reduction potential. The similarity is anything but coincidental, because all of these are really specific instances of free energy changes. If we write



we really don’t mean that a proton has literally been removed from the acid AH. In the gas phase at least, this would require the input of approximately 1200 kJ/mol! What we really mean is that the proton has been *transferred* to a suitable acceptor molecule, usually water:



The appropriate free energy relationship is of course

$pK_a = \frac{\Delta G^\circ}{2.303 RT}$ (3.30)

Similarly, in the case of an oxidation-reduction reaction



we don’t really mean that A oxidizes independently. What we really mean (and what is much more likely in biochemical systems) is that the electron is transferred to a suitable acceptor:



and the relevant free energy relationship is

$\Delta \mathcal{E}_o = \frac{-\Delta G^\circ}{n\mathcal{F}}$ (3.32)

where n is the number of equivalents of electrons transferred, and $\mathcal{F}$ is **Faraday’s constant**.

Similarly, the release of free energy that occurs upon the hydrolysis of ATP and other “high-energy phosphates” can be treated quantitatively in terms of *group transfer*. It is common to write for the hydrolysis of ATP



The free energy change, which we henceforth call the *group transfer potential*, is given by

$$\Delta G^\circ = -RT \ln K_{\text{eq}} \tag{3.34}$$

where K_{eq} is the equilibrium constant for the group transfer, which is normally written as

$$K_{\text{eq}} = \frac{[\text{ADP}][\text{P}_i]}{[\text{ATP}][\text{H}_2\text{O}]} \tag{3.35}$$

Even this set of equations represents an approximation, because ATP, ADP, and P_i all exist in solutions as a mixture of ionic species. This problem is discussed in a later section. For now, it is enough to note that the free energy changes listed in Table 3.3 are the group transfer potentials observed for transfers to water.

Phosphoric Acid Anhydrides

ATP contains two *pyrophosphoryl* or *phosphoric acid anhydride* linkages, as shown in Figure 3.9. Other common biomolecules possessing phosphoric acid anhydride linkages include ADP, GTP, GDP and the other nucleoside triphosphates, sugar nucleotides such as UDP-glucose, and inorganic pyrophosphate itself. All exhibit large negative free energies of hydrolysis, as shown in Table 3.3. The chemical reasons for the large negative $\Delta G^\circ'$ values for the hydrolysis reactions include destabilization of the reactant due to bond strain caused by electrostatic repulsion, stabilization of the products by ionization and resonance, and entropy factors due to hydrolysis and subsequent ionization.

Destabilization Due to Electrostatic Repulsion

Electrostatic repulsion in the reactants is best understood by comparing these phosphoric anhydrides with other reactive anhydrides, such as acetic anhydride. As shown in Figure 3.10a, the electronegative carbonyl oxygen atoms withdraw electrons from the $\text{C}=\text{O}$ bonds, producing partial negative charges on the oxygens and partial positive charges on the carbonyl carbons. Each of

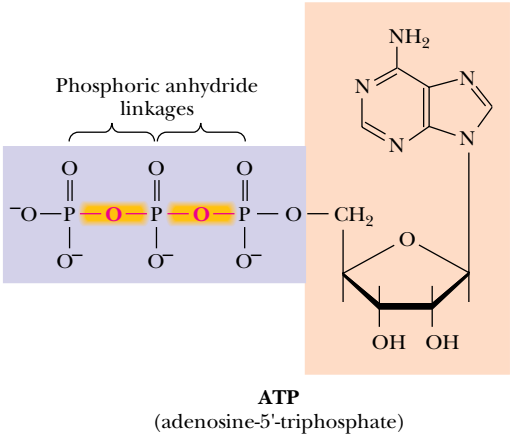


FIGURE 3.9 • The triphosphate chain of ATP contains two pyrophosphate linkages, both of which release large amounts of energy upon hydrolysis.

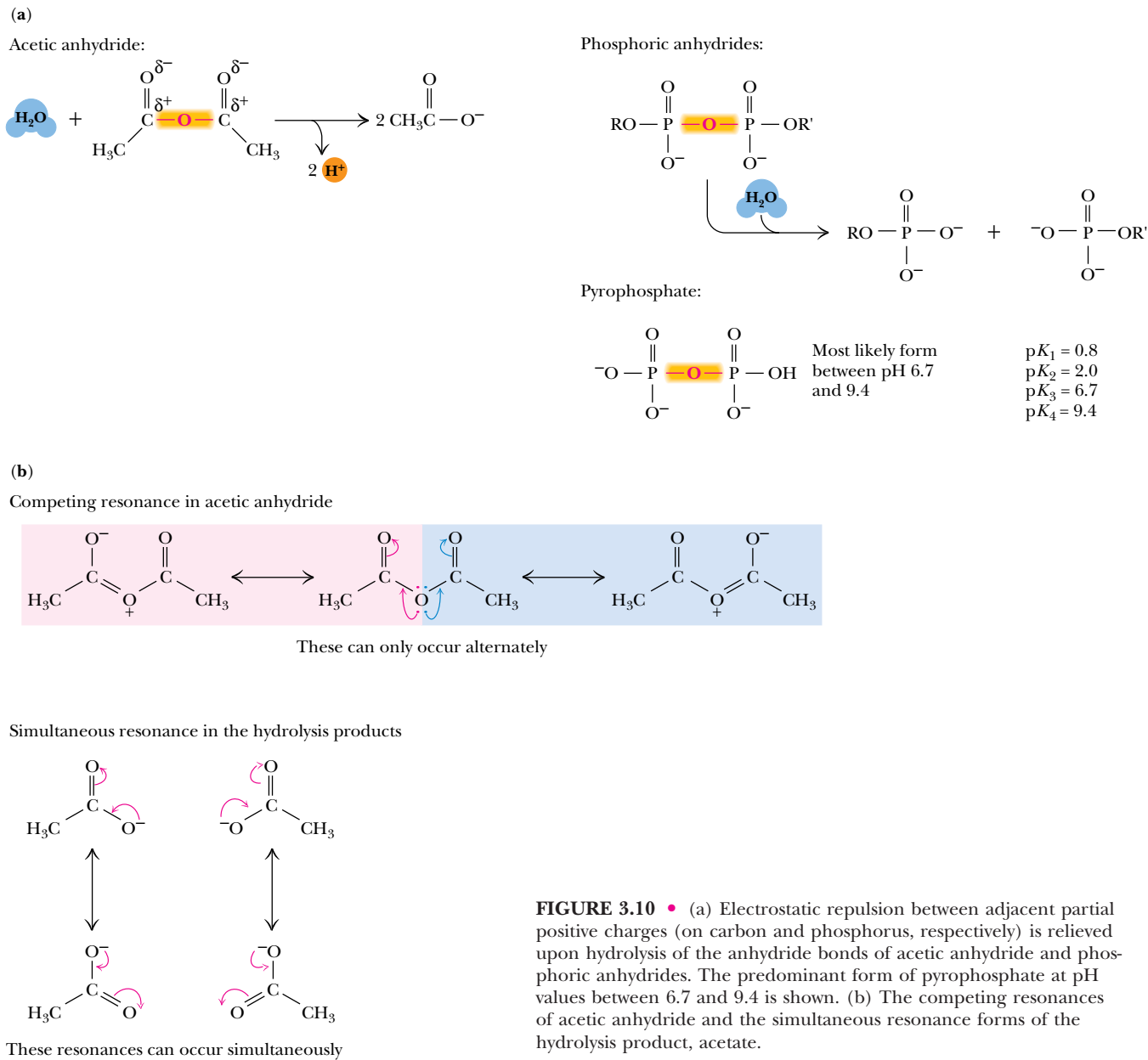


FIGURE 3.10 • (a) Electrostatic repulsion between adjacent partial positive charges (on carbon and phosphorus, respectively) is relieved upon hydrolysis of the anhydride bonds of acetic anhydride and phosphoric anhydrides. The predominant form of pyrophosphate at pH values between 6.7 and 9.4 is shown. (b) The competing resonances of acetic anhydride and the simultaneous resonance forms of the hydrolysis product, acetate.

these electrophilic carbonyl carbons is further destabilized by the other acetyl group, which is also electron-withdrawing in nature. As a result, acetic anhydride is unstable with respect to the products of hydrolysis.

The situation with phosphoric anhydrides is similar. The phosphorus atoms of the pyrophosphate anion are electron-withdrawing and destabilize PP_i with respect to its hydrolysis products. Furthermore, the reverse reaction, reformation of the anhydride bond from the two anionic products, requires that the electrostatic repulsion between these anions be overcome (see following).

Stabilization of Hydrolysis Products by Ionization and Resonance

The pyrophosphate moiety possesses three negative charges at pH values above 7.5 or so (note the pK_a values, Figure 3.10a). The hydrolysis products, two mole-

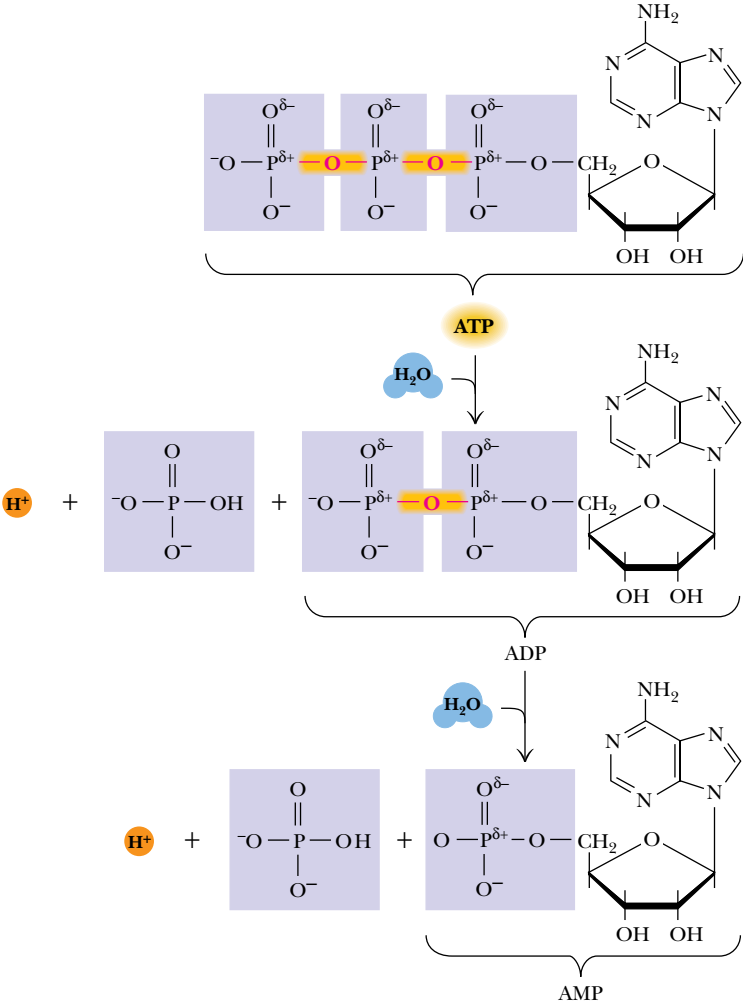
cules of inorganic phosphate, both carry about two negative charges, at pH values above 7.2. The increased ionization of the hydrolysis products helps to stabilize the electrophilic phosphorus nuclei.

Resonance stabilization in the products is best illustrated by the reactant anhydrides (Figure 3.10b). The unpaired electrons of the bridging oxygen atoms in acetic anhydride (and phosphoric anhydride) cannot participate in resonance structures with both electrophilic centers at once. This **competing resonance** situation is relieved in the product acetate or phosphate molecules.

Entropy Factors Arising from Hydrolysis and Ionization

For the phosphoric anhydrides, and for most of the high-energy compounds discussed here, there is an additional “entropic” contribution to the free energy of hydrolysis. Most of the hydrolysis reactions of Table 3.3 result in an increase in the number of molecules in solution. As shown in Figure 3.11, the hydrolysis of ATP (as pH values above 7) creates three species—ADP, inorganic phosphate (P_i), and a hydrogen ion—from only two reactants (ATP and H_2O). The entropy of the solution increases because the more particles, the more disordered the system.³ (This effect is ionization-dependent because, at low pH, the

FIGURE 3.11 • Hydrolysis of ATP to ADP (and/or of ADP to AMP) leads to relief of electrostatic repulsion.



³Imagine the “disorder” created by hitting a crystal with a hammer and breaking it into many small pieces.

hydrogen ion created in many of these reactions simply protonates one of the phosphate oxygens, and one fewer “particle” results from the hydrolysis.)

A Comparison of the Free Energy of Hydrolysis of ATP, ADP, and AMP

The concepts of destabilization of reactants and stabilization of products described for pyrophosphate also apply for ATP and other phosphoric anhydrides (Figure 3.11). ATP and ADP are destabilized relative to the hydrolysis products by electrostatic repulsion, competing resonance, and entropy. AMP, on the other hand, is a phosphate ester (not an anhydride) possessing only a single phosphoryl group and is not markedly different from the product inorganic phosphate in terms of electrostatic repulsion and resonance stabilization. Thus, the ΔG° for hydrolysis of AMP is much smaller than the corresponding values for ATP and ADP.

Phosphoric–Carboxylic Anhydrides

The mixed anhydrides of phosphoric and carboxylic acids, frequently called acyl phosphates, are also energy-rich. Two biologically important acyl phosphates are acetyl phosphate and 1,3-bisphosphoglycerate. Hydrolysis of these species yields acetate and 3-phosphoglycerate, respectively, in addition to inorganic phosphate (Figure 3.12). Once again, the large $\Delta G^{\circ'}$ values indicate that the reactants are destabilized relative to products. This arises from bond strain, which can be traced to the partial positive charges on the carbonyl carbon and phosphorus atoms of these structures. The energy stored in the mixed anhydride bond (which is required to overcome the charge-charge repulsion) is released upon hydrolysis. Increased resonance possibilities in the products relative to the reactants also contribute to the large negative $\Delta G^{\circ'}$ values. The

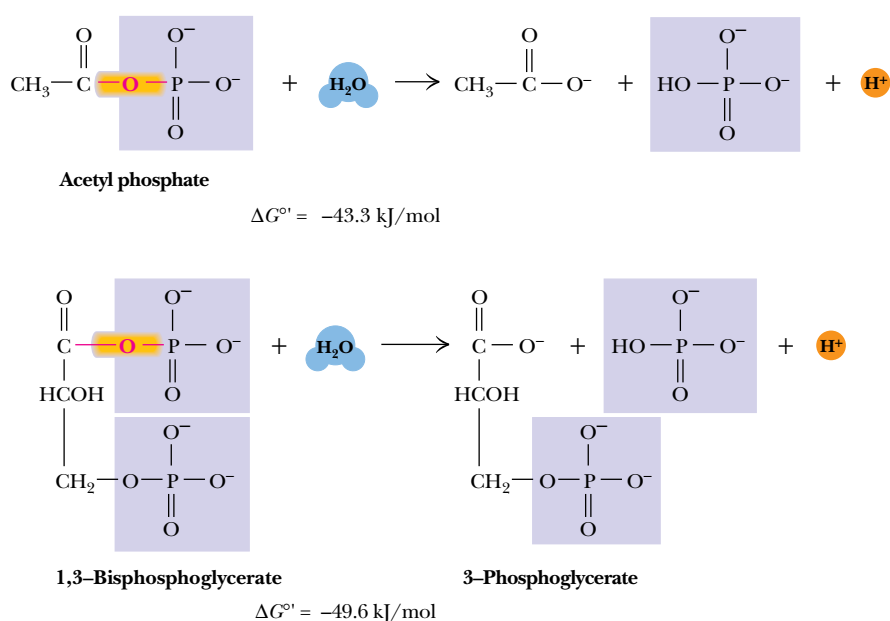


FIGURE 3.12 • The hydrolysis reactions of acetyl phosphate and 1,3-bisphosphoglycerate.

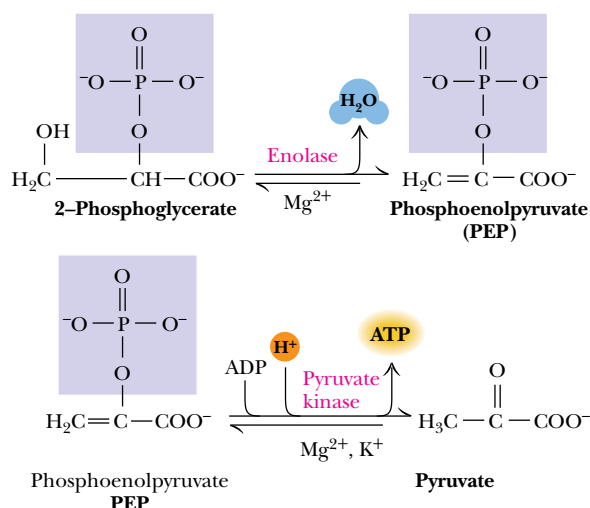


FIGURE 3.13 • Phosphoenolpyruvate (PEP) is produced by the enolase reaction (in glycolysis; see Chapter 19) and in turn drives the phosphorylation of ADP to form ATP in the pyruvate kinase reaction.

value of $\Delta G^{\circ'}$ depends on the pK_a values of the starting anhydride and the product phosphoric and carboxylic acids, and of course also on the pH of the medium.

Enol Phosphates

The largest value of $\Delta G^\circ'$ in Table 3.3 belongs to *phosphoenolpyruvate* or *PEP*, an example of an enolic phosphate. This molecule is an important intermediate in carbohydrate metabolism and, due to its large negative $\Delta G^\circ'$, it is a potent phosphorylating agent. PEP is formed via dehydration of 2-phosphoglycerate by enolase during fermentation and glycolysis. PEP is subsequently transformed into pyruvate upon transfer of its phosphate to ADP by pyruvate kinase (Figure 3.13). The very large negative value of $\Delta G^\circ'$ for the latter reaction is to a large extent the result of a secondary reaction of the *enol* form of pyruvate. Upon hydrolysis, the unstable enolic form of pyruvate immediately converts to the keto form with a resulting large negative $\Delta G^\circ'$ (Figure 3.14). Together, the hydrolysis and subsequent *tautomerization* result in an overall $\Delta G^\circ'$ of -62.2 kJ/mol .

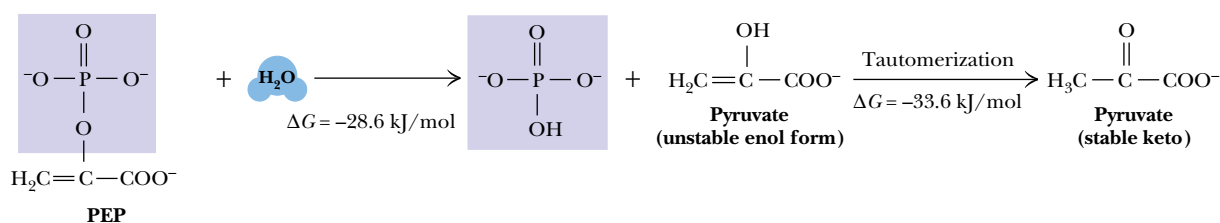


FIGURE 3.14 • Hydrolysis and the subsequent tautomerization account for the very large $\Delta G^{\circ'}$ of PEP.

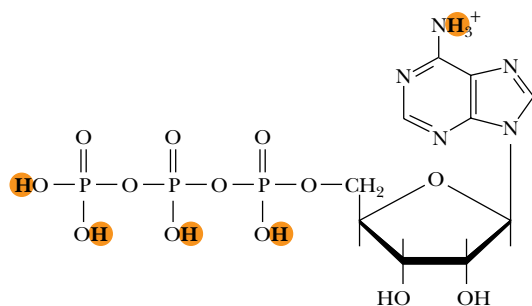


FIGURE 3.15 • Adenosine 5'-triphosphate (ATP).

Color indicates the locations of the five dissociable protons of ATP.

3.7 • Complex Equilibria Involved in ATP Hydrolysis

So far, as in Equation (3.33), the hydrolyses of ATP and other high-energy phosphates have been portrayed as simple processes. The situation in a real biological system is far more complex, owing to the operation of several ionic equilibria. First, ATP, ADP, and the other species in Table 3.3 can exist in several different ionization states that must be accounted for in any quantitative analysis. Second, phosphate compounds bind a variety of divalent and monovalent cations with substantial affinity, and the various metal complexes must also be considered in such analyses. Consideration of these special cases makes the quantitative analysis far more realistic. The importance of these multiple equilibria in group transfer reactions is illustrated for the hydrolysis of ATP, but the principles and methods presented are general and can be applied to any similar hydrolysis reaction.

The Multiple Ionization States of ATP and the pH Dependence of ΔG°

ATP has five dissociable protons, as indicated in Figure 3.15. Three of the protons on the triphosphate chain dissociate at very low pH. The adenine ring amino group exhibits a pK_a of 4.06, whereas the last proton to dissociate from the triphosphate chain possesses a pK_a of 6.95. At higher pH values, ATP is completely deprotonated. ADP and phosphoric acid also undergo multiple ionizations. These multiple ionizations make the equilibrium constant for ATP hydrolysis more complicated than the simple expression in Equation (3.35). Multiple ionizations must also be taken into account when the pH dependence of ΔG° is considered. The calculations are beyond the scope of this text, but Figure 3.16 shows the variation of ΔG° as a function of pH. The free energy of hydrolysis is nearly constant from pH 4 to pH 6. At higher values of pH, ΔG° varies linearly with pH, becoming more negative by 5.7 kJ/mol for every pH unit of increase at 37°C. Because the pH of most biological tissues and fluids is near neutrality, the effect on ΔG° is relatively small, but it must be taken into account in certain situations.

The Effect of Metal Ions on the Free Energy of Hydrolysis of ATP

Most biological environments contain substantial amounts of divalent and monovalent metal ions, including Mg^{2+} , Ca^{2+} , Na^+ , K^+ , and so on. What effect do metal ions have on the equilibrium constant for ATP hydrolysis and the

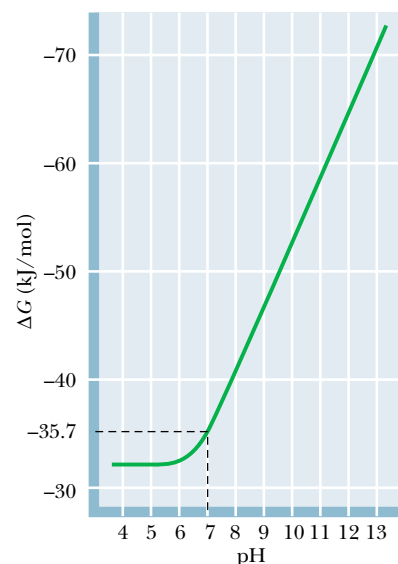


FIGURE 3.16 • The pH dependence of the free energy of hydrolysis of ATP. Because pH varies only slightly in biological environments, the effect on ΔG is usually small.

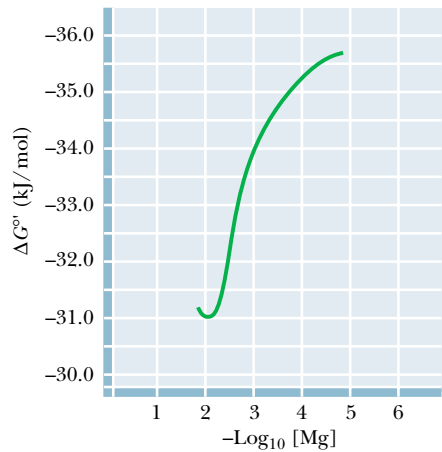


FIGURE 3.17 • The free energy of hydrolysis of ATP as a function of total Mg^{2+} ion concentration at 38°C and $\text{pH } 7.0$. (Adapted from Gwynn, R. W., and Veech, R. L., 1973. *The equilibrium constants of the adenosine triphosphate hydrolysis and the adenosine triphosphate-citrate lyase reactions*. Journal of Biological Chemistry 248:6966–6972.)

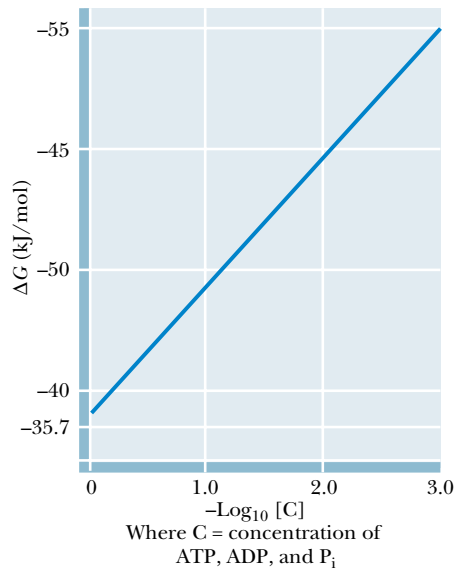


FIGURE 3.18 • The free energy of hydrolysis of ATP as a function of concentration at 38°C , $\text{pH } 7.0$. The plot follows the relationship described in Equation (3.36), with the concentrations $[C]$ of ATP, ADP, and P_i assumed to be equal.

associated free energy change? Figure 3.17 shows the change in $\Delta G^\circ'$ with pMg (that is, $-\log_{10}[\text{Mg}^{2+}]$) at $\text{pH } 7.0$ and 38°C . The free energy of hydrolysis of ATP at zero Mg^{2+} is -35.7 kJ/mol , and at 5 mM free Mg^{2+} (the minimum in the plot) the $\Delta G_{\text{obs}}^\circ$ is approximately -31 kJ/mol . Thus, in most real biological environments (with pH near 7 and Mg^{2+} concentrations of 5 mM or more) the free energy of hydrolysis of ATP is altered more by metal ions than by protons. A widely used “consensus value” for $\Delta G^\circ'$ of ATP in biological systems is -30.5 kJ/mol (Table 3.3). This value, cited in the 1976 *Handbook of Biochemistry and Molecular Biology* (3rd ed., *Physical and Chemical Data*, Vol. 1, pp. 296–304, Boca Raton, FL: CRC Press), was determined in the presence of “excess Mg^{2+} .” *This is the value we use for metabolic calculations in the balance of this text.*

The Effect of Concentration on the Free Energy of Hydrolysis of ATP

Through all these calculations of the effect of pH and metal ions on the ATP hydrolysis equilibrium, we have assumed “standard conditions” with respect to concentrations of all species except for protons. The levels of ATP, ADP, and other high-energy metabolites never even begin to approach the standard state of 1 M . In most cells, the concentrations of these species are more typically 1 to 5 mM or even less. Earlier, we described the effect of concentration on equilibrium constants and free energies in the form of Equation (3.12). For the present case, we can rewrite this as

$$\Delta G = \Delta G^\circ + RT \ln \frac{[\Sigma \text{ADP}][\Sigma \text{P}_i]}{[\Sigma \text{ATP}]} \tag{3.36}$$

where the terms in brackets represent the sum (Σ) of the concentrations of all the ionic forms of ATP, ADP, and P_i .

It is clear that changes in the concentrations of these species can have large effects on ΔG . The concentrations of ATP, ADP, and P_i may, of course, vary rather independently in real biological environments, but if, for the sake of some model calculations, we assume that all three concentrations are equal, then the effect of concentration on ΔG is as shown in Figure 3.18. The free energy of hydrolysis of ATP, which is -35.7 kJ/mol at 1 M , becomes -49.4 kJ/mol at 5 mM (that is, the concentration for which $\text{pC} = -2.3$ in Figure 3.18). At 1 mM ATP, ADP, and P_i , the free energy change becomes even more negative at -53.6 kJ/mol . *Clearly, the effects of concentration are much greater than the effects of protons or metal ions under physiological conditions.*

Does the “concentration effect” change ATP’s position in the energy hierarchy (in Table 3.3)? Not really. All the other high- and low-energy phosphates experience roughly similar changes in concentration under physiological conditions and thus similar changes in their free energies of hydrolysis. The roles of the very high-energy phosphates (PEP, 1,3-bisphosphoglycerate, and creatine phosphate) in the synthesis and maintenance of ATP in the cell are considered in our discussions of metabolic pathways. In the meantime, several of the problems at the end of this chapter address some of the more interesting cases.

3.8 • The Daily Human Requirement for ATP

We can end this discussion of ATP and the other important high-energy compounds in biology by discussing the daily metabolic consumption of ATP by humans. An approximate calculation gives a somewhat surprising and impressive result. Assume that the average adult human consumes approximately

11,700 kJ (2800 kcal, that is, 2,800 Calories) per day. Assume also that the metabolic pathways leading to ATP synthesis operate at a thermodynamic efficiency of approximately 50%. Thus, of the 11,700 kJ a person consumes as food, about 5,860 kJ end up in the form of synthesized ATP. As indicated earlier, the hydrolysis of 1 mole of ATP yields approximately 50 kJ of free energy under cellular conditions. This means that the body cycles through $5860/50 = 117$ moles of ATP each day. The disodium salt of ATP has a molecular weight of 551 g/mol, so that an average person hydrolyzes about

$$(117 \text{ moles}) \frac{551 \text{ g}}{\text{mole}} = 64,467 \text{ g of ATP per day}$$

The average adult human, with a typical weight of 70 kg or so, thus consumes approximately 65 kilograms of ATP per day, an amount nearly equal to his/her own body weight! Fortunately, we have a highly efficient recycling system for ATP/ADP utilization. The energy released from food is stored transiently in the form of ATP. Once ATP energy is used and ADP and phosphate are released, our bodies recycle it to ATP through intermediary metabolism, so that it may be reused. The typical 70-kg body contains only about 50 grams of ATP/ADP total. Therefore, each ATP molecule in our bodies must be recycled nearly 1300 times each day! Were it not for this fact, at current commercial prices of about \$10 per gram, our ATP “habit” would cost approximately \$650,000 per day! In these terms, the ability of biochemistry to sustain the marvelous activity and vigor of organisms gains our respect and fascination.

PROBLEMS

1. An enzymatic hydrolysis of fructose-1-P,



was allowed to proceed to equilibrium at 25°C. The original concentration of fructose-1-P was 0.2 M, but when the system had reached equilibrium the concentration of fructose-1-P was only 6.52×10^{-5} M. Calculate the equilibrium constant for this reaction and the free energy of hydrolysis of fructose-1-P.

2. The equilibrium constant for some process $A \rightleftharpoons B$ is 0.5 at 20°C and 10 at 30°C. Assuming that ΔH° is independent of temperature, calculate ΔH° for this reaction. Determine ΔG° and ΔS° at 20° and at 30°C. Why is it important in this problem to assume that ΔH° is independent of temperature?

3. The standard-state free energy of hydrolysis for acetyl phosphate is $\Delta G^\circ = -42.3$ kJ/mol.



Calculate the free energy change for acetyl phosphate hydrolysis in a solution of 2 mM acetate, 2 mM phosphate, and 3 nM acetyl phosphate.

4. Define a state function. Name three thermodynamic quantities that are state functions and three that are not.
5. ATP hydrolysis at pH 7.0 is accompanied by release of a hydrogen ion to the medium



If the $\Delta G^{\circ'}$ for this reaction is -30.5 kJ/mol, what is ΔG° (that is, the free energy change for the same reaction with all components, including H^+ , at a standard state of 1 M)?

6. For the process $A \rightleftharpoons B$, $K_{\text{eq}}(\text{AB})$ is 0.02 at 37°C. For the process $B \rightleftharpoons C$, $K_{\text{eq}}(\text{BC}) = 1000$ at 37°C.

- a. Determine $K_{\text{eq}}(\text{AC})$, the equilibrium constant for the overall process $A \rightleftharpoons C$, from $K_{\text{eq}}(\text{AB})$ and $K_{\text{eq}}(\text{BC})$.

- b. Determine standard-state free energy changes for all three processes, and use $\Delta G^\circ(\text{AC})$ to determine $K_{\text{eq}}(\text{AC})$. Make sure that this value agrees with that determined in part a of this problem.

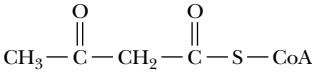
7. Draw all possible resonance structures for creatine phosphate and discuss their possible effects on resonance stabilization of the molecule.

8. Write the equilibrium constant, K_{eq} , for the hydrolysis of creatine phosphate and calculate a value for K_{eq} at 25°C from the value of $\Delta G^{\circ'}$ in Table 3.3.

9. Imagine that creatine phosphate, rather than ATP, is the universal energy carrier molecule in the human body. Repeat the calculation presented in Section 3.8, calculating the weight of creatine phosphate that would need to be consumed each day by a typical adult human if creatine phosphate could not be recycled. If recycling of creatine phosphate were possible, and if the typical adult human body contained 20 grams of creatine phosphate, how many times would each creatine phosphate molecule need to be turned over or recycled each day? Repeat the calculation assuming that glycerol-3-phosphate is the universal energy carrier, and that the body contains 20 grams of glycerol-3-phosphate.

10. Calculate the free energy of hydrolysis of ATP in a rat liver cell in which the ATP, ADP, and P_i concentrations are 3.4, 1.3, and 4.8 mM, respectively.

- 11.** Hexokinase catalyzes the phosphorylation of glucose from ATP, yielding glucose-6-P and ADP. Using the values of Table 3.3, calculate the standard-state free energy change and equilibrium constant for the hexokinase reaction.
- 12.** Would you expect the free energy of hydrolysis of aceto-acetyl-coenzyme A (see diagram) to be greater than, equal to, or less than that of acetyl-coenzyme A? Provide a chemical rationale for your answer.



FURTHER READING

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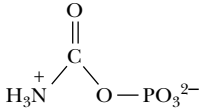
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- 13.** Consider carbamoyl phosphate, a precursor in the biosynthesis of pyrimidines:



Based on the discussion of high-energy phosphates in this chapter, would you expect carbamoyl phosphate to possess a high free energy of hydrolysis? Provide a chemical rationale for your answer.

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Chapter 4

Amino Acids



All objects have mirror images. Like many biomolecules, amino acids exist in mirror-image forms (stereoisomers) that are not superimposable. Only the L-isomers of amino acids commonly occur in nature. (The Mirror of Venus (1898), Sir Edward Burne-Jones/Museu Calouste Gulbenkian Lisbon/The Bridgeman Art Library)

Proteins are the indispensable agents of biological function, and **amino acids** are the building blocks of proteins. The stunning diversity of the thousands of proteins found in nature arises from the intrinsic properties of only 20 commonly occurring amino acids. These features include (1) the capacity to polymerize, (2) novel acid–base properties, (3) varied structure and chemical functionality in the amino acid side chains, and (4) chirality. This chapter describes each of these properties, laying a foundation for discussions of protein structure (Chapters 5 and 6), enzyme function (Chapters 14–16), and many other subjects in later chapters.

To hold, as 'twere, the mirror up to nature.

WILLIAM SHAKESPEARE, *Hamlet*

OUTLINE

- 4.1 • Amino Acids: Building Blocks of Proteins
- 4.2 • Acid–Base Chemistry of Amino Acids
- 4.3 • Reactions of Amino Acids
- 4.4 • Optical Activity and Stereochemistry of Amino Acids
- 4.5 • Spectroscopic Properties of Amino Acids
- 4.6 • Separation and Analysis of Amino Acid Mixtures

4.1 • Amino Acids: Building Blocks of Proteins

Structure of a Typical Amino Acid

The structure of a single typical amino acid is shown in Figure 4.1. Central to this structure is the tetrahedral alpha (α) carbon (C_α), which is covalently linked to both the amino group and the carboxyl group. Also bonded to this α -carbon is a hydrogen and a variable side chain. It is the side chain, the so-called R group, that gives each amino acid its identity. The detailed acid–base properties of amino acids are discussed in the following sections. It is sufficient for now to realize that, in neutral solution (pH 7), the carboxyl group exists as —COO^- and the amino group as —NH_3^+ . Because the resulting amino acid contains one positive and one negative charge, it is a neutral molecule called a **zwitterion**. Amino acids are also *chiral* molecules. With four different groups attached to it, the α -carbon is said to be *asymmetric*. The two possible configurations for the α -carbon constitute nonidentical mirror image isomers or *enantiomers*. Details of amino acid stereochemistry are discussed in Section 4.4.

Amino Acids Can Join via Peptide Bonds

The crucial feature of amino acids that allows them to polymerize to form peptides and proteins is the existence of their two identifying chemical groups: the amino (—NH_3^+) and carboxyl (—COO^-) groups, as shown in Figure 4.2. The amino and carboxyl groups of amino acids can react in a head-to-tail fashion, eliminating a water molecule and forming a covalent amide linkage, which, in the case of peptides and proteins, is typically referred to as a **peptide bond**. The equilibrium for this reaction in aqueous solution favors peptide bond hydrolysis. For this reason, biological systems as well as peptide chemists in the laboratory must carry out peptide bond formation in an indirect manner or with energy input.

Iteration of the reaction shown in Figure 4.2 produces **polypeptides** and **proteins**. The remarkable properties of proteins, which we shall discover and come to appreciate in later chapters, all depend in one way or another on the unique properties and chemical diversity of the 20 common amino acids found in proteins.

Common Amino Acids

The structures and abbreviations for the 20 amino acids commonly found in proteins are shown in Figure 4.3. All the amino acids except proline have both free α -amino and free α -carboxyl groups (Figure 4.1). There are several ways to classify the common amino acids. The most useful of these classifications is based on the polarity of the side chains. Thus, the structures shown in Figure 4.3 are grouped into the following categories: (1) nonpolar or hydrophobic

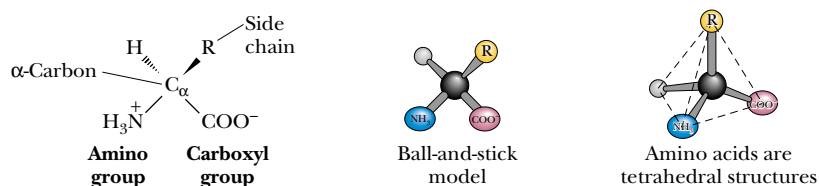


FIGURE 4.1 • Anatomy of an amino acid. Except for proline and its derivatives, all of the amino acids commonly found in proteins possess this type of structure.

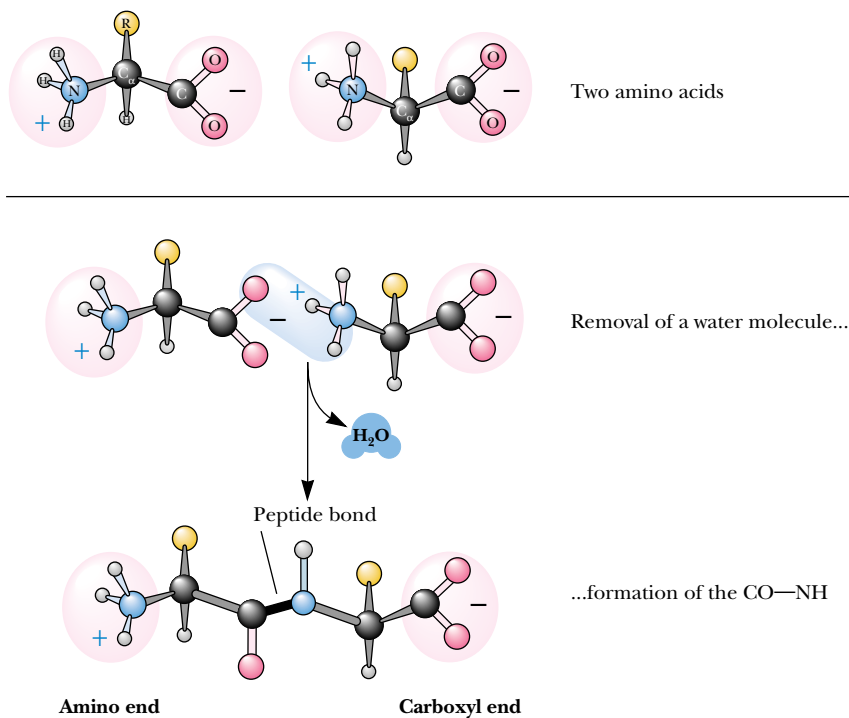


FIGURE 4.2 • The α -COOH and α -NH₃⁺ groups of two amino acids can react with the resulting loss of a water molecule to form a covalent amide bond. (Irving Geis.)

amino acids, (2) neutral (uncharged) but polar amino acids, (3) acidic amino acids (which have a net negative charge at pH 7.0), and (4) basic amino acids (which have a net positive charge at neutral pH). In later chapters, the importance of this classification system for predicting protein properties becomes clear. Also shown in Figure 4.3 are the three-letter and one-letter codes used to represent the amino acids. These codes are useful when displaying and comparing the sequences of proteins in shorthand form. (Note that several of the one-letter abbreviations are phonetic in origin: arginine = “Rginine” = R, phenylalanine = “Fenylalanine” = F, aspartic acid = “asparDic” = D.)

Nonpolar Amino Acids

The nonpolar amino acids (Figure 4.3a) include all those with alkyl chain R groups (alanine, valine, leucine, and isoleucine), as well as proline (with its unusual cyclic structure), methionine (one of the two sulfur-containing amino acids), and two aromatic amino acids, phenylalanine and tryptophan. Tryptophan is sometimes considered a borderline member of this group because it can interact favorably with water via the N–H moiety of the indole ring. Proline, strictly speaking, is not an amino acid but rather an α -imino acid.

Polar, Uncharged Amino Acids

The polar, uncharged amino acids (Figure 4.3b) except for glycine contain R groups that can form hydrogen bonds with water. Thus, these amino acids are usually more soluble in water than the nonpolar amino acids. Several exceptions should be noted. Tyrosine displays the lowest solubility in water of the 20 common amino acids (0.453 g/L at 25°C). Also, proline is very soluble in water, and alanine and valine are about as soluble as arginine and serine. The amide groups of asparagine and glutamine; the hydroxyl groups of tyrosine, threonine, and serine; and the sulfhydryl group of cysteine are all good hydrogen

(Text continues on page 86.)

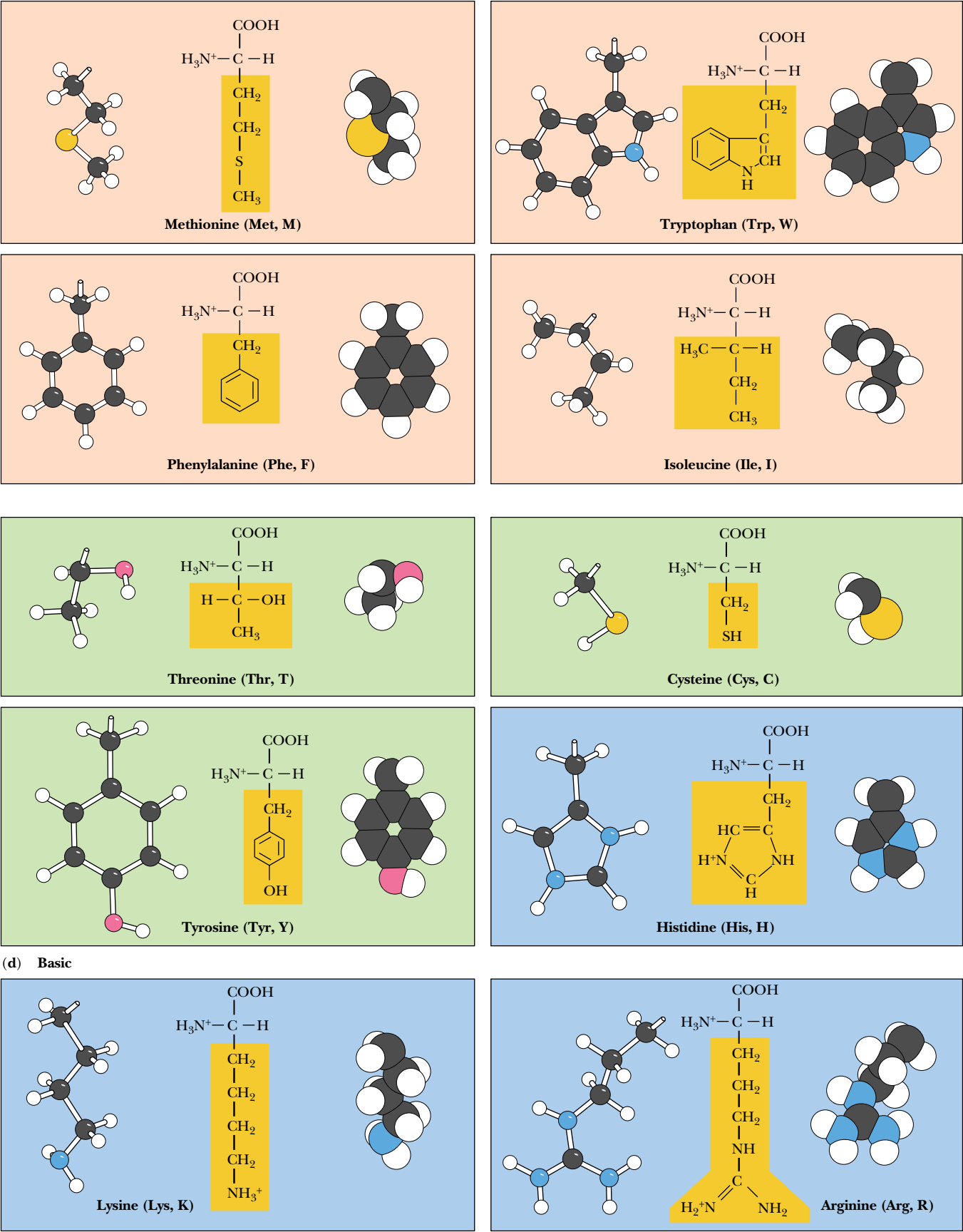
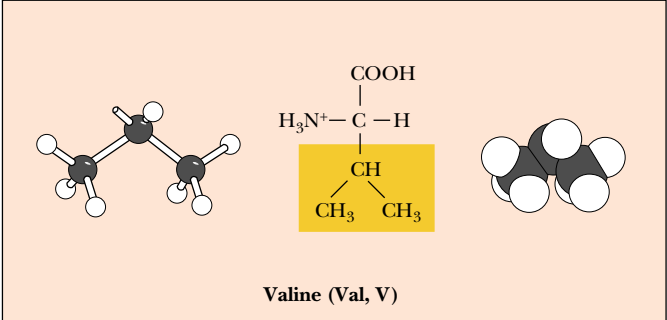
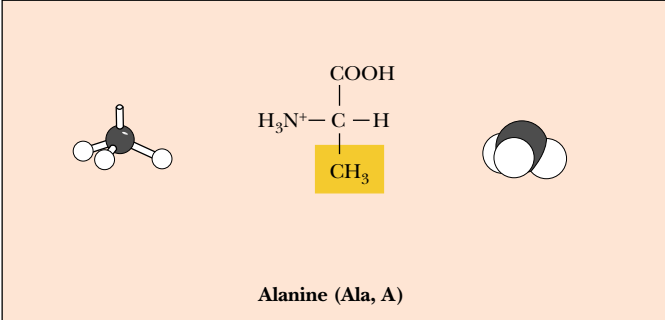
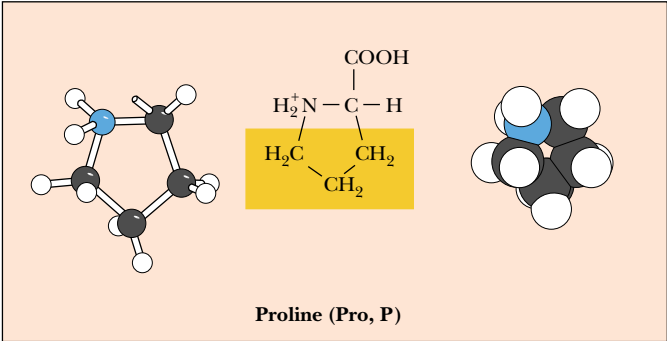
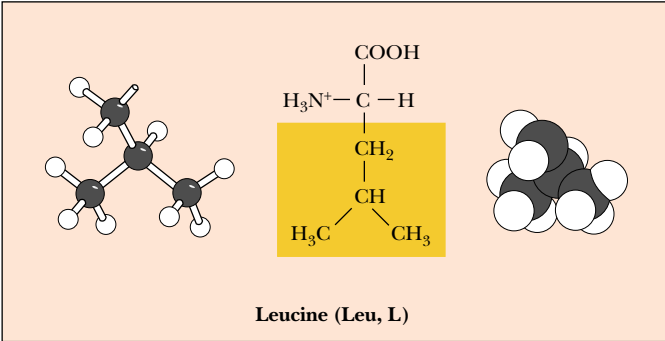
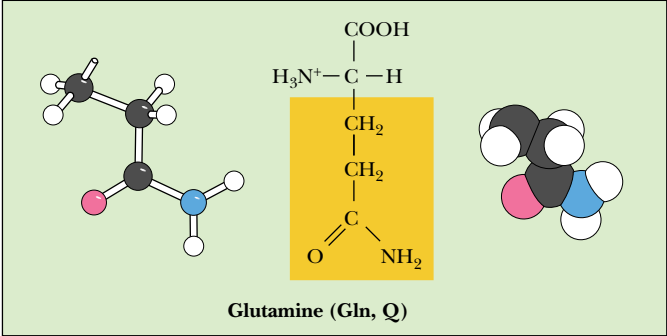
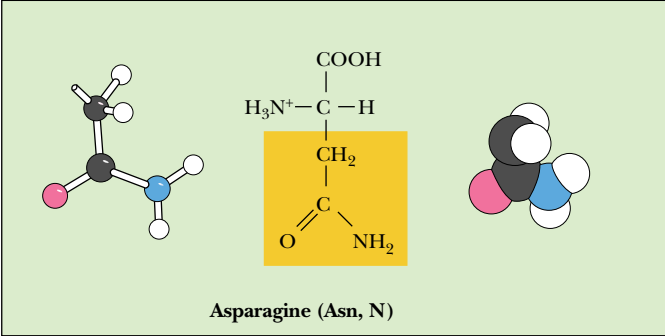
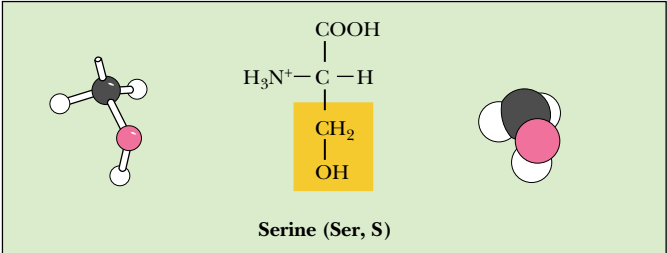
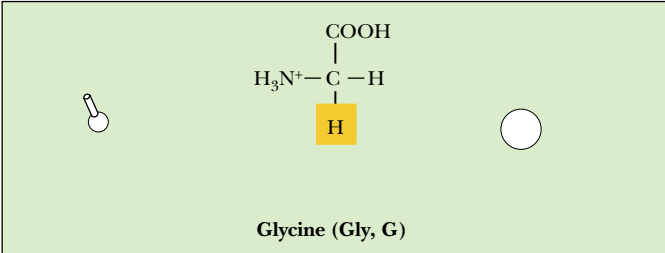


FIGURE 4.3 • The 20 amino acids that are the building blocks of most proteins can be classified as (a) nonpolar (hydrophobic), (b) polar, neutral, (c) acidic, or (d) basic.

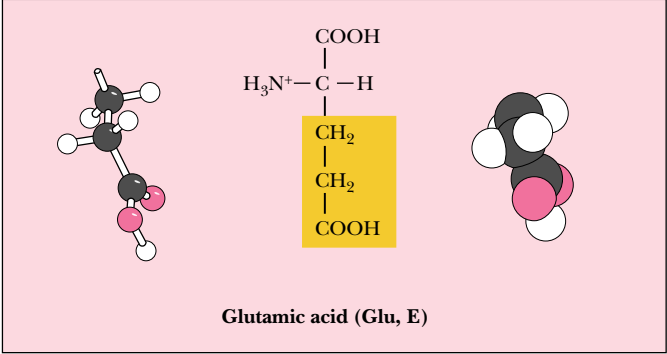
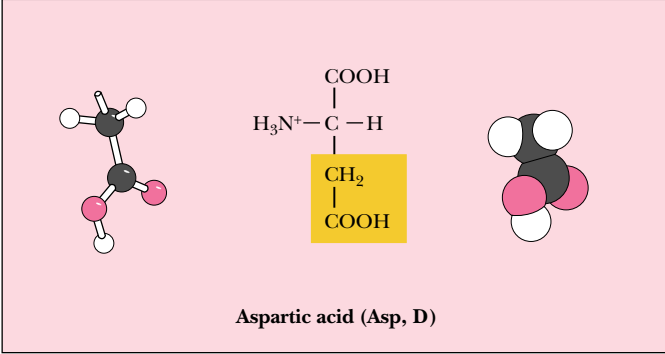
(a) Nonpolar (hydrophobic)



(b) Polar, uncharged



(c) Acidic



Also shown are the one-letter and three-letter codes used to denote amino acids. For each amino acid, the ball-and-stick (*left*) and space-filling (*right*) models show only the side chain. (Irving Geis)

bond-forming moieties. Glycine, the simplest amino acid, has only a single hydrogen for an R group, and this hydrogen is not a good hydrogen bond former. Glycine's solubility properties are mainly influenced by its polar amino and carboxyl groups, and thus glycine is best considered a member of the polar, uncharged group. It should be noted that tyrosine has significant nonpolar characteristics due to its aromatic ring and could arguably be placed in the nonpolar group (Figure 4.3a). However, with a pK_a of 10.1, tyrosine's phenolic hydroxyl is a charged, polar entity at high pH.

Acidic Amino Acids

There are two acidic amino acids—aspartic acid and glutamic acid—whose R groups contain a carboxyl group (Figure 4.3c). These side chain carboxyl groups are weaker acids than the α -COOH group, but are sufficiently acidic to exist as $-\text{COO}^-$ at neutral pH. Aspartic acid and glutamic acid thus have a net negative charge at pH 7. These negatively charged amino acids play several important roles in proteins. Many proteins that bind metal ions for structural or functional purposes possess metal binding sites containing one or more aspartate and glutamate side chains. Carboxyl groups may also act as nucleophiles in certain enzyme reactions and may participate in a variety of electrostatic bonding interactions. The acid–base chemistry of such groups is considered in detail in Section 4.2.

Basic Amino Acids

Three of the common amino acids have side chains with net positive charges at neutral pH: histidine, arginine, and lysine (Figure 4.3d). The ionized group of histidine is an imidazolium, that of arginine is a guanidinium, and lysine contains a protonated alkyl amino group. The side chains of the latter two amino acids are fully protonated at pH 7, but histidine, with a side chain pK_a of 6.0, is only 10% protonated at pH 7. With a pK_a near neutrality, histidine side chains play important roles as proton donors and acceptors in many enzyme reactions. Histidine-containing peptides are important biological buffers, as discussed in Chapter 2. Arginine and lysine side chains, which are protonated under physiological conditions, participate in electrostatic interactions in proteins.

Uncommon Amino Acids

Several amino acids occur only rarely in proteins (Figure 4.4). These include **hydroxylysine** and **hydroxyproline**, which are found mainly in the collagen and gelatin proteins, and **thyroxine** and **3,3',5-triiodothyronine**, iodinated amino acids that are found only in thyroglobulin, a protein produced by the thyroid gland. (Thyroxine and 3,3',5-triiodothyronine are produced by iodination of tyrosine residues in thyroglobulin in the thyroid gland. Degradation of thyroglobulin releases these two iodinated amino acids, which act as hormones to regulate growth and development.) Certain muscle proteins contain methylated amino acids, including **methylhistidine**, **ϵ -N-methyllysine**, and **ϵ -N,N,N-trimethyllysine** (Figure 4.4). **γ -Carboxyglutamic acid** is found in several proteins involved in blood clotting, and **pyroglutamic acid** is found in a unique light-driven proton-pumping protein called bacteriorhodopsin, which is discussed elsewhere in this book. Certain proteins involved in cell growth and regulation are reversibly phosphorylated on the $-\text{OH}$ groups of serine, threonine, and tyrosine residues. **Aminoadipic acid** is found in proteins isolated from corn. Finally, **N-methylarginine** and **N-acetyllysine** are found in histone proteins associated with chromosomes.

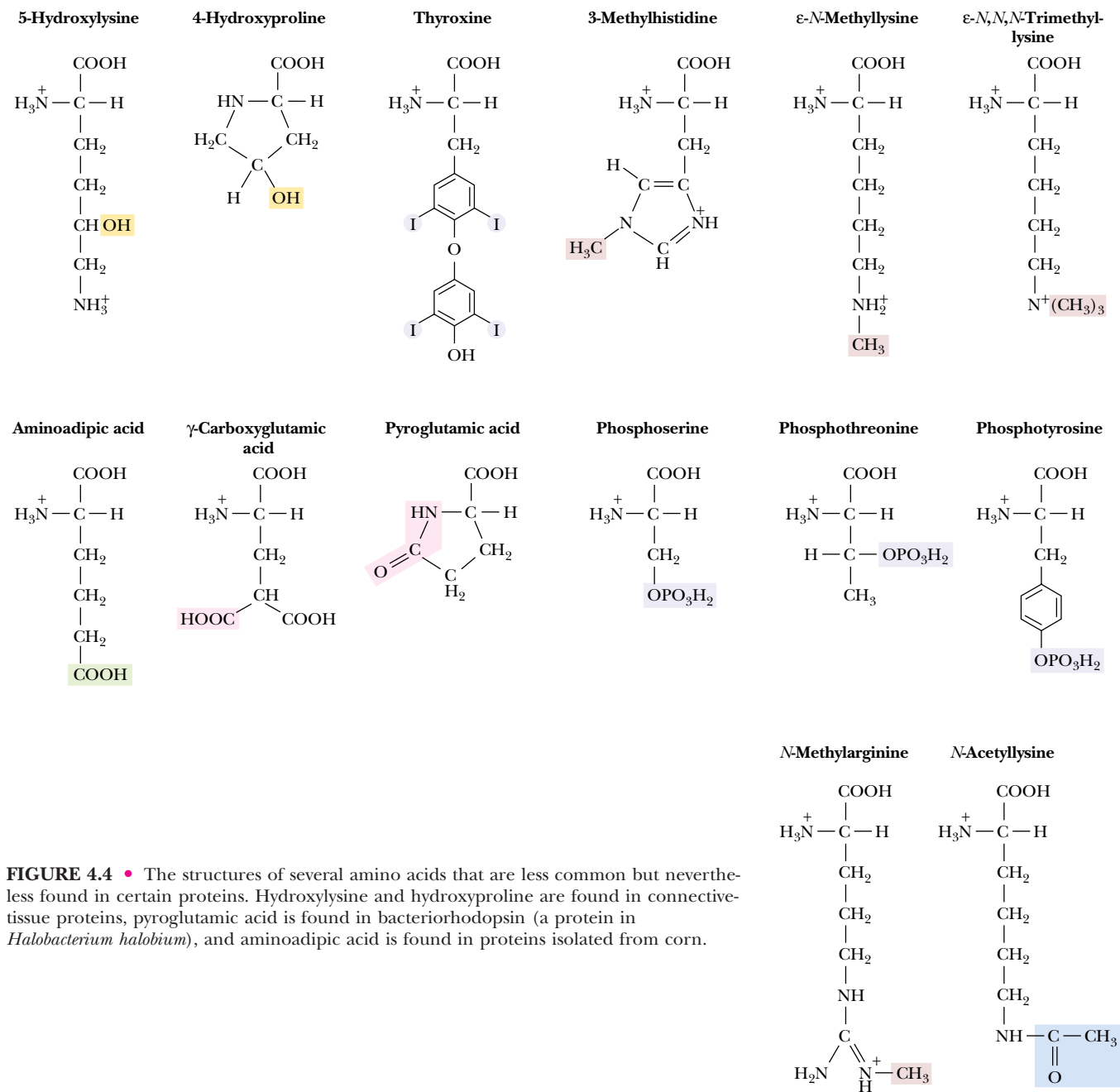


FIGURE 4.4 • The structures of several amino acids that are less common but nevertheless found in certain proteins. Hydroxylysine and hydroxyproline are found in connective-tissue proteins, pyroglutamic acid is found in bacteriorhodopsin (a protein in *Halobacterium halobium*), and aminoadipic acid is found in proteins isolated from corn.

Amino Acids Not Found in Proteins

Certain amino acids and their derivatives, although not found in proteins, nonetheless are biochemically important. A few of the more notable examples are shown in Figure 4.5. **γ-Aminobutyric acid**, or **GABA**, is produced by the decarboxylation of glutamic acid and is a potent neurotransmitter. **Histamine**, which is synthesized by decarboxylation of histidine, and **serotonin**, which is derived from tryptophan, similarly function as neurotransmitters and regulators. **β-Alanine** is found in nature in the peptides carnosine and anserine and is a component of pantothenic acid (a vitamin), which is a part of coenzyme A. **Epinephrine** (also known as **adrenaline**), derived from tyrosine, is an important hormone. **Penicillamine** is a constituent of the penicillin antibiotics. **Ornithine**, **betaine**, **homocysteine**, and **homoserine** are important metabolic intermediates. **Citrulline** is the immediate precursor of arginine.

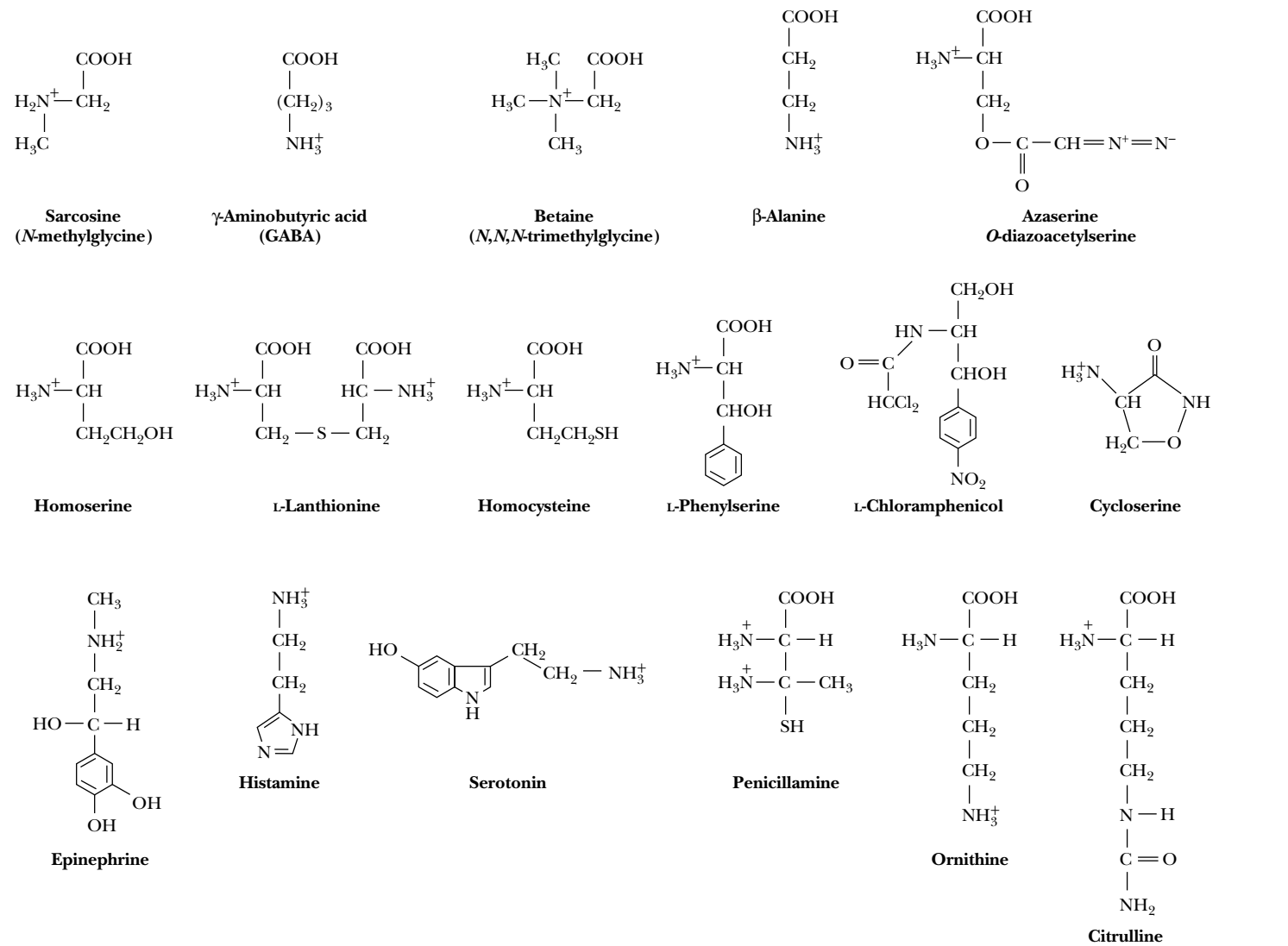


FIGURE 4.5 • The structures of some amino acids that are not normally found in proteins but that perform other important biological functions. Epinephrine, histamine, and serotonin, although not amino acids, are derived from and closely related to amino acids.

4.2 • Acid–Base Chemistry of Amino Acids

Amino Acids Are Weak Polyprotic Acids

From a chemical point of view, the common amino acids are all weak polyprotic acids. The ionizable groups are not strongly dissociating ones, and the degree of dissociation thus depends on the pH of the medium. All the amino acids contain at least two dissociable hydrogens.

Consider the acid–base behavior of glycine, the simplest amino acid. At low pH, both the amino and carboxyl groups are protonated and the molecule has a net positive charge. If the counterion in solution is a chloride ion, this form is referred to as **glycine hydrochloride**. If the pH is increased, the carboxyl group is the first to dissociate, yielding the neutral zwitterionic species Gly⁰ (Figure 4.6). Further increase in pH eventually results in dissociation of the amino group to yield the negatively charged **glycinate**. If we denote these

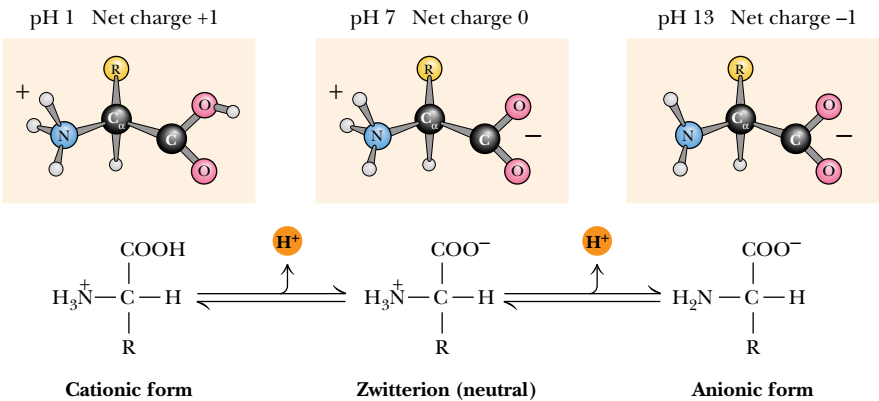
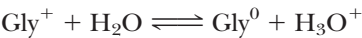


FIGURE 4.6 • The ionic forms of the amino acids, shown without consideration of any ionizations on the side chain. The cationic form is the low pH form, and the titration of the cationic species with base yields the zwitterion and finally the anionic form. (*Irving Geis*)

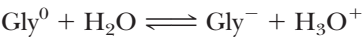
three forms as Gly⁺, Gly⁰, and Gly[−], we can write the first dissociation of Gly⁺ as



and the dissociation constant *K*₁ as

$$K_1 = \frac{[\text{Gly}^0][\text{H}_3\text{O}^+]}{[\text{Gly}^+]}$$

Values for *K*₁ for the common amino acids are typically 0.4 to 1.0 × 10^{−2} *M*, so that typical values of p*K*₁ center on values of 2.0 to 2.4 (see Table 4.1). In a similar manner, we can write the second dissociation reaction as



p <i>K</i> _a Values of Common Amino Acids			
Amino Acid	α-COOH p <i>K</i> _a	α-NH ₃ ⁺ p <i>K</i> _a	R group p <i>K</i> _a
Alanine	2.4	9.7	
Arginine	2.2	9.0	12.5
Asparagine	2.0	8.8	
Aspartic acid	2.1	9.8	3.9
Cysteine	1.7	10.8	8.3
Glutamic acid	2.2	9.7	4.3
Glutamine	2.2	9.1	
Glycine	2.3	9.6	
Histidine	1.8	9.2	6.0
Isoleucine	2.4	9.7	
Leucine	2.4	9.6	
Lysine	2.2	9.0	10.5
Methionine	2.3	9.2	
Phenylalanine	1.8	9.1	
Proline	2.1	10.6	
Serine	2.2	9.2	~13
Threonine	2.6	10.4	~13
Tryptophan	2.4	9.4	
Tyrosine	2.2	9.1	10.1
Valine	2.3	9.6	

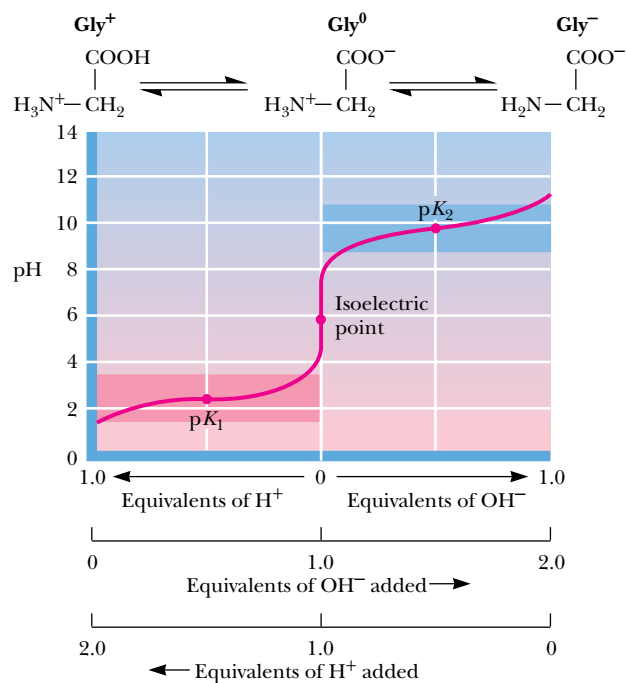


FIGURE 4.7 • Titration of glycine, a simple amino acid. The isoelectric point, pI, the pH where the molecule has a net charge of 0, is defined as $(\text{p}K_1 + \text{p}K_2)/2$.

and the dissociation constant K_2 as

$$K_2 = \frac{[\text{Gly}^-][\text{H}_3\text{O}^+]}{[\text{Gly}^0]}$$

Typical values for $\text{p}K_2$ are in the range of 9.0 to 9.8. At physiological pH, the α -carboxyl group of a simple amino acid (with no ionizable side chains) is completely dissociated, whereas the α -amino group has not really begun its dissociation. The titration curve for such an amino acid is shown in Figure 4.7.

EXAMPLE

What is the pH of a glycine solution in which the $\alpha\text{-NH}_3^+$ group is one-third dissociated?

SOLUTION

The appropriate Henderson–Hasselbalch equation is

$$\text{pH} = \text{p}K_a + \log_{10} \frac{[\text{Gly}^-]}{[\text{Gly}^0]}$$

If the α -amino group is one-third dissociated, there is one part Gly⁻ for every two parts Gly⁰. The important $\text{p}K_a$ is the $\text{p}K_a$ for the amino group. The glycine α -amino group has a $\text{p}K_a$ of 9.6. The result is

$$\begin{aligned} \text{pH} &= 9.6 + \log_{10} (1/2) \\ \text{pH} &= 9.3 \end{aligned}$$

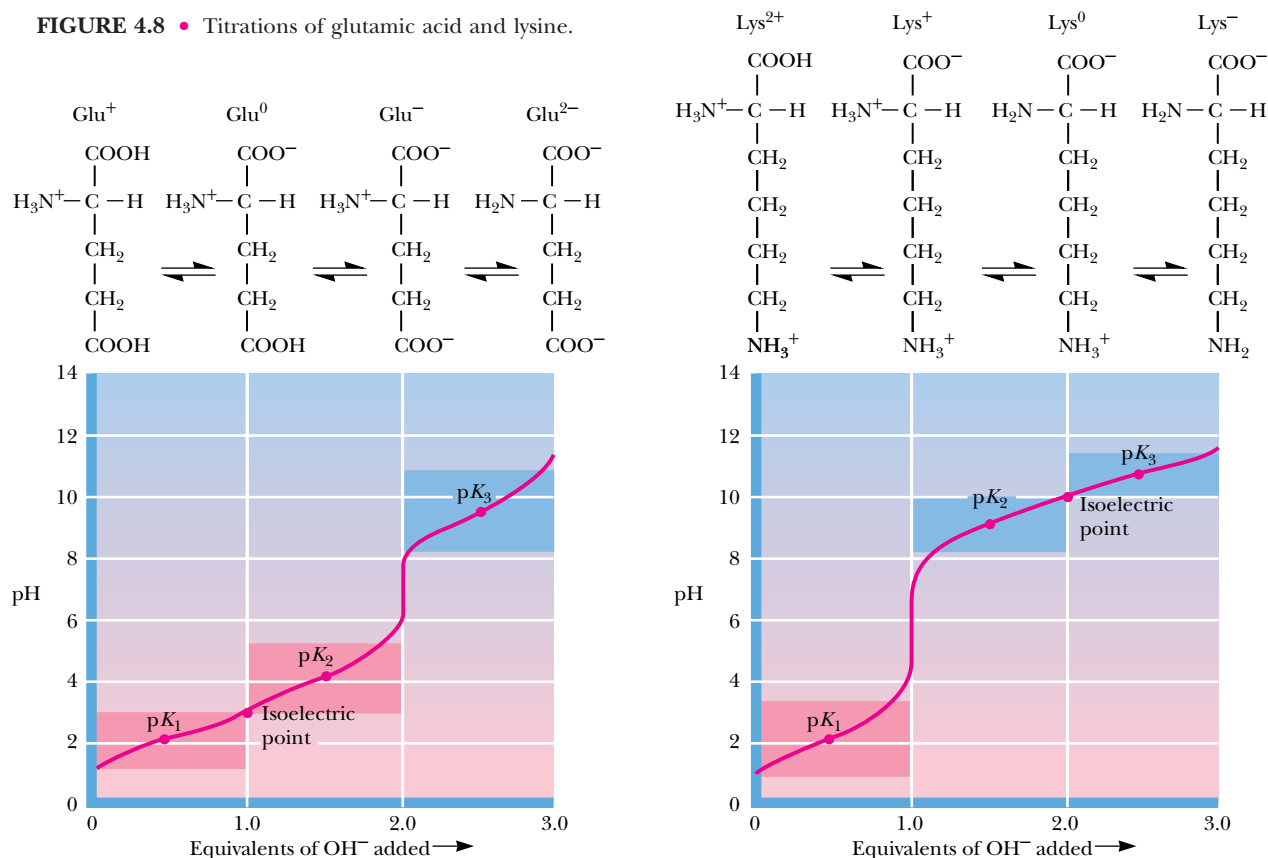
Note that the dissociation constants of both the α -carboxyl and α -amino groups are affected by the presence of the other group. The adjacent α -amino group makes the α -COOH group more acidic (that is, it lowers the $\text{p}K_a$) so

that it gives up a proton more readily than simple alkyl carboxylic acids. Thus, the pK_1 of 2.0 to 2.1 for α -carboxyl groups of amino acids is substantially lower than that of acetic acid ($pK_a = 4.76$), for example. What is the chemical basis for the low pK_a of the α -COOH group of amino acids? The α -NH₃⁺ (ammonium) group is strongly electron-withdrawing, and the positive charge of the amino group exerts a strong field effect and stabilizes the carboxylate anion. (The effect of the α -COO[−] group on the pK_a of the α -NH₃⁺ group is the basis for Problem 4 at the end of this chapter.)

Ionization of Side Chains

As we have seen, the side chains of several of the amino acids also contain dissociable groups. Thus, aspartic and glutamic acids contain an additional carboxyl function, and lysine possesses an aliphatic amino function. Histidine contains an ionizable imidazolium proton, and arginine carries a guanidinium function. Typical pK_a values of these groups are shown in Table 4.1. The β -carboxyl group of aspartic acid and the γ -carboxyl side chain of glutamic acid exhibit pK_a values intermediate to the α -COOH on the one hand and typical aliphatic carboxyl groups on the other hand. In a similar fashion, the ϵ -amino group of lysine exhibits a pK_a that is higher than the α -amino group but similar to that for a typical aliphatic amino group. These intermediate values for side-chain pK_a values reflect the slightly diminished effect of the α -carbon dissociable groups that lie several carbons removed from the side-chain functional groups. Figure 4.8 shows typical titration curves for glutamic acid and lysine, along with the ionic species that predominate at various points in the

FIGURE 4.8 • Titrations of glutamic acid and lysine.

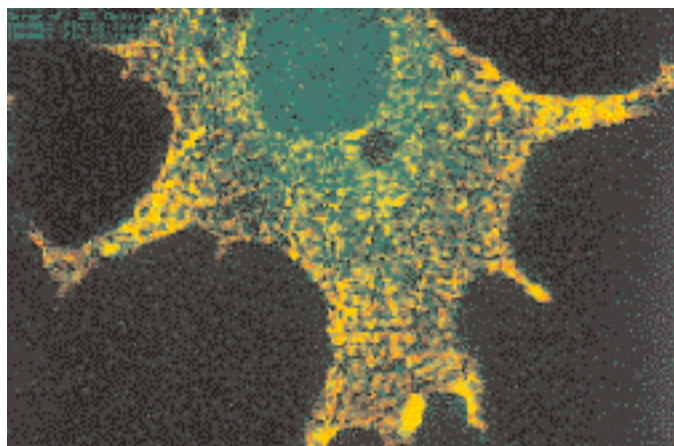
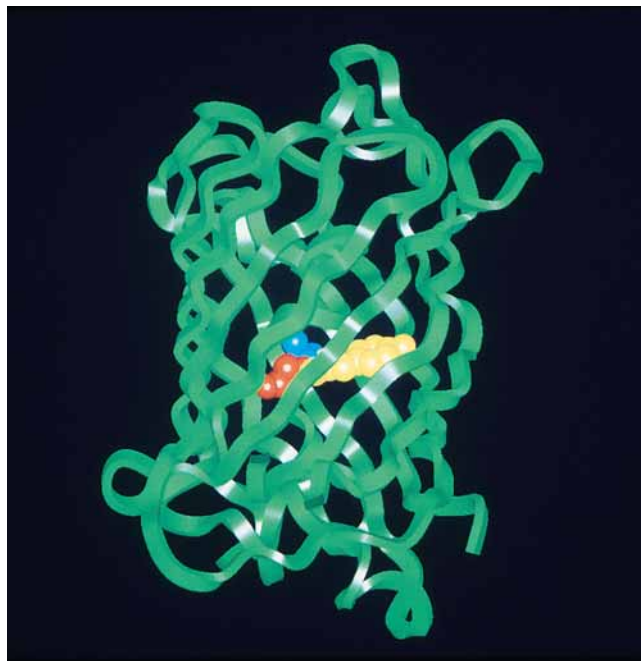
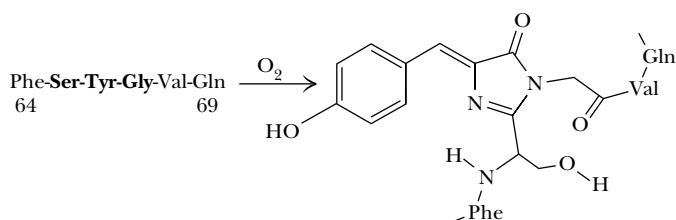


CRITICAL DEVELOPMENTS IN BIOCHEMISTRY

Green Fluorescent Protein—The “Light Fantastic” from Jellyfish to Gene Expression

Aequorea victoria, a species of jellyfish found in the northwest Pacific Ocean, contains a **green fluorescent protein (GFP)** that works together with another protein, **aequorin**, to provide a defense mechanism for the jellyfish. When the jellyfish is attacked or shaken, aequorin produces a blue light. This light energy is captured by GFP, which then emits a bright green flash that presumably blinds or startles the attacker. Remarkably, the fluorescence of GFP occurs without the assistance of a **prosthetic group**—a “helper molecule” that would mediate GFP’s fluorescence. Instead, the light-transducing capability of GFP is the result of a reaction between three amino acids in the protein itself. As shown below, adjacent **serine**, **tyrosine**, and **glycine** in the sequence of the protein react to form the pigment complex—termed a **chromophore**. No enzymes are required; the reaction is autocatalytic.

Because the light-transducing talents of GFP depend only on the protein itself (upper photo, chromophore highlighted), GFP has quickly become a darling of genetic engineering laboratories. The promoter of any gene whose cellular expression is of interest can be fused to the DNA sequence coding for GFP. Telltale green fluorescence tells the researcher when this fused gene has been expressed (see lower photo and also Chapter 13).



Autocatalytic oxidation of GFP amino acids leads to the chromophore shown on the left. The green fluorescence requires further interactions of the chromophore with other parts of the protein.

Boxer, S.G., 1997. Another green revolution. *Nature* **383**:484–485.

titration. The only other side-chain groups that exhibit any significant degree of dissociation are the *para*-OH group of tyrosine and the —SH group of cysteine. The pK_a of the cysteine sulfhydryl is 8.32, so that it is about 12% dissociated at pH 7. The tyrosine *para*-OH group is a very weakly acidic group, with a pK_a of about 10.1. This group is essentially fully protonated and uncharged at pH 7.

4.3 • Reactions of Amino Acids

Carboxyl and Amino Group Reactions

The α -carboxyl and α -amino groups of all amino acids exhibit similar chemical reactivity. The side chains, however, exhibit specific chemical reactivities, depending on the nature of the functional groups. Whereas all of these reactivities are important in the study and analysis of isolated amino acids, it is the characteristic behavior of the side chain that governs the reactivity of amino acids incorporated into proteins. There are three reasons to consider these reactivities. Proteins can be chemically modified in very specific ways by taking advantage of the chemical reactivity of certain amino acid side chains. The detection and quantification of amino acids and proteins often depend on reactions that are specific to one or more amino acids and that result in color, radioactivity, or some other quantity that can be easily measured. Finally and most importantly, the biological functions of proteins depend on the behavior and reactivity of specific R groups.

The carboxyl groups of amino acids undergo all the simple reactions common to this functional group. Reaction with ammonia and primary amines yields unsubstituted and substituted amides, respectively (Figure 4.9a,b). Esters

CARBOXYL GROUP REACTIONS

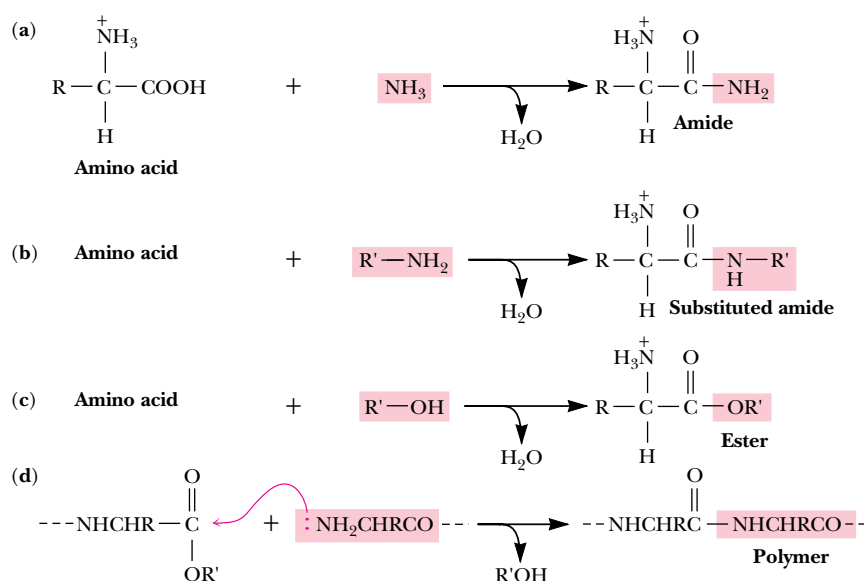
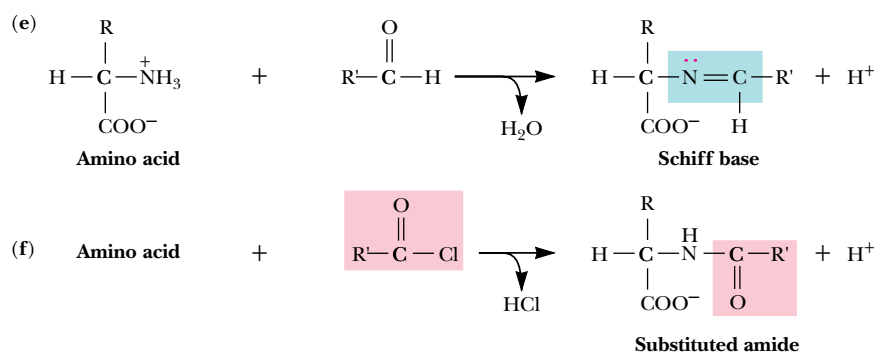


FIGURE 4.9 • Typical reactions of the common amino acids (see text for details).

AMINO GROUP REACTIONS



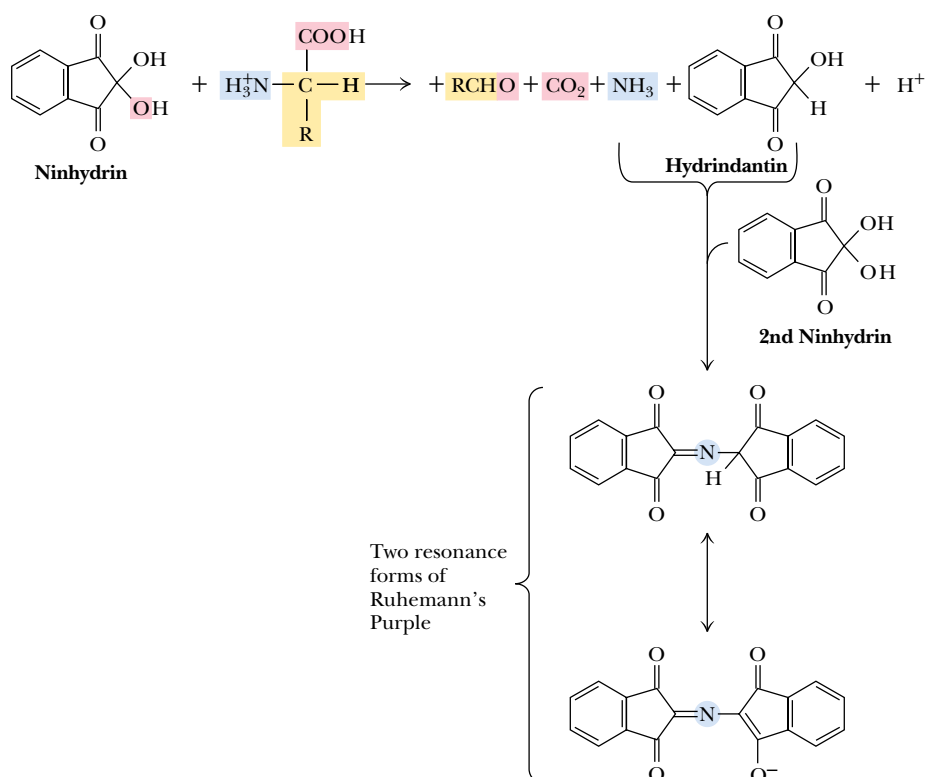


FIGURE 4.10 • The pathway of the ninhydrin reaction, which produces a colored product called “Ruhemann’s Purple” that absorbs light at 570 nm. Note that the reaction involves and consumes two molecules of ninhydrin.

and acid chlorides are also readily formed. Esterification proceeds in the presence of the appropriate alcohol and a strong acid (Figure 4.9c). Polymerization can occur by repetition of the reaction shown in Figure 4.9d. Free amino groups may react with aldehydes to form Schiff bases (Figure 4.9e) and can be acylated with acid anhydrides and acid halides (Figure 4.9f).

The Ninhydrin Reaction

Amino acids can be readily detected and quantified by reaction with ninhydrin. As shown in Figure 4.10, *ninhydrin*, or triketohydrindene hydrate, is a strong oxidizing agent and causes the oxidative deamination of the α -amino function. The products of the reaction are the resulting aldehyde, ammonia, carbon dioxide, and hydrindantin, a reduced derivative of ninhydrin. The ammonia produced in this way can react with the hydrindantin and another molecule of ninhydrin to yield a purple product (Ruhemann’s Purple) that can be quantified spectrophotometrically at 570 nm. The appearance of CO_2 can also be monitored. Indeed, CO_2 evolution is diagnostic of the presence of an α -amino acid. α -Imino acids, such as proline and hydroxyproline, give bright yellow ninhydrin products with absorption maxima at 440 nm, allowing these to be distinguished from the α -amino acids. Because amino acids are one of the components of human skin secretions, the ninhydrin reaction was once used extensively by law enforcement and forensic personnel for fingerprint detection. (Fingerprints as old as 15 years can be successfully identified using the ninhydrin reaction.) More sensitive fluorescent reagents are now used routinely for this purpose.

Specific Reactions of Amino Acid Side Chains

A number of reactions of amino acids have become important in recent years because they are essential to the degradation, sequencing, and chemical synthesis of peptides and proteins. These reactions are discussed in Chapter 5.

CRITICAL DEVELOPMENTS IN BIOCHEMISTRY

Discovery of Optically Active Molecules and Determination of Absolute Configuration

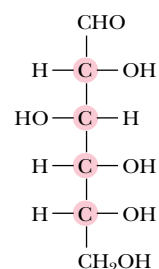
The optical activity of quartz and certain other materials was first discovered by Jean-Baptiste Biot in 1815 in France, and in 1848 a young chemist in Paris named Louis Pasteur made a related and remarkable discovery. Pasteur noticed that preparations of optically inactive sodium ammonium tartrate contained two visibly different kinds of crystals that were mirror images of each other. Pasteur carefully separated the two types of crystals, dissolved them each in water, and found that each solution was *optically active*. Even more intriguing, the specific rotations of these two solutions were equal in magnitude and of opposite sign. Because these differences in optical rotation were apparent properties of the dissolved molecules, Pasteur eventually proposed that the molecules themselves were mirror images of each other, just like their respective crystals. Based on this and other related evidence, in 1847 van't Hoff and LeBel proposed the tetrahedral arrangement of valence bonds to carbon.

In 1888, Emil Fischer decided that it should be possible to determine the *relative* configuration of (+)-glucose, a six-carbon sugar with four asymmetric centers (see figure). Because each of the four C could be either of two configurations, glucose conceivably could exist in any one of 16 possible isomeric structures. It took three years to complete the solution of an elaborate chemical and logical puzzle. By 1891, Fischer had reduced his puzzle to a choice between two enantiomeric structures. (Methods for determining *absolute* configuration were not yet available, so Fischer made a simple guess, selecting the structure shown in the figure.) For this remarkable feat, Fischer received the Nobel Prize in chemistry in 1902. Sadly, Fischer, a brilliant but troubled chemist, later committed suicide.

The absolute choice between Fischer's two enantiomeric possibilities would not be made for a long time. In 1951, J.M. Bijvoet in Utrecht, the Netherlands, used a new X-ray diffraction tech-

nique to determine the absolute configuration of (among other things) the sodium rubidium salt of (+)-tartaric acid. Because the tartaric acid configuration could be related to that of glyceraldehyde and because sugar and amino acid configurations could all be related to glyceraldehyde, it became possible to determine the absolute configuration of sugars and the common amino acids. The absolute configuration of tartaric acid determined by Bijvoet turned out to be the configuration that, up to then, had only been assumed. This meant that Emil Fischer's arbitrary guess 60 years earlier had been correct.

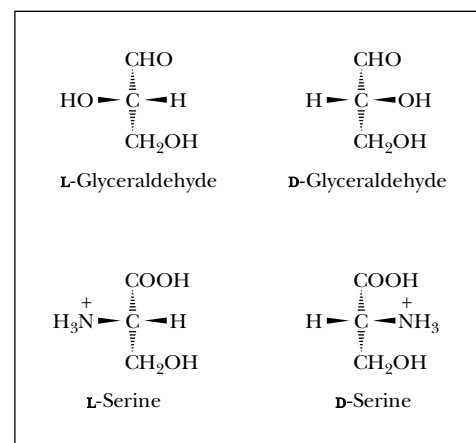
It was M.A. Rosanoff, a chemist and instructor at New York University, who first proposed (in 1906) that the isomers of glyceraldehyde be the standards for denoting the stereochemistry of sugars and other molecules. Later, when experiments showed that the configuration of (+)-glyceraldehyde was related to (+)-glucose, (+)-glyceraldehyde was given the designation *D*. Emil Fischer rejected the **Rosanoff convention**, but it was universally accepted. Ironically, this nomenclature system is often mistakenly referred to as the **Fischer convention**.



The absolute configuration of (+)-glucose.

chiral center can be more easily, completely, and unambiguously described with (*R,S*) notation. Several amino acids, including isoleucine, threonine, hydroxyproline, and hydroxylysine, have two chiral centers. In the (*R,S*) system, L-threonine is (2*S*,3*R*)-threonine. A chemical compound with *n* chiral centers can exist in 2^{*n*}-isomeric structures, and the four amino acids just listed can thus each take on four different isomeric configurations. This amounts to two pairs of enantiomers. Isomers that differ in configuration at only one of the asymmetric centers are non-mirror image isomers or **diastereomers**. The four stereo-

FIGURE 4.13 • The configuration of the common L-amino acids can be related to the configuration of L(-)-glyceraldehyde as shown. These drawings are known as Fischer projections. The horizontal lines of the Fischer projections are meant to indicate bonds coming out of the page from the central carbon, and vertical lines represent bonds extending behind the page from the central carbon atom.



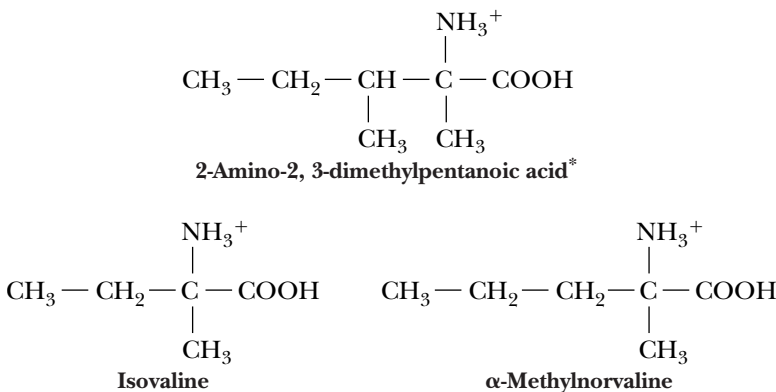
The Murchison Meteorite—Discovery of Extraterrestrial Handedness

The predominance of L-amino acids in biological systems is one of life’s most intriguing features. Prebiotic syntheses of amino acids would be expected to produce equal amounts of L- and D-enantiomers. Some kind of enantiomeric selection process must have intervened to select L-amino acids over their D-counterparts as the constituents of proteins. Was it random chance that chose L- over D-isomers?

Analysis of carbon compounds—even amino acids—from extraterrestrial sources might provide deeper insights into this mystery. John Cronin and Sandra Pizzarello have examined the enantiomeric distribution of unusual amino acids obtained from the Murchison meteorite, which struck the earth on September 28, 1969, near Murchison, Australia. (By selecting unusual amino

acids for their studies, Cronin and Pizzarello ensured that they were examining materials that were native to the meteorite and not earth-derived contaminants.) Four α-dialkyl amino acids—α-methylisoleucine, α-methylalloisoleucine, α-methylnorvaline, and isovaline—were found to have an L-enantiomeric excess of 2 to 9%.

This may be the first demonstration that a natural L-enantiomer enrichment occurs in certain cosmological environments. Could these observations be relevant to the emergence of L-enantiomers as the dominant amino acids on the earth? And, if so, could there be life elsewhere in the universe that is based upon the same amino acid handedness?

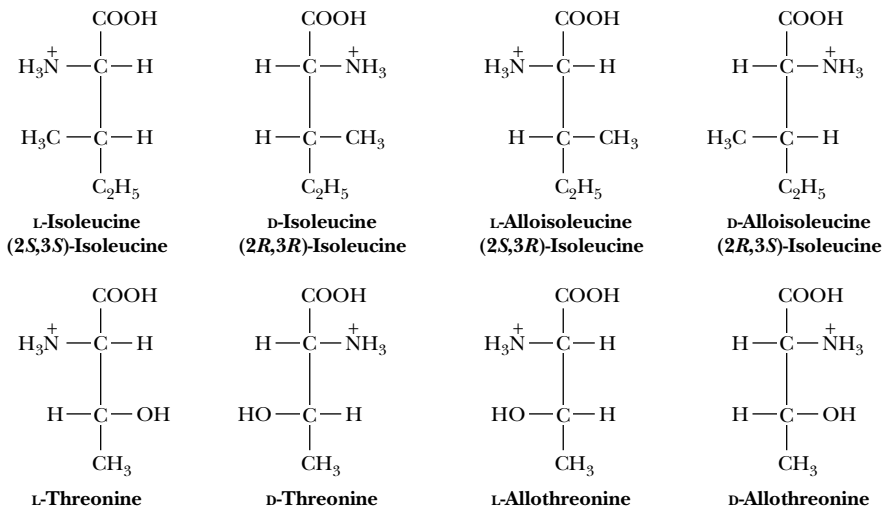


Amino acids found in the Murchison meteorite

*The four stereoisomers of this amino acid include the D- and L-forms of α-methylisoleucine and α-methylalloisoleucine. Cronin, J.R., and Pizzarello, S., 1997. Enantiomeric excesses in meteoritic amino acids. *Science* **275**:951–955.

isomers of isoleucine are shown in Figure 4.14. The isomer obtained from digests of natural proteins is arbitrarily designated L-isoleucine. In the (R,S) system, L-isoleucine is (2S,3S)-isoleucine. Its diastereomer is referred to as L-allo-isoleucine. The D-enantiomeric pair of isomers is named in a similar manner.

FIGURE 4.14 • The stereoisomers of isoleucine and threonine. The structures at the far left are the naturally occurring isomers.



CRITICAL DEVELOPMENTS IN BIOCHEMISTRY

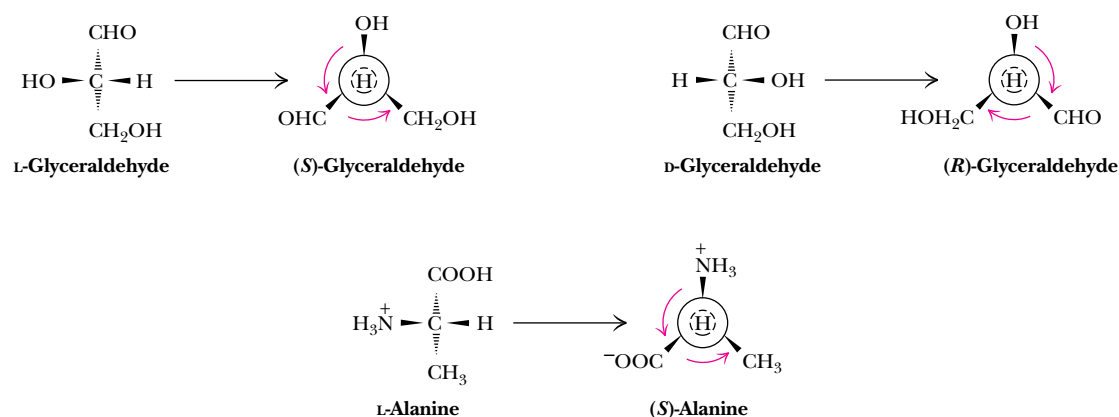
Rules for Description of Chiral Centers in the (R,S) System

Naming a chiral center in the (*R,S*) system is accomplished by viewing the molecule from the chiral center to the atom with the lowest priority. If the other three atoms facing the viewer then decrease in priority in a clockwise direction, the center is said to have the (*R*) configuration (where *R* is from the Latin *rectus* meaning “right”). If the three atoms in question decrease in priority in a counterclockwise fashion, the chiral center is of the (*S*) configuration (where *S* is from the Latin *sinistrus* meaning “left”). If two of the atoms coordinated to a chiral center are identical,

the atoms bound to these two are considered for priorities. For such purposes, the priorities of certain functional groups found in amino acids and related molecules are in the following order:



From this, it is clear that D-glyceraldehyde is (*R*)-glyceraldehyde and L-alanine is (*S*)-alanine (see figure). Interestingly, the α -carbon configuration of all the L-amino acids *except for cysteine* is (*S*). Cysteine, by virtue of its thiol group, is in fact (*R*)-cysteine.



4.5 • Spectroscopic Properties of Amino Acids

One of the most important and exciting advances in modern biochemistry has been the application of **spectroscopic methods**, which measure the absorption and emission of energy of different frequencies by molecules and atoms. Spectroscopic studies of proteins, nucleic acids, and other biomolecules are providing many new insights into the structure and dynamic processes in these molecules.

Ultraviolet Spectra

Many details of the structure and chemistry of the amino acids have been elucidated or at least confirmed by spectroscopic measurements. None of the amino acids absorbs light in the visible region of the electromagnetic spectrum. Several of the amino acids, however, do absorb **ultraviolet** radiation, and all absorb in the **infrared** region. The absorption of energy by electrons as they rise to higher energy states occurs in the ultraviolet/visible region of the energy spectrum. Only the aromatic amino acids phenylalanine, tyrosine, and tryptophan exhibit significant ultraviolet absorption above 250 nm, as shown in Figure 4.15. These strong absorptions can be used for spectroscopic determinations of protein concentration. The aromatic amino acids also exhibit relatively weak fluorescence, and it has recently been shown that tryptophan can exhibit *phos-*

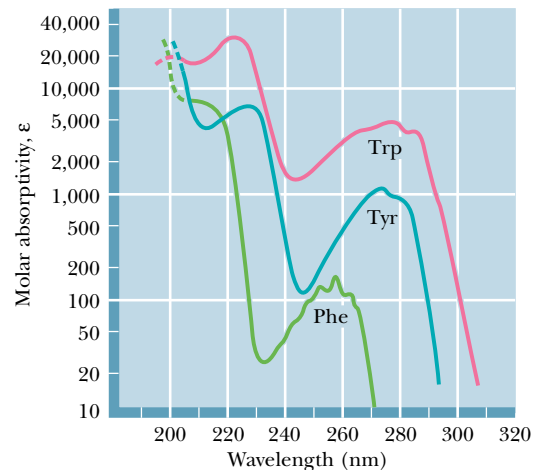


FIGURE 4.15 • The ultraviolet absorption spectra of the aromatic amino acids at pH 6. (From Wetlauffer, D.B., 1962. *Ultraviolet spectra of proteins and amino acids*. *Advances in Protein Chemistry* 17:303–390.)

phorescence—a relatively long-lived emission of light. These fluorescence and phosphorescence properties are especially useful in the study of protein structure and dynamics (see Chapter 6).

Nuclear Magnetic Resonance Spectra

The development in the 1950s of **nuclear magnetic resonance** (NMR), a spectroscopic technique that involves the absorption of radio frequency energy by certain nuclei in the presence of a magnetic field, played an important part in the chemical characterization of amino acids and proteins. Several important principles rapidly emerged from these studies. First, the **chemical shift**¹ of amino acid protons depends on their particular chemical environment and thus on the state of ionization of the amino acid. Second, the change in electron density during a titration is transmitted throughout the carbon chain in the aliphatic amino acids and the aliphatic portions of aromatic amino acids, as evidenced by changes in the chemical shifts of relevant protons. Finally, the magnitude of the **coupling constants** between protons on adjacent carbons depends in some cases on the ionization state of the amino acid. This apparently reflects differences in the preferred conformations in different ionization states. Proton NMR spectra of two amino acids are shown in Figure 4.16. Because they are highly sensitive to their environment, the chemical shifts of individual NMR signals can detect the pH-dependent ionizations of amino acids. Figure 4.17 shows the ¹³C chemical shifts occurring in a titration of lysine. Note that the chemical shifts of the carboxyl C, C_α, and C_β carbons of lysine are sensitive to dissociation of the nearby α-COOH and α-NH₃⁺ protons (with pK_a values of about 2 and 9, respectively), whereas the C_δ and C_ε carbons are sensitive to dissociation of the ε-NH₃⁺ group. Such measurements have been very useful for studies of the ionization behavior of amino acid residues in pro-

¹The chemical shift for any NMR signal is the difference in resonant frequency between the observed signal and a suitable reference signal. If two nuclei are magnetically coupled, the NMR signals of these nuclei split, and the separation between such split signals, known as the coupling constant, is likewise dependent on the structural relationship between the two nuclei.

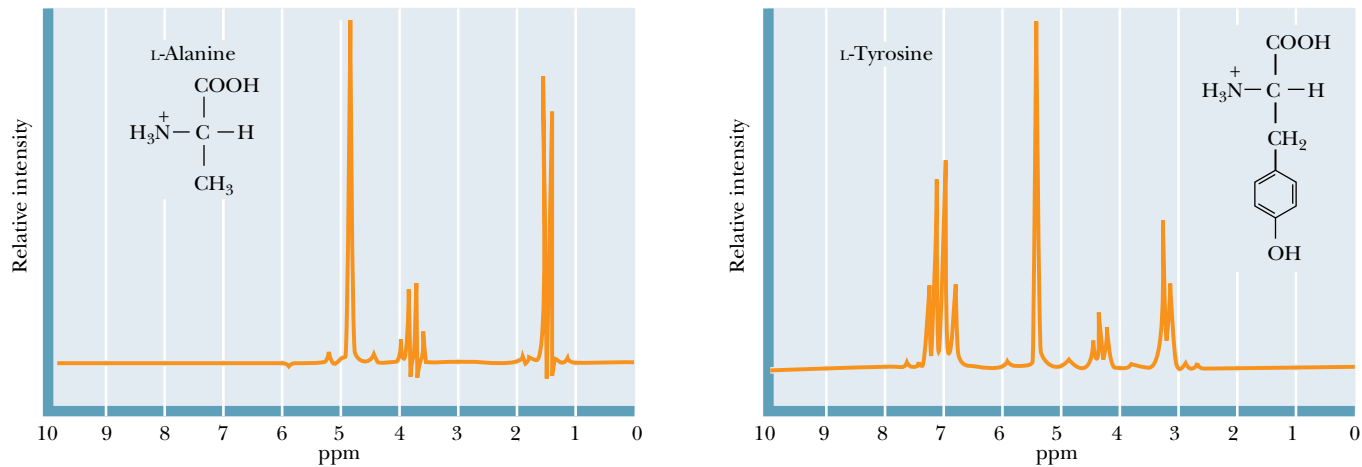


FIGURE 4.16 • Proton NMR spectra of several amino acids. Zero on the chemical shift scale is defined by the resonance of tetramethylsilane (TMS). (Adapted from Aldrich Library of NMR Spectra.)

teins. More sophisticated NMR measurements at very high magnetic fields are also used to determine the three-dimensional structures of peptides and even small proteins.

4.6 • Separation and Analysis of Amino Acid Mixtures

Chromatographic Methods

The purification and analysis of individual amino acids from complex mixtures was once a very difficult process. Today, however, the biochemist has a wide variety of methods available for the separation and analysis of amino acids, or for that matter, any of the other biological molecules and macromolecules we

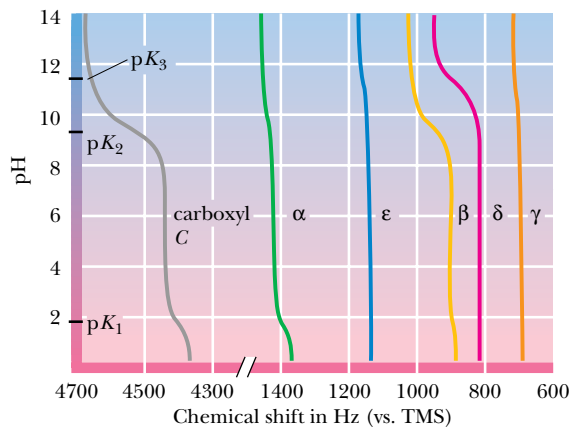


FIGURE 4.17 • A plot of chemical shifts versus pH for the carbons of lysine. Changes in chemical shift are most pronounced for atoms near the titrating groups. Note the correspondence between the pK_a values and the particular chemical shift changes. All chemical shifts are defined relative to tetramethylsilane (TMS). (From Suprenant, H., et al., 1980. Journal of Magnetic Resonance 40:231–243.)

encounter. All of these methods take advantage of the relative differences in the physical and chemical characteristics of amino acids, particularly ionization behavior and solubility characteristics. The methods important for amino acids include separations based on **partition** properties (the tendency to associate with one solvent or phase over another) and separations based on **electrical charge**. In all of the partition methods discussed here, the molecules of interest are allowed (or forced) to flow through a medium consisting of two phases—solid–liquid, liquid–liquid, or gas–liquid. In all of these methods, the molecules must show a preference for associating with one or the other phase. In this manner, the molecules partition, or distribute themselves, between the two phases in a manner based on their particular properties. The ratio of the concentrations of the amino acid (or other species) in the two phases is designated the *partition coefficient*.

In 1903, a separation technique based on repeated partitioning between phases was developed by Mikhail Tswett for the separation of plant pigments (carotenes and chlorophylls). Tswett, a Russian botanist, poured solutions of the pigments through columns of finely divided alumina and other solid media, allowing the pigments to partition between the liquid solvent and the solid support. Owing to the colorful nature of the pigments thus separated, Tswett called his technique **chromatography**. This term is now applied to a wide variety of separation methods, regardless of whether the products are colored or not. The success of all chromatography techniques depends on the repeated microscopic partitioning of a solute mixture between the available phases. The more frequently this partitioning can be made to occur within a given time span or over a given volume, the more efficient is the resulting separation. Chromatographic methods have advanced rapidly in recent years, due in part to the development of sophisticated new solid-phase materials. Methods important for amino acid separations include ion exchange chromatography, gas chromatography (GC), and high-performance liquid chromatography (HPLC).

Ion Exchange Chromatography

The separation of amino acids and other solutes is often achieved by means of **ion exchange chromatography**, in which the molecule of interest is *exchanged* for another ion onto and off of a charged solid support. In a typical procedure, solutes in a liquid phase, usually water, are passed through columns filled with a porous solid phase, usually a bed of synthetic resin particles, containing charged groups. Resins containing positive charges attract negatively charged solutes and are referred to as *anion exchangers*. Solid supports possessing negative charges attract positively charged species and are referred to as *cation exchangers*. Several typical cation and anion exchange resins with different types of charged groups are shown in Figure 4.18. The strength of the acidity or basicity of these groups and their number per unit volume of resin determine the type and strength of binding of an exchanger. Fully ionized acidic groups such as sulfonic acids result in an exchanger with a negative charge which binds cations very strongly. Weakly acidic or basic groups yield resins whose charge (and binding capacity) depends on the pH of the eluting solvent. The choice of the appropriate resin depends on the strength of binding desired. The bare charges on such solid phases must be counterbalanced by oppositely charged ions in solution (“counterions”). Washing a cation exchange resin, such as Dowex-50, which has strongly acidic phenyl-SO₃[−] groups, with a NaCl solution results in the formation of the so-called sodium form of the resin (see Figure 4.19). When the mixture whose separation is desired is added to the column, the positively charged solute molecules displace the Na⁺ ions and bind to the

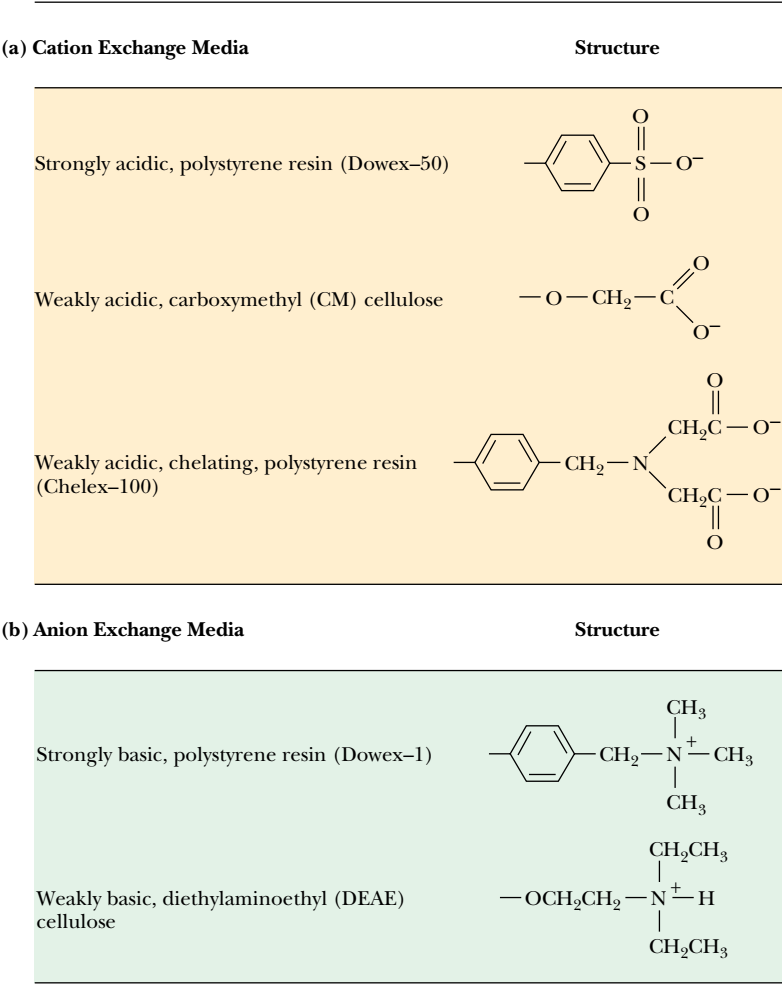


FIGURE 4.18 • Cation (a) and anion (b) exchange resins commonly used for biochemical separations.

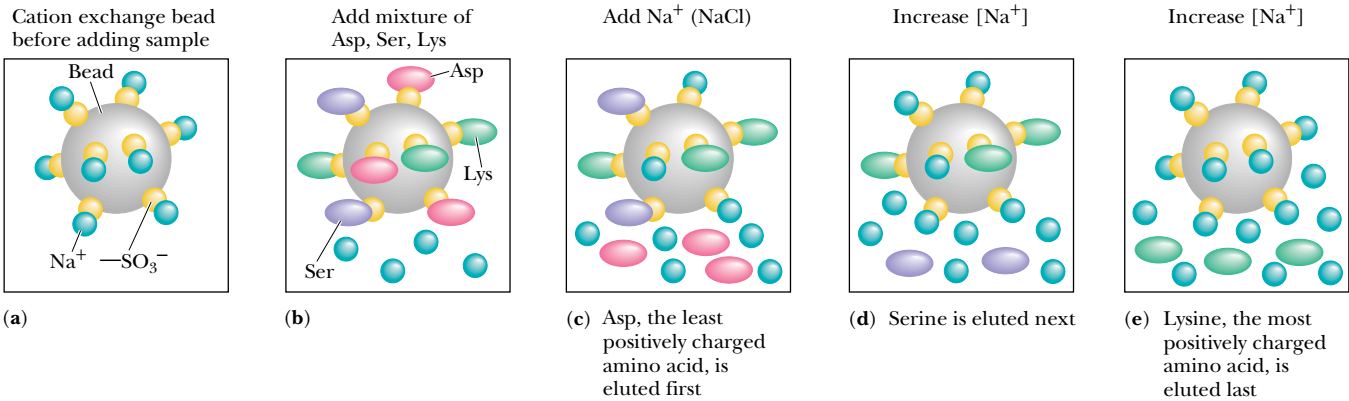


FIGURE 4.19 • Operation of a cation exchange column, separating a mixture of Asp, Ser, and Lys. (a) The cation exchange resin in the beginning, Na⁺ form. (b) A mixture of Asp, Ser, and Lys is added to the column containing the resin. (c) A gradient of the eluting salt (e.g., NaCl) is added to the column. Asp, the least positively charged amino acid, is eluted first. (d) As the salt concentration increases, Ser is eluted. (e) As the salt concentration is increased further, Lys, the most positively charged of the three amino acids, is eluted last.

resin. A gradient of an appropriate salt is then applied to the column, and the solute molecules are competitively (and sequentially) displaced (eluted) from the column by the rising concentration of cations in the gradient, in an order that is inversely related to their affinities for the column. The separation of a mixture of amino acids on such a column is shown in Figures 4.19 and 4.20. Figure 4.21, taken from a now-classic 1958 paper by Stanford Moore, Darrel Spackman, and William Stein, shows a typical separation of the common amino acids. The events occurring in this separation are essentially those depicted in Figures 4.19 and 4.20. The amino acids are applied to the column at low pH (4.25), under which conditions the acidic amino acids (aspartate and glutamate, among others) are weakly bound and the basic amino acids, such as arginine and lysine, are tightly bound. Sodium citrate solutions, at two different concentrations and three different values of pH, are used to elute the amino acids gradually from the column.

A typical HPLC chromatogram using precolumn derivatization of amino acids with *o*-phthalaldehyde (OPA) is shown in Figure 4.22. HPLC has rapidly become the chromatographic technique of choice for most modern biochemists. The very high resolution, excellent sensitivity, and high speed of this technique usually outweigh the disadvantage of relatively low capacity.

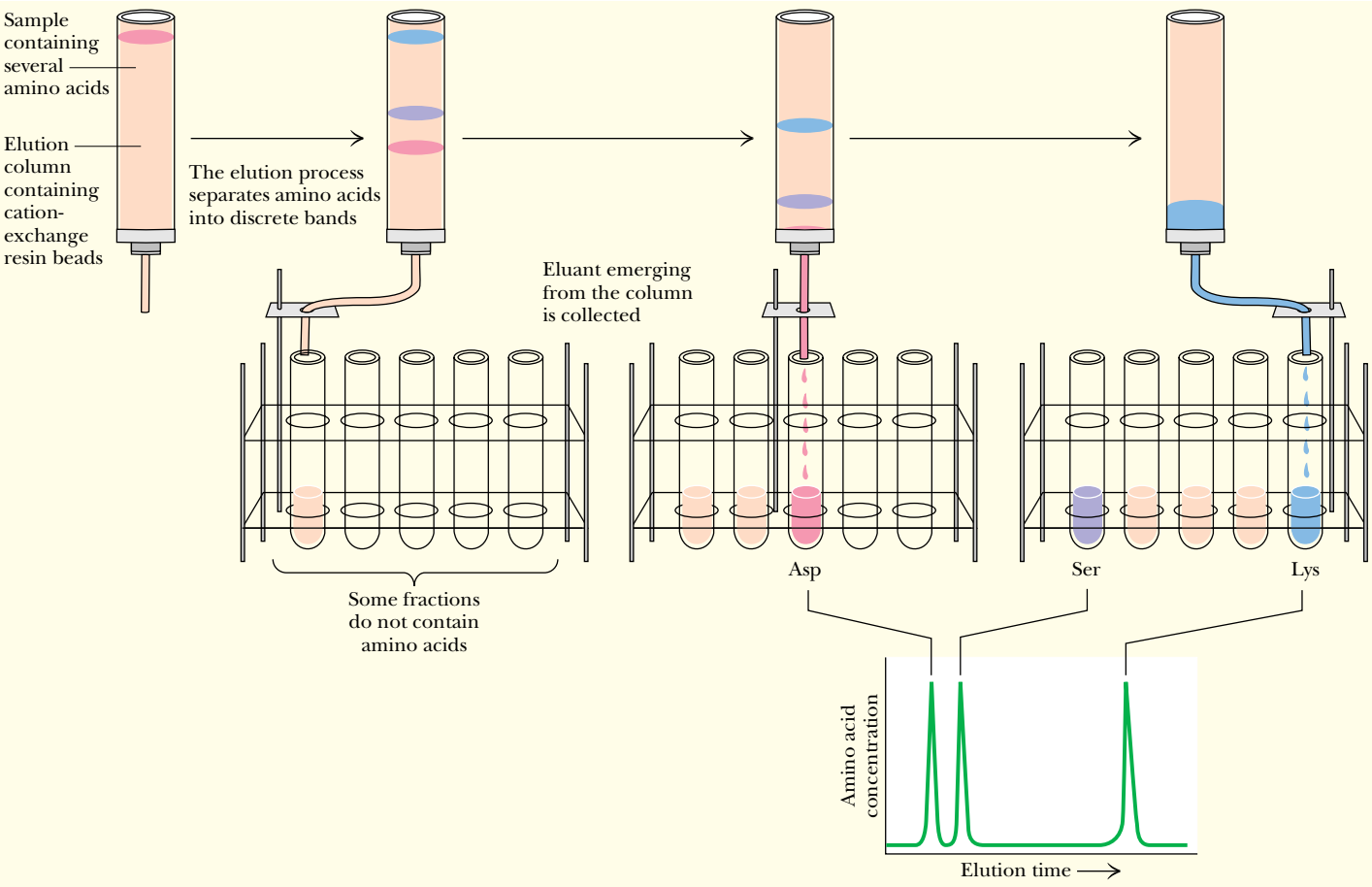


FIGURE 4.20 • The separation of amino acids on a cation exchange column.

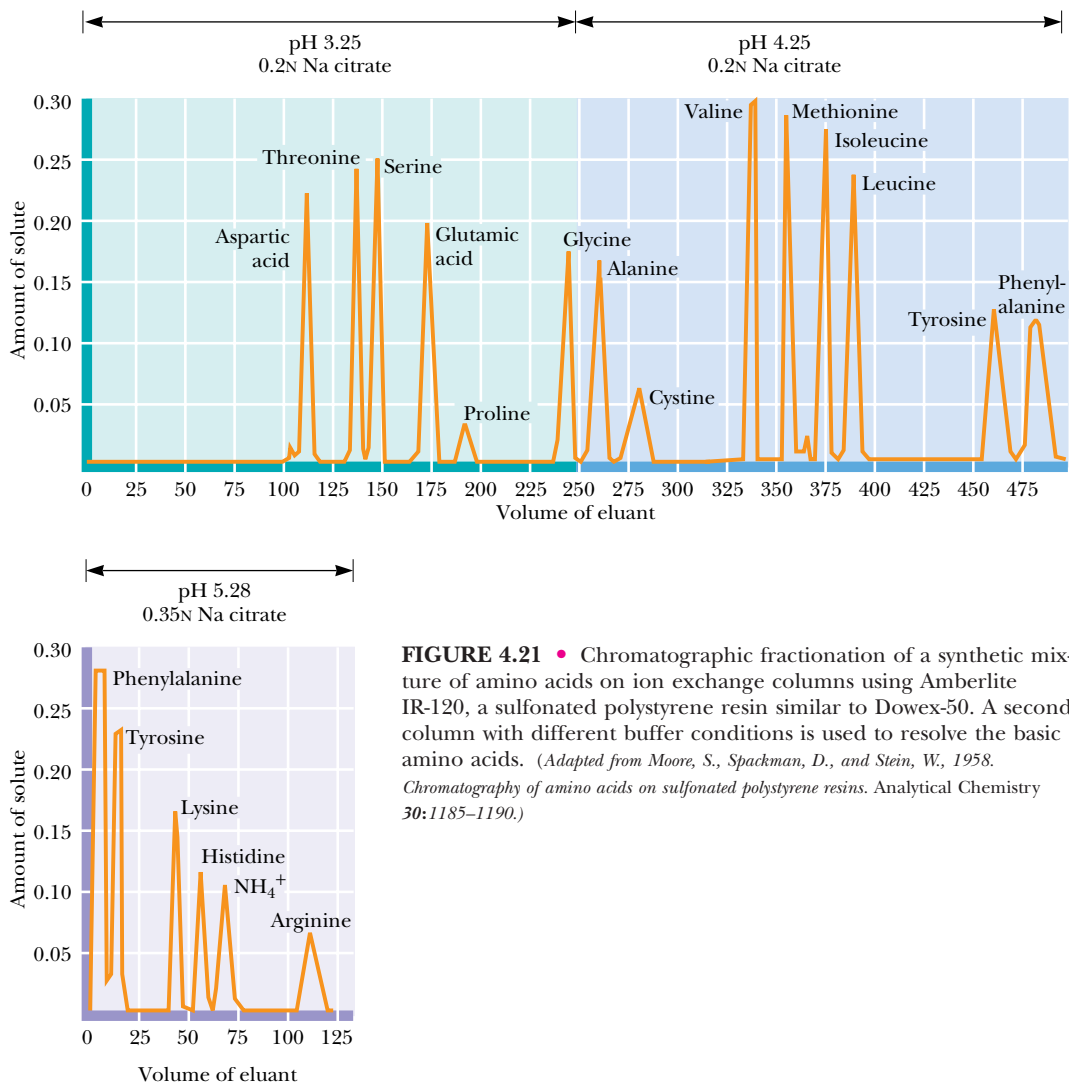


FIGURE 4.21 • Chromatographic fractionation of a synthetic mixture of amino acids on ion exchange columns using Amberlite IR-120, a sulfonated polystyrene resin similar to Dowex-50. A second column with different buffer conditions is used to resolve the basic amino acids. (Adapted from Moore, S., Spackman, D., and Stein, W., 1958. *Chromatography of amino acids on sulfonated polystyrene resins*. Analytical Chemistry 30:1185–1190.)

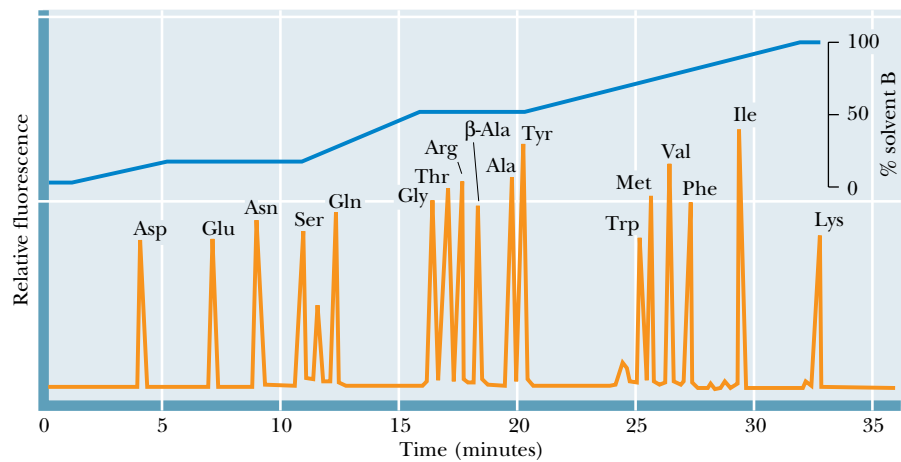


FIGURE 4.22 • HPLC chromatogram of amino acids employing precolumn derivatization with OPA. Chromatography was carried out on an Ultrasphere ODS column using a complex tetrahydrofuran:methanol:0.05 M sodium acetate (pH 5.9) 1:19:80 to methanol:0.05 M sodium acetate (pH 5.9) 4:1 gradient at a flow rate of 1.7 mL/min. (Adapted from Jones, B. N., Pääbo, S., and Stein, S., 1981. *Amino acid analysis and enzymic sequence determination of peptides by an improved o-phthalaldehyde precolumn labeling procedure*. Journal of Liquid Chromatography 4:56–586.)

PROBLEMS

- Without consulting chapter figures, draw Fischer projection formulas for glycine, aspartate, leucine, isoleucine, methionine, and threonine.
- Without reference to the text, give the one-letter and three-letter abbreviations for asparagine, arginine, cysteine, lysine, proline, tyrosine, and tryptophan.
- Write equations for the ionic dissociations of alanine, glutamate, histidine, lysine, and phenylalanine.
- How is the pK_a of the $\alpha\text{-NH}_3^+$ group affected by the presence on an amino acid of the $\alpha\text{-COO}^-$?
- Draw an appropriate titration curve for aspartic acid, labeling the axes and indicating the equivalence points and the pK_a values.
- Calculate the concentrations of all ionic species in a 0.25 M solution of histidine at pH 2, pH 6.4, and pH 9.3.
- Calculate the pH at which the γ -carboxyl group of glutamic acid is two-thirds dissociated.
- Calculate the pH at which the ϵ -amino group of lysine is 20% dissociated.
- Calculate the pH of a 0.3 M solution of (a) leucine hydrochloride, (b) sodium leucinate, and (c) isoelectric leucine.
- Quantitative measurements of optical activity are usually expressed in terms of the specific rotation, $[\alpha]_D^{25}$, defined as

$$[\alpha]_D^{25} = \frac{\text{Measured rotation in degrees} \times 100}{(\text{Optical path in dm}) \times (\text{conc. in g/mL})}$$

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For any measurement of optical rotation, the wavelength of the light used and the temperature must both be specified. In this case, D refers to the “D line” of sodium at 589 nm and 25 refers to a measurement temperature of 25°C. Calculate the concentration of a solution of L-arginine that rotates the incident light by 0.35° in an optical path length of 1 dm (decimeter).

- Absolute configurations of the amino acids are referenced to D- and L-glyceraldehyde on the basis of chemical transformations that can convert the molecule of interest to either of these reference isomeric structures. In such reactions, the stereochemical consequences for the asymmetric centers must be understood for each reaction step. Propose a sequence of reactions that would demonstrate that L(–)-serine is stereochemically related to L(–)-glyceraldehyde.
- Describe the stereochemical aspects of the structure of cystine, the structure that is a disulfide-linked pair of cysteines.
- Draw a simple mechanism for the reaction of a cysteine sulfhydryl group with iodoacetamide.
- Describe the expected elution pattern for a mixture of aspartate, histidine, isoleucine, valine, and arginine on a column of Dowex-50.
- Assign (*R,S*) nomenclature to the threonine isomers of Figure 4.14.

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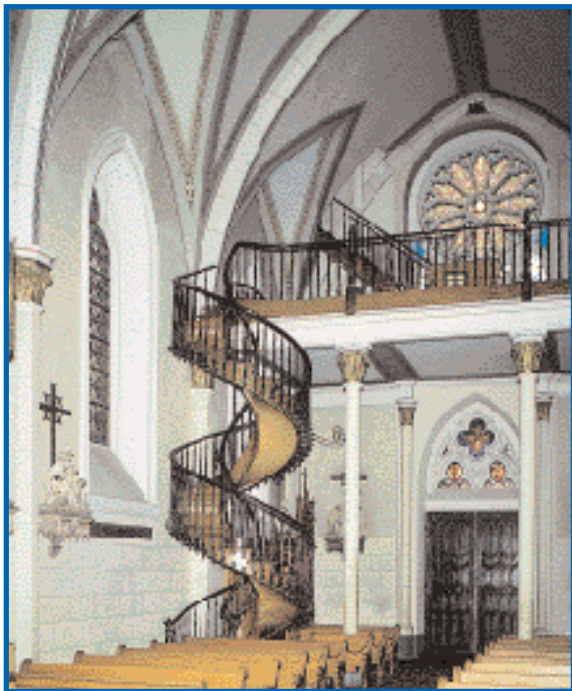
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Chapter 5

Proteins: Their Biological Functions and Primary Structure



Although helices are uncommon in manmade architecture, they are a common structural theme in biological macromolecules—proteins, nucleic acids, and even polysaccharides.

(Loretto Chapel, Santa Fe, NM/ © Sarbo)

Proteins are a diverse and abundant class of biomolecules, constituting more than 50% of the dry weight of cells. This diversity and abundance reflect the central role of proteins in virtually all aspects of cell structure and function. An extraordinary diversity of cellular activity is possible only because of the versatility inherent in proteins, each of which is specifically tailored to its biological role. The pattern by which each is tailored resides within the genetic information of cells, encoded in a specific sequence of nucleotide bases in DNA.

... by small and simple things are great things brought to pass.

ALMA 37.6, *The Book of Mormon*

OUTLINE

- 5.1 • Proteins Are Linear Polymers of Amino Acids
- 5.2 • Architecture of Protein Molecules
- 5.3 • The Many Biological Functions of Proteins
- 5.4 • Some Proteins Have Chemical Groups Other Than Amino Acids
- 5.5 • Reactions of Peptides and Proteins
- 5.6 • Purification of Protein Mixtures
- 5.7 • The Primary Structure of a Protein: Determining the Amino Acid Sequence
- 5.8 • Nature of Amino Acid Sequences
- 5.9 • Synthesis of Polypeptides in the Laboratory

Each such segment of encoded information defines a gene, and expression of the gene leads to synthesis of the specific protein encoded by it, endowing the cell with the functions unique to that particular protein. Proteins are the agents of biological function; they are also the expressions of genetic information.

5.1 • Proteins Are Linear Polymers of Amino Acids

Chemically, proteins are unbranched polymers of amino acids linked head to tail, from carboxyl group to amino group, through formation of covalent **peptide bonds**, a type of amide linkage (Figure 5.1).

Peptide bond formation results in the release of H₂O. The peptide “backbone” of a protein consists of the repeated sequence —N—C_α—C—, where the N represents the amide nitrogen, the C_α is the α-carbon atom of an amino acid in the polymer chain, and the final C is the carbonyl carbon of the amino acid, which in turn is linked to the amide N of the next amino acid down the line. The geometry of the peptide backbone is shown in Figure 5.2. Note that the carbonyl oxygen and the amide hydrogen are *trans* to each other in this figure. This conformation is favored energetically because it results in less steric hindrance between nonbonded atoms in neighboring amino acids. Because the α-carbon atom of the amino acid is a chiral center (in all amino acids except glycine), the polypeptide chain is inherently asymmetric. Only L-amino acids are found in proteins.

The Peptide Bond Has Partial Double Bond Character

The peptide linkage is usually portrayed by a single bond between the carbonyl carbon and the amide nitrogen (Figure 5.3a). Therefore, in principle, rotation may occur about any covalent bond in the polypeptide backbone because all three kinds of bonds (N—C_α, C_α—C_o, and the C_o—N peptide bond) are single bonds. In this representation, the C and N atoms of the peptide grouping are both in planar *sp*² hybridization and the C and O atoms are linked by a π bond, leaving the nitrogen with a lone pair of electrons in a 2*p* orbital. However, another resonance form for the peptide bond is feasible in which the C and N atoms participate in a π bond, leaving a lone *e*[−] pair on the oxygen (Figure 5.3b). This structure prevents free rotation about the C_o—N peptide bond because it becomes a double bond. The real nature of the peptide bond lies somewhere between these extremes; that is, it has partial double bond character, as represented by the intermediate form shown in Figure 5.3c.

Peptide bond resonance has several important consequences. First, it restricts free rotation around the peptide bond and leaves the peptide backbone with only two degrees of freedom per amino acid group: rotation around

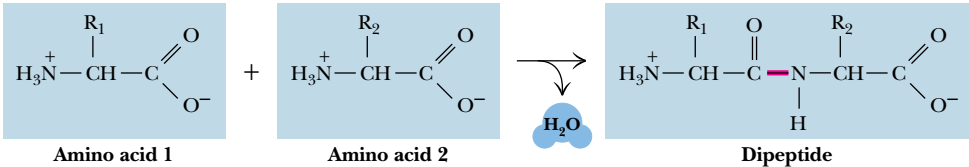


FIGURE 5.1 • Peptide formation is the creation of an amide bond between the carboxyl group of one amino acid and the amino group of another amino acid. R₁ and R₂ represent the R groups of two different amino acids.

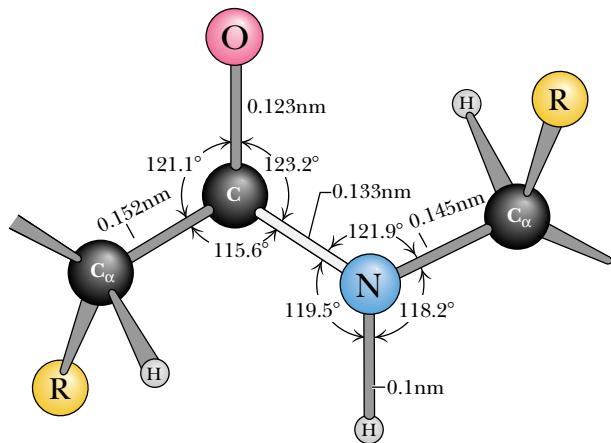


FIGURE 5.2 • The peptide bond is shown in its usual *trans* conformation of carbonyl O and amide H. The C_α atoms are the α-carbons of two adjacent amino acids joined in peptide linkage. The dimensions and angles are the average values observed by crystallographic analysis of amino acids and small peptides. The peptide bond is the light gray bond between C and N. (Adapted from Ramachandran, G. N., et al., 1974. *Biochimica Biophysica Acta* 359:298–302.)

the N—C_α bond and rotation around the C_α—C_o bond.¹ Second, the six atoms composing the peptide bond group tend to be coplanar, forming the so-called **amide plane** of the polypeptide backbone (Figure 5.4). Third, the C_o—N bond length is 0.133 nm, which is shorter than normal C—N bond lengths (for example, the C_α—N bond of 0.145 nm) but longer than typical C=N bonds (0.125 nm). The peptide bond is estimated to have 40% double-bond character.

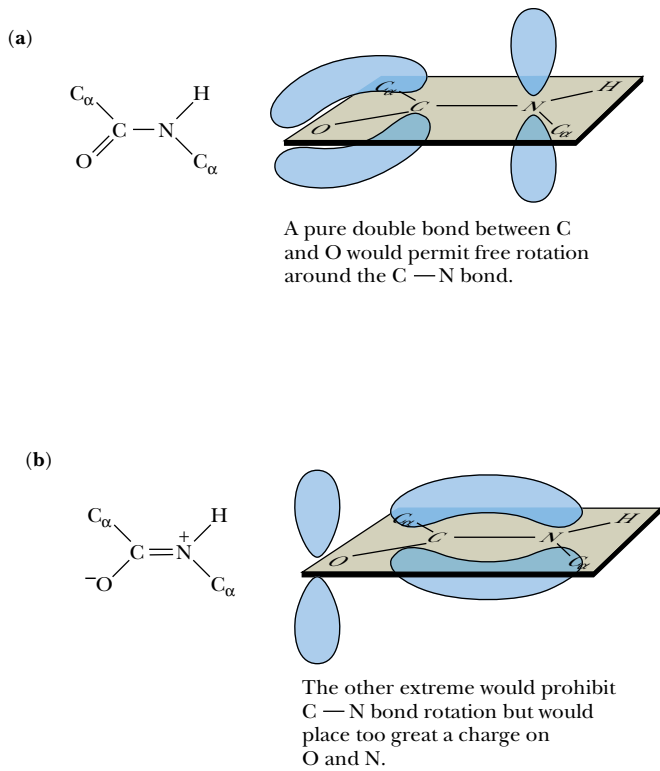
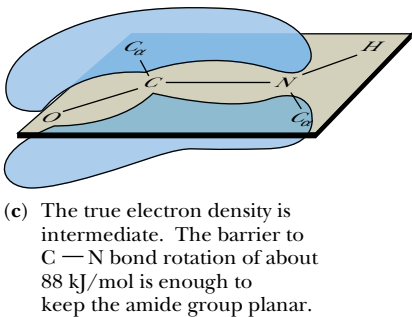
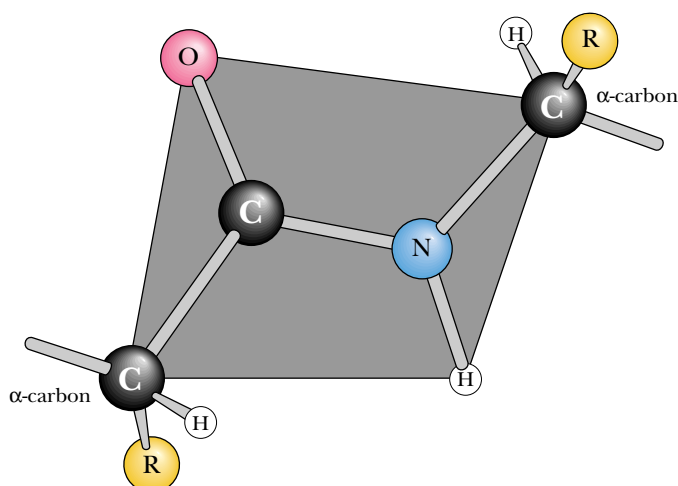


FIGURE 5.3 • The partial double bond character of the peptide bond. Resonance interactions among the carbon, oxygen, and nitrogen atoms of the peptide group can be represented by two resonance extremes (a and b). (a) The usual way the peptide atoms are drawn. (b) In an equally feasible form, the peptide bond is now a double bond; the amide N bears a positive charge and the carbonyl O has a negative charge. (c) The actual peptide bond is best described as a resonance hybrid of the forms in (a) and (b). Significantly, all of the atoms associated with the peptide group are coplanar, rotation about C_o—N is restricted, and the peptide is distinctly polar. (Irving Geis)



¹The angle of rotation about the N—C_α bond is designated ϕ , phi, whereas the C_α—C_o angle of rotation is designated ψ , psi.

FIGURE 5.4 • The coplanar relationship of the atoms in the amide group is highlighted as an imaginary shaded plane lying between two successive α -carbon atoms in the peptide backbone.



The Polypeptide Backbone Is Relatively Polar

Peptide bond resonance also causes the peptide backbone to be relatively polar. As shown in Figure 5.3b, the amide nitrogen represents a protonated or positively charged form, and the carbonyl oxygen becomes a negatively charged atom in the double-bonded resonance state. In actuality, the hybrid state of the partially double-bonded peptide arrangement gives a net positive charge of 0.28 on the amide N and an equivalent net negative charge of 0.28 on the carbonyl O. The presence of these partial charges means that the peptide bond has a permanent dipole. Nevertheless, the peptide backbone is relatively unreactive chemically, and protons are gained or lost by the peptide groups only at extreme pH conditions.

Peptide Classification

Peptide is the name assigned to short polymers of amino acids. Peptides are classified by the number of amino acid units in the chain. Each unit is called an **amino acid residue**, the word *residue* denoting what is left after the release of H_2O when an amino acid forms a peptide link upon joining the peptide chain. **Dipeptides** have two amino acid residues, tripeptides have three, tetrapeptides four, and so on. After about 12 residues, this terminology becomes cumbersome, so peptide chains of more than 12 and less than about 20 amino acid residues are usually referred to as **oligopeptides**, and, when the chain exceeds several dozen amino acids in length, the term **polypeptide** is used. The distinctions in this terminology are not precise.

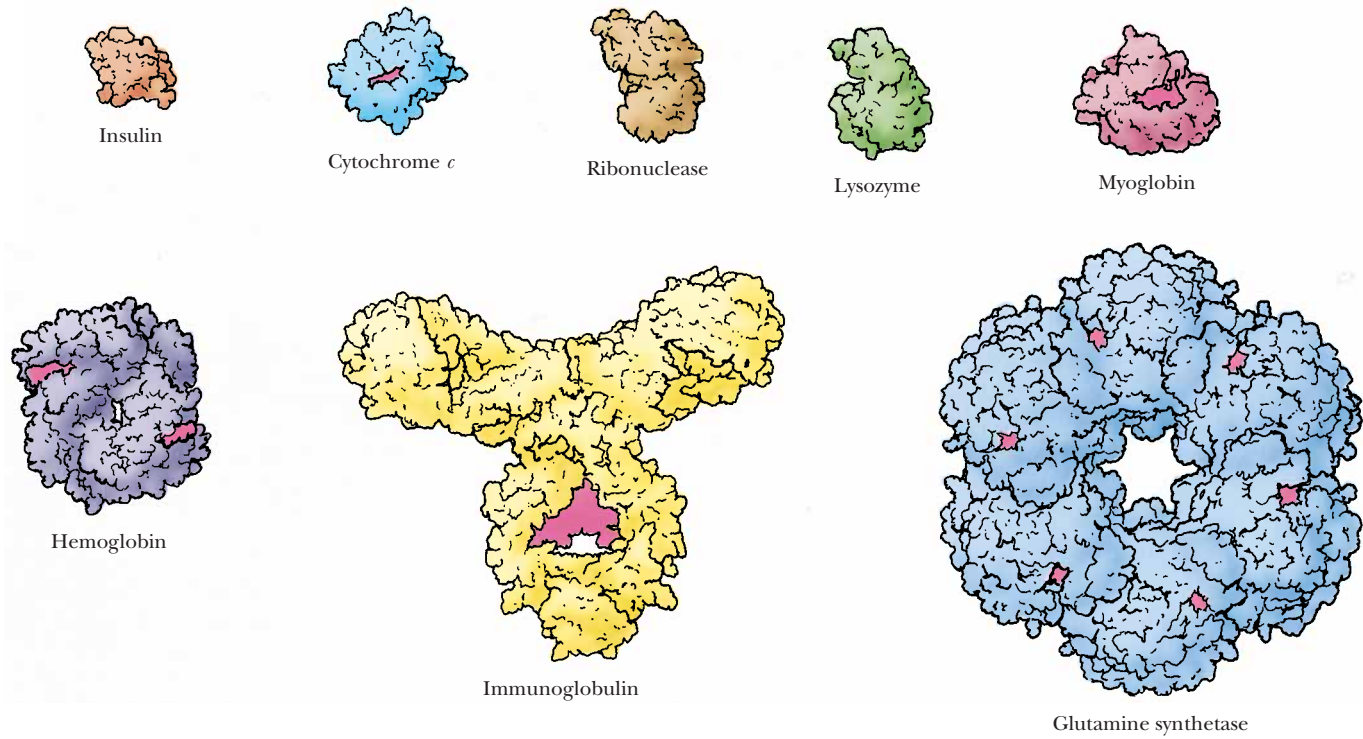
Proteins Are Composed of One or More Polypeptide Chains

The terms *polypeptide* and *protein* are used interchangeably in discussing single polypeptide chains. The term **protein** broadly defines molecules composed of one or more polypeptide chains. Proteins having only one polypeptide chain are **monomeric proteins**. Proteins composed of more than one polypeptide chain are **multimeric proteins**. Multimeric proteins may contain only one kind of polypeptide, in which case they are **homomultimeric**, or they may be composed of several different kinds of polypeptide chains, in which instance they are **heteromultimeric**. Greek letters and subscripts are used to denote the polypeptide composition of multimeric proteins. Thus, an α_2 -type protein is a dimer of identical polypeptide subunits, or a **homodimer**. Hemoglobin (Table 5.1) consists of four polypeptides of two different kinds; it is an $\alpha_2\beta_2$ heteromultimer.

Table 5.1

Size of Protein Molecules*

Protein	M _r	Number of Residues per Chain	Subunit Organization
Insulin (bovine)	5,733	21 (A) 30 (B)	$\alpha\beta$
Cytochrome <i>c</i> (equine)	12,500	104	α_1
Ribonuclease A (bovine pancreas)	12,640	124	α_1
Lysozyme (egg white)	13,930	129	α_1
Myoglobin (horse)	16,980	153	α_1
Chymotrypsin (bovine pancreas)	22,600	13 (α) 132 (β) 97 (γ)	$\alpha\beta\gamma$
Hemoglobin (human)	64,500	141 (α) 146 (β)	$\alpha_2\beta_2$
Serum albumin (human)	68,500	550	α_1
Hexokinase (yeast)	96,000	200	α_4
γGlobulin (horse)	149,900	214 (α) 446 (β)	$\alpha_2\beta_2$
Glutamate dehydrogenase (liver)	332,694	500	α_6
Myosin (rabbit)	470,000	1800 (heavy, <i>h</i>) 190 (α) 149 (α') 160 (β)	$h_2\alpha_1\alpha'_2\beta_2$
Ribulose biphosphate carboxylase (spinach)	560,000	475 (α) 123 (β)	$\alpha_8\beta_8$
Glutamine synthetase (<i>E. coli</i>)	600,000	468	α_{12}



*Illustrations of selected proteins listed in Table 5.1 are drawn to constant scale.
Adapted from Goodsell and Olson, 1993. *Trends in Biochemical Sciences* 18:65–68.

Polypeptide chains of proteins range in length from about 100 amino acids to 1800, the number found in each of the two polypeptide chains of myosin, the contractile protein of muscle. However, titin, another muscle protein, has nearly 27,000 amino acid residues and a molecular weight of 2.8×10^6 . The average molecular weight of polypeptide chains in eukaryotic cells is about 31,700, corresponding to about 270 amino acid residues. Table 5.1 is a representative list of proteins according to size. The molecular weights (M_r) of proteins can be estimated by a number of physicochemical methods such as polyacrylamide gel electrophoresis or ultracentrifugation (see Chapter Appendix). Precise determinations of protein molecular masses are best obtained by simple calculations based on knowledge of their amino acid sequence. No simple generalizations correlate the size of proteins with their functions. For instance, the same function may be fulfilled in different cells by proteins of different molecular weight. The *Escherichia coli* enzyme responsible for glutamine synthesis (a protein known as *glutamine synthetase*) has a molecular weight of 600,000, whereas the analogous enzyme in brain tissue has a molecular weight of just 380,000.

Acid Hydrolysis of Proteins

Peptide bonds of proteins are hydrolyzed by either strong acid or strong base. Because acid hydrolysis proceeds without racemization and with less destruction of certain amino acids (Ser, Thr, Arg, and Cys) than alkaline treatment, it is the method of choice in analysis of the amino acid composition of proteins and polypeptides. Typically, samples of a protein are hydrolyzed with 6 N HCl at 110°C for 24, 48, and 72 hr in sealed glass vials. Tryptophan is destroyed by acid and must be estimated by other means to determine its contribution to the total amino acid composition. The OH-containing amino acids serine and threonine are slowly destroyed, but the data obtained for the three time points (24, 48, and 72 hr) allow extrapolation to zero time to estimate the original Ser and Thr content (Figure 5.5). In contrast, peptide bonds involving hydrophobic residues such as valine and isoleucine are only slowly hydrolyzed in acid. Another complication arises because the β - and γ -amide linkages in asparagine (Asn) and glutamine (Gln) are acid labile. The amino nitrogen is released as free ammonium, and all of the Asn and Gln residues of the protein become aspartic acid (Asp) and glutamic acid (Glu), respectively. The

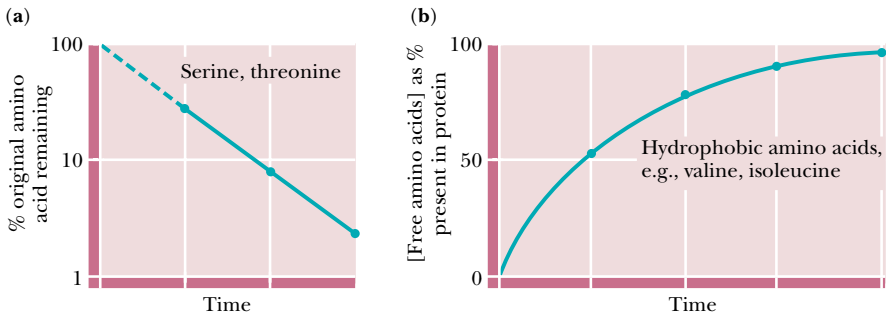


FIGURE 5.5 • (a) The hydroxy amino acids serine and threonine are slowly destroyed during the course of protein hydrolysis for amino acid composition analysis. Extrapolation of the data back to time zero allows an accurate estimation of the amount of these amino acids originally present in the protein sample. (b) Peptide bonds involving hydrophobic amino acid residues such as valine and isoleucine resist hydrolysis by HCl. With time, these amino acids are released and their free concentrations approach a limiting value that can be approximated with reliability.

amount of ammonium released during acid hydrolysis gives an estimate of the total number of Asn and Gln residues in the original protein, but not the amounts of either. Accordingly, the concentrations of Asp and Glu determined in amino acid analysis are expressed as Asx and Glx, respectively. Because the relative contributions of [Asn + Asp] or [Gln + Glu] cannot be derived from the data, this information must be obtained by alternative means.

Amino Acid Analysis of Proteins

The complex amino acid mixture in the hydrolysate obtained after digestion of a protein in 6 *N* HCl can be separated into the component amino acids by either ion exchange chromatography (see Chapter 4) or by reversed-phase high-pressure liquid chromatography (HPLC) (see Chapter Appendix). The amount of each amino acid can then be determined. In ion exchange chromatography, the amino acids are separated and then quantified following reaction with ninhydrin (so-called postcolumn derivatization). In HPLC, the amino acids are converted to phenylthiohydantoin (PTH) derivatives via reaction with Edman's reagent (see Figure 5.19) prior to chromatography (precursor derivatization). Both of these methods of separation and analysis are fully automated in instruments called **amino acid analyzers**. Analysis of the amino acid composition of a 30-kD protein by these methods requires less than 1 hour and only 6 μ g (0.2 nmol) of the protein.

Table 5.2 gives the amino acid composition of several selected proteins: ribonuclease A, alcohol dehydrogenase, myoglobin, histone H3, and collagen. Each of the 20 naturally occurring amino acids is usually represented at least once in a polypeptide chain. However, some small proteins may not have a representative of every amino acid. Note that ribonuclease (12.6 kD, 124 amino acid residues) does not contain any tryptophan. Amino acids almost never occur in equimolar ratios in proteins, indicating that proteins are not composed of repeating arrays of amino acids. There are a few exceptions to this rule. Collagen, for example, contains large proportions of glycine and proline, and much of its structure is composed of (Gly-*x*-Pro) repeating units, where *x* is any amino acid. Other proteins show unusual abundances of various amino acids. For example, histones are rich in positively charged amino acids such as arginine and lysine. Histones are a class of proteins found associated with the anionic phosphate groups of eukaryotic DNA.

Amino acid analysis itself does not directly give the number of residues of each amino acid in a polypeptide, but it does give amounts from which the percentages or ratios of the various amino acids can be obtained (Table 5.2). If the molecular weight *and* the exact amount of the protein analyzed are known (or the number of amino acid residues per molecule is known), the molar ratios of amino acids in the protein can be calculated. Amino acid analysis provides no information on the order or sequence of amino acid residues in the polypeptide chain. Because the polypeptide chain is unbranched, it has only two ends, an amino-terminal or **N-terminal end** and a carboxyl-terminal or **C-terminal end**.

The Sequence of Amino Acids in Proteins

The unique characteristic of each protein is the distinctive sequence of amino acid residues in its polypeptide chain(s). Indeed, it is the **amino acid sequence** of proteins that is encoded by the nucleotide sequence of DNA. This amino acid sequence, then, is a form of genetic information. By convention, the amino acid sequence is read from the N-terminal end of the polypeptide chain through to the C-terminal end. As an example, every molecule of ribonucle-

Table 5.2

Amino Acid Composition of Some Selected Proteins

Values expressed are percent representation of each amino acid.

Amino Acid	Proteins*				
	RNase	ADH	Mb	Histone H3	Collagen
Ala	6.9	7.5	9.8	13.3	11.7
Arg	3.7	3.2	1.7	13.3	4.9
Asn	7.6	2.1	2.0	0.7	1.0
Asp	4.1	4.5	5.0	3.0	3.0
Cys	6.7	3.7	0	1.5	0
Gln	6.5	2.1	3.5	5.9	2.6
Glu	4.2	5.6	8.7	5.2	4.5
Gly	3.7	10.2	9.0	5.2	32.7
His	3.7	1.9	7.0	1.5	0.3
Ile	3.1	6.4	5.1	5.2	0.8
Leu	1.7	6.7	11.6	8.9	2.1
Lys	7.7	8.0	13.0	9.6	3.6
Met	3.7	2.4	1.5	1.5	0.7
Phe	2.4	4.8	4.6	3.0	1.2
Pro	4.5	5.3	2.5	4.4	22.5
Ser	12.2	7.0	3.9	3.7	3.8
Thr	6.7	6.4	3.5	7.4	1.5
Trp	0	0.5	1.3	0	0
Tyr	4.0	1.1	1.3	2.2	0.5
Val	7.1	10.4	4.8	4.4	1.7
Acidic	8.4	10.2	13.7	8.1	7.5
Basic	15.0	13.1	21.8	24.4	8.8
Aromatic	6.4	6.4	7.2	5.2	1.7
Hydrophobic	18.0	30.7	27.6	23.0	6.5

*Proteins are as follows:
RNase: Bovine ribonuclease A, an enzyme; 124 amino acid residues. Note that RNase lacks tryptophan.
ADH: Horse liver alcohol dehydrogenase, an enzyme; dimer of identical 374 amino acid polypeptide chains. The amino acid composition of ADH is reasonably representative of the norm for water-soluble proteins.
Mb: Sperm whale myoglobin, an oxygen-binding protein; 153 amino acid residues. Note that Mb lacks cysteine.
Histone H3: Histones are DNA-binding proteins found in chromosomes; 135 amino acid residues. Note the very basic nature of this protein due to its abundance of Arg and Lys residues. It also lacks tryptophan.
Collagen: Collagen is an extracellular structural protein; 1052 amino acid residues. Collagen has an unusual amino acid composition; it is about one-third glycine and is rich in proline. Note that it also lacks Cys and Trp and is deficient in aromatic amino acid residues in general.

ase A from bovine pancreas has the same amino acid sequence, beginning with N-terminal lysine at position 1 and ending with C-terminal valine at position 124 (Figure 5.6). Given the possibility of any of the 20 amino acids at each position, the number of unique amino acid sequences is astronomically large. The astounding sequence variation possible within polypeptide chains provides

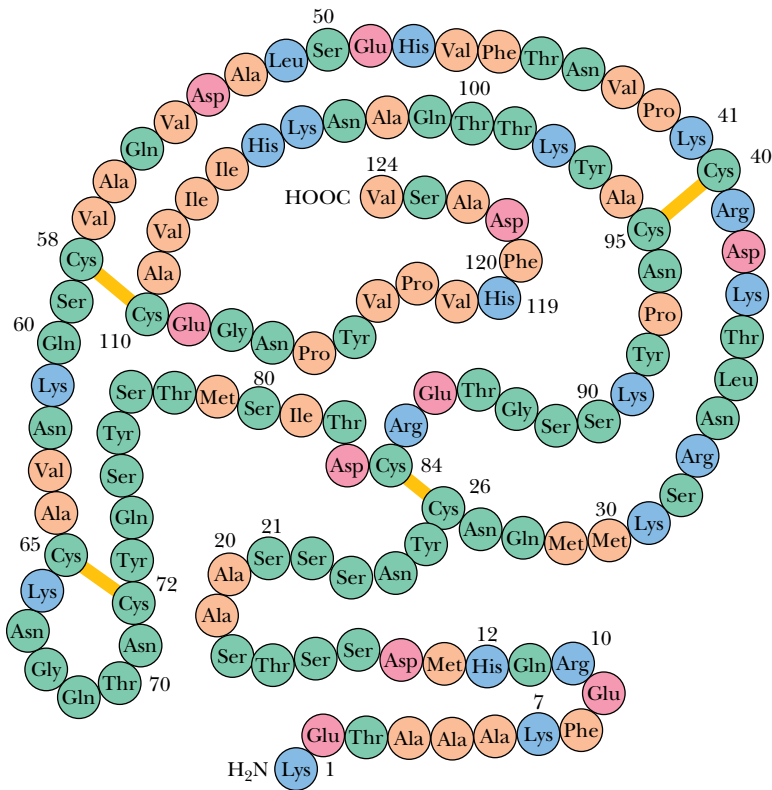


FIGURE 5.6 • Bovine pancreatic ribonuclease A contains 124 amino acid residues, none of which are tryptophan. Four intrachain disulfide bridges (S—S) form cross-links in this polypeptide between Cys²⁶ and Cys⁸⁴, Cys⁴⁰ and Cys⁹⁵, Cys⁵⁸ and Cys¹¹⁰, and Cys⁶⁵ and Cys⁷². These disulfides are depicted by yellow bars.



a key insight into the incredible functional diversity of protein molecules in biological systems, which is discussed shortly.

5.2 • Architecture of Protein Molecules

Protein Shape

As a first approximation, proteins can be assigned to one of three global classes on the basis of shape and solubility: fibrous, globular, or membrane (Figure 5.7). **Fibrous proteins** tend to have relatively simple, regular linear structures. These proteins often serve structural roles in cells. Typically, they are insoluble in water or in dilute salt solutions. In contrast, **globular proteins** are roughly spherical in shape. The polypeptide chain is compactly folded so that hydrophobic amino acid side chains are in the interior of the molecule and the hydrophilic side chains are on the outside exposed to the solvent, water. Consequently, globular proteins are usually very soluble in aqueous solutions. Most soluble proteins of the cell, such as the cytosolic enzymes, are globular in shape. **Membrane proteins** are found in association with the various membrane systems of cells. For interaction with the nonpolar phase within membranes, membrane proteins have hydrophobic amino acid side chains oriented outward. As such, membrane proteins are insoluble in aqueous solutions but can be solubilized in solutions of detergents. Membrane proteins characteristically have fewer hydrophilic amino acids than cytosolic proteins.

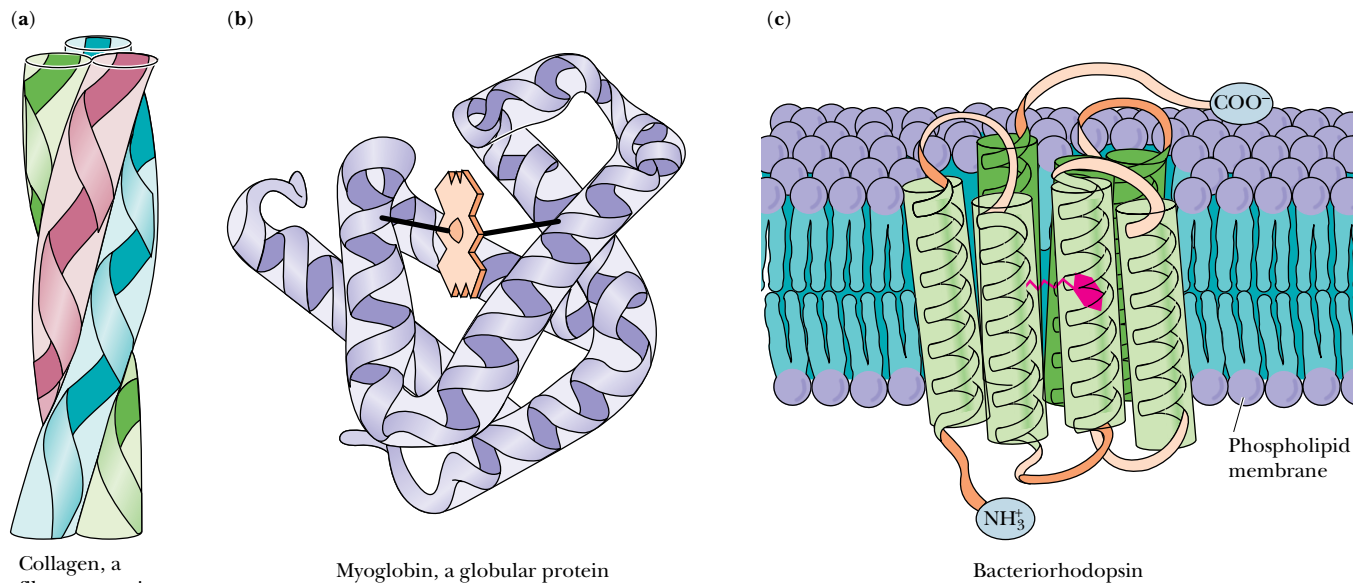


FIGURE 5.7 • (a) Proteins having structural roles in cells are typically fibrous and often water insoluble. Collagen is a good example. Collagen is composed of three polypeptide chains that intertwine. (b) Soluble proteins serving metabolic functions can be characterized as compactly folded globular molecules, such as myoglobin. The folding pattern puts hydrophilic amino acid side chains on the outside and buries hydrophobic side chains in the interior, making the protein highly water soluble. (c) Membrane proteins fold so that hydrophobic amino acid side chains are exposed in their membrane-associated regions. The portions of membrane proteins extending into or exposed at the aqueous environments are hydrophilic in character, like soluble proteins. Bacteriorhodopsin is a typical membrane protein; it binds the light-absorbing pigment, *cis*-retinal, shown here in red.

(a, b, Irving Geis)

A DEEPER LOOK

The Virtually Limitless Number of Different Amino Acid Sequences

Given 20 different amino acids, a polypeptide chain of n residues can have any one of 20^n possible sequence arrangements. To portray this, consider the number of tripeptides possible if there were only three different amino acids, A, B, and C (tripeptide = $3 = n$; $n^3 = 3^3 = 27$):

AAA	BBB	CCC
AAB	BBA	CCA
AAC	BBC	CCB
ABA	BAB	CBC
ACA	BCB	CAC
ABC	BAA	CBA
ACB	BCC	CAB
ABB	BAC	CBB
ACC	BCA	CAA

For a polypeptide chain of 100 residues in length, a rather modest size, the number of possible sequences is 20^{100} , or because $20 = 10^{1.3}$, 10^{130} unique possibilities. These numbers are more than astronomical! Because an average protein molecule of 100 residues would have a mass of 13,800 daltons (average molecular mass of an amino acid residue = 138), 10^{130} such molecules would have a mass of 1.38×10^{134} daltons. The mass of the observable universe is estimated to be 10^{80} proton masses (about 10^{80} daltons). Thus, the universe lacks enough material to make just one molecule of each possible polypeptide sequence for a protein only 100 residues in length.

The Levels of Protein Structure

The architecture of protein molecules is quite complex. Nevertheless, this complexity can be resolved by defining various levels of structural organization.

Primary Structure

The amino acid sequence is the **primary (1°) structure** of a protein, such as that shown in Figure 5.6, for example.

Secondary Structure

Through hydrogen bonding interactions between adjacent amino acid residues (discussed in detail in Chapter 6), the polypeptide chain can arrange itself into characteristic helical or pleated segments. These segments constitute structural conformities, so-called **regular structures**, that extend along one dimension, like the coils of a spring. Such architectural features of a protein are designated **secondary (2°) structures** (Figure 5.8). Secondary structures are just one of the higher levels of structure that represent the three-dimensional arrangement of the polypeptide in space.

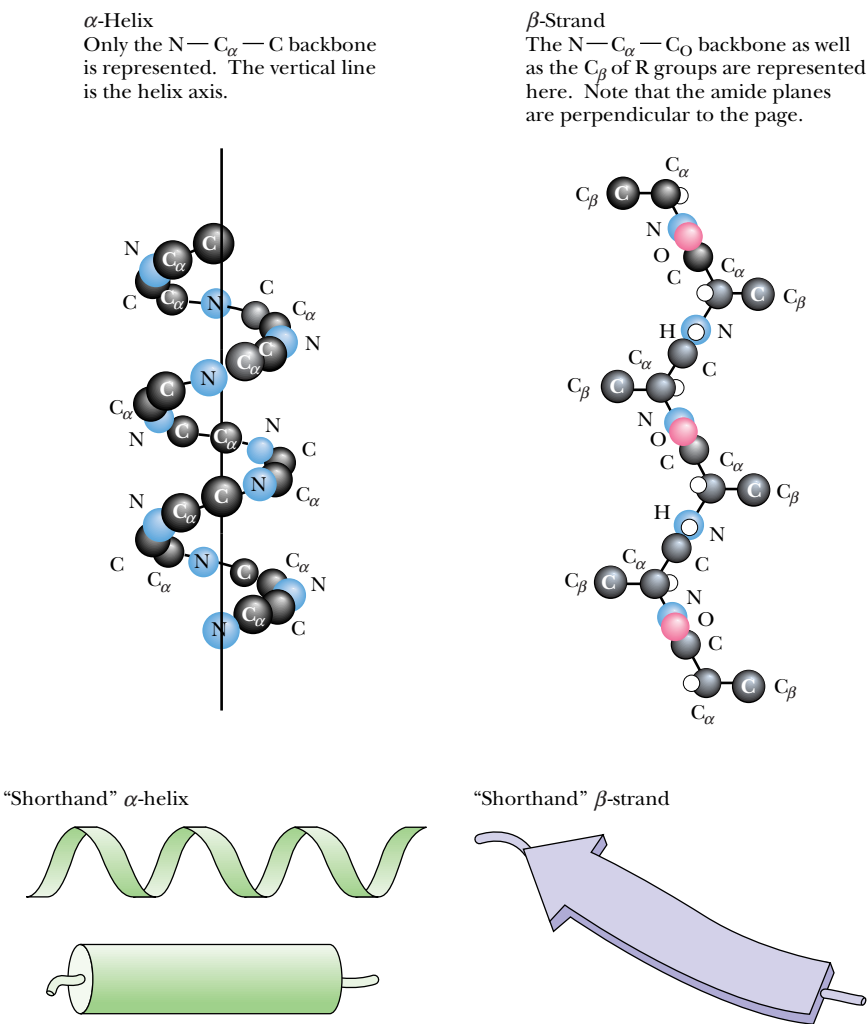
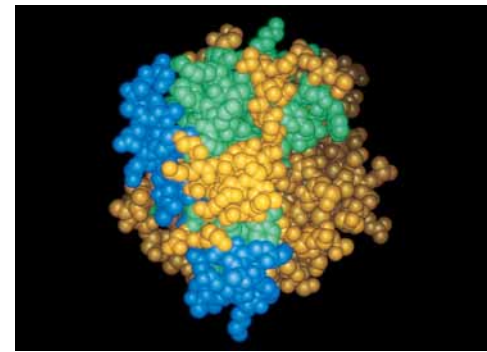


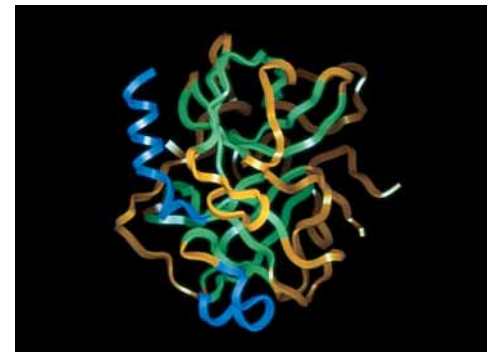
FIGURE 5.8 • Two structural motifs that arrange the primary structure of proteins into a higher level of organization predominate in proteins: the α -helix and the β -pleated strand. Atomic representations of these secondary structures are shown here, along with the symbols used by structural chemists to represent them: the flat, helical ribbon for the α -helix and the flat, wide arrow for β -structures. Both of these structures owe their stability to the formation of hydrogen bonds between N—H and O=C functions along the polypeptide backbone (see Chapter 6).

(a) Chymotrypsin primary structure

H₂N-CGVPAIQPVL₁₀SGL[SR]IVNGE₂₀EAVPGSWPWQ₃₀VSLQDKTGFH₄₀GGSLINEN₅₀WVVTAAHCGV₆₀TTSDVVVAGE₇₀FDQGSSEKI₈₀QKLKIA
KVFK₉₀NSKYNSLTIN₁₀₀NDITLLKLST₁₁₀AASFSQTVSA₁₂₀VCLPSASDDF₁₃₀AAGTTCVTTG₁₄₀WGLTRY[TN]AN₁₅₀LPSDRLQQASL₁₆₀PLLSNTNCK
K₁₇₀YWGTKIKDAM₁₈₀ICAGASGVSS₁₉₀CMGDSGGPLV₂₀₀CKKNGAWTLV₂₁₀GIVSWGSSTC₂₂₀STSTPGVYAR₂₃₀VTALVNWVQQ₂₄₀TLAAN-COOH



Chymotrypsin space-filling model



Chymotrypsin ribbon

(b) Chymotrypsin tertiary structure

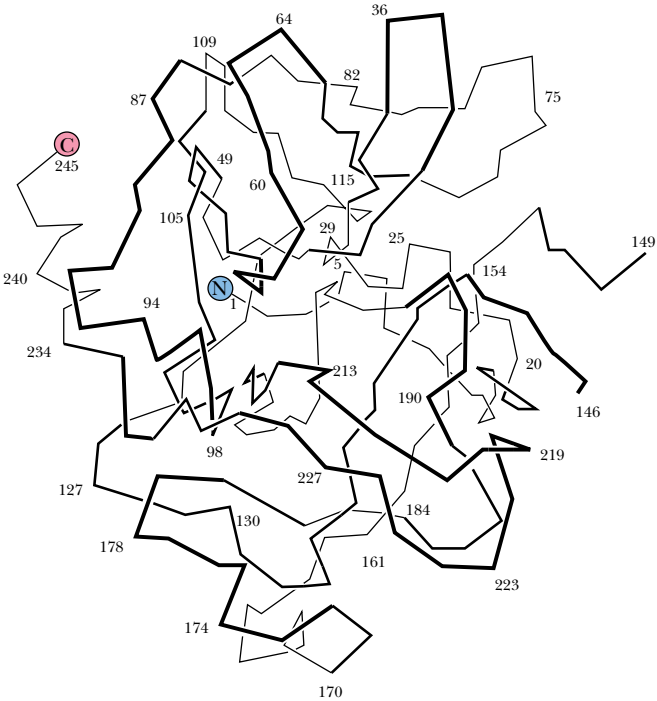


FIGURE 5.9 • Folding of the polypeptide chain into a compact, roughly spherical conformation creates the tertiary level of protein structure. (a) The primary structure and (b) a representation of the tertiary structure of chymotrypsin, a proteolytic enzyme, are shown here. The tertiary representation in (b) shows the course of the chymotrypsin folding pattern by successive numbering of the amino acids in its sequence. (Residues 14 and 15 and 147 and 148 are missing because these residues are removed when chymotrypsin is formed from its larger precursor, chymotrypsinogen.) The ribbon diagram depicts the three-dimensional track of the polypeptide in space.



Tertiary Structure

When the polypeptide chains of protein molecules bend and fold in order to assume a more compact three-dimensional shape, a **tertiary (3°) level of structure** is generated (Figure 5.9). It is by virtue of their tertiary structure that proteins adopt a globular shape. A globular conformation gives the lowest surface-to-volume ratio, minimizing interaction of the protein with the surrounding environment.

Quaternary Structure

Many proteins consist of two or more interacting polypeptide chains of characteristic tertiary structure, each of which is commonly referred to as a **subunit** of the protein. Subunit organization constitutes another level in the hierarchy of protein structure, defined as the protein’s **quaternary (4°) structure** (Figure 5.10). Questions of quaternary structure address the various kinds of subunits within a protein molecule, the number of each, and the ways in which they interact with one another.

Whereas the primary structure of a protein is determined by the covalently linked amino acid residues in the polypeptide backbone, secondary and higher

orders of structure are determined principally by noncovalent forces such as hydrogen bonds and ionic, van der Waals, and hydrophobic interactions. It is important to emphasize that *all the information necessary for a protein molecule to achieve its intricate architecture is contained within its 1° structure*, that is, within the amino acid sequence of its polypeptide chain(s). Chapter 6 presents a detailed discussion of the 2°, 3°, and 4° structure of protein molecules.

Protein Conformation

The overall three-dimensional architecture of a protein is generally referred to as its **conformation**. This term is not to be confused with **configuration**, which denotes the geometric possibilities for a particular set of atoms (Figure 5.11). In going from one configuration to another, covalent bonds must be broken and rearranged. In contrast, the conformational possibilities of a molecule are achieved without breaking any covalent bonds. In proteins, rotations about each of the single bonds along the peptide backbone have the potential to alter the course of the polypeptide chain in three-dimensional space. These rotational possibilities create many possible orientations for the protein chain, referred to as its *conformational possibilities*. Of the great number of theoretical conformations a given protein might adopt, only a very few are favored energetically under physiological conditions. At this time, the rules that direct the folding of protein chains into energetically favorable conformations are still not entirely clear; accordingly, they are the subject of intensive contemporary research.

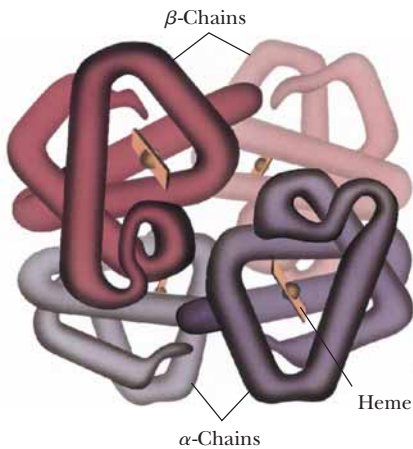


FIGURE 5.10 • Hemoglobin, which consists of two α and two β polypeptide chains, is an example of the quaternary level of protein structure. In this drawing, the β -chains are the two uppermost polypeptides and the two α -chains are the lower half of the molecule. The two closest chains (darkest colored) are the β_2 -chain (*upper left*) and the α_1 -chain (*lower right*). The heme groups of the four globin chains are represented by rectangles with spheres (the heme iron atom). Note the symmetry of this macromolecular arrangement. (*Irving Geis*)

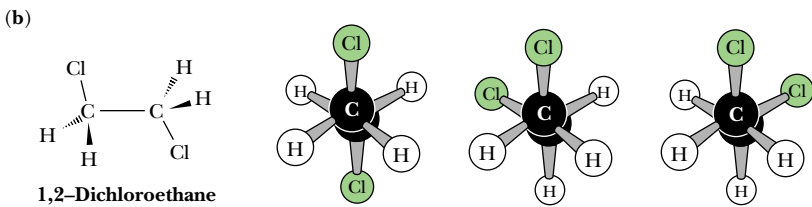
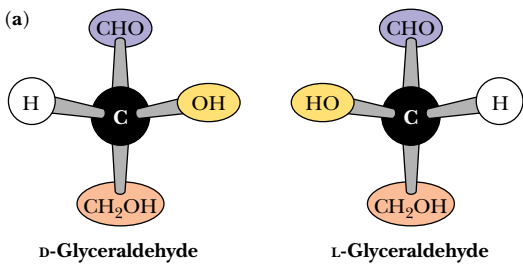
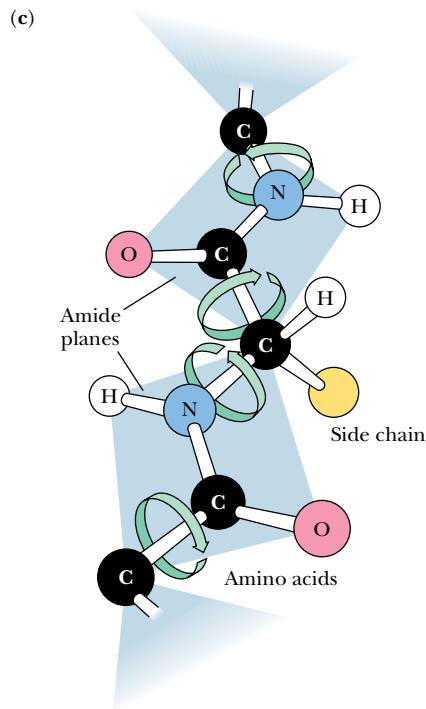


FIGURE 5.11 • Configuration and conformation are *not* synonymous. (a) Rearrangements between configurational alternatives of a molecule can be achieved only by breaking and remaking bonds, as in the transformation between the D- and L-configurations of glyceraldehyde. No possible rotational reorientation of bonds linking the atoms of D-glyceraldehyde yields geometric identity with L-glyceraldehyde, even though they are mirror images of each other. (b) The intrinsic free rotation around single covalent bonds creates a great variety of three-dimensional conformations, even for relatively simple molecules. Consider 1,2-dichloroethane. Viewed end-on in a Newman projection, three principal rotational orientations or conformations predominate. Steric repulsion between eclipsed and partially eclipsed conformations keeps the possibilities at a reasonable number. (c) Imagine the conformational possibilities for a protein in which two of every three bonds along its backbone are freely rotating single bonds.



Later we return to an analysis of the 1° structure of proteins and the methodology used in determining the amino acid sequence of polypeptide chains, but let's first consider the extraordinary variety and functional diversity of these most interesting macromolecules.

5.3 • The Many Biological Functions of Proteins

Proteins are the agents of biological function. Virtually every cellular activity is dependent on one or more particular proteins. Thus, a convenient way to classify the enormous number of proteins is by the biological roles they fill. Table 5.3 summarizes the classification of proteins by function and gives examples of representative members of each class.

Enzymes

By far the largest class of proteins is enzymes. More than 3000 different enzymes are listed in *Enzyme Nomenclature*, the standard reference volume on enzyme classification. **Enzymes** are catalysts that accelerate the rates of biological reactions. Each enzyme is very specific in its function and acts only in a particular metabolic reaction. Virtually every step in metabolism is catalyzed by an enzyme. The catalytic power of enzymes far exceeds that of synthetic catalysts. Enzymes can enhance reaction rates in cells as much as 10¹⁶ times the uncatalyzed rate. Enzymes are systematically classified according to the nature of the reaction that they catalyze, such as the transfer of a phosphate group (*phosphotransferase*) or an oxidation–reduction (*oxidoreductase*). The formal names of enzymes come from the particular reaction within the class that they catalyze, as in ATP:D-fructose-6-phosphate 1-phosphotransferase and alcohol:NAD⁺ oxidoreductase. Often, enzymes have common names in addition to their formal names. ATP:D-fructose-6-phosphate 1-phosphotransferase is more commonly known as *phosphofructokinase* (*kinase* is a common name given to ATP-dependent phosphotransferases). Similarly, alcohol:NAD⁺ oxidoreductase is casually referred to as *alcohol dehydrogenase*. The reactions catalyzed by these two enzymes are shown in Figure 5.12. Other enzymes are known by trivial names that have historical roots, such as *catalase* (systematic name, hydrogen-peroxide:hydrogen-peroxide oxidoreductase), and sometimes these trivial names have descriptive connotations as well, as in *malic enzyme* (systematic name, L-malate:NADP⁺ oxidoreductase).

FIGURE 5.12 • Enzymes are classified according to the specific biological reaction that they catalyze. Cells contain thousands of different enzymes. Two common examples drawn from carbohydrate metabolism are phosphofructokinase (PFK), or, more precisely, ATP:D-fructose-6-phosphate 1-phosphotransferase, and alcohol dehydrogenase (ADH), or alcohol:NAD⁺ oxidoreductase, which catalyze the reactions shown here.

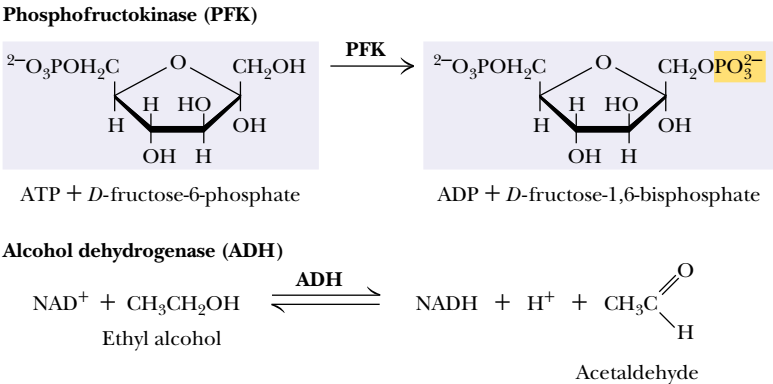


Table 5.3

Biological Functions of Proteins and Some Representative Examples	
Functional Class	Examples
Enzymes	Ribonuclease
	Trypsin
	Phosphofructokinase
	Alcohol dehydrogenase
	Catalase
	“Malic” enzyme
Regulatory proteins	Insulin
	Somatotropin
	Thyrotropin
	<i>lac</i> repressor
	NF1 (nuclear factor 1)
	Catabolite activator protein (CAP)
Transport proteins	API
	Hemoglobin
	Serum albumin
Storage proteins	Glucose transporter
	Ovalbumin
	Casein
	Zein
	Phaseolin
Contractile and motile proteins	Ferritin
	Actin
	Myosin
	Tubulin
	Dynein
Structural proteins	Kinesin
	α -Keratin
	Collagen
	Elastin
	Fibroin
Scaffold proteins	Proteoglycans
	Grb 2
	crk
	shc
	stat
Protective and exploitive proteins	IRS-1
	Immunoglobulins
	Thrombin
	Fibrinogen
	Antifreeze proteins
	Snake and bee venom proteins
Exotic proteins	Diphtheria toxin
	Ricin
	Monellin
	Resilin
	Glue proteins

Regulatory Proteins

A number of proteins do not perform any obvious chemical transformation but nevertheless can regulate the ability of other proteins to carry out their physiological functions. Such proteins are referred to as **regulatory proteins**. A well-known example is *insulin*, the hormone regulating glucose metabolism in animals. Insulin is a relatively small protein (5.7 kD) and consists of two polypeptide chains held together by disulfide cross-bridges. Other hormones that are also proteins include pituitary *somatotropin* (21 kD) and *thyrotropin* (28 kD), which stimulates the thyroid gland. Another group of regulatory proteins is involved in the regulation of gene expression. These proteins characteristically act by binding to DNA sequences that are adjacent to coding regions of genes, either activating or inhibiting the transcription of genetic information into RNA. Examples include **repressors**, which, because they block transcription, are considered negative control elements.

A prokaryotic representative is *lac repressor* (37 kD), which controls expression of the enzyme system responsible for the metabolism of lactose (milk sugar); a mammalian example is *NF1* (*nuclear factor 1*, 60 kD), which inhibits transcription of the gene encoding the β -globin polypeptide chain of hemoglobin. Positively acting control elements are also known. For example, the *E. coli* catabolite gene activator protein (**CAP**) (44 kD), under appropriate metabolic conditions, can bind to specific sites along the *E. coli* chromosome and increase the rate of transcription of adjacent genes. The mammalian *AP1* is a heterodimeric transcription factor composed of one polypeptide from the *Jun* family of gene-regulatory proteins and one polypeptide from the *Fos* family of gene-regulatory proteins. AP1 activates expression of the β -globin gene. These various DNA-binding regulatory proteins often possess characteristic structural features, such as helix-turn-helix, leucine zipper, and zinc finger motifs (see Chapter 31).

Transport Proteins

A third class of proteins is the **transport proteins**. These proteins function to transport specific substances from one place to another. One type of transport is exemplified by the transport of oxygen from the lungs to the tissues by *hemoglobin* (Figure 5.13a) or by the transport of fatty acids from adipose tissue to various organs by the blood protein *serum albumin*. A very different type is the transport of metabolites across permeability barriers such as cell membranes, as mediated by specific membrane proteins. These *membrane transport proteins* take up metabolite molecules on one side of a membrane, transport them across the membrane, and release them on the other side. Examples include the transport proteins responsible for the uptake of essential nutrients into the cell, such as glucose or amino acids (Figure 5.13b). All naturally occurring membrane transport proteins studied thus far form channels in the membrane through which the transported substances are passed.

Storage Proteins

Proteins whose biological function is to provide a reservoir of an essential nutrient are called **storage proteins**. Because proteins are amino acid polymers and because nitrogen is commonly a limiting nutrient for growth, organisms have exploited proteins as a means to provide sufficient nitrogen in times of need. For example, *ovalbumin*, the protein of egg white, provides the developing bird embryo with a source of nitrogen during its isolation within the egg. *Casein* is the most abundant protein of milk and thus the major nitrogen source for

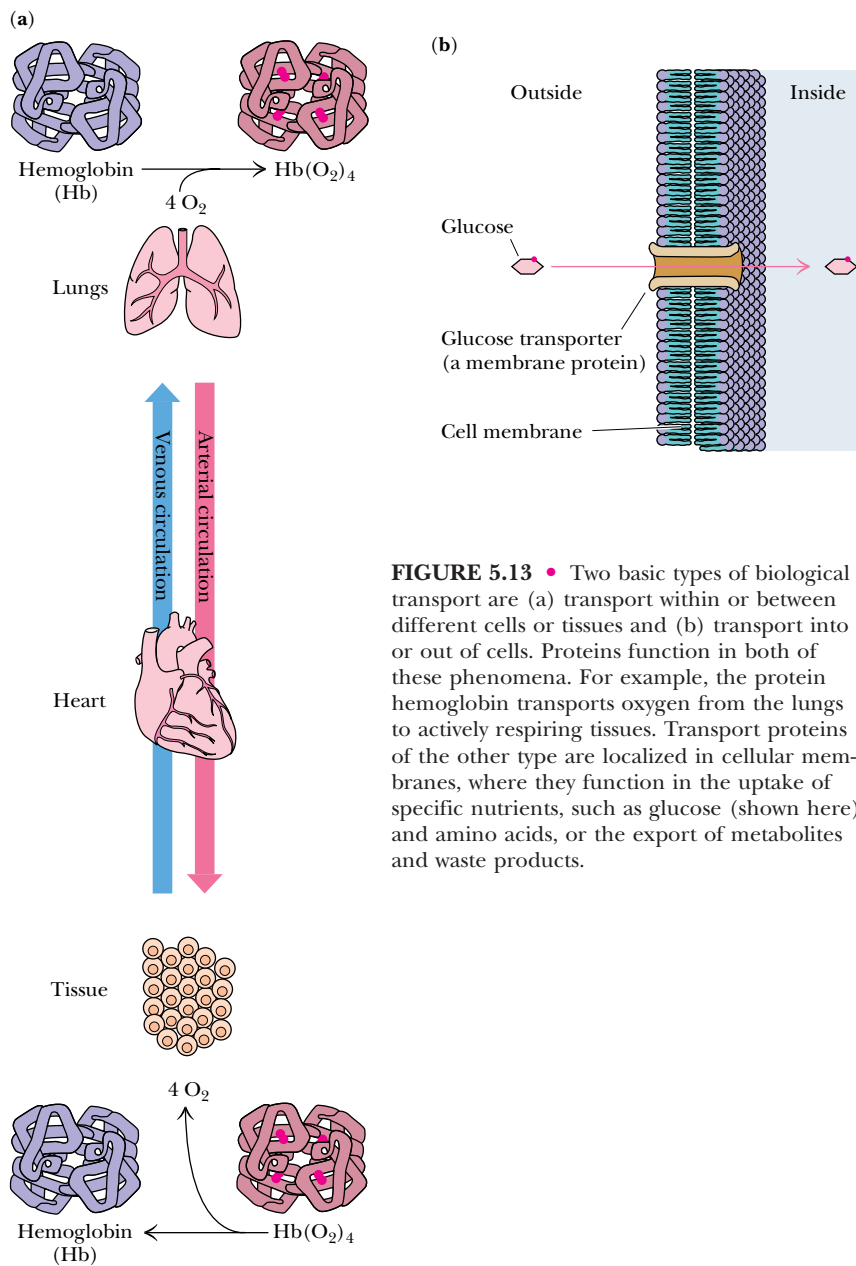


FIGURE 5.13 • Two basic types of biological transport are (a) transport within or between different cells or tissues and (b) transport into or out of cells. Proteins function in both of these phenomena. For example, the protein hemoglobin transports oxygen from the lungs to actively respiring tissues. Transport proteins of the other type are localized in cellular membranes, where they function in the uptake of specific nutrients, such as glucose (shown here) and amino acids, or the export of metabolites and waste products.

mammalian infants. The seeds of higher plants often contain as much as 60% storage protein to make the germinating seed nitrogen-sufficient during this crucial period of plant development. In corn (*Zea mays* or *maize*), a family of low molecular weight proteins in the kernel called *zeins* serve this purpose; peas (the seeds of *Phaseolus vulgaris*) contain a storage protein called *phaseolin*. The use of proteins as a reservoir of nitrogen is more efficient than storing an equivalent amount of amino acids. Not only is the osmotic pressure minimized, but the solvent capacity of the cell is taxed less in solvating one molecule of a polypeptide than in dissolving, for example, 100 molecules of free amino acids. Proteins can also serve to store nutrients other than the more obvious elements composing amino acids (N, C, H, O, and S). As an example, *ferritin* is a protein found in animal tissues that binds iron, retaining this essential metal so

that it is available for the synthesis of important iron-containing proteins such as hemoglobin. One molecule of ferritin (460 kD) binds as many as 4500 atoms of iron (35% by weight).

Contractile and Motile Proteins

Certain proteins endow cells with unique capabilities for movement. Cell division, muscle contraction, and cell motility represent some of the ways in which cells execute motion. The **contractile** and **motile proteins** underlying these motions share a common property: they are filamentous or polymerize to form filaments. Examples include *actin* and *myosin*, the filamentous proteins forming the contractile systems of cells, and *tubulin*, the major component of microtubules (the filaments involved in the mitotic spindle of cell division as well as in flagella and cilia). Another class of proteins involved in movement includes *dynein* and *kinesin*, so-called **motor proteins** that drive the movement of vesicles, granules, and organelles along microtubules serving as established cytoskeletal “tracks.”

Structural Proteins

An apparently passive but very important role of proteins is their function in creating and maintaining biological structures. **Structural proteins** provide strength and protection to cells and tissues. Monomeric units of structural proteins typically polymerize to generate long fibers (as in hair) or protective sheets of fibrous arrays, as in cowhide (leather). α -*Keratins* are insoluble fibrous proteins making up hair, horns, and fingernails. *Collagen*, another insoluble fibrous protein, is found in bone, connective tissue, tendons, cartilage, and hide, where it forms inelastic fibrils of great strength. One-third of the total protein in a vertebrate animal is collagen. A structural protein having elastic properties is, appropriately, *elastin*, an important component of ligaments. Because of the way elastin monomers are cross-linked in forming polymers, elastin can stretch in two dimensions. Certain insects make a structurally useful protein, *fibroin* (a β -keratin), the major constituent of cocoons (silk) and spider webs. An important protective barrier for animal cells is the **extracellular matrix** containing *collagen* and *proteoglycans*, covalent protein–polysaccharide complexes that cushion and lubricate.

Scaffold Proteins (Adapter Proteins)

Some proteins play a recently discovered role in the complex pathways of cellular response to hormones and growth factors. These proteins, the **scaffold** or **adapter proteins**, have a modular organization in which specific parts (**modules**) of the protein’s structure recognize and bind certain structural elements in other proteins through **protein–protein interactions**. For example, *SH2* modules bind to proteins in which a tyrosine residue has become phosphorylated on its phenolic —OH, and *SH3* modules bind to proteins having a characteristic grouping of proline residues. Others include *PH* modules, which bind to membranes, and *PDZ*-containing proteins, which bind specifically to the C-terminal amino acid of certain proteins. Because scaffold proteins typically possess several of these different kinds of modules, they can act as a scaffold onto which a set of different proteins is assembled into a multiprotein complex. Such assemblages are typically involved in coordinating and communicating the many intracellular responses to hormones or other signalling molecules (Figure 5.14). **Anchoring** (or **targeting**) **proteins** are proteins that bind other proteins, causing them to associate with other structures in the cell. A family of *anchoring proteins*, known as *AKAP* or *A kinase anchoring proteins*, exists in

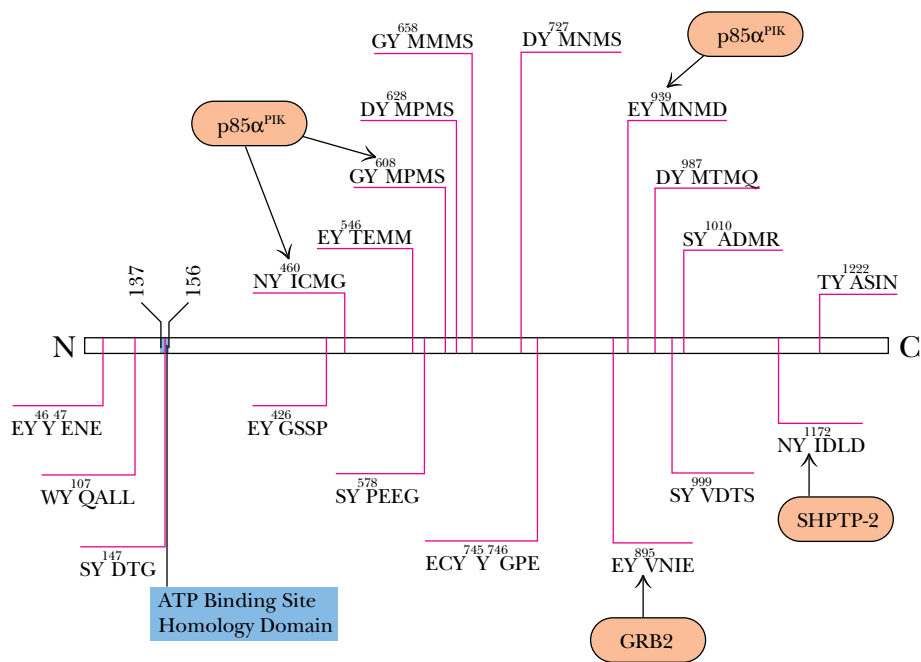


FIGURE 5.14 • Diagram of the N→C sequence organization of the adapter protein *insulin receptor substrate-1 (IRS-1)* showing the various amino acid sequences (in one-letter code) that contain tyrosine (Y) residues that are potential sites for phosphorylation. The other adapter proteins that recognize various of these sites are shown as *Grb2*, *SHPTP-2*, and *p85α^{PIK}*. Insulin binding to the insulin receptor activates the enzymatic activity that phosphorylates these Tyr residues on IRS-1. (Adapted from White, M. F., and Kahn, C. R., 1994. *Journal of Biological Chemistry* 269:1–4.)

which specific AKAP members bind the regulatory enzyme *protein kinase A* (PKA) to particular subcellular compartments. For example, AKAP100 targets PKA to the endoplasmic reticulum, whereas AKAP79 targets PKA to the plasma membrane.

Protective and Exploitive Proteins

In contrast to the passive protective nature of some structural proteins, another group can be more aptly classified as **protective** or **exploitive proteins** because of their biologically active role in cell defense, protection, or exploitation. Prominent among the protective proteins are the *immunoglobulins* or *antibodies* produced by the lymphocytes of vertebrates. Antibodies have the remarkable ability to “ignore” molecules that are an intrinsic part of the host organism, yet they can specifically recognize and neutralize “foreign” molecules resulting from the invasion of the organism by bacteria, viruses, or other infectious agents. Another group of protective proteins is the blood-clotting proteins, *thrombin* and *fibrinogen*, which prevent the loss of blood when the circulatory system is damaged. Arctic and Antarctic fishes have *antifreeze proteins* to protect their blood against freezing in the below-zero temperatures of high-latitude seas. In addition, various proteins serve defensive or exploitive roles for organisms, including the lytic and neurotoxic proteins of snake and bee venoms and toxic plant proteins, such as *ricin*, whose apparent purpose is to thwart predation by herbivores. Another class of exploitive proteins includes the toxins produced by bacteria, such as diphtheria toxin and cholera toxin.

Exotic Proteins

Some proteins display rather exotic functions that do not quite fit the previous classifications. *Monellin*, a protein found in an African plant, has a very sweet taste and is being considered as an artificial sweetener for human consumption. *Resilin*, a protein having exceptional elastic properties, is found in

the hinges of insect wings. Certain marine organisms such as mussels secrete *glue proteins*, allowing them to attach firmly to hard surfaces. It is worth repeating that the great diversity of function in proteins, as reflected in this survey, is attained using just 20 amino acids.

5.4 • Some Proteins Have Chemical Groups Other Than Amino Acids

Many proteins consist of only amino acids and contain no other chemical groups. The enzyme ribonuclease and the contractile protein actin are two such examples. Such proteins are called **simple proteins**. However, many other proteins contain various chemical constituents as an integral part of their structure. These proteins are termed **conjugated proteins** (Table 5.4). If the non-protein part is crucial to the protein's function, it is referred to as a **prosthetic group**. If the nonprotein moiety is not covalently linked to the protein, it can usually be removed by denaturing the protein structure. However, if the conjugate is covalently joined to the protein, it may be necessary to carry out acid hydrolysis of the protein into its component amino acids in order to release it. Conjugated proteins are typically classified according to the chemical nature of their nonamino acid component; a representative selection of them is given here and in Table 5.4. (Note that comparisons of Tables 5.3 and 5.4 reveal two distinctly different ways of considering the nature of proteins—function versus chemistry.)

GLYCOPROTEINS. Glycoproteins are proteins that contain carbohydrate. Proteins destined for an extracellular location are characteristically glycoproteins. For example, fibronectin and proteoglycans are important components of the extracellular matrix that surrounds the cells of most tissues in animals. Immunoglobulin G molecules are the principal antibody species found circulating free in the blood plasma. Many membrane proteins are glycosylated on their extracellular segments.

LIPOPROTEINS. Blood plasma lipoproteins are prominent examples of the class of proteins conjugated with lipid. The plasma lipoproteins function primarily in the transport of lipids to sites of active membrane synthesis. Serum levels of *low density lipoproteins* (LDLs) are often used as a clinical index of susceptibility to vascular disease.

NUCLEOPROTEINS. Nucleoprotein conjugates have many roles in the storage and transmission of genetic information. Ribosomes are the sites of protein synthesis. Virus particles and even chromosomes are protein–nucleic acid complexes.

PHOSPHOPROTEINS. These proteins have phosphate groups esterified to the hydroxyls of serine, threonine, or tyrosine residues. Casein, the major protein of milk, contains many phosphates and serves to bring essential phosphorus to the growing infant. Many key steps in metabolism are regulated between states of activity or inactivity, depending on the presence or absence of phosphate groups on proteins, as we shall see in Chapter 15. Glycogen phosphorylase *a* is one well-studied example.

METALLOPROTEINS. Metalloproteins are either metal storage forms, as in the case of ferritin, or enzymes in which the metal atom participates in a catalytically important manner. We encounter many examples throughout this book of the vital metabolic functions served by metalloenzymes.

Table 5.4

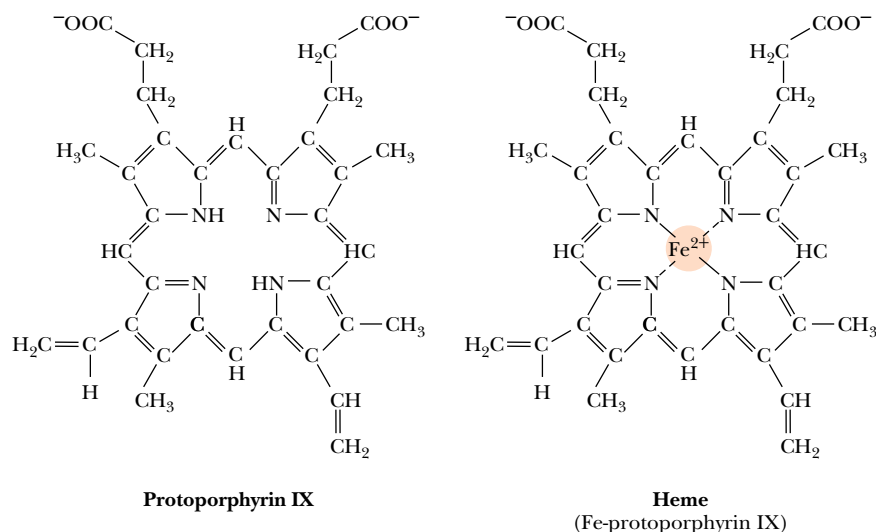
Representative Conjugated Proteins

Class	Prosthetic Group	Percent by Weight (approx.)
<i>Glycoproteins contain carbohydrate</i>		
Fibronectin		
γ-Globulin		
Proteoglycan		
<i>Lipoproteins contain lipid</i>		
Blood plasma lipoproteins:		
High density lipoprotein (HDL) (α-lipoprotein)	Triacylglycerols, phospholipids, cholesterol	75
Low density lipoprotein (LDL) (β-lipoprotein)	Triacylglycerols, phospholipids, cholesterol	67
<i>Nucleoprotein complexes contain nucleic acid</i>		
Ribosomes	RNA	60
Tobacco mosaic virus	RNA	5
Adenovirus	DNA	
HIV-1 (AIDS virus)	RNA	
<i>Phosphoproteins contain phosphate</i>		
Casein	Phosphate groups	
Glycogen phosphorylase <i>a</i>	Phosphate groups	
<i>Metalloproteins contain metal atoms</i>		
Ferritin	Iron	35
Alcohol dehydrogenase	Zinc	
Cytochrome oxidase	Copper and iron	
Nitrogenase	Molybdenum and iron	
Pyruvate carboxylase	Manganese	
<i>Hemoproteins contain heme</i>		
Hemoglobin		
Cytochrome <i>c</i>		
Catalase		
Nitrate reductase		
Ammonium oxidase		
<i>Flavoproteins contain flavin</i>		
Succinate dehydrogenase	FAD	
NADH dehydrogenase	FMN	
Dihydroorotate dehydrogenase	FAD and FMN	
Sulfite reductase	FAD and FMN	

HEMOPROTEINS. These proteins are actually a subclass of metalloproteins because their prosthetic group is **heme**, the name given to iron protoporphyrin IX (Figure 5.15). Because heme-containing proteins enjoy so many prominent biological functions, they are considered a class by themselves.

FLAVOPROTEINS. *Flavin* is an essential substance for the activity of a number of important oxidoreductases. We discuss the chemistry of flavin and its derivatives, FMN and FAD, in the chapter on electron transport and oxidative phosphorylation (Chapter 21).

FIGURE 5.15 • Heme consists of protoporphyrin IX and an iron atom. Protoporphyrin, a highly conjugated system of double bonds, is composed of four 5-membered heterocyclic rings (pyrroles) fused together to form a tetrapyrrole macrocycle. The specific isomeric arrangement of methyl, vinyl, and propionate side chains shown is protoporphyrin IX. Coordination of an atom of ferrous iron (Fe^{2+}) by the four pyrrole nitrogen atoms yields heme.



5.5 • Reactions of Peptides and Proteins

The chemical properties of peptides and proteins are most easily considered in terms of the chemistry of their component functional groups. That is, they possess reactive amino and carboxyl termini and they display reactions characteristic of the chemistry of the R groups of their component amino acids. These reactions are familiar to us from Chapter 4 and from the study of organic chemistry and need not be repeated here.

5.6 • Purification of Protein Mixtures

Cells contain thousands of different proteins. A major problem for protein chemists is to purify a chosen protein so that they can study its specific properties in the absence of other proteins. Proteins have been separated and purified on the basis of their two prominent physical properties: size and electrical charge. A more direct approach is to employ **affinity purification** strategies that take advantage of the biological function or similar specific recognition properties of a protein (see Table 5.5 and Chapter Appendix).

Separation Methods

Separation methods based on size include size exclusion chromatography, ultrafiltration, and ultracentrifugation (see Chapter Appendix). The ionic properties of peptides and proteins are determined principally by their complement of amino acid side chains. Furthermore, the ionization of these groups is pH-dependent.

A variety of procedures exploit electrical charge as a means of discriminating between proteins, including ion exchange chromatography (see Chapter 4), electrophoresis (see Chapter Appendix), and solubility. Proteins tend to be least soluble at their **isoelectric point**, the pH value at which the sum of their positive and negative electrical charges is zero. At this pH, electrostatic repulsion between protein molecules is minimal and they are more likely to coalesce and precipitate out of solution. Ionic strength also profoundly influences protein solubility. Most globular proteins tend to become increas-

DEEPER LOOK

Estimation of Protein Concentrations in Solutions of Biological Origin

Biochemists are often interested in knowing the protein concentration in various preparations of biological origin. Such quantitative analysis is not straightforward. Cell extracts are complex mixtures that typically contain protein molecules of many different molecular weights, so the results of protein estimations cannot be expressed on a molar basis. Also, aside from the rather unreactive repeating peptide backbone, little common chemical identity is seen among the many proteins found in cells that might be readily exploited for exact chemical analysis. Most of their chemical properties vary with their amino acid composition, for example, nitrogen or sulfur content or the presence of aromatic, hydroxyl, or other functional groups.

Lowry Procedure

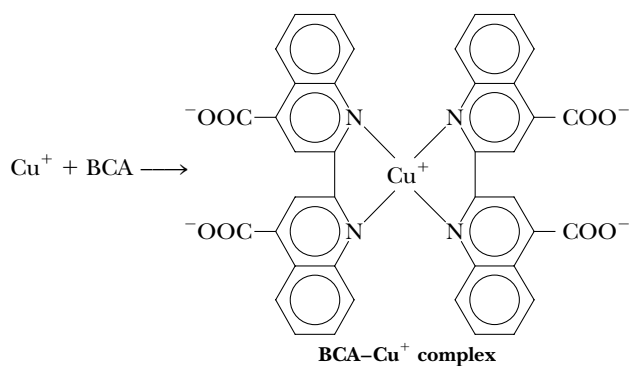
A method that has been the standard of choice for many years is the **Lowry procedure**. This method uses Cu^{2+} ions along with *Folin–Ciocalteu reagent*, a combination of phosphomolybdic and phosphotungstic acid complexes that react with Cu^+ . Cu^+ is generated from Cu^{2+} by readily oxidizable protein components, such as cysteine or the phenols and indoles of tyrosine and tryptophan. Although the precise chemistry of the Lowry method remains uncertain, the Cu^+ reaction with the Folin reagent gives intensely colored products measurable spectrophotometrically.

BCA Method

Recently, a reagent that reacts more efficiently with Cu^+ than Folin–Ciocalteu reagent has been developed for protein assays. *Bicinchoninic acid (BCA)* forms a purple complex with Cu^+ in alkaline solution.

Assays Based on Dye Binding

Several other protocols for protein estimation enjoy prevalent usage in biochemical laboratories. The **Bradford assay** is a rapid and reliable technique that uses a dye called *Coomassie Brilliant Blue G-250*, which undergoes a change in its color upon non-covalent binding to proteins. The binding is quantitative and less sensitive to variations in the protein’s amino acid composition. The color change is easily measured by a spectrophotometer. A similar, very sensitive method capable of quantifying nanogram amounts of protein is based on the shift in color of colloidal gold upon binding to proteins.



ingly soluble as the ionic strength is raised. This phenomenon, the salting-in of proteins, is attributed to the diminishment by the salt ions of electrostatic attractions between the protein molecules. Such electrostatic interactions would otherwise lead to precipitation. However, as the salt concentration reaches high levels (greater than 1 M), the effect may reverse so that the protein is salted out of solution. In such cases, the numerous salt ions begin to compete with the protein for waters of solvation, and, as they win out, the protein becomes insoluble. The solubility properties of a typical protein are shown in Figure 5.16.

Although the side chains of most nonpolar amino acids in soluble proteins are usually buried in the interior of the protein away from contact with the aqueous solvent, a portion of them is exposed at the protein’s surface, giving

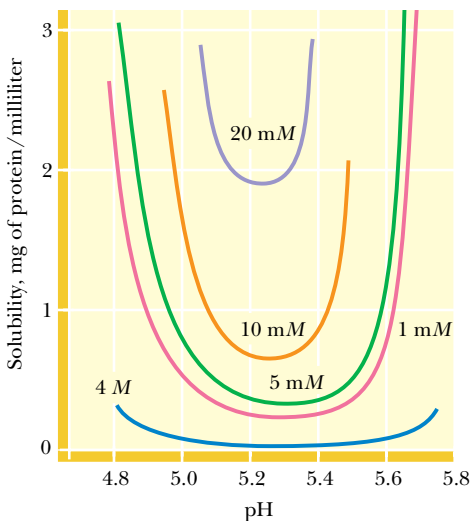


FIGURE 5.16 • The solubility of most globular proteins is markedly influenced by pH and ionic strength. This figure shows the solubility of a typical protein as a function of pH and various salt concentrations.

Table 5.5

Example of a Protein Purification Scheme: Purification of the Enzyme Xanthine Dehydrogenase from a Fungus					
Fraction	Volume (mL)	Total Protein (mg)	Total Activity*	Specific Activity [†]	Percent Recovery [‡]
1. Crude extract	3,800	22,800	2,460	0.108	100
2. Salt precipitate	165	2,800	1,190	0.425	48
3. Ion exchange chromatography	65	100	720	7.2	29
4. Molecular sieve chromatography	40	14.5	555	38.3	23
5. Immunoaffinity chromatography [§]	6	1.8	275	152	11

*The relative enzymatic activity of each fraction in catalyzing the xanthine dehydrogenase reaction is cited as arbitrarily defined units.

[†]The specific activity is the total activity of the fraction divided by the total protein in the fraction. This value gives an indication of the increase in purity attained during the course of the purification as the samples become enriched for xanthine dehydrogenase protein.

[‡]The percent recovery of total activity is a measure of the yield of the desired product, xanthine dehydrogenase.

[§]The last step in the procedure is an affinity method in which antibodies specific for xanthine dehydrogenase are covalently coupled to a chromatography matrix and packed into a glass tube to make a chromatographic column through which fraction 4 is passed. The enzyme is bound by this immunoaffinity matrix while other proteins pass freely out. The enzyme is then recovered by passing a strong salt solution through the column, which dissociates the enzyme–antibody complex.

Adapted from Lyon, E. S., and Garrett, R. H., 1978. *Journal of Biological Chemistry*. **253**:2604–2614.

it a partially hydrophobic character. *Hydrophobic interaction chromatography* is a protein purification technique that exploits this hydrophobicity (see Chapter Appendix).

A Typical Protein Purification Scheme

Most purification procedures for a particular protein are developed in an empirical manner, the overriding principle being purification of the protein to a homogeneous state with acceptable yield. Table 5.5 presents a summary of a purification scheme for a selected protein. Note that the **specific activity** of the protein (the enzyme xanthine dehydrogenase) in the immuno-affinity purified fraction (fraction 5) has been increased 152/0.108, or 1407 times the specific activity in the crude extract (fraction 1). Thus, xanthine dehydrogenase in fraction 5 versus fraction 1 is enriched more than 1400-fold by the purification procedure.

5.7 • The Primary Structure of a Protein: Determining the Amino Acid Sequence

In 1953, Frederick Sanger of Cambridge University in England reported the amino acid sequences of the two polypeptide chains composing the protein insulin (Figure 5.17). Not only was this a remarkable achievement in analytical chemistry but it helped to demystify speculation about the chemical nature of proteins. Sanger’s results clearly established that all of the molecules of a given protein have a fixed amino acid composition, a defined amino acid sequence, and therefore an invariant molecular weight. In short, proteins are well defined chemically. Today, the amino acid sequences of some 100,000 different proteins are known. Although many sequences have been determined from application of the principles first established by Sanger, most are now deduced from knowledge of the nucleotide sequence of the gene that encodes the protein.

Protein Sequencing Strategy

The usual strategy for determining the amino acid sequence of a protein involves eight basic steps:

1. If the protein contains more than one polypeptide chain, the chains are separated and purified.
2. Intrachain S—S (disulfide) cross-bridges between cysteine residues in the polypeptide chain are cleaved. (If these disulfides are interchain linkages, then step 2 precedes step 1.)
3. The amino acid composition of each polypeptide chain is determined.
4. The N-terminal and C-terminal residues are identified.
5. Each polypeptide chain is cleaved into smaller fragments, and the amino acid composition and sequence of each fragment are determined.
6. Step 5 is repeated, using a different cleavage procedure to generate a different and therefore overlapping set of peptide fragments.
7. The overall amino acid sequence of the protein is reconstructed from the sequences in overlapping fragments.
8. The positions of S—S cross-bridges formed between cysteine residues are located.

Each of these steps is discussed in greater detail in the following sections.

Step 1. Separation of Polypeptide Chains

If the protein of interest is a **heteromultimer** (composed of more than one type of polypeptide chain), then the protein must be dissociated and its component polypeptide subunits must be separated from one another and sequenced individually. Subunit associations in multimeric proteins are typically maintained solely by noncovalent forces, and therefore most multimeric proteins can usually be dissociated by exposure to pH extremes, 8 M urea, 6 M guanidinium hydrochloride, or high salt concentrations. (All of these treatments disrupt polar interactions such as hydrogen bonds both within the protein molecule and between the protein and the aqueous solvent.) Once dissociated, the individual polypeptides can be isolated from one another on the basis of differences in size and/or charge. Occasionally, heteromultimers are linked together by interchain S—S bridges. In such instances, these cross-links must be cleaved prior to dissociation and isolation of the individual chains. The methods described under step 2 are applicable for this purpose.

Step 2. Cleavage of Disulfide Bridges

A number of methods exist for cleaving disulfides (Figure 5.18). An important consideration is to carry out these cleavages so that the original or even new S—S links do not form. Oxidation of a disulfide by performic acid results in the formation of two equivalents of cysteic acid (Figure 5.18a). Because these cysteic acid side chains are ionized SO_3^- groups, electrostatic repulsion (as well as altered chemistry) prevents S—S recombination. Alternatively, sulfhydryl compounds such as 2-mercaptoethanol readily reduce S—S bridges to regenerate two cysteine—SH side chains (Figure 5.18b). However, these SH groups recombine to re-form either the original disulfide link or, if other free Cys—SHs are available, new disulfide links. To prevent this, S—S reduction must be followed by treatment with alkylating agents such as iodoacetate or 3-bromopropylamine, which modify the SH groups and block disulfide bridge formation (Figure 5.18a).

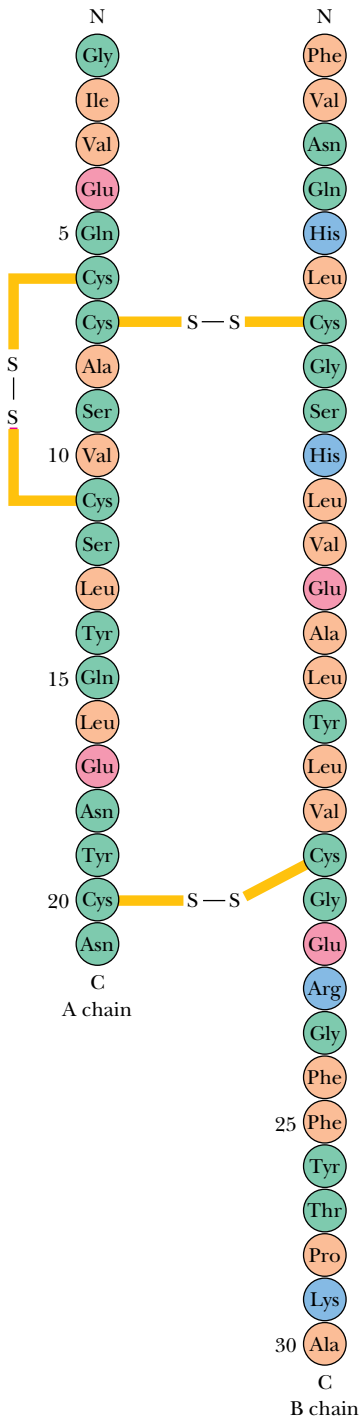


FIGURE 5.17 • The hormone insulin consists of two polypeptide chains, A and B, held together by two disulfide cross-bridges (S—S). The A chain has 21 amino acid residues and an intrachain disulfide; the B polypeptide contains 30 amino acids. The sequence shown is for bovine insulin.

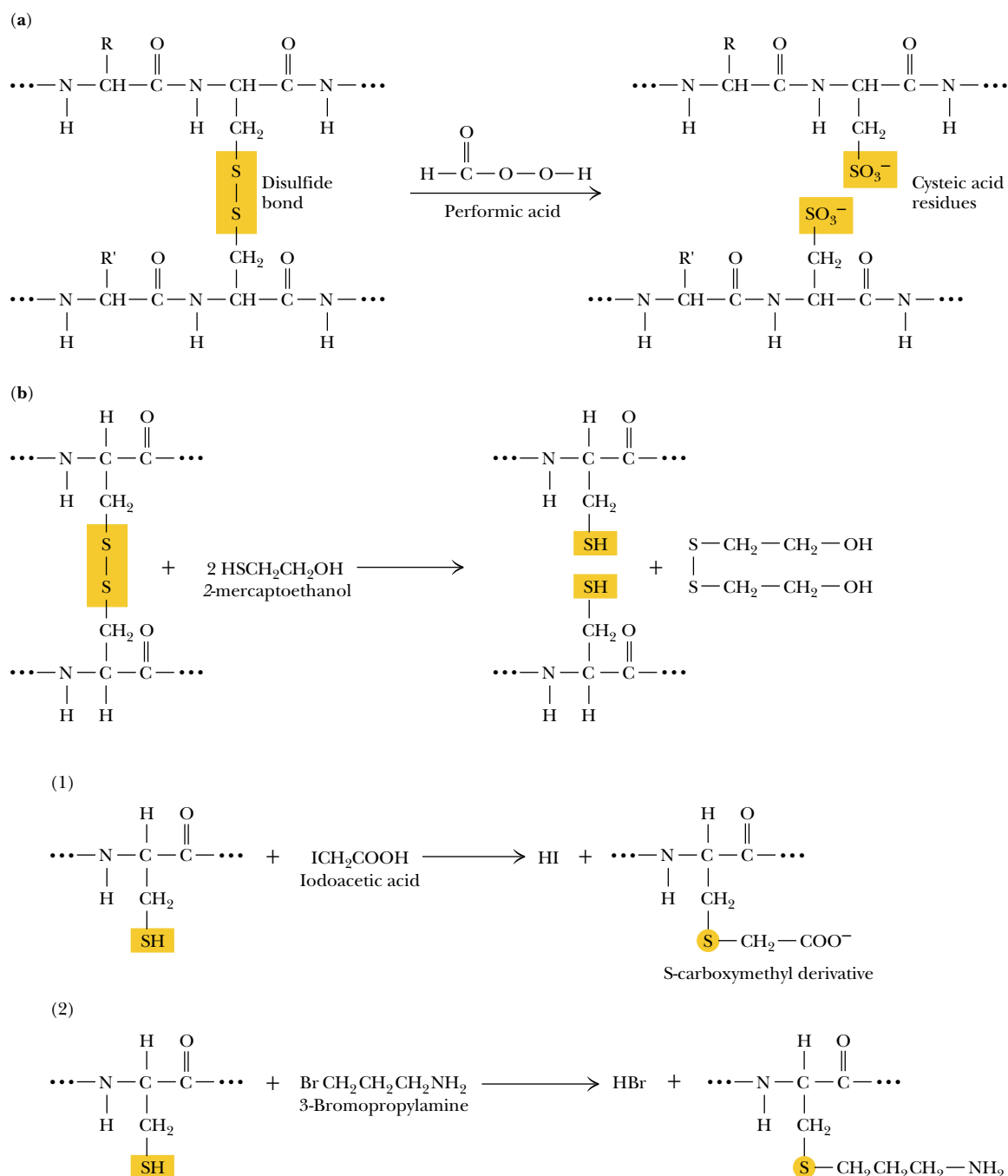


FIGURE 5.18 • Methods for cleavage of disulfide bonds in proteins. (a) Oxidative cleavage by reaction with performic acid. (b) Reductive cleavage with sulfhydryl compounds. Disulfide bridges can be broken by reduction of the S—S link with sulfhydryl agents such as 2-mercaptoethanol or dithiothreitol. Because reaction between the newly reduced —SH groups to re-establish disulfide bonds is a likelihood, S—S reduction must be followed by —SH modification: (1) alkylation with iodoacetate (ICH₂COOH) or (2) modification with 3-bromopropylamine (Br—(CH₂)₃—NH₂).

Step 3. Analysis of Amino Acid Composition

The standard protocol for analysis of the amino acid composition of proteins is discussed in Section 5.1. Results of such analyses allow the researcher to anticipate which methods of polypeptide fragmentation might be useful for the protein.

Step 4. Identification of the N- and C-Terminal Residues

End-group analysis reveals several things. First, it identifies the N- and C-terminal residues in the polypeptide chain. Second, it can be a clue to the number of ends in the protein. That is, if the protein consists of two or more different polypeptide chains, then more than one end group may be discovered, alerting the investigator to the presence of multiple polypeptides.

A. N-Terminal Analysis

The amino acid residing at the N-terminal end of a protein can be identified in a number of ways; one method, **Edman degradation**, has become the procedure of choice. This method is preferable because it allows the sequential identification of a series of residues beginning at the N-terminus (Figure 5.19). In weakly basic solutions, phenylisothiocyanate, or **Edman's reagent** (phenyl—N=C=S), combines with the free amino terminus of a protein (Figure 5.19), which can be excised from the end of the polypeptide chain and recovered as a phenylthiohydantoin (PTH) derivative. This PTH derivative can be identified by chromatographic methods. Importantly, in this procedure, the rest of the polypeptide chain remains intact and can be subjected to further rounds of Edman degradation to identify successive amino acid residues in the chain. Often, the carboxyl terminus of the polypeptide under analysis is coupled to an insoluble matrix, allowing the polypeptide to be easily recovered by filtration following each round of Edman reaction. Thus, Edman reaction not only identifies the N-terminus of proteins but can also reveal additional information regarding sequence. Automated instruments (so-called *Edman sequencers*) have been designed to carry out the reaction cycle of the Edman procedure. In practical terms, as many as 50 cycles of reaction can be accomplished

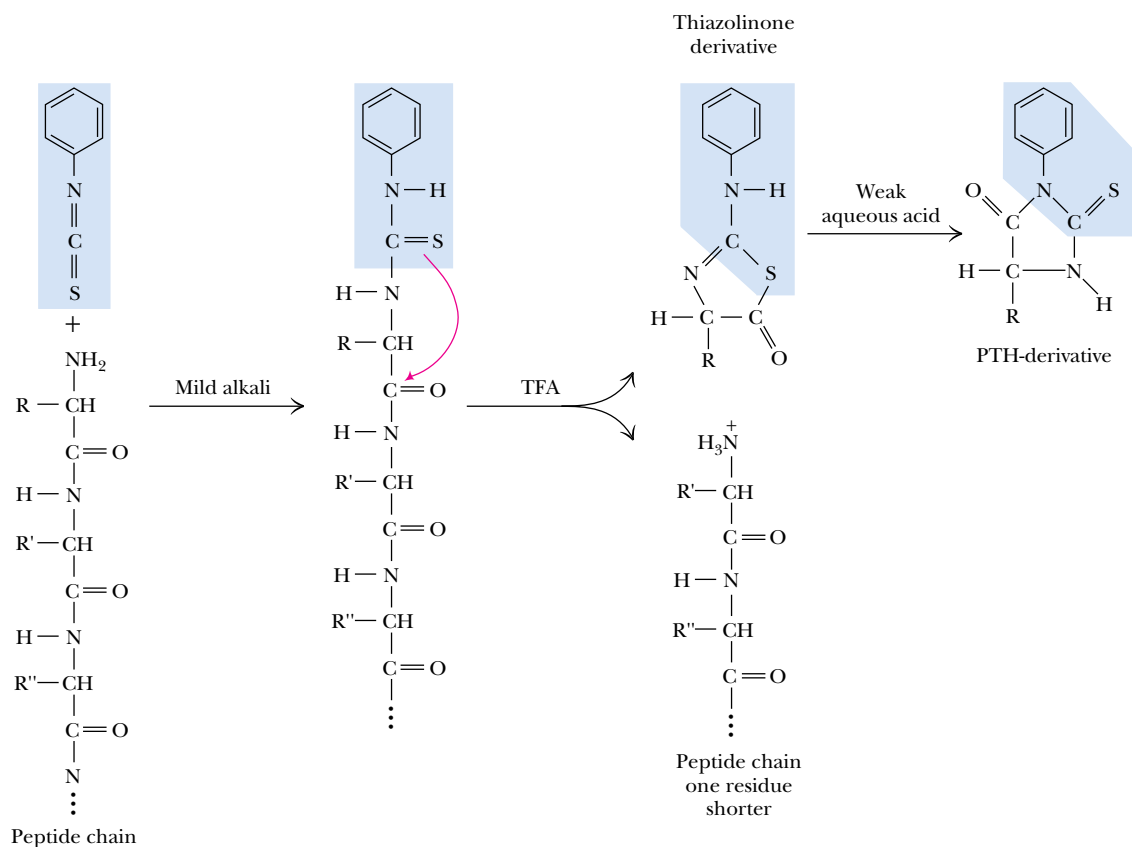


FIGURE 5.19 • N-Terminal analysis using Edman's reagent, phenylisothiocyanate. Phenylisothiocyanate combines with the N-terminus of a peptide under mildly alkaline conditions to form a phenylthiocarbamoyl substitution. Upon treatment with TFA (trifluoroacetic acid), this cyclizes to release the N-terminal amino acid residue as a thiazolinone derivative, but the other peptide bonds are not hydrolyzed. Organic extraction and treatment with aqueous acid yield the N-terminal amino acid as a phenylthiohydantoin (PTH) derivative.

on 50 pmol (about 0.1 μg) of a polypeptide 100 to 200 residues long, generating the sequential order of the first 50 amino acid residues in the protein. The efficiency with larger proteins is less; a typical 2000-amino acid protein provides only 10 to 20 cycles of reaction.

B. C-Terminal Analysis

For the identification of the C-terminal residue of polypeptides, an enzymatic approach is commonly used.

ENZYMATIC ANALYSIS WITH CARBOXYPEPTIDASES. Carboxypeptidases are enzymes that cleave amino acid residues from the C-termini of polypeptides in a successive fashion. Four carboxypeptidases are in general use: A, B, C, and Y. *Carboxypeptidase A* (from bovine pancreas) works well in hydrolyzing the C-terminal peptide bond of all residues except proline, arginine, and lysine. The analogous enzyme from hog pancreas, *carboxypeptidase B*, is effective only when Arg or Lys are the C-terminal residues. Thus, a mixture of carboxypeptidases A and B liberates any C-terminal amino acid except proline. *Carboxypeptidase C* from citrus leaves and *carboxypeptidase Y* from yeast act on any C-terminal residue. Because the nature of the amino acid residue at the end often determines the rate at which it is cleaved and because these enzymes remove residues successively, care must be taken in interpreting results. Carboxypeptidase Y cleavage has been adapted to an automated protocol analogous to that used in Edman sequencers.



Steps 5 and 6. Fragmentation of the Polypeptide Chain

The aim at this step is to produce fragments useful for sequence analysis. The cleavage methods employed are usually enzymatic, but proteins can also be fragmented by specific or nonspecific chemical means (such as partial acid hydrolysis). Proteolytic enzymes offer an advantage in that they may hydrolyze only specific peptide bonds, and this specificity immediately gives information about the peptide products. As a first approximation, fragments produced upon cleavage should be small enough to yield their sequences through end-group analysis and Edman degradation, yet not so small that an over-abundance of products must be resolved before analysis. However, the determination of total sequences for proteins predates the Edman procedure, and alternative approaches obviously exist.

A. Trypsin

The digestive enzyme *trypsin* is the most commonly used reagent for specific proteolysis. Trypsin is specific in hydrolyzing only peptide bonds in which the carbonyl function is contributed by an arginine or a lysine residue. That is, trypsin cleaves on the C-side of Arg or Lys, generating a set of peptide fragments having Arg or Lys at their C-termini. The number of smaller peptides resulting from trypsin action is equal to the total number of Arg and Lys residues in the protein *plus one*—the protein's C-terminal peptide fragment (Figure 5.20).

B. Chymotrypsin

Chymotrypsin shows a strong preference for hydrolyzing peptide bonds formed by the carboxyl groups of the aromatic amino acids, phenylalanine, tyrosine, and tryptophan. However, over time chymotrypsin also hydrolyzes amide bonds involving amino acids other than Phe, Tyr, or Trp. Peptide bonds having leucine-donated carboxyls become particularly susceptible. Thus, the specificity

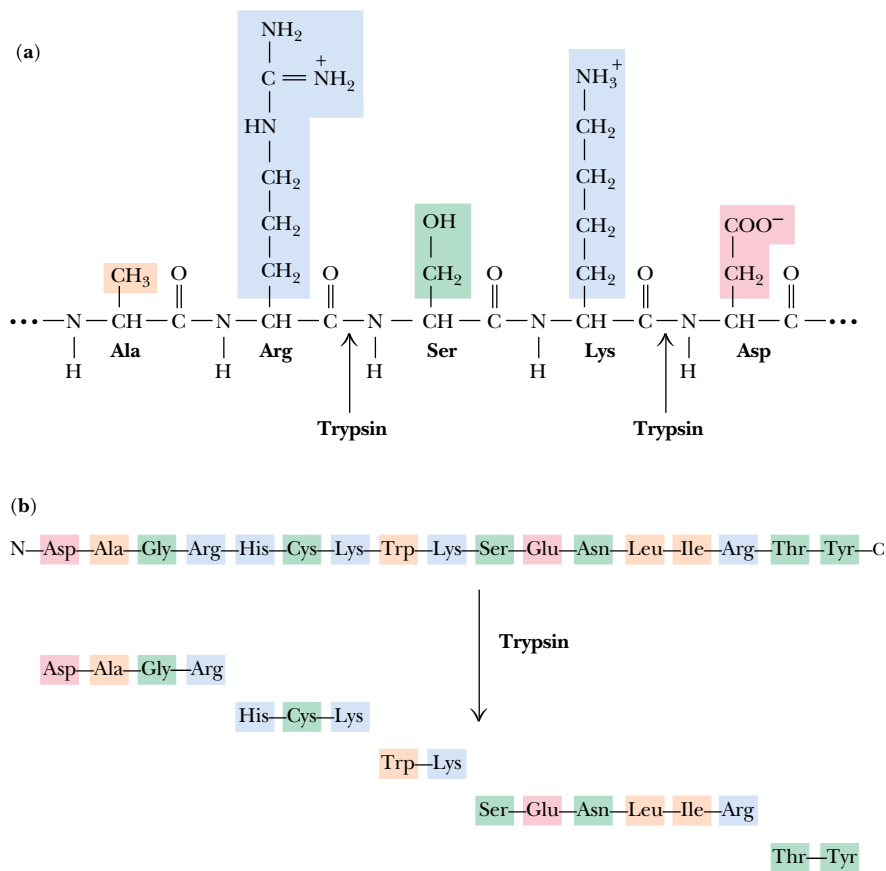


FIGURE 5.20 • Trypsin is a proteolytic enzyme, or *protease*, that specifically cleaves only those peptide bonds in which arginine or lysine contributes the carbonyl function. The products of the reaction are a mixture of peptide fragments with C-terminal Arg or Lys residues *and* a single peptide derived from the polypeptide’s C-terminal end.

of chymotrypsin is only relative. Because chymotrypsin produces a very different set of products than trypsin, treatment of separate samples of a protein with these two enzymes generates fragments whose sequences overlap. Resolution of the order of amino acid residues in the fragments yields the amino acid sequence in the original protein.

C. Relatively Nonspecific Endopeptidases

A number of other *endopeptidases* (proteases that cleave peptide bonds within the interior of a polypeptide chain) are also used in sequence investigations. These include *clostripain*, which acts only at Arg residues, *endopeptidase Lys-C*, which cleaves only at Lys residues, and *staphylococcal protease*, which acts at the acidic residues, Asp and Glu. Other, relatively nonspecific endopeptidases are handy for digesting large tryptic or chymotryptic fragments. *Pepsin*, *papain*, *subtilisin*, *thermolysin*, and *elastase* are some examples. Papain is the active ingredient in meat tenderizer and in soft contact lens cleaner as well as in some laundry detergents. The abundance of papain in papaya, and a similar protease (bromelain) in pineapple, causes the hydrolysis of gelatin and prevents the preparation of Jell-O[®] containing either of these fresh fruits. Cooking these fruits thermally denatures their proteolytic enzymes so that they can be used in gelatin desserts.

D. Cyanogen Bromide

Several highly specific chemical methods of proteolysis are available, the most widely used being *cyanogen bromide (CNBr)* cleavage. CNBr acts upon methio-

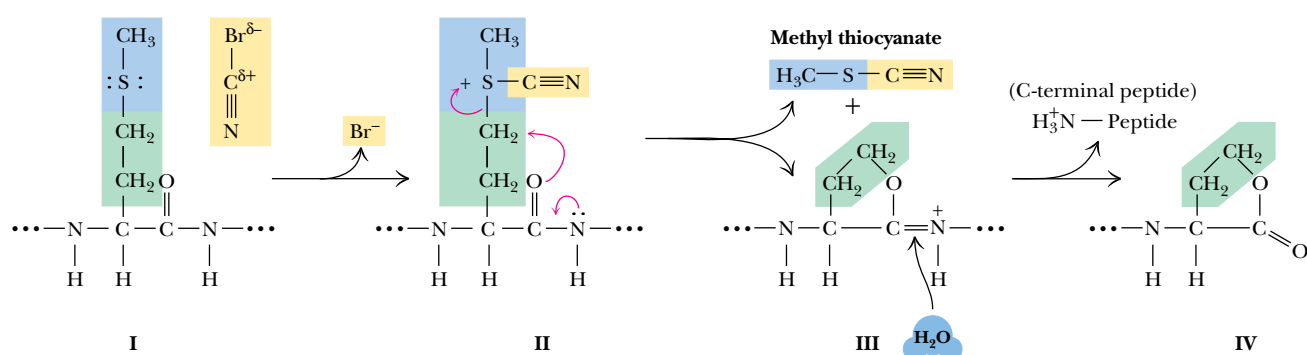
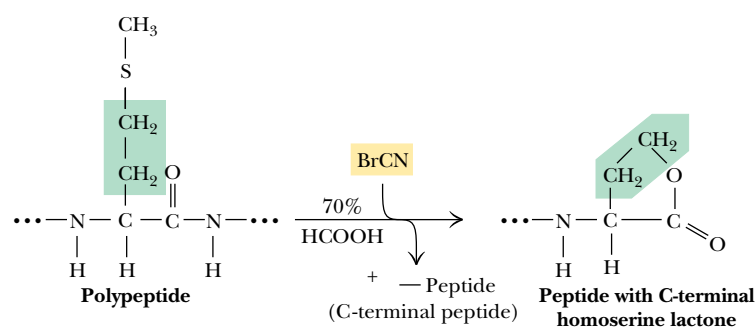


FIGURE 5.21 • Cyanogen bromide (CNBr) is a highly selective reagent for cleavage of peptides only at methionine residues. (I) The reaction occurs in 70% formic acid via nucleophilic attack of the Met S atom on the $\text{—C}\equiv\text{N}$ carbon atom, with displacement of Br^- . (II) The cyano intermediate undergoes nucleophilic attack by the Met carbonyl oxygen atom on the R group, (III) resulting in formation of the cyclic derivative, which is unstable in aqueous solution. (IV) Hydrolysis ensues, producing cleavage of the Met peptide bond and release of peptide fragments, with C-terminal homoserine lactone residues where Met residues once were. One peptide does not have a C-terminal homoserine lactone: the original C-terminal end of the polypeptide.

OVERALL REACTION:



nine residues (Figure 5.21). The nucleophilic sulfur atom of Met reacts with CNBr, yielding a sulfonium ion that undergoes a rapid intramolecular rearrangement to form a cyclic iminolactone. Water readily hydrolyzes this iminolactone, cleaving the polypeptide and generating peptide fragments having C-terminal homoserine lactone residues at the former Met positions.

Other Methods of Fragmentation

A number of other chemical methods give specific fragmentation of polypeptides, including cleavage at asparagine–glycine bonds by hydroxylamine (NH_2OH) at pH 9 and selective hydrolysis at aspartyl–prolyl bonds under mildly acidic conditions. Table 5.6 summarizes the various procedures described here for polypeptide cleavage. These methods are only a partial list of the arsenal of reactions available to protein chemists. Cleavage products generated by these procedures must be isolated and individually analyzed with respect to amino acid composition, end-group identity, and amino acid sequence to accumulate the information necessary to reconstruct the protein's total amino acid sequence. In the past, sequence was often deduced from exhaustive study of the amino acid composition and end-group analysis of small, overlapping peptides. Peptide sequencing today is most commonly done by Edman degradation of relatively large peptides.

Sequence Determination by Mass Spectrometry (MS)

Mass spectrometers exploit the difference in the mass-to-charge (m/z) ratio of ionized atoms or molecules to separate them from each other. The m/z ratio of a molecule is also a highly characteristic property that can be used for determining chemical and structural information. Further, molecules can be fragmented in distinctive ways in mass spectrometers, and the fragments that arise also provide quite specific structural information about the molecule. The basic

Table 5.6

Specificity of Representative Polypeptide Cleavage Procedures Used in Sequence Analysis

Method	Peptide Bond on Carboxyl (C) or Amino (N) Side of Susceptible Residue	Susceptible Residue(s)
<i>Proteolytic enzymes</i>		
Trypsin	C	Arg or Lys
Chymotrypsin	C	Phe, Trp, or Tyr; Leu
Clostripain	C	Arg
Staphylococcal protease	C	Asp or Glu
<i>Chemical methods</i>		
Cyanogen bromide	C	Met
NH ₂ OH	Asn-Gly bonds	
pH 2.5, 40°C	Asp-Pro bonds	

Table 5.7

Macromolecular Ionization Methods in Mass Spectrometry

Electrospray ionization (ESI-MS)	A solution of macromolecules is sprayed in the form of fine droplets from a glass capillary under the influence of a strong electrical field. The droplets pick up charge as they exit the capillary; evaporation of the solvent leaves highly charged molecules.
Fast-atom bombardment (FAB-MS)	A high-energy beam of inert gas molecules (argon or xenon) is directed at a solid sample, knocking molecules into the gas phase and ionizing them.
Laser ionization (LIMS)	A laser pulse is used to knock material from the surface of a solid sample; the laser pulse creates a microplasma that ionizes molecules in the sample.
Matrix-assisted desorption ionization (MALDI)	MALDI is a LIMS method capable of vaporizing and ionizing large biological molecules such as proteins or DNA. The biological molecules are dispersed in a solid matrix that serves as a carrier. Nicotinic acid is a commonly used matrix substance.

operation of a mass spectrometer is to (1) evaporate and ionize molecules in a vacuum, creating gas-phase ions, (2) separate the ions in space and/or time based on their *m/z* ratios, and (3) measure the amount of ions with specific *m/z* ratios. Because proteins (as well as nucleic acids and carbohydrates) decompose upon heating, rather than evaporating, attempts to ionize such molecules for MS analysis require innovative approaches (Table 5.7). Figure 5.22 illus-

FIGURE 5.22 • The three principal steps in electrospray mass spectrometry (ES-MS). (a) Small, highly charged droplets are formed by electrostatic dispersion of a protein solution through a glass capillary subjected to a high electric field; (b) protein ions are desorbed from the droplets into the gas phase (assisted by evaporation of the droplets in a stream of hot N₂ gas); and (c) analysis of the protein ions in a mass spectrometer. (Adapted from Figure 1 in Mann, M., and Wilm, M., 1995. Trends in Biochemical Sciences 20:219–224.)

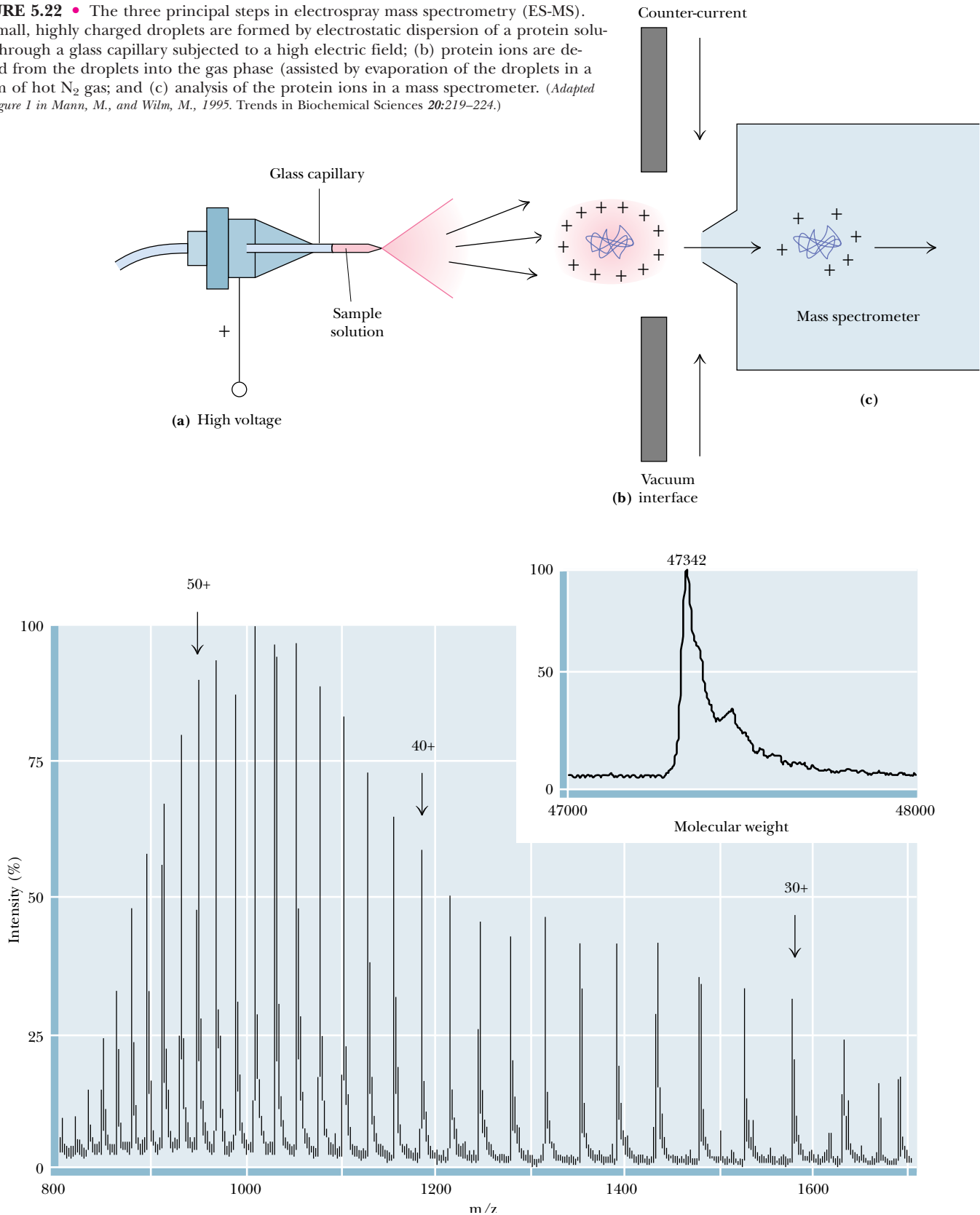


FIGURE 5.23 • Electrospray mass spectrum of the protein, aerolysin K. The attachment of many protons per protein molecule (from less than 30 to more than 50 here) leads to a series of m/z peaks for this single protein. The inset shows a computer analysis of the data from this series of peaks that generates a single peak at the correct molecular mass of the protein. (Adapted from Figure 2 in Mann, M., and Wilm, M., 1995. Trends in Biochemical Sciences 20:219–224.)

trates the basic features of electrospray mass spectrometry (ES-MS). In this technique, proteins pick up, on average, about one positive charge (proton) per kilodalton, leading to the spectrum of m/z ratios for a single protein species (Figure 5.23). Computer algorithms can convert these data into a single spectrum having a peak at the correct protein mass (inset, Figure 5.23).

SEQUENCING BY TANDEM MASS SPECTROMETRY. *Tandem MS* (or *MS/MS*) allows sequencing of proteins by hooking two mass spectrometers in tandem. The first mass spectrometer is used to separate oligopeptides from a protein digest and then to select in turn each of these oligopeptides for further analysis. A selected ionized oligopeptide is directed toward the second mass spectrometer; on the way, this oligopeptide is fragmented by collision with helium or argon gas molecules, and the collection of fragments is analyzed by the second mass spectrometer (Figure 5.24). Fragmentation occurs primarily in the peptide bonds

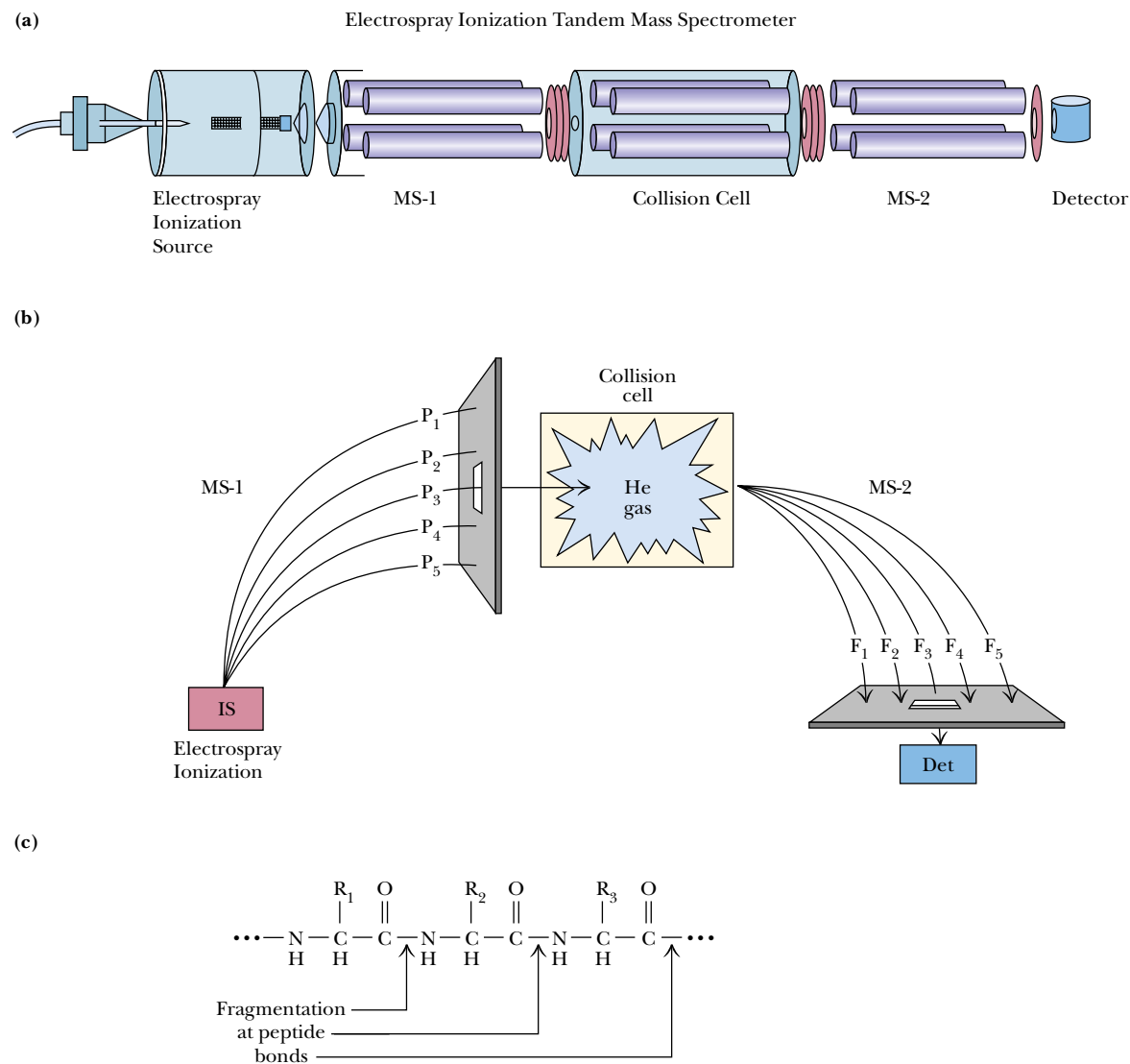


FIGURE 5.24 • Tandem mass spectrometry. (a) Configuration used in tandem MS. (b) Schematic description of tandem MS: tandem MS involves electrospray ionization of a protein digest (IS in this figure), followed by selection of a single peptide ion mass for collision with inert gas molecules (He) and mass analysis of the fragment ions resulting from the collisions. (c) Fragmentation usually occurs at peptide bonds, as indicated. (I: Adapted from Yates, J. R., 1996. *Methods in Enzymology* 271:351–376; II: Adapted from Gillette-Castro, B. L., and Stults, J. T., 1996. *Methods in Enzymology* 271:427–447.)

Step 8. Location of Disulfide Cross-Bridges

Strictly speaking, the disulfide bonds formed between cysteine residues in a protein are not a part of its primary structure. Nevertheless, information about their location can be obtained by procedures used in sequencing, provided the disulfides are not broken prior to cleaving the polypeptide chain. Because these covalent bonds are stable under most conditions used in the cleavage of polypeptides, intact disulfides link the peptide fragments containing their specific cysteinyl residues and thus these linked fragments can be isolated and identified within the protein digest.

An effective way to isolate these fragments is through **diagonal electrophoresis** (Figure 5.26) (the basic technique of *electrophoresis* is described in

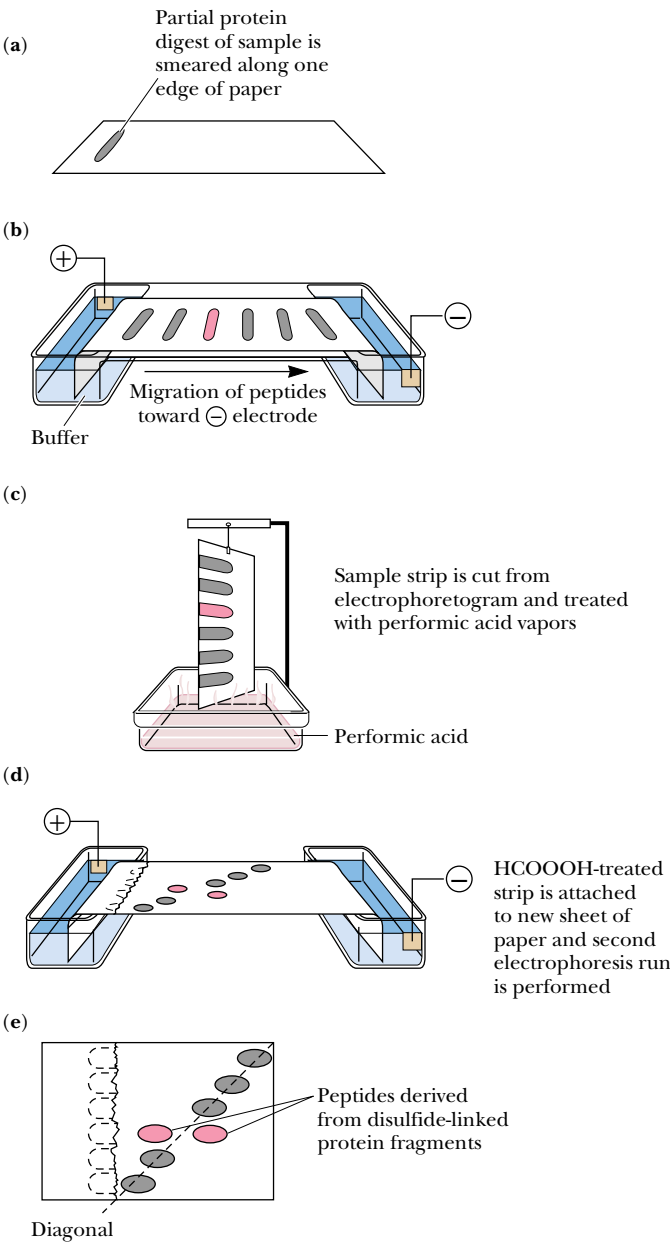


FIGURE 5.26 • Disulfide bridges typically are cleaved prior to determining the primary structure of a polypeptide. Consequently, the positions of disulfide links are not obvious from the sequence data. To determine their location, a sample of the polypeptide with intact S—S bonds can be fragmented and the sites of any disulfides can be elucidated from fragments that remain linked. *Diagonal electrophoresis* is a technique for identifying such fragments. (a) A protein digest in which any disulfide bonds remain intact and link their respective Cys-containing peptides is streaked along the edge of a filter paper and (b) subjected to electrophoresis. (c) A strip cut from the edge of the paper is then exposed to performic acid fumes to oxidize any disulfide bridges. (d) Then the paper strip is attached to a new filter paper so that a second electrophoresis can be run in a direction perpendicular to the first. (e) Peptides devoid of disulfides experience no mobility change, and thus their pattern of migration defines a diagonal. Peptides that had disulfides migrate off this diagonal and can be easily identified, isolated, and sequenced to reveal the location of cysteic acid residues formerly involved in disulfide bridges.

the Chapter Appendix). Peptides that were originally linked by disulfides now migrate as distinct species following disulfide cleavage and are obvious by their location off the diagonal (Figure 5.26e). These cysteic acid-containing peptides are then isolated from the paper and sequenced. From this information, the positions of the disulfides in the protein can be stipulated.

Sequence Databases

A database of protein sequences collected by protein chemists can be found in the *Atlas of Protein Sequence and Structure*. However, most protein sequence information has been derived from translating the nucleotide sequences of genes into codons and, thus, amino acid sequences (see Chapter 13). Sequencing the order of nucleotides in cloned genes is a more rapid, efficient, and informative process than determining the amino acid sequences of proteins. A number of electronic databases containing continuously updated sequence information are readily accessible by personal computer. Prominent among these are PIR (Protein Identification Resource Protein Sequence Database), GenBank (Genetic Sequence Data Bank), and EMBL (European Molecular Biology Laboratory Data Library).

5.8 • Nature of Amino Acid Sequences

With a knowledge of the methodology in hand, let's review the results of amino acid composition and sequence studies on proteins. Table 5.8 lists the relative frequencies of the amino acids in various proteins. It is very unusual for a globular protein to have an amino acid composition that deviates substantially from these values. Apparently, these abundances reflect a distribution of amino acid polarities that is optimal for protein stability in an aqueous milieu. Membrane proteins have relatively more hydrophobic and fewer ionic amino acids, a condition consistent with their location. Fibrous proteins may show compositions that are atypical with respect to these norms, indicating an underlying relationship between the composition and the structure of these proteins.

Proteins have unique amino acid sequences, and it is this uniqueness of sequence that ultimately gives each protein its own particular personality. Because the number of possible amino acid sequences in a protein is astronomically large, the probability that two proteins will, by chance, have similar amino acid sequences is negligible. Consequently, sequence similarities between proteins imply evolutionary relatedness.

Homologous Proteins from Different Organisms Have Homologous Amino Acid Sequences

Proteins sharing a significant degree of sequence similarity are said to be **homologous**. Proteins that perform the same function in different organisms are also referred to as homologous. For example, the oxygen transport protein, hemoglobin, serves a similar role and has a similar structure in all vertebrates. The study of the amino acid sequences of homologous proteins from different organisms provides very strong evidence for their evolutionary origin within a common ancestor. Homologous proteins characteristically have polypeptide chains that are nearly identical in length, and their sequences share identity in direct correlation to the relatedness of the species from which they are derived.

Table 5.8
Frequency of Occurrence of Amino Acid Residues in Proteins

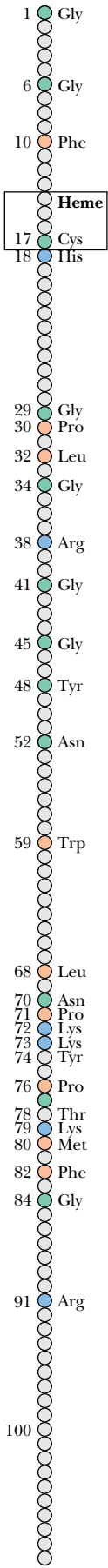
Amino Acid		M _r *	Occurrence in Proteins (%) [†]
Alanine	Ala A	71.1	9.0
Arginine	Arg R	156.2	4.7
Asparagine	Asn N	114.1	4.4
Aspartic acid	Asp D	115.1	5.5
Cysteine	Cys C	103.1	2.8
Glutamine	Gln Q	128.1	3.9
Glutamic acid	Glu E	129.1	6.2
Glycine	Gly G	57.1	7.5
Histidine	His H	137.2	2.1
Isoleucine	Ile I	113.2	4.6
Leucine	Leu L	113.2	7.5
Lysine	Lys K	128.2	7.0
Methionine	Met M	131.2	1.7
Phenylalanine	Phe F	147.2	3.5
Proline	Pro P	97.1	4.6
Serine	Ser S	87.1	7.1
Threonine	Thr T	101.1	6.0
Tryptophan	Trp W	186.2	1.1
Tyrosine	Tyr Y	163.2	3.5
Valine	Val V	99.1	6.9

*Molecular weight of amino acid *minus* that of water.
[†]Frequency of occurrence of each amino acid residue in the polypeptide chains of 207 unrelated proteins of known sequence.
Values from Klapper, M. H., 1977. *Biochemical and Biophysical Research Communications* 78:1018–1024.

Cytochrome c

The electron transport protein, cytochrome *c*, found in the mitochondria of all eukaryotic organisms, provides the best-studied example of homology. The polypeptide chain of cytochrome *c* from most species contains slightly more than 100 amino acids and has a molecular weight of about 12.5 kD. Amino acid sequencing of cytochrome *c* from more than 40 different species has revealed that there are 28 positions in the polypeptide chain where the same amino acid residues are always found (Figure 5.27). These **invariant residues** apparently serve roles crucial to the biological function of this protein, and thus substitutions of other amino acids at these positions cannot be tolerated.

FIGURE 5.27 • Cytochrome *c* is a small protein consisting of a single polypeptide chain of 104 residues in terrestrial vertebrates, 103 or 104 in fishes, 107 in insects, 107 to 109 in fungi and yeasts, and 111 or 112 in green plants. Analysis of the sequence of cytochrome *c* from more than 40 different species reveals that 28 residues are invariant. These invariant residues are scattered irregularly along the polypeptide chain, except for a cluster between residues 70 and 80. All cytochrome *c* polypeptide chains have a cysteine residue at position 17, and all but one have another Cys at position 14. These Cys residues serve to link the heme prosthetic group of cytochrome *c* to the protein, a role explaining their invariable presence.



	Chimpanzee	Sheep	Rattlesnake	Carp	Snail	Moth	Yeast	Cauliflower	Parsnip
Human	0	10	14	18	29	31	44	44	43
Chimpanzee		10	14	18	29	31	44	44	43
Sheep			20	11	24	27	44	46	46
Rattlesnake				26	28	33	47	45	43
Carp					26	26	44	47	46
Garden snail						28	48	51	50
Tobacco hornworm moth							44	44	41
Baker's yeast (iso-1)								47	47
Cauliflower									13

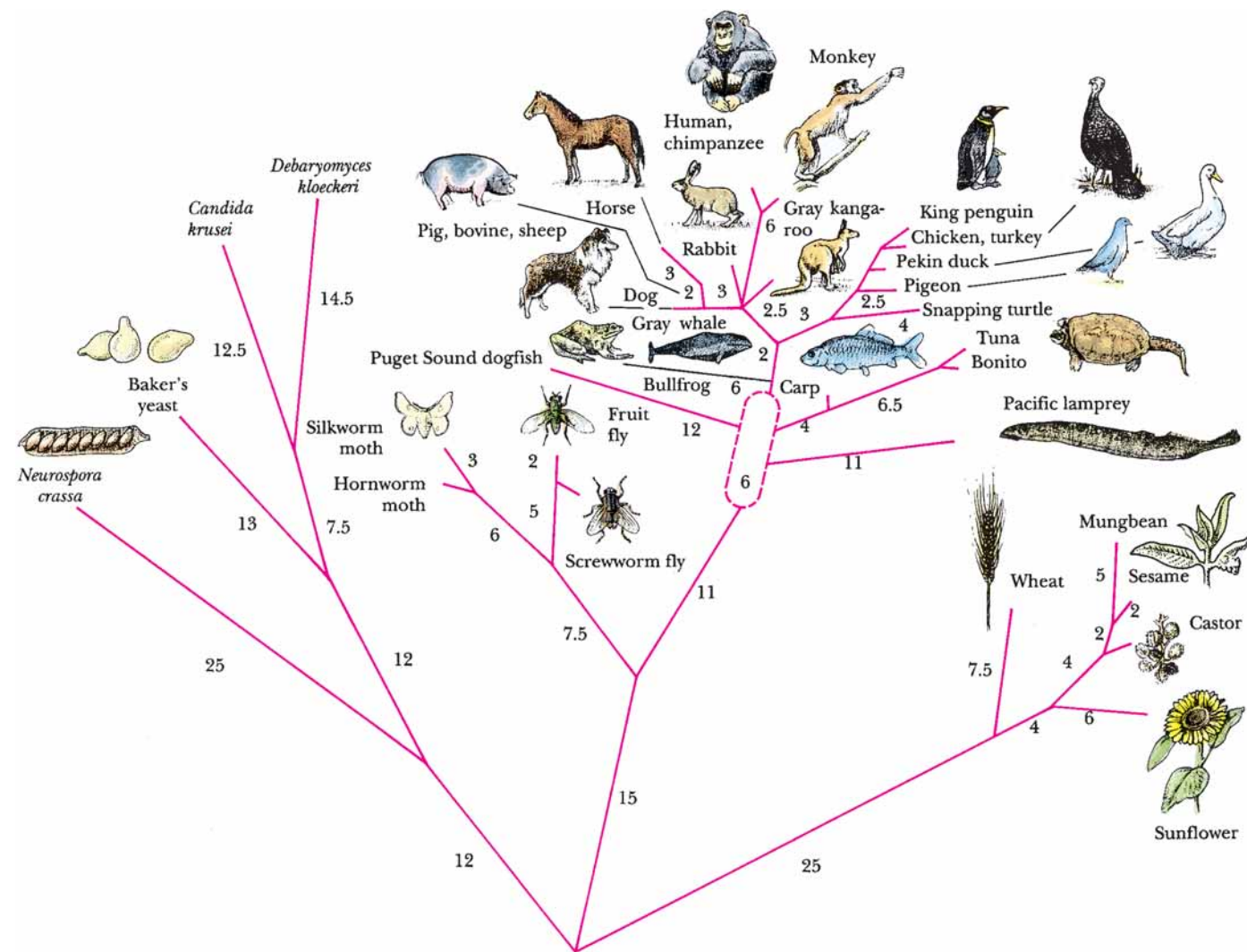
FIGURE 5.28 • The number of amino acid differences among the cytochrome *c* sequences of various organisms can be compared. The numbers bear a direct relationship to the degree of relatedness between the organisms. Each of these species has a cytochrome *c* of at least 104 residues, so any given pair of species has more than half its residues in common. (Adapted from Creighton, T. E., 1983. Proteins: Structure and Molecular Properties. San Francisco: W. H. Freeman and Co.)

Furthermore, as shown in Figure 5.28, the number of amino acid differences between two cytochrome *c* sequences is proportional to the phylogenetic difference between the species from which they are derived. The cytochrome *c* in humans and in chimpanzees is identical; human and another mammalian (sheep) cytochrome *c* differ at 10 residues. The human cytochrome *c* sequence has 14 variant residues from a reptile sequence (rattlesnake), 18 from a fish (carp), 29 from a mollusc (snail), 31 from an insect (moth), and more than 40 from yeast or higher plants (cauliflower).

The Phylogenetic Tree for Cytochrome c

Figure 5.29 displays a **phylogenetic tree** (a diagram illustrating the evolutionary relationships among a group of organisms) constructed from the sequences of cytochrome *c*. The tips of the branches are occupied by contemporary species whose sequences have been determined. The tree has been deduced by computer analysis of these sequences to find the minimum number of mutational changes connecting the branches. Other computer methods can be used to infer potential ancestral sequences represented by *nodes*, or branch points, in the tree. Such analysis ultimately suggests a primordial cytochrome *c* sequence lying at the base of the tree. Evolutionary trees constructed in this manner, that is, solely on the basis of amino acid differences occurring in the primary sequence of one selected protein, show remarkable agreement with phylogenetic relationships derived from more classic approaches and have given rise to the field of *molecular evolution*.

FIGURE 5.29 • This phylogenetic tree depicts the evolutionary relationships among organisms as determined by the similarity of their cytochrome *c* amino acid sequences. The numbers along the branches give the amino acid changes between a species and a hypothetical progenitor. Note that extant species are located only at the tips of branches. Below, the sequence of human cytochrome *c* is compared with an inferred ancestral sequence represented by the base of the tree. Uncertainties are denoted by question marks. (Adapted from Creighton, T. E., 1983. Proteins: Structure and Molecular Properties. San Francisco: W. H. Freeman and Co.)



Ancestral cytochrome c	1	10	20
Human cytochrome c	-Pro-Ala-Gly-Asp-?-Lys-Lys-Gly-Ala-Lys-Ile-Phe-Lys-Thr-?-Cys-Ala-Gln-Cys-His-Thr-Val-Glu-?-Gly-Gly-?-		
	Gly-Asp-Val-Glu-Lys-Gly-Lys-Lys-Ile-Phe-Ile-Met-Lys-Cys-Ser-Gln-Cys-His-Thr-Val-Glu-Lys-Gly-Gly-Lys-		
	30	40	50
	-His-Lys-Val-Gly-Pro-Asn-Leu-His-Gly-Leu-Phe-Gly-Arg-Lys-?-Gly-Gln-Ala-?-Gly-Tyr-Ser-Tyr-Thr-Asp-		
	-His-Lys-Thr-Gly-Pro-Asn-Leu-His-Gly-Leu-Phe-Gly-Arg-Lys-Thr-Gly-Gln-Ala-Pro-Gly-Tyr-Ser-Tyr-Thr-Ala-		
	60	70	
	-Ala-Asn-Lys-Asn-Lys-Gly-?-?-Trp-?-Glu-Asn-Thr-Leu-Phe-Glu-Tyr-Leu-Glu-Asn-Pro-Lys-Lys-Tyr-Ile-		
	-Ala-Asn-Lys-Asn-Lys-Gly-Ile-Ile-Trp-Gly-Glu-Asp-Thr-Leu-Met-Gln-Tyr-Leu-Glu-Asn-Pro-Lys-Lys-Tyr-Pro-		
	80	90	100
	-Pro-Gly-Thr-Lys-Met-?-Phe-?-Gly-Leu-Lys-Lys-?-?-Asp-Arg-Ala-Asp-Leu-Ile-Ala-Tyr-Leu-Lys-?-		
	-Pro-Gly-Thr-Lys-Met-Ile-Phe-Val-Gly-Ile-Lys-Lys-Lys-Glu-Glu-Arg-Ala-Asp-Leu-Ile-Ala-Tyr-Leu-Lys-Lys-		
	-Ala-Thr-Ala		
	-Ala-Thr-Asn-Glu		

identities, whereas human α -globin and human β -globin have 64 residues in common. The relatedness suggests an evolutionary sequence of events in which chance mutations led to amino acid substitutions and divergence in primary structure. The ancestral myoglobin gene diverged first, after duplication of a primordial globin gene had given rise to its progenitor and an ancestral hemoglobin gene (Figure 5.31). Subsequently, the ancestral hemoglobin gene duplicated to generate the progenitors of the present-day α -globin and β -globin genes. The ability to bind O₂ via a heme prosthetic group is retained by all three of these polypeptides.

Serine Proteases

Whereas the globins provide an example of gene duplication giving rise to a set of proteins in which the biological function has been highly conserved, other sets of proteins united by strong sequence homology show more divergent biological functions. **Trypsin**, **chymotrypsin** (see Section 5.7), and **elastase** are members of a class of proteolytic enzymes called **serine proteases** because of the central role played by specific serine residues in their catalytic activity. **Thrombin**, an essential enzyme in blood clotting, is also a serine protease. These enzymes show sufficient sequence homology to conclude that they arose via duplication of a progenitor serine protease gene, even though their substrate preferences are now quite different.

Apparently Different Proteins May Share a Common Ancestry

A more remarkable example of evolutionary relatedness is inferred from sequence homology between hen egg white **lysozyme** and human milk **α -lactalbumin**, proteins of quite different biological activity and origin. Lysozyme (129 residues) and α -lactalbumin (123 residues) are identical at 48 positions. Lysozyme hydrolyzes the polysaccharide wall of bacterial cells, whereas α -lactalbumin regulates milk sugar (lactose) synthesis in the mammary gland. Although both proteins act in reactions involving carbohydrates, their functions show little similarity otherwise. Nevertheless, their tertiary structures are strikingly similar (Figure 5.32). It is conceivable that many proteins are related in this way, but time and the course of evolutionary change erased most evidence of their common ancestry. In an interesting contrast to this case, the proteins *actin* and *hexokinase* share essentially no sequence homology, yet they have very similar three-dimensional structures, even though their biological roles and physical properties are quite different. Actin forms a filamentous polymer that is a principal component of the contractile apparatus in muscle; hexokinase is a cytosolic enzyme that catalyzes the first reaction in glucose catabolism.

Mutant Proteins

Given a large population of individuals, a considerable number of sequence variants can be found for a protein. These variants are a consequence of **mutations** in a gene (base substitutions in DNA) that have arisen naturally within the population. Gene mutations lead to mutant forms of the protein in which the amino acid sequence is altered at one or more positions. Many of these mutant forms are “neutral” in that the functional properties of the protein are unaffected by the amino acid substitution. Others may be nonfunctional (if loss of function is not lethal to the individual), and still others may display a range of aberrations between these two extremes. The severity of the effects on function depends on the nature of the amino acid substitution and its role in the protein. These conclusions are exemplified by the more than 300 human

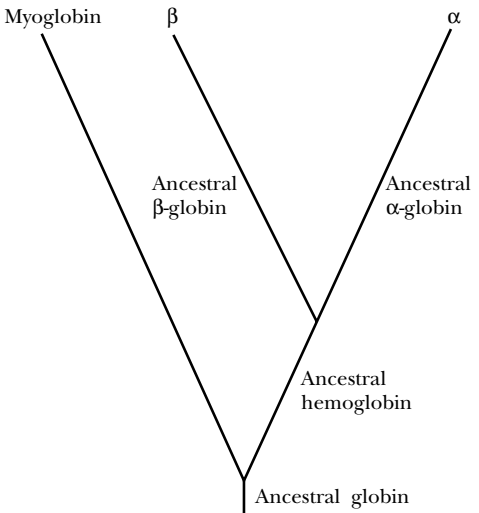


FIGURE 5.31 • This evolutionary tree is inferred from the homology between the amino acid sequences of the α -globin, β -globin, and myoglobin chains. Duplication of an ancestral globin gene allowed the divergence of the myoglobin and ancestral hemoglobin genes. Another gene duplication event subsequently gave rise to ancestral α and β forms, as indicated. Gene duplication is an important evolutionary force in creating diversity.

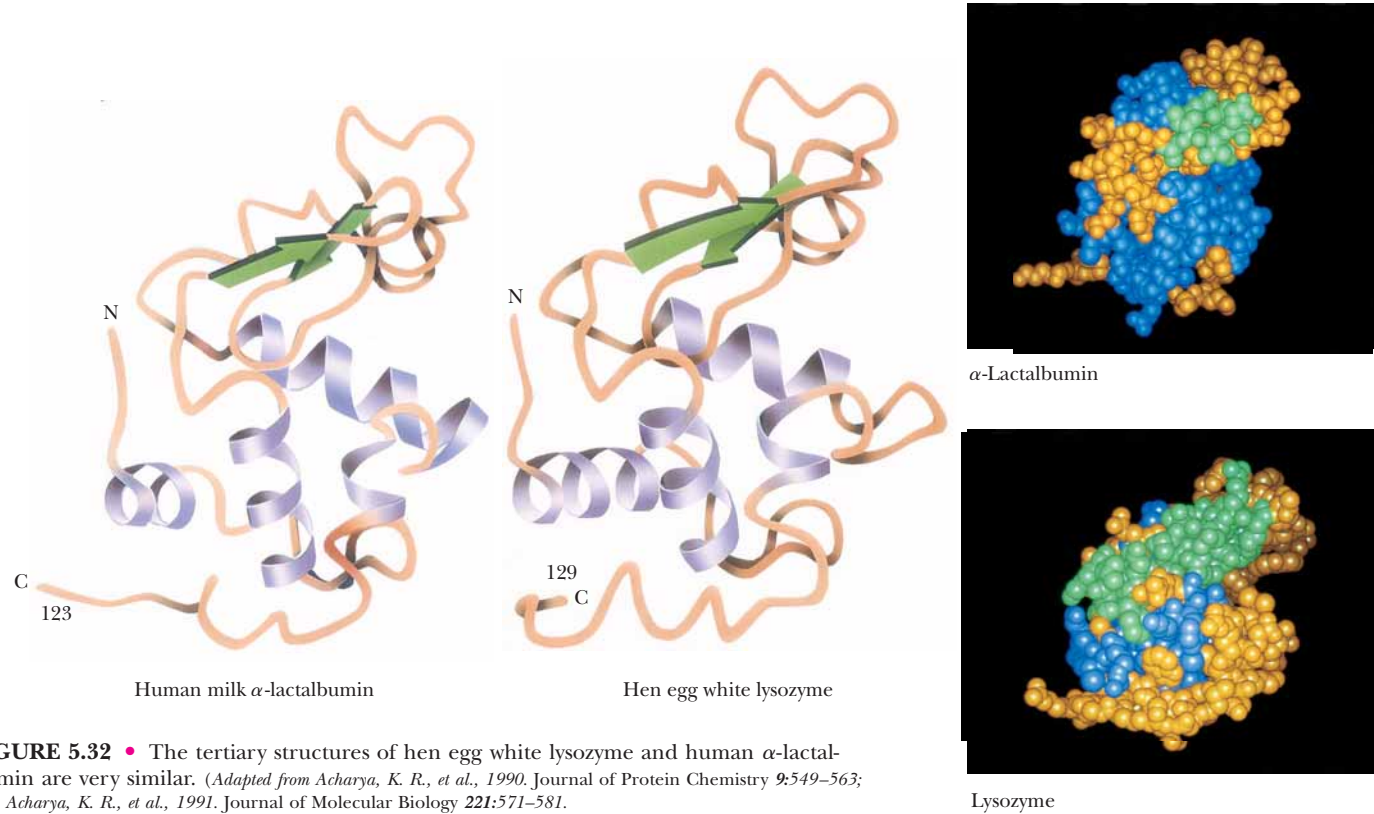


FIGURE 5.32 • The tertiary structures of hen egg white lysozyme and human α -lactalbumin are very similar. (Adapted from Acharya, K. R., et al., 1990. Journal of Protein Chemistry 9:549–563; and Acharya, K. R., et al., 1991. Journal of Molecular Biology 221:571–581.



Table 5.9
Some Pathological Sequence Variants of Human Hemoglobin

Abnormal Hemoglobin*	Normal Residue and Position	Substitution
<i>Alpha chain</i>		
Torino	Phenylalanine 43	Valine
M _{Boston}	Histidine 58	Tyrosine
Chesapeake	Arginine 92	Leucine
G _{Georgia}	Proline 95	Leucine
Tarrant	Aspartate 126	Asparagine
Suresnes	Arginine 141	Histidine
<i>Beta chain</i>		
S	Glutamate 6	Valine
Riverdale–Bronx	Glycine 24	Arginine
Genova	Leucine 28	Proline
Zurich	Histidine 63	Arginine
M _{Milwaukee}	Valine 67	Glutamate
M _{Hyde Park}	Histidine 92	Tyrosine
Yoshizuka	Asparagine 108	Aspartate
Hiroshima	Histidine 146	Aspartate

*Hemoglobin variants are often given the geographical name of their origin.
Adapted from Dickerson, R. E., and Geis, I., 1983. *Hemoglobin: Structure, Function, Evolution and Pathology*. Menlo Park, CA: Benjamin-Cummings Publishing Co.

hemoglobin variants that have been discovered to date. Some of these are listed in Table 5.9.

A variety of effects on the hemoglobin molecule are seen in these mutants, including alterations in oxygen affinity, heme affinity, stability, solubility, and subunit interactions between the α -globin and β -globin polypeptide chains. Some variants show no apparent changes, whereas others, such as HbS, sickle-cell hemoglobin (see Chapter 15), result in serious illness. This diversity of response indicates that some amino acid changes are relatively unimportant, whereas others drastically alter one or more functions of a protein.

5.9 • Synthesis of Polypeptides in the Laboratory

Chemical synthesis of peptides and polypeptides of defined sequence can be carried out in the laboratory. Formation of peptide bonds linking amino acids together is not a chemically complex process, but making a specific peptide can be challenging because various functional groups present on side chains of amino acids may also react under the conditions used to form peptide bonds. Furthermore, if correct sequences are to be synthesized, the α -COOH group of residue x must be linked to the α -NH₂ group of neighboring residue y in a way that prevents reaction of the amino group of x with the carboxyl group of y . Ingenious synthetic strategies are required to circumvent these technical problems. In essence, any functional groups to be excluded from reaction must be blocked while the desired coupling reactions proceed. Also, the blocking groups must be removable later under conditions in which the newly formed peptide bonds are stable. These limitations mean that addition of each amino acid requires several steps. Further, all reactions must proceed with high yield if peptide recoveries are to be acceptable. Peptide formation between amino and carboxyl groups is not spontaneous under normal conditions (see Chapter 4), so one or the other of these groups must be activated to facilitate the reaction. Despite these difficulties, biologically active peptides and polypeptides have been recreated by synthetic organic chemistry. Milestones include the pioneering synthesis of the nonapeptide posterior pituitary hormones oxytocin and vasopressin by du Vigneaud in 1953, and in later years, the blood pressure-regulating hormone bradykinin (9 residues), melanocyte-stimulating hormone (24 residues), adrenocorticotropin (39 residues), insulin (21 A-chain and 30 B-chain residues), and ribonuclease A (124 residues).

Solid Phase Peptide Synthesis

Bruce Merrifield and his collaborators found a clever solution to the problem of recovering intermediate products in the course of a synthesis. The carboxyl-terminal residues of synthesized peptide chains were covalently anchored to an insoluble resin particle large enough to be removed from reaction mixtures simply by filtration. After each new residue was added successively at the free amino-terminus, the elongated product was recovered by filtration and readied for the next synthetic step. Because the growing peptide chain was coupled to an insoluble resin bead, the method is called **solid phase synthesis**. The procedure is detailed in Figure 5.33. This cyclic process has been automated and computer controlled so that the reactions take place in a small cup with reagents being pumped in and removed as programmed. The 124-residue-long bovine pancreatic ribonuclease A sequence was synthesized, and the final product was enzymatically active as an RNase.

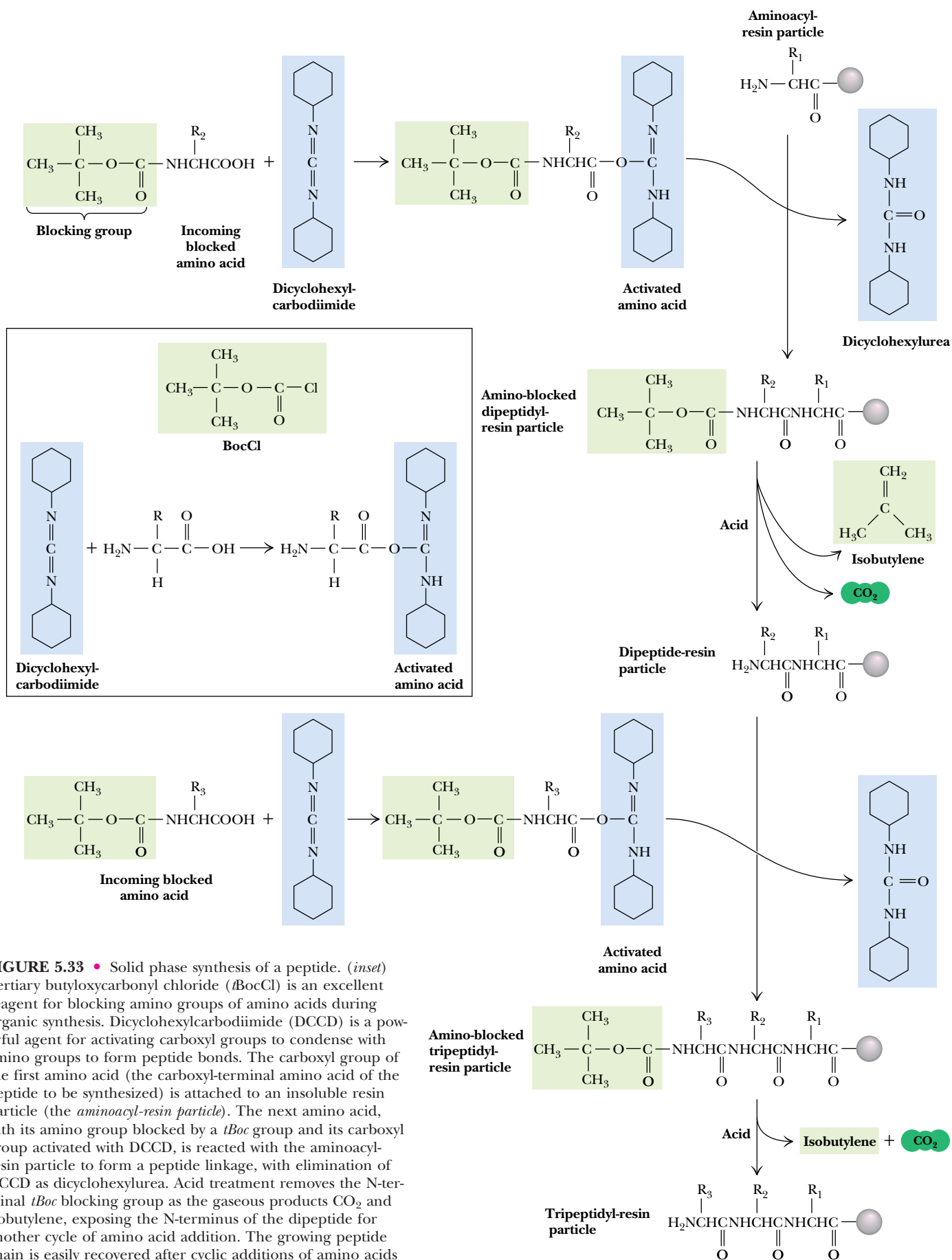


FIGURE 5.33 • Solid phase synthesis of a peptide. (*inset*) Tertiary butyloxycarbonyl chloride (*t*BocCl) is an excellent reagent for blocking amino groups of amino acids during organic synthesis. Dicyclohexylcarbodiimide (DCCD) is a powerful agent for activating carboxyl groups to condense with amino groups to form peptide bonds. The carboxyl group of the first amino acid (the carboxyl-terminal amino acid of the peptide to be synthesized) is attached to an insoluble resin particle (the *aminoacyl-resin particle*). The next amino acid, with its amino group blocked by a *t*Boc group and its carboxyl group activated with DCCD, is reacted with the aminoacyl-resin particle to form a peptide linkage, with elimination of DCCD as dicyclohexylurea. Acid treatment removes the N-terminal *t*Boc blocking group as the gaseous products CO₂ and isobutylene, exposing the N-terminus of the dipeptide for another cycle of amino acid addition. The growing peptide chain is easily recovered after cyclic additions of amino acids simply by filtering or centrifuging the reaction mixture.

PROBLEMS

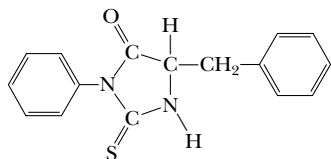
1. The element molybdenum (atomic weight 95.95) constitutes 0.08% of the weight of nitrate reductase. If the molecular weight of nitrate reductase is 240,000, what is its likely quaternary structure?

2. Amino acid analysis of an oligopeptide seven residues long gave

Asp Leu Lys Met Phe Tyr

The following facts were observed:

- Trypsin treatment had no apparent effect.
- The phenylthiohydantoin released by Edman degradation was



c. Brief chymotrypsin treatment yielded several products, including a dipeptide and a tetrapeptide. The amino acid composition of the tetrapeptide was Leu, Lys, and Met.

d. Cyanogen bromide treatment yielded a dipeptide, a tetrapeptide, and free Lys.

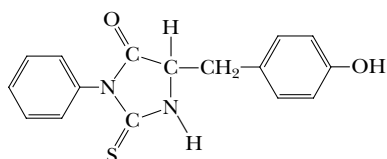
What is the amino acid sequence of this heptapeptide?

3. Amino acid analysis of another heptapeptide gave

Asp Glu Leu Lys
Met Tyr Trp NH_4^+

The following facts were observed:

- Trypsin had no effect.
- The phenylthiohydantoin released by Edman degradation was



c. Brief chymotrypsin treatment yielded several products, including a dipeptide and a tetrapeptide. The amino acid composition of the tetrapeptide was Glx, Leu, Lys, and Met.

d. Cyanogen bromide treatment yielded a tetrapeptide that had a net positive charge at pH 7 and a tripeptide that had a zero net charge at pH 7.

What is the amino acid sequence of this heptapeptide?

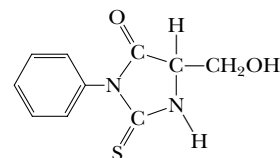
4. Amino acid analysis of a decapeptide revealed the presence of the following products:

NH_4^+ Asp Glu Tyr Arg
Met Pro Lys Ser Phe

The following facts were observed:

- Neither carboxypeptidase A or B treatment of the decapeptide had any effect.
- Trypsin treatment yielded two tetrapeptides and free Lys.
- Clostripain treatment yielded a tetrapeptide and a hexapeptide.
- Cyanogen bromide treatment yielded an octapeptide and a dipeptide of sequence NP (using the one-letter codes).
- Chymotrypsin treatment yielded two tripeptides and a tetrapeptide. The N-terminal chymotryptic peptide had a net charge of -1 at neutral pH and a net charge of -3 at pH 12.

f. One cycle of Edman degradation gave the PTH derivative



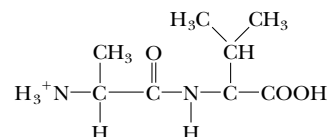
What is the amino acid sequence of this decapeptide?

5. Analysis of the blood of a catatonic football fan revealed large concentrations of a psychotoxic octapeptide. Amino acid analysis of this octapeptide gave the following results:

2 Ala 1 Arg 1 Asp 1 Met 2 Tyr 1 Val 1 NH_4^+

The following facts were observed:

a. Partial acid hydrolysis of the octapeptide yielded a dipeptide of the structure



b. Chymotrypsin treatment of the octapeptide yielded two tetrapeptides, each containing an alanine residue.

c. Trypsin treatment of one of the tetrapeptides yielded two dipeptides.

d. Cyanogen bromide treatment of another sample of the same tetrapeptide yielded a tripeptide and free Tyr.

e. End-group analysis of the other tetrapeptide gave Asp.

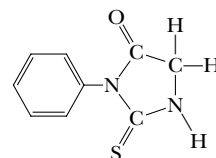
What is the amino acid sequence of this octapeptide?

6. Amino acid analysis of an octapeptide revealed the following composition:

2 Arg 1 Gly 1 Met 1 Trp 1 Tyr 1 Phe 1 Lys

The following facts were observed:

a. Edman degradation gave



b. CNBr treatment yielded a pentapeptide and a tripeptide containing phenylalanine.

c. Chymotrypsin treatment yielded a tetrapeptide containing a C-terminal indole amino acid, and two dipeptides.

d. Trypsin treatment yielded a tetrapeptide, a dipeptide, and free Lys and Phe.

e. Clostripain yielded a pentapeptide, a dipeptide, and free Phe.

What is the amino acid sequence of this octapeptide?

7. Amino acid analysis of an octapeptide gave the following results:

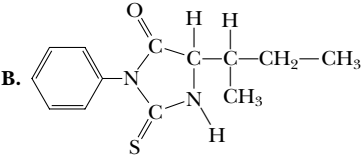
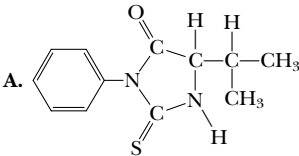
1 Ala 1 Arg 1 Asp 1 Gly 3 Ile 1 Val 1 NH_4^+

The following facts were observed:

- Trypsin treatment yielded a pentapeptide and a tripeptide.
- Chemical reaction of the free α -COOH and subsequent acid hydrolysis yielded 2-aminopropanol.
- Partial acid hydrolysis of the tryptic pentapeptide yielded,

among other products, two dipeptides, each of which contained C-terminal isoleucine. One of these dipeptides migrated as an anionic species upon electrophoresis at neutral pH.

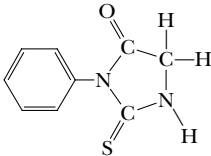
d. The tryptic tripeptide was degraded in an Edman sequenator, yielding first **A**, then **B**:



What is an amino acid sequence of the octapeptide? Four sequences are possible, but only one suits the authors. Why?

8. An octapeptide consisting of 2 Gly, 1 Lys, 1 Met, 1 Pro, 1 Arg, 1 Trp, and 1 Tyr was subjected to sequence studies. The following was found:

a. Edman degradation yielded



b. Upon treatment with carboxypeptidases A, B, and C, only carboxypeptidase C had any effect.

c. Trypsin treatment gave two tripeptides and a dipeptide.

d. Chymotrypsin treatment gave two tripeptides and a dipeptide.

Acid hydrolysis of the dipeptide yielded only Gly.

e. Cyanogen bromide treatment yielded two tetrapeptides.

f. Clostripain treatment gave a pentapeptide and a tripeptide.

What is the amino acid sequence of this octapeptide?

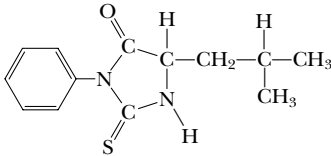
9. Amino acid analysis of an oligopeptide containing nine residues revealed the presence of the following amino acids:

Arg Cys Gly Leu Met Pro Tyr Val

The following was found:

a. Carboxypeptidase A treatment yielded no free amino acid.

b. Edman analysis of the intact oligopeptide released



c. Neither trypsin nor chymotrypsin treatment of the nonapeptide released smaller fragments. However, combined trypsin and chymotrypsin treatment liberated free Arg.

d. CNBr treatment of the eight-residue fragment left after combined trypsin and chymotrypsin action yielded a six-residue fragment containing Cys, Gly, Pro, Tyr, and Val; and a dipeptide.

e. Treatment of the six-residue fragment with β -mercaptoethanol yielded two tripeptides. Brief Edman analysis of the tripeptide mixture yielded only PTH-Cys. (The sequence of each tripeptide, as read from the N-terminal end, is alphabetical if the one-letter designation for amino acids is used.)

What is the amino acid sequence of this nonapeptide?

10. Describe the synthesis of the dipeptide Lys-Ala by Merrifield’s solid phase chemical method of peptide synthesis. What pitfalls might be encountered if you attempted to add a leucine residue to Lys-Ala to make a tripeptide?

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Appendix to Chapter 5

Protein Techniques¹

Size Exclusion Chromatography

Size exclusion chromatography is also known as *gel filtration chromatography* or *molecular sieve chromatography*. In this method, fine, porous beads are packed into a chromatography column. The beads are composed of dextran polymers (*Sephadex*), agarose (*Sephacryl* or *BioGel P*). The pore sizes of these beads approximate the dimensions of macromolecules. The total bed volume (Figure 5A.1) of the packed chromatography column, V_t , is equal to the volume outside the porous beads (V_o) plus the volume inside the beads (V_i) plus the volume actually occupied by the bead material (V_g): $V_t = V_o + V_i + V_g$. (V_g is typically less than 1% of V_t and can be conveniently ignored in most applications.)

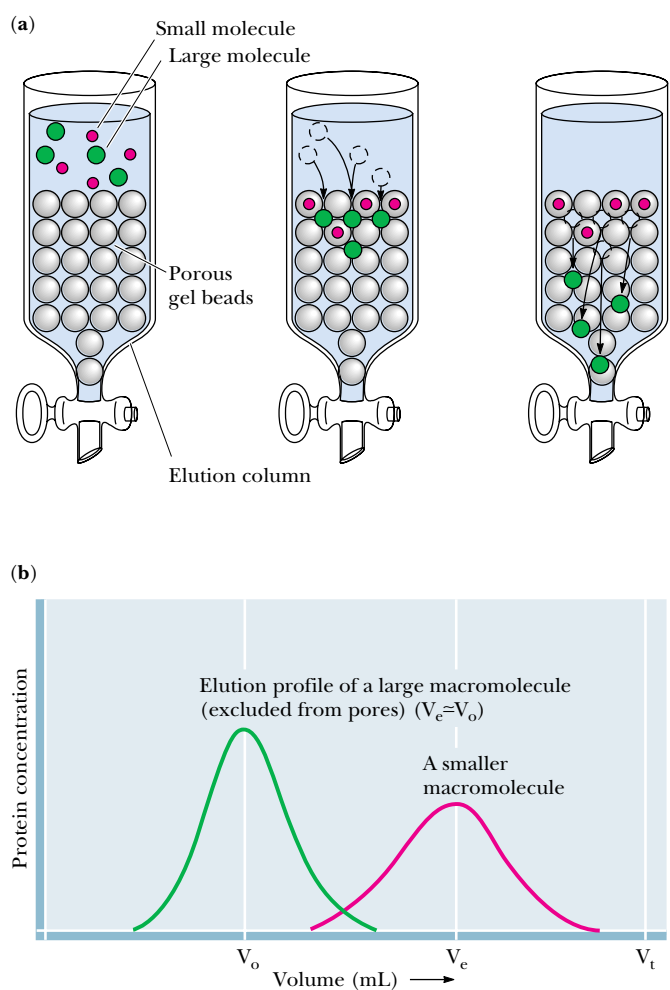


FIGURE 5A.1 • A gel filtration chromatography column. Larger molecules are excluded from the gel beads and emerge from the column sooner than smaller molecules, whose migration is retarded because they can enter the beads.

¹Although this appendix is entitled *Protein Techniques*, these methods are also applicable to other macromolecules such as nucleic acids.

As a solution of molecules is passed through the column, the molecules passively distribute between V_o and V_i , depending on their ability to enter the pores (that is, their size). If a molecule is too large to enter at all, it is totally excluded from V_i and emerges first from the column at an elution volume, V_e , equal to V_o (Figure 5A.1). If a particular molecule can enter the pores in the gel, its distribution is given by the *distribution coefficient*, K_D :

$$K_D = (V_e - V_o) / V_i$$

where V_e is the molecule's characteristic elution volume (Figure 5A.1). The chromatography run is complete when a volume of solvent equal to V_t has passed through the column.

Dialysis and Ultrafiltration

If a solution of protein is separated from a bathing solution by a semipermeable membrane, small molecules and ions can pass through the semipermeable membrane to equilibrate between the protein solution and the bathing solution, called the *dialysis bath* or *dialysate* (Figure 5A.2). This method is useful for removing small molecules from macromolecular solutions or for altering the composition of the protein-containing solution.

Ultrafiltration is an improvement on the dialysis principle. Filters having pore sizes over the range of biomolecular dimensions are used to filter solutions to select for molecules in a particular size range. Because the pore sizes in these filters are microscopic, high pressures are often required to force the solution through the filter. This technique is useful for concentrating dilute solutions of macromolecules. The concentrated protein can then be diluted into the solution of choice.

Electrophoresis

Electrophoretic techniques are based on the movement of ions in an electrical field. An ion of charge q experiences a force F given by $F = Eq/d$, where E is the voltage (or *electrical potential*) and d is the distance between the electrodes. In a vacuum, F would cause the molecule to accelerate. In solution, the molecule experiences *frictional drag*, F_f , due to the solvent:

$$F_f = 6\pi r\eta v$$

where r is the radius of the charged molecule, η is the viscosity of the solution, and v is the velocity at which the charged molecule is moving. So, the velocity of the charged molecule is proportional to its charge q and the voltage E , but inversely proportional to the viscosity of the medium η and d , the distance between the electrodes.

Generally, electrophoresis is carried out *not* in free solution but in a porous support matrix such as polyacrylamide or agarose, which retards the movement of molecules according to their dimensions relative to the size of the pores in the matrix.

SDS-Polyacrylamide Gel Electrophoresis (SDS-PAGE)

SDS is sodium dodecylsulfate (sodium lauryl sulfate) (Figure 5A.3). The hydrophobic tail of dodecylsulfate interacts strongly with polypeptide chains. The number of SDS molecules bound by a polypeptide is proportional to the length (number of amino acid residues) of the polypeptide. Each dodecylsulfate contributes two negative charges. Collectively, these charges overwhelm any intrinsic charge that the protein might have. SDS is also a detergent that

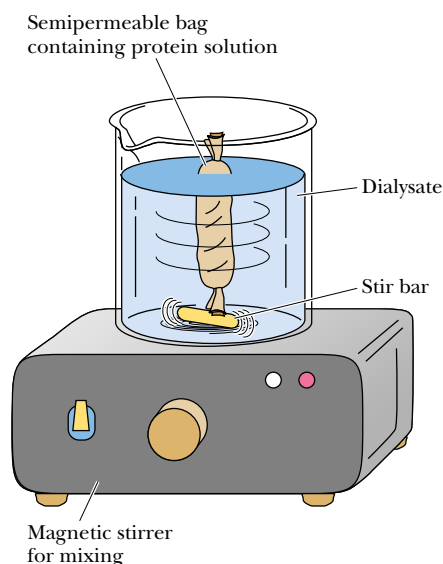


FIGURE 5A.2 • A dialysis experiment. The solution of macromolecules to be dialyzed is placed in a semipermeable membrane bag, and the bag is immersed in a bathing solution. A magnetic stirrer gently mixes the solution to facilitate equilibrium of diffusible solutes between the dialysate and the solution contained in the bag.

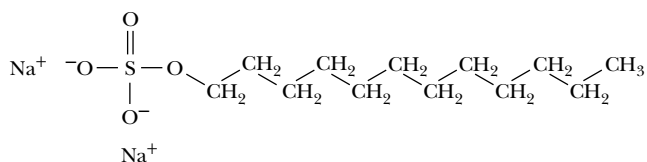


FIGURE 5A.3 • The structure of sodium dodecylsulfate (SDS).

disrupts protein folding (protein 3° structure). SDS-PAGE is usually run in the presence of sulfhydryl-reducing agents such as β -mercaptoethanol so that any disulfide links between polypeptide chains are broken. The electrophoretic mobility of proteins upon SDS-PAGE is inversely proportional to the logarithm of the protein’s molecular weight (Figure 5A.4). SDS-PAGE is often used to determine the molecular weight of a protein.

Isoelectric Focusing

Isoelectric focusing is an electrophoretic technique for separating proteins according to their *isoelectric points* (pIs). A solution of *ampholytes* (amphoteric electrolytes) is first electrophoresed through a gel, usually contained in a small tube. The migration of these substances in an electric field establishes a pH gradient in the tube. Then a protein mixture is applied to the gel and electrophoresis is resumed. As the protein molecules move down the gel, they experience the pH gradient and migrate to a position corresponding to their respective pIs. At its pI, a protein has no net charge and thus moves no farther.

Two-Dimensional Gel Electrophoresis

This separation technique uses isoelectric focusing in one dimension and SDS-PAGE in the second dimension to resolve protein mixtures. The proteins in a mixture are first separated according to pI by isoelectric focusing in a polyacrylamide gel in a tube. The gel is then removed and laid along the top of an SDS-PAGE slab, and the proteins are electrophoresed into the SDS polyacrylamide gel, where they are separated according to size (Figure 5A.5). The gel slab can then be stained to reveal the locations of the individual proteins. Using this powerful technique, researchers have the potential to visualize and construct catalogs of virtually *all* the proteins present in particular cell types. The **ExPASy** Molecular Biology World Wide Web server of the University of Geneva (located at URL: <http://expasy.hcuge.ch/>) provides access to a two-dimensional polyacrylamide gel electrophoresis database named **SWISS-2DPAGE**. This database contains information on proteins, identified as spots on two-dimensional electrophoresis gels, from many different cell and tissue types.

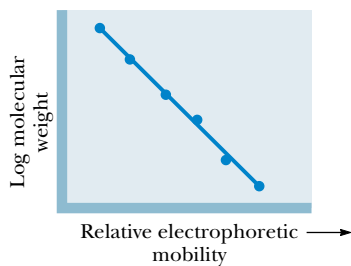


FIGURE 5A.4 • A plot of the relative electrophoretic mobility of proteins in SDS-PAGE versus the log of the molecular weights of the individual polypeptides.

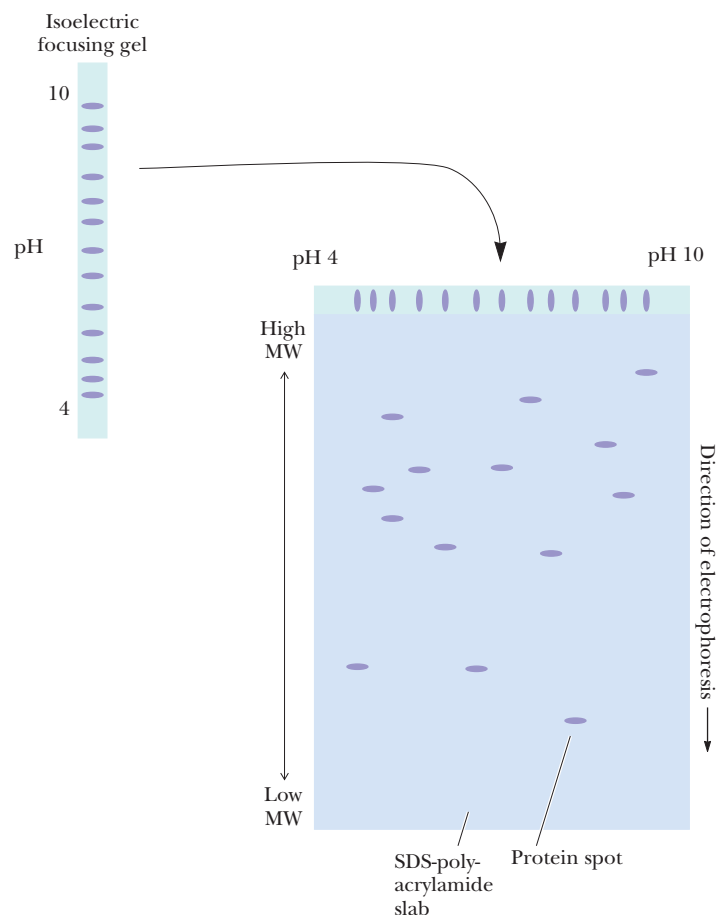


FIGURE 5A.5 • A two-dimensional electrophoresis separation. A mixture of macromolecules is first separated according to charge by isoelectric focusing in a tube gel. The gel containing separated molecules is then placed on top of an SDS-PAGE slab, and the molecules are electrophoresed into the SDS-PAGE gel, where they are separated according to size.

Affinity Chromatography

Recently, *affinity purification* strategies for proteins have been developed to exploit the biological function of the target protein. In most instances, proteins carry out their biological activity through binding or complex formation with specific small biomolecules, or *ligands*, as in the case of an enzyme binding its substrate. If this small molecule can be immobilized through covalent attachment to an insoluble matrix, such as a chromatographic medium like cellulose or polyacrylamide, then the protein of interest, in displaying affinity for its ligand, becomes bound and immobilized itself. It can then be removed from contaminating proteins in the mixture by simple means such as filtration and washing the matrix. Finally, the protein is dissociated or eluted from the matrix by the addition of high concentrations of the free ligand in solution. Figure 5A.6 depicts the protocol for such an *affinity chromatography* scheme. Because this method of purification relies on the biological specificity of the protein of interest, it is a very efficient procedure and proteins can be purified several thousandfold in a single step.

FIGURE 5A.6 • Diagram illustrating affinity chromatography.

Hydrophobic Interaction Chromatography (HIC)

Hydrophobic interaction chromatography (HIC) exploits the hydrophobic nature of proteins in purifying them. Proteins are passed over a chromatographic column packed with a support matrix to which hydrophobic groups are covalently linked. *Phenyl Sepharose*[®], an agarose support matrix to which phenyl groups are affixed, is a prime example of such material. In the presence of high salt concentrations, proteins bind to the phenyl groups by virtue of hydrophobic interactions. Proteins in a mixture can be differentially eluted from the phenyl groups by lowering the salt concentration or by adding solvents such as polyethylene glycol to the elution fluid.

High-Performance Liquid Chromatography

The principles exploited in *high-performance* (or high-pressure) *liquid chromatography (HPLC)* are the same as those used in the common chromatographic methods such as ion exchange chromatography or size exclusion chromatography. Very high resolution separations can be achieved quickly and with high sensitivity in HPLC using automated instrumentation. *Reverse-phase HPLC* is a widely used chromatographic procedure for the separation of nonpolar solutes. In reverse-phase HPLC, a solution of nonpolar solutes is chromatographed on a column having a nonpolar liquid immobilized on an inert matrix; this nonpolar liquid serves as the *stationary phase*. A more polar liquid that serves as the *mobile phase* is passed over the matrix, and solute molecules are eluted in proportion to their solubility in this more polar liquid.

Ultracentrifugation

Centrifugation methods separate macromolecules on the basis of their characteristic densities. Particles tend to “fall” through a solution if the density of the solution is less than the density of the particle. The velocity of the particle through the medium is proportional to the difference in density between the particle and the solution. The tendency of any particle to move through a solution under centrifugal force is given by the *sedimentation coefficient, S*:

$$S = (\rho_p - \rho_m) V / f$$

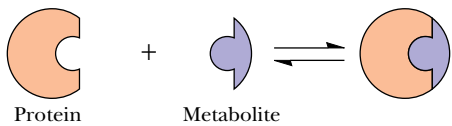
where ρ_p is the density of the particle or macromolecule, ρ_m is the density of the medium or solution, V is the volume of the particle, and f is the frictional coefficient, given by

$$f = F_f / v$$

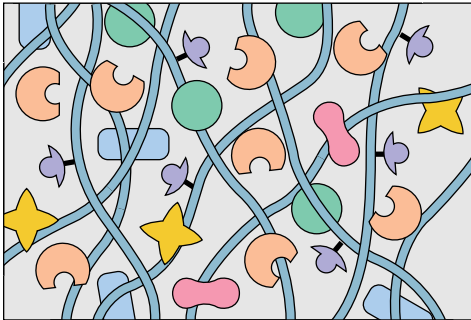
where v is the velocity of the particle and F_f is the frictional drag. Nonspherical molecules have larger frictional coefficients and thus smaller sedimentation coefficients. The smaller the particle and the more its shape deviates from spherical, the more slowly that particle sediments in a centrifuge.

Centrifugation can be used either as a preparative technique for separating and purifying macromolecules and cellular components or as an analytical technique to characterize the hydrodynamic properties of macromolecules such as proteins and nucleic acids.

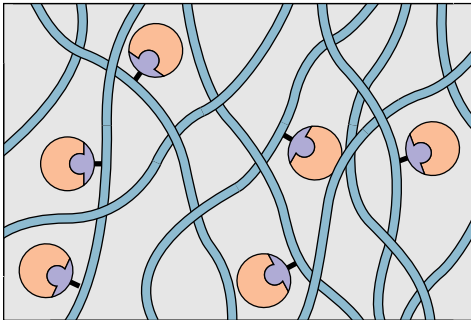
A protein interacts with a metabolite. The metabolite is thus a ligand which binds specifically to this protein



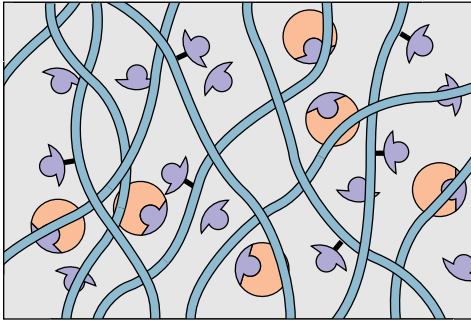
The metabolite can be immobilized by covalently coupling it to an insoluble matrix such as an agarose polymer. Cell extracts containing many individual proteins may be passed through the matrix.



Specific protein binds to ligand. All other unbound material is washed out of the matrix.



Adding an excess of free metabolite that will compete for the bound protein dissociates the protein from the chromatographic matrix. The protein passes out of the column complexed with free metabolite.



Purifications of proteins as much as 1000-fold or more are routinely achieved in a single affinity chromatographic step like this.

Growing in size and complexity
Living things, masses of atoms, DNA,
protein
Dancing a pattern ever more intricate.
Out of the cradle onto the dry land
Here it is standing
Atoms with consciousness
Matter with curiosity.
Stands at the sea
Wonders at wondering
I
A universe of atoms
An atom in the universe.

RICHARD P. FEYNMAN (1918 – 1988)
From “*The Value of Science*” in Edward Hutchings, Jr.
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OUTLINE

- 6.1 • Forces Influencing Protein Structure
- 6.2 • Role of the Amino Acid Sequence in Protein Structure
- 6.3 • Secondary Structure in Proteins
- 6.4 • Protein Folding and Tertiary Structure
- 6.5 • Subunit Interactions and Quaternary Structure

Chapter 6

Proteins: Secondary, Tertiary, and Quaternary Structure



Like the Greek sea god Proteus, who could assume different forms, proteins act through changes in conformation. Proteins (from the Greek proteios, meaning “primary”) are the primary agents of biological function. (“Proteus, Old Man of the Sea, Roman period mosaic, from Thessalonika, 1st century A.D. National Archaeological Museum, Athens/Ancient Art and Architecture Collection Ltd./Bridgeman Art Library, London/New York.)

Nearly all biological processes involve the specialized functions of one or more protein molecules. Proteins function to produce other proteins, control all aspects of cellular metabolism, regulate the movement of various molecular and ionic species across membranes, convert and store cellular energy, and carry out many other activities. Essentially all of the information required to initiate, conduct, and regulate each of these functions must be contained in

the structure of the protein itself. The previous chapter described the details of primary protein structure. However, proteins do not normally exist as fully extended polypeptide chains but rather as compact, folded structures, and the function of a given protein is rarely if ever dependent only on the amino acid sequence. Instead, the ability of a particular protein to carry out its function in nature is normally determined by its overall three-dimensional shape or *conformation*. This native, folded structure of the protein is dictated by several factors: (1) interactions with solvent molecules (normally water), (2) the pH and ionic composition of the solvent, and most important, (3) the sequence of the protein. The first two of these effects are intuitively reasonable, but the third, the role of the amino acid sequence, may not be. In ways that are just now beginning to be understood, the primary structure facilitates the development of short-range interactions among adjacent parts of the sequence and also long-range interactions among distant parts of the sequence. Although the resulting overall structure of the complete protein molecule may at first look like a disorganized and random arrangement, it is in nearly all cases a delicate and sophisticated balance of numerous forces that combine to determine the protein's unique conformation. This chapter considers the details of protein structure and the forces that maintain these structures.

6.1 • Forces Influencing Protein Structure

Several different kinds of noncovalent interactions are of vital importance in protein structure. Hydrogen bonds, hydrophobic interactions, electrostatic bonds, and van der Waals forces are all noncovalent in nature, yet are extremely important influences on protein conformations. The stabilization free energies afforded by each of these interactions may be highly dependent on the local environment within the protein, but certain generalizations can still be made.

Hydrogen Bonds

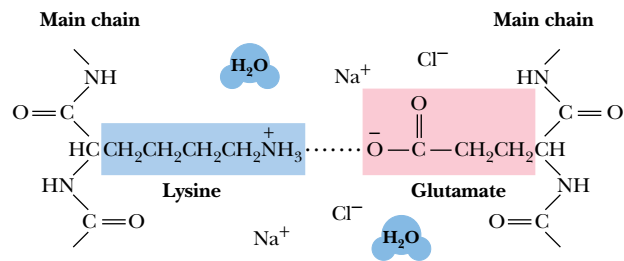
Hydrogen bonds are generally made wherever possible within a given protein structure. In most protein structures that have been examined to date, component atoms of the peptide backbone tend to form hydrogen bonds with one another. Furthermore, side chains capable of forming H bonds are usually located on the protein surface and form such bonds primarily with the water solvent. Although each hydrogen bond may contribute an average of only about 12 kJ/mol in stabilization energy for the protein structure, the number of H-bonds formed in the typical protein is very large. For example, in α -helices, the C=O and N—H groups of every residue participate in H bonds. The importance of H bonds in protein structure cannot be overstated.

Hydrophobic Interactions

Hydrophobic “bonds,” or, more accurately, *interactions*, form because nonpolar side chains of amino acids and other nonpolar solutes prefer to cluster in a nonpolar environment rather than to intercalate in a polar solvent such as water. The forming of hydrophobic bonds minimizes the interaction of nonpolar residues with water and is therefore highly favorable. Such clustering is entropically driven. The side chains of the amino acids in the interior or core of the protein structure are almost exclusively hydrophobic. Polar amino acids are almost never found in the interior of a protein, but the protein surface may consist of both polar and nonpolar residues.

intercalate • to insert between

FIGURE 6.1 • An electrostatic interaction between the ϵ -amino group of a lysine and the γ -carboxyl group of a glutamate residue.



Electrostatic Interactions

Ionic interactions arise either as electrostatic attractions between opposite charges or repulsions between like charges. Chapter 4 discusses the ionization behavior of amino acids. Amino acid side chains can carry positive charges, as in the case of lysine, arginine, and histidine, or negative charges, as in aspartate and glutamate. In addition, the NH₂-terminal and COOH-terminal residues of a protein or peptide chain usually exist in ionized states and carry positive or negative charges, respectively. All of these may experience electrostatic interactions in a protein structure. Charged residues are normally located on the protein surface, where they may interact optimally with the water solvent. It is energetically unfavorable for an ionized residue to be located in the hydrophobic core of the protein. Electrostatic interactions between charged groups on a protein surface are often complicated by the presence of salts in the solution. For example, the ability of a positively charged lysine to attract a nearby negative glutamate may be weakened by dissolved NaCl (Figure 6.1). The Na⁺ and Cl⁻ ions are highly mobile, compact units of charge, compared to the amino acid side chains, and thus compete effectively for charged sites on the protein. In this manner, electrostatic interactions among amino acid residues on protein surfaces may be damped out by high concentrations of salts. Nevertheless, these interactions are important for protein stability.

Van der Waals Interactions

Both attractive forces and repulsive forces are included in van der Waals interactions. The attractive forces are due primarily to instantaneous dipole-induced dipole interactions that arise because of fluctuations in the electron charge distributions of adjacent nonbonded atoms. Individual van der Waals interactions are weak ones (with stabilization energies of 4.0 to 1.2 kJ/mol), but many such interactions occur in a typical protein, and, by sheer force of numbers, they can represent a significant contribution to the stability of a protein. Peter Privalov and George Makhatadze have shown that, for pancreatic ribonuclease A, hen egg white lysozyme, horse heart cytochrome *c*, and sperm whale myoglobin, van der Waals interactions between tightly packed groups in the interior of the protein are a major contribution to protein stability.

6.2 • Role of the Amino Acid Sequence in Protein Structure

It can be inferred from the first section of this chapter that many different forces work together in a delicate balance to determine the overall three-dimensional structure of a protein. These forces operate both within the protein structure itself and between the protein and the water solvent. How, then, does nature dictate the manner of protein folding to generate the three-dimensional

structure that optimizes and balances these many forces? *All of the information necessary for folding the peptide chain into its “native” structure is contained in the amino acid sequence of the peptide.* This principle was first appreciated by C. B. Anfinsen and F. White, whose work in the early 1960s dealt with the chemical denaturation and subsequent renaturation of bovine pancreatic ribonuclease. Ribonuclease was first denatured with urea and mercaptoethanol, a treatment that cleaved the four covalent disulfide (S—S) cross-bridges in the protein. Subsequent air oxidation permitted random formation of disulfide cross-bridges, most of which were incorrect. Thus, the air-oxidized material showed little enzymatic activity. However, treatment of these inactive preparations with small amounts of mercaptoethanol allowed a reshuffling of the disulfide bonds and permitted formation of significant amounts of active native enzyme. In such experiments, the only road map for the protein, that is, the only “instructions” it has, are those directed by its primary structure, the linear sequence of its amino acid residues.

Just how proteins recognize and interpret the information that is stored in the polypeptide sequence is not well understood yet. It may be assumed that certain loci along the peptide chain act as nucleation points, which initiate folding processes that eventually lead to the correct structures. Regardless of how this process operates, it must take the protein correctly to the final native structure, without getting trapped in a local energy-minimum state which, although stable, may be different from the native state itself. A long-range goal of many researchers in the protein structure field is the prediction of three-dimensional conformation from the amino acid sequence. As the details of secondary and tertiary structure are described in this chapter, the complexity and immensity of such a prediction will be more fully appreciated. This area is perhaps the greatest uncharted frontier remaining in molecular biology.

6.3 • Secondary Structure in Proteins

Any discussion of protein folding and structure must begin with the *peptide bond*, the fundamental structural unit in all proteins. As we saw in Chapter 5, the resonance structures experienced by a peptide bond constrain the oxygen, carbon, nitrogen, and hydrogen atoms of the peptide group, as well as the adjacent α -carbons, to all lie in a plane. The resonance stabilization energy of this planar structure is approximately 88 kJ/mol, and substantial energy is required to twist the structure about the C—N bond. A twist of θ degrees involves a twist energy of $88 \sin^2 \theta$ kJ/mol.

Consequences of the Amide Plane

The planarity of the peptide bond means that there are only two degrees of freedom per residue for the peptide chain. Rotation is allowed about the bond linking the α -carbon and the carbon of the peptide bond and also about the bond linking the nitrogen of the peptide bond and the adjacent α -carbon. As shown in Figure 6.2, each α -carbon is the joining point for two planes defined by peptide bonds. The angle about the C_α —N bond is denoted by the Greek letter ϕ (phi) and that about the C_α —C_O is denoted by ψ (psi). For either of these bond angles, a value of 0° corresponds to an orientation with the amide plane bisecting the H—C _{α} —R (sidechain) plane and a *cis* configuration of the main chain around the rotating bond in question (Figure 6.3). In any case, the entire path of the peptide backbone in a protein is known if the ϕ and ψ rotation angles are all specified. Some values of ϕ and ψ are not allowed due to steric interference between nonbonded atoms. As shown in Figure 6.4, values

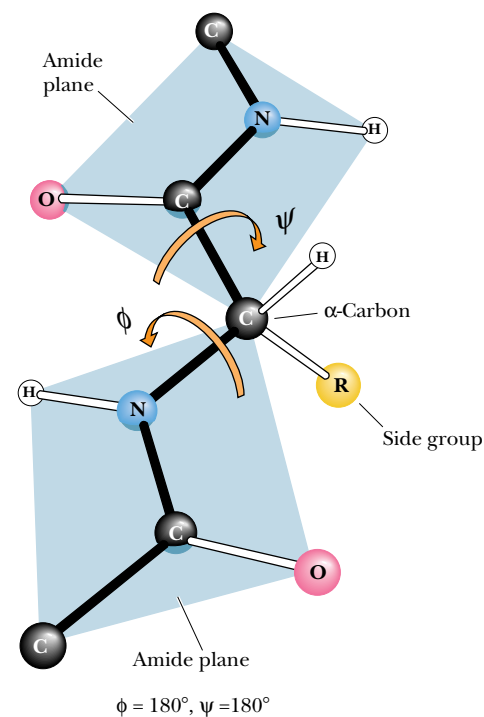


FIGURE 6.2 • The amide or peptide bond planes are joined by the tetrahedral bonds of the α -carbon. The rotation parameters are ϕ and ψ . The conformation shown corresponds to $\phi = 180^\circ$ and $\psi = 180^\circ$. Note that positive values of ϕ and ψ correspond to clockwise rotation as viewed from C _{α} . Starting from 0° , a rotation of 180° in the clockwise direction ($+180^\circ$) is equivalent to a rotation of 180° in the counterclockwise direction (-180°). (Irving Geis)

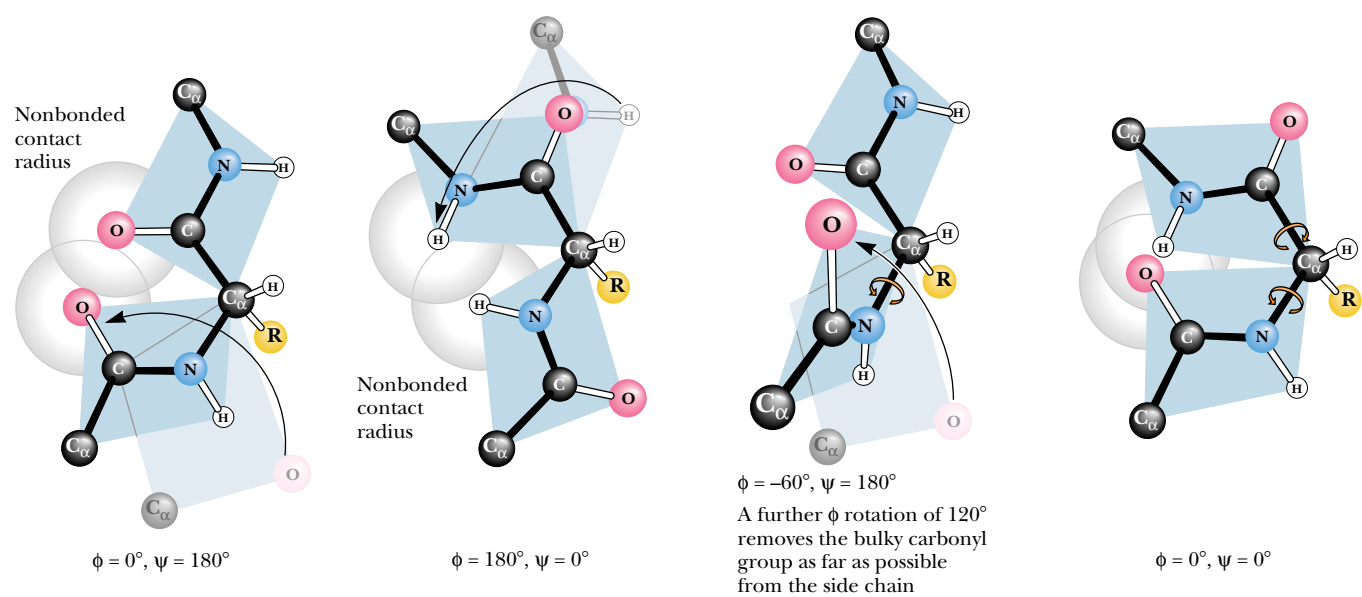


FIGURE 6.3 • Many of the possible conformations about an α -carbon between two peptide planes are forbidden because of steric crowding. Several noteworthy examples are shown here.

Note: The formal IUPAC-IUB Commission on Biochemical Nomenclature convention for the definition of the torsion angles ϕ and ψ in a polypeptide chain (*Biochemistry* 9:3471–3479, 1970) is different from that used here, where the C_α atom serves as the point of reference for both rotations, but the result is the same. (*Irving Geis*)

of $\phi = 180^\circ$ and $\psi = 0^\circ$ are not allowed because of the forbidden overlap of the N—H hydrogens. Similarly, $\phi = 0^\circ$ and $\psi = 180^\circ$ are forbidden because of unfavorable overlap between the carbonyl oxygens.

G. N. Ramachandran and his coworkers in Madras, India, first showed that it was convenient to plot ϕ values against ψ values to show the distribution of allowed values in a protein or in a family of proteins. A typical **Ramachandran plot** is shown in Figure 6.4. Note the clustering of ϕ and ψ values in a few regions of the plot. Most combinations of ϕ and ψ are sterically forbidden, and the corresponding regions of the Ramachandran plot are sparsely populated. The combinations that are sterically allowed represent the subclasses of structure described in the remainder of this section.

The Alpha-Helix

The discussion of hydrogen bonding in Section 6.1 pointed out that the carbonyl oxygen and amide hydrogen of the peptide bond could participate in H bonds either with water molecules in the solvent or with other H-bonding groups in the peptide chain. In nearly all proteins, the carbonyl oxygens and the amide protons of many peptide bonds participate in H bonds that link one peptide group to another, as shown in Figure 6.5. These structures tend to form in cooperative fashion and involve substantial portions of the peptide chain. Structures resulting from these interactions constitute **secondary structure** for proteins (see Chapter 5). When a number of hydrogen bonds form between portions of the peptide chain in this manner, two basic types of structures can result: α -*helices* and β -*pleated sheets*.

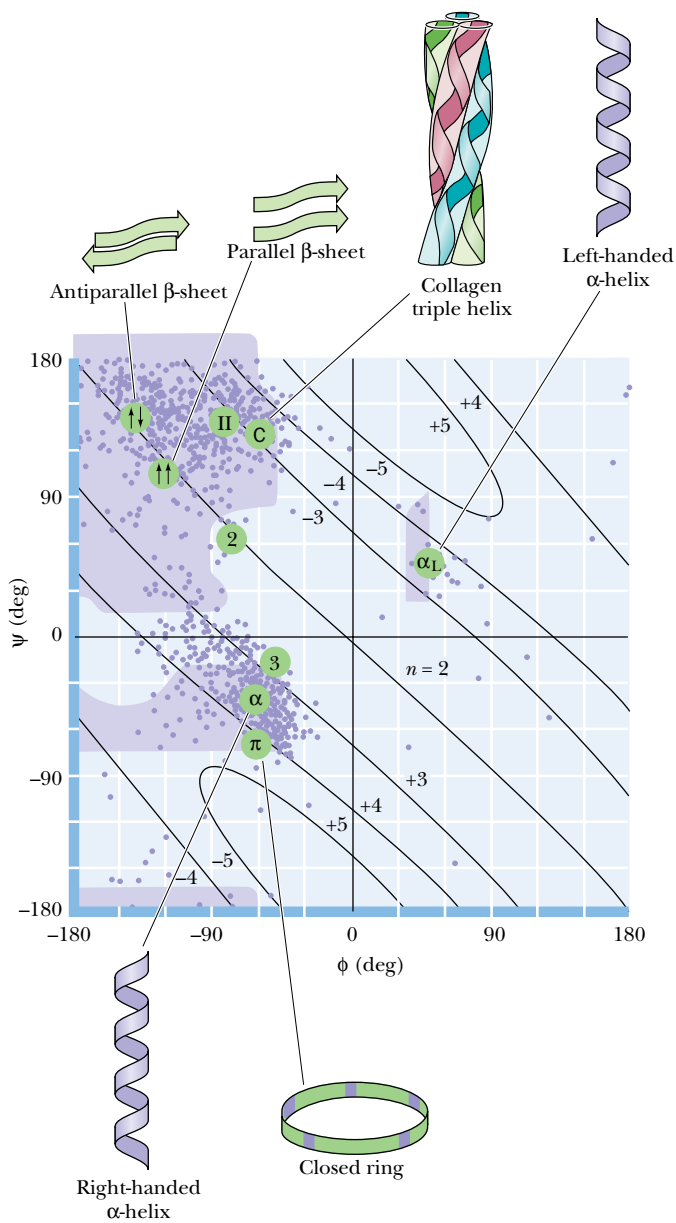
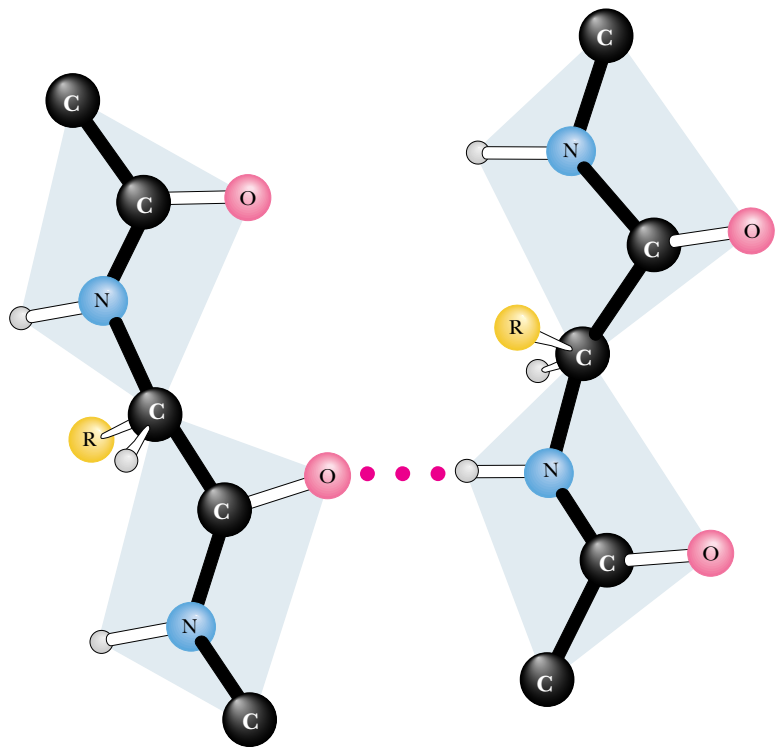


FIGURE 6.4 • A Ramachandran diagram showing the sterically reasonable values of the angles ϕ and ψ . The shaded regions indicate particularly favorable values of these angles. Dots in purple indicate actual angles measured for 1000 residues (excluding glycine, for which a wider range of angles is permitted) in eight proteins. The lines running across the diagram (numbered +5 through 2 and -5 through -3) signify the number of amino acid residues per turn of the helix; “+” means right-handed helices; “-” means left-handed helices. (After Richardson, J. S., 1981, *Advances in Protein Chemistry* 34:167–339.)

FIGURE 6.5 • A hydrogen bond between the amide proton and carbonyl oxygen of adjacent peptide groups.



DEEPER LOOK

Knowing What the Right Hand and Left Hand Are Doing

Certain conventions related to peptide bond angles and the “handedness” of biological structures are useful in any discussion of protein structure. To determine the ϕ and ψ angles between peptide planes, viewers should imagine themselves at the C_α carbon looking outward and should imagine starting from the $\phi = 0^\circ$, $\psi = 0^\circ$ conformation. From this perspective, positive values of ϕ correspond to clockwise rotations about the C_α —N bond of the plane that includes the adjacent N—H group. Similarly, positive values of ψ correspond to clockwise

rotations about the C_α —C bond of the plane that includes the adjacent C=O group.

Biological structures are often said to exhibit “right-hand” or “left-hand” twists. For all such structures, the sense of the twist can be ascertained by holding the structure in front of you and looking along the polymer backbone. If the twist is clockwise as one proceeds outward and through the structure, it is said to be right-handed. If the twist is counterclockwise, it is said to be left-handed.

Evidence for helical structures in proteins was first obtained in the 1930s in studies of fibrous proteins. However, there was little agreement at that time about the exact structure of these helices, primarily because there was also lack of agreement about interatomic distances and bond angles in peptides. In 1951, Linus Pauling, Robert Corey, and their colleagues at the California Institute of Technology summarized a large volume of crystallographic data in a set of dimensions for polypeptide chains. (A summary of data similar to what they reported is shown in Figure 5.2.) With these data in hand, Pauling, Corey, and their colleagues proposed a new model for a helical structure in proteins, which they called the **α -helix**. The report from Caltech was of particular interest to Max Perutz in Cambridge, England, a crystallographer who was also interested in protein structure. By taking into account a critical but previously ignored feature of the X-ray data, Perutz realized that the α -helix existed in keratin, a protein from hair, and also in several other proteins. Since then, the α -helix has proved to be a fundamentally important peptide structure. Several representations of the α -helix are shown in Figure 6.6. One turn of the helix represents 3.6 amino acid residues. (A single turn of the α -helix involves 13 atoms from the O to the H of the H bond. For this reason, the α -helix is sometimes referred to as the 3.6_{13} helix.) This is in fact the feature that most confused crystallographers before the Pauling and Corey α -helix model. Crystallographers were so accustomed to finding twofold, threefold, sixfold, and similar integral axes in simpler molecules that the notion of a nonintegral number of units per turn was never taken seriously before Pauling and Corey’s work.

Each amino acid residue extends **1.5 Å (0.15 nm)** along the helix axis. With **3.6 residues per turn**, this amounts to $3.6 \times 1.5 \text{ Å}$ or **5.4 Å (0.54 nm)** of travel along the helix axis per turn. This is referred to as the translation distance or the **pitch** of the helix. If one ignores side chains, the helix is about 6 Å in diameter. The side chains, extending outward from the core structure of the helix, are removed from steric interference with the polypeptide backbone. As can be seen in Figure 6.6, *each peptide carbonyl is hydrogen bonded to the peptide N—H group four residues farther up the chain*. Note that all of the H bonds lie parallel to the helix axis and that all of the carbonyl groups are pointing in one direction along the helix axis while the N—H groups are pointing in the opposite direction. Recall that the entire path of the peptide backbone can be known if the ϕ and ψ twist angles are specified for each residue. The α -helix is formed if the values of ϕ are approximately -60° and the values of ψ are in the range of -45 to -50° . Figure 6.7 shows the structures of two proteins that contain α -helical segments. The number of residues involved in a given α -helix

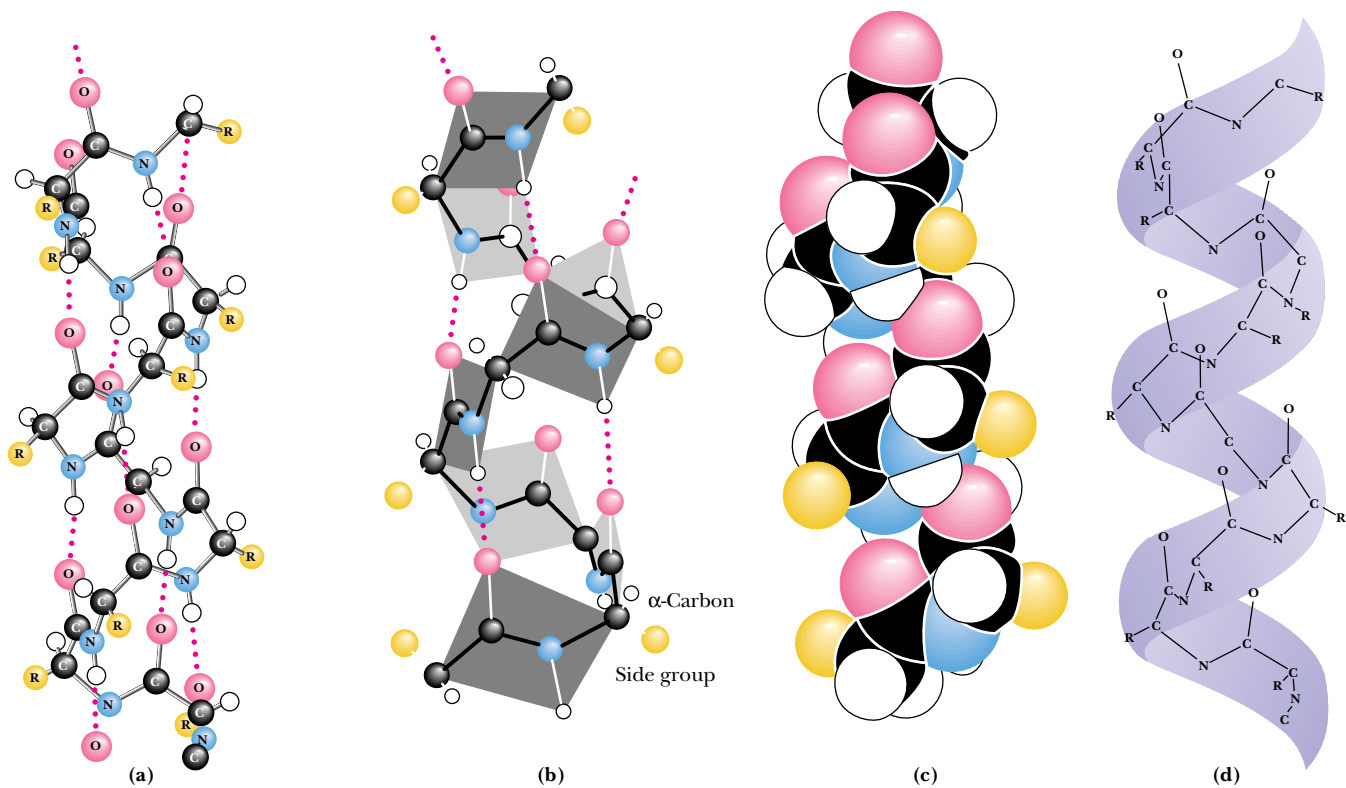


FIGURE 6.6 • Four different graphic representations of the α -helix. (a) As it originally appeared in Pauling’s 1960 *The Nature of the Chemical Bond*. (b) Showing the arrangement of peptide planes in the helix. (c) A space-filling computer graphic presentation. (d) A “ribbon structure” with an inlaid stick figure, showing how the ribbon indicates the path of the polypeptide backbone. (*Irving Geis*)

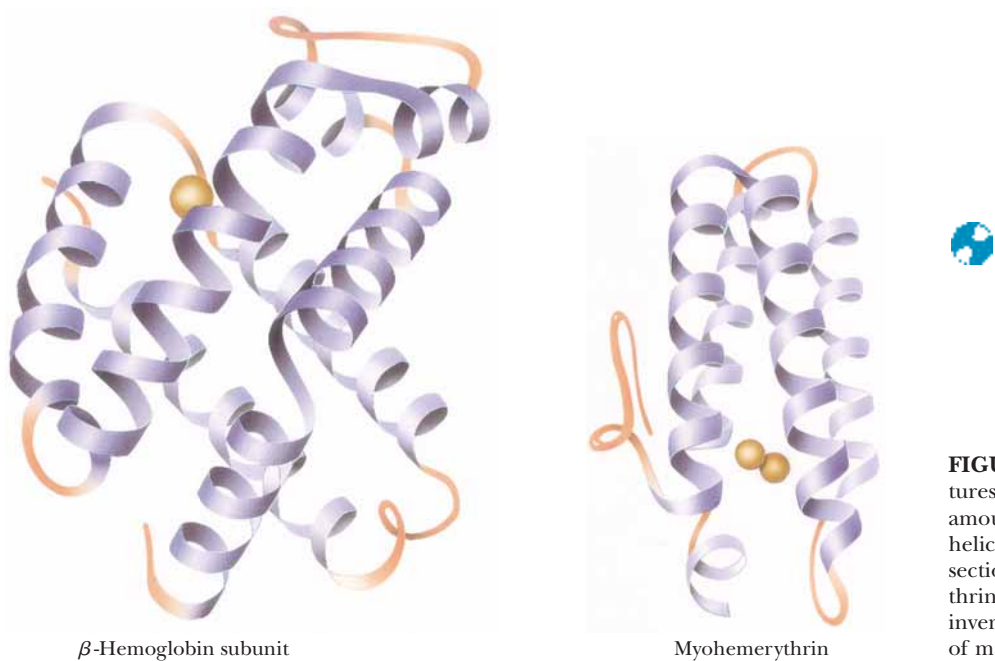
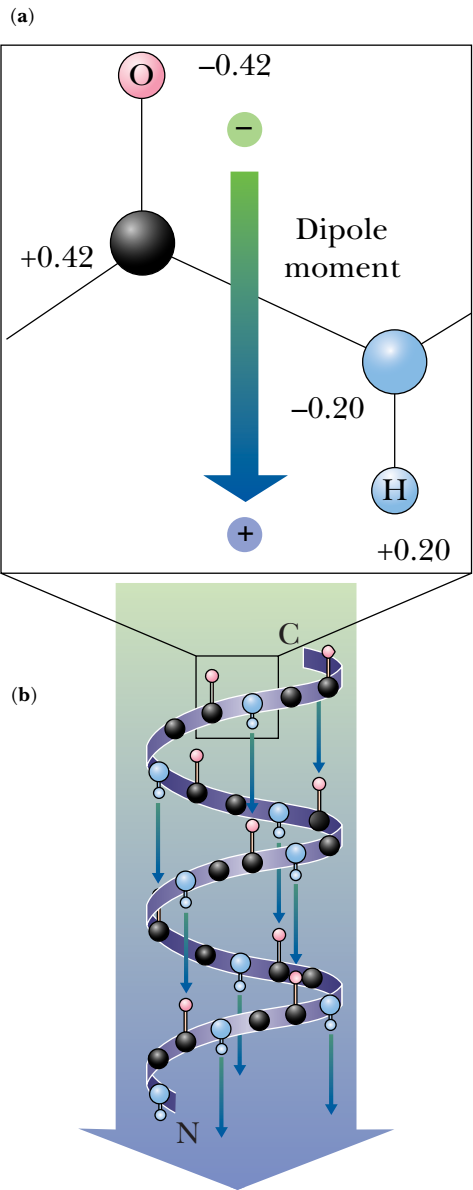


FIGURE 6.7 • The three-dimensional structures of two proteins that contain substantial amounts of α -helix in their structures. The helices are represented by the regularly coiled sections of the ribbon drawings. Myohemerythrin is the oxygen-carrying protein in certain invertebrates, including *Sipunculids*, a phylum of marine worm. (*Jane Richardson*)

FIGURE 6.8 • The arrangement of N—H and C=O groups (each with an individual dipole moment) along the helix axis creates a large net dipole for the helix. Numbers indicate fractional charges on respective atoms.



varies from helix to helix and from protein to protein. On average, there are about 10 residues per helix. Myoglobin, one of the first proteins in which α -helices were observed, has eight stretches of α -helix that form a box to contain the heme prosthetic group. The structures of the α and β subunits of hemoglobin are strikingly similar, with only a few differences at the C- and N-termini and on the surfaces of the structure that contact or interact with the other subunits of this multisubunit protein.

As shown in Figure 6.6, all of the hydrogen bonds point in the same direction along the α -helix axis. Each peptide bond possesses a dipole moment that arises from the polarities of the N—H and C=O groups, and, because these groups are all aligned along the helix axis, the helix itself has a substantial dipole moment, with a partial positive charge at the N-terminus and a partial negative charge at the C-terminus (Figure 6.8). Negatively charged ligands (e.g., phosphates) frequently bind to proteins near the N-terminus of an α -helix. By contrast, positively charged ligands are only rarely found to bind near the C-terminus of an α -helix.

In a typical α -helix of 12 (or n) residues, there are 8 (or $n - 4$) hydrogen bonds. As shown in Figure 6.9, the first 4 amide hydrogens and the last 4 carbonyl oxygens cannot participate in helix H-bonds. Also, nonpolar residues sit-

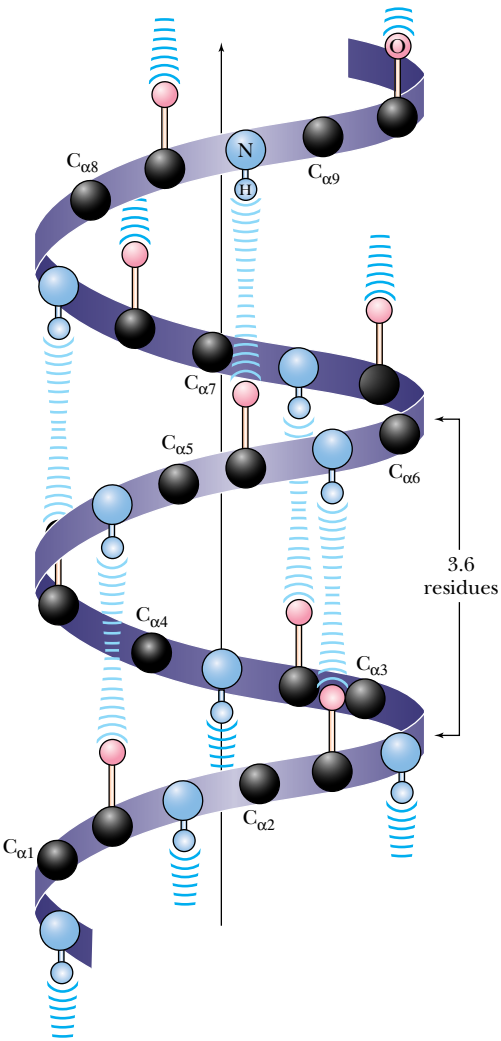


FIGURE 6.9 • Four N—H groups at the N-terminal end of an α -helix and four C=O groups at the C-terminal end cannot participate in hydrogen bonding. The formation of H-bonds with other nearby donor and acceptor groups is referred to as **helix capping**. Capping may also involve appropriate hydrophobic interactions that accommodate nonpolar side chains at the ends of helical segments.

uated near the helix termini can be exposed to solvent. Proteins frequently compensate for these problems by **helix capping**—providing H-bond partners for the otherwise bare N—H and C=O groups and folding other parts of the protein to foster hydrophobic contacts with exposed nonpolar residues at the helix termini.

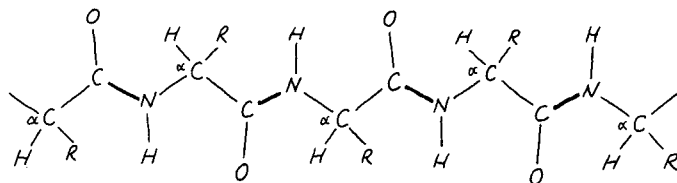
Careful studies of the **polyamino acids**, polymers in which all the amino acids are identical, have shown that certain amino acids tend to occur in α -helices, whereas others are less likely to be found in them. Polyisoleucine and polyalanine, for example, readily form α -helical structures. In contrast, polyaspartic acid and polyglutamic acid, which are highly negatively charged at pH 7.0, form only random structures because of strong charge repulsion between the R groups along the peptide chain. At pH 1.5 to 2.5, however, where the side chains are protonated and thus uncharged, these latter species spontaneously form α -helical structures. In similar fashion, polylysine is a random coil at pH values below about 11, where repulsion of positive charges prevents helix formation. At pH 12, where polylysine is a neutral peptide chain, it readily forms an α -helix.

CRITICAL DEVELOPMENTS IN BIOCHEMISTRY

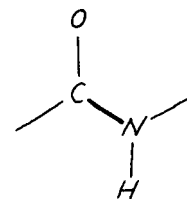
In Bed with a Cold, Pauling Stumbles onto the α -Helix and a Nobel Prize¹

As high technology continues to transform the modern biochemical laboratory, it is interesting to reflect on Linus Pauling's discovery of the α -helix. It involved only a piece of paper, a pencil, scissors, and a sick Linus Pauling, who had tired of reading detective novels. The story is told in the excellent book *The Eighth Day of Creation* by Horace Freeland Judson:

From the spring of 1948 through the spring of 1951 . . . rivalry sputtered and blazed between Pauling's lab and (Sir Lawrence) Bragg's — over protein. The prize was to propose and verify in nature a general three-dimensional structure for the polypeptide chain. Pauling was working up from the simpler structures of components. In January 1948, he went to Oxford as a visiting professor for two terms, to lecture on the chemical bond and on molecular structure and biological specificity. "In Oxford, it was April, I believe, I caught cold. I went to bed, and read detective stories for a day, and got bored, and thought why don't I have a crack at that problem of alpha keratin." Confined, and still fingering the polypeptide chain in his mind, Pauling called for paper, pencil, and straightedge and attempted to reduce the problem to an almost Euclidean purity. "I took a sheet of paper — I still have this sheet of paper — and drew, rather roughly, the way that I thought a polypeptide chain would look if it were spread out into a plane." The repetitious herringbone of the chain he could stretch across the paper as simply as this —



— putting in lengths and bond angles from memory. . . . He knew that the peptide bond, at the carbon-to-nitrogen link, was always rigid:



And this meant that the chain could turn corners only at the alpha carbons. . . . "I creased the paper in parallel creases through the alpha carbon atoms, along the chain, have tetrahedral value. And then I looked to see if I could form hydrogen bonds from one part of the chain to the next." He saw that if he folded the strip like a chain of paper dolls into a helix, and if he got the pitch of the screw right, hydrogen bonds could be shown to form, N—H $\cdots$ O—C, three or four knuckles apart along the backbone, holding the helix in shape. After several tries, changing the angle of the parallel creases in order to adjust the pitch of the helix, he found one where the hydrogen bonds would drop into place, connecting the turns, as straight lines of the right length. He had a model.

¹The discovery of the α -helix structure was only one of many achievements that led to Pauling's Nobel Prize in chemistry in 1954. The official citation for the prize was "for his research into the nature of the chemical bond and its application to the elucidation of the structure of complex substances."

Table 6.1

Helix-Forming and Helix-Breaking Behavior of the Amino Acids

Amino Acid		Helix Behavior*	
A	Ala	H	(I)
C	Cys	Variable	
D	Asp	Variable	
E	Glu	H	
F	Phe	H	
G	Gly	I	(B)
H	His	H	(I)
I	Ile	H	(C)
K	Lys	Variable	
L	Leu	H	
M	Met	H	
N	Asn	C	(I)
P	Pro	B	
Q	Gln	H	(I)
R	Arg	H	(I)
S	Ser	C	(B)
T	Thr	Variable	
V	Val	Variable	
W	Trp	H	(C)
Y	Tyr	H	(C)

*H = helix former; I = indifferent; B = helix breaker; C = random coil; () = secondary tendency.

The tendencies of the amino acids to stabilize or destabilize α -helices are different in typical proteins than in polyamino acids. The occurrence of the common amino acids in helices is summarized in Table 6.1. Notably, proline (and hydroxyproline) act as helix breakers due to their unique structure, which fixes the value of the $C_{\alpha}-N-C$ bond angle. Helices can be formed from either D- or L-amino acids, but a given helix must be composed entirely of amino acids of one configuration. α -Helices cannot be formed from a mixed copolymer of D- and L-amino acids. An α -helix composed of D-amino acids is left-handed.

Other Helical Structures

There are several other far less common types of helices found in proteins. The most common of these is the 3_{10} helix, which contains 3.0 residues per turn (with 10 atoms in the ring formed by making the hydrogen bond three residues up the chain). It normally extends over shorter stretches of sequence than the α -helix. Other helical structures include the 2_7 ribbon and the π -helix, which has 4.4 residues and 16 atoms per turn and is thus called the 4.4_{16} helix.

The Beta-Pleated Sheet

Another type of structure commonly observed in proteins also forms because of local, cooperative formation of hydrogen bonds. That is the pleated sheet, or β -structure, often called the **β -pleated sheet**. This structure was also first postulated by Pauling and Corey in 1951 and has now been observed in many natural proteins. A β -pleated sheet can be visualized by laying thin, pleated strips of paper side by side to make a “pleated sheet” of paper (Figure 6.10). Each strip of paper can then be pictured as a single peptide strand in which the peptide backbone makes a zigzag pattern along the strip, with the α -carbons lying at the folds of the pleats. The pleated sheet can exist in both par-

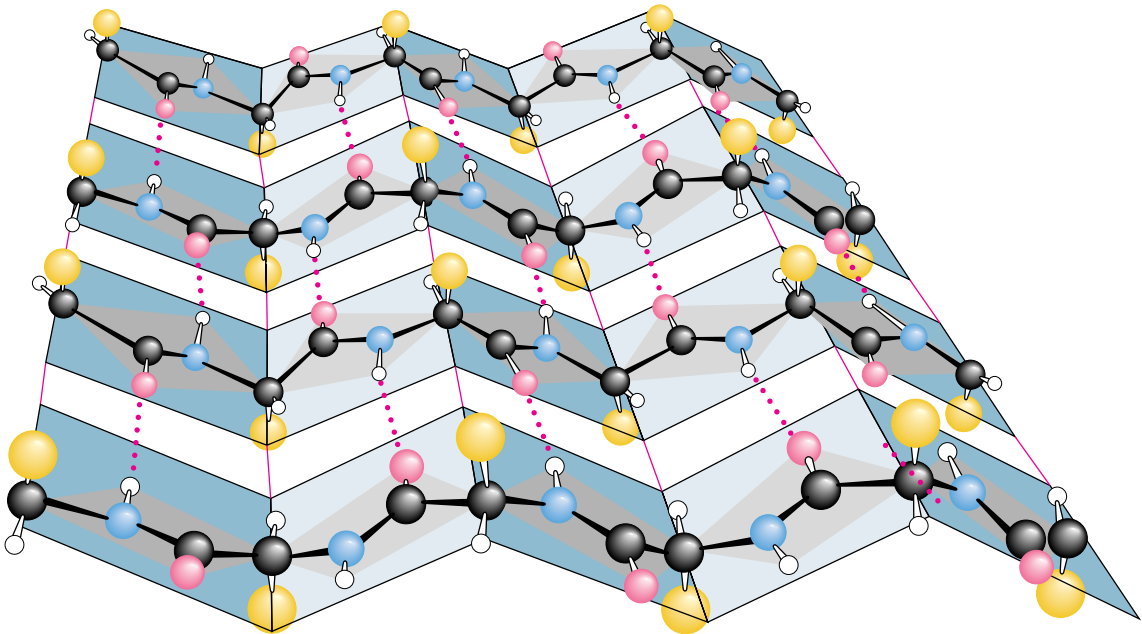


FIGURE 6.10 • A “pleated sheet” of paper with an antiparallel β -sheet drawn on it. (Irving Geis)

allel and antiparallel forms. In the **parallel β -pleated sheet**, adjacent chains run in the same direction ($N \rightarrow C$ or $C \rightarrow N$). In the **antiparallel β -pleated sheet**, adjacent strands run in opposite directions.

Each single strand of the β -sheet structure can be pictured as a twofold helix, that is, a helix with two residues per turn. The arrangement of successive amide planes has a pleated appearance due to the tetrahedral nature of the C_α atom. It is important to note that the hydrogen bonds in this structure are essentially *interstrand* rather than *intrastrand*. The peptide backbone in the β -sheet is in its most extended conformation (sometimes called the **ϵ -conformation**). The optimum formation of H bonds in the parallel pleated sheet results in a slightly less extended conformation than in the antiparallel sheet. The H bonds thus formed in the parallel β -sheet are bent significantly. The distance between residues is 0.347 nm for the antiparallel pleated sheet, but only 0.325 nm for the parallel pleated sheet. Figure 6.11 shows examples of both parallel and antiparallel β -pleated sheets. Note that the side chains in the pleated sheet are oriented perpendicular or normal to the plane of the sheet, extending out from the plane on alternating sides.

Parallel β -sheets tend to be more regular than antiparallel β -sheets. The range of ϕ and ψ angles for the peptide bonds in parallel sheets is much smaller than that for antiparallel sheets. Parallel sheets are typically large structures; those composed of less than five strands are rare. Antiparallel sheets, however, may consist of as few as two strands. Parallel sheets characteristically distribute

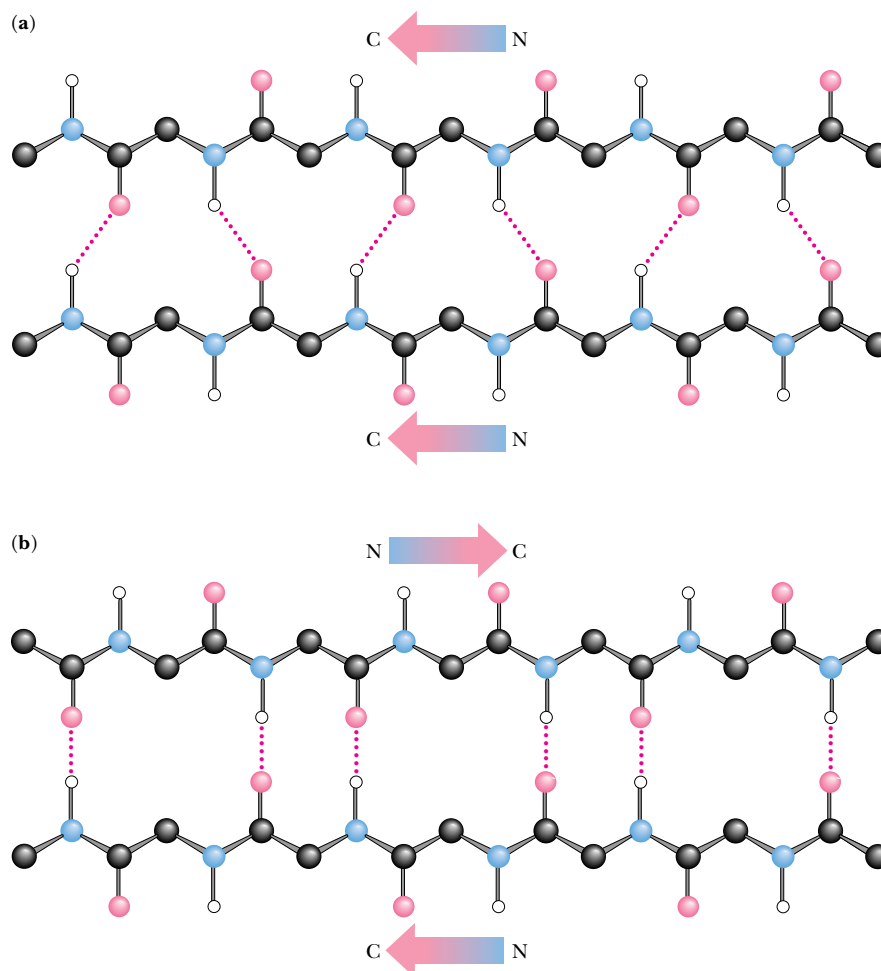


FIGURE 6.11 • The arrangement of hydrogen bonds in (a) parallel and (b) antiparallel β -pleated sheets.

hydrophobic side chains on both sides of the sheet, while antiparallel sheets are usually arranged with all their hydrophobic residues on one side of the sheet. This requires an alternation of hydrophilic and hydrophobic residues in the primary structure of peptides involved in antiparallel β -sheets because alternate side chains project to the same side of the sheet (see Figure 6.10).

Antiparallel pleated sheets are the fundamental structure found in silk, with the polypeptide chains forming the sheets running parallel to the silk fibers. The silk fibers thus formed have properties consistent with those of the β -sheets that form them. They are quite flexible but cannot be stretched or extended to any appreciable degree. Antiparallel structures are also observed in many other proteins, including immunoglobulin G, superoxide dismutase from bovine erythrocytes, and concanavalin A. Many proteins, including carbonic anhydrase, egg lysozyme, and glyceraldehyde phosphate dehydrogenase, possess both α -helices and β -pleated sheet structures within a single polypeptide chain.

The Beta-Turn

Most proteins are globular structures. The polypeptide chain must therefore possess the capacity to bend, turn, and reorient itself to produce the required compact, globular structures. A simple structure observed in many proteins is the **β -turn** (also known as the *tight turn* or *β -bend*), in which the peptide chain forms a tight loop with the carbonyl oxygen of one residue hydrogen-bonded with the amide proton of the residue three positions down the chain. This H bond makes the β -turn a relatively stable structure. As shown in Figure 6.12, the β -turn allows the protein to reverse the direction of its peptide chain. This figure shows the two major types of β -turns, but a number of less common types are also found in protein structures. Certain amino acids, such as proline and glycine, occur frequently in β -turn sequences, and the particular conformation of the β -turn sequence depends to some extent on the amino acids composing it. Due to the absence of a side chain, glycine is sterically the most adaptable of the amino acids, and it accommodates conveniently to other steric constraints in the β -turn. Proline, however, has a cyclic structure and a fixed ϕ angle, so, to some extent, it forces the formation of a β -turn, and in many cases this facilitates the turning of a polypeptide chain upon itself. Such bends promote formation of antiparallel β -pleated sheets.

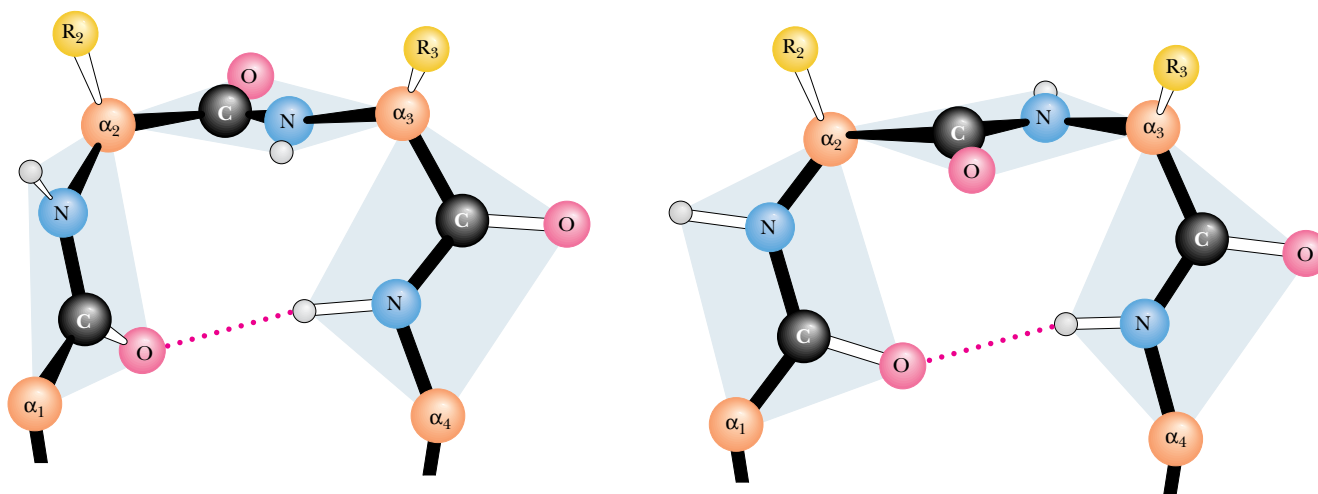


FIGURE 6.12 • The structures of two kinds of β -turns (also called tight turns or β -bends) (Irving Geis)

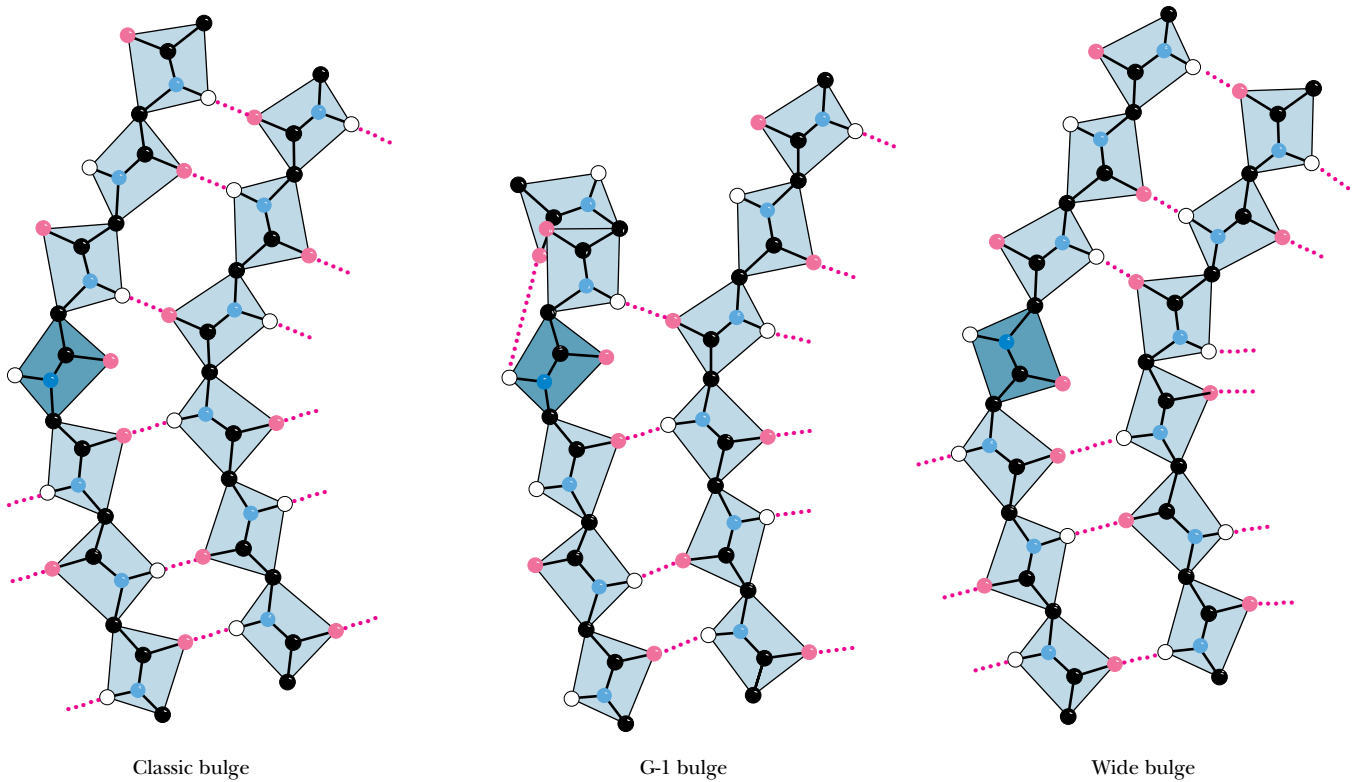


FIGURE 6.13 • Three different kinds of β -bulge structures involving a pair of adjacent polypeptide chains. (Adapted from Richardson, J. S., 1981. *Advances in Protein Chemistry* 34:167–339.)

The Beta-Bulge

One final secondary structure, the **β -bulge**, is a small piece of nonrepetitive structure that can occur by itself, but most often occurs as an irregularity in antiparallel β -structures. A β -bulge occurs between two normal β -structure hydrogen bonds and comprises two residues on one strand and one residue on the opposite strand. Figure 6.13 illustrates typical β -bulges. The extra residue on the longer side, which causes additional backbone length, is accommodated partially by creating a bulge in the longer strand and partially by forcing a slight bend in the β -sheet. Bulges thus cause changes in the direction of the polypeptide chain, but to a lesser degree than tight turns do. Over 100 examples of β -bulges are known in protein structures.

The secondary structures we have described here are all found commonly in proteins in nature. In fact, it is hard to find proteins that do not contain one or more of these structures. The energetic (mostly H-bond) stabilization afforded by α -helices, β -pleated sheets, and β -turns is important to proteins, and they seize the opportunity to form such structures wherever possible.

6.4 • Protein Folding and Tertiary Structure

The folding of a single polypeptide chain in three-dimensional space is referred to as its **tertiary structure**. As discussed in Section 6.2, all of the information needed to fold the protein into its native tertiary structure is contained within the primary structure of the peptide chain itself. With this in mind, it was disappointing to the biochemists of the 1950s when the early protein structures did not reveal the governing principles in any particular detail. It soon became apparent that the proteins knew how they were supposed to fold into tertiary

shapes, even if the biochemists did not. Vigorous work in many laboratories has slowly brought important principles to light.

First, secondary structures—helices and sheets—form whenever possible as a consequence of the formation of large numbers of hydrogen bonds. Second, α -helices and β -sheets often associate and pack close together in the protein. No protein is stable as a single-layer structure, for reasons that become apparent later. There are a few common methods for such packing to occur. Third, because the peptide segments between secondary structures in the protein tend to be short and direct, the peptide does not execute complicated twists and knots as it moves from one region of a secondary structure to another. A consequence of these three principles is that protein chains are usually folded so that the secondary structures are arranged in one of a few common patterns. For this reason, there are families of proteins that have similar tertiary structure, with little apparent evolutionary or functional relationship among them. Finally, proteins generally fold so as to form the most stable structures possible. The stability of most proteins arises from (1) the formation of large numbers of intramolecular hydrogen bonds and (2) the reduction in the surface area accessible to solvent that occurs upon folding.

Fibrous Proteins

In Chapter 5, we saw that proteins can be grouped into three large classes based on their structure and solubility: *fibrous proteins*, *globular proteins*, and *membrane proteins*. Fibrous proteins contain polypeptide chains organized approximately parallel along a single axis, producing long fibers or large sheets. Such proteins tend to be mechanically strong and resistant to solubilization in water and dilute salt solutions. Fibrous proteins often play a structural role in nature (see Chapter 5).

α -Keratin

As their name suggests, the structure of the **α -keratins** is dominated by α -helical segments of polypeptide. The amino acid sequence of α -keratin subunits is composed of central α -helix-rich rod domains about 311 to 314 residues in length, flanked by nonhelical N- and C-terminal domains of varying size and composition (Figure 6.14a). The structure of the central rod domain of a typical α -keratin is shown in Figure 6.14b. It consists of four helical strands arranged as twisted pairs of two-stranded **coiled coils**. X-ray diffraction patterns show that these structures resemble α -helices, but with a pitch of 0.51 nm rather than the expected 0.54 nm. This is consistent with a tilt of the helix relative to the long axis of the fiber, as in the two-stranded “rope” in Figure 6.14.

The primary structure of the central rod segments of α -keratin consists of quasi-repeating seven-residue segments of the form $(a-b-c-d-e-f-g)_n$. These units are not true repeats, but residues *a* and *d* are usually nonpolar amino acids. In α -helices, with 3.6 residues per turn, these nonpolar residues are arranged in an inclined row or stripe that twists around the helix axis. These nonpolar residues would make the helix highly unstable if they were exposed to solvent, but the association of hydrophobic strips on two coiled coils to form the two-stranded rope effectively buries the hydrophobic residues and forms a highly stable structure (Figure 6.14). The helices clearly sacrifice some stability in assuming this twisted conformation, but they gain stabilization energy from the packing of side chains between the helices. In other forms of keratin, covalent disulfide bonds form between cysteine residues of adjacent molecules, making the overall structure rigid, inextensible, and insoluble—important properties for structures such as claws, fingernails, hair, and horns in animals. How and

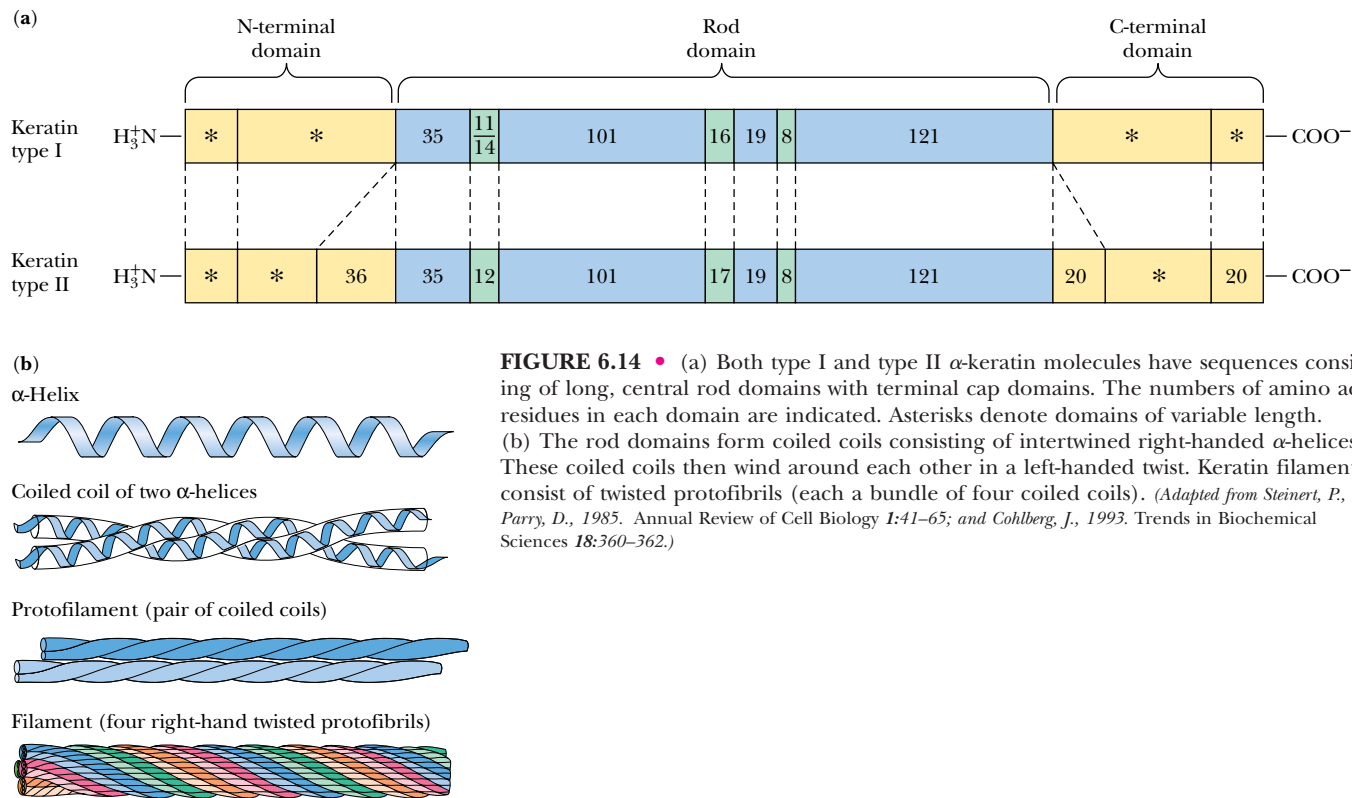


FIGURE 6.14 • (a) Both type I and type II α-keratin molecules have sequences consisting of long, central rod domains with terminal cap domains. The numbers of amino acid residues in each domain are indicated. Asterisks denote domains of variable length. (b) The rod domains form coiled coils consisting of intertwined right-handed α-helices. These coiled coils then wind around each other in a left-handed twist. Keratin filaments consist of twisted protofibrils (each a bundle of four coiled coils). (Adapted from Steinert, P., and Parry, D., 1985. Annual Review of Cell Biology 1:41–65; and Cohlberg, J., 1993. Trends in Biochemical Sciences 18:360–362.)

where these disulfides form determines the amount of curling in hair and wool fibers. When a hairstylist creates a permanent wave (simply called a “permanent”) in a hair salon, disulfides in the hair are first reduced and cleaved, then reorganized and reoxidized to change the degree of curl or wave. In contrast, a “set” that is created by wetting the hair, setting it with curlers, and then drying it represents merely a rearrangement of the hydrogen bonds between helices and between fibers. (On humid or rainy days, the hydrogen bonds in curled hair may rearrange, and the hair becomes “frizzy.”)

Fibroin and β-Keratin: β-Sheet Proteins

The **fibroin** proteins found in silk fibers represent another type of fibrous protein. These are composed of stacked antiparallel β-sheets, as shown in Figure 6.15. In the polypeptide sequence of silk proteins, there are large stretches in which every other residue is a glycine. As previously mentioned, the residues of a β-sheet extend alternately above and below the plane of the sheet. As a result, the glycines all end up on one side of the sheet and the other residues (mainly alanines and serines) compose the opposite surface of the sheet. Pairs of β-sheets can then pack snugly together (glycine surface to glycine surface or alanine–serine surface to alanine–serine surface). The β-keratins found in bird feathers are also made up of stacked β-sheets.

Collagen: A Triple Helix

Collagen is a rigid, inextensible fibrous protein that is a principal constituent of connective tissue in animals, including tendons, cartilage, bones, teeth, skin, and blood vessels. The high tensile strength of collagen fibers in these struc-

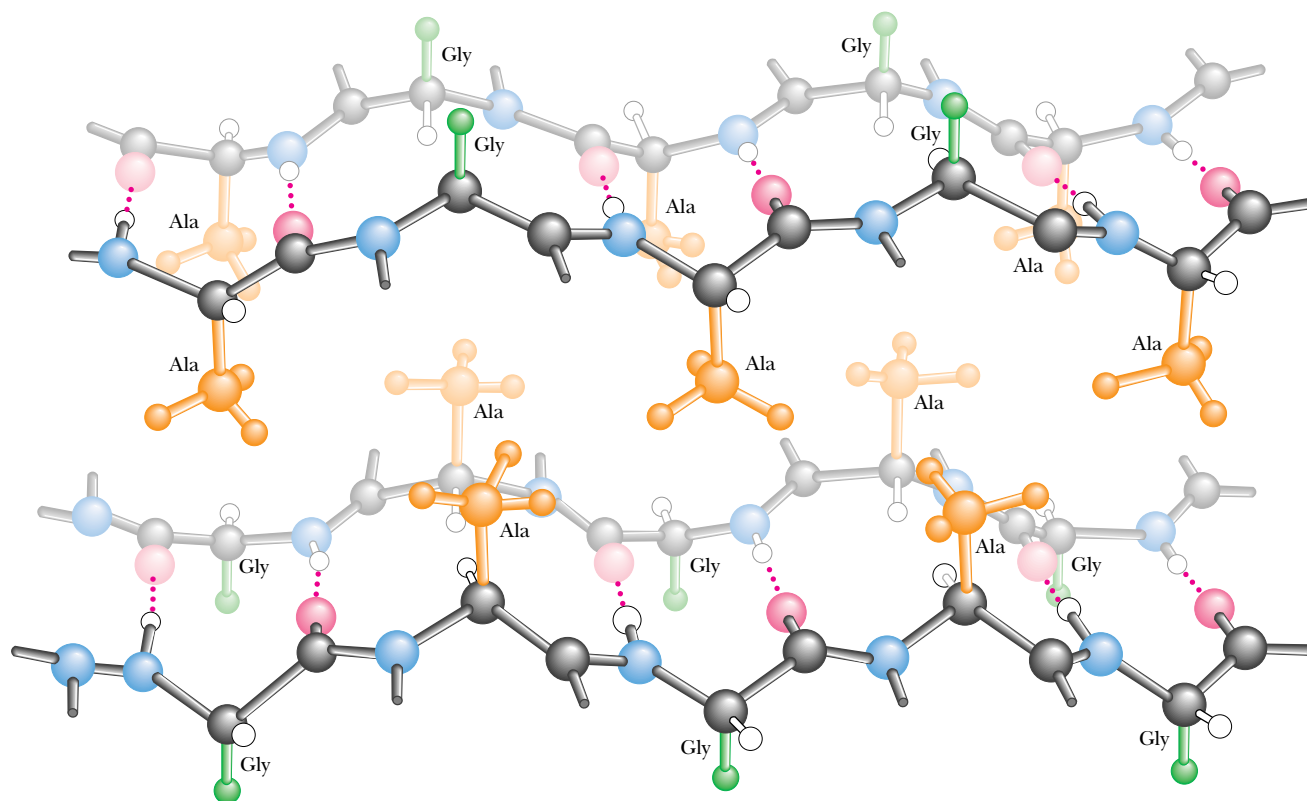


FIGURE 6.15 • Silk fibroin consists of a unique stacked array of β -sheets. The primary structure of fibroin molecules consists of long stretches of alternating glycine and alanine or serine residues. When the sheets stack, the more bulky alanine and serine residues on one side of a sheet interdigitate with similar residues on an adjoining sheet. Glycine hydrogens on the alternating faces interdigitate in a similar manner, but with a smaller intersheet spacing. (Irving Geis)

tures makes possible the various animal activities such as running and jumping that put severe stresses on joints and skeleton. Broken bones and tendon and cartilage injuries to knees, elbows, and other joints involve tears or hyper-extensions of the collagen matrix in these tissues.

The basic structural unit of collagen is **tropocollagen**, which has a molecular weight of 285,000 and consists of three intertwined polypeptide chains, each about 1000 amino acids in length. Tropocollagen molecules are about 300 nm long and only about 1.4 nm in diameter. Several kinds of collagen have been identified. *Type I collagen*, which is the most common, consists of two identical peptide chains designated $\alpha 1(I)$ and one different chain designated $\alpha 2(I)$. Type I collagen predominates in bones, tendons, and skin. *Type II collagen*, found in cartilage, and *type III collagen*, found in blood vessels, consist of three identical polypeptide chains.

Collagen has an amino acid composition that is unique and is crucial to its three-dimensional structure and its characteristic physical properties. Nearly one residue out of three is a glycine, and the proline content is also unusually high. Three unusual modified amino acids are also found in collagen: 4-hydroxyproline (Hyp), 3-hydroxyproline, and 5-hydroxylysine (Hyl) (Figure 6.16). Proline and Hyp together compose up to 30% of the residues of collagen. Interestingly, these three amino acids are formed from normal proline and lysine *after* the collagen polypeptides are synthesized. The modifications are

DEEPER LOOK

Charlotte’s Web Revisited: Helix–Sheet Composites in Spider Dragline Silk

E. B. White’s endearing story *Charlotte’s Web* centers around the web-spinning feats of Charlotte the spider. Although the intricate designs of spiderwebs are eye- (and fly-) catching, it might be argued that the composition of web silk itself is even more remarkable. Spider silk is synthesized in special glands in the spider’s abdomen. The silk strands produced by these glands are both strong and elastic. *Dragline silk* (that from which the spider hangs) has a tensile strength of 200,000 psi (pounds per square inch)—stronger than steel and similar to Kevlar, the synthetic material used in bulletproof vests! This same silk fiber is also flexible enough to withstand strong winds and other natural stresses. This combination of strength and flexibility derives from the *composite nature* of spider silk. As keratin protein is extruded from

the spider’s glands, it endures shearing forces that break the H bonds stabilizing keratin α -helices. These regions then form microcrystalline arrays of β -sheets. These microcrystals are surrounded by the keratin strands, which adopt a highly disordered state composed of α -helices and random coil structures. The β -sheet microcrystals contribute strength, and the disordered array of helix and coil make the silk strand flexible. The resulting silk strand resembles modern human-engineered composite materials. Certain tennis racquets, for example, consist of fiberglass polymers impregnated with microcrystalline graphite. The fiberglass provides flexibility, and the graphite crystals contribute strength. Modern high technology, for all its sophistication, is merely imitating nature—and Charlotte’s web—after all.

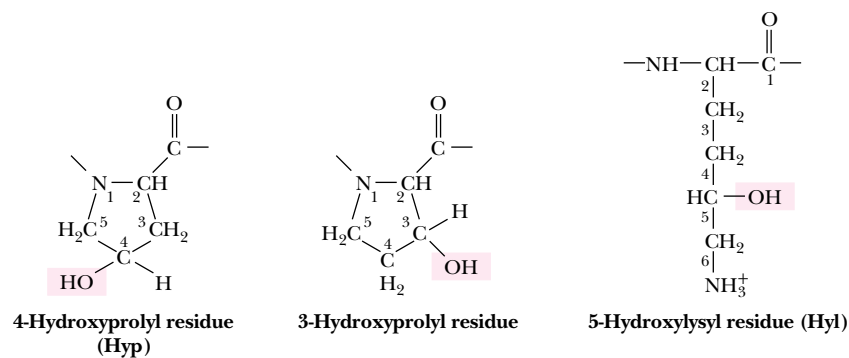
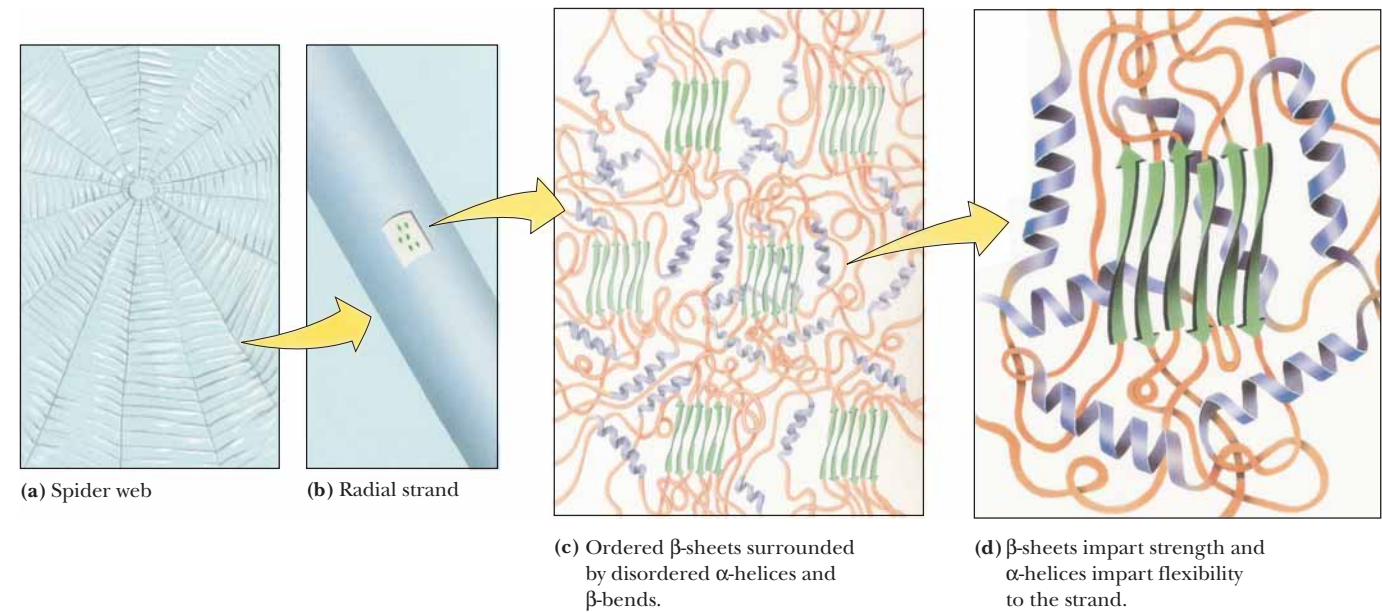


FIGURE 6.16 • The hydroxylated residues typically found in collagen.

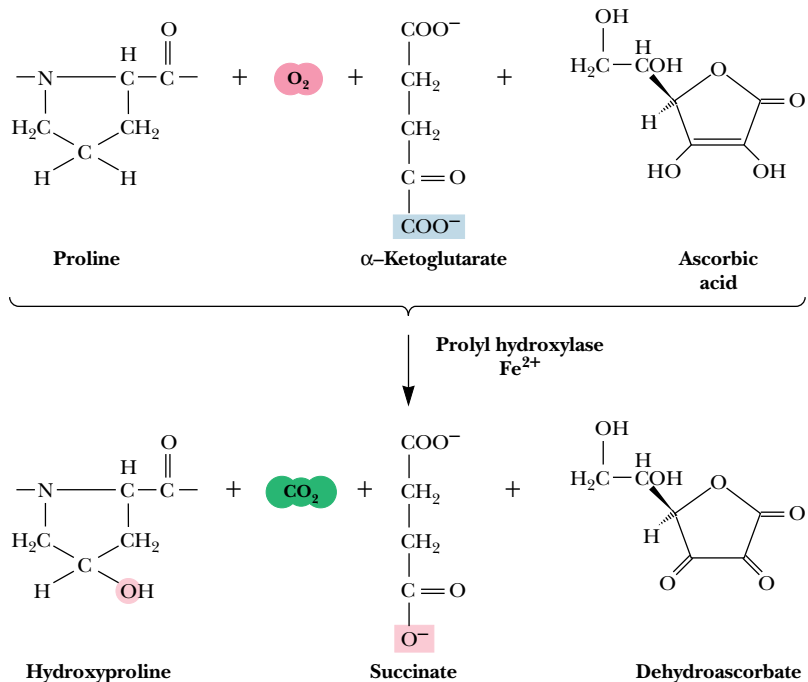


FIGURE 6.17 • Hydroxylation of proline residues is catalyzed by prolyl hydroxylase. The reaction requires α -ketoglutarate and ascorbic acid (vitamin C).

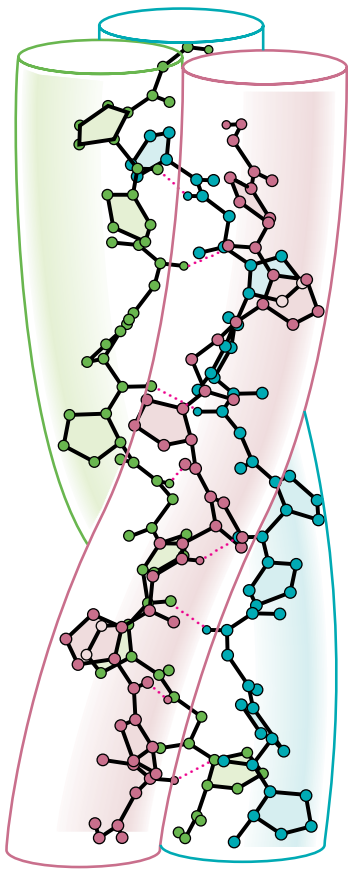


FIGURE 6.18 • Poly(Gly-Pro-Pro), a collagen-like right-handed triple helix composed of three left-handed helical chains. (Adapted from Miller, M. H., and Scheraga, H. A., 1976, Calculation of the structures of collagen models. Role of interchain interactions in determining the triple-helical coiled-coil conformation. I. Poly(glycyl-prolyl-prolyl). Journal of Polymer Science Symposium 54:171–200.)

effected by two enzymes: *prolyl hydroxylase* and *lysyl hydroxylase*. The prolyl hydroxylase reaction (Figure 6.17) requires molecular oxygen, α -ketoglutarate, and ascorbic acid (vitamin C) and is activated by Fe^{2+} . The hydroxylation of lysine is similar. These processes are referred to as **posttranslational modifications** because they occur after genetic information from DNA has been *translated* into newly formed protein.

Because of their high content of glycine, proline, and hydroxyproline, collagen fibers are incapable of forming traditional structures such as α -helices and β -sheets. Instead, collagen polypeptides intertwine to form a unique **triple helix**, with each of the three strands arranged in a helical fashion (Figure 6.18). Compared to the α -helix, the collagen helix is much more extended, with a rise per residue along the triple helix axis of 2.9 Å, compared to 1.5 Å for the α -helix. There are about 3.3 residues per turn of each of these helices. *The triple helix is a structure that forms to accommodate the unique composition and sequence of collagen.* Long stretches of the polypeptide sequence are repeats of a Gly-x-y motif, where x is frequently Pro and y is frequently Pro or Hyp. In the triple helix, every third residue faces or contacts the crowded center of the structure. This area is so crowded that only Gly can fit, and thus every third residue must be a Gly (as observed). Moreover, the triple helix is a *staggered* structure, such that Gly residues from the three strands stack along the center of the triple helix and the Gly from one strand lies adjacent to an x residue from the second strand and to a y from the third. This allows the N—H of each Gly residue to hydrogen bond with the C=O of the adjacent x residue. The triple helix structure is further stabilized and strengthened by the formation of interchain H bonds involving hydroxyproline.

Collagen types I, II, and III form strong, organized **fibrils**, consisting of staggered arrays of tropocollagen molecules (Figure 6.19). The periodic

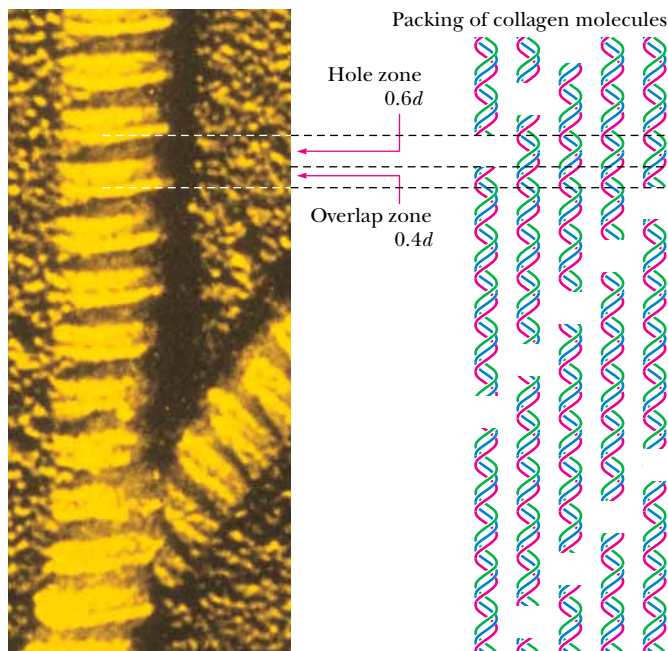


FIGURE 6.19 • In the electron microscope, collagen fibers exhibit alternating light and dark bands. The dark bands correspond to the 40-nm gaps or “holes” between pairs of aligned collagen triple helices. The repeat distance, d , for the light- and dark-banded pattern is 68 nm. The collagen molecule is 300 nm long, which corresponds to $4.41d$. The molecular repeat pattern of five staggered collagen molecules corresponds to $5d$. (J. Gross, Biozentrum/Science Photo Library)

arrangement of triple helices in a head-to-tail fashion results in banded patterns in electron micrographs. The banding pattern typically has a periodicity (repeat distance) of 68 nm. Because collagen triple helices are 300 nm long, 40-nm gaps occur between adjacent collagen molecules in a row along the long axis of the fibrils and the pattern repeats every five rows ($5 \times 68 \text{ nm} = 340 \text{ nm}$). The 40-nm gaps are referred to as *hole regions*, and they are important in at least two ways. First, sugars are found covalently attached to 5-hydroxylysine residues in the hole regions of collagen (Figure 6.20). The occurrence of carbohydrate in the hole region has led to the proposal that it plays a role in organizing fibril assembly. Second, the hole regions may play a role in bone formation. Bone consists of microcrystals of **hydroxyapatite**, $\text{Ca}_5(\text{PO}_4)_3\text{OH}$, embedded in a matrix of collagen fibrils. When new bone tissue forms, the formation of new hydroxyapatite crystals occurs at intervals of 68 nm. The hole regions of collagen fibrils may be the sites of nucleation for the mineralization of bone.

The collagen fibrils are further strengthened and stabilized by the formation of both *intramolecular* (within a tropocollagen molecule) and *intermolecular* (between tropocollagen molecules in the fibril) cross-links. Intramolecular cross-links are formed between lysine residues in the (nonhelical) N-terminal region of tropocollagen in a unique pair of reactions shown in Figure 6.21. The enzyme *lysyl oxidase* catalyzes the formation of aldehyde groups at the lysine side chains in a copper-dependent reaction. The aldehyde groups of two such side chains then link covalently in a spontaneous nonenzymatic *aldol condensation*. The intermolecular cross-linking of tropocollagens involves the formation of a unique **hydroxypyridinium** structure from one lysine and two hydroxy-

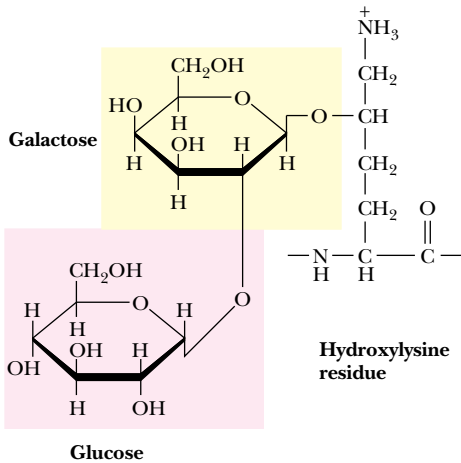


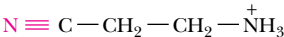
FIGURE 6.20 • A disaccharide of galactose and glucose is covalently linked to the 5-hydroxyl group of hydroxylysines in collagen by the combined action of the enzymes galactosyl transferase and glucosyl transferase.

HUMAN BIOCHEMISTRY

Collagen-Related Diseases

Collagen provides an ideal case study of the molecular basis of physiology and disease. For example, the nature and extent of collagen cross-linking depends on the age and function of the tissue. Collagen from young animals is predominantly un-cross-linked and can be extracted in soluble form, whereas collagen from older animals is highly cross-linked and thus insoluble. The loss of flexibility of joints with aging is probably due in part to increased cross-linking of collagen.

Several serious and debilitating diseases involving collagen abnormalities are known. **Lathyrism** occurs in animals due to the regular consumption of seeds of *Lathyrus odoratus*, the sweet pea, and involves weakening and abnormalities in blood vessels, joints, and bones. These conditions are caused by **β -aminopropionitrile** (see figure), which covalently inactivates lysyl oxidase and leads to greatly reduced intramolecular cross-linking of collagen in affected animals (or humans).



β -Aminopropionitrile (present in sweet peas) covalently inactivates lysyl oxidase, preventing intramolecular cross-linking of collagen and causing abnormalities in joints, bones, and blood vessels.

Scurvy results from a dietary vitamin C deficiency and involves the inability to form collagen fibrils properly. This is the result of reduced activity of prolyl hydroxylase, which is vitamin C-dependent, as previously noted. Scurvy leads to lesions in the skin and blood vessels, and, in its advanced stages, it can lead to grotesque disfiguration and eventual death. Although rare in the modern world, it was a disease well known to sea-faring explorers in earlier times who did not appreciate the importance of fresh fruits and vegetables in the diet.

A number of rare genetic diseases involve collagen abnormalities, including *Marfan's syndrome* and the *Ehlers-Danlos syndromes*, which result in hyperextensible joints and skin. The formation of *atherosclerotic plaques*, which cause arterial blockages in advanced stages, is due in part to the abnormal formation of collagenous structures in blood vessels.

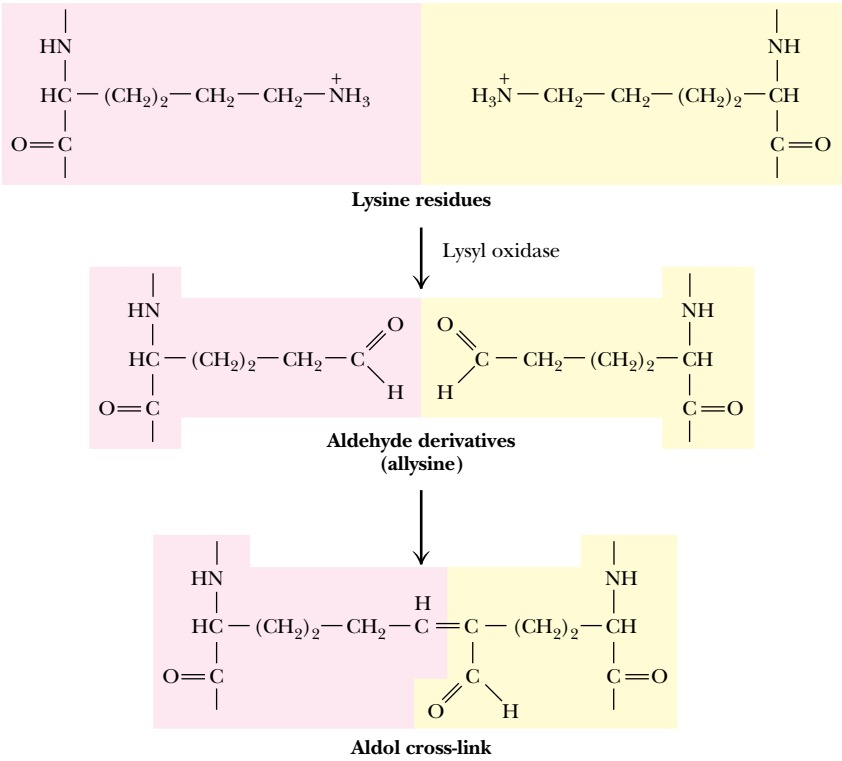


FIGURE 6.21 • Collagen fibers are stabilized and strengthened by Lys–Lys cross-links. Aldehyde moieties formed by lysyl oxidase react in a spontaneous nonenzymatic aldol reaction.

lysine residues (Figure 6.22). These cross-links form between the N-terminal region of one tropocollagen and the C-terminal region of an adjacent tropocollagen in the fibril.

Globular Proteins

Fibrous proteins, although interesting for their structural properties, represent only a small percentage of the proteins found in nature. **Globular proteins**, so named for their approximately spherical shape, are far more numerous.

Helices and Sheets in Globular Proteins

Globular proteins exist in an enormous variety of three-dimensional structures, but nearly all contain substantial amounts of the α -helices and β -sheets that form the basic structures of the simple fibrous proteins. For example, myoglobin, a small, globular, oxygen-carrying protein of muscle (17 kD, 153 amino acid residues), contains eight α -helical segments, each containing 7 to 26 amino acid residues. These are arranged in an apparently irregular (but invariant) fashion (see Figure 5.7). The space between the helices is filled efficiently and tightly with (mostly hydrophobic) amino acid side chains. Most of the polar side chains in myoglobin (and in most other globular proteins) face the outside of the protein structure and interact with solvent water. Myoglobin's structure is unusual because most globular proteins contain a relatively small amount of α -helix. A more typical globular protein (Figure 6.23) is *bovine ribonuclease A*, a small protein (14.6 kD, 129 residues) that contains a few short helices, a broad section of antiparallel β -sheet, a few β -turns, and several peptide segments without defined secondary structure.

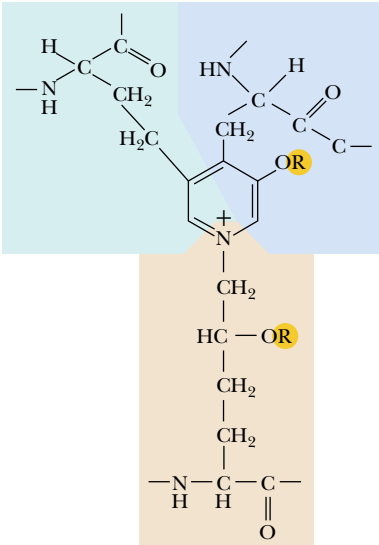


FIGURE 6.22 • The hydroxypyridinium structure formed by the cross-linking of a Lys and two hydroxy Lys residues.

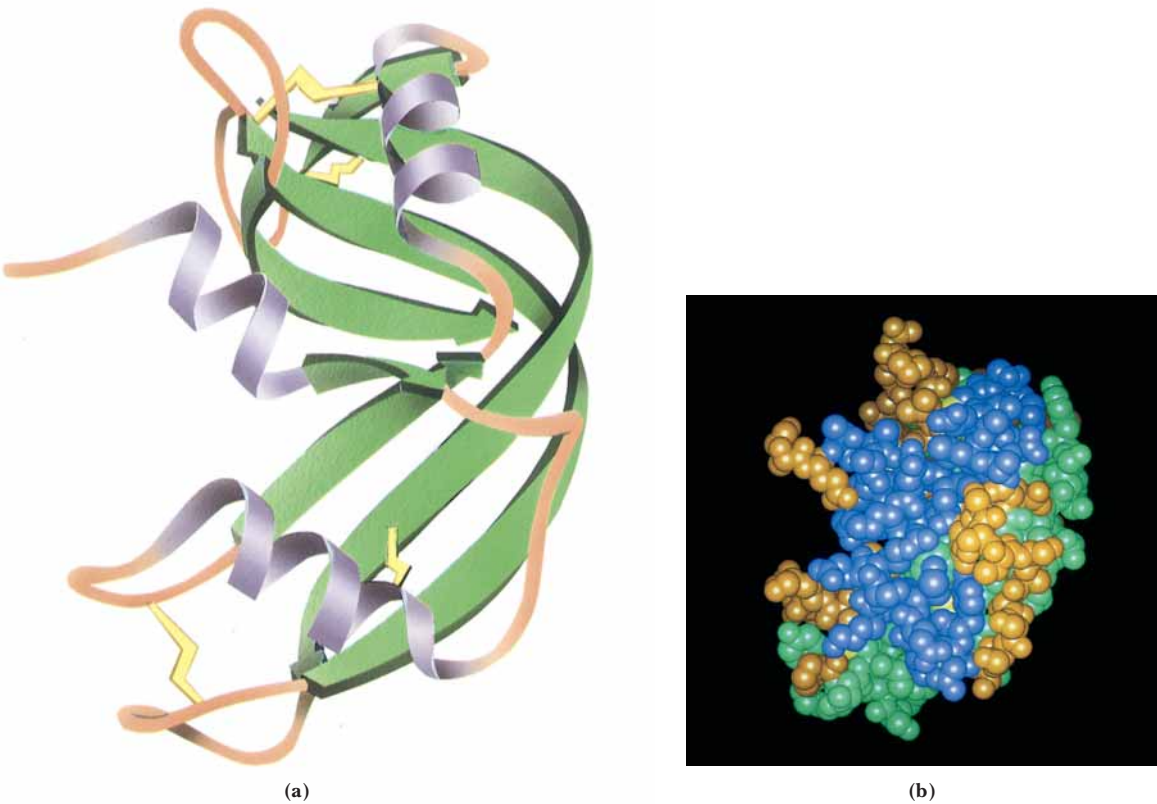


FIGURE 6.23 • The three-dimensional structure of bovine ribonuclease A, showing the α -helices as ribbons. (Jane Richardson)

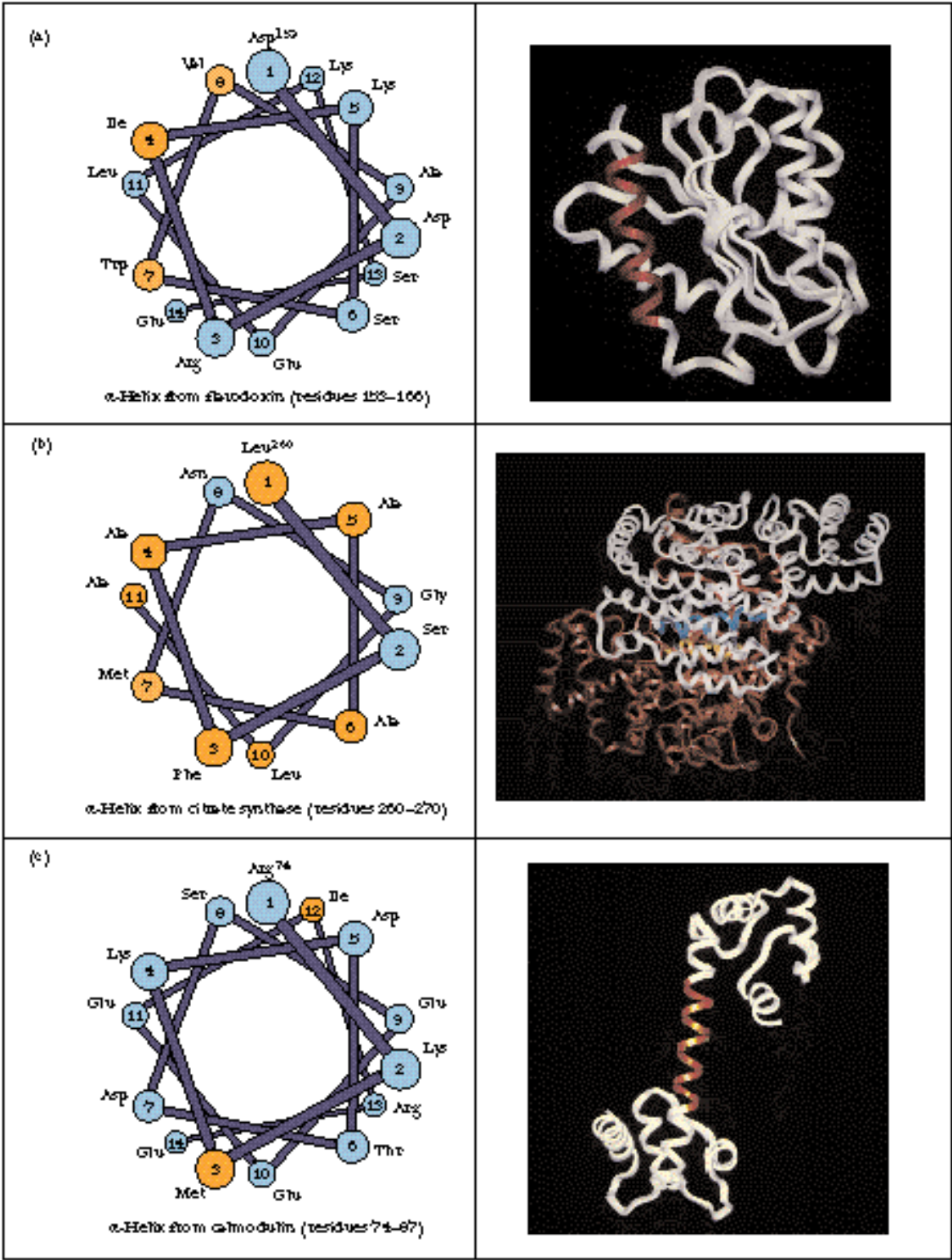


FIGURE 6.24 • (a) The alpha helix consisting of residues 153–166 (red) in flavodoxin from *Anabaena* is a surface helix and is amphipathic. (b) The two helices (yellow and blue) in the interior of the citrate synthase dimer (residues 260–270 in each monomer) are mostly hydrophobic. (c) The exposed helix (residues 74–87—red) of calmodulin is entirely accessible to solvent and consists mainly of polar and charged residues.

Why should the cores of most globular and membrane proteins consist almost entirely of α -helices and β -sheets? The reason is that the highly polar N—H and C=O moieties of the peptide backbone must be neutralized in the hydrophobic core of the protein. The extensively H-bonded nature of α -helices and β -sheets is ideal for this purpose, and these structures effectively stabilize the polar groups of the peptide backbone in the protein core.

In globular protein structures, it is common for one face of an α -helix to be exposed to the water solvent, with the other face toward the hydrophobic interior of the protein. The outward face of such an **amphiphilic helix** consists mainly of polar and charged residues, whereas the inward face contains mostly nonpolar, hydrophobic residues. A good example of such a surface helix is that of residues 153 to 166 of **flavodoxin** from *Anabaena* (Figure 6.24). Note that the **helical wheel presentation** of this helix readily shows that one face contains four hydrophobic residues and that the other is almost entirely polar and charged.

Less commonly, an α -helix can be completely buried in the protein interior or completely exposed to solvent. **Citrate synthase** is a dimeric protein in which α -helical segments form part of the subunit–subunit interface. As shown in Figure 6.24, one of these helices (residues 260 to 270) is highly hydrophobic and contains only two polar residues, as would befit a helix in the protein core. On the other hand, Figure 6.24 also shows the solvent-exposed helix (residues 74 to 87) of **calmodulin**, which consists of 10 charged residues, 2 polar residues, and only 2 nonpolar residues.

Packing Considerations

The secondary and tertiary structures of myoglobin and ribonuclease A illustrate the importance of packing in tertiary structures. Secondary structures pack closely to one another and also intercalate with (insert between) extended polypeptide chains. If the sum of the van der Waals volumes of a protein's constituent amino acids is divided by the volume occupied by the protein, packing densities of 0.72 to 0.77 are typically obtained. This means that, even with close packing, approximately 25% of the total volume of a protein is not occupied by protein atoms. Nearly all of this space is in the form of very small cavities. Cavities the size of water molecules or larger do occasionally occur, but they make up only a small fraction of the total protein volume. It is likely that such cavities provide flexibility for proteins and facilitate conformation changes and a wide range of protein dynamics (discussed later).

Ordered, Nonrepetitive Structures

In any protein structure, the segments of the polypeptide chain that cannot be classified as defined secondary structures, such as helices or sheets, have been traditionally referred to as *coil* or *random coil*. Both these terms are misleading. Most of these segments are neither coiled nor random, in any sense of the words. These structures are every bit as highly organized and stable as the defined secondary structures. They are just more variable and difficult to describe. These so-called coil structures are strongly influenced by side-chain interactions. Few of these interactions are well understood, but a number of interesting cases have been described. In his early studies of myoglobin structure, John Kendrew found that the —OH group of threonine or serine often forms a hydrogen bond with a backbone NH at the beginning of an α -helix. The same stabilization of an α -helix by a serine is observed in the three-dimensional structure of pancreatic trypsin inhibitor (Figure 6.25). Also in this same structure, an asparagine residue adjacent to a β -strand is found to form H bonds that stabilize the β -structure.

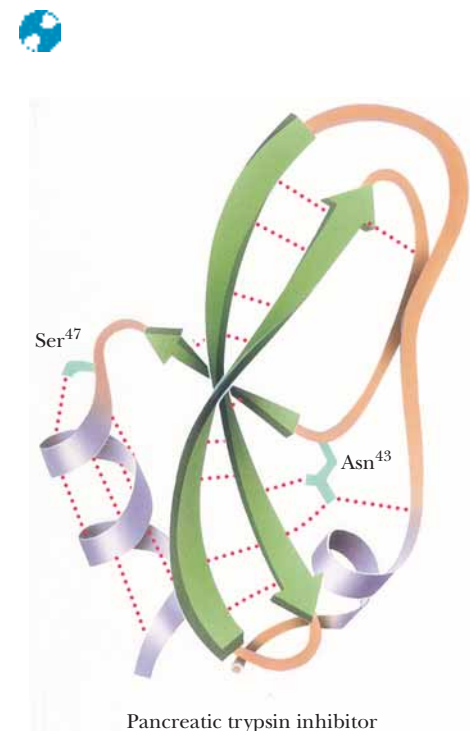


FIGURE 6.25 • The three-dimensional structure of bovine pancreatic trypsin inhibitor. Note the stabilization of the α -helix by a hydrogen bond to Ser⁴⁷ and the stabilization of the β -sheet by Asn⁴³.

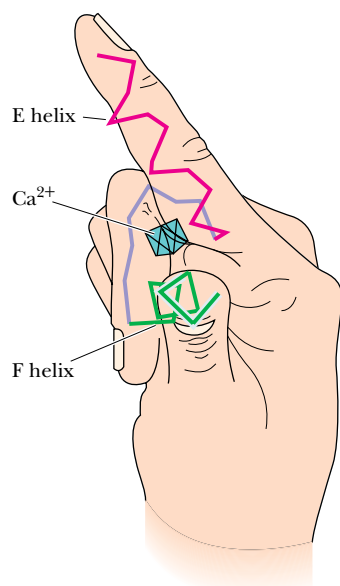


FIGURE 6.26 • A representation of the so-called E-F hand structure, which forms calcium-binding sites in a variety of proteins. The stick drawing shows the peptide backbone of the E-F hand motif. The “E” helix extends along the index finger, a loop traces the approximate arrangement of the curled middle finger, and the “F” helix extends outward along the thumb. A calcium ion (Ca^{2+}) snuggles into the pocket created by the two helices and the loop. Kretsinger and coworkers originally assigned letters alphabetically to the helices in parvalbumin, a protein from carp. The E-F hand derives its name from the letters assigned to the helices at one of the Ca^{2+} -binding sites.

Nonrepetitive but well-defined structures of this type form many important features of enzyme active sites. In some cases, a particular arrangement of “coil” structure providing a specific type of functional site recurs in several functionally related proteins. The peptide loop that binds iron–sulfur clusters in both ferredoxin and high potential iron protein is one example. Another is the central loop portion of the *E-F hand structure* that binds a calcium ion in several calcium-binding proteins, including calmodulin, carp parvalbumin, troponin C, and the intestinal calcium-binding protein. This loop, shown in Figure 6.26, connects two short α -helices. The calcium ion nestles into the pocket formed by this structure.

Flexible, Disordered Segments

In addition to nonrepetitive but well-defined structures, which exist in all proteins, genuinely disordered segments of polypeptide sequence also occur. These sequences either do not show up in electron density maps from X-ray crystallographic studies or give diffuse or ill-defined electron densities. These segments either undergo actual motion in the protein crystals themselves or take on many alternate conformations in different molecules within the protein crystal. Such behavior is quite common for long, charged side chains on the surface of many proteins. For example, 16 of the 19 lysine side chains in myoglobin have uncertain orientations beyond the δ -carbon, and five of these are disordered beyond the β -carbon. Similarly, a majority of the lysine residues are disordered in trypsin, rubredoxin, ribonuclease, and several other proteins. Arginine residues, however, are usually well ordered in protein structures. For the four proteins just mentioned, 70% of the arginine residues are highly ordered, compared to only 26% of the lysines.

Motion in Globular Proteins

Although we have distinguished between well-ordered and disordered segments of the polypeptide chain, it is important to realize that even well-ordered side chains in a protein undergo motion, sometimes quite rapid. These motions should be viewed as momentary oscillations about a single, highly stable conformation. *Proteins are thus best viewed as dynamic structures.* The allowed motions may be motions of individual atoms, groups of atoms, or even whole sections of the protein. Furthermore, they may arise from either thermal energy or specific, triggered conformational changes in the protein. **Atomic fluctuations** such as vibrations typically are random, very fast, and usually occur over small distances (less than 0.5 Å), as shown in Table 6.2. These motions arise from the kinetic energy within the protein and are a function of temperature. These very fast motions can be modeled by molecular dynamics calculations and studied by X-ray diffraction.

A class of slower motions, which may extend over larger distances, is **collective motions**. These are movements of groups of atoms covalently linked in such a way that the group moves as a unit. Such groups range in size from a few atoms to hundreds of atoms. Whole structural domains within a protein may be involved, as in the case of the flexible antigen-binding domains of immunoglobulins, which move as relatively rigid units to selectively bind separate antigen molecules. Such motions are of two types—(1) those that occur quickly but infrequently, such as tyrosine ring flips, and (2) those that occur slowly, such as *cis-trans* isomerizations of prolines. These collective motions also arise from thermal energies in the protein and operate on a time scale of 10^{-12} to 10^{-3} sec. These motions can be studied by nuclear magnetic resonance (NMR) and fluorescence spectroscopy.

Table 6.2

Motion and Fluctuations in Proteins

Type of Motion	Spatial Displacement (Å)	Characteristic Time (sec)	Source of Energy
Atomic vibrations	0.01–1	10^{-15} – 10^{-11}	Kinetic energy
Collective motions	0.01–5 or more	10^{-12} – 10^{-3}	Kinetic energy
1. Fast: Tyr ring flips; methyl group rotations 2. Slow: hinge bending between domains			
Triggered conformation changes	0.5–10 or more	10^{-9} – 10^3	Interactions with triggering agent

Adapted from Petsko and Ringe (1984).

Conformational changes involve motions of groups of atoms (individual side chains, for example) or even whole sections of proteins. These motions occur on a time scale of 10^{-9} to 10^3 sec, and the distances covered can be as large as 1 nm. These motions may occur in response to specific stimuli or arise from specific interactions within the protein, such as hydrogen bonding, electrostatic interactions, and ligand binding. More will be said about conformational changes when enzyme catalysis and regulation are discussed (see Chapters 14 and 15).

Forces Driving the Folding of Globular Proteins

As already pointed out, the driving force for protein folding and the resulting formation of a tertiary structure is the formation of the most stable structure possible. Two forces are at work here. The peptide chain must both (1) satisfy the constraints inherent in its own structure and (2) fold so as to “bury” the hydrophobic side chains, minimizing their contact with solvent. The polypeptide itself does not usually form simple straight chains. Even in chain segments where helices and sheets are not formed, an extended peptide chain, being composed of L-amino acids, has a tendency to twist slightly in a right-handed direction. As shown in Figure 6.27, this tendency is apparently the basis for the formation of a variety of tertiary structures having a right-handed sense. Principal among these are the right-handed twists in arrays of β -sheets and right-handed cross-overs in parallel β -sheet arrays. Right-handed twisted β -sheets are found at the center of a number of proteins and provide an extended, highly stable structural core. Phosphoglycerate mutase, adenylate kinase, and carbonic anhydrase, among others, exist as smoothly twisted planes or saddle-shaped structures. Triose phosphate isomerase, soybean trypsin inhibitor, and domain 1 of pyruvate kinase contain right-handed twisted cylinders or barrel structures at their cores.

Connections between β -strands are of two types—hairpins and cross-overs. **Hairpins**, as shown in Figure 6.27, connect adjacent antiparallel β -strands. **Cross-overs** are necessary to connect adjacent (or nearly adjacent) parallel β -strands. Nearly all cross-over structures are right-handed. Only in subtilisin and phosphoglucosomerase have isolated left-handed cross-overs been identi-

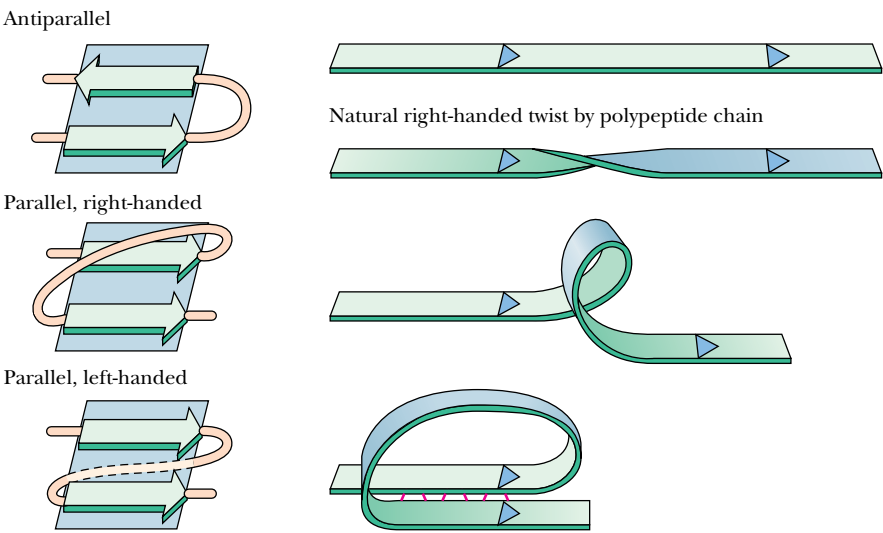


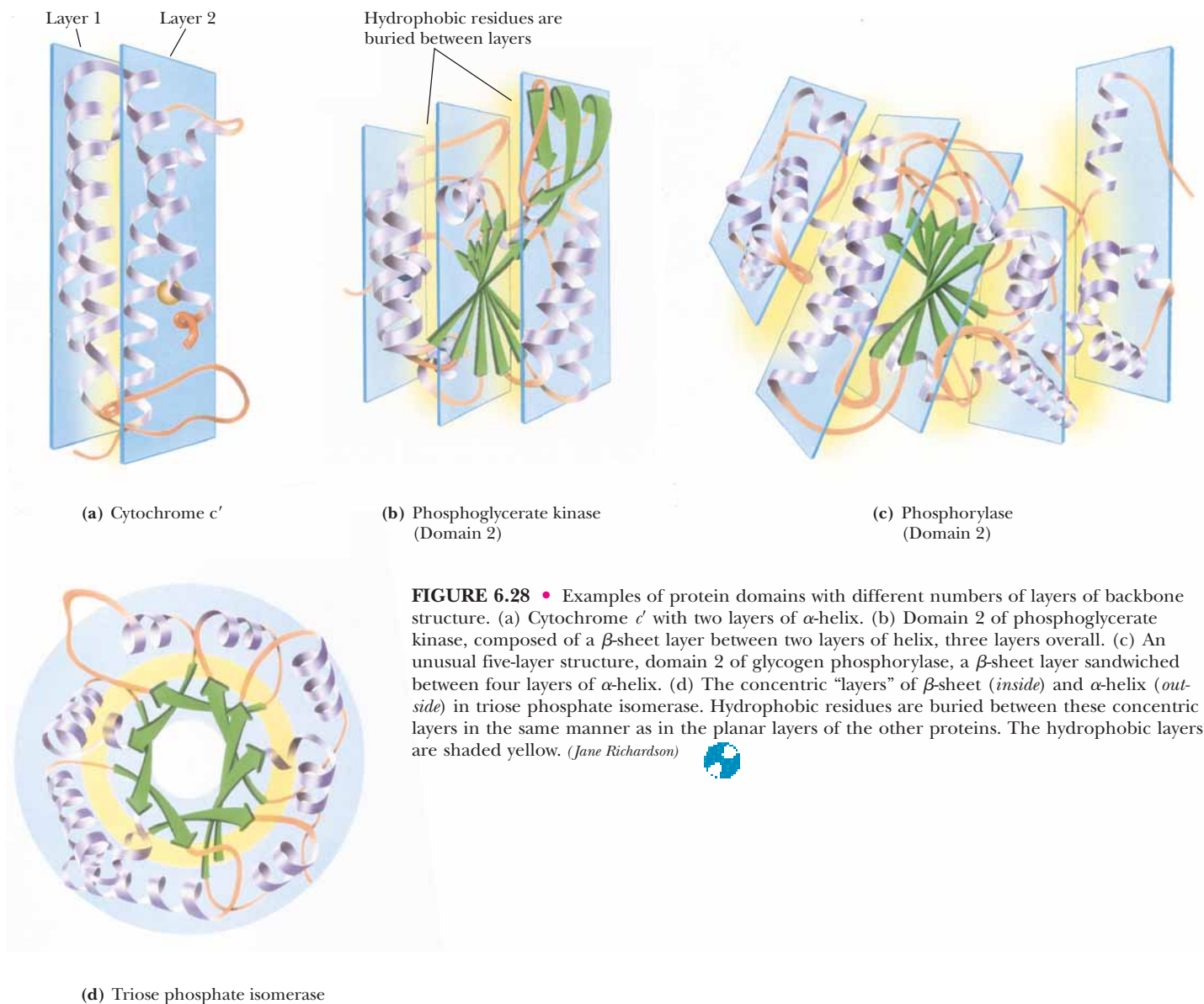
FIGURE 6.27 • The natural right-handed twist exhibited by polypeptide chains, and the variety of structures that arise from this twist.

fied. In many cross-over structures, the cross-over connection itself contains an α -helical segment. This is referred to as a **$\beta\alpha\beta$ -loop**. As shown in Figure 6.27, the strong tendency in nature to form right-handed cross-overs, the wide occurrence of α -helices in the cross-over connection, and the right-handed twists of β -sheets can all be understood as arising from the tendency of an extended polypeptide chain of L-amino acids to adopt a right-handed twist structure. This is a chiral effect. Proteins composed of D-amino acids would tend to adopt left-handed twist structures.

The second driving force that affects the folding of polypeptide chains is the need to bury the hydrophobic residues of the chain, protecting them from solvent water. From a topological viewpoint, then, all globular proteins must have an “inside” where the hydrophobic core can be arranged and an “outside” toward which the hydrophilic groups must be directed. The sequestration of hydrophobic residues away from water is the dominant force in the arrangement of secondary structures and nonrepetitive peptide segments to form a given tertiary structure. Globular proteins can be classified mainly on the basis of the particular kind of core or backbone structure they use to accomplish this goal. The term *hydrophobic core*, as used here, refers to a region in which hydrophobic side chains cluster together, away from the solvent. *Backbone* refers to the polypeptide backbone itself, excluding the particular side chains. Globular proteins can be pictured as consisting of “layers” of backbone, with hydrophobic core regions between them. Over half the known globular protein structures have two layers of backbone (separated by one hydrophobic core). Roughly one-third of the known structures are composed of three backbone layers and two hydrophobic cores. There are also a few known four-layer structures and one known five-layer structure. A few structures are not easily classified in this way, but it is remarkable that most proteins fit into one of these classes. Examples of each are presented in Figure 6.28.

Classification of Globular Proteins

In addition to classification based on layer structure, proteins can be grouped according to the type and arrangement of secondary structure. There are four such broad groups: antiparallel α -helix, parallel or mixed β -sheet, antiparallel β -sheet, and the small metal- and disulfide-rich proteins.



It is important to note that the similarities of tertiary structure within these groups do not necessarily reflect similar or even related functions. Instead, **functional homology** usually depends on structural similarities on a smaller and more intimate scale.

Antiparallel α -Helix Proteins

Antiparallel α -helix proteins are structures heavily dominated by α -helices. The simplest way to pack helices is in an antiparallel manner, and most of the proteins in this class consist of bundles of antiparallel helices. Many of these exhibit a slight (15°) left-handed twist of the helix bundle. Figure 6.29 shows a representative sample of antiparallel α -helix proteins. Many of these are regular, uniform structures, but in a few cases (uteroglobin, for example) one of the helices is tilted away from the bundle. Tobacco mosaic virus protein has small, highly

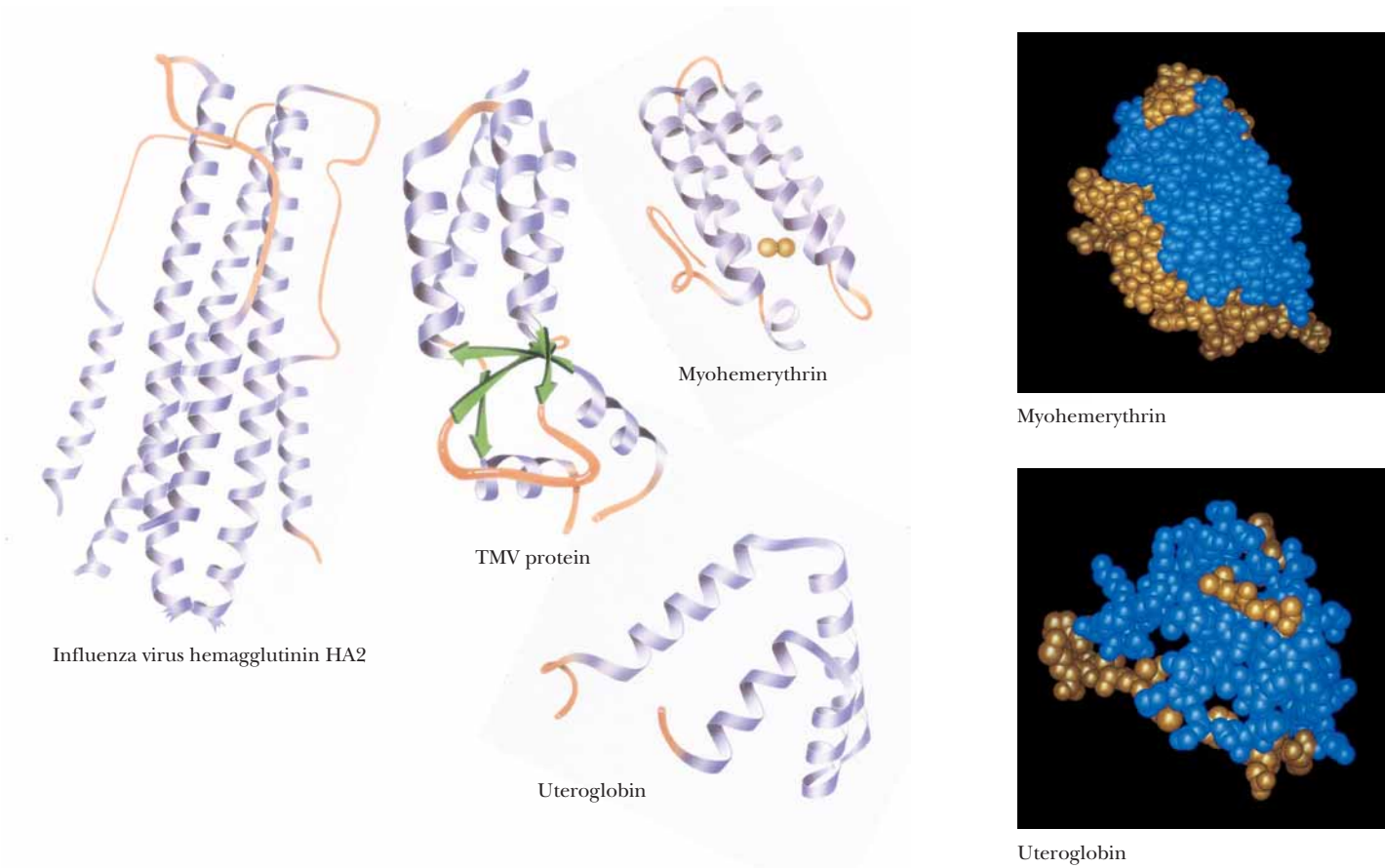


FIGURE 6.29 • Several examples of antiparallel α -proteins. (Jane Richardson)

twisted antiparallel β -sheets on one end of the helix bundle with two additional helices on the other side of the sheet. Notice in Figure 6.29 that most of the antiparallel α -helix proteins are made up of four-helix bundles.

The so-called globin proteins are an important group of α -helical proteins. These include hemoglobins and myoglobins from many species. The globin structure can be viewed as two layers of helices, with one of these layers perpendicular to the other and the polypeptide chain moving back and forth between the layers.

Parallel or Mixed β -Sheet Proteins

The second major class of protein structures contains structures based around **parallel or mixed β -sheets**. Parallel β -sheet arrays, as previously discussed, distribute hydrophobic side chains on both sides of the sheet. This means that neither side of parallel β -sheets can be exposed to solvent. Parallel β -sheets are thus typically found as core structures in proteins, with little access to solvent.

Another important parallel β -array is the eight-stranded **parallel β -barrel**, exemplified in the structures of triose phosphate isomerase and pyruvate kinase (Figure 6.30). Each β -strand in the barrel is flanked by an antiparallel α -helix. The α -helices thus form a larger cylinder of parallel helices concentric with the β -barrel. Both cylinders thus formed have a right-handed twist. Another parallel β -structure consists of an internal twisted wall of parallel or mixed β -sheet protected on both sides by helices or other substructures. This structure is called the **doubly wound parallel β -sheet** because the structure can be

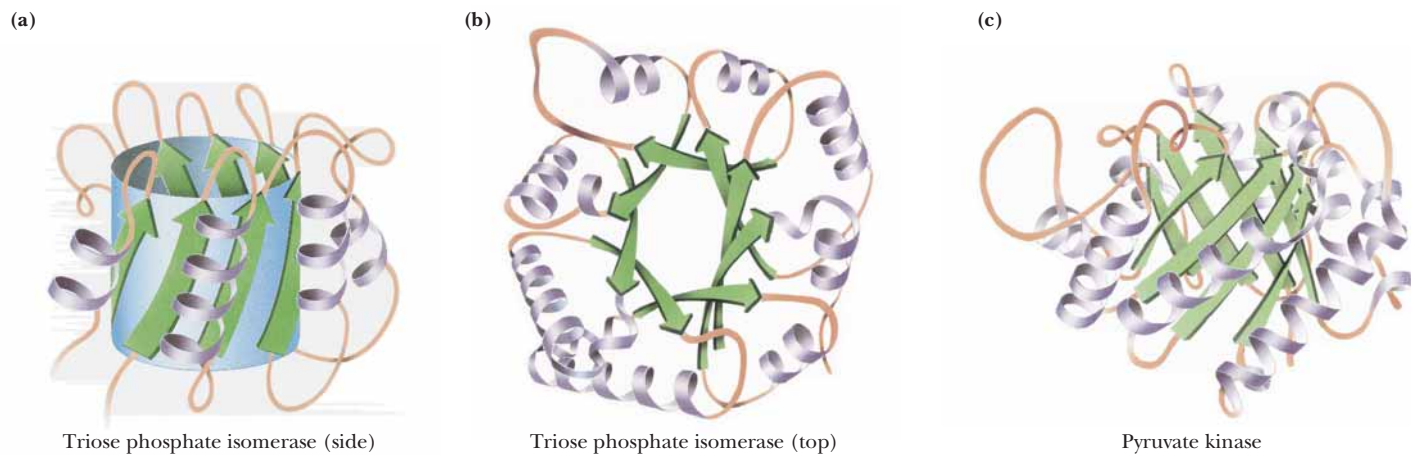


FIGURE 6.30 • Parallel β -array proteins—the eight-stranded β -barrels of triose phosphate isomerase (a, *side view*, and b, *top view*) and (c) pyruvate kinase. (*Jane Richardson*)



imagined to have been wound by strands beginning in the middle and going outward in opposite directions. The essence of this structure is shown in Figure 6.31. Whereas the barrel structures have four layers of backbone structure, the doubly wound sheet proteins have three major layers and thus two hydrophobic core regions.

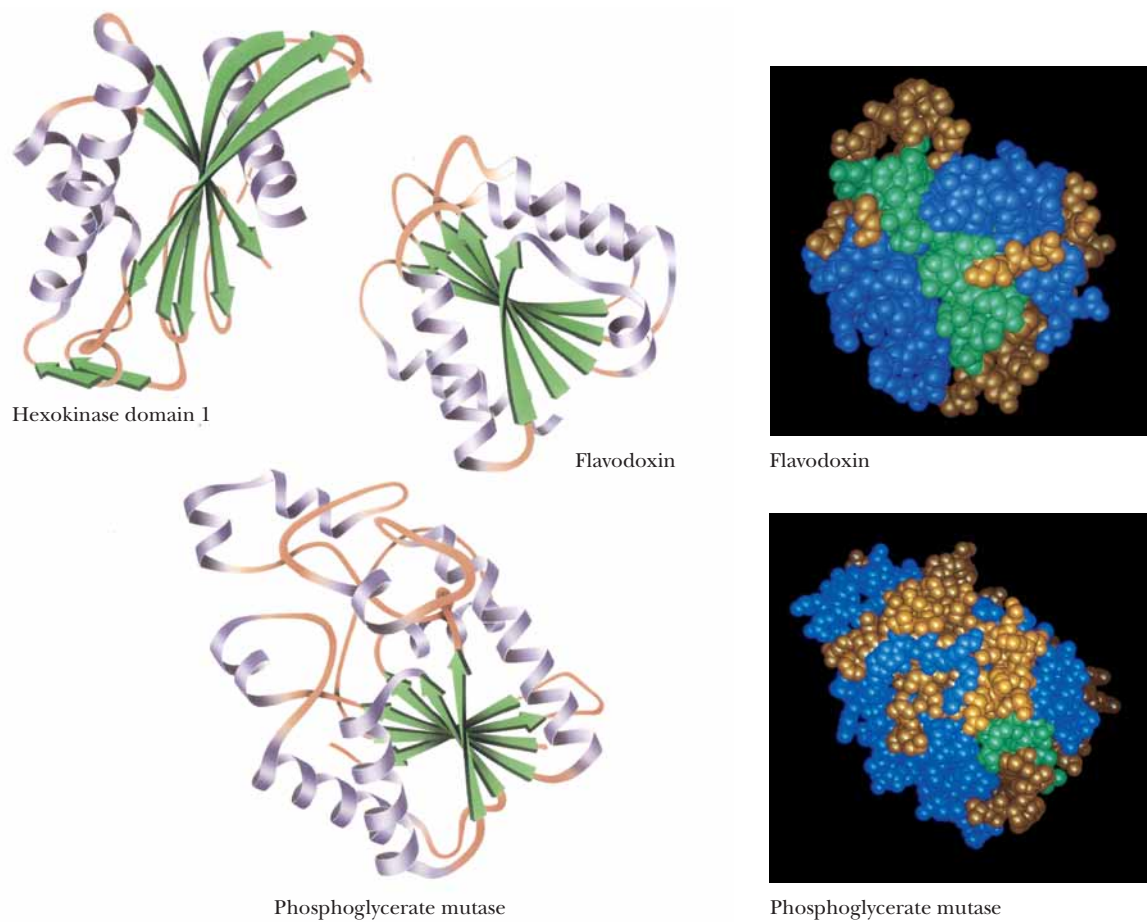
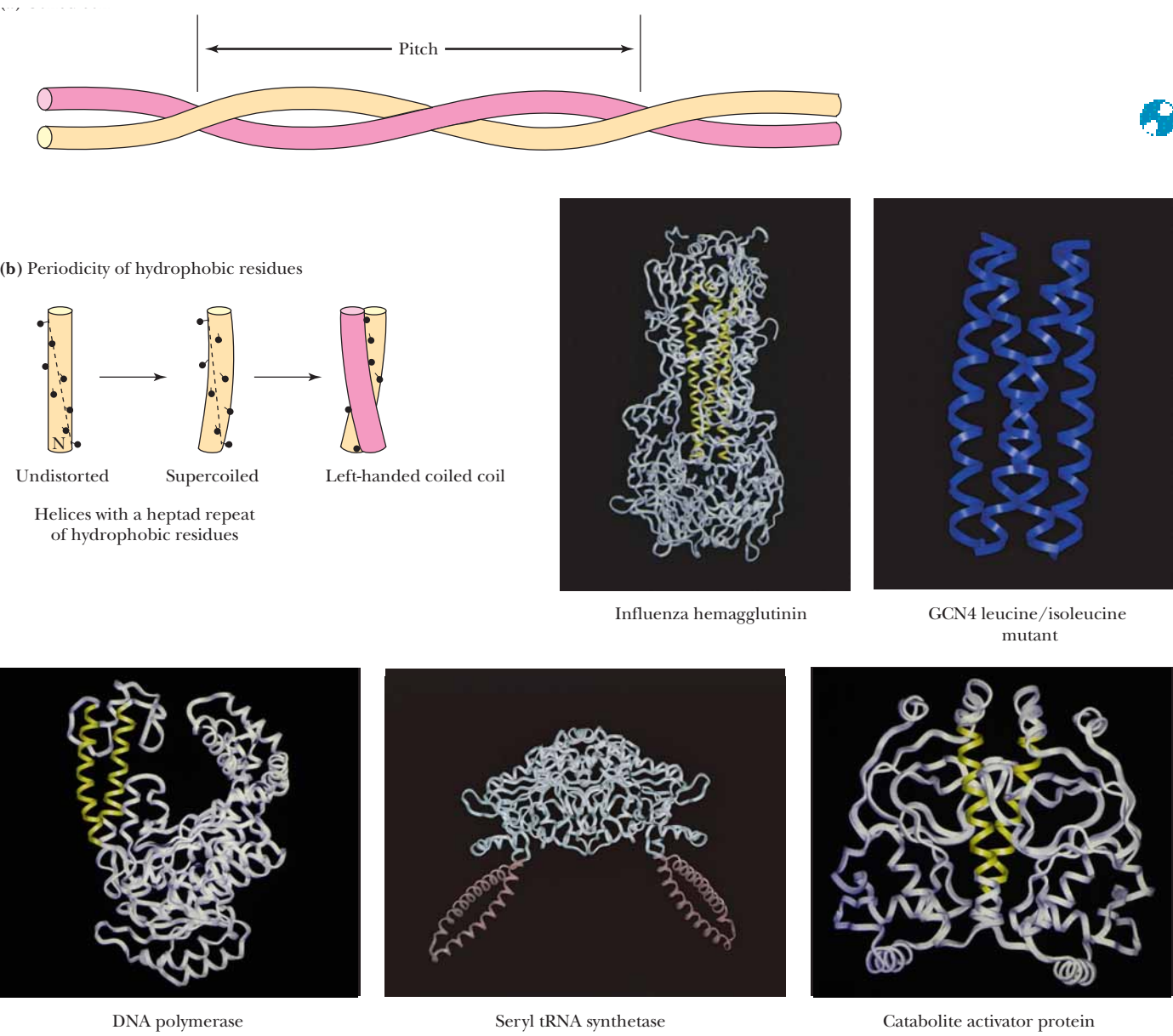


FIGURE 6.31 • Several typical doubly wound parallel β -sheet proteins. (*Jane Richardson*)

The Coiled Coil Motif in Proteins

The **coiled coil** motif was first identified in 1953 by Linus Pauling, Robert Corey, and Francis Crick as the main structural element of fibrous proteins such as keratin and myosin. Since that time, many proteins have been found to contain one or more coiled coil segments or domains. A coiled coil is a bundle of α -helices that are wound into a superhelix. Two, three, or four helical segments may be found in the bundle, and they may be arranged parallel or antiparallel to one another. Coiled coils are characterized by a distinctive and regular packing of side chains in the core of the bundle. This regular meshing of side chains requires

that they occupy equivalent positions turn after turn. This is not possible for undistorted α -helices, which have 3.6 residues per turn. The positions of side chains on their surface shift continuously along the helix surface (see figure). However, giving the right-handed α -helix a left-handed twist reduces the number of residues per turn to 3.5, and, because 3.5 times 2 equals 7.0, the positions of the side chains repeat after two turns (seven residues). Thus, a **heptad repeat** pattern in the peptide sequence is diagnostic of a coiled coil structure. The figure shows a sampling of coiled coil structures (highlighted in color) in various proteins.



Antiparallel β -Sheet Proteins

Another important class of tertiary protein conformations is the **antiparallel β -sheet** structures. Antiparallel β -sheets, which usually arrange hydrophobic residues on just one side of the sheet, can exist with one side exposed to solvent. The minimal structure for an antiparallel β -sheet protein is thus a two-layered structure, with hydrophobic faces of the two sheets juxtaposed and the opposite faces exposed to solvent. Such domains consist of β -sheets arranged in a cylinder or barrel shape. These structures are usually less symmetric than the singly wound parallel barrels and are not as efficiently hydrogen bonded, but they occur much more frequently in nature. Barrel structures tend to be either all parallel or all antiparallel and usually consist of even numbers of β -strands. Good examples of antiparallel structures include soybean trypsin inhibitor, rubredoxin, and domain 2 of papain (Figure 6.32). Topology diagrams of antiparallel β -sheet barrels reveal that many of them arrange the polypeptide sequence in an interlocking pattern reminiscent of patterns found on ancient Greek vases (Figure 6.33) and are thus referred to as a *Greek key topology*. Several of these, including concanavalin A and γ -crystallin, contain an extra swirl in the Greek key pattern (see Figure 6.33). Antiparallel arrangements of β -strands can also form sheets as well as barrels. Glyceraldehyde-3-phosphate dehydrogenase, *Streptomyces* subtilisin inhibitor, and glutathione reductase are examples of single-sheet, double-layered topology (Figure 6.34).

Metal- and Disulfide-Rich Proteins

Other than the structural classes just described and a few miscellaneous structures that do not fit nicely into these categories, there is only one other major class of protein tertiary structures—the small metal-rich and disulfide-rich structures. These proteins or fragments of proteins are usually small (<100 residues), and their conformations are heavily influenced by their high content of either liganded metals or disulfide bonds. The structures of disulfide-rich proteins are unstable if their disulfide bonds are broken. Figure 6.35 shows several representative disulfide-rich proteins, including insulin, phospholipase A₂, and crambin (from the seeds of *Crambe abyssinica*), as well as several metal-rich proteins, including ferredoxin and high potential iron protein (HiPIP). The structures of some of these proteins bear a striking resemblance to structural classes that have already been discussed. For example, phospholipase A₂

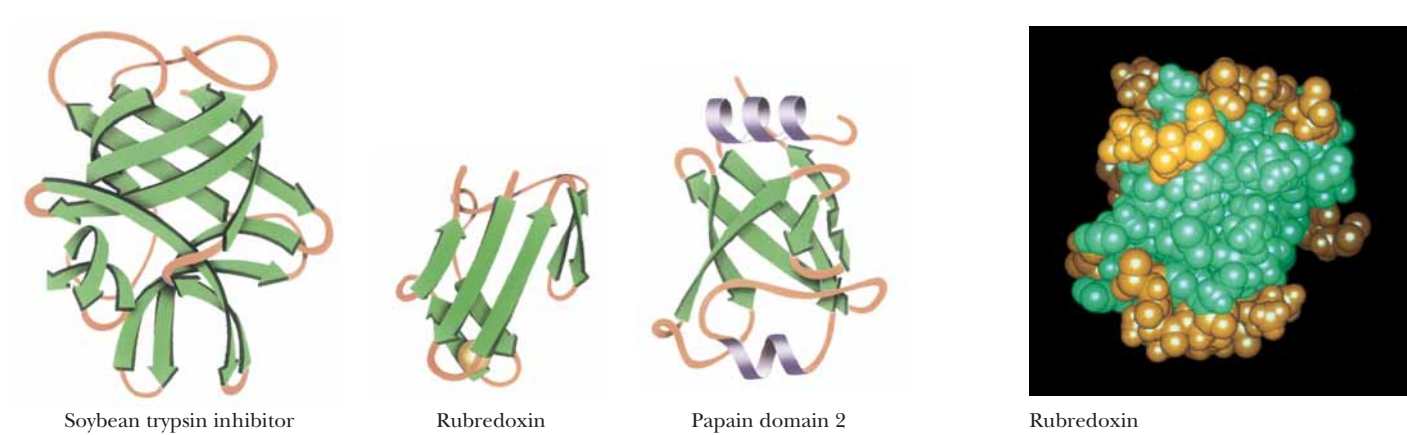


FIGURE 6.32 • Examples of antiparallel β -sheet structures in proteins. (Jane Richardson)

FIGURE 6.33 • Examples of the so-called Greek key antiparallel β -barrel structure in proteins.

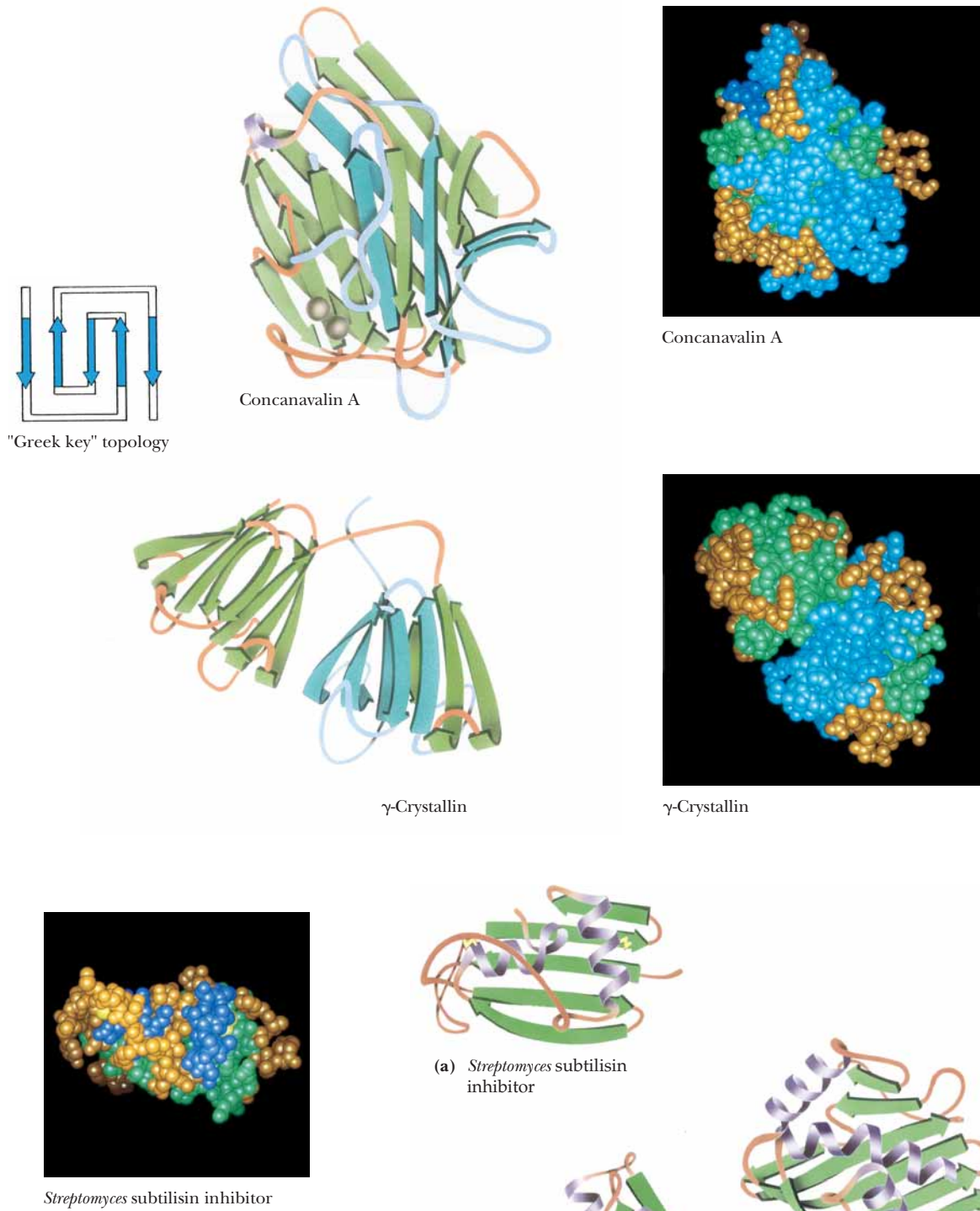


FIGURE 6.34 • Sheet structures formed from antiparallel arrangements of β -strands. (a) *Streptomyces subtilisin inhibitor*, (b) glutathione reductase domain 3, and (c) the second domain of glyceraldehyde-3-phosphate dehydrogenase represent minimal antiparallel β -sheet domain structures. In each of these cases, an antiparallel β -sheet is largely exposed to solvent on one face and covered by helices and random coils on the other face. (*Jane Richardson*)

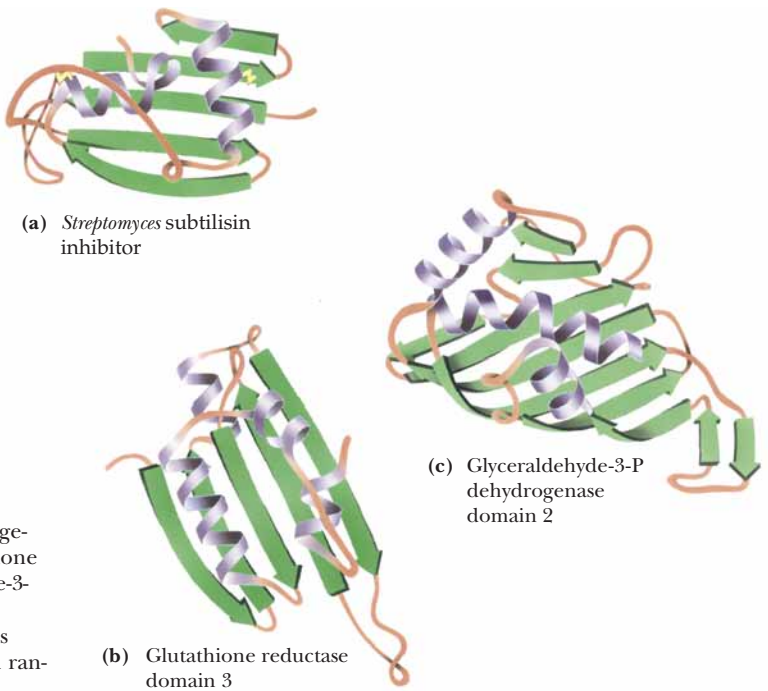
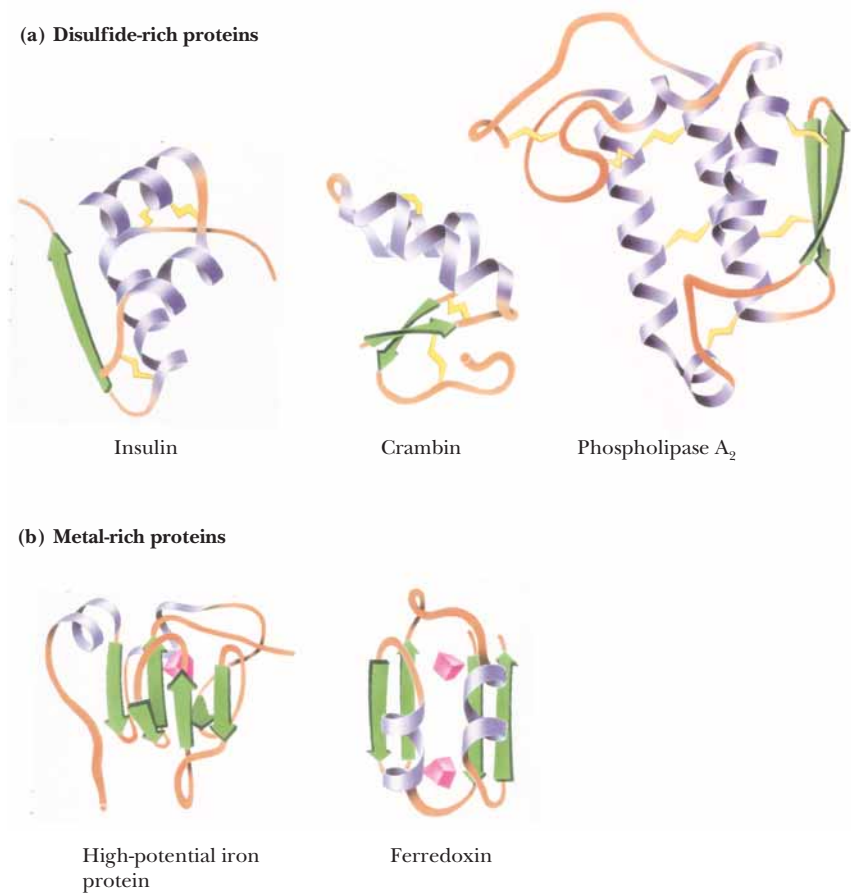


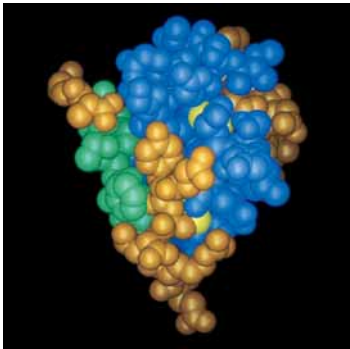
FIGURE 6.35 • Examples of the (a) disulfide-rich and (b) metal-rich proteins. (Jane Richardson)



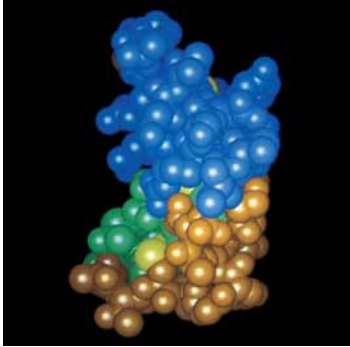
is a distorted α -helix cluster, whereas HiPIP is a distorted β -barrel structure. Others among this class (such as insulin and crambin), however, are not easily likened to any of the standard structure classes.

Molecular Chaperones: Proteins That Help Fold Globular Proteins

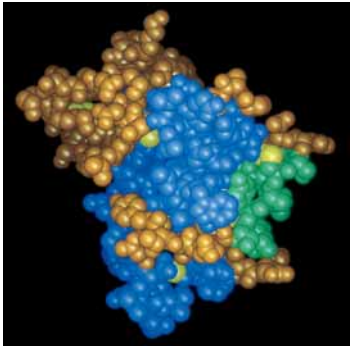
The landmark experiments by Christian Anfinsen on the refolding of ribonuclease clearly show that the refolding of a denatured protein *in vitro* can be a spontaneous process. As noted above, this refolding is driven by the small Gibbs free energy difference between the unfolded and folded states. It has also been generally assumed that all the information necessary for the correct folding of a polypeptide chain is contained in the primary structure and requires no additional molecular factors. However, the folding of proteins in the cell is a different matter. The highly concentrated protein matrix in the cell may adversely affect the folding process by causing aggregation of some unfolded or partially folded proteins. Also, it may be necessary to accelerate slow steps in the folding process or to suppress or reverse incorrect or premature folding. Recent studies have uncovered a family of proteins, known as **molecular chaperones**,



Insulin



Crambin



Phospholipase A₂



CRITICAL DEVELOPMENTS IN BIOCHEMISTRY

Thermodynamics of the Folding Process in Globular Proteins

Section 6.1 considered the noncovalent binding energies that stabilize a protein structure. However, the folding of a protein depends ultimately on the difference in Gibbs free energy (ΔG) between the folded (F) and unfolded (U) states at some temperature T :

$$\Delta G = G_F - G_U = \Delta H - T\Delta S$$

$$= (H_F - H_U) - T(S_F - S_U)$$

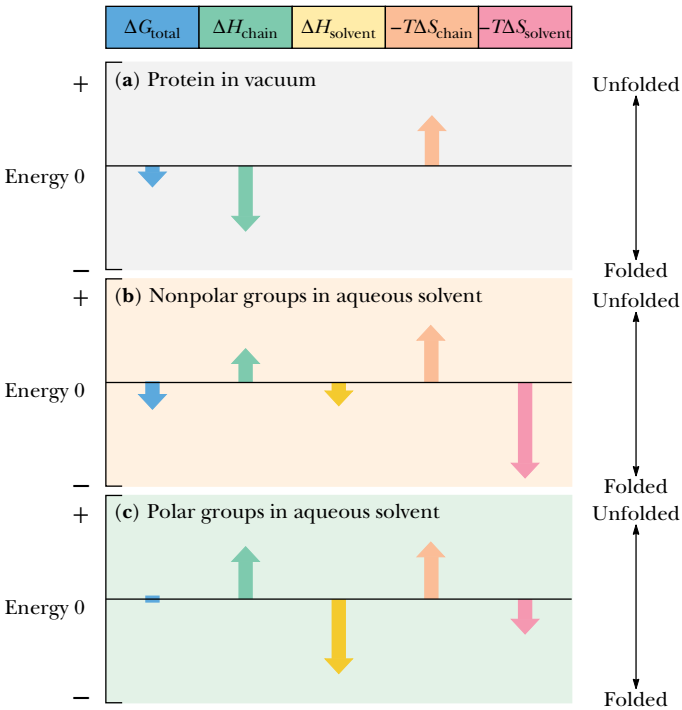
In the unfolded state, the peptide chain and its R groups interact with solvent water, and any measurement of the free energy change upon folding must consider contributions to the enthalpy change (ΔH) and the entropy change (ΔS) both for the polypeptide chain and for the solvent:

$$\Delta G_{\text{total}} = \Delta H_{\text{chain}} + \Delta H_{\text{solvent}} - T\Delta S_{\text{chain}} - T\Delta S_{\text{solvent}}$$

If each of the four terms on the right side of this equation is understood, the thermodynamic basis for protein folding should be clear. A summary of the signs and magnitudes of these quantities for a typical protein is shown in the accompanying figure. The folded protein is a highly ordered structure compared to the unfolded state, so ΔS_{chain} is a negative number and thus $-T\Delta S_{\text{chain}}$ is a positive quantity in the equation. The other terms depend on the nature of the particular ensemble of R groups. The nature of ΔH_{chain} depends on both residue–residue interactions and residue–solvent interactions. Nonpolar groups in the folded protein interact mainly with one another via weak van der Waals forces. Interactions between nonpolar groups and water in the unfolded state are stronger because the polar water molecules induce dipoles in the nonpolar groups, producing a significant electrostatic interaction. As a result, ΔH_{chain} is positive for nonpolar groups and favors the unfolded state. $\Delta H_{\text{solvent}}$ for nonpolar groups, however, is negative and favors the folded state. This is because folding allows many water molecules to interact (favorably) with one another rather than (less favorably) with the nonpolar side chains. The magnitude of ΔH_{chain} is smaller than that of $\Delta H_{\text{solvent}}$, but both these terms are small and usually do not dominate the folding process. However, $\Delta S_{\text{solvent}}$ for nonpolar

groups is large and positive and strongly favors the folded state. This is because nonpolar groups force order upon the water solvent in the unfolded state.

For polar side chains, ΔH_{chain} is positive and $\Delta H_{\text{solvent}}$ is negative. Because solvent molecules are ordered to some extent around polar groups, $\Delta S_{\text{solvent}}$ is small and positive. As shown in the figure, ΔG_{total} for the polar groups of a protein is near zero. Comparison of all the terms considered here makes it clear that *the single largest contribution to the stability of a folded protein is $\Delta S_{\text{solvent}}$ for the nonpolar residues.*



that appear to be essential for the correct folding of certain polypeptide chains *in vivo*, for their assembly into oligomers, and for preventing inappropriate liaisons with other proteins during their synthesis, folding, and transport. Many of these proteins were first identified as **heat shock proteins**, which are induced in cells by elevated temperature or other stress. The most thoroughly studied proteins are **hsp70**, a 70-kD heat shock protein, and the so-called **chaperonins**, also known as **cpn60s** or **hsp60s**, a class of 60-kD heat shock proteins. A well-characterized **hsp60** chaperonin is **GroEL**, an *E. coli* protein that has been shown to affect the folding of several proteins.

The way in which molecular chaperones interact with polypeptides during the folding process is not completely understood. What *is* clear is that chaperones bind effectively to the exposed hydrophobic regions of partially folded structures. These folding intermediates are less compact than the native folded proteins. They contain large amounts of secondary and even some tertiary

structure, but they undergo relatively large conformational fluctuations. It is possible that chaperone proteins recognize exposed helices or other secondary structure elements on their target proteins. This initial interaction may then allow the chaperone to guide or regulate the subsequent events of folding (Figure 6.36).

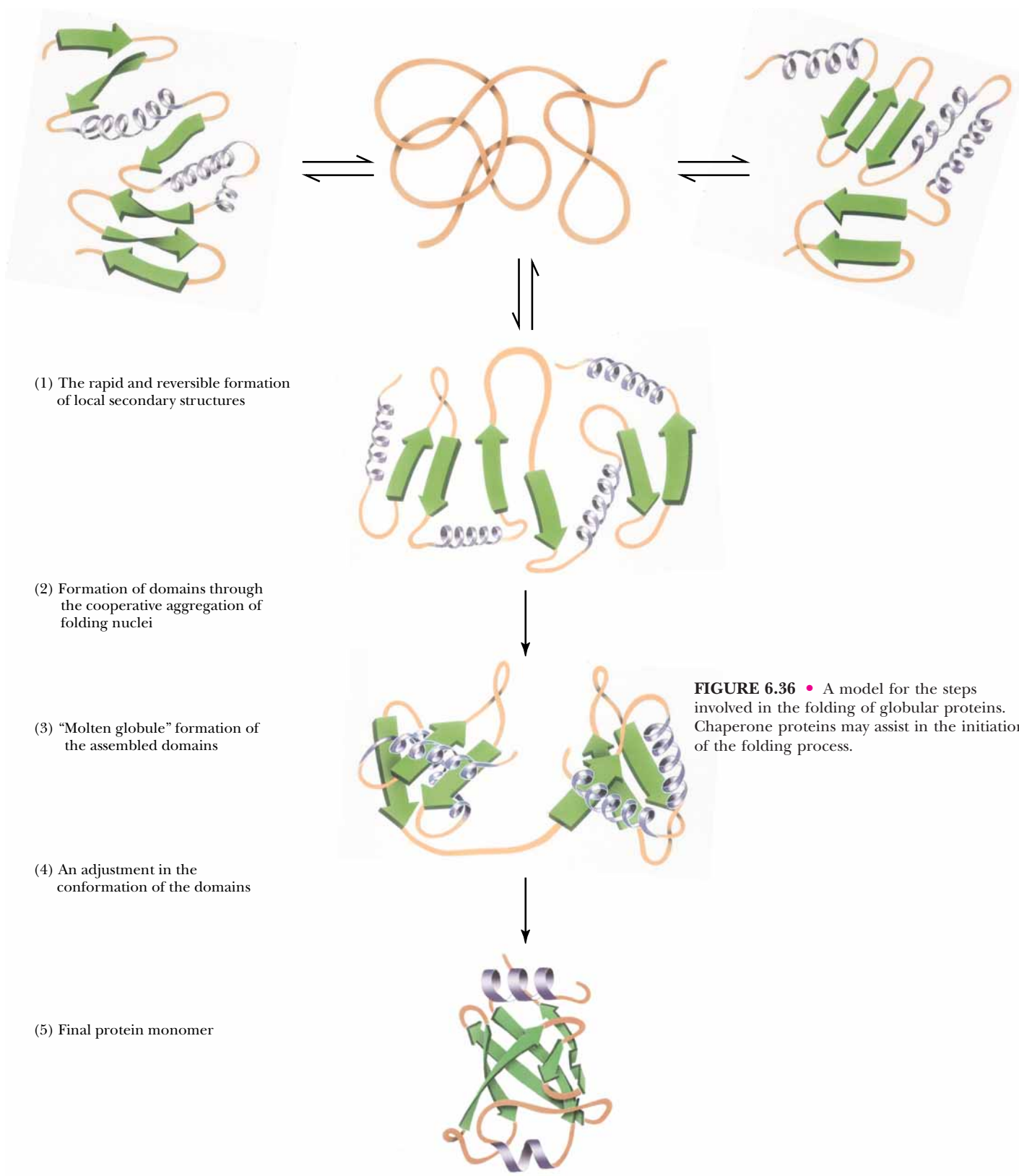


FIGURE 6.36 • A model for the steps involved in the folding of globular proteins. Chaperone proteins may assist in the initiation of the folding process.

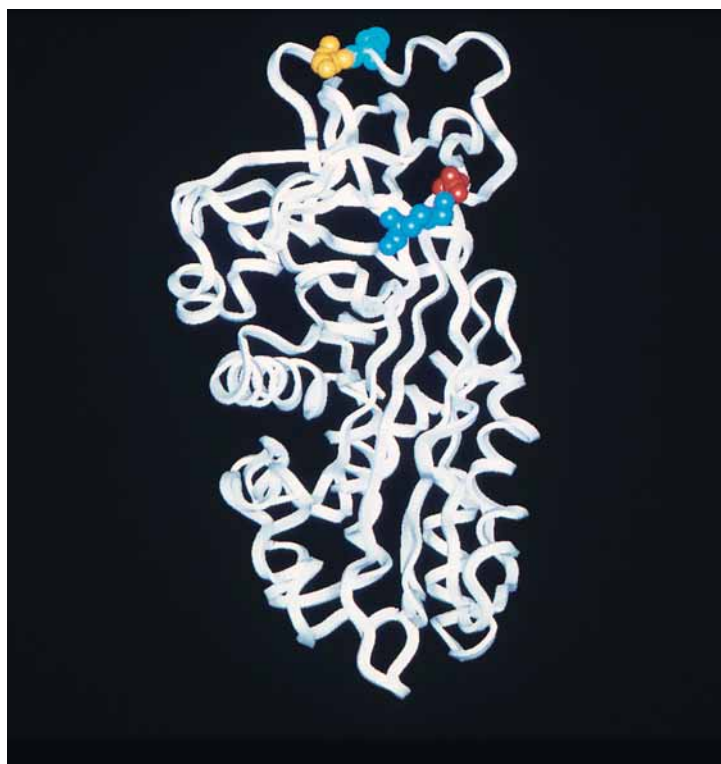
HUMAN BIOCHEMISTRY

A Mutant Protein That Folds Slowly Can Cause Emphysema and Liver Damage

Lungs enable animals to acquire oxygen from the air and to give off CO_2 produced in respiration. Exchange of oxygen and CO_2 occurs in the alveoli—air sacs surrounded by capillaries that connect the pulmonary veins and arteries. The walls of alveoli consist of the elastic protein **elastin**. Inhalation expands the alveoli and exhalation compresses them. A pair of human lungs contains 300 million alveoli, and the total area of the alveolar walls in contact with capillaries is about 70 m^2 —an area about the size of a tennis court! White blood cells naturally secrete **elastase**—a serine protease—which can attack and break down the elastin of the alveolar walls. However, **α_1 -antitrypsin**—a 52-kD protein belonging to the **serpin** (*serine protease inhibitor*) family—normally binds to elastase, preventing alveolar damage. The structural gene for α_1 -antitrypsin is extremely polymorphic (i.e., occurs as many different sequence variants), and several versions of this gene encode a protein that is poorly secreted into the circulation. Deficiency of α_1 -antitrypsin in the blood can lead to destruction of the alveolar walls by white cell elastase, resulting in **emphysema**—a condition in which the alveoli are destroyed, leaving large air sacs that cannot be compressed during exhalation.

α_1 -Antitrypsin normally adopts a highly ordered tertiary structure composed of three β -sheets and eight α -helices (see figure). Elastase and other serine proteases interact with a reactive, inhibitory site involving two amino acids—Met³⁵⁸ and Ser³⁵⁹—on the so-called **reactive-center loop**. Formation of a tight complex between elastase and α_1 -antitrypsin renders the elastase inactive. The most common α_1 -antitrypsin deficiency involves the so-called **Z-variant** of the protein, in which lysine is substituted for glutamate at position 342 (Glu³⁴²→Lys). Residue 342 lies at the amino-terminal base of the **reactive-center loop**, and glutamate at this position normally forms a crucial salt bridge with Lys²⁹⁰ on an adjacent strand of sheet A (see figure). In normal α_1 -antitrypsin, the reactive-center loop is fully exposed and can interact readily with elastase. However, in the Z-variant, the Glu³⁴²→Lys substitution destabilizes sheet A, separating the strands slightly and allowing the reactive-center loop of one molecule to insert into the β -sheet of another. Repetition of this anomalous association of α_1 -antitrypsin molecules results in “loop-sheet” polymerization and the formation of large protein aggregates. α_1 -Antitrypsin is synthesized in hepatocytes of the liver and is normally secreted into the circulation. Accumulation of Z-variant protein aggregates in the endoplasmic reticulum of liver cells leads to deficiency of circulating α_1 -antitrypsin. In some cases, the accumulation of these protein aggregates can also cause liver damage.

Myeong-Hee Yu and coworkers at the Korea Institute of Science and Technology have studied the folding kinetics of normal and Z-variant α_1 -antitrypsin and have found that the Z-variant of α_1 -antitrypsin folds identically to—but much more slowly than—normal α_1 -antitrypsin. Newly synthesized Z-variant protein, incubated for 5 hours at 30°C , eventually adopts a native and active conformation and can associate tightly with elastase. However, incubation of the Z-variant at 37°C results in loop-sheet polymerization and self-aggregation of the protein. These results imply that emphysema arising in individuals carrying the Z-variant of α_1 -antitrypsin is due to the slow folding kinetics of the protein rather than the adoption of an altered three-dimensional structure.



α_1 -Antitrypsin. Note Met³⁵⁸ (blue) and Ser³⁵⁹ (yellow) at top, as well as Glu³⁴² (red) and Lys²⁹⁰ (blue—upper right).

Protein Domains: Nature's Modular Strategy for Protein Design

Now that many thousands of proteins have been sequenced (more than 100,000 sequences are known), it has become obvious that certain protein sequences that give rise to distinct structural domains are used over and over again in modular fashion. These **protein modules** may occur in a wide variety of pro-

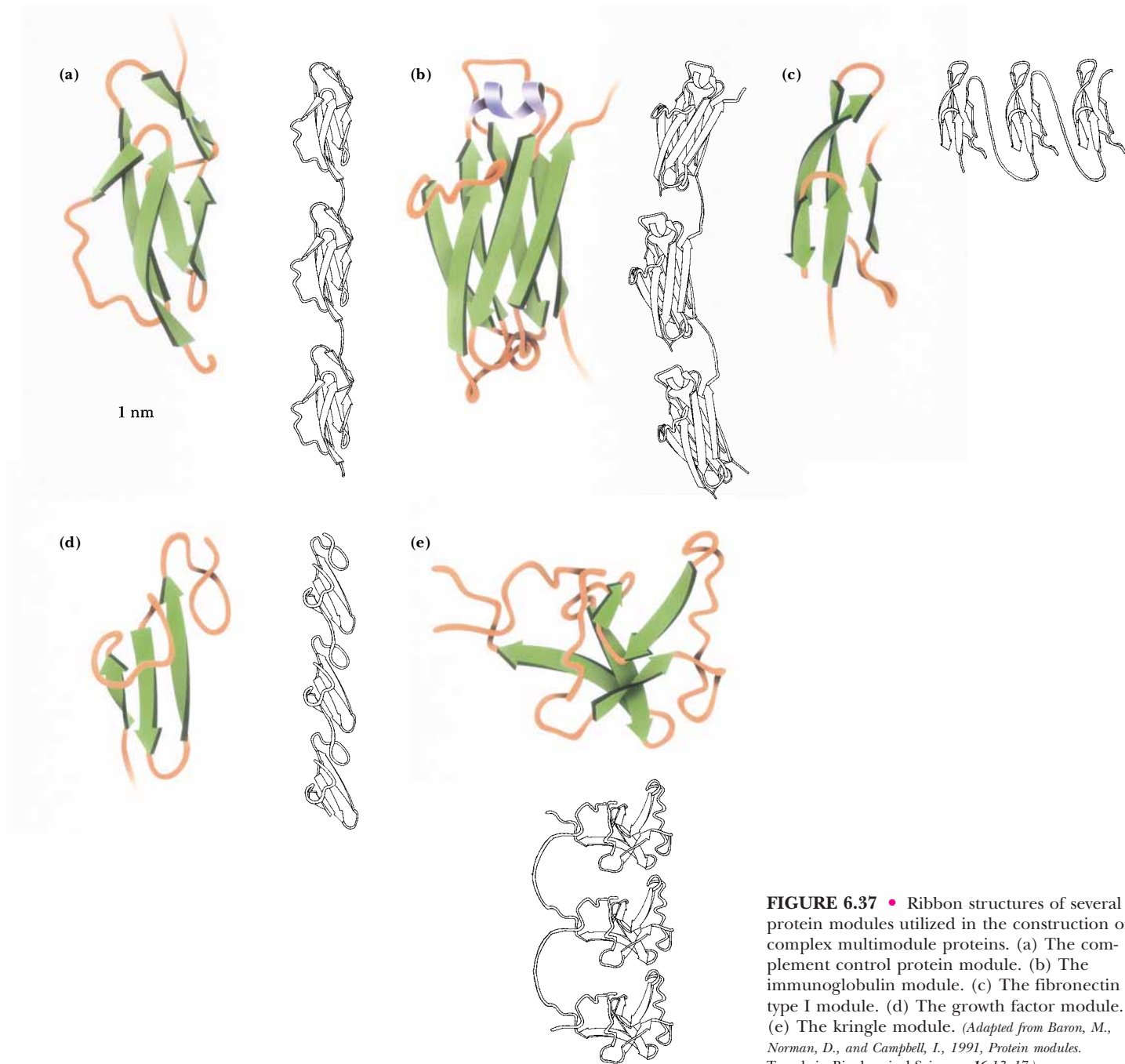
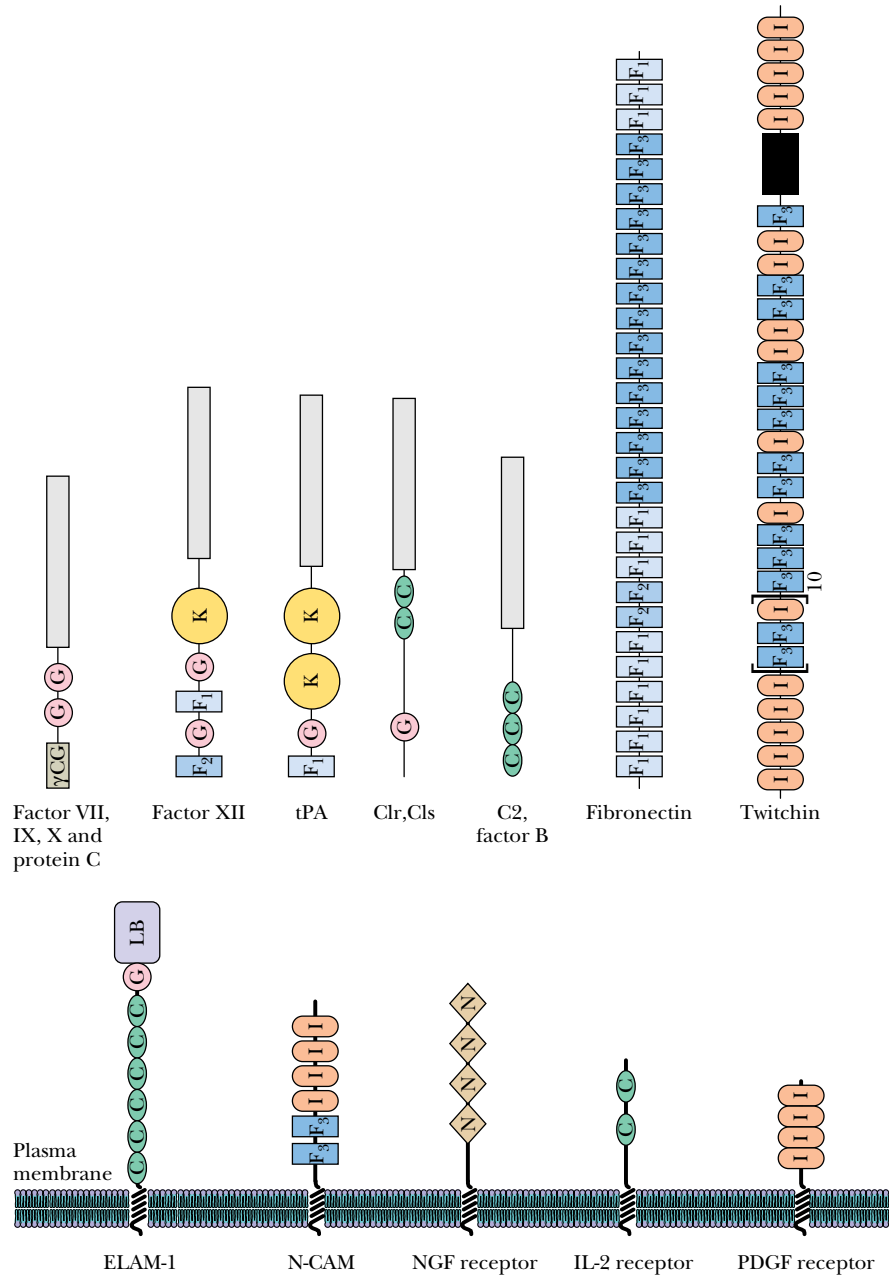


FIGURE 6.37 • Ribbon structures of several protein modules utilized in the construction of complex multimodule proteins. (a) The complement control protein module. (b) The immunoglobulin module. (c) The fibronectin type I module. (d) The growth factor module. (e) The kringle module. (Adapted from Baron, M., Norman, D., and Campbell, I., 1991, *Protein modules*. Trends in Biochemical Sciences 16:13–17.)

teins, often being used for different purposes, or they may be used repeatedly in the same protein. Figure 6.37 shows the tertiary structures of five protein modules, and Figure 6.38 presents several proteins that contain versions of these modules. These modules typically contain about 40 to 100 amino acids and often adopt a stable tertiary structure when isolated from their parent protein. One of the best-known examples of a protein module is the **immunoglobulin module**, which has been found not only in immunoglobulins but also in a wide variety of cell surface proteins, including cell adhesion molecules and growth factor receptors, and even in *twitchin*, an intracellular protein found in muscle. It is likely that more protein modules will be identified. (The role of protein modules in signal transduction is discussed in Chapter 34.)

FIGURE 6.38 • A sampling of proteins that consist of mosaics of individual protein modules. The modules shown include γ CG, a module containing γ -carboxyglutamate residues; G, an epidermal growth-factor–like module; K, the “kringle” domain, named for a Danish pastry; C, which is found in complement proteins; F1, F2, and F3, first found in fibronectin; I, the immunoglobulin superfamily domain; N, found in some growth factor receptors; E, a module homologous to the calcium-binding E-F hand domain; and LB, a lectin module found in some cell surface proteins. (Adapted from Baron, M., Norman, D., and Campbell, I., 1991, *Protein modules*. Trends in Biochemical Sciences **16**:13–17.)



How Do Proteins Know How to Fold?

Christian Anfinsen’s experiments demonstrated that proteins can fold reversibly. A corollary result of Anfinsen’s work is that the native structures of at least some globular proteins are thermodynamically stable states. But the matter of how a given protein achieves such a stable state is a complex one. Cyrus Levinthal pointed out in 1968 that so many conformations are possible for a typical protein that the protein does not have sufficient time to reach its most stable conformational state by sampling all the possible conformations. This argument, termed “Levinthal’s paradox,” goes as follows: consider a protein of 100 amino acids. Assume that there are only two conformational possibilities per amino acid, or $2^{100} = 1.27 \times 10^{30}$ possibilities. Allow 10^{-13} sec for

the protein to test each conformational possibility in search of the overall energy minimum:

$$(10^{-13} \text{ sec})(1.27 \times 10^{30}) = 1.27 \times 10^{17} \text{ sec} = 4 \times 10^9 \text{ years}$$

Levinthal’s paradox led protein chemists to hypothesize that proteins must fold by specific “folding pathways,” and many research efforts have been devoted to the search for these pathways.

Implicit in the presumption of folding pathways is the existence of intermediate, partially folded conformational states. The notion of intermediate states on the pathway to a tertiary structure raises the possibility that segments of a protein might independently adopt local and well-defined secondary structures (α -helices and β -sheets). The tendency of a peptide segment to prefer a particular secondary structure depends in turn on its amino acid composition and sequence.

Surveys of the frequency with which various residues appear in helices and sheets show that some residues, such as alanine, glutamate, and methionine, occur much more frequently in α -helices than do others. In contrast, glycine and proline are the least likely residues to be found in an α -helix. Likewise, certain residues, including valine, isoleucine, and the aromatic amino acids, are more likely to be found in β -sheets than other residues, and aspartate, glutamate, and proline are much less likely to be found in β -sheets.

Such observations have led to many efforts to predict the occurrence of secondary structure in proteins from knowledge of the peptide sequence. Such **predictive algorithms** consider the composition of short segments of a polypeptide. If these segments are rich in residues that are found frequently in helices or sheets, then that segment is judged likely to adopt the corresponding secondary structure. The predictive algorithm designed by Peter Chou and Gerald Fasman in 1974 used data like that in Figure 6.39 to classify the 20 amino acids for their α -helix-forming and β -sheet-forming **propensities**, P_α and P_β (Table 6.3). These residues are classified as strong helix formers (H_α), helix formers (h_α), weak helix formers (I_α), indifferent helix formers (i_α), helix breakers (b_α), and strong helix breakers (B_α). Similar classes were established by Chou

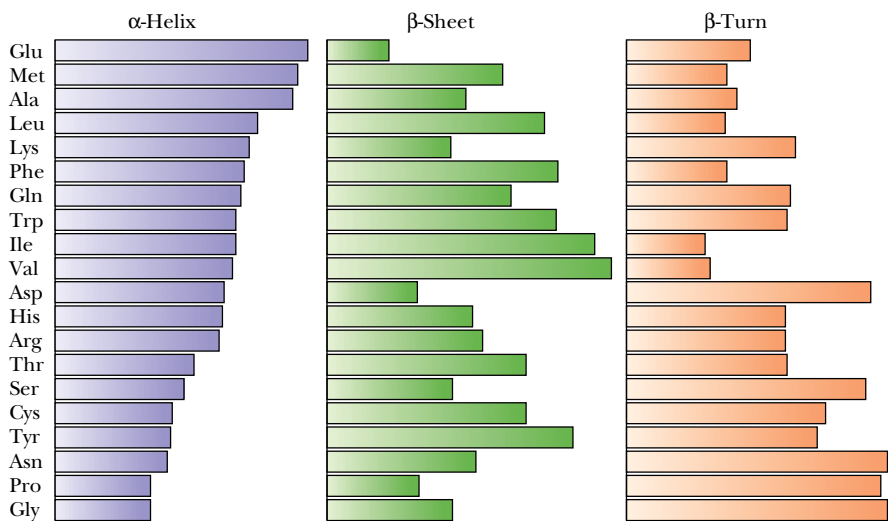


FIGURE 6.39 • Relative frequencies of occurrence of amino acid residues in α -helices, β -sheets, and β -turns in proteins of known structure. (Adapted from Bell, J. E., and Bell, E. T., 1988, *Proteins and Enzymes*, Englewood Cliffs, NJ: Prentice-Hall.)

Table 6.3

Chou–Fasman Helix and Sheet Propensities (P_α and P_β) of the Amino Acids

Amino Acid	P_α	Helix Classification	P_β	Sheet Classification
A Ala	1.42	H $_\alpha$	0.83	i $_\beta$
C Cys	0.70	i $_\alpha$	1.19	h $_\beta$
D Asp	1.01	I $_\alpha$	0.54	B $_\beta$
E Glu	1.51	H $_\alpha$	0.37	B $_\beta$
F Phe	1.13	h $_\alpha$	1.38	h $_\beta$
G Gly	0.57	B $_\alpha$	0.75	b $_\beta$
H His	1.00	I $_\alpha$	0.87	h $_\beta$
I Ile	1.08	h $_\alpha$	1.60	H $_\beta$
K Lys	1.16	h $_\alpha$	0.74	b $_\beta$
L Leu	1.21	H $_\alpha$	1.30	h $_\beta$
M Met	1.45	H $_\alpha$	1.05	h $_\beta$
N Asn	0.67	b $_\alpha$	0.89	i $_\beta$
P Pro	0.57	B $_\alpha$	0.55	B $_\beta$
Q Gln	1.11	h $_\alpha$	1.10	h $_\beta$
R Arg	0.98	i $_\alpha$	0.93	i $_\beta$
S Ser	0.77	i $_\alpha$	0.75	b $_\beta$
T Thr	0.83	i $_\alpha$	1.19	h $_\beta$
V Val	1.06	h $_\alpha$	1.70	H $_\beta$
W Trp	1.08	h $_\alpha$	1.37	h $_\beta$
Y Tyr	0.69	b $_\alpha$	1.47	H $_\beta$

Source: Chou, P. Y., and Fasman, G. D., 1978. *Annual Review of Biochemistry* 47:258.

and Fasman for β -sheet-forming ability. Such algorithms are only modestly successful in predicting the occurrence of helices and sheets in proteins.

George Rose and Rajgopal Srinivasan at Johns Hopkins University have taken a different, and very successful, approach to the prediction of protein structures. They begin by assuming that protein folding is both *local* and *hierarchical*. “Local” in this context means that each amino acid’s folding behavior is influenced by other residues nearby in the sequence. “Hierarchical” means that folded structures develop from the smallest structural units and work up to more and more complex entities. These and other assumptions are the basis for a computer program called **LINUS**—for Local Independently Nucleated Units of Structure—which Rose and Srinivasan have used to generate remarkably accurate predicted structures for a number of small proteins. LINUS considers groups of three amino acids in a sequence—for example, residues 2, 3, and 4 in a sequence of 50. The initial assumption is that this group of amino acids will (randomly) adopt one of four possible structures—helix, sheet, turn, or “loop” (any tertiary structure that is not helix, sheet, or turn). The program then asks whether this assumed “ministructure” is energetically suited to the six amino acids on either side. The program then moves on to the next set of three residues—amino acids 3, 4, and 5 in the present case—and randomly selects one of the four secondary structures for this unit, evaluates the energetic consequences in terms of the six residues on either side, and then continues in like manner to groups 4, 5, and 6, then 5, 6, and 7, and so on to the end of the protein. The program carries out this random selection and testing

routine a total of 5000 times for the entire protein. Once this is done, the program analyzes all the trials, looking for local groups of amino acids that seem to prefer one of the four conformations 70% of the time. Such groups are then held in those conformations while the program repeats the entire process, this time comparing energetic preferences with respect to 12 amino acids on either side of selected groups of three residues, then 18 amino acids on either side, 24, 32, and so on up to 48 residues. The results of calculations with LINUS on several proteins whose structures are known are shown in Figure 6.40.

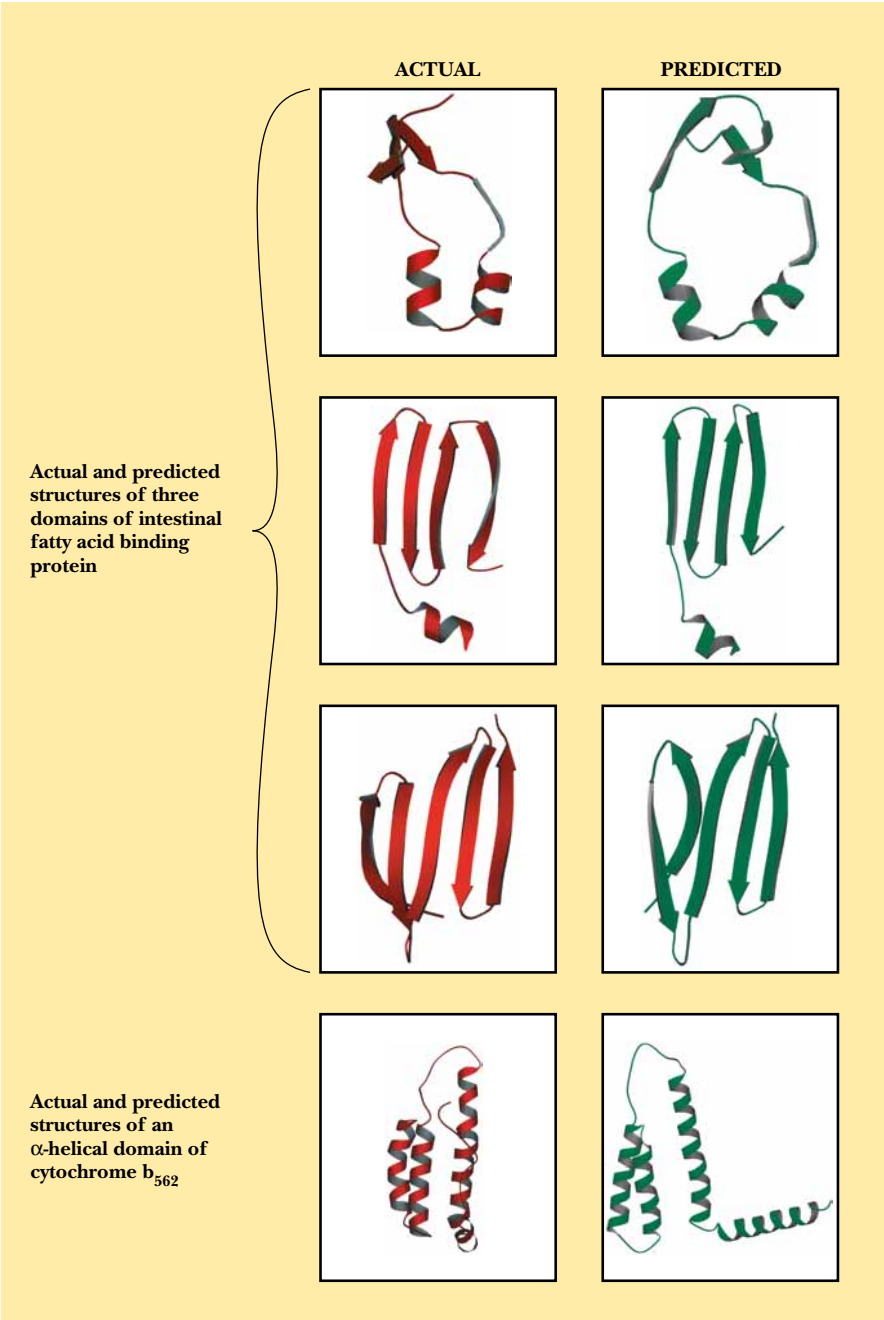


FIGURE 6.40 • A comparison of the structures of four protein domains and predictions of these structures by the program LINUS by Rose and Srinivasan. (Professor George Rose/Johns Hopkins University)

6.5 • Subunit Interactions and Quaternary Structure

Many proteins exist in nature as **oligomers**, complexes composed of (often symmetric) noncovalent assemblies of two or more monomer subunits. In fact, subunit association is a common feature of macromolecular organization in biology. Most intracellular enzymes are oligomeric and may be composed either of a single type of monomer subunit (*homomultimers*) or of several different kinds of subunits (*heteromultimers*). The simplest case is a protein composed of identical subunits. Liver alcohol dehydrogenase, shown in Figure 6.41, is such a protein. More complicated proteins may have several different subunits in one, two, or more copies. Hemoglobin, for example, contains two each of two different subunits and is referred to as an $\alpha_2\beta_2$ -complex. An interesting counterpoint to these relatively simple cases is made by the proteins that form polymeric structures. Tubulin is an $\alpha\beta$ -dimeric protein that polymerizes to form microtubules of the formula $(\alpha\beta)_n$. The way in which separate folded monomeric protein subunits associate to form the oligomeric protein constitutes the **quaternary structure** of that protein. Table 6.4 lists several proteins and their subunit compositions (see also Table 5.1). Clearly, proteins with two to four subunits dominate the list, but many cases of higher numbers exist.

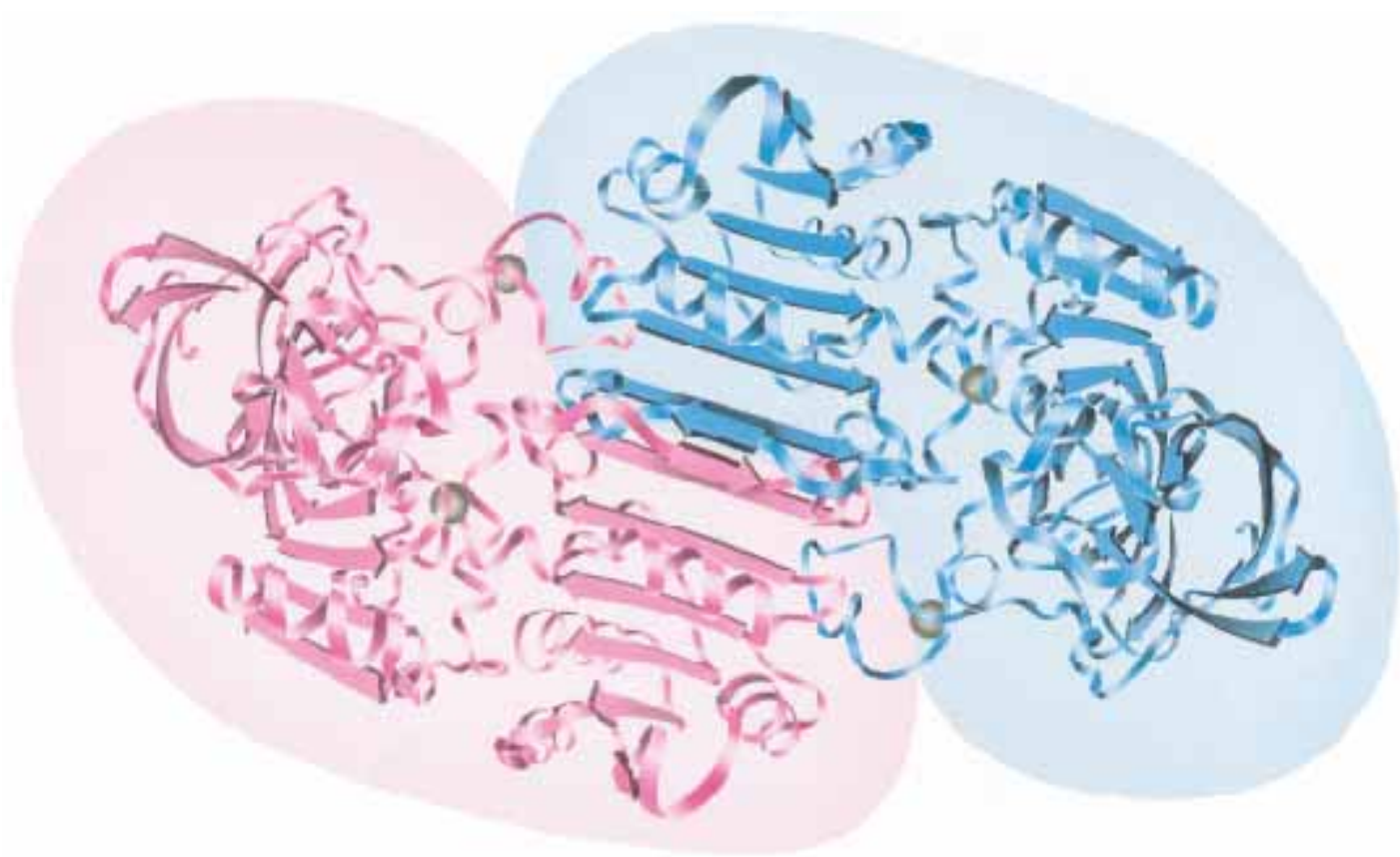


FIGURE 6.41 • The quaternary structure of liver alcohol dehydrogenase. Within each subunit is a six-stranded parallel sheet. Between the two subunits is a two-stranded antiparallel sheet. The point in the center is a C_2 symmetry axis. (*Jane Richardson*)

The subunits of an oligomeric protein typically fold into apparently independent globular conformations and then interact with other subunits. The particular surfaces at which protein subunits interact are similar in nature to the interiors of the individual subunits. These interfaces are closely packed and involve both polar and hydrophobic interactions. Interacting surfaces must therefore possess complementary arrangements of polar and hydrophobic groups.

Oligomeric associations of protein subunits can be divided into those between **identical** subunits and those between **nonidentical** subunits. Interactions among identical subunits can be further distinguished as either **isologous** or **heterologous**. In isologous interactions, the interacting surfaces are identical, and the resulting structure is necessarily dimeric and closed, with a twofold axis of symmetry (Figure 6.42). If any additional interactions occur to form a trimer or tetramer, these must use different interfaces on the protein’s surface. Many proteins, including concanavalin and prealbumin, form tetramers by means of two sets of isologous interactions, one of which is shown in Figure 6.43. Such structures possess three different twofold axes of symmetry. In contrast, heterologous associations among subunits involve nonidentical interfaces. These surfaces must be complementary, but they are generally not symmetric. As shown in Figure 6.43, heterologous interactions are necessarily open-ended. This can give rise either to a closed cyclic structure, if geometric constraints exist, or to large polymeric assemblies. The closed cyclic structures are far more common and include the trimers of aspartate transcarbamoylase catalytic subunits and the tetramers of neuraminidase and hemerythrin.

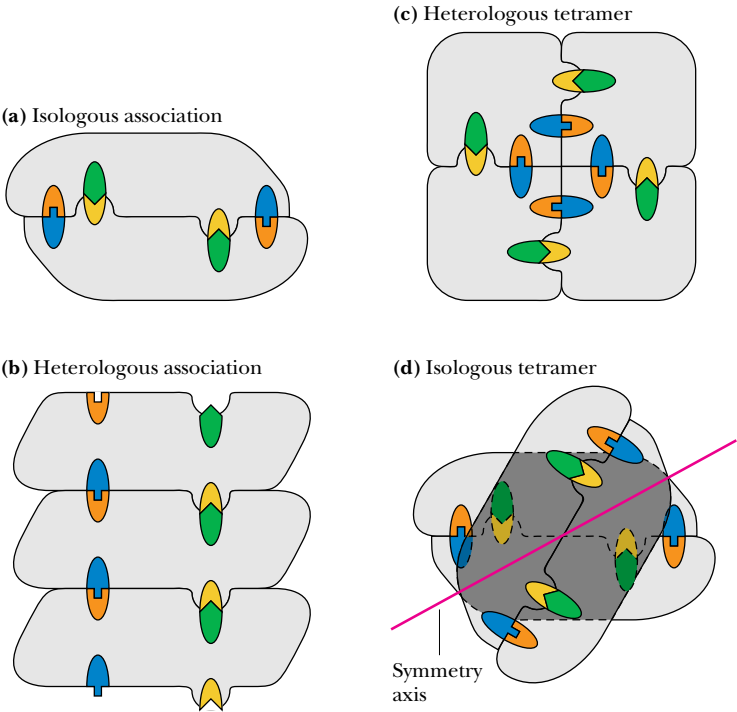


FIGURE 6.42 • Isologous and heterologous associations between protein subunits. (a) An isologous interaction between two subunits with a twofold axis of symmetry perpendicular to the plane of the page. (b) A heterologous interaction that could lead to the formation of a long polymer. (c) A heterologous interaction leading to a closed structure—a tetramer. (d) A tetramer formed by two sets of isologous interactions.

Table 6.4
Aggregation Symmetries of Globular Proteins

Protein	Number of Subunits
Alcohol dehydrogenase	2
Immunoglobulin	4
Malate dehydrogenase	2
Superoxide dismutase	2
Triose phosphate isomerase	2
Glycogen phosphorylase	2
Alkaline phosphatase	2
6-Phosphogluconate dehydrogenase	2
Wheat germ agglutinin	2
Phosphoglucosomerase	2
Tyrosyl-tRNA synthetase	2
Glutathione reductase	2
Aldolase	3
Bacteriochlorophyll protein	3
TMV protein disc	17
Concanavalin A	4
Glyceraldehyde-3-phosphate dehydrogenase	4
Lactate dehydrogenase	4
Prealbumin	4
Pyruvate kinase	4
Phosphoglycerate mutase	4
Hemoglobin	2 + 2
Insulin	6
Aspartate transcarbamoylase	6 + 6
Glutamine synthetase	12
Apo ferritin	24
Coat of tomato bushy stunt virus	180

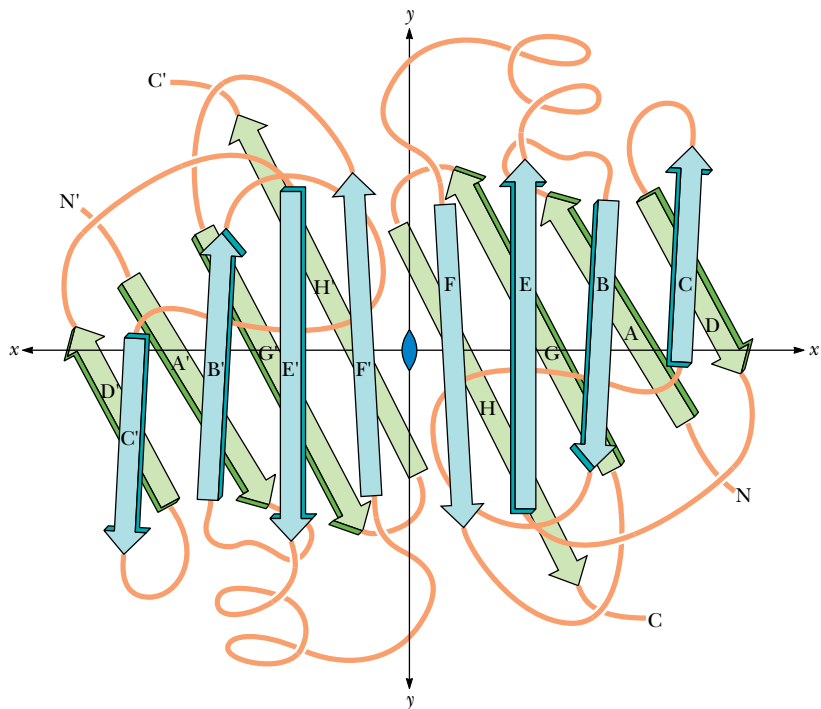


FIGURE 6.43 • The polypeptide backbone of the prealbumin dimer. The monomers associate in a manner that continues the β -sheets. A tetramer is formed by isologous interactions between the side chains extending outward from sheet D'A'G'H'HGAD in both dimers, which pack together nearly at right angles to one another. (Jane Richardson)

Symmetry of Quaternary Structures

One useful way to consider quaternary interactions in proteins involves the symmetry of these interactions. Globular protein subunits are always asymmetric objects. All of the polypeptide's α -carbons are asymmetric, and the polypeptide nearly always folds to form a low-symmetry structure. (The long helical arrays formed by some synthetic polypeptides are an exception.) Thus, protein subunits do not have mirror reflection planes, points, or axes of inversion. The only symmetry operation possible for protein subunits is a rotation. The most common symmetries observed for multisubunit proteins are cyclic symmetry and dihedral symmetry. In **cyclic symmetry**, the subunits are arranged around a single rotation axis, as shown in Figure 6.44. If there are two subunits, the axis is referred to as a *twofold rotation axis*. Rotating the quaternary structure 180° about this axis gives a structure identical to the original one. With three subunits arranged about a threefold rotation axis, a rotation of 120° about that axis gives an identical structure. **Dihedral symmetry** occurs when a structure possesses at least one twofold rotation axis perpendicular to another n -fold rotation axis. This type of subunit arrangement (Figure 6.44) occurs in concanavalin A (where $n = 2$) and in insulin (where $n = 3$). Higher symmetry groups, including the tetrahedral, octahedral, and icosahedral symmetries, are much less common among multisubunit proteins, partly because of the large number of asymmetric subunits required to assemble truly symmetric tetrahedra and other high symmetry groups. For example, a truly symmetric tetrahedral protein structure would require 12 identical monomers arranged in triangles, as shown in Figure 6.45. Simple four-subunit tetrahedra of protein monomers, which actually possess dihedral symmetry, are more common in biological systems.

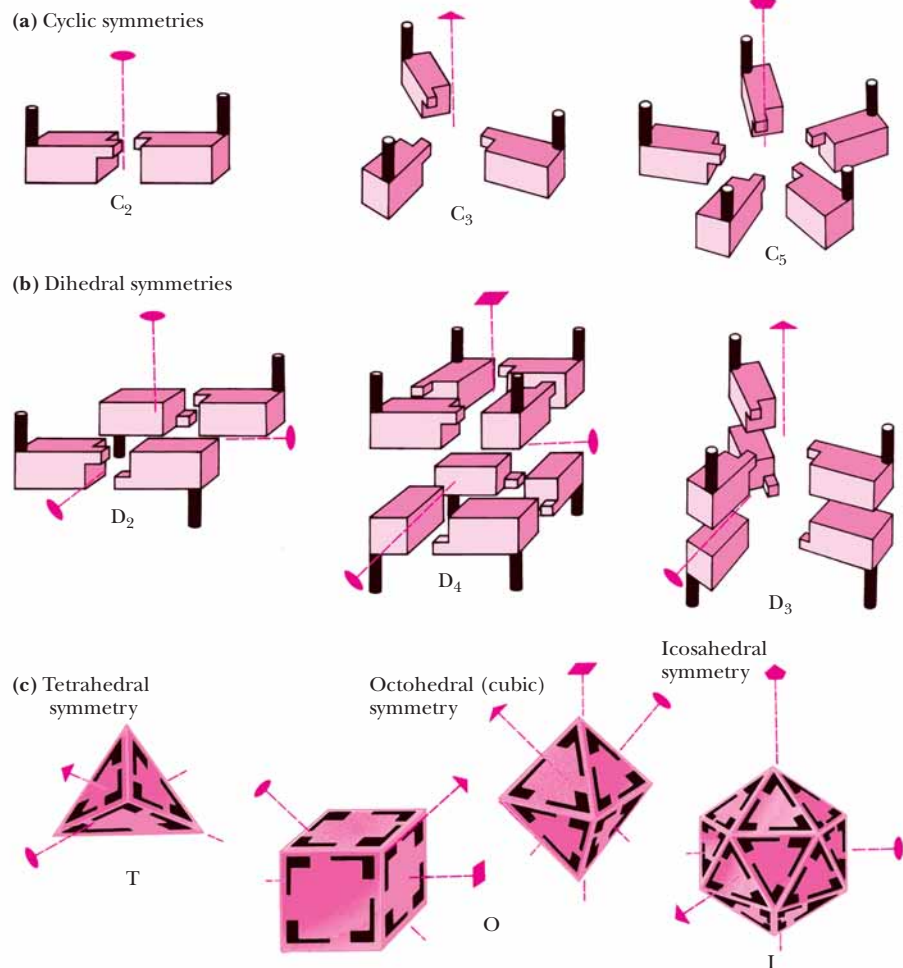


FIGURE 6.44 • Several possible symmetric arrays of identical protein subunits, including (a) cyclic symmetry, (b) dihedral symmetry, and (c) cubic symmetry, including examples of tetrahedral (T), octahedral (O), and icosahedral (I) symmetry. (Irving Geis)

Forces Driving Quaternary Association

The forces that stabilize quaternary structure have been evaluated for a few proteins. Typical dissociation constants for simple two-subunit associations range from 10^{-8} to 10^{-16} M. These values correspond to free energies of association of about 50 to 100 kJ/mol at 37°C. Dimerization of subunits is accompanied by both favorable and unfavorable energy changes. The favorable interactions include van der Waals interactions, hydrogen bonds, ionic bonds, and hydrophobic interactions. However, considerable entropy loss occurs when subunits interact. When two subunits move as one, three translational degrees of freedom are lost for one subunit because it is constrained to move with the other one. In addition, many peptide residues at the subunit interface, which were previously free to move on the protein surface, now have their movements restricted by the subunit association. This unfavorable energy of association is in the range of 80 to 120 kJ/mol for temperatures of 25 to 37°C. Thus, to achieve stability, the dimerization of two subunits must involve approximately 130 to 220 kJ/mol of favorable interactions.¹ Van der Waals interactions at protein interfaces are numerous, often running to several hundred for a typical monomer–monomer association. This would account for about 150 to 200

¹For example, 130 kJ/mol of favorable interaction minus 80 kJ/mol of unfavorable interaction equals a net free energy of association of 50 kJ/mol.

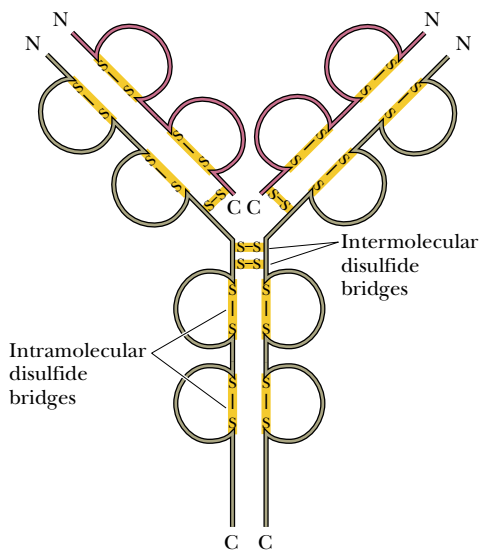


FIGURE 6.45 • Schematic drawing of an immunoglobulin molecule showing the intramolecular and intermolecular disulfide bridges. (A space-filling model of the same molecule is shown in Figure 1.11.)

kJ/mol of favorable free energy of association. However, when solvent is removed from the protein surface to form the subunit–subunit contacts, nearly as many van der Waals associations are lost as are made. One subunit is simply trading water molecules for peptide residues in the other subunit. As a result, the energy of subunit association due to van der Waals interactions actually contributes little to the stability of the dimer. Hydrophobic interactions, however, are generally very favorable. For many proteins, the subunit association process effectively buries as much as 20 nm² of surface area previously exposed to solvent, resulting in as much as 100 to 200 kJ/mol of favorable hydrophobic interactions. Together with whatever polar interactions occur at the protein–protein interface, this is sufficient to account for the observed stabilization that occurs when two protein subunits associate.

An additional and important factor contributing to the stability of subunit associations for some proteins is the formation of disulfide bonds between different subunits. All antibodies are $\alpha_2\beta_2$ -tetramers composed of two heavy chains (53 to 75 kD) and two relatively light chains (23 kD). In addition to *intrasubunit* disulfide bonds (four per heavy chain, two per light chain), two *intersubunit* disulfide bridges hold the two heavy chains together and a disulfide bridge links each of the two light chains to a heavy chain (Figure 6.45).

Modes and Models for Quaternary Structures

When a protein is composed of only one kind of polypeptide chain, the manner in which the subunits interact and the arrangement of the subunits to produce the quaternary structure are usually simple matters. Sometimes, however, the same protein derived from several different species can exhibit different modes of quaternary interactions. Hemerythrin, the oxygen-carrying protein in certain species of marine invertebrates, is composed of a compact arrangement of four antiparallel α -helices. It is capable of forming dimers, trimers, tetramers, octamers, and even higher aggregates (Figure 6.46).

When two or more distinct peptide chains are involved, the nature of their interactions can be quite complicated. Multimeric proteins with more than one kind of subunit often display different affinities between different pairs of sub-

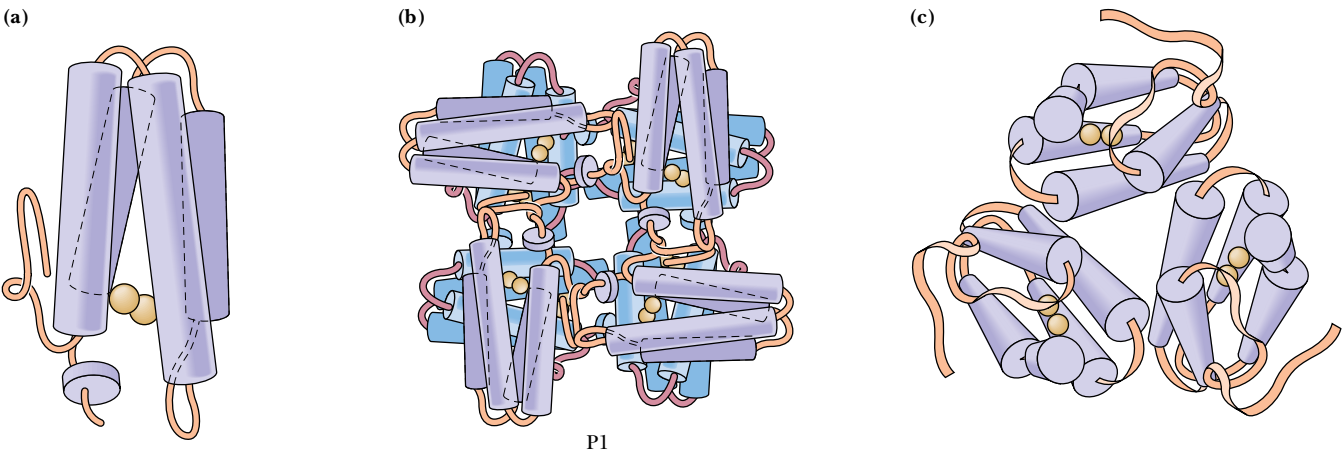


FIGURE 6.46 • The oligomeric states of hemerythrin in various marine worms. (a) The hemerythrin in *Thermiste zostericola* crystallized as a monomer; (b) the octameric hemerythrin crystallized from *Phascolopsis gouldii*; (c) the trimeric hemerythrin crystallized from *Siphonossoma* collected in mangrove swamps in Fiji.

DEEPER LOOK

Immunoglobulins—All the Features of Protein Structure Brought Together

The immunoglobulin structure in Figure 6.45 represents the confluence of all the details of protein structure that have been thus far discussed. As for all proteins, the primary structure determines other aspects of structure. There are numerous elements of secondary structure, including β -sheets and tight turns. The tertiary structure consists of 12 distinct domains, and the protein adopts a heterotetrameric quaternary structure. To make matters more interesting, both intrasubunit and intersubunit disulfide linkages act to stabilize the discrete domains and to stabilize the tetramer itself.

One more level of sophistication awaits. As discussed in

Chapter 29, the amino acid sequences of both light and heavy immunoglobulin chains are not constant! Instead, the primary structure of these chains is highly variable in the N-terminal regions (first 108 residues). Heterogeneity of amino acid sequence leads to variations in the conformation of these variable regions. This variation accounts for antibody diversity and the ability of antibodies to recognize and bind a virtually limitless range of antigens. This full potential of antibody:antigen recognition enables organisms to mount immunologic responses to almost any antigen that might challenge the organism.

units. Whereas strongly denaturing solvents may dissociate the protein entirely into monomers, more subtle denaturing conditions may dissociate the oligomeric structure in a carefully controlled stepwise manner. Hemoglobin is a good example. Strong denaturants dissociate hemoglobin into α - and β -monomers. Using mild denaturing conditions, however, it is possible to dissociate hemoglobin almost completely into $\alpha\beta$ -dimers, with few or no free monomers occurring. In this sense, hemoglobin behaves functionally like a two-subunit protein, with each “subunit” composed of an $\alpha\beta$ -dimer.

Open Quaternary Structures and Polymerization

All of the quaternary structures we have considered to this point have been **closed** structures, with a limited capacity to associate. Many proteins in nature associate to form **open** heterologous structures, which can polymerize more or less indefinitely, creating structures that are both esthetically attractive and functionally important to the cells or tissue in which they exist. One such protein is **tubulin**, the $\alpha\beta$ -dimeric protein that polymerizes into long, tubular structures, which are the structural basis of cilia flagella and the cytoskeletal matrix. The *microtubule* thus formed (Figure 6.47) may be viewed as consisting of 13 parallel filaments arising from end-to-end aggregation of the tubulin dimers. Human immunodeficiency virus, HIV, the causative agent of AIDS (also discussed in Chapter 16), is enveloped by a spherical shell composed of hundreds of coat protein subunits, a large-scale quaternary association.

Structural and Functional Advantages of Quaternary Association

There are several important reasons for protein subunits to associate in oligomeric structures.

Stability

One general benefit of subunit association is a favorable reduction of the protein’s surface-to-volume ratio. The surface-to-volume ratio becomes smaller as the radius of any particle or object becomes larger. (This is because surface area is a function of the radius squared and volume is a function of the radius cubed.) Because interactions within the protein usually tend to stabilize the protein energetically and because the interaction of the protein surface with

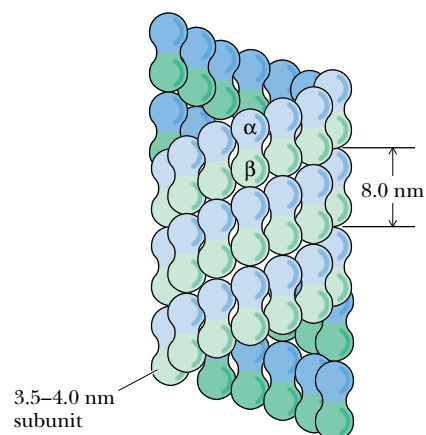


FIGURE 6.47 • The structure of a typical microtubule, showing the arrangement of the α - and β -monomers of the tubulin dimer.

solvent water is often energetically unfavorable, decreased surface-to-volume ratios usually result in more stable proteins. Subunit association may also serve to shield hydrophobic residues from solvent water. Subunits that recognize either themselves or other subunits avoid any errors arising in genetic translation by binding mutant forms of the subunits less tightly.

Genetic Economy and Efficiency

Oligomeric association of protein monomers is genetically economical for an organism. Less DNA is required to code for a monomer that assembles into a homomultimer than for a large polypeptide of the same molecular mass. Another way to look at this is to realize that virtually all of the information that determines oligomer assembly and subunit–subunit interaction is contained in the genetic material needed to code for the monomer. For example, HIV protease, an enzyme that is a dimer of identical subunits, performs a catalytic function similar to homologous cellular enzymes that are single polypeptide chains of twice the molecular mass (Chapter 16).

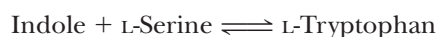
Bringing Catalytic Sites Together

Many enzymes (see Chapters 14 to 16) derive at least some of their catalytic power from oligomeric associations of monomer subunits. This can happen in several ways. The monomer may not constitute a complete enzyme active site. Formation of the oligomer may bring all the necessary catalytic groups together to form an active enzyme. For example, the active sites of bacterial glutamine synthetase are formed from pairs of adjacent subunits. The dissociated monomers are inactive.

Oligomeric enzymes may also carry out different but related reactions on different subunits. Thus, tryptophan synthase is a tetramer consisting of pairs of different subunits, $\alpha_2\beta_2$. Purified α -subunits catalyze the following reaction:



and the β -subunits catalyze this reaction:



Indole, the product of the α -reaction and the reactant for the β -reaction, is passed directly from the α -subunit to the β -subunit and cannot be detected as a free intermediate.

Cooperativity

There is another, more important reason for monomer subunits to associate into oligomeric complexes. Most oligomeric enzymes regulate catalytic activity by means of subunit interactions, which may give rise to cooperative phenomena. Multisubunit proteins typically possess multiple binding sites for a given ligand. If the binding of ligand at one site changes the affinity of the protein for ligand at the other binding sites, the binding is said to be **cooperative**. Increases in affinity at subsequent sites represent positive cooperativity, whereas decreases in affinity correspond to negative cooperativity. The points of contact between protein subunits provide a mechanism for communication between the subunits. This in turn provides a way in which the binding of ligand to one subunit can influence the binding behavior at the other subunits. Such cooperative behavior, discussed in greater depth in Chapter 15, is the underlying mechanism for regulation of many biological processes.

Faster-Acting Insulin: Genetic Engineering Solves a Quaternary Structure Problem

Insulin is a peptide hormone, secreted by the pancreas, that regulates glucose metabolism in the body. Insufficient production of insulin or failure of insulin to stimulate target sites in liver, muscle, and adipose tissue leads to the serious metabolic disorder known as *diabetes mellitus*. Diabetes afflicts millions of people worldwide. Diabetic individuals typically exhibit high levels of glucose in the blood, but insulin injection therapy allows diabetic individuals to maintain normal levels of blood glucose.

Insulin is composed of two peptide chains covalently linked by disulfide bonds (Figures 5.17 and 6.35). This “monomer” of insulin is the active form that binds to receptors in target cells. However, in solution, insulin spontaneously forms dimers, which themselves aggregate to form hexamers. The surface of the insulin molecule that self-associates to form hexamers is also the surface that binds to insulin receptors in target cells. Thus, hexamers of insulin are inactive.

Insulin released from the pancreas is monomeric and acts

rapidly at target tissues. However, when insulin is administered (by injection) to a diabetic patient, the insulin hexamers dissociate slowly, and the patient’s blood glucose levels typically drop slowly (over a period of several hours).

In 1988, G. Dodson showed that insulin could be genetically engineered to prefer the monomeric (active) state. Dodson and his colleagues used recombinant DNA technology (discussed in Chapter 13) to produce insulin with an aspartate residue replacing a proline at the contact interface between adjacent subunits. The negative charge on the Asp side chain creates electrostatic repulsion between subunits and increases the dissociation constant for the hexamer $\rightleftharpoons$ monomer equilibrium. Injection of this mutant insulin into test animals produced more rapid decreases in blood glucose than did ordinary insulin. The Danish pharmaceutical company Novo is conducting clinical trials of the mutant insulin, which may eventually replace ordinary insulin in treatment of diabetes.

PROBLEMS

- The central rod domain of a keratin protein is approximately 312 residues in length. What is the length (in Å) of the keratin rod domain? If this same peptide segment were a true α -helix, how long would it be? If the same segment were a β -sheet, what would its length be?
- A teenager can grow 4 inches in a year during a “growth spurt.” Assuming that the increase in height is due to vertical growth of collagen fibers (in bone), calculate the number of collagen helix turns synthesized per minute.
- Discuss the potential contributions to hydrophobic and van der Waals interactions and ionic and hydrogen bonds for the side chains of Asp, Leu, Tyr, and His in a protein.
- Figure 6.40 shows that Pro is the amino acid least commonly found in α -helices but most commonly found in β -turns. Discuss the reasons for this behavior.
- For flavodoxin in Figure 6.32, identify the right-handed cross-overs and the left-handed cross-overs in the parallel β -sheet.
- Choose any three regions in the Ramachandran plot and discuss the likelihood of observing that combination of ϕ and ψ in a

- peptide or protein. Defend your answer using suitable molecular models of a peptide.
- A new protein of unknown structure has been purified. Gel filtration chromatography reveals that the native protein has a molecular weight of 240,000. Chromatography in the presence of 6 *M* guanidine hydrochloride yields only a peak for a protein of *M*_r 60,000. Chromatography in the presence of 6 *M* guanidine hydrochloride and 10 mM β -mercaptoethanol yields peaks for proteins of *M*_r 34,000 and 26,000. Explain what can be determined about the structure of this protein from these data.
 - Two polypeptides, A and B, have similar tertiary structures, but A normally exists as a monomer, whereas B exists as a tetramer, B₄. What differences might be expected in the amino acid composition of A versus B?
 - The hemagglutinin protein in influenza virus contains a remarkably long α -helix, with 53 residues.
 - How long is this α -helix (in nm)?
 - How many turns does this helix have?
 - Each residue in an α -helix is involved in two H bonds. How many H bonds are present in this helix?

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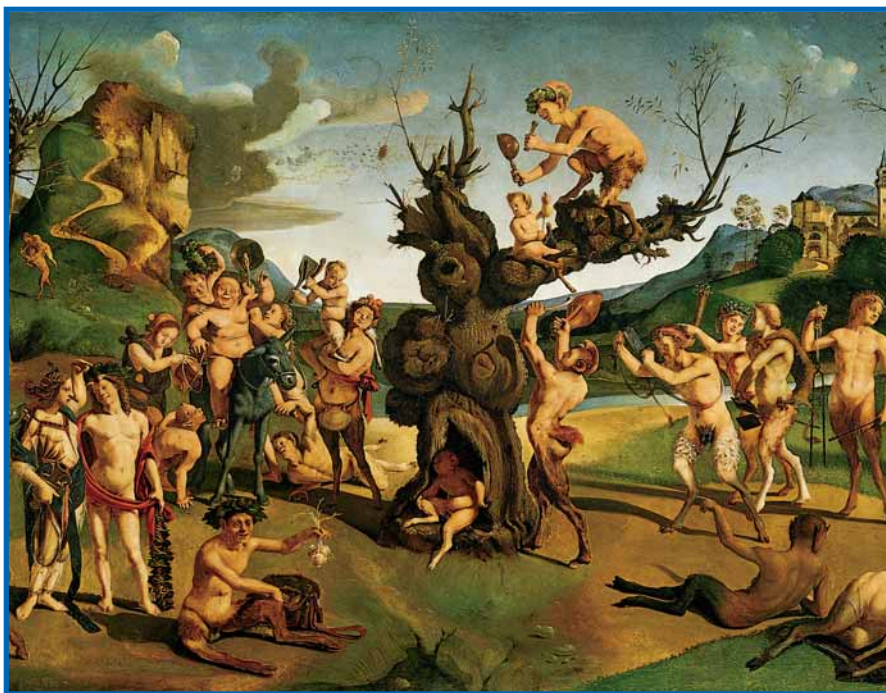
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Chapter 7

Carbohydrates



“The Discovery of Honey”—Piero di Cosimo (1462). (Courtesy of the Worcester Art Museum)

Carbohydrates are the single most abundant class of organic molecules found in nature. The name *carbohydrate* arises from the basic molecular formula $(\text{CH}_2\text{O})_n$, which can be rewritten $(\text{C} \cdot \text{H}_2\text{O})_n$ to show that these substances are hydrates of carbon, where $n = 3$ or more. Carbohydrates constitute a versatile class of molecules. Energy from the sun captured by green plants, algae, and some bacteria during photosynthesis (see Chapter 22) is stored in the form of carbohydrates. In turn, carbohydrates are the metabolic precursors of virtually all other biomolecules. Breakdown of carbohydrates provides the energy that sustains animal life. In addition, carbohydrates are covalently linked with a variety of other molecules. Carbohydrates linked to lipid molecules, or **glycolipids**, are common components of biological membranes. Proteins that have covalently linked carbohydrates are called **glycoproteins**. These two classes of biomolecules, together called **glycoconjugates**, are important components of cell walls and extracellular structures in plants, animals, and bacteria. In addition to the structural roles such molecules play, they also serve in a variety of

Sugar in the gourd and honey in the horn,
I never was so happy since the hour I was born.

Turkey in the Straw, stanza 6 (classic American folk tune)

OUTLINE

- 7.1 • Carbohydrate Nomenclature
- 7.2 • Monosaccharides
- 7.3 • Oligosaccharides
- 7.4 • Polysaccharides

processes involving *recognition* between cell types or recognition of cellular structures by other molecules. Recognition events are important in normal cell growth, fertilization, transformation of cells, and other processes.

All of these functions are made possible by the characteristic chemical features of carbohydrates: (1) the existence of at least one and often two or more asymmetric centers, (2) the ability to exist either in linear or ring structures, (3) the capacity to form polymeric structures via *glycosidic* bonds, and (4) the potential to form multiple hydrogen bonds with water or other molecules in their environment.

7.1 • Carbohydrate Nomenclature

Carbohydrates are generally classified into three groups: **monosaccharides** (and their derivatives), **oligosaccharides**, and **polysaccharides**. The monosaccharides are also called **simple sugars** and have the formula $(\text{CH}_2\text{O})_n$. Monosaccharides cannot be broken down into smaller sugars under mild conditions. Oligosaccharides derive their name from the Greek word *oligo*, meaning “few,” and consist of from two to ten simple sugar molecules. Disaccharides are common in nature, and trisaccharides also occur frequently. Four- to six-sugar-unit oligosaccharides are usually bound covalently to other molecules, including glycoproteins. As their name suggests, polysaccharides are polymers of the simple sugars and their derivatives. They may be either linear or branched polymers and may contain hundreds or even thousands of monosaccharide units. Their molecular weights range up to 1 million or more.

7.2 • Monosaccharides

Classification

Monosaccharides consist typically of three to seven carbon atoms and are described either as **aldoses** or **ketoses**, depending on whether the molecule contains an aldehyde function or a ketone group. The simplest aldose is glyceraldehyde, and the simplest ketose is dihydroxyacetone (Figure 7.1). These two simple sugars are termed **trioses** because they each contain three carbon atoms. The structures and names of a family of aldoses and ketoses with three, four, five, and six carbons are shown in Figure 7.2 and 7.3. *Hexoses* are the most abundant sugars in nature. Nevertheless, sugars from all these classes are important in metabolism.

Monosaccharides, either aldoses or ketoses, are often given more detailed generic names to describe both the important functional groups and the total number of carbon atoms. Thus, one can refer to *aldotetroses* and *ketotetroses*, *aldopentoses* and *ketopentoses*, *aldohexoses* and *ketohexoses*, and so on. Sometimes the ketone-containing monosaccharides are named simply by inserting the letters -ul- into the simple generic terms, such as *tetrols*, *pentuloses*, *hexuloses*, *heptuloses*, and so on. The simplest monosaccharides are water-soluble, and most taste sweet.

Stereochemistry

Aldoses with at least three carbons and ketoses with at least four carbons contain **chiral centers** (Chapter 4). The nomenclature for such molecules must specify the **configuration** about each asymmetric center, and drawings of these molecules must be based on a system that clearly specifies these configurations.

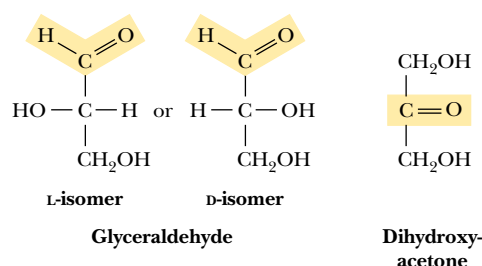


FIGURE 7.1 • Structure of a simple aldose (glyceraldehyde) and a simple ketose (dihydroxyacetone).

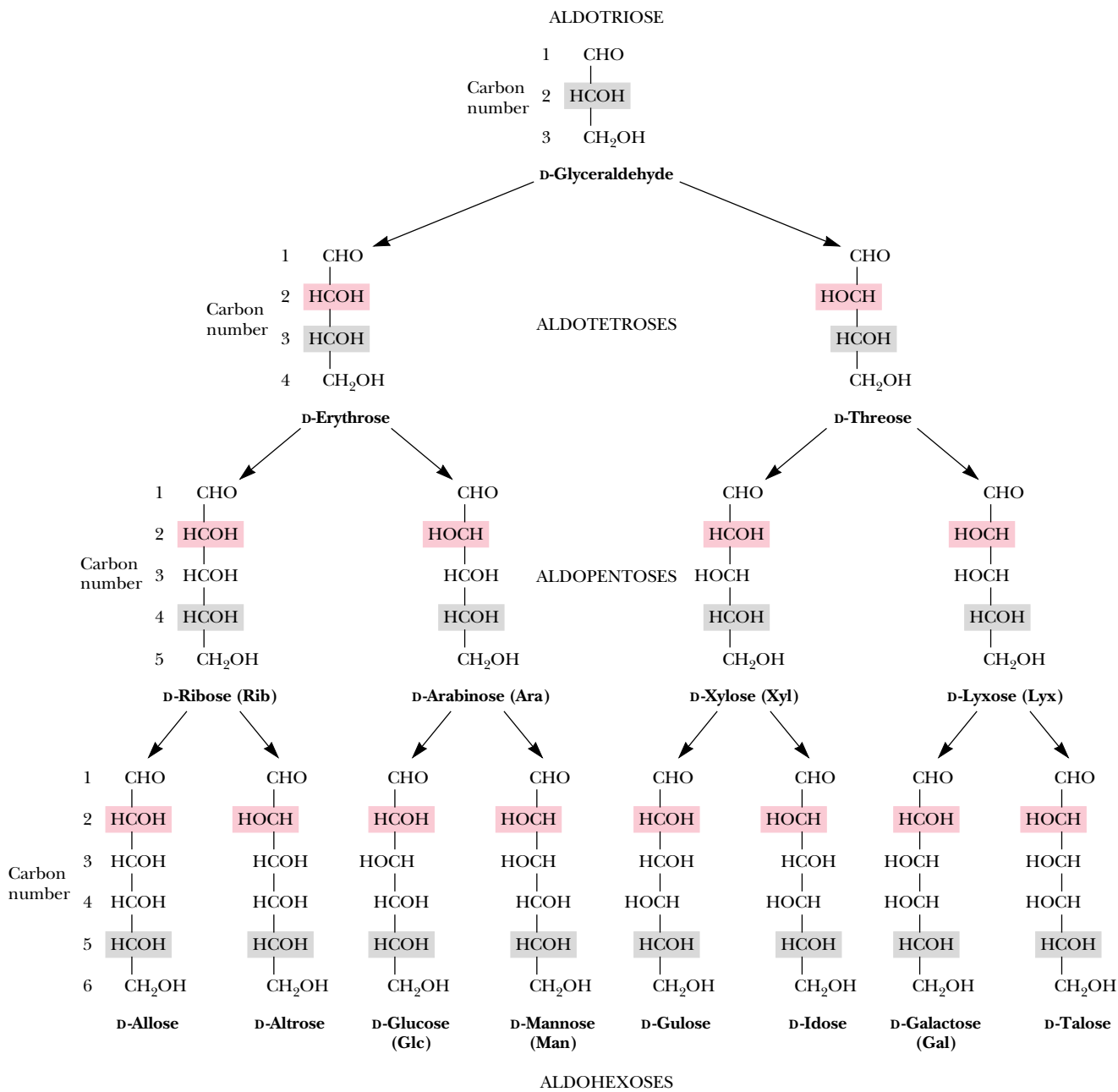


FIGURE 7.2 • The structure and stereochemical relationships of D-aldoses having three to six carbons. The configuration in each case is determined by the highest numbered asymmetric carbon (shown in gray). In each row, the “new” asymmetric carbon is shown in red.

As noted in Chapter 4, the **Fischer projection** system is used almost universally for this purpose today. The structures shown in Figures 7.2 and 7.3 are Fischer projections. For monosaccharides with two or more asymmetric carbons, the prefix D or L refers to the configuration of the highest numbered asymmetric carbon (the asymmetric carbon farthest from the carbonyl carbon). A monosaccharide is designated D if the hydroxyl group on the highest numbered asymmetric carbon is drawn to the right in a Fischer projection, as in D-glyceraldehyde (Figure 7.1). Note that the designation D or L merely relates the



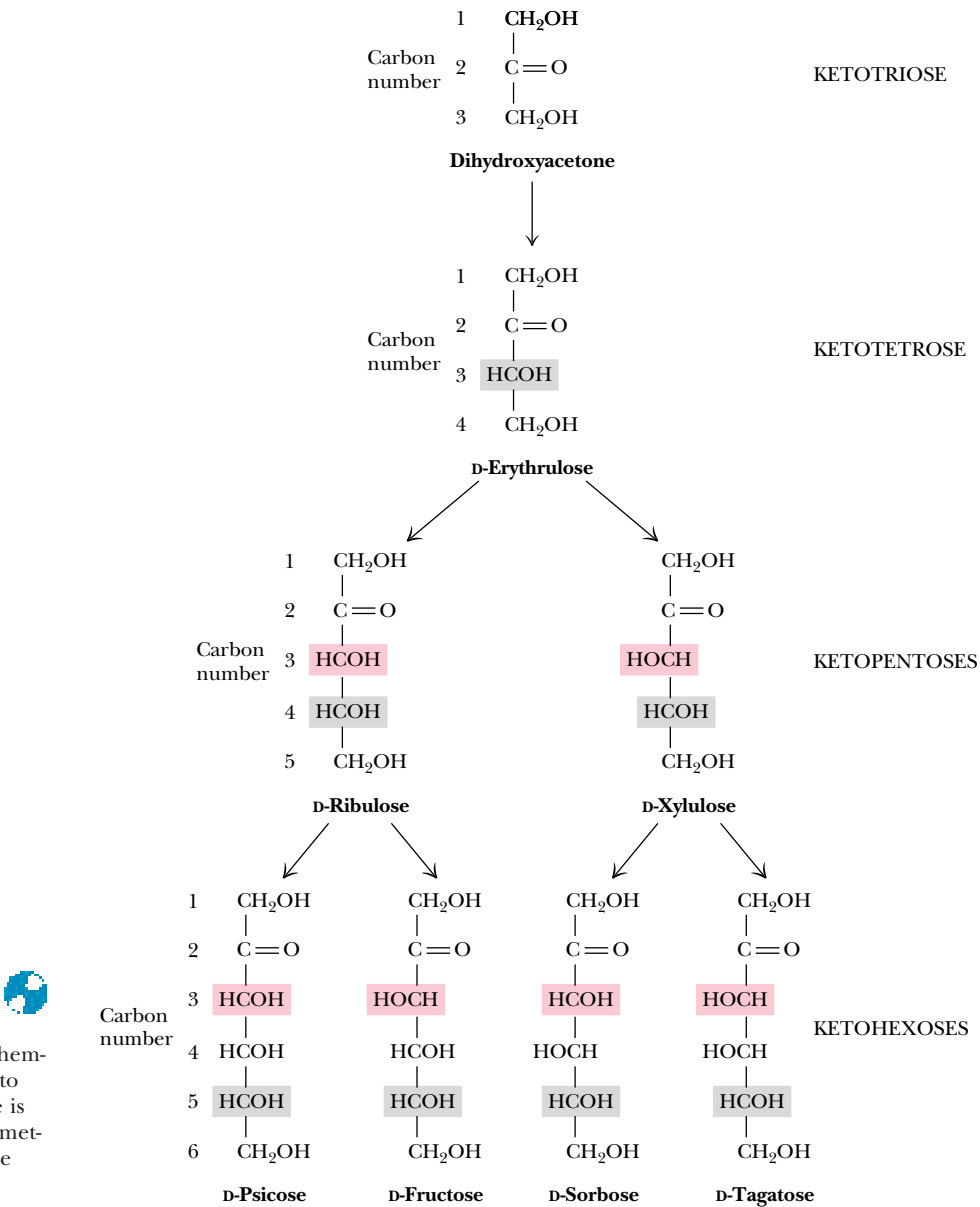


FIGURE 7.3 • The structure and stereochemical relationships of D-ketoses having three to six carbons. The configuration in each case is determined by the highest numbered asymmetric carbon (shown in gray). In each row, the “new” asymmetric carbon is shown in red.

configuration of a given molecule to that of glyceraldehyde and does *not* specify the sign of rotation of plane-polarized light. If the sign of optical rotation is to be specified in the name, the Fischer convention of D or L designations may be used along with a + (plus) or – (minus) sign. Thus, D-glucose (Figure 7.2) may also be called D(+)-glucose because it is dextrorotatory, whereas D-fructose (Figure 7.3), which is levorotatory, can also be named D(–)-fructose.

All of the structures shown in Figures 7.2 and 7.3 are D-configurations, and the D-forms of monosaccharides predominate in nature, just as L-amino acids do. These preferences, established in apparently random choices early in evolution, persist uniformly in nature because of the stereospecificity of the enzymes that synthesize and metabolize these small molecules.

L-Monosaccharides do exist in nature, serving a few relatively specialized roles. L-Galactose is a constituent of certain polysaccharides, and L-arabinose is a constituent of bacterial cell walls.

According to convention, the D- and L-forms of a monosaccharide are *mirror images* of each other, as shown in Figure 7.4 for fructose. Stereoisomers that are mirror images of each other are called **enantiomers**, or sometimes *enantiomeric pairs*. For molecules that possess two or more chiral centers, more than two stereoisomers can exist. Pairs of isomers that have opposite configurations at one or more of the chiral centers but that are not mirror images of each other are called **diastereomers** or *diastereomeric pairs*. Any two structures in a given row in Figures 7.2 and 7.3 are diastereomeric pairs. Two sugars that differ in configuration at *only one* chiral center are described as **epimers**. For example, D-mannose and D-talose are epimers and D-glucose and D-mannose are epimers, whereas D-glucose and D-talose are *not* epimers but merely diastereomers.

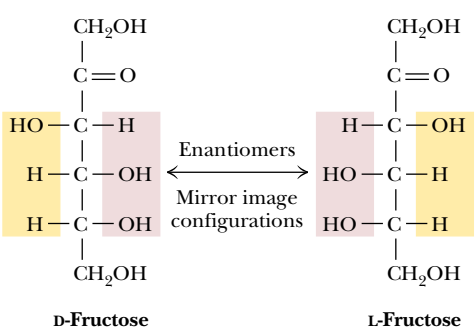


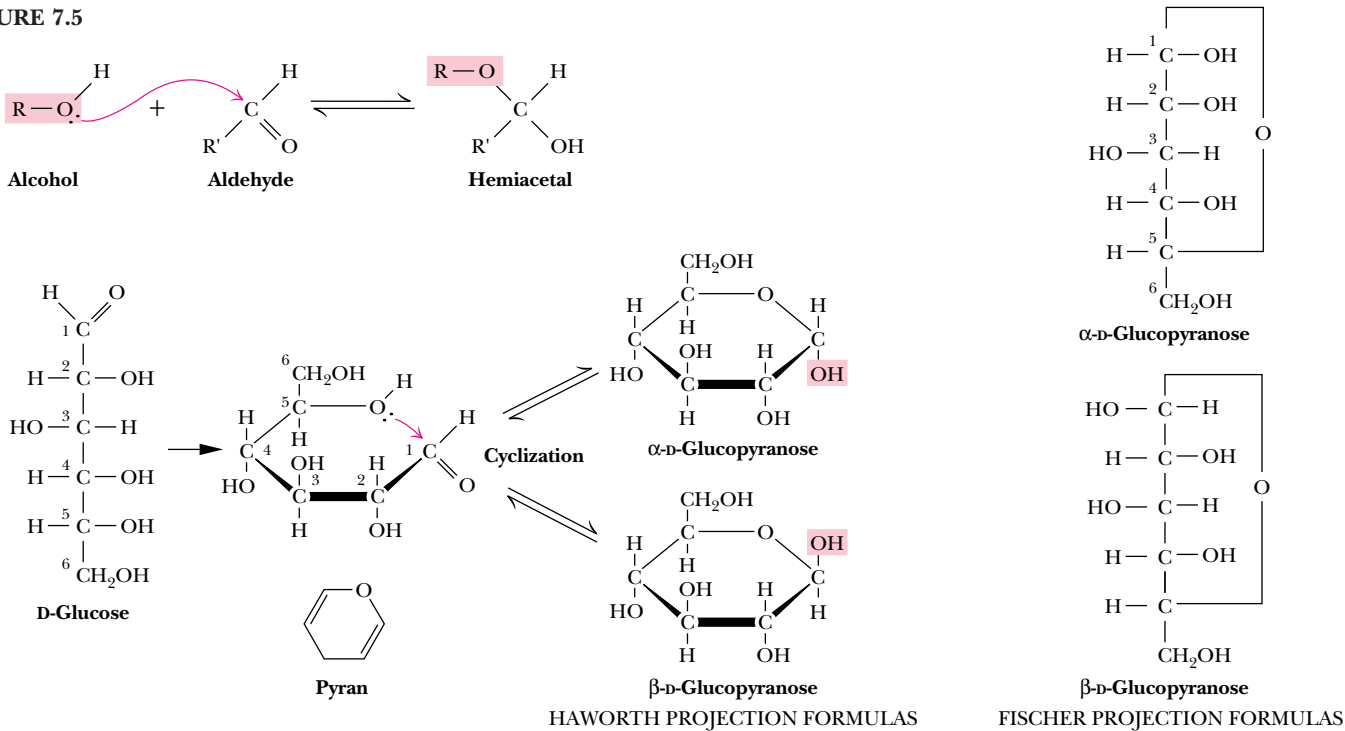
FIGURE 7.4 • D-Fructose and L-fructose, an enantiomeric pair. Note that changing the configuration only at C₅ would change D-fructose to L-sorbose.



Cyclic Structures and Anomeric Forms

Although Fischer projections are useful for presenting the structures of particular monosaccharides and their stereoisomers, they ignore one of the most interesting facets of sugar structure—the ability to form cyclic structures with formation of an additional asymmetric center. Alcohols react readily with aldehydes to form **hemiacetals** (Figure 7.5). The British carbohydrate chemist Sir Norman Haworth showed that the linear form of glucose (and other aldohexoses) could undergo a similar *intramolecular* reaction to form a *cyclic hemiacetal*. The resulting six-membered, oxygen-containing ring is similar to *pyran* and is designated a **pyranose**. The reaction is catalyzed by acid (H⁺) or base (OH[−]) and is readily reversible.

FIGURE 7.5



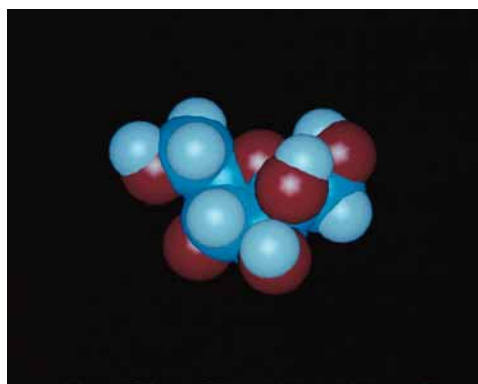
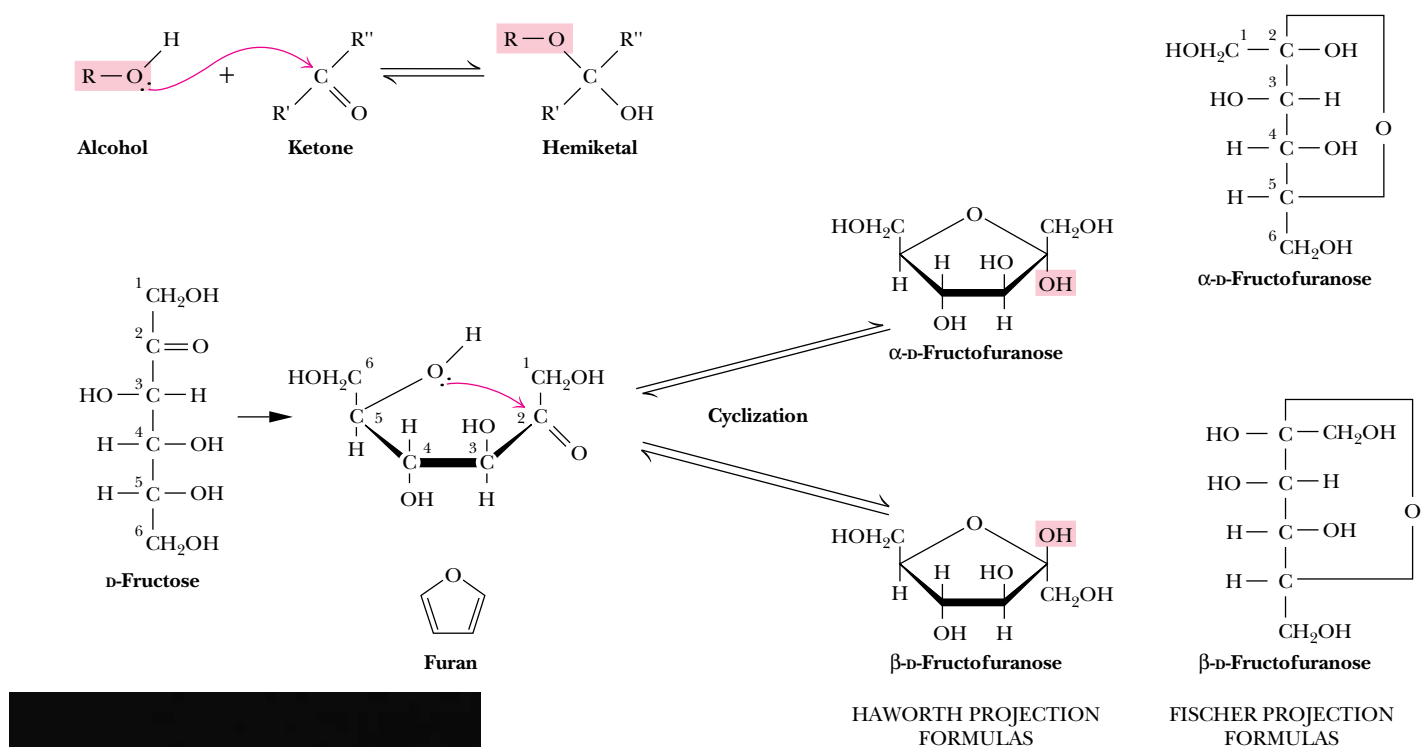


FIGURE 7.6



In a similar manner, ketones can react with alcohols to form **hemiketals**. The analogous intramolecular reaction of a ketose sugar such as fructose yields a *cyclic hemiketal* (Figure 7.6). The five-membered ring thus formed is reminiscent of *furan* and is referred to as a **furanose**. The cyclic pyranose and furanose forms are the preferred structures for monosaccharides in aqueous solution. At equilibrium, the linear aldehyde or ketone structure is only a minor component of the mixture (generally much less than 1%).

When hemiacetals and hemiketals are formed, the carbon atom that carried the carbonyl function becomes an asymmetric carbon atom. Isomers of monosaccharides that differ only in their configuration about that carbon atom are called **anomers**, designated as α or β , as shown in Figure 7.5, and the carbonyl carbon is thus called the **anomeric carbon**. When the hydroxyl group at the anomeric carbon is on the *same side* of a Fischer projection as the oxygen atom at the highest numbered asymmetric carbon, the configuration at the anomeric carbon is α , as in α -D-glucose. When the anomeric hydroxyl is on the *opposite side* of the Fischer projection, the configuration is β , as in β -D-glucopyranose (Figure 7.5).

The addition of this asymmetric center upon hemiacetal and hemiketal formation alters the optical rotation properties of monosaccharides, and the original assignment of the α and β notations arose from studies of these properties. Early carbohydrate chemists frequently observed that the optical rotation of glucose (and other sugar) solutions could change with time, a process called **mutarotation**. This indicated that a structural change was occurring. It was eventually found that α -D-glucose has a specific optical rotation, $[\alpha]_D^{20}$, of 112.2° , and that β -D-glucose has a specific optical rotation of 18.7° . Mutarotation involves interconversion of α and β forms of the monosaccharide with intermediate formation of the linear aldehyde or ketone, as shown in Figures 7.5 and 7.6.

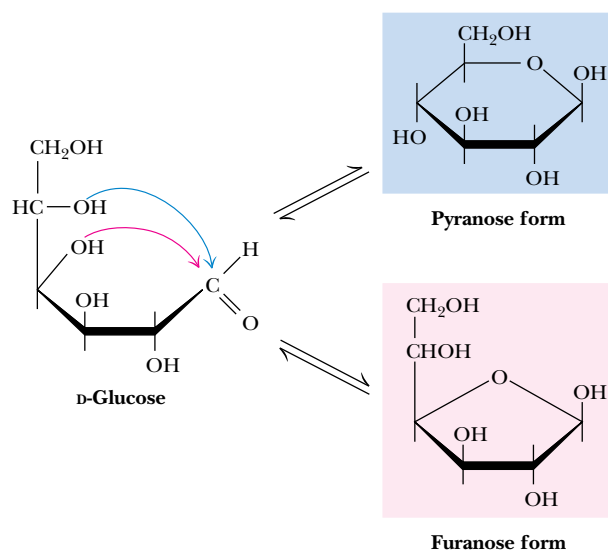


FIGURE 7.7 • D-Glucose can cyclize in two ways, forming either furanose or pyranose structures.

Haworth Projections

Another of Haworth's lasting contributions to the field of carbohydrate chemistry was his proposal to represent pyranose and furanose structures as hexagonal and pentagonal rings lying perpendicular to the plane of the paper, with thickened lines indicating the side of the ring closest to the reader. Such **Haworth projections**, which are now widely used to represent saccharide structures (Figures 7.5 and 7.6), show substituent groups extending either above or below the ring. Substituents drawn to the left in a Fischer projection are drawn above the ring in the corresponding Haworth projection. Substituents drawn to the right in a Fischer projection are below the ring in a Haworth projection. Exceptions to these rules occur in the formation of furanose forms of pentoses and the formation of furanose or pyranose forms of hexoses. In these cases, the structure must be redrawn with a rotation about the carbon whose hydroxyl group is involved in the formation of the cyclic form (Figures 7.7 and 7.8) in order to orient the appropriate hydroxyl group for ring formation. This is

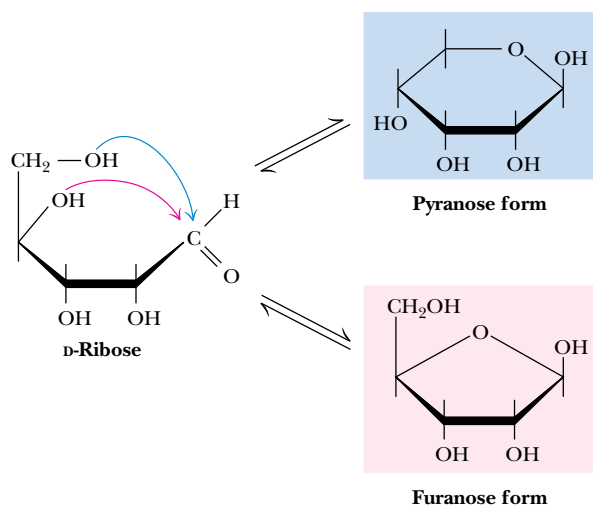


FIGURE 7.8 • D-Ribose and other five-carbon saccharides can form either furanose or pyranose structures.

merely for illustrative purposes and involves no change in configuration of the saccharide molecule.

The rules previously mentioned for assignment of α - and β -configurations can be readily applied to Haworth projection formulas. For the D-sugars, the anomeric hydroxyl group is below the ring in the α -anomer and above the ring in the β -anomer. For L-sugars, the opposite relationship holds.

As Figures 7.7 and 7.8 imply, in most monosaccharides there are two or more hydroxyl groups which can react with an aldehyde or ketone at the other end of the molecule to form a hemiacetal or hemiketal. Consider the possibilities for glucose, as shown in Figure 7.7. If the C-4 hydroxyl group reacts with the aldehyde of glucose, a five-membered ring is formed, whereas if the C-5 hydroxyl reacts, a six-membered ring is formed. The C-6 hydroxyl does not react effectively because a seven-membered ring is too strained to form a stable hemiacetal. The same is true for the C-2 and C-3 hydroxyls, and thus five- and six-membered rings are by far the most likely to be formed from six-membered monosaccharides. D-Ribose, with five carbons, readily forms either five-membered rings (α - or β -D-ribofuranose) or six-membered rings (α - or β -D-ribopyranose) (Figure 7.8). In general, aldoses and ketoses with five or more carbons can form *either* furanose or pyranose rings, and the more stable form depends on structural factors. The nature of the substituent groups on the carbonyl and hydroxyl groups and the configuration about the asymmetric carbon will determine whether a given monosaccharide prefers the pyranose or furanose structure. In general, the pyranose form is favored over the furanose ring for aldohexose sugars, although, as we shall see, furanose structures are more stable for ketohexoses.

Although Haworth projections are convenient for display of monosaccharide structures, they do not accurately portray the conformations of pyranose and furanose rings. Given C—C—C tetrahedral bond angles of 109° and C—O—C angles of 111° , neither pyranose nor furanose rings can adopt true planar structures. Instead, they take on puckered conformations, and in the case of pyranose rings, the two favored structures are the **chair conformation** and the **boat conformation**, shown in Figure 7.9. Note that the ring substituents

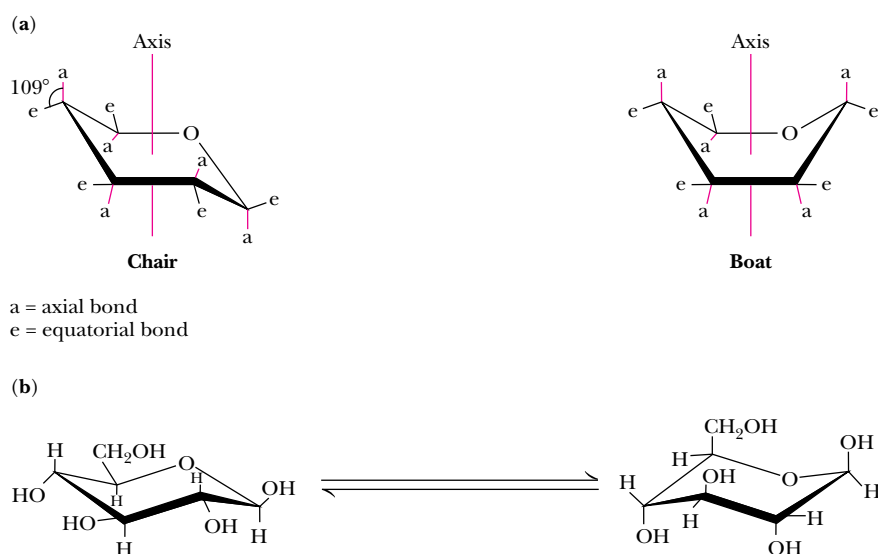


FIGURE 7.9 • (a) Chair and boat conformations of a pyranose sugar. (b) Two possible chair conformations of β -D-glucose.

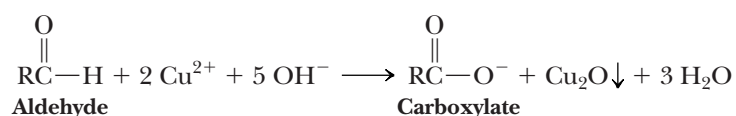
in these structures can be **equatorial**, which means approximately coplanar with the ring, or **axial**, that is, parallel to an axis drawn through the ring as shown. Two general rules dictate the conformation to be adopted by a given saccharide unit. First, bulky substituent groups on such rings are more stable when they occupy equatorial positions rather than axial positions, and second, chair conformations are slightly more stable than boat conformations. For a typical pyranose, such as β -D-glucose, there are two possible chair conformations (Figure 7.9). Of all the D-aldohexoses, β -D-glucose is the only one that can adopt a conformation with all its bulky groups in an equatorial position. With this advantage of stability, it may come as no surprise that β -D-glucose is the most widely occurring organic group in nature and the central hexose in carbohydrate metabolism.

Derivatives of Monosaccharides

A variety of chemical and enzymatic reactions produce **derivatives** of the simple sugars. These modifications produce a diverse array of saccharide derivatives. Some of the most common derivations are discussed here.

Sugar Acids

Sugars with free anomeric carbon atoms are reasonably good reducing agents and will reduce hydrogen peroxide, ferricyanide, certain metals (Cu^{2+} and Ag^+), and other oxidizing agents. Such reactions convert the sugar to a **sugar acid**. For example, addition of alkaline CuSO_4 (called *Fehling's solution*) to an aldose sugar produces a red cuprous oxide (Cu_2O) precipitate:



and converts the aldose to an **aldonic acid**, such as **gluconic acid** (Figure 7.10). Formation of a precipitate of red Cu_2O constitutes a positive test for an aldehyde. Carbohydrates that can reduce oxidizing agents in this way are referred to as **reducing sugars**. By quantifying the amount of oxidizing agent reduced by a sugar solution, one can accurately determine the concentration of the sugar. *Diabetes mellitus* is a condition that causes high levels of glucose in urine and blood, and frequent analysis of reducing sugars in diabetic patients is an important part of the diagnosis and treatment of this disease. Over-the-counter kits for the easy and rapid determination of reducing sugars have made this procedure a simple one for diabetics.

Monosaccharides can be oxidized enzymatically at C-6, yielding **uronic acids**, such as **D-glucuronic** and **L-iduronic** acids (Figure 7.10). L-Iduronic acid is similar to D-glucuronic acid, except for having an opposite configuration at C-5. Oxidation at both C-1 and C-6 produces **aldaric acids**, such as **D-glucaric acid**.

Sugar Alcohols

Sugar alcohols, another class of sugar derivative, can be prepared by the mild reduction (with NaBH_4 or similar agents) of the carbonyl groups of aldoses and ketoses. Sugar alcohols, or **alditols**, are designated by the addition of *-itol* to the name of the parent sugar (Figure 7.11). The alditols are linear molecules that cannot cyclize in the manner of aldoses. Nonetheless, alditols are characteristically sweet tasting, and **sorbitol**, **mannitol**, and **xylitol** are widely used to sweeten sugarless gum and mints. Sorbitol buildup in the eyes of dia-

FIGURE 7.10

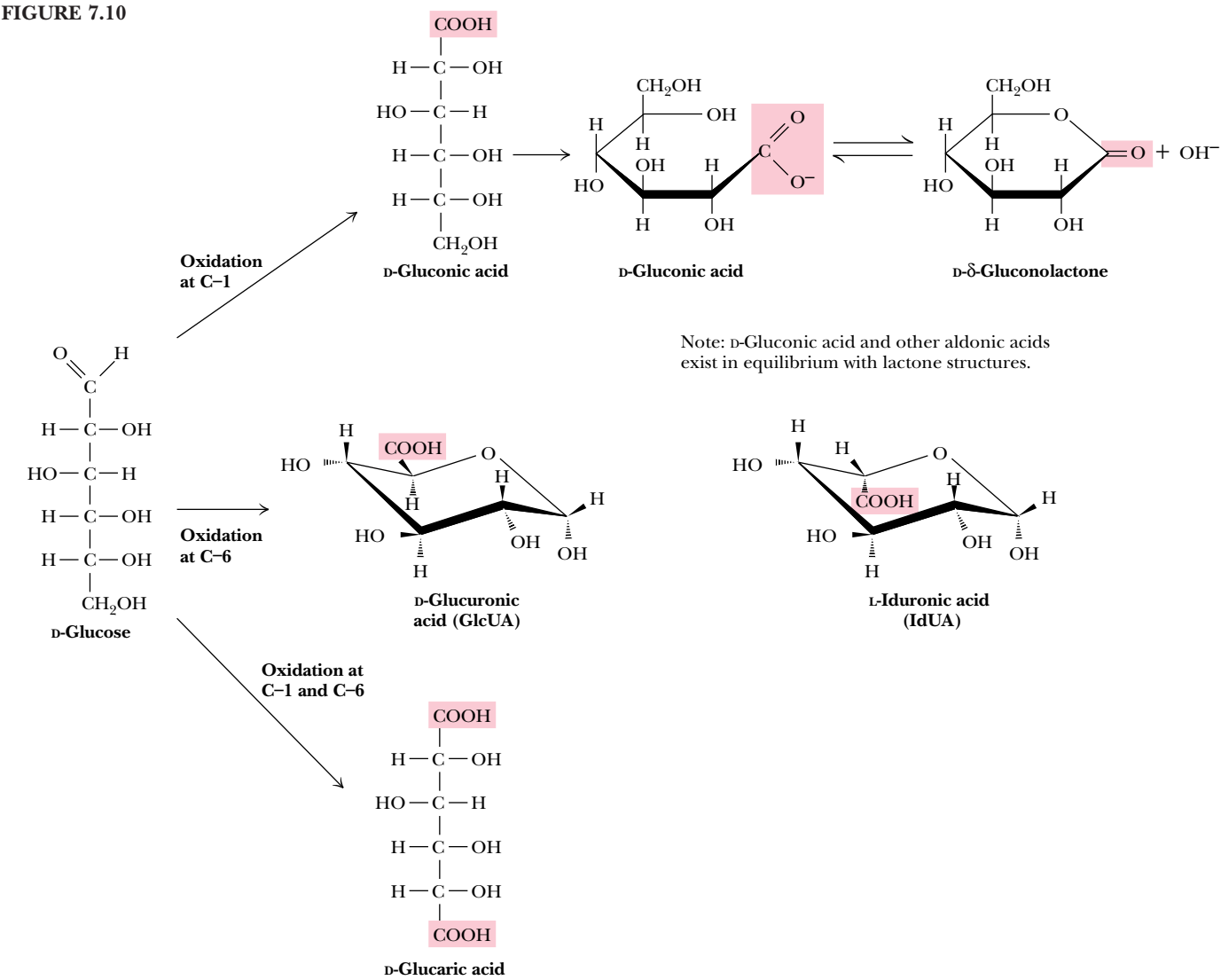
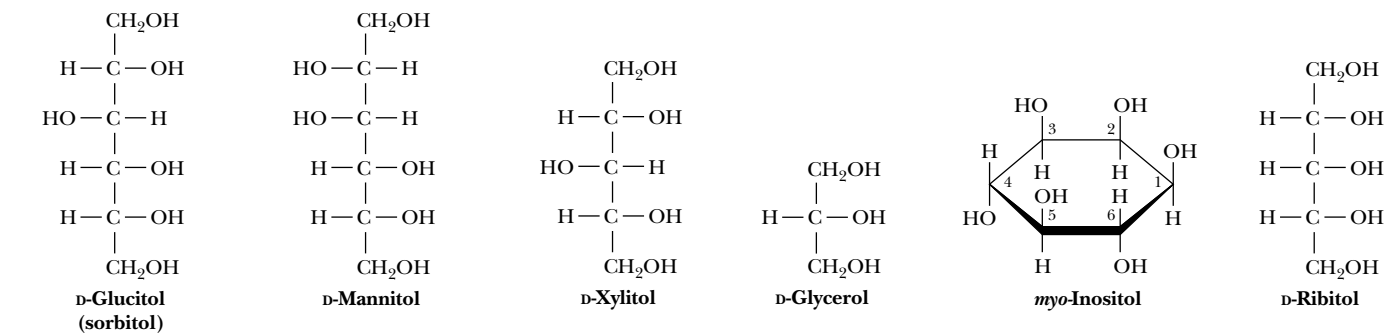


FIGURE 7.11 • Structures of some sugar alcohols.



betics is implicated in cataract formation. **Glycerol** and **myo-inositol**, a cyclic alcohol, are components of lipids (see Chapter 8). There are nine different stereoisomers of inositol; the one shown in Figure 7.11 was first isolated from heart muscle and thus has the prefix *myo-* for muscle. **Ribitol** is a constituent of flavin coenzymes (see Chapter 20).

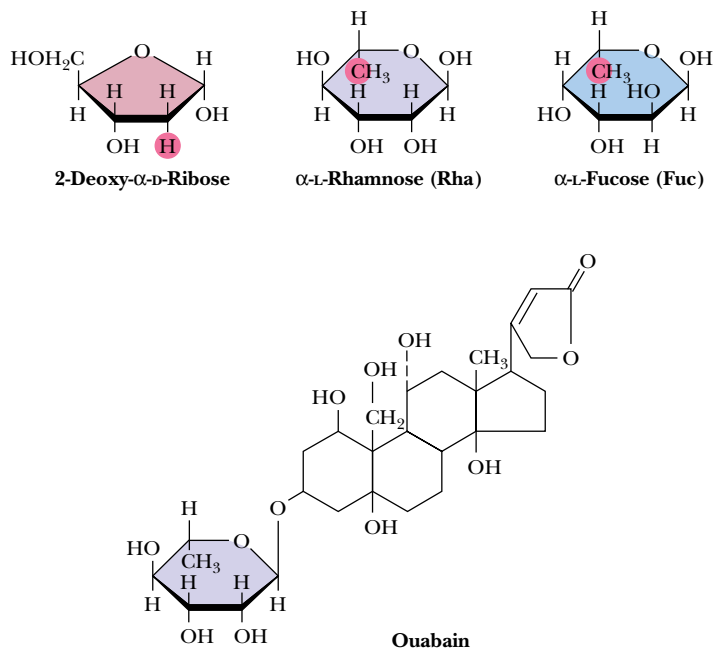


FIGURE 7.12 • Several deoxy sugars and ouabain, which contains α-L-rhamnose (Rha). Hydrogen atoms highlighted in red are “deoxy” positions.

Deoxy Sugars

The **deoxy sugars** are monosaccharides with one or more hydroxyl groups replaced by hydrogens. 2-Deoxy-D-ribose (Figure 7.12), whose systematic name is 2-deoxy-D-erythropentose, is a constituent of DNA in all living things (see Chapter 11). Deoxy sugars also occur frequently in glycoproteins and polysaccharides. L-Fucose and L-rhamnose, both 6-deoxy sugars, are components of some cell walls, and rhamnose is a component of **ouabain**, a highly toxic *cardiac glycoside* found in the bark and root of the ouabaio tree. Ouabain is used by the East African Somalis as an arrow poison. The sugar moiety is not the toxic part of the molecule (see Chapter 10).

Sugar Esters

Phosphate esters of glucose, fructose, and other monosaccharides are important metabolic intermediates, and the ribose moiety of nucleotides such as ATP and GTP is phosphorylated at the 5'-position (Figure 7.13).

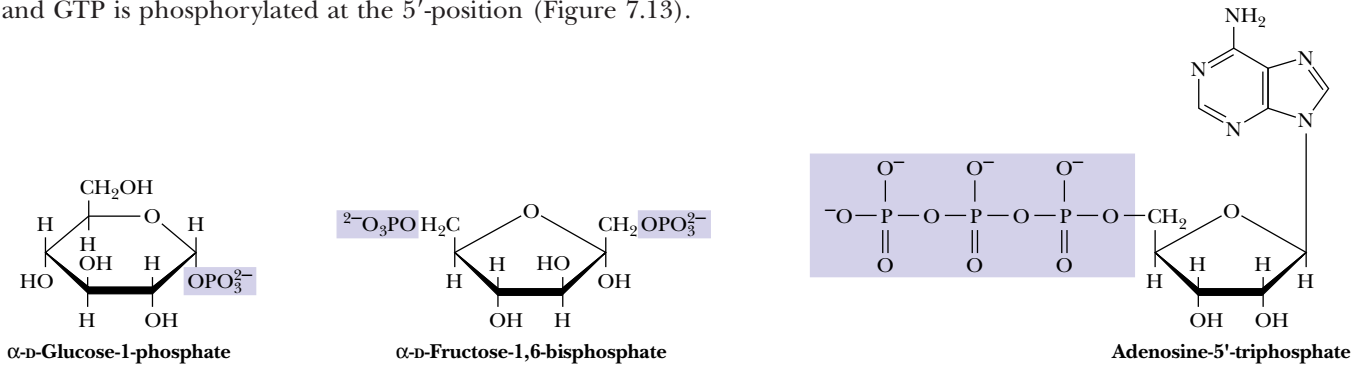


FIGURE 7.13 • Several sugar esters important in metabolism.

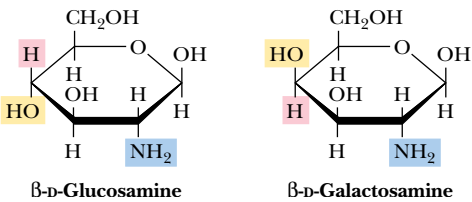
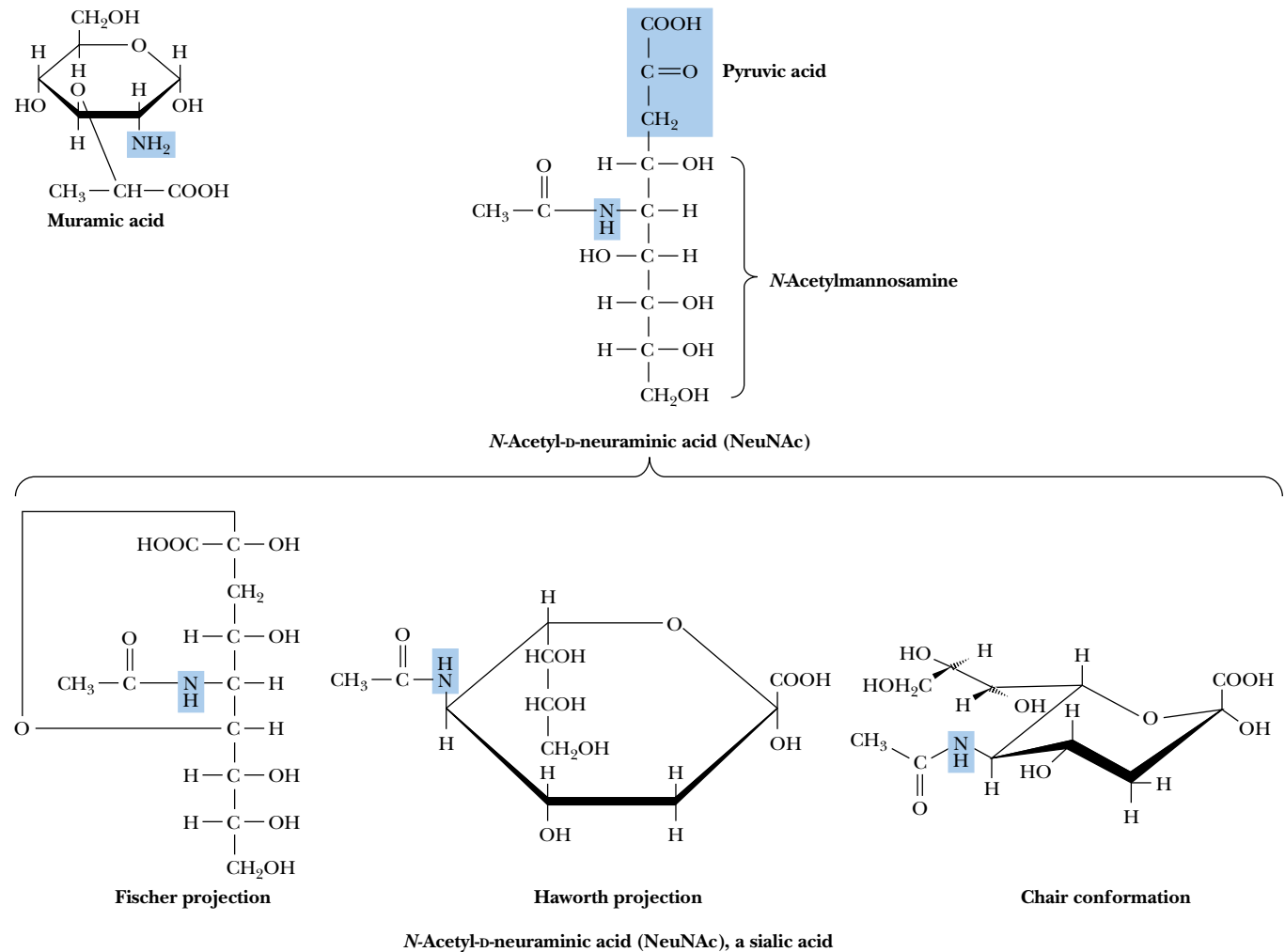


FIGURE 7.14 • Structures of D-glucosamine and D-galactosamine.

Amino Sugars

Amino sugars, including D-glucosamine and D-galactosamine (Figure 7.14), contain an amino group (instead of a hydroxyl group) at the C-2 position. They are found in many oligo- and polysaccharides, including *chitin*, a polysaccharide in the exoskeletons of crustaceans and insects.

Muramic acid and neuraminic acid, which are components of the polysaccharides of cell membranes of higher organisms and also bacterial cell walls, are glucosamines linked to three-carbon acids at the C-1 or C-3 positions. In muramic acid (thus named as an *amine* isolated from bacterial cell wall polysaccharides; *murus* is Latin for “wall”), the hydroxyl group of a lactic acid moiety makes an ether linkage to the C-3 of glucosamine. Neuraminic acid (an *amine* isolated from *neural* tissue) forms a C—C bond between the C-1 of *N*-acetylmannosamine and the C-3 of pyruvic acid (Figure 7.15). The *N*-acetyl and *N*-glycolyl derivatives of neuraminic acid are collectively known as **sialic acids** and are distributed widely in bacteria and animal systems.



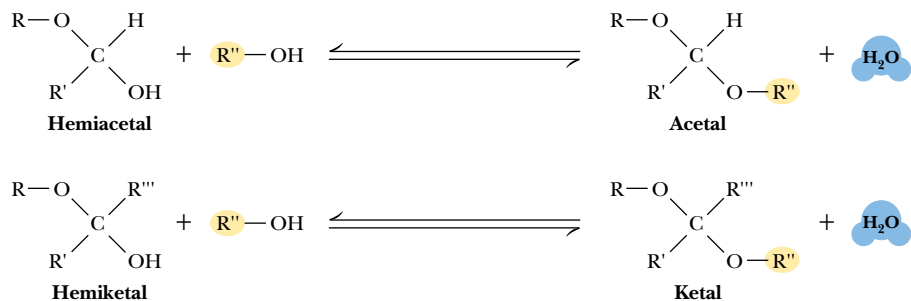


FIGURE 7.16 • Acetals and ketals can be formed from hemiacetals and hemiketals, respectively.

Acetals, Ketals, and Glycosides

Hemiacetals and hemiketals can react with alcohols in the presence of acid to form **acetals** and **ketals**, as shown in Figure 7.16. This reaction is another example of a *dehydration synthesis* and is similar in this respect to the reactions undergone by amino acids to form peptides and nucleotides to form nucleic acids. The pyranose and furanose forms of monosaccharides react with alcohols in this way to form **glycosides** with retention of the α - or β -configuration at the C-1 carbon. The new bond between the anomeric carbon atom and the oxygen atom of the alcohol is called a **glycosidic bond**. Glycosides are named according to the parent monosaccharide. For example, *methyl- β -D-glucoside* (Figure 7.17) can be considered a derivative of β -D-glucose.

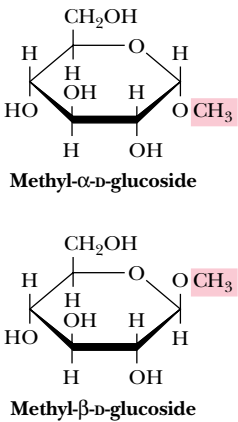


FIGURE 7.17 • The anomeric forms of methyl-D-glucoside.

7.3 • Oligosaccharides

Given the relative complexity of oligosaccharides and polysaccharides in higher organisms, it is perhaps surprising that these molecules are formed from relatively few different monosaccharide units. (In this respect, the oligo- and polysaccharides are similar to proteins; both form complicated structures based on a small number of different building blocks.) Monosaccharide units include the hexoses glucose, fructose, mannose, and galactose and the pentoses ribose and xylose.

Disaccharides

The simplest oligosaccharides are the **disaccharides**, which consist of two monosaccharide units linked by a glycosidic bond. As in proteins and nucleic acids, each individual unit in an oligosaccharide is termed a *residue*. The disaccharides shown in Figure 7.18 are all commonly found in nature, with sucrose, maltose, and lactose being the most common. Each is a mixed acetal, with one hydroxyl group provided intramolecularly and one hydroxyl from the other monosaccharide. Except for sucrose, each of these structures possesses one free unsubstituted anomeric carbon atom, and thus each of these disaccharides is a reducing sugar. The end of the molecule containing the free anomeric carbon is called the **reducing end**, and the other end is called the **nonreducing end**. In the case of sucrose, both of the anomeric carbon atoms are substituted, that is, neither has a free —OH group. The substituted anomeric carbons cannot be converted to the aldehyde configuration and thus cannot participate in

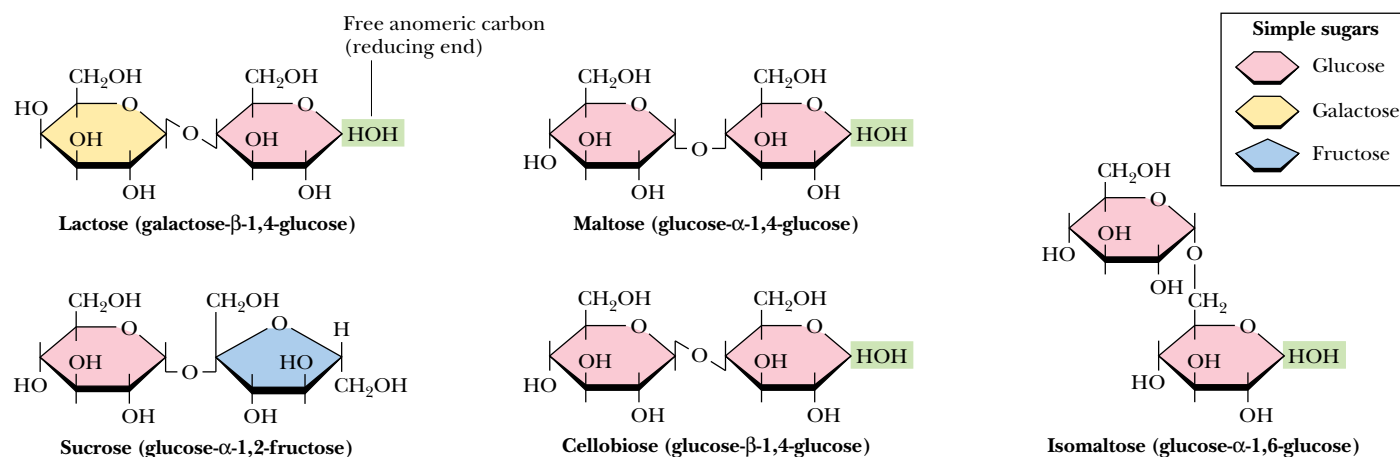
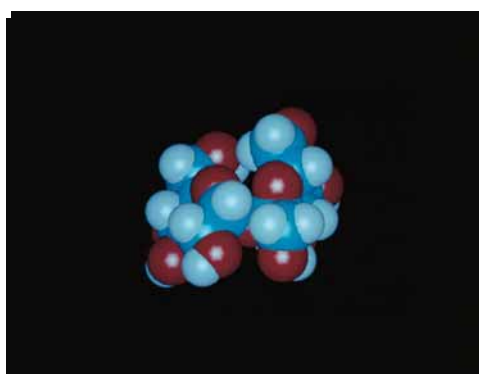


FIGURE 7.18 • The structures of several important disaccharides. Note that the notation —HOH means that the configuration can be either α or β . If the —OH group is above the ring, the configuration is termed β . The configuration is α if the —OH group is below the ring as shown. Also note that sucrose has no free anomeric carbon atoms.



Sucrose

the oxidation–reduction reactions characteristic of reducing sugars. Thus, sucrose is *not* a reducing sugar.

Maltose, isomaltose, and cellobiose are all **homodisaccharides** because they each contain only one kind of monosaccharide, namely, glucose. **Maltose** is produced from starch (a polymer of α -D-glucose produced by plants) by the action of amylase enzymes and is a component of malt, a substance obtained by allowing grain (particularly barley) to soften in water and germinate. The enzyme **diastase**, produced during the germination process, catalyzes the hydrolysis of starch to maltose. Maltose is used in beverages (malted milk, for example), and because it is fermented readily by yeast, it is important in the brewing of beer. In both maltose and cellobiose, the glucose units are **1 $\rightarrow$ 4 linked**, meaning that the C-1 of one glucose is linked by a glycosidic bond to the C-4 oxygen of the other glucose. The only difference between them is in the configuration at the glycosidic bond. Maltose exists in the α -configuration, whereas cellobiose is β . **Isomaltose** is obtained in the hydrolysis of some polysaccharides (such as dextran), and **cellobiose** is obtained from the acid hydrolysis of cellulose. Isomaltose also consists of two glucose units in a glycosidic bond, but in this case, C-1 of one glucose is linked to C-6 of the other, and the configuration is α .

The complete structures of these disaccharides can be specified in shorthand notation by using abbreviations for each monosaccharide, α or β , to denote configuration, and appropriate numbers to indicate the nature of the linkage. Thus, cellobiose is Glc β 1-4Glc, whereas isomaltose is Glc α 1-6Glc. Often the glycosidic linkage is written with an arrow so that cellobiose and isomaltose would be Glc β 1 $\rightarrow$ 4Glc and Glc α 1 $\rightarrow$ 6Glc, respectively. Because the linkage carbon on the first sugar is always C-1, a newer trend is to drop the 1- or

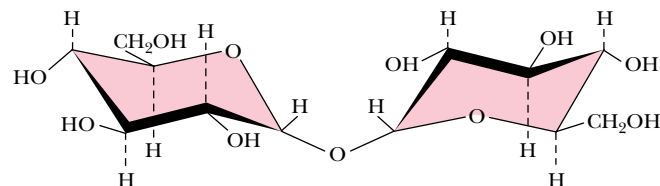
A DEEPER LOOK

Trehalose—A Natural Protectant for Bugs

Insects use an open circulatory system to circulate **hemolymph** (insect blood). The “blood sugar” is not glucose but rather **trehalose**, an unusual, nonreducing disaccharide (see Figure). Trehalose is found typically in organisms that are naturally subject to temperature variations and other environmental stresses—bacterial spores, fungi, yeast, and many insects. (Interestingly, honeybees do not have trehalose in their hemolymph, perhaps because they practice a colonial, rather than solitary, lifestyle. Bee colonies maintain a rather constant temperature of 18°C, protecting the residents from large temperature changes.)

What might explain this correlation between trehalose utilization and environmentally stressful lifestyles? Konrad Bloch* suggests that trehalose may act as a natural cryoprotectant. Freezing and thawing of biological tissues frequently causes irreversible structural changes, destroying biological activity. High concentrations of polyhydroxy compounds, such as sucrose and glycerol, can protect biological materials from such damage.

Trehalose is particularly well-suited for this purpose and has been shown to be superior to other polyhydroxy compounds, especially at low concentrations. Support for this novel idea comes from studies by P. A. Attfield,[†] which show that trehalose levels in the yeast *Saccharomyces cerevisiae* increase significantly during exposure to high salt and high growth temperatures—the same conditions that elicit the production of heat-shock proteins!



*Bloch, K., 1994. *Blondes in Venetian Paintings, the Nine-Banded Armadillo, and Other Essays in Biochemistry*. New Haven: Yale University Press.

[†]Attfield, P. A., 1987. Trehalose accumulates in *Saccharomyces cerevisiae* during exposure to agents that induce heat shock responses. *FEBS Letters* **225**:259.

1→ and describe these simply as Glcβ4Glc and Glcα6Glc, respectively. More complete names can also be used, however, so that maltose would be O-α-D-glucopyranosyl-(1→4)-D-glucopyranose. Cellobiose, because of its β-glycosidic linkage, is formally O-β-D-glucopyranosyl-(1→4)-D-glucopyranose.

β-D-lactose (O-β-D-Galactopyranosyl-(1→4)-D-glucopyranose) (Figure 7.18) is the principal carbohydrate in milk and is of critical nutritional importance to mammals in the early stages of their lives. It is formed from D-galactose and D-glucose via a β(1→4) link, and because it has a free anomeric carbon, it is capable of mutarotation and is a reducing sugar. It is an interesting quirk of nature that lactose cannot be absorbed directly into the bloodstream. It must first be broken down into galactose and glucose by **lactase**, an intestinal enzyme that exists in young, nursing mammals but is not produced in significant quantities in the mature mammal. Most humans, with the exception of certain groups in Africa and northern Europe, produce only low levels of lactase. For most individuals, this is not a problem, but some cannot tolerate lactose and experience intestinal pain and diarrhea upon consumption of milk.

Sucrose, in contrast, is a disaccharide of almost universal appeal and tolerance. Produced by many higher plants and commonly known as *table sugar*, it is one of the products of photosynthesis and is composed of fructose and glucose. Sucrose has a specific optical rotation, $[\alpha]_D^{20}$, of +66.5°, but an equimolar mixture of its component monosaccharides has a net negative rotation ($[\alpha]_D^{20}$ of glucose is +52.5° and of fructose is −92°). Sucrose is hydrolyzed by the enzyme **invertase**, so named for the inversion of optical rotation accompanying this reaction. Sucrose is also easily hydrolyzed by dilute acid, apparently because the fructose in sucrose is in the relatively unstable furanose form. Although sucrose and maltose are important to the human diet, they are not taken up directly in the body. In a manner similar to lactose, they are first hydrolyzed by **sucrase** and **maltase**, respectively, in the human intestine.

A D E E P E R L O O K

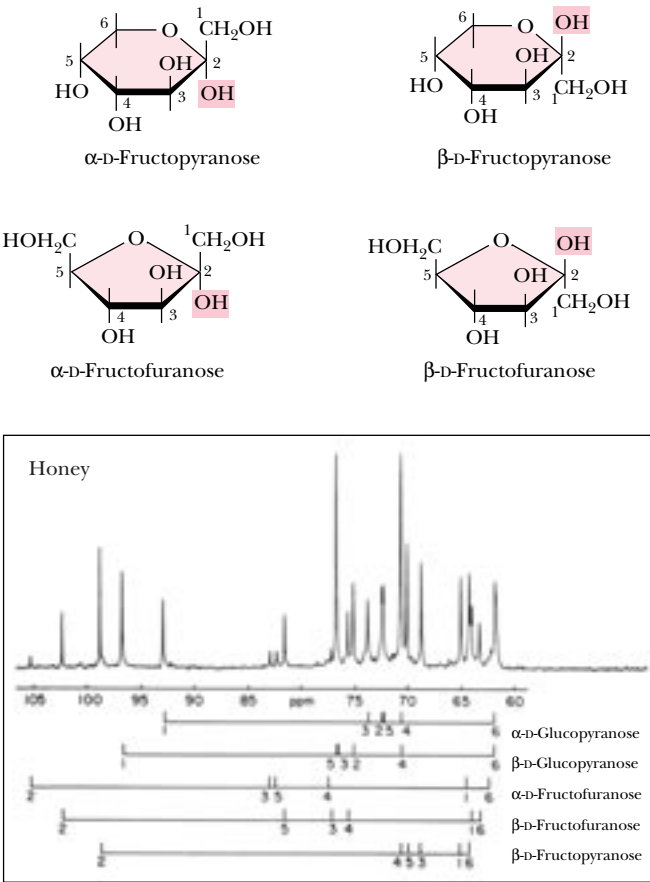
Honey—An Ancestral Carbohydrate Treat

Honey, the first sweet known to humankind, is the only sweetening agent that can be stored and used exactly as produced in nature. Bees process the nectar of flowers so that their final product is able to survive long-term storage at ambient temperature. Used as a ceremonial material and medicinal agent in earliest times, honey was not regarded as a food until the Greeks and Romans. Only in modern times have cane and beet sugar surpassed honey as the most frequently used sweetener. What is the chemical nature of this magical, viscous substance?

The bees’ processing of honey consists of (1) reducing the water content of the nectar (30 to 60%) to the self-preserving range of 15 to 19%, (2) hydrolyzing the significant amount of sucrose in nectar to glucose and fructose by the action of the enzyme invertase, and (3) producing small amounts of gluconic acid from glucose by the action of the enzyme **glucose oxidase**. Most of the sugar in the final product is glucose and fructose,

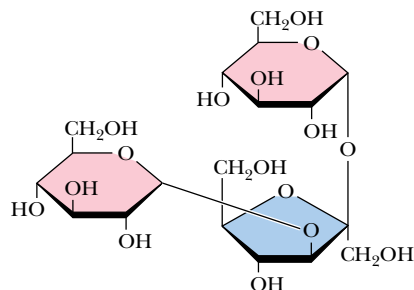
and the final product is supersaturated with respect to these monosaccharides. Honey actually consists of an emulsion of microscopic glucose hydrate and fructose hydrate crystals in a thick syrup. Sucrose accounts for only about 1% of the sugar in the final product, with fructose at about 38% and glucose at 31% by weight.

The figure shows a ¹³C nuclear magnetic resonance spectrum of honey from a mixture of wildflowers in southeastern Pennsylvania. Interestingly, five major hexose species contribute to this spectrum. Although most textbooks show fructose exclusively in its furanose form, the predominant form of fructose (67% of total fructose) is β-D-fructopyranose, with the β- and α-fructofuranose forms accounting for 27% and 6% of the fructose, respectively. In polysaccharides, fructose invariably prefers the furanose form, but free fructose (and crystalline fructose) is predominantly β-fructopyranose.

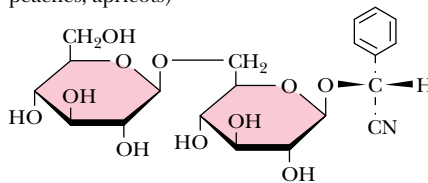


White, J. W., 1978. Honey. *Advances in Food Research* 24:287–374.
Prince, R. C., Gunson, D. E., Leigh, J. S., and McDonald, G. G., 1982. The predominant form of fructose is a pyranose, not a furanose ring. *Trends in Biochemical Sciences* 7:239–240.

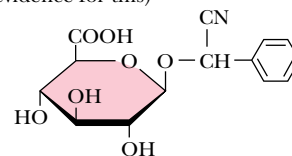
Melezitose (a constituent of honey)



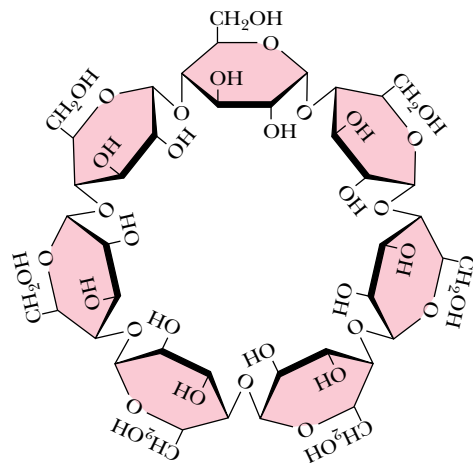
Amygdalin (occurs in seeds of *Rosaceae*, glycoside of bitter almonds, in kernels of cherries, peaches, apricots)



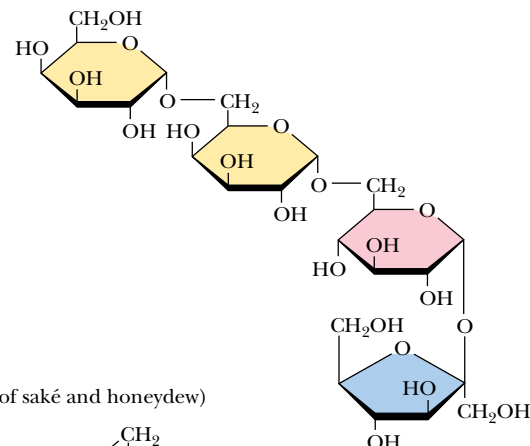
Laetrile (claimed to be an anticancer agent, but there is no rigorous scientific evidence for this)



Cycloheptaamylose (a breakdown product of starch; useful in chromatographic separations)



Stachyose (a constituent of many plants: white jasmine, yellow lupine, soybeans, lentils, etc.; causes flatulence since humans cannot digest it)



Dextrantriose (a constituent of saké and honeydew)

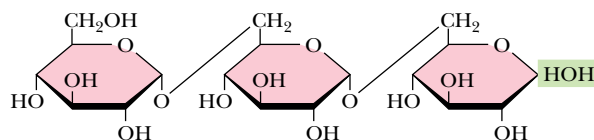
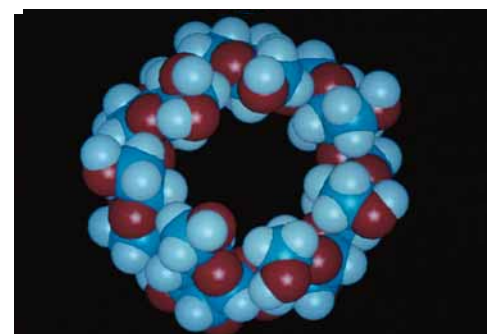


FIGURE 7.19 • The structures of some interesting oligosaccharides.

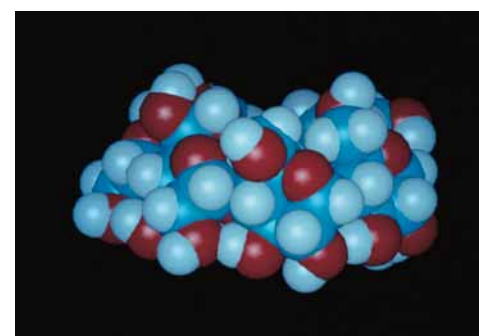
Higher Oligosaccharides

In addition to the simple disaccharides, many other oligosaccharides are found in both prokaryotic and eukaryotic organisms, either as naturally occurring substances or as hydrolysis products of natural materials. Figure 7.19 lists a number of simple oligosaccharides, along with descriptions of their origins and interesting features. Several are constituents of the sweet nectars or saps exuded or extracted from plants and trees. One particularly interesting and useful group of oligosaccharides is the **cycloamyloses**. These oligosaccharides are cyclic structures, and in solution they form molecular “pockets” of various diameters. These pockets are surrounded by the chiral carbons of the saccharides themselves and are able to form stereospecific inclusion complexes with chiral molecules that can fit into the pockets. Thus, mixtures of stereoisomers of small organic molecules can be separated into pure isomers on columns of **cycloheptaamylose**, for example.

Stachyose is typical of the oligosaccharide components found in substantial quantities in beans, peas, bran, and whole grains. These oligosaccharides are not digested by stomach enzymes, but *are* metabolized readily by bacteria in the intestines. This is the source of the flatulence that often accompanies the consumption of such foods. Commercial products are now available that assist in the digestion of the gas-producing components of these foods. These products contain an enzyme that hydrolyzes the culprit oligosaccharides in the stomach before they become available to intestinal microorganisms.



Cycloheptaamylose



Cycloheptaamylose (side view)

Another notable glycoside is **amygdalin**, which occurs in bitter almonds and in the kernels or pits of cherries, peaches, and apricots. Hydrolysis of this substance and subsequent oxidation yields **laetrile**, which has been claimed by some to have anticancer properties. There is no scientific evidence for these claims, and the U.S. Food and Drug Administration has never approved laetrile for use in the United States.

Oligosaccharides also occur widely as components (via glycosidic bonds) of *antibiotics* derived from various sources. Figure 7.20 shows the structures of a few representative carbohydrate-containing antibiotics. Some of these antibiotics also show antitumor activity. One of the most important of this type is **bleomycin A₂**, which is used clinically against certain tumors.

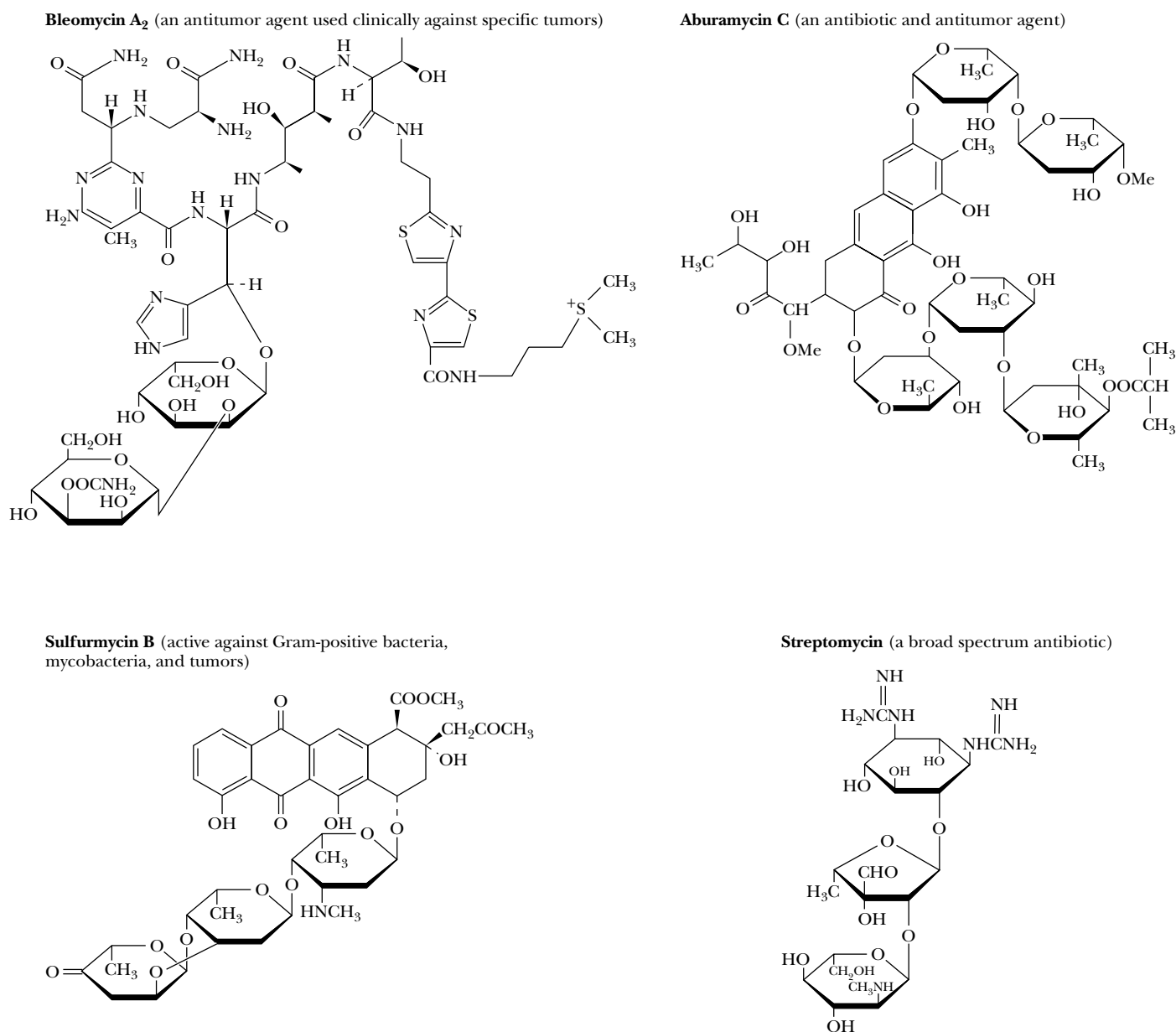


FIGURE 7.20 • Some antibiotics are oligosaccharides or contain oligosaccharide groups.

7.4 • Polysaccharides

Structure and Nomenclature

By far the majority of carbohydrate material in nature occurs in the form of polysaccharides. By our definition, polysaccharides include not only those substances composed only of glycosidically linked sugar residues but also molecules that contain polymeric saccharide structures linked via covalent bonds to amino acids, peptides, proteins, lipids, and other structures.

Polysaccharides, also called **glycans**, consist of monosaccharides and their derivatives. If a polysaccharide contains only one kind of monosaccharide molecule, it is a **homopolysaccharide**, or **homoglycan**, whereas those containing more than one kind of monosaccharide are **heteropolysaccharides**. The most common constituent of polysaccharides is D-glucose, but D-fructose, D-galactose, L-galactose, D-mannose, L-arabinose, and D-xylose are also common. Common monosaccharide derivatives in polysaccharides include the amino sugars (D-glucosamine and D-galactosamine), their derivatives (*N*-acetylneuraminic acid and *N*-acetylmuramic acid), and simple sugar acids (glucuronic and iduronic acids). Homopolysaccharides are often named for the sugar unit they contain, so that glucose homopolysaccharides are called **glucans**, while mannose homopolysaccharides are **mannans**. Other homopolysaccharide names are just as obvious: *galacturonans*, *arabinans*, and so on. Homopolysaccharides of uniform linkage type are often named by including notation to denote ring size and linkage type. Thus, cellulose is a (1→4)-β-D-glucopyranan. Polysaccharides differ not only in the nature of their component monosaccharides but also in the length of their chains and in the amount of chain branching that occurs. Although a given sugar residue has only one anomeric carbon and thus can form only one glycosidic linkage with hydroxyl groups on other molecules, each sugar residue carries several hydroxyls, one or more of which may be an acceptor of glycosyl substituents (Figure 7.21). This ability to form branched structures distinguishes polysaccharides from proteins and nucleic acids, which occur only as linear polymers.

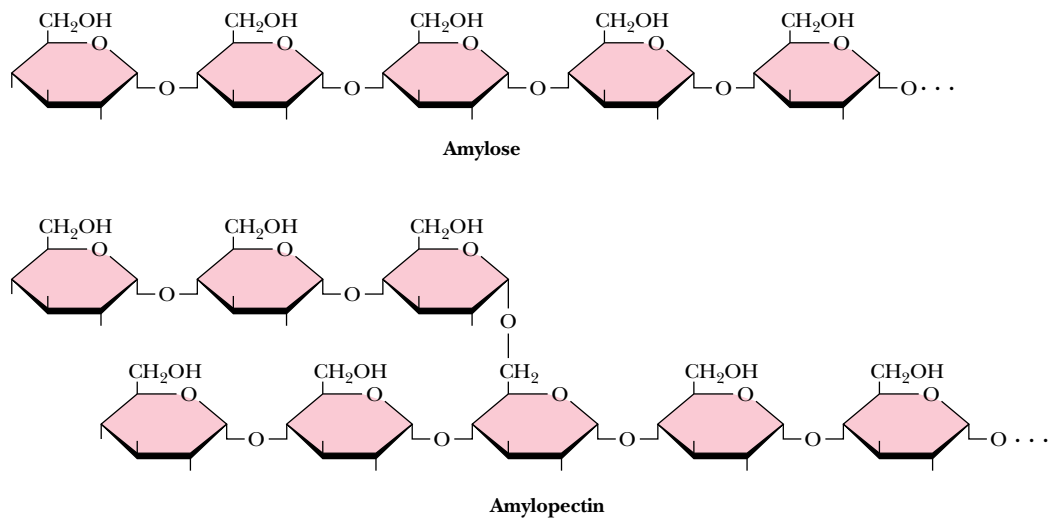


FIGURE 7.21 • Amylose and amylopectin are the two forms of starch. Note that the linear linkages are α(1→4), but the branches in amylopectin are α(1→6). Branches in polysaccharides can involve any of the hydroxyl groups on the monosaccharide components. Amylopectin is a highly branched structure, with branches occurring every 12 to 30 residues.

Polysaccharide Functions

The functions of many individual polysaccharides cannot be assigned uniquely, and some of their functions may not yet be appreciated. Traditionally, biochemistry textbooks have listed the functions of polysaccharides as storage materials, structural components, or protective substances. Thus, *starch*, *glycogen*, and other storage polysaccharides, as readily metabolizable food, provide energy reserves for cells. *Chitin* and *cellulose* provide strong support for the skeletons of arthropods and green plants, respectively. Mucopolysaccharides, such as the *hyaluronic acids*, form protective coats on animal cells. In each of these cases, the relevant polysaccharide is either a homopolymer or a polymer of small repeating units. Recent research indicates, however, that oligosaccharides and polysaccharides with varied structures may also be involved in much more sophisticated tasks in cells, including a variety of cellular recognition and intercellular communication events, as discussed later.

Storage Polysaccharides

Storage polysaccharides are an important carbohydrate form in plants and animals. It seems likely that organisms store carbohydrates in the form of polysaccharides rather than as monosaccharides to lower the osmotic pressure of the sugar reserves. Because osmotic pressures depend only on *numbers of molecules*, the osmotic pressure is greatly reduced by formation of a few polysaccharide molecules out of thousands (or even millions) of monosaccharide units.

Starch

By far the most common storage polysaccharide in plants is **starch**, which exists in two forms: **α -amylose** and **amylopectin**, the structures of which are shown in Figure 7.21. Most forms of starch in nature are 10 to 30% α -amylose and 70 to 90% amylopectin. Typical cornstarch produced in the United States is about 25% α -amylose and 75% amylopectin. α -Amylose is composed of linear chains of D-glucose in $\alpha(1\rightarrow4)$ linkages. The chains are of varying length, having molecular weights from several thousand to half a million. As can be seen from the structure in Figure 7.21, the chain has a reducing end and a nonreducing end. Although poorly soluble in water, α -amylose forms micelles in which the polysaccharide chain adopts a helical conformation (Figure 7.22). Iodine reacts with α -amylose to give a characteristic blue color, which arises from the insertion of iodine into the middle of the hydrophobic amylose helix.

In contrast to α -amylose, amylopectin, the other component of typical starches, is a highly branched chain of glucose units (Figure 7.21). Branches occur in these chains every 12 to 30 residues. The average branch length is between 24 and 30 residues, and molecular weights of amylopectin molecules can range up to 100 million. The linear linkages in amylopectin are $\alpha(1\rightarrow4)$, whereas the branch linkages are $\alpha(1\rightarrow6)$. As is the case for α -amylose, amylopectin forms micellar suspensions in water; iodine reacts with such suspensions to produce a red-violet color.

Starch is stored in plant cells in the form of granules in the stroma of plastids (plant cell organelles) of two types: **chloroplasts**, in which photosynthesis takes place, and **amyloplasts**, plastids that are specialized starch accumulation bodies. When starch is to be mobilized and used by the plant that stored it, it must be broken down into its component monosaccharides. Starch is split into its monosaccharide elements by stepwise phosphorolytic cleavage of glucose units, a reaction catalyzed by **starch phosphorylase** (Figure 7.23). This is formally an $\alpha(1\rightarrow4)$ -glucan phosphorylase reaction, and at each step, the prod-

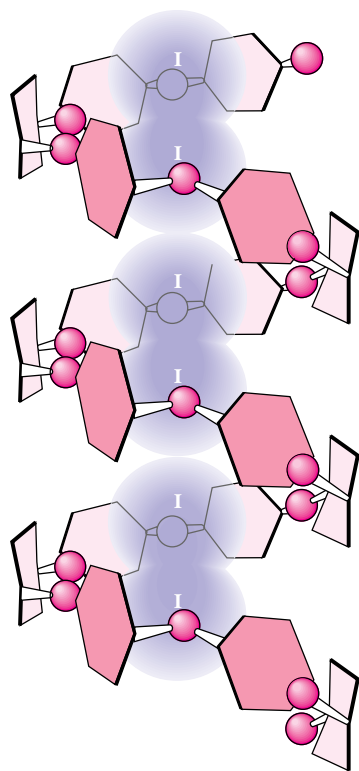


FIGURE 7.22 • Suspensions of amylose in water adopt a helical conformation. Iodine (I_2) can insert into the middle of the amylose helix to give a blue color that is characteristic and diagnostic for starch.

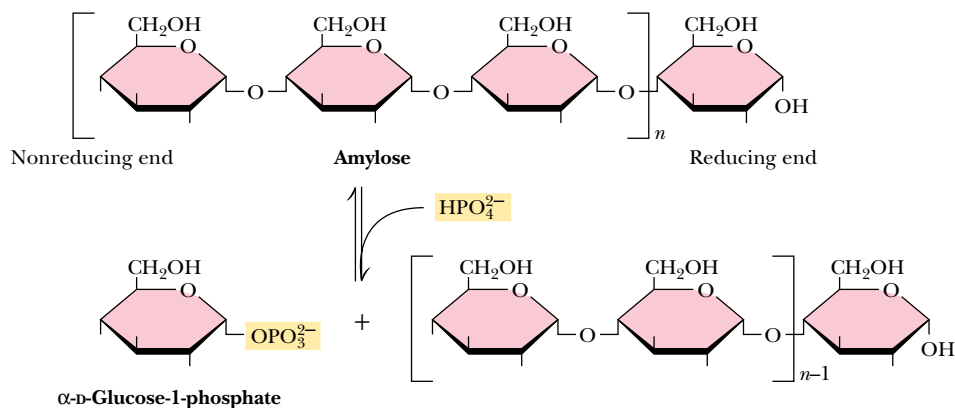


FIGURE 7.23 • The starch phosphorylase reaction cleaves glucose residues from amylose, producing α -D-glucose-1-phosphate.

ucts are one molecule of glucose-1-phosphate and a starch molecule with one less glucose unit. In α -amylose, this process continues all along the chain until the end is reached. However, the $\alpha(1\rightarrow6)$ branch points of amylopectin are not susceptible to cleavage by phosphorylase, and thorough digestion of amylopectin by phosphorylase leaves a *limit dextrin*, which must be attacked by an $\alpha(1\rightarrow6)$ -glucosidase to cleave the $1\rightarrow6$ branch points and allow complete hydrolysis of the remaining $1\rightarrow4$ linkages. Glucose-1-phosphate units are thus delivered to the plant cell, suitable for further processing in glycolytic pathways (see Chapter 19).

In animals, digestion and use of plant starches begins in the mouth with **salivary α -amylase** ($\alpha(1\rightarrow4)$ -glucan 4-glucanohydrolase), the major enzyme secreted by the salivary glands. Although the capability of making and secreting salivary α -amylases is widespread in the animal world, some animals (such as cats, dogs, birds, and horses) do not secrete them. Salivary α -amylase is an **endoamylase** that splits $\alpha(1\rightarrow4)$ glycosidic linkages only within the chain. Raw starch is not very susceptible to salivary endoamylase. However, when suspensions of starch granules are heated, the granules swell, taking up water and causing the polymers to become more accessible to enzymes. Thus, cooked starch is more digestible. In the stomach, salivary α -amylase is inactivated by the lower pH, but pancreatic secretions also contain α -amylase. **β -Amylase**, an enzyme absent in animals but prevalent in plants and microorganisms, cleaves disaccharide (maltose) units from the termini of starch chains and is an **exoamylase**. Neither α -amylase nor β -amylase, however, can cleave the $\alpha(1\rightarrow6)$ branch points of amylopectin, and once again, $\alpha(1\rightarrow6)$ -glucosidase is required to cleave at the branch points and allow complete hydrolysis of starch amylopectin.

Glycogen

The major form of storage polysaccharide in animals is **glycogen**. Glycogen is found mainly in the liver (where it may amount to as much as 10% of liver mass) and skeletal muscle (where it accounts for 1 to 2% of muscle mass). Liver glycogen consists of granules containing highly branched molecules, with $\alpha(1\rightarrow6)$ branches occurring every 8 to 12 glucose units. Like amylopectin, glycogen yields a red-violet color with iodine. Glycogen can be hydrolyzed by both α - and β -amylases, yielding glucose and maltose, respectively, as products and can also be hydrolyzed by **glycogen phosphorylase**, an enzyme present in liver and muscle tissue, to release glucose-1-phosphate.

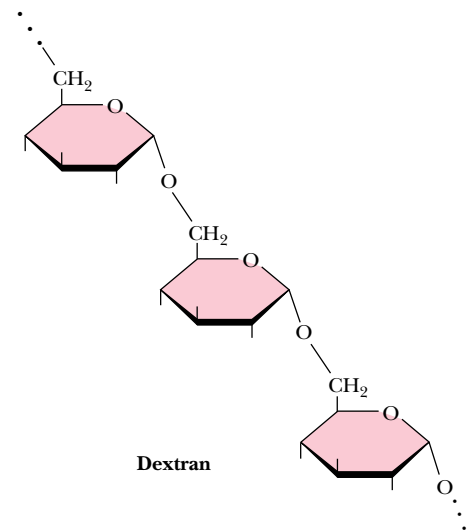


FIGURE 7.24 • Dextran is a branched polymer of D-glucose units. The main chain linkage is $\alpha(1\rightarrow6)$, but $1\rightarrow2$, $1\rightarrow3$, or $1\rightarrow4$ branches can occur.

Dextran

Another important family of storage polysaccharides are the **dextrans**, which are $\alpha(1\rightarrow6)$ -linked polysaccharides of D-glucose with branched chains found in yeast and bacteria (Figure 7.24). Because the main polymer chain is $\alpha(1\rightarrow6)$ linked, the repeating unit is *isomaltose*, $\text{Glc}\alpha(1\rightarrow6)\text{-Glc}$. The branch points may be $1\rightarrow2$, $1\rightarrow3$, or $1\rightarrow4$ in various species. The degree of branching and the average chain length between branches depend on the species and strain of the organism. Bacteria growing on the surfaces of teeth produce extracellular accumulations of dextrans, an important component of *dental plaque*. Bacterial dextrans are frequently used in research laboratories as the support medium for column chromatography of macromolecules. Dextran chains cross-linked with epichlorohydrin yield the structure shown in Figure 7.25. These preparations (known by various trade names such as Sephadex and Bio-gel) are extremely hydrophilic and swell to form highly hydrated gels in water. Depending on the degree of cross-linking and the size of the gel particle, these materials form gels containing from 50 to 98% water. Dextran can also be cross-linked with other agents, forming gels with slightly different properties.

Structural Polysaccharides

Cellulose

The **structural polysaccharides** have properties that are dramatically different from those of the storage polysaccharides, even though the compositions of these two classes are similar. The structural polysaccharide **cellulose** is the most

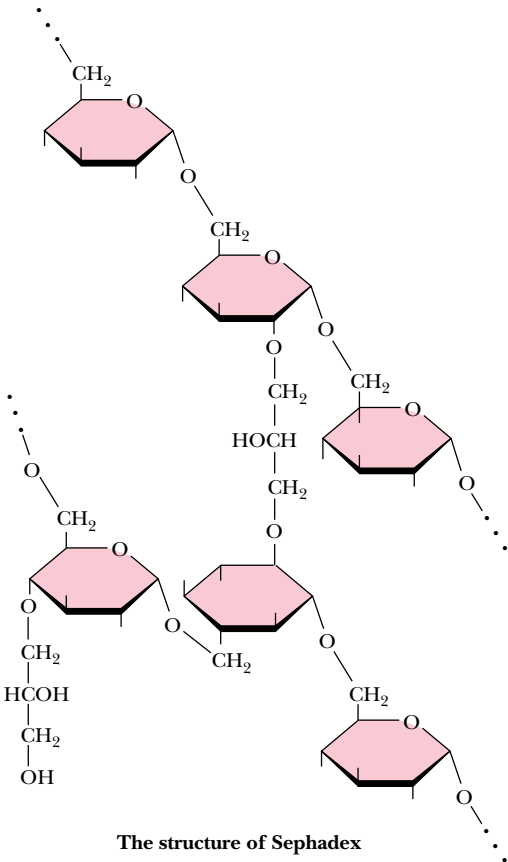


FIGURE 7.25 • Sephadex gels are formed from dextran chains cross-linked with epichlorohydrin. The degree of cross-linking determines the chromatographic properties of Sephadex gels. Sephacryl gels are formed by cross-linking of dextran polymers with *N,N'*-methylene bisacrylamide.

FIGURE 7.27 • The structure of cellulose, showing the hydrogen bonds (blue) between the sheets, which strengthen the structure. Intrachain hydrogen bonds are in red and interchain hydrogen bonds are in green.

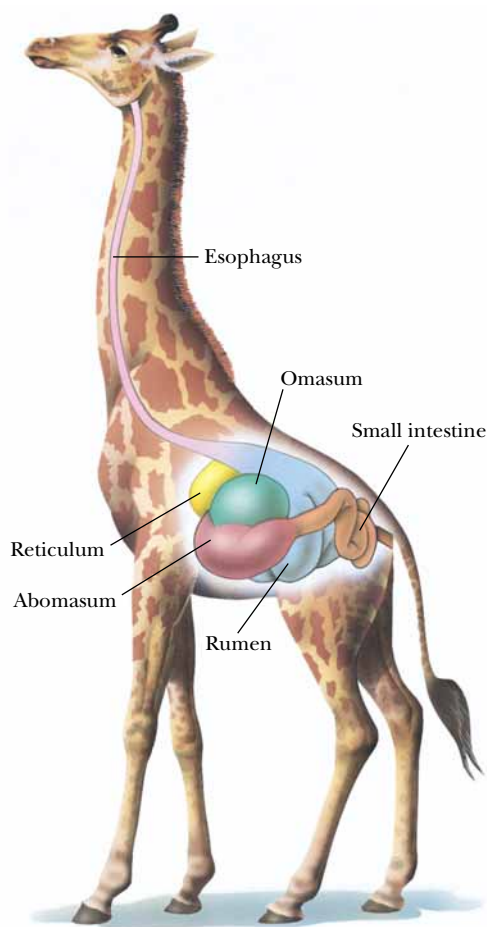
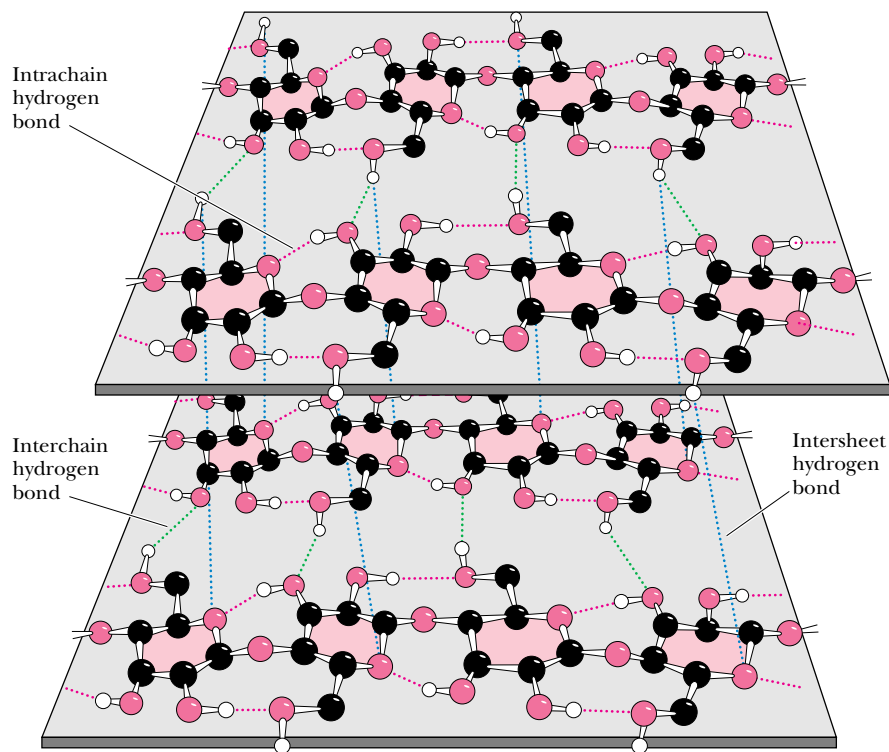


FIGURE 7.28 • Giraffes, cattle, deer, and camels are ruminant animals that are able to metabolize cellulose, thanks to bacterial cellulase in the rumen, a large first compartment in the stomach of a ruminant.

Chitin

A polysaccharide that is similar to cellulose, both in its biological function and its primary, secondary, and tertiary structure, is **chitin**. Chitin is present in the cell walls of fungi and is the fundamental material in the exoskeletons of crustaceans, insects, and spiders. The structure of chitin, an extended ribbon, is identical to cellulose, except that the —OH group on each C-2 is replaced by —NHCOCH₃, so that the repeating units are *N*-acetyl-*D*-glucosamines in $\beta(1\rightarrow4)$ linkage. Like cellulose (Figure 7.27), the chains of chitin form extended ribbons (Figure 7.29) and pack side by side in a crystalline, strongly hydrogen-bonded form. One significant difference between cellulose and chitin is whether the chains are arranged in **parallel** (all the reducing ends together at one end of a packed bundle and all the nonreducing ends together at the other end) or **antiparallel** (each sheet of chains having the chains arranged oppositely from the sheets above and below). Natural cellulose seems to occur only in parallel arrangements. Chitin, however, can occur in three forms, sometimes all in the same organism. α -*Chitin* is an all-parallel arrangement of the chains, whereas β -*chitin* is an antiparallel arrangement. In δ -*chitin*, the structure is thought to involve pairs of parallel sheets separated by single antiparallel sheets.

Chitin is the earth's second most abundant carbohydrate polymer (after cellulose), and its ready availability and abundance offer opportunities for industrial and commercial applications. Chitin-based coatings can extend the shelf life of fruits, and a chitin derivative that binds to iron atoms in meat has been found to slow the reactions that cause rancidity and flavor loss. Without such a coating, the iron in meats activates oxygen from the air, forming reactive free radicals that attack and oxidize polyunsaturated lipids, causing most of the flavor loss associated with rancidity. Chitin-based coatings coordinate the iron atoms, preventing their interaction with oxygen.

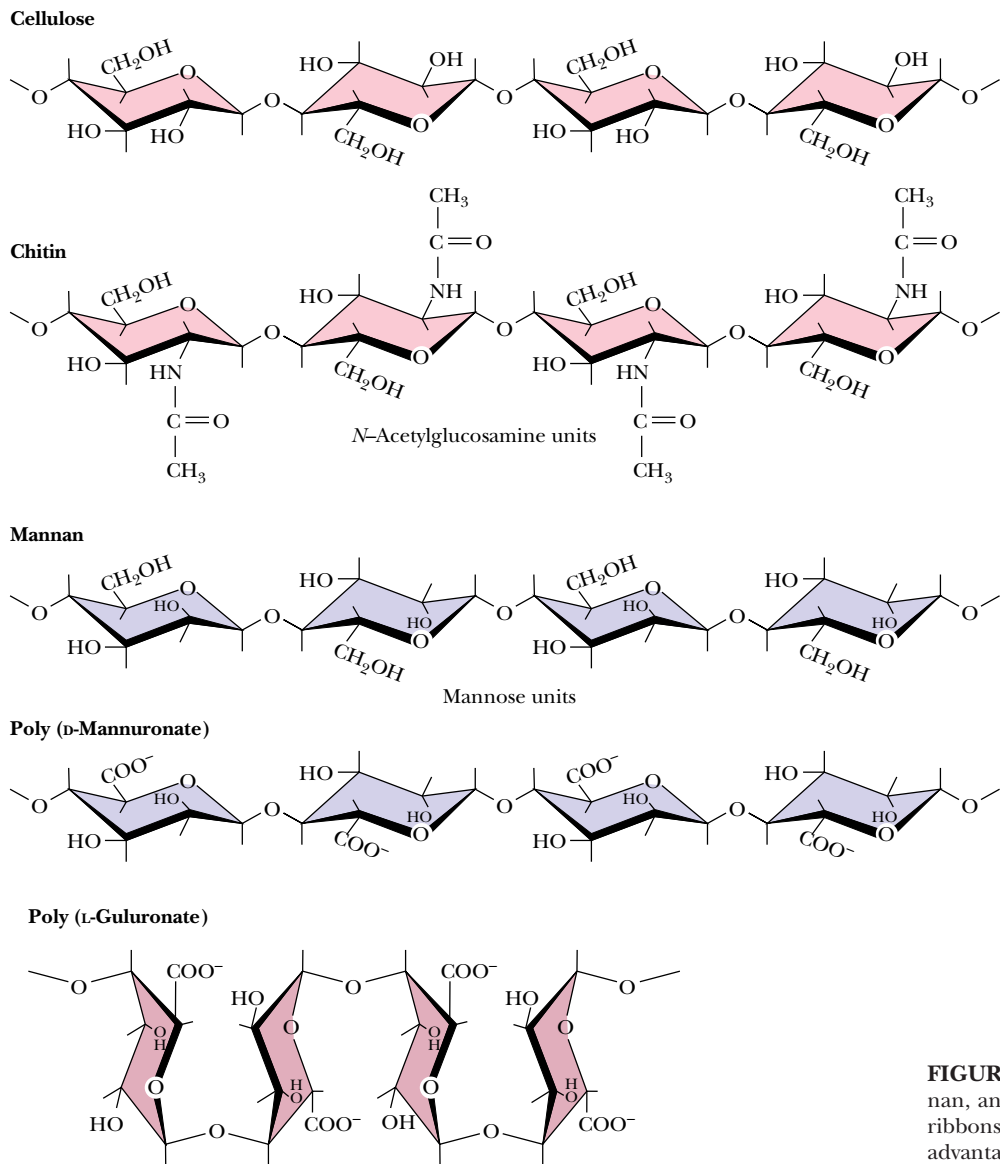


FIGURE 7.29 • Like cellulose, chitin, mannan, and poly(D-mannuronate) form extended ribbons and pack together efficiently, taking advantage of multiple hydrogen bonds.

Alginates

A family of novel extended ribbon structures that bind metal ions, particularly calcium, in their structure are the **alginate** polysaccharides of marine brown algae (*Phaeophyceae*). These include **poly(β-D-mannuronate)** and **poly(α-L-guluronate)**, which are (1→4) linked chains formed from β-mannuronic acid and α-L-guluronic acid, respectively. Both of these homopolymers are found together in most marine alginates, although to widely differing extents, and mixed chains containing both monomer units are also found. As shown in Figure 7.29, the conformation of poly(β-D-mannuronate) is similar to that of cellulose. In the solid state, the free form of the polymer exists in cellulose-like form. However, complexes of the polymer with cations (such as lithium, sodium, potassium, and calcium) adopt a threefold helix structure, presumably to accommodate the bound cations. For poly(α-L-guluronate) (Figure 7.29), the axial-axial configuration of the glycosidic linkage leads to a distinctly buckled ribbon with limited flexibility. Cooperative interactions between such buckled ribbons can only

A D E E P E R L O O K

Billiard Balls, Exploding Teeth, and Dynamite— The Colorful History of Cellulose

Although humans cannot digest it and most people's acquaintance with cellulose is limited to comfortable cotton clothing, cellulose has enjoyed a colorful and varied history of utilization. In 1838, Théophile Pelouze in France found that paper or cotton could be made explosive if dipped in concentrated nitric acid. Christian Schönbein, a professor of chemistry at the University of Basel, prepared "nitrocotton" in 1845 by dipping cotton in a mixture of nitric and sulfuric acids and then washing the material to remove excess acid. In 1860, Major E. Schultze of the Prussian army used the same material, now called **guncotton**, as a propellant replacement for gunpowder, and its preparation in brass cartridges soon made it popular for this purpose. The only problem was that it was too explosive and could detonate unpredictably in factories where it was produced. The entire town of Faversham, England, was destroyed in such an accident. In 1868, Alfred Nobel mixed guncotton with ether and alcohol, thus preparing **nitro-cellulose**, and in turn mixed this with nitroglycerine and sawdust to produce **dynamite**. Nobel's income from dynamite and also

from his profitable development of the Russian oil fields in Baku eventually formed the endowment for the Nobel Prizes.

In 1869, concerned over the precipitous decline (from hunting) of the elephant population in Africa, the billiard ball manufacturers Phelan and Collander offered a prize of \$10,000 for production of a substitute for ivory. Brothers Isaiah and John Hyatt in Albany, New York, produced a substitute for ivory by mixing guncotton with camphor, then heating and squeezing it to produce **celluloid**. This product found immediate uses well beyond billiard balls. It was easy to shape, strong, and resilient, and it exhibited a high tensile strength. Celluloid was used eventually to make dolls, combs, musical instruments, fountain pens, piano keys, and a variety of other products. The Hyatt brothers eventually formed the Albany Dental Company to make false teeth from celluloid. Because camphor was used in their production, the company advertised that their teeth smelled "clean," but, as reported in the *New York Times* in 1875, the teeth also occasionally exploded!

Portions adapted from Burke, J., 1996. *The Pinball Effect: How Renaissance Water Gardens Made the Carburator Possible and Other Journeys Through Knowledge*. New York: Little, Brown, & Company.

be strong if the interstices are filled effectively with water molecules or metal ions. Figure 7.30 shows a molecular model of a Ca^{2+} -induced dimer of poly(α -L-guluronate).

Agarose

An important polysaccharide mixture isolated from marine red algae (*Rhodophyceae*) is **agar**, which consists of two components, **agarose** and **agaropectin**. Agarose (Figure 7.31) is a chain of alternating D-galactose and 3,6-anhydro-L-galactose, with side chains of 6-methyl-D-galactose. Agaropectin is similar, but contains in addition sulfate ester side chains and D-glucuronic acid. The three-dimensional structure of agarose is a double helix with a threefold screw axis, as shown in Figure 7.31. The central cavity is large enough to accommodate water molecules. Agarose and agaropectin readily form gels containing large amounts (up to 99.5%) of water. Agarose can be processed to remove most of the charged groups, yielding a material (trade name Sepharose) useful for purification of macromolecules in gel exclusion chromatography. Pairs of chains form double helices that subsequently aggregate in bundles to form a stable gel, as shown in Figure 7.32.

Glycosaminoglycans

A class of polysaccharides known as **glycosaminoglycans** is involved in a variety of extracellular (and sometimes intracellular) functions. Glycosaminoglycans consist of linear chains of repeating disaccharides in which one of the monosaccharide units is an amino sugar and one (or both) of the monosaccharide units contains at least one negatively charged sulfate or carboxylate group. The repeating disaccharide structures found commonly in glycosaminoglycans are

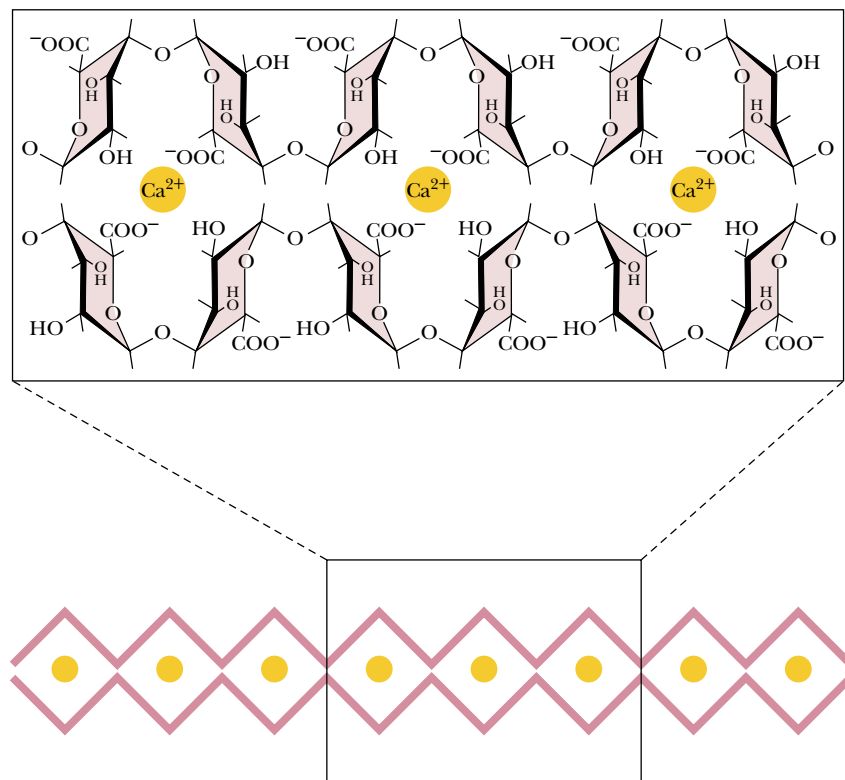


FIGURE 7.30 • Poly(α-L-guluronate) strands dimerize in the presence of Ca²⁺, forming a structure known as an “egg carton.”

shown in Figure 7.33. **Heparin**, with the highest net negative charge of the disaccharides shown, is a natural anticoagulant substance. It binds strongly to *antithrombin III* (a protein involved in terminating the clotting process) and inhibits blood clotting. **Hyaluronate** molecules may consist of as many as 25,000 disaccharide units, with molecular weights of up to 10⁷. Hyaluronates are impor-

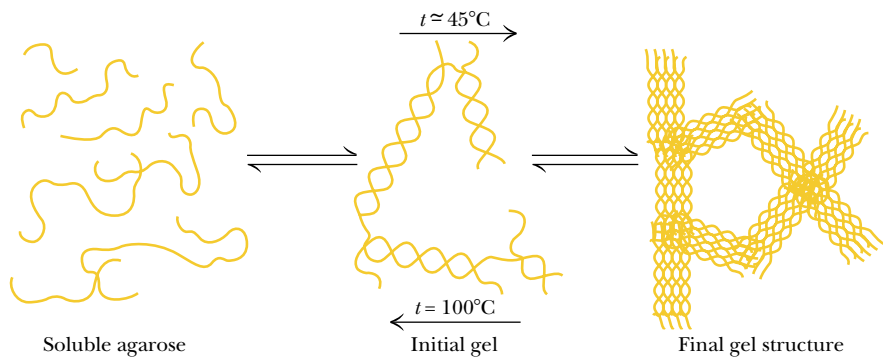


FIGURE 7.32 • The ability of agarose to assemble in complex bundles to form gels in aqueous solution makes it useful in numerous chromatographic procedures, including gel exclusion chromatography and electrophoresis. Cells grown in culture can be embedded in stable agarose gel “threads” so that their metabolic and physiological properties can be studied.

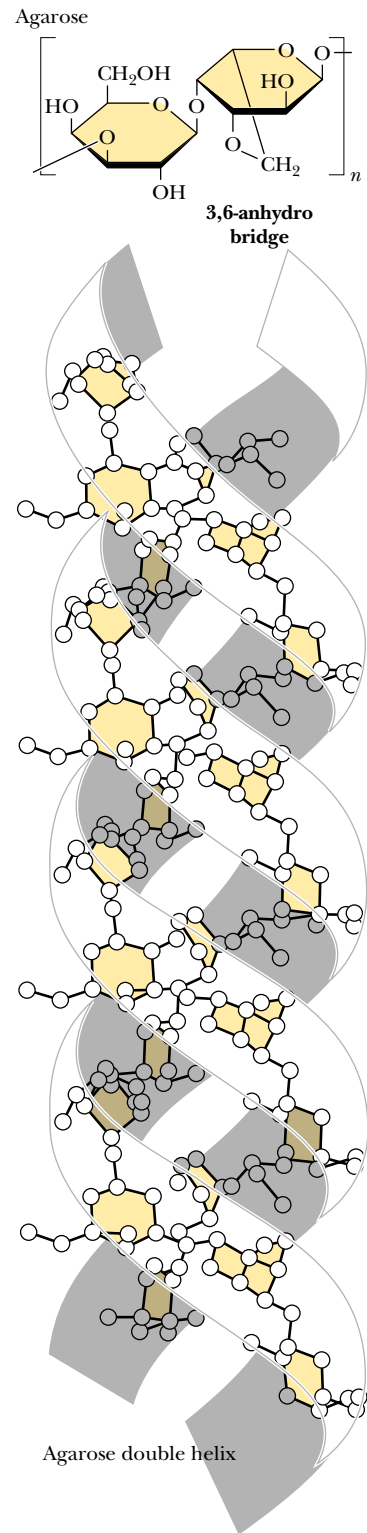


FIGURE 7.31 • The favored conformation of agarose in water is a double helix with a three-fold screw axis.

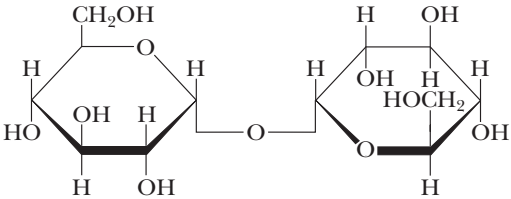
FIGURE 7.33 • Glycosaminoglycans are formed from repeating disaccharide arrays. Glycosaminoglycans are components of the proteoglycans.

D-Glucuronate N-Acetyl-D-galactosamine-4-sulfate	D-Glucuronate-2-sulfate N-Sulfo-D-glucosamine-6-sulfate
Chondroitin-4-sulfate	Heparin
D-Glucuronate N-Acetyl-D-galactosamine-6-sulfate	D-Glucuronate N-Acetyl-D-glucosamine
Chondroitin-6-sulfate	Hyaluronate
L-Iduronate N-Acetyl-D-galactosamine-4-sulfate	D-Galactose N-Acetyl-D-glucosamine-6-sulfate
Dermatan sulfate	Keratan sulfate

tant components of the vitreous humor in the eye and of *synovial fluid*, the lubricant fluid of joints in the body. The **chondroitins** and **keratan sulfate** are found in tendons, cartilage, and other connective tissue, whereas **dermatan sulfate**, as its name implies, is a component of the extracellular matrix of skin. Glycosaminoglycans are fundamental constituents of *proteoglycans* (discussed later).

PROBLEMS

- 1. Draw Haworth structures for the two possible isomers of D-altrose (Figure 7.2) and D-psicose (Figure 7.3).
- 2. Give the systematic name for stachyose (Figure 7.19).
- 3. Trehalose, a disaccharide produced in fungi, has the following structure:



- a. What is the systematic name for this disaccharide?
- b. Is trehalose a reducing sugar? Explain.

- 4. Draw a Fischer projection structure for L-sorbose (D-sorbose is shown in Figure 7.3).
- 5. α -D-Glucose has a specific rotation, $[\alpha]_D^{20}$, of $+112.2^\circ$, whereas β -D-glucose has a specific rotation of $+18.7^\circ$. What is the composition of a mixture of α -D- and β -D-glucose, which has a specific rotation of 83.0° ?
- 6. A 0.2-g sample of amylopectin was analyzed to determine the fraction of the total glucose residues that are branch points in the structure. The sample was exhaustively methylated and then digested, yielding $50\text{ }\mu\text{mol}$ of 2,3-dimethylglucose and $0.4\text{ }\mu\text{mol}$ of 1, 2, 3, 6-tetramethylglucose.
 - a. What fraction of the total residues are branch points?
 - b. How many reducing ends does this amylopectin have?

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A feast of fat things, a feast of wines on the lees.

ISAIAH 25:6

OUTLINE

- 8.1 • Fatty Acids
- 8.2 • Triacylglycerols
- 8.3 • Glycerophospholipids
- 8.4 • Sphingolipids
- 8.5 • Waxes
- 8.6 • Terpenes
- 8.7 • Steroids

Chapter 8

Lipids



“The mighty whales which swim in a sea of water, and have a sea of oil swimming in them.”
Herman Melville, “Extracts.” *Moby Dick*. New York: Penguin Books, 1972. (Humpback whale
(*Megaptera novaeangliae*) breaching, Cape Cod, MA; photo © Steven Morello/Peter Arnold, Inc.)

Lipids are a class of biological molecules defined by low solubility in water and high solubility in nonpolar solvents. As molecules that are largely hydrocarbon in nature, lipids represent highly reduced forms of carbon and, upon oxidation in metabolism, yield large amounts of energy. Lipids are thus the molecules of choice for metabolic energy storage.

The lipids found in biological systems are either **hydrophobic** (containing only nonpolar groups) or **amphipathic**, which means they possess both polar and nonpolar groups. The hydrophobic nature of lipid molecules allows membranes to act as effective barriers to more polar molecules. In this chapter, we discuss the chemical and physical properties of the various classes of lipid molecules. The following chapter considers membranes, whose properties depend intimately on their lipid constituents.

8.1 • Fatty Acids

A **fatty acid** is composed of a long hydrocarbon chain (“tail”) and a terminal carboxyl group (or “head”). The carboxyl group is normally ionized under physiological conditions. Fatty acids occur in large amounts in biological systems, but rarely in the free, uncomplexed state. They typically are esterified to glycerol or other backbone structures. Most of the fatty acids found in nature have an even number of carbon atoms (usually 14 to 24). Certain marine organisms, however, contain substantial amounts of fatty acids with odd numbers of carbon atoms. Fatty acids are either **saturated** (all carbon–carbon bonds are single bonds) or **unsaturated** (with one or more double bonds in the hydrocarbon chain). If a fatty acid has a single double bond, it is said to be **monounsaturated**, and if it has more than one, **polyunsaturated**. Fatty acids can be named or described in at least three ways, as listed in Table 8.1. For example, a fatty acid composed of an 18-carbon chain with no double bonds can be called by its systematic name (**octadecanoic acid**), its common name (stearic acid), or its shorthand notation, in which the number of carbons is followed by a colon and the number of double bonds in the molecule (18:0 for stearic acid). The structures of several fatty acids are given in Figure 8.1. **Stearic acid** (18:0) and **palmitic acid** (16:0) are the most common saturated fatty acids in nature.

Free rotation around each of the carbon–carbon bonds makes saturated fatty acids extremely flexible molecules. Owing to steric constraints, however, the fully extended conformation (Figure 8.1) is the most stable for saturated fatty acids. Nonetheless, the degree of stabilization is slight, and (as will be seen) saturated fatty acid chains adopt a variety of conformations.

Unsaturated fatty acids are slightly more abundant in nature than saturated fatty acids, especially in higher plants. The most common unsaturated fatty acid

Table 8.1
Common Biological Fatty Acids

Number of Carbons	Common Name	Systematic Name	Symbol	Structure
Saturated fatty acids				
12	Lauric acid	Dodecanoic acid	12:0	$\text{CH}_3(\text{CH}_2)_{10}\text{COOH}$
14	Myristic acid	Tetradecanoic acid	14:0	$\text{CH}_3(\text{CH}_2)_{12}\text{COOH}$
16	Palmitic acid	Hexadecanoic acid	16:0	$\text{CH}_3(\text{CH}_2)_{14}\text{COOH}$
18	Stearic acid	Octadecanoic acid	18:0	$\text{CH}_3(\text{CH}_2)_{16}\text{COOH}$
20	Arachidic acid	Eicosanoic acid	20:0	$\text{CH}_3(\text{CH}_2)_{18}\text{COOH}$
22	Behenic acid	Docosanoic acid	22:0	$\text{CH}_3(\text{CH}_2)_{20}\text{COOH}$
24	Lignoceric acid	Tetracosanoic acid	24:0	$\text{CH}_3(\text{CH}_2)_{22}\text{COOH}$
Unsaturated fatty acids (all double bonds are <i>cis</i>)				
16	Palmitoleic acid	9-Hexadecenoic acid	16:1	$\text{CH}_3(\text{CH}_2)_5\text{CH}=\text{CH}(\text{CH}_2)_7\text{COOH}$
18	Oleic acid	9-Octadecenoic acid	18:1	$\text{CH}_3(\text{CH}_2)_7\text{CH}=\text{CH}(\text{CH}_2)_7\text{COOH}$
18	Linoleic acid	9,12-Octadecadienoic acid	18:2	$\text{CH}_3(\text{CH}_2)_4(\text{CH}=\text{CHCH}_2)_2(\text{CH}_2)_6\text{COOH}$
18	α -Linolenic acid	9,12,15-Octadecatrienoic acid	18:3	$\text{CH}_3\text{CH}_2(\text{CH}=\text{CHCH}_2)_3(\text{CH}_2)_6\text{COOH}$
18	γ -Linolenic acid	6,9,12-Octadecatrienoic acid	18:3	$\text{CH}_3(\text{CH}_2)_4(\text{CH}=\text{CHCH}_2)_3(\text{CH}_2)_3\text{COOH}$
20	Arachidonic acid	5,8,11,14-Eicosatetraenoic acid	20:4	$\text{CH}_3(\text{CH}_2)_4(\text{CH}=\text{CHCH}_2)_4(\text{CH}_2)_2\text{COOH}$
24	Nervonic acid	15-Tetracosenoic acid	24:1	$\text{CH}_3(\text{CH}_2)_7\text{CH}=\text{CH}(\text{CH}_2)_{13}\text{COOH}$

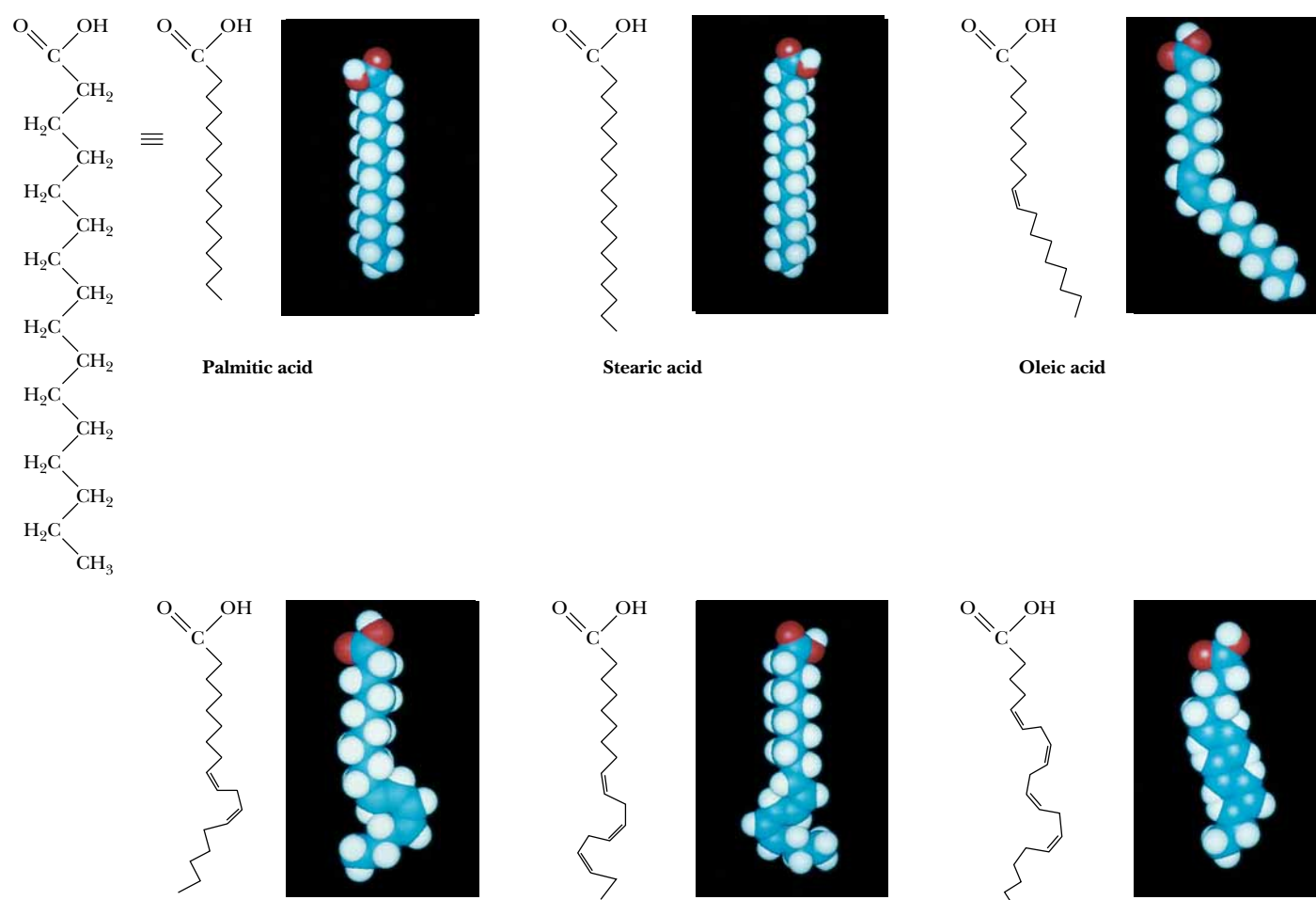


FIGURE 8.1 • The structures of some typical fatty acids. Note that most natural fatty acids contain an even number of carbon atoms and that the double bonds are nearly always *cis* and rarely conjugated.

is **oleic acid**, or 18:1(9), with the number in parentheses indicating that the double bond is between carbons 9 and 10. The number of double bonds in an unsaturated fatty acid varies typically from one to four, but, in the fatty acids found in most bacteria, this number rarely exceeds one.

The double bonds found in fatty acids are nearly always in the *cis* configuration. As shown in Figure 8.1, this causes a bend or “kink” in the fatty acid chain. This bend has very important consequences for the structure of biological membranes. Saturated fatty acid chains can pack closely together to form ordered, rigid arrays under certain conditions, but unsaturated fatty acids prevent such close packing and produce flexible, fluid aggregates.

Some fatty acids are not synthesized by mammals and yet are necessary for normal growth and life. These *essential fatty acids* include **linoleic** and **γ -linolenic acids**. These must be obtained by mammals in their diet (specifically from plant sources). **Arachidonic acid**, which is not found in plants, can only be synthesized by mammals from linoleic acid. At least one function of the essential fatty acids is to serve as a precursor for the synthesis of **eicosanoids**, such as

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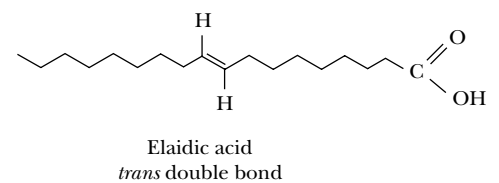
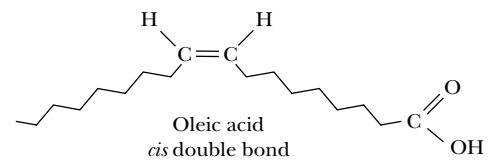
Fatty Acids in Food: Saturated Versus Unsaturated

Fats consumed in the modern human diet vary widely in their fatty acid compositions. The table below provides a brief summary. The incidence of cardiovascular disease is correlated with diets high in saturated fatty acids. By contrast, a diet that is relatively higher in unsaturated fatty acids (especially polyunsaturated fatty acids) may reduce the risk of heart attacks and strokes. Corn oil, abundant in the United States and high in (polyunsaturated) linoleic acid, is an attractive dietary choice. *Margarine* made from corn, safflower, or sunflower oils is much lower in saturated fatty acids than is butter, which is made from milk fat. However, margarine may present its own health risks. Its fatty acids contain *trans*-double bonds (introduced by the hydrogenation process), which may also contribute to cardiovascular disease. (Margarine was invented by a French chemist, H. Mège Mouriès, who won a prize from Napoleon III in 1869 for developing a substitute for butter.)

Although vegetable oils usually contain a higher proportion of unsaturated fatty acids than do animal oils and fats, several plant oils are actually high in saturated fats. Palm oil is low in polyunsaturated fatty acids and particularly high in (saturated) palmitic acid (whence the name *palmitic*). Coconut oil is particularly high in lauric and myristic acids (both saturated) and contains very few unsaturated fatty acids.

Some of the fatty acids found in the diets of developed nations (often 1 to 10 g of daily fatty acid intake) are *trans* fatty acids—fatty acids with one or more double bonds in the *trans* configuration. Some of these derive from dairy fat and ruminant meats, but the bulk are provided by partially hydrogenated vegetable or fish

oils. Substantial evidence now exists to indicate that *trans* fatty acids may have deleterious health consequences. Numerous studies have shown that *trans* fatty acids raise plasma LDL cholesterol levels when exchanged for *cis*-unsaturated fatty acids in the diet and may also lower HDL cholesterol levels and raise triglyceride levels. The effects of *trans* fatty acids on LDL, HDL, and cholesterol levels are similar to those of saturated fatty acids, and diets aimed at reducing the risk of coronary heart disease should be low in both *trans* and saturated fatty acids.



Structure of *cis* and *trans* monounsaturated C₁₈ fatty acids.

Fatty Acid Compositions of Some Dietary Lipids*					
Source	Lauric and Myristic	Palmitic	Stearic	Oleic	Linoleic
Beef	5	24–32	20–25	37–43	2–3
Milk		25	12	33	3
Coconut	74	10	2	7	—
Corn		8–12	3–4	19–49	34–62
Olive		9	2	84	4
Palm		39	4	40	8
Safflower		6	3	13	78
Soybean		9	6	20	52
Sunflower		6	1	21	66

Data from *Merck Index*, 10th ed. Rahway, NJ: Merck and Co.; and Wilson, et al., 1967, *Principles of Nutrition*, 2nd ed. New York: Wiley.

*Values are percentages of total fatty acids.

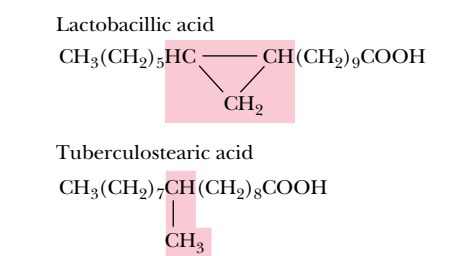


FIGURE 8.2 • Structures of two unusual fatty acids: lactobacillic acid, a fatty acid containing a cyclopropane ring, and tuberculostearic acid, a branched-chain fatty acid.

prostaglandins, a class of compounds that exert hormone-like effects in many physiological processes (discussed in Chapter 25).

In addition to unsaturated fatty acids, several other modified fatty acids are found in nature. Microorganisms, for example, often contain branched-chain fatty acids, such as **tuberculostearic acid** (Figure 8.2). When these fatty acids are incorporated in membranes, the methyl group constitutes a local structural perturbation in a manner similar to the double bonds in unsaturated fatty acids (see Chapter 9). Some bacteria also synthesize fatty acids containing cyclic structures such as cyclopropane, cyclopropene, and even cyclopentane rings.

8.2 • Triacylglycerols

A significant number of the fatty acids in plants and animals exist in the form of **triacylglycerols** (also called **triglycerides**). Triacylglycerols are a major energy reserve and the principal neutral derivatives of glycerol found in animals. These molecules consist of a glycerol esterified with three fatty acids (Figure 8.3). If all three fatty acid groups are the same, the molecule is called a simple triacylglycerol. Examples include **tristearoylglycerol** (common name *tristearin*) and **trioleoylglycerol** (*triolein*). Mixed triacylglycerols contain two or three different fatty acids. Triacylglycerols in animals are found primarily in the adipose tissue (body fat), which serves as a depot or storage site for lipids. Monoacylglycerols and diacylglycerols also exist, but are far less common than the triacylglycerols. Most natural plant and animal fat is composed of mixtures of simple and mixed triacylglycerols.

Acylglycerols can be hydrolyzed by heating with acid or base or by treatment with lipases. Hydrolysis with alkali is called **saponification** and yields salts of free fatty acids and glycerol. This is how **soap** (a metal salt of an acid derived from fat) was made by our ancestors. One method used potassium hydroxide (*potash*) leached from wood ashes to hydrolyze animal fat (mostly triacylglycerols). (The tendency of such soaps to be precipitated by Mg^{2+} and Ca^{2+} ions in hard water makes them less useful than modern detergents.) When the fatty acids esterified at the first and third carbons of glycerol are different, the sec-

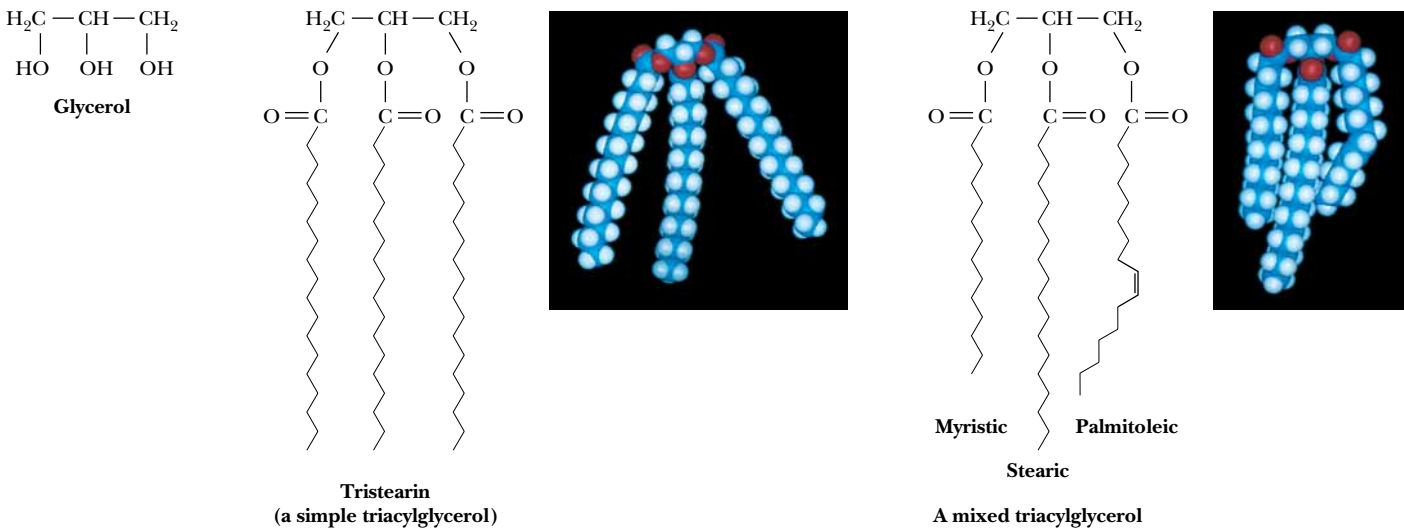


FIGURE 8.3 • Triacylglycerols are formed from glycerol and fatty acids.

A DEEPER LOOK

Polar Bears Use Triacylglycerols to Survive Long Periods of Fasting

The polar bear is magnificently adapted to thrive in its harsh Arctic environment. Research by Malcolm Ramsey (at the University of Saskatchewan in Canada) and others has shown that polar bears eat only during a few weeks out of the year and then fast for periods of 8 months or more, consuming no food or water during that time. Eating mainly in the winter, the adult polar bear feeds almost exclusively on seal blubber (largely composed of triacylglycerols), thus building up its own triacylglycerol reserves. Through the Arctic summer, the polar bear maintains normal physical activity, roaming over long distances, but relies entirely on its body fat for sustenance, burning as much as 1 to 1.5 kg of fat per day. It neither urinates nor defecates for extended periods. All the water needed to sustain life is provided from the metabolism of triacylglycerides (because oxidation of fatty acids yields carbon dioxide and water).

Ironically, the word *Arctic* comes from the ancient Greeks, who understood that the northernmost part of the earth lay under the stars of the constellation Ursa Major, the Great Bear. Although unaware of the polar bear, they called this region *Arktikós*, which means “the country of the great bear.”



(Thomas D. Mangelsen/Images of Nature)

ond carbon is asymmetric. The various acylglycerols are normally soluble in benzene, chloroform, ether, and hot ethanol. Although triacylglycerols are insoluble in water, mono- and diacylglycerols readily form organized structures in water (discussed later), owing to the polarity of their free hydroxyl groups.

Triacylglycerols are rich in highly reduced carbons and thus yield large amounts of energy in the oxidative reactions of metabolism. Complete oxidation of 1 g of triacylglycerols yields about 38 kJ of energy, whereas proteins and carbohydrates yield only about 17 kJ/g. Also, their hydrophobic nature allows them to aggregate in highly anhydrous forms, whereas polysaccharides and proteins are highly hydrated. For these reasons, triacylglycerols are the molecules of choice for energy storage in animals. Body fat (mainly triacylglycerols) also provides good insulation. Whales and Arctic mammals rely on body fat for both insulation and energy reserves.

8.3 • Glycerophospholipids

A 1,2-diacylglycerol that has a phosphate group esterified at carbon atom 3 of the glycerol backbone is a **glycerophospholipid**, also known as a *phosphoglyceride* or a *glycerol phosphatide* (Figure 8.4). These lipids form one of the largest classes of natural lipids and one of the most important. They are essential components of cell membranes and are found in small concentrations in other parts of the cell. It should be noted that all glycerophospholipids are members of the broader class of lipids known as **phospholipids**.

The numbering and nomenclature of glycerophospholipids present a dilemma in that the number 2 carbon of the glycerol backbone of a phos-

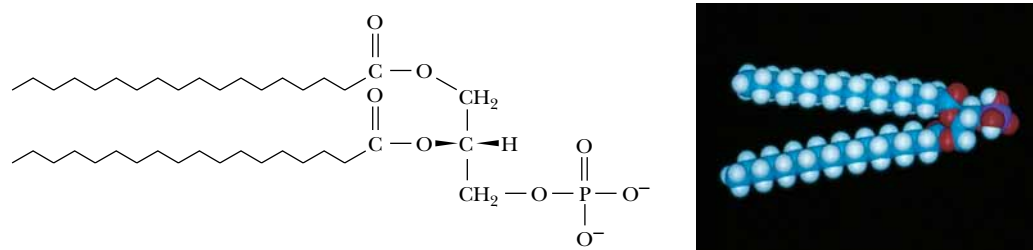


FIGURE 8.4 • Phosphatidic acid, the parent compound for glycerophospholipids.

pholipid is asymmetric. It is possible to name these molecules either as D- or L-isomers. Thus, glycerol phosphate itself can be referred to either as D-glycerol-1-phosphate or as L-glycerol-3-phosphate (Figure 8.5). Instead of naming the glycerol phosphatides in this way, biochemists have adopted the *stereospecific numbering* or *sn*-system. In this system, the *pro-S* position of a prochiral atom is denoted as the *1-position*, the prochiral atom as the *2-position*, and so on. When this scheme is used, the prefix *sn*-precedes the molecule name (glycerol phosphate in this case) and distinguishes this nomenclature from other approaches. In this way, the glycerol phosphate in natural phosphoglycerides is named *sn*-glycerol-3-phosphate.

The Most Common Phospholipids

Phosphatidic acid, the parent compound for the glycerol-based phospholipids (Figure 8.4), consists of *sn*-glycerol-3-phosphate, with fatty acids esterified at the 1- and 2-positions. Phosphatidic acid is found in small amounts in most natural systems and is an important intermediate in the biosynthesis of the more common glycerophospholipids (Figure 8.6). In these compounds, a

A DEEPER LOOK

Prochirality

If a tetrahedral center in a molecule has two identical substituents, it is referred to as **prochiral** since, if either of the like substituents is converted to a different group, the tetrahedral center then becomes chiral. Consider glycerol: the central carbon of glycerol is prochiral since replacing either of the $\text{—CH}_2\text{OH}$ groups would make the central carbon chiral. Nomenclature for prochiral centers is based on the (*R,S*) system (in Chapter 3). To name the otherwise identical substituents of a prochiral center, imagine

increasing slightly the priority of one of them (by substituting a deuterium for a hydrogen, for example) as shown: the resulting molecule has an (*S*)-configuration about the (now chiral) central carbon atom. The group that contains the deuterium is thus referred to as the *pro-S* group. As a useful exercise, you should confirm that labeling the other CH_2OH group with a deuterium produces the (*R*)-configuration at the central carbon, so that this latter CH_2OH group is the *pro-R* substituent.



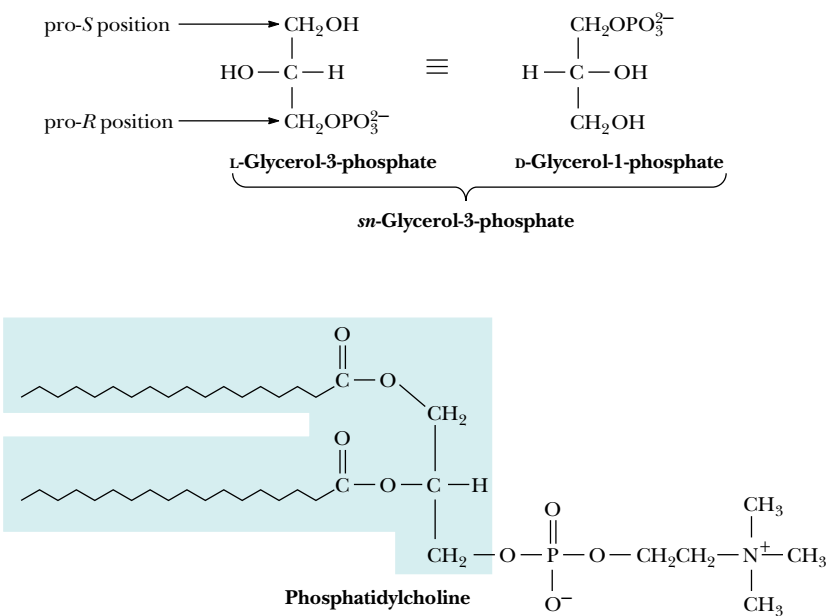
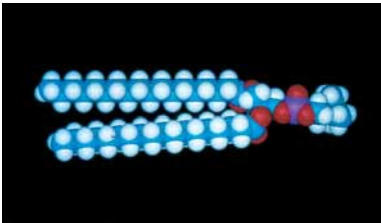


FIGURE 8.5 • The absolute configuration of *sn*-glycerol-3-phosphate. The pro-*(R)* and pro-*(S)* positions of the parent glycerol are also indicated.



GLYCEROLIPIDS WITH OTHER HEAD GROUPS:

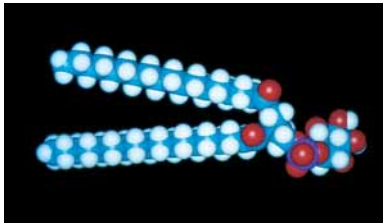
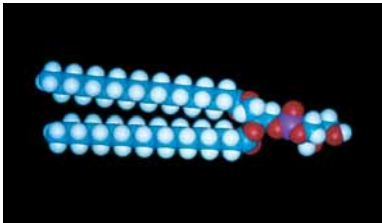
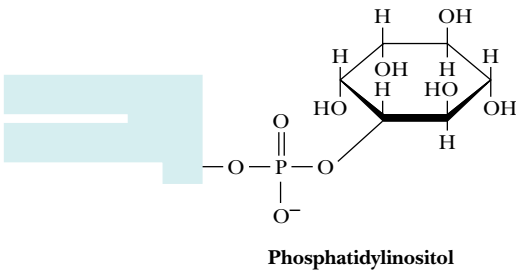
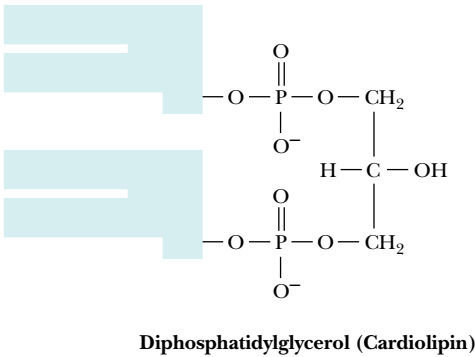
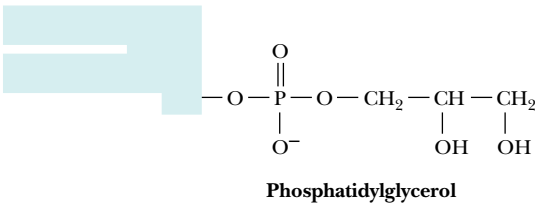
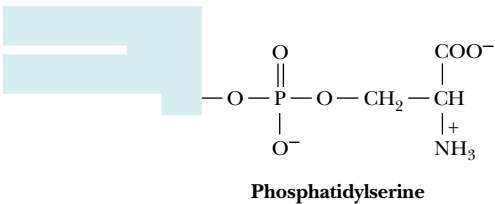
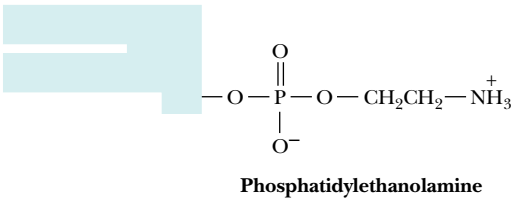


FIGURE 8.6 • Structures of several glycerophospholipids and space-filling models of phosphatidylcholine, phosphatidylglycerol, and phosphatidylinositol.

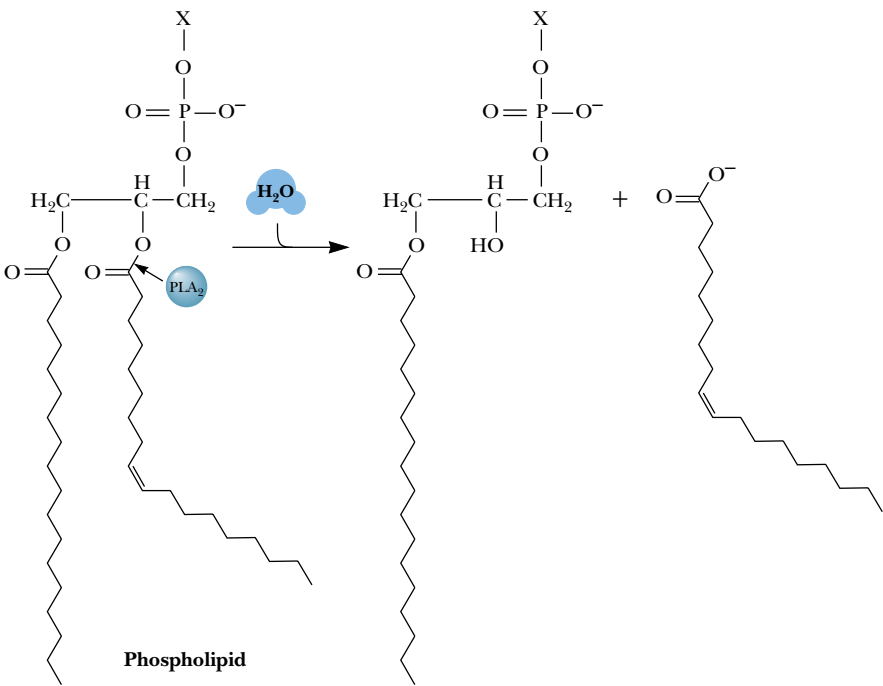
variety of polar groups are esterified to the phosphoric acid moiety of the molecule. The phosphate, together with such esterified entities, is referred to as a “head” group. Phosphatides with choline or ethanolamine are referred to as **phosphatidylcholine** (known commonly as **lecithin**) or **phosphatidylethanolamine**, respectively. These phosphatides are two of the most common constituents of biological membranes. Other common *head groups* found in phosphatides include glycerol, serine, and inositol (Figure 8.6). Another kind of glycerol phosphatide found in many tissues is **diphosphatidylglycerol**. First observed in heart tissue, it is also called **cardiolipin**. In cardiolipin, a phosphatidylglycerol is esterified through the C-1 hydroxyl group of the glycerol moiety of the head group to the phosphoryl group of another phosphatidic acid molecule.

A D E E P E R L O O K

Glycerophospholipid Degradation: One of the Effects of Snake Venoms

The venoms of poisonous snakes contain (among other things) a class of enzymes known as **phospholipases**, enzymes that cause the breakdown of phospholipids. For example, the venoms of the eastern diamondback rattlesnake (*Crotalus adamanteus*) and the Indian cobra (*Naja naja*) both contain phospholipase A₂, which catalyzes the hydrolysis of fatty acids at the C-2 position of glycerophospholipids.

The phospholipid breakdown product of this reaction, *lysolecithin*, acts as a detergent and dissolves the membranes of red blood cells, causing them to rupture. Indian cobras kill several thousand people each year.



Eastern diamondback rattlesnake. (Dr. E. R. Degginger)



Indian cobra. (Dr. E. R. Degginger)



FIGURE 8.7 • A space-filling model of 1-stearoyl-2-oleoyl-phosphatidylcholine.

Phosphatides exist in many different varieties, depending on the fatty acids esterified to the glycerol group. As we shall see, the nature of the fatty acids can greatly affect the chemical and physical properties of the phosphatides and the membranes that contain them. In most cases, glycerol phosphatides have a saturated fatty acid at position 1 and an unsaturated fatty acid at position 2 of the glycerol. Thus, **1-stearoyl-2-oleoyl-phosphatidylcholine** (Figure 8.7) is a common constituent in natural membranes, but **1-linoleoyl-2-palmitoylphosphatidylcholine** is not.

Both structural and functional strategies govern the natural design of the many different kinds of glycerophospholipid head groups and fatty acids. The structural roles of these different glycerophospholipid classes are described in Chapter 9. Certain phospholipids, including phosphatidylinositol and phosphatidylcholine, participate in complex cellular signaling events. These roles, appreciated only in recent years, are described in Chapter 34.

Ether Glycerophospholipids

Ether glycerophospholipids possess an ether linkage instead of an acyl group at the C-1 position of glycerol (Figure 8.8). One of the most versatile biochemical signal molecules found in mammals is **platelet activating factor**, or **PAF**, a unique ether glycerophospholipid (Figure 8.9). The alkyl group at C-1 of PAF is typically a 16-carbon chain, but the acyl group at C-2 is a 2-carbon acetate unit. By virtue of this acetate group, PAF is much more water-soluble

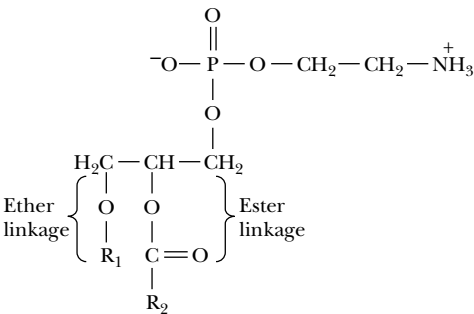


FIGURE 8.8 • A 1-alkyl 2-acyl-phosphatidylethanolamine (an ether glycerophospholipid).

DEEPER LOOK

Platelet Activating Factor: A Potent Glyceroether Mediator

Platelet activating factor (PAF) was first identified by its ability (at low levels) to cause platelet aggregation and dilation of blood vessels, but it is now known to be a potent mediator in inflammation, allergic responses, and shock. PAF effects are observed at tissue concentrations as low as 10^{-12} M. PAF causes a dramatic inflammation of air passages and induces asthma-like symptoms in laboratory animals. **Toxic-shock syndrome** occurs when fragments of destroyed bacteria act as toxins and induce the synthesis of PAF. This results in a drop in blood pressure and a reduced

volume of blood pumped by the heart, which leads to shock and, in severe cases, death.

Beneficial effects have also been attributed to PAF. In reproduction, PAF secreted by the fertilized egg is instrumental in the implantation of the egg in the uterine wall. PAF is produced in significant quantities in the lungs of the fetus late in pregnancy and may stimulate the production of fetal lung surfactant, a protein–lipid complex that prevents collapse of the lungs in a newborn infant.

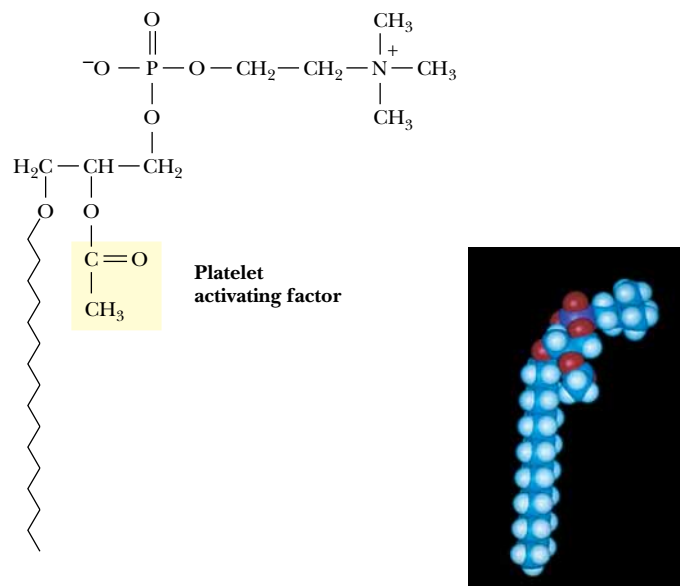


FIGURE 8.9 • The structure of 1-alkyl 2-acetyl-phosphatidylcholine, also known as platelet activating factor or PAF.

than other lipids, allowing PAF to function as a soluble messenger in signal transduction.

Plasmalogens are ether glycerophospholipids in which the alkyl moiety is *cis*- α,β -unsaturated (Figure 8.10). Common plasmalogen head groups include choline, ethanolamine, and serine. These lipids are referred to as phosphatidyl choline, phosphatidyl ethanolamine, and phosphatidyl serine.

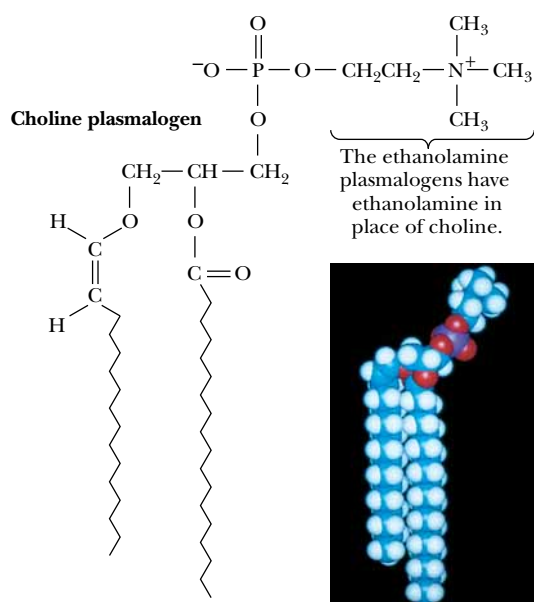


FIGURE 8.10 • The structure and a space-filling model of a choline plasmalogen.

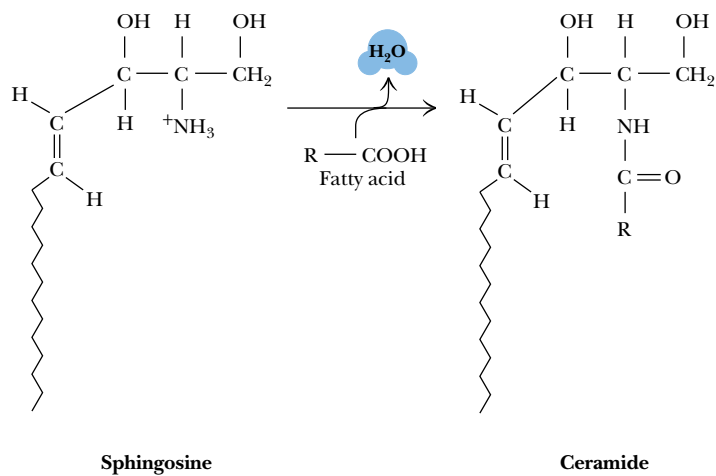


FIGURE 8.11 • Formation of an amide linkage between a fatty acid and sphingosine produces a ceramide.

8.4 • Sphingolipids

Sphingolipids represent another class of lipids found frequently in biological membranes. An 18-carbon amino alcohol, **sphingosine** (Figure 8.11), forms the backbone of these lipids rather than glycerol. Typically, a fatty acid is joined to a sphingosine via an amide linkage to form a **ceramide**. **Sphingomyelins** represent a phosphorus-containing subclass of sphingolipids and are especially important in the nervous tissue of higher animals. A **sphingomyelin** is formed by the esterification of a phosphorylcholine or a phosphorylethanolamine to the 1-hydroxy group of a ceramide (Figure 8.12).

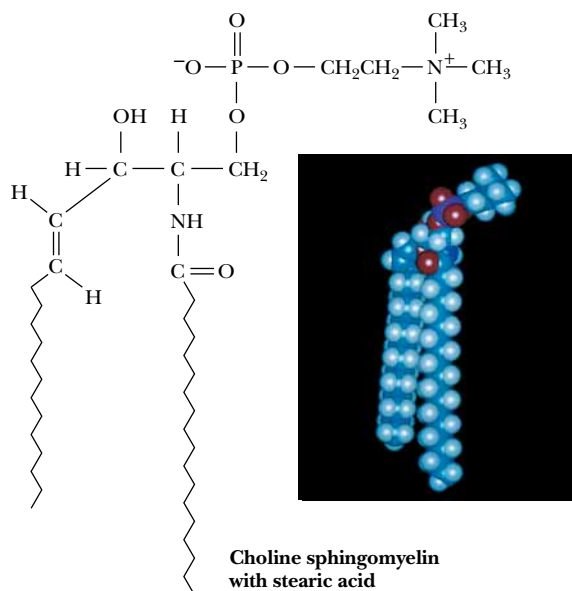


FIGURE 8.12 • A structure and a space-filling model of a choline sphingomyelin formed from stearic acid.

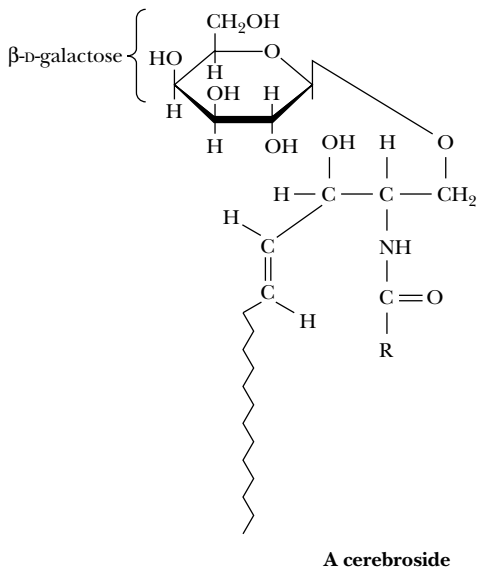


FIGURE 8.13 • The structure of a cerebroside. Note the sphingosine backbone.

There is another class of ceramide-based lipids which, like the sphingomyelins, are important components of muscle and nerve membranes in animals. These are the **glycosphingolipids**, and they consist of a ceramide with one or more sugar residues in a β -glycosidic linkage at the 1-hydroxyl moiety. The neutral glycosphingolipids contain only neutral (uncharged) sugar residues. When a single glucose or galactose is bound in this manner, the molecule is a **cerebroside** (Figure 8.13). Another class of lipids is formed when a sulfate is esterified at the 3-position of the galactose to make a **sulfatide**. **Gangliosides** (Figure 8.14) are more complex glycosphingolipids that consist of a ceramide backbone with three or more sugars esterified, one of these being a **sialic acid** such as **N-acetylneuraminic acid**. These latter compounds are referred to as *acidic glycosphingolipids*, and they have a net negative charge at neutral pH.

The glycosphingolipids have a number of important cellular functions, despite the fact that they are present only in small amounts in most membranes. Glycosphingolipids at cell surfaces appear to determine, at least in part, certain elements of tissue and organ specificity. Cell–cell recognition and tissue immunity appear to depend upon specific glycosphingolipids. Gangliosides are present in nerve endings and appear to be important in nerve impulse transmission. A number of genetically transmitted diseases involve the accumulation of specific glycosphingolipids due to an absence of the enzymes needed for their degradation. Such is the case for ganglioside G_{M2} in the brains of *Tay-Sachs disease* victims, a rare but fatal disease characterized by a red spot on the retina, gradual blindness, and loss of weight, especially in infants and children.

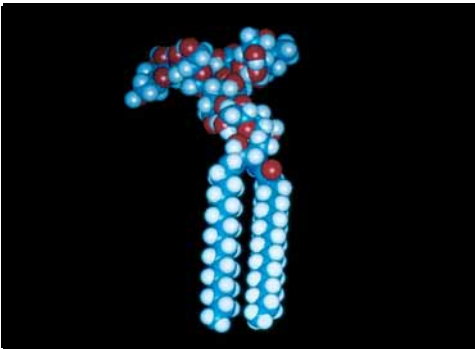
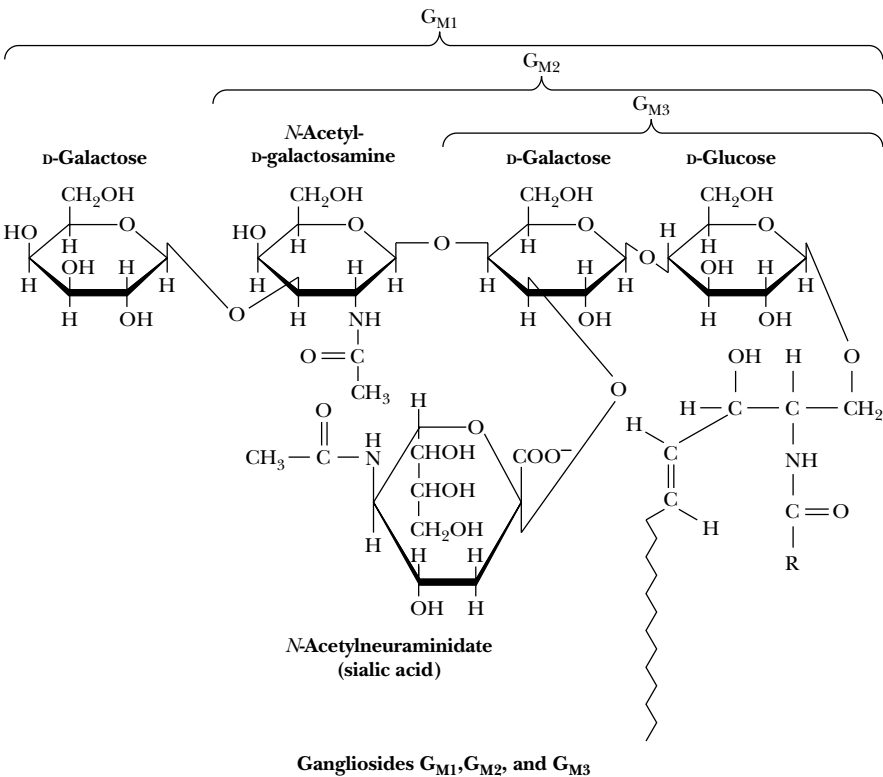


FIGURE 8.14 • The structures of several important gangliosides. Also shown is a space-filling model of ganglioside G_{M1} .

A D E E P E R L O O K

Moby Dick and Spermaceti: A Valuable Wax from Whale Oil

When oil from the head of the sperm whale is cooled, **spermaceti**, a translucent wax with a white, pearly luster, crystallizes from the mixture. Spermaceti, which makes up 11% of whale oil, is composed mainly of the wax **cetyl palmitate**:



as well as smaller amounts of cetyl alcohol:



Spermaceti and cetyl palmitate have been widely used in the making of cosmetics, fragrant soaps, and candles.

In the literary classic *Moby Dick*, Herman Melville describes Ishmael's impressions of spermaceti, when he muses that the waxes "discharged all their opulence, like fully ripe grapes their wine; as I snuffed that uncontaminated aroma—literally and truly like the smell of spring violets."^{*}

^{*}Melville, H., *Moby Dick*, Octopus Books, London, 1984, p. 205 (Adapted from *Chemistry in Moby Dick*, Waddell, T. G., and Sanderlin, R. R. (1986), *Journal of Chemical Education* **63**:1019–1020.)

8.5 • Waxes

Waxes are esters of long-chain alcohols with long-chain fatty acids. The resulting molecule can be viewed (in analogy to the glycerolipids) as having a weakly polar head group (the ester moiety itself) and a long, nonpolar tail (the hydrocarbon chains) (Figure 8.15). Fatty acids found in waxes are usually saturated. The alcohols found in waxes may be saturated or unsaturated and may include sterols, such as cholesterol (see later section). Waxes are water-insoluble due to the weakly polar nature of the ester group. As a result, this class of molecules confers water-repellant character to animal skin, to the leaves of certain plants, and to bird feathers. The glossy surface of a polished apple results from a wax coating. **Carnauba wax**, obtained from the fronds of a species of palm tree in Brazil, is a particularly hard wax used for high gloss finishes, such as in automobile wax, boat wax, floor wax, and shoe polish. **Lanolin**, a component of wool wax, is used as a base for pharmaceutical and cosmetic products because it is rapidly assimilated by human skin.

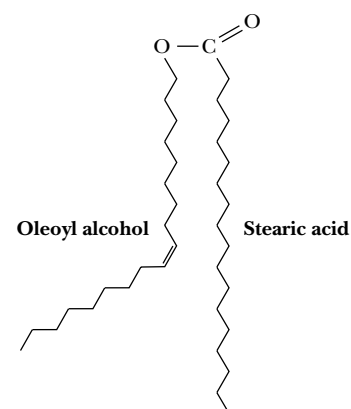


FIGURE 8.15 • An example of a wax. Oleoyl alcohol is esterified to stearic acid in this case.

8.6 • Terpenes

The **terpenes** are a class of lipids formed from combinations of two or more molecules of 2-methyl-1,3-butadiene, better known as **isoprene** (a five-carbon unit that is abbreviated C₅). A **monoterpene** (C₁₀) consists of two isoprene units, a **sesquiterpene** (C₁₅) consists of three isoprene units, a **diterpene** (C₂₀) has four isoprene units, and so on. Isoprene units can be linked in terpenes to form straight chain or cyclic molecules, and the usual method of linking isoprene units is head to tail (Figure 8.16). Monoterpenes occur in all higher plants, while sesquiterpenes and diterpenes are less widely known. Several examples of these classes of terpenes are shown in Figure 8.17. The **triterpenes** are C₃₀ terpenes and include **squalene** and **lanosterol**, two of the precursors of cholesterol and other steroids (discussed later). **Tetraterpenes** (C₄₀) are less common but include the carotenoids, a class of colorful photosynthetic pigments. β -Carotene is the precursor of vitamin A, while lycopene, similar to β -carotene but lacking the cyclopentene rings, is a pigment found in tomatoes.

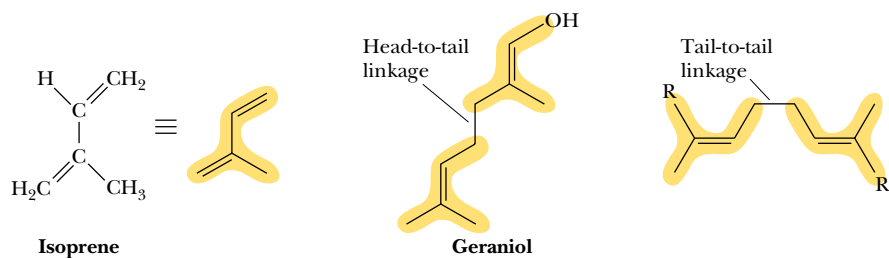
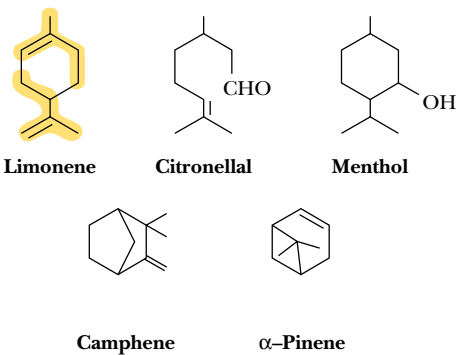


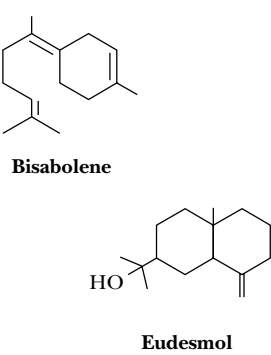
FIGURE 8.16 • The structure of isoprene (2-methyl-1,3-butadiene) and the structure of head-to-tail and tail-to-tail linkages. Isoprene itself can be formed by distillation of natural rubber, a linear head-to-tail polymer of isoprene units.

Long-chain polyisoprenoid molecules with a terminal alcohol moiety are called **polyprenols**. The **dolichols**, one class of prenyls (Figure 8.18), consist of 16 to 22 isoprene units and, in the form of dolichyl phosphates, function to carry carbohydrate units in the biosynthesis of glycoproteins in animals. Prenyl groups serve to *anchor* certain proteins to biological membranes (discussed in Chapter 9).

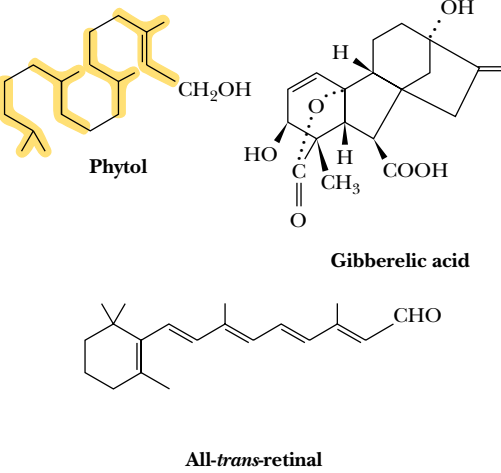
MONOTERPENES



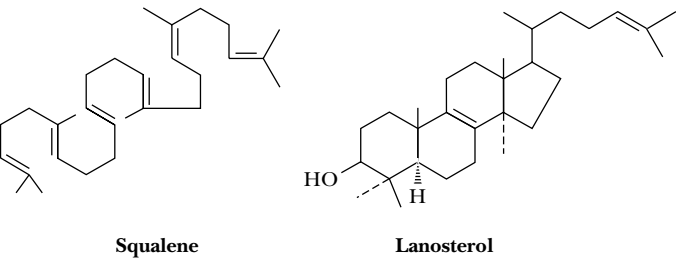
SESQUITERPENES



DITERPENES



TRITERPENES



TETRATERPENES

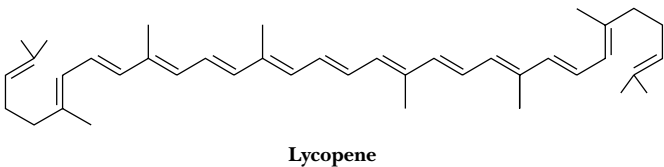


FIGURE 8.17 • Many monoterpenes are readily recognized by their characteristic flavors or odors (limonene in lemons; citronellal in roses, geraniums, and some perfumes; pinene in turpentine; and menthol from peppermint, used in cough drops and nasal inhalers). The diterpenes, which are C₂₀ terpenes, include retinal (the essential light-absorbing pigment in rhodopsin, the photoreceptor protein of the eye), phytol (a constituent of chlorophyll), and the gibberellins (potent plant hormones). The triterpene lanosterol is a constituent of wool fat. Lycopene is a carotenoid found in ripe fruit, especially tomatoes.

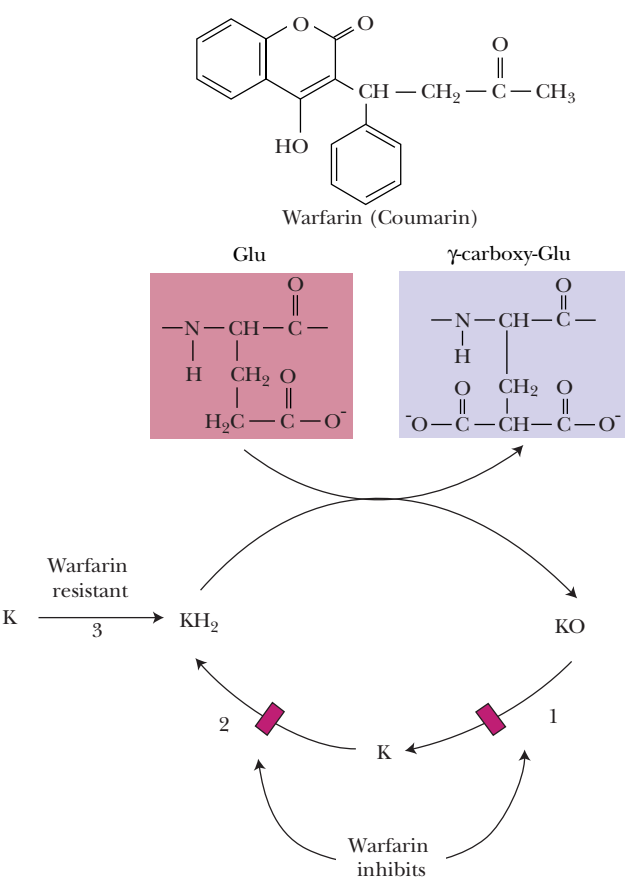
HUMAN BIOCHEMISTRY

Coumarin or Warfarin—Agent of Life or Death

The isoprene-derived molecule whose structure is shown here is known alternately as **Coumarin** and **warfarin**. By the former name, it is a widely prescribed anticoagulant. By the latter name, it is a component of rodent poisons. How can the same chemical species be used for such disparate purposes? The key to both uses lies in its ability to act as an antagonist of vitamin K in the body.

Vitamin K stimulates the carboxylation of glutamate residues on certain proteins, including some proteins in the blood-clotting cascade (including **prothrombin**, **Factor VII**, **Factor IX**, and **Factor X**, which undergo a Ca^{2+} -dependent conformational change in the course of their biological activity, as well as **protein C** and **protein S**, two regulatory proteins). Carboxylation of these coagulation factors is catalyzed by a carboxylase that requires the reduced form of vitamin K (vitamin KH_2), molecular oxygen, and carbon dioxide. KH_2 is oxidized to vitamin K epoxide, which is recycled to KH_2 by the enzymes **vitamin K epoxide reductase** (1) and **vitamin K reductase** (2, 3). Coumarin/warfarin exerts its anticoagulant effect by inhibiting vitamin K epoxide reductase and possibly also vitamin K reductase. This inhibition depletes vitamin KH_2 and reduces the activity of the carboxylase.

Coumarin/warfarin, given at a typical dosage of 4 to 5 mg/day, prevents the deleterious formation in the bloodstream of small blood clots and thus reduces the risk of heart attacks and strokes for individuals whose arteries contain sclerotic plaques. Taken in much larger doses, as for example in rodent poisons, Coumarin/warfarin can cause massive hemorrhages and death.



8.7 • Steroids

Cholesterol

A large and important class of terpene-based lipids is the **steroids**. This molecular family, whose members effect an amazing array of cellular functions, is based on a common structural motif of three six-membered rings and one five-membered ring all fused together. **Cholesterol** (Figure 8.19) is the most common steroid in animals and the precursor for all other animal steroids. The numbering system for cholesterol applies to all such molecules. Many steroids contain methyl groups at positions 10 and 13 and an 8- to 10-carbon alkyl side chain at position 17. The polyprenyl nature of this compound is particularly evident in the side chain. Many steroids contain an oxygen at C-3, either a hydroxyl group in sterols or a carbonyl group in other steroids. Note also that the carbons at positions 10 and 13 and the alkyl group at position 17 are nearly always oriented on the same side of the steroid nucleus, the β -orientation. Alkyl groups that extend from the other side of the steroid backbone are in an α -orientation.

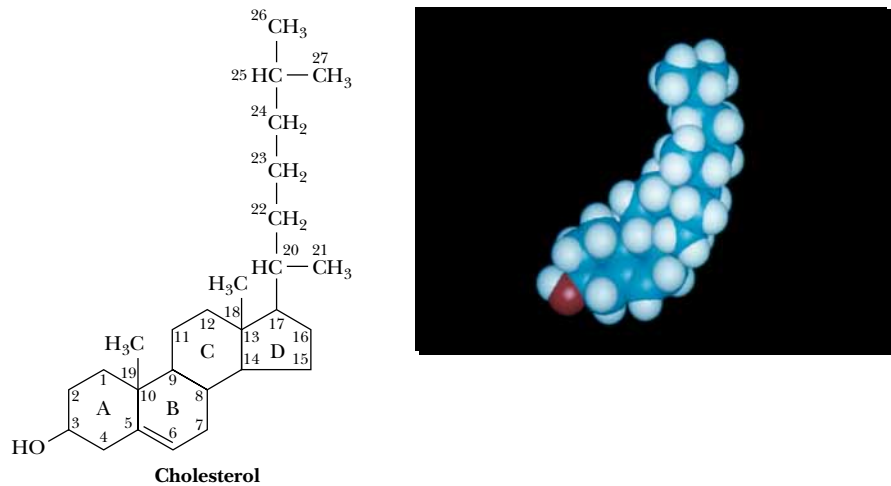


FIGURE 8.19 • The structure of cholesterol, shown with steroid ring designations and carbon numbering.

Cholesterol is a principal component of animal cell plasma membranes, and much smaller amounts of cholesterol are found in the membranes of intra-cellular organelles. The relatively rigid fused ring system of cholesterol and the weakly polar alcohol group at the C-3 position have important consequences for the properties of plasma membranes. Cholesterol is also a component of *lipoprotein complexes* in the blood, and it is one of the constituents of *plaques* that form on arterial walls in *atherosclerosis*.

Steroid Hormones

Steroids derived from cholesterol in animals include five families of hormones (the androgens, estrogens, progestins, glucocorticoids and mineralocorticoids) and bile acids (Figure 8.20). **Androgens** such as **testosterone** and **estrogens** such as **estradiol** mediate the development of sexual characteristics and sexual function in animals. The **progestins** such as **progesterone** participate in control of

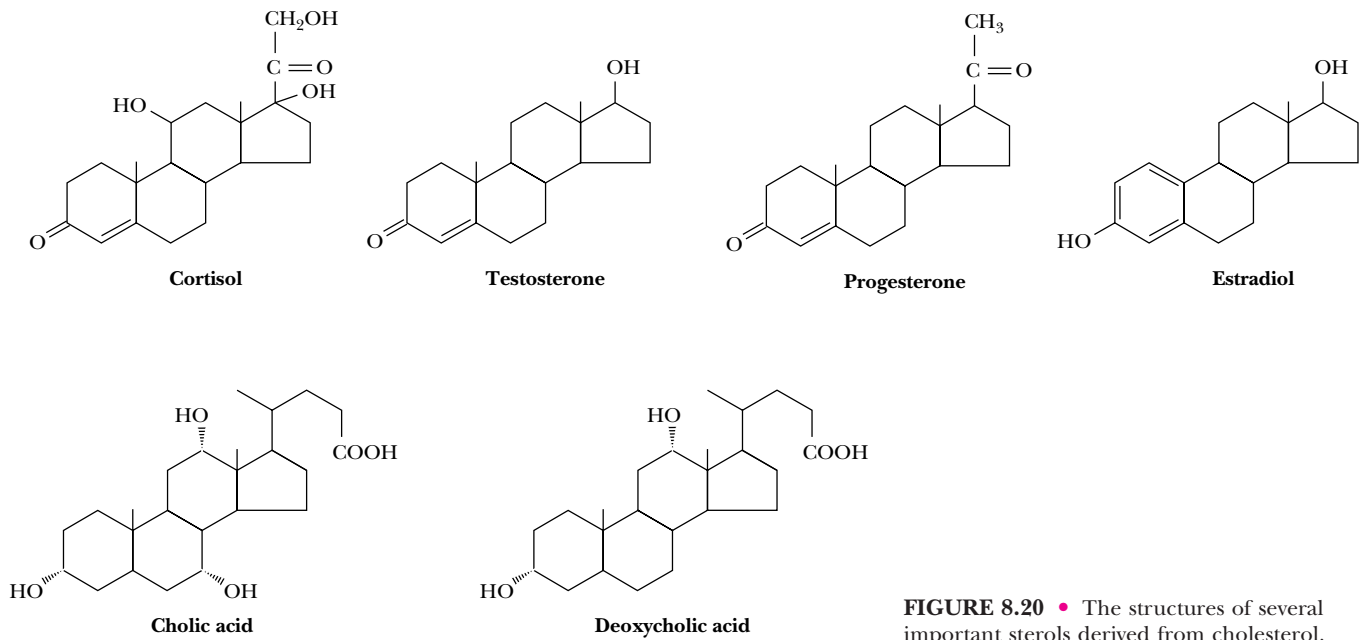


FIGURE 8.20 • The structures of several important sterols derived from cholesterol.

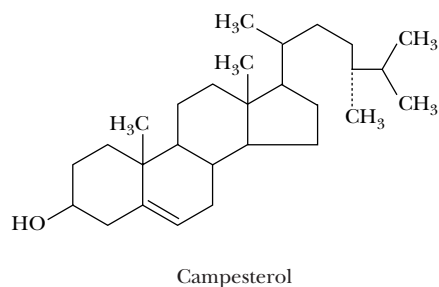
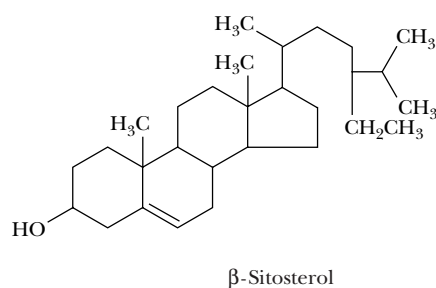
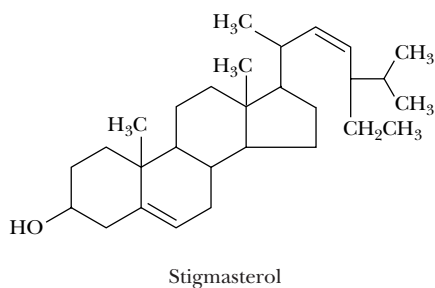
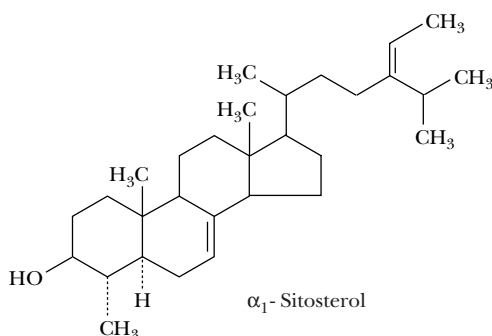
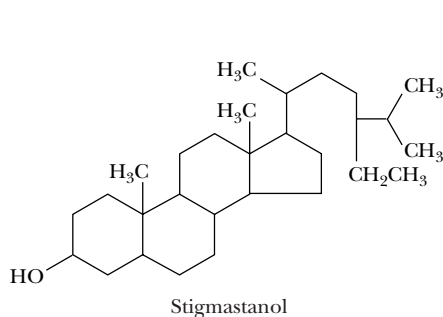
the menstrual cycle and pregnancy. **Glucocorticoids** (**cortisol**, for example) participate in the control of carbohydrate, protein, and lipid metabolism, whereas the **mineralocorticoids** regulate salt (Na^+ , K^+ , and Cl^-) balances in tissues. The **bile acids** (including **cholic** and **deoxycholic acid**) are detergent molecules secreted in bile from the gallbladder that assist in the absorption of dietary lipids in the intestine.

HUMAN BIOCHEMISTRY

Plant Sterols—Natural Cholesterol Fighters

Dietary guidelines for optimal health call for reducing the intake of cholesterol. One strategy for doing so involves the plant sterols, including sitosterol, stigmasterol, stigmasterol, and campesterol, shown in the figure. Despite their structural similarity to cholesterol, minor isomeric differences and/or the presence of methyl and ethyl groups in the side chains of these substances result in their poor absorption by intestinal mucosal cells. Interestingly, although plant sterols are not effectively absorbed by the body, they nonetheless are highly effective in blocking the absorption of cholesterol itself by intestinal cells.

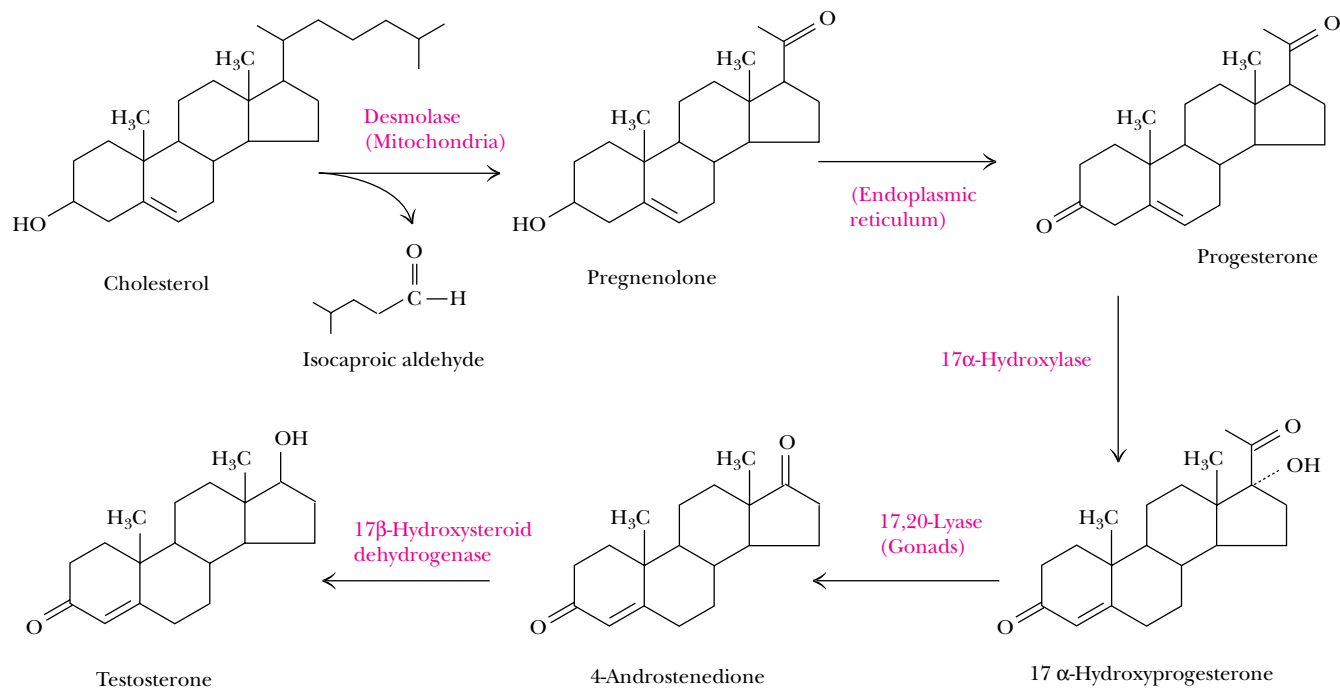
The practical development of plant sterol drugs as cholesterol-lowering agents will depend both on structural features of the sterols themselves and on the form of the administered agent. For example, the unsaturated sterol sitosterol is poorly absorbed in the human intestine, whereas sitostanol, the saturated analog, is almost totally unabsorbable. In addition, there is evidence that plant sterols administered in a soluble, micellar form (see page 261 for a description of micelles) are more effective in blocking cholesterol absorption than plant sterols administered in a solid, crystalline form.



17 β -Hydroxysteroid Dehydrogenase 3 Deficiency

Testosterone, the principal male sex steroid hormone, is synthesized in five steps from cholesterol, as shown below. In the last step, five isozymes catalyze the 17 β -hydroxysteroid dehydrogenase reactions that interconvert 4-androstenedione and testosterone. Defects in the synthesis or action of testosterone can impair the development of the male phenotype during embryogenesis and cause the disorders of human sexuality termed male pseudohermaphroditism. Specifically, mutations in isozyme 3 of the 17 β -hydroxysteroid dehydrogenase in the fetal testis impair the for-

mation of testosterone and give rise to genetic males with female external genitalia and blind-ending vaginas. Such individuals are typically raised as females but virilize at puberty, due to an increase in serum testosterone, and develop male hair growth patterns. Fourteen different mutations of 17 β -hydroxysteroid dehydrogenase 3 have been identified in 17 affected families in the United States, the Middle East, Brazil, and Western Europe. These families account for about 45% of the patients with this disorder reported in scientific literature.



PROBLEMS

- Draw the structures of (a) all the possible triacylglycerols that can be formed from glycerol with stearic and arachidonic acid, and (b) all the phosphatidylserine isomers that can be formed from palmitic and linolenic acids.
- Describe in your own words the structural features of
 - a ceramide, and how it differs from a cerebroside.
 - a phosphatidylethanolamine, and how it differs from a phosphatidylcholine.
 - an ether glycerophospholipid, and how it differs from a plasmalogen.
 - a ganglioside, and how it differs from a cerebroside.
 - testosterone, and how it differs from estradiol.
- From your memory of the structures, name
 - the glycerophospholipids that carry a net positive charge.
 - the glycerophospholipids that carry a net negative charge.
 - the glycerophospholipids that have zero net charge.

- Compare and contrast two individuals, one of whose diet consists largely of meats containing high levels of cholesterol, and the other of whose diet is rich in plant sterols. Are their risks of cardiovascular disease likely to be similar or different? Explain your reasoning.
- James G. Watt, Secretary of the Interior (1981–1983) in Ronald Reagan's first term, provoked substantial controversy by stating publicly that trees cause significant amounts of air pollution. Based on your reading of this chapter, evaluate Watt's remarks.
- In a departure from his usual and highly popular westerns, author Louis L'Amour wrote a novel in 1987, *Best of the Breed* (Bantam Press), in which a military pilot of native American ancestry is shot down over the former Soviet Union and is forced to use the survival skills of his ancestral culture to escape his enemies. On the rare occasions when he is able to trap and kill an animal for food, he selectively eats the fat, not the meat. Based on your reading of this chapter, what is his reasoning for doing so?

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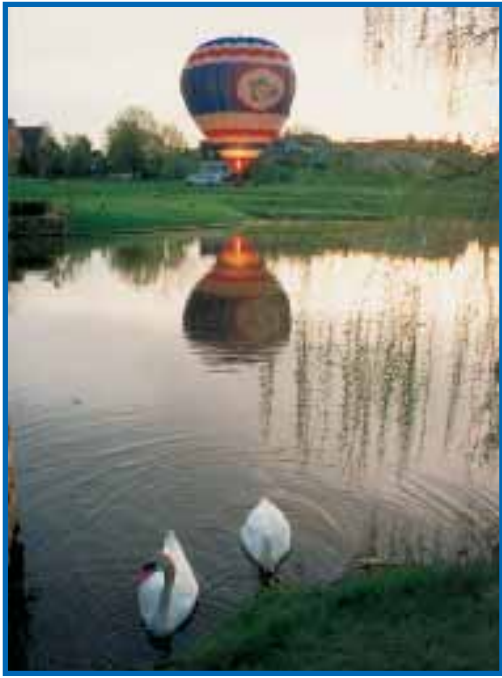
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Chapter 9

Membranes and Cell Surfaces



Membranes are thin envelopes that define the volume of cells, like this hot-air balloon is defined in space by its colorful envelope. (Boar's Head Inn Balloon, Charlottesville, VA, by Larry Swank)

Membranes serve a number of essential cellular functions. They constitute the boundaries of cells and intracellular organelles, and they provide a surface where many important biological reactions and processes occur. Membranes have proteins that mediate and regulate the transport of metabolites, macromolecules, and ions. Hormones and many other biological signal molecules and regulatory agents exert their effects via interactions with membranes. Photosynthesis, electron transport, oxidative phosphorylation, muscle contraction, and electrical activity all depend on membranes and membrane proteins. Thirty percent of the genes of at least one organism, *Mycoplasma genitalium* (whose entire genome has been sequenced), are thought to encode membrane proteins.

Surgeons must be very careful
When they take the knife!
Underneath their fine incisions
Stirs the Culprit—Life!

EMILY DICKINSON
Poem 108, 1859

OUTLINE

- 9.1 • Membranes
- 9.2 • Structure of Membrane Proteins
- 9.3 • Membrane and Cell-Surface Polysaccharides
- 9.4 • Glycoproteins
- 9.5 • Proteoglycans

Biological membranes are uniquely organized arrays of lipids and proteins (either of which may be decorated with carbohydrate groups). The lipids found in biological systems are often **amphipathic**, signifying that they possess both polar and nonpolar groups. The hydrophobic nature of lipid molecules allows membranes to act as effective barriers to polar molecules. The polar moieties of amphipathic lipids typically lie at the surface of membranes, where they interact with water. Proteins interact with the lipids of membranes in a variety of ways. Some proteins associate with membranes via electrostatic interactions with polar groups on the membrane surface, whereas other proteins are embedded to various extents in the hydrophobic core of the membrane. Other proteins are *anchored* to membranes via covalently bound lipid molecules that associate strongly with the hydrophobic membrane core.

This chapter discusses the composition, structure, and dynamic processes of biological membranes.

9.1 • Membranes

Cells make use of many different types of membranes. All cells have a cytoplasmic membrane, or *plasma membrane*, that functions (in part) to separate the cytoplasm from the surroundings. In the early days of biochemistry, the plasma membrane was not accorded many functions other than this one of partition. We now know that the plasma membrane is also responsible for (1) the exclusion of certain toxic ions and molecules from the cell, (2) the accumulation of cell nutrients, and (3) energy transduction. It functions in (4) cell locomotion, (5) reproduction, (6) signal transduction processes, and (7) interactions with molecules or other cells in the vicinity.

Even the plasma membranes of prokaryotic cells (bacteria) are complex (Figure 9.1). With no intracellular organelles to divide and organize the work, bacteria carry out processes either at the plasma membrane or in the cyto-

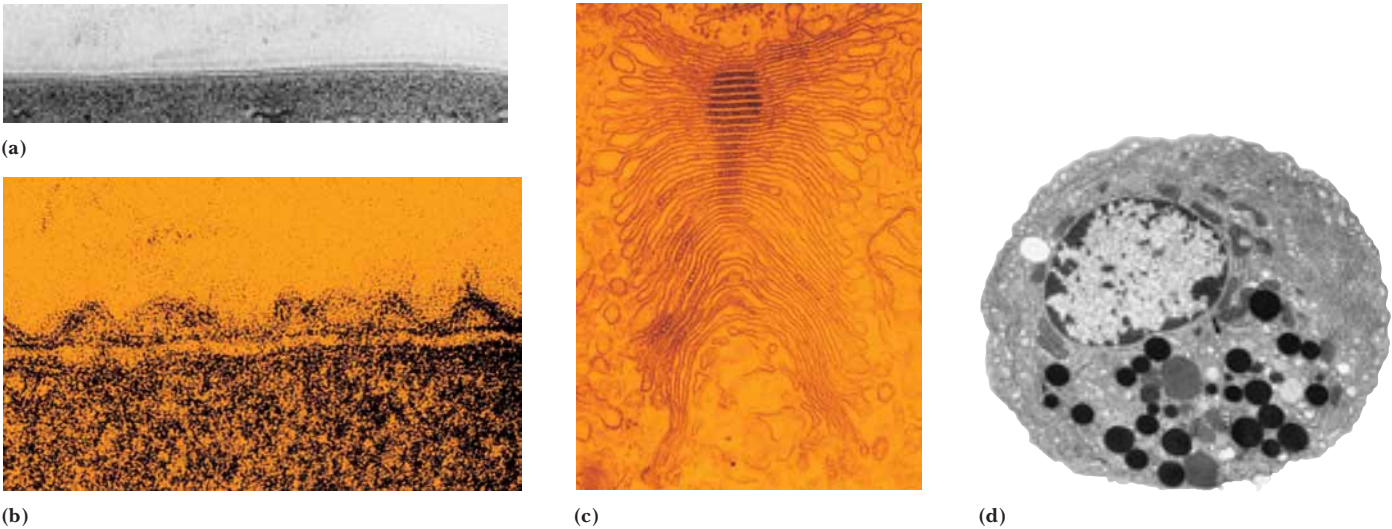


FIGURE 9.1 • Electron micrographs of several different membrane structures: (a) *Menoidium*, a protozoan; (b) Gram-negative envelope of *Aquaspirillum serpens*; (c) Golgi apparatus; (d) pancreatic acinar cell. (a, T. T. Beveridge/Visuals Unlimited; b, © Cabisco/Visuals Unlimited; c, d, © D. W. Fawcett/Photo Researchers, Inc.)

plasm itself. Eukaryotic cells, however, contain numerous intracellular organelles that perform specialized tasks. Nucleic acid biosynthesis is handled in the nucleus; mitochondria are the site of electron transport, oxidative phosphorylation, fatty acid oxidation, and the tricarboxylic acid cycle; and secretion of proteins and other substances is handled by the endoplasmic reticulum and the Golgi apparatus. This partitioning of labor is not the only contribution of the membranes in these cells. Many of the processes occurring in these organelles (or in the prokaryotic cell) actively involve membranes. Thus, some of the enzymes involved in nucleic acid metabolism are membrane-associated. The electron transfer chain and its associated system for ATP synthesis are embedded in the mitochondrial membrane. Many enzymes responsible for aspects of lipid biosynthesis are located in the endoplasmic reticulum membrane.

Spontaneously Formed Lipid Structures

Monolayers and Micelles

Amphipathic lipids spontaneously form a variety of structures when added to aqueous solution. All these structures form in ways that minimize contact between the hydrophobic lipid chains and the aqueous milieu. For example, when small amounts of a fatty acid are added to an aqueous solution, a monolayer is formed at the air–water interface, with the polar head groups in contact with the water surface and the hydrophobic tails in contact with the air (Figure 9.2). Few lipid molecules are found as monomers in solution.

Further addition of fatty acid eventually results in the formation of micelles. **Micelles** formed from an amphipathic lipid in water position the hydrophobic tails in the center of the lipid aggregation with the polar head groups facing outward. Amphipathic molecules that form micelles are characterized by a unique **critical micelle concentration**, or **CMC**. Below the CMC, individual lipid molecules predominate. Nearly all the lipid added above the CMC, however, spontaneously forms micelles. Micelles are the preferred form of aggregation in water for detergents and soaps. Some typical CMC values are listed in Figure 9.3.

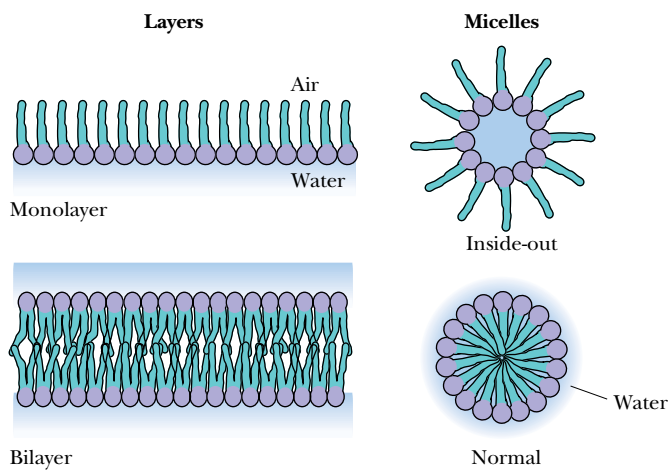


FIGURE 9.2 • Several spontaneously formed lipid structures.

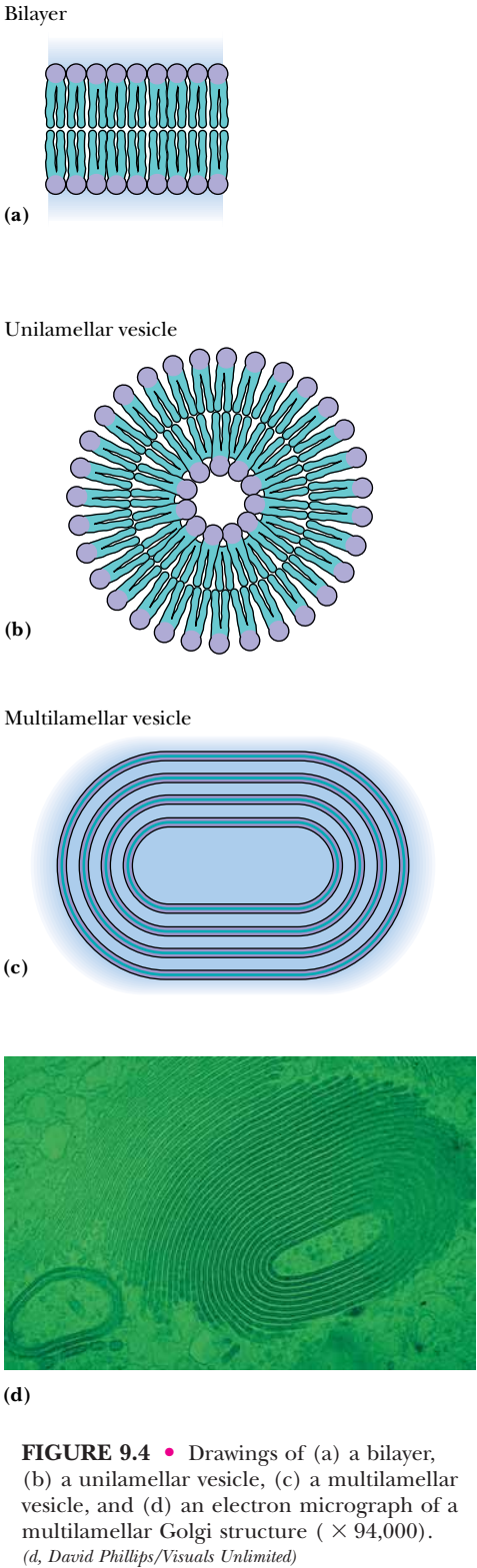


FIGURE 9.4 • Drawings of (a) a bilayer, (b) a unilamellar vesicle, (c) a multilamellar vesicle, and (d) an electron micrograph of a multilamellar Golgi structure (× 94,000). (d, David Phillips/Visuals Unlimited)

Structure	M _r	CMC	Micelle M _r
Triton X-100 	625	0.24 mM	90–95,000
Octyl glucoside 	292	25 mM	
C₁₂E₈ (Dodecyl octaoxyethylene ether) $C_{12}H_{25}-(OCH_2CH_2)_8-OH$	538	0.071 mM	

FIGURE 9.3 • The structures of some common detergents and their physical properties. Micelles formed by detergents can be quite large. Triton X-100, for example, typically forms micelles with a total molecular mass of 90 to 95 kD. This corresponds to approximately 150 molecules of Triton X-100 per micelle.

Lipid Bilayers

Lipid bilayers consist of back-to-back arrangements of monolayers (Figure 9.2). Phospholipids prefer to form bilayer structures in aqueous solution because their pairs of fatty acyl chains do not pack well in the interior of a micelle. Phospholipid bilayers form rapidly and spontaneously when phospholipids are added to water, and they are stable structures in aqueous solution. As opposed to micelles, which are small, self-limiting structures of a few hundred molecules, bilayers may form spontaneously over large areas (10⁸ nm² or more). Because exposure of the edges of the bilayer to solvent is highly unfavorable, extensive bilayers normally wrap around themselves and form closed vesicles (Figure 9.4). The nature and integrity of these vesicle structures are very much dependent on the lipid composition. Phospholipids can form either *unilamellar vesicles* (with a single lipid bilayer) known as *liposomes*, or *multilamellar vesicles*. These latter structures are reminiscent of the layered structure of onions. Multilamellar vesicles were discovered by Sir Alex Bangham and are sometimes referred to as “Bangosomes” in his honor.

Liposomes are highly stable structures that can be subjected to manipulations such as gel filtration chromatography and dialysis. With such methods, it is possible to prepare liposomes having different inside and outside solution compositions. Liposomes can be used as drug and enzyme delivery systems in therapeutic applications. For example, liposomes can be used to introduce contrast agents into the body for diagnostic imaging procedures, including *computerized tomography (CT)* and *magnetic resonance imaging (MRI)* (Figure 9.5). Liposomes can fuse with cells, mixing their contents with the intracellular medium. If methods can be developed to target liposomes to selected cell populations, it may be possible to deliver drugs, therapeutic enzymes, and contrast agents to particular kinds of cells (such as cancer cells).

That vesicles and liposomes form at all is a consequence of the amphipathic nature of the phospholipid molecule. Ionic interactions between the

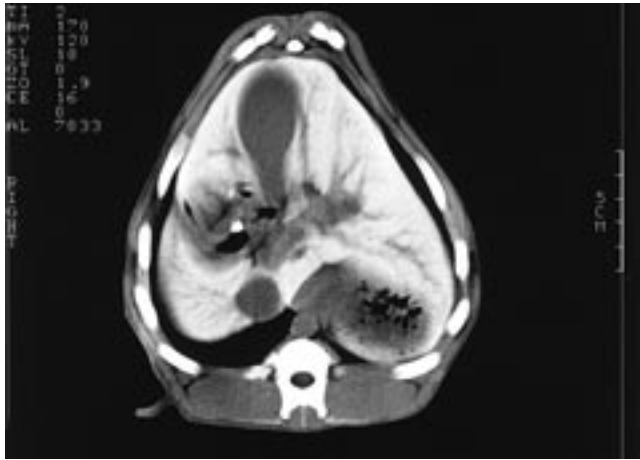


FIGURE 9.5 • A computerized tomography (CT) image of the upper abdomen of a dog, following administration of liposome-encapsulated iodine, a contrast agent that improves the light/dark contrast of objects in the image. The spine is the bright white object at the bottom and the other bright objects on the periphery are ribs. The liver (white) occupies most of the abdominal space. The gallbladder (bulbous object at the center top) and blood vessels appear dark in the image. The liposomal iodine contrast agent has been taken up by Kupffer cells, which are distributed throughout the liver, except in tumors. The dark object in the lower right is a large tumor. None of these anatomical features would be visible in a CT image in the absence of the liposomal iodine contrast agent. (Courtesy of Walter Perkins, The Liposome Co., Inc., Princeton, NJ, and Brigham and Women's Hospital, Boston, MA)

polar head groups and water are maximized, whereas hydrophobic interactions (see Chapter 2) facilitate the association of hydrocarbon chains in the interior of the bilayer. The formation of vesicles results in a favorable increase in the entropy of the solution, because the water molecules are not required to order themselves around the lipid chains. It is important to consider for a moment the physical properties of the bilayer membrane, which is the basis of vesicles and also of natural membranes. Bilayers have a polar surface and a nonpolar core. This hydrophobic core provides a substantial barrier to ions and other polar entities. The rates of movement of such species across membranes are thus quite slow. However, this same core also provides a favorable environment for nonpolar molecules and hydrophobic proteins. We will encounter numerous cases of hydrophobic molecules that interact with membranes and regulate biological functions in some way by binding to or embedding themselves in membranes.

Fluid Mosaic Model

In 1972, S. J. Singer and G. L. Nicolson proposed the **fluid mosaic model** for membrane structure, which suggested that membranes are dynamic structures composed of proteins and phospholipids. In this model, the phospholipid bilayer is a *fluid* matrix, in essence, a two-dimensional solvent for proteins. Both lipids and proteins are capable of rotational and lateral movement.

Singer and Nicolson also pointed out that proteins can be associated with the surface of this bilayer or embedded in the bilayer to varying degrees (Figure 9.6). They defined two classes of membrane proteins. The first, called **peripheral proteins** (or **extrinsic proteins**), includes those that do not penetrate the bilayer to any significant degree and are associated with the membrane by virtue

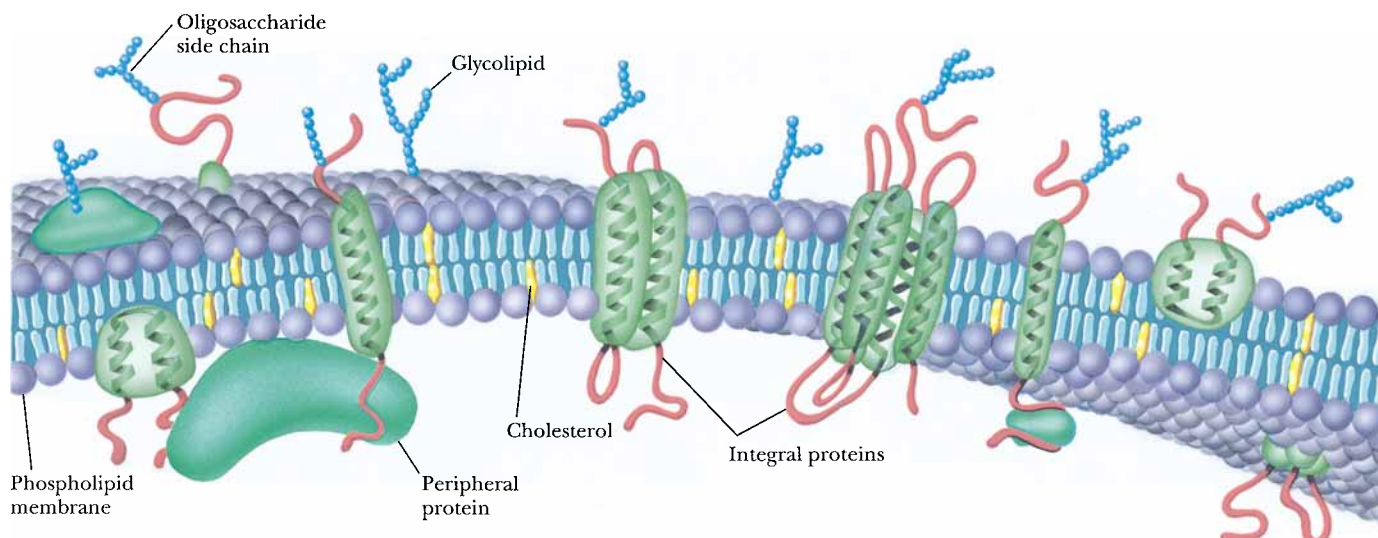


FIGURE 9.6 • The fluid mosaic model of membrane structure proposed by S. J. Singer and G. L. Nicolson. In this model, the lipids and proteins are assumed to be mobile, so that they can move rapidly and laterally in the plane of the membrane. Transverse motion may also occur, but it is much slower.

of ionic interactions and hydrogen bonds between the membrane surface and the surface of the protein. Peripheral proteins can be dissociated from the membrane by treatment with salt solutions or by changes in pH (treatments that disrupt hydrogen bonds and ionic interactions). **Integral proteins** (or **intrinsic proteins**), in contrast, possess hydrophobic surfaces that can readily penetrate the lipid bilayer itself, as well as surfaces that prefer contact with the aqueous medium. These proteins can either insert into the membrane or extend all the way across the membrane and expose themselves to the aqueous solvent on both sides. Singer and Nicolson also suggested that a portion of the bilayer lipid interacts in specific ways with integral membrane proteins and that these interactions might be important for the function of certain membrane proteins. Because of these intimate associations with membrane lipid, integral proteins can only be removed from the membrane by agents capable of breaking up the hydrophobic interactions within the lipid bilayer itself (such as detergents and organic solvents). The fluid mosaic model has become the paradigm for modern studies of membrane structure and function.

Membrane Bilayer Thickness

The Singer–Nicolson model suggested a value of approximately 5 nm for membrane thickness, the same thickness as a lipid bilayer itself. Low angle X-ray diffraction studies in the early 1970s showed that many natural membranes were approximately 5 nm in thickness and that the interiors of these membranes were low in electron density. This is consistent with the arrangement of bilayers having the hydrocarbon tails (low in electron density) in the interior of the membrane. The outside edges of these same membranes were shown to be of high electron density, which is consistent with the arrangement of the polar lipid head groups on the outside surfaces of the membrane.

Hydrocarbon Chain Orientation in the Bilayer

An important aspect of membrane structure is the orientation or ordering of lipid molecules in the bilayer. In the bilayers sketched in Figures 9.2 and 9.4, the long axes of the lipid molecules are portrayed as being perpendicular (or

normal) to the plane of the bilayer. In fact, the hydrocarbon tails of phospholipids may tilt and bend and adopt a variety of orientations. Typically, the portions of a lipid chain near the membrane surface lie most nearly perpendicular to the membrane plane, and lipid chain ordering decreases toward the end of the chain (toward the middle of the bilayer).

Membrane Bilayer Mobility

The idea that lipids and proteins could move rapidly in biological membranes was a relatively new one when the fluid mosaic model was proposed. Many of the experiments designed to test this hypothesis involved the use of specially designed probe molecules. The first experiment demonstrating protein lateral movement in the membrane was described by L. Frye and M. Edidin in 1970. In this experiment, human cells and mouse cells were allowed to fuse together. Frye and Edidin used fluorescent antibodies to determine whether integral membrane proteins from the two cell types could move and intermingle in the newly formed, fused cells. The antibodies specific for human cell proteins were labeled with rhodamine, a red fluorescent marker, and the antibodies specific for mouse cell proteins were labeled with fluorescein, a green fluorescent marker. When both types of antibodies were added to newly fused cells, the binding pattern indicated that integral membrane proteins from the two cell types had moved laterally and were dispersed throughout the surface of the fused cell (Figure 9.7). This clearly demonstrated that integral membrane proteins possess significant lateral mobility.

Just how fast can proteins move in a biological membrane? Many membrane proteins can move laterally across a membrane at a rate of a few microns per minute. On the other hand, some integral membrane proteins are much more restricted in their lateral movement, with diffusion rates of about 10 nm/sec or even slower. These latter proteins are often found to be anchored to the *cytoskeleton* (Chapter 17), a complex latticelike structure that maintains the cell's shape and assists in the controlled movement of various substances through the cell.

Lipids also undergo rapid lateral motion in membranes. A typical phospholipid can diffuse laterally in a membrane at a linear rate of several microns per second. At that rate, a phospholipid could travel from one end of a bacterial cell to the other in less than a second or traverse a typical animal cell in a few minutes. On the other hand, *transverse* movement of lipids (or proteins) from one face of the bilayer to the other is much slower (and much less likely). For example, it can take as long as several days for half the phospholipids in a bilayer vesicle to “flip” from one side of the bilayer to the other.

Membranes Are Asymmetric Structures

Biological membranes are **asymmetric** structures. There are several kinds of asymmetry to consider. Both the lipids and the proteins of membranes exhibit lateral and transverse asymmetries. **Lateral asymmetry** arises when lipids or proteins of particular types cluster in the plane of the membrane.

Lipids Exhibit Lateral Membrane Asymmetry

Lipids in model systems are often found in asymmetric clusters (see Figure 9.8). Such behavior is referred to as a **phase separation**, which arises either spontaneously or as the result of some extraneous influence. Phase separations can be induced in model membranes by divalent cations, which interact with negatively charged moieties on the surface of the bilayer. For example, Ca^{2+} induces phase separations in membranes formed from phosphatidylserine (PS)

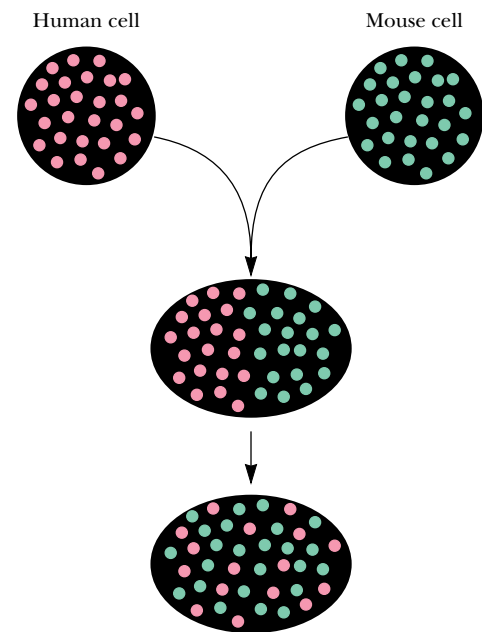


FIGURE 9.7 • The Frye–Edidin experiment. Human cells with membrane antigens for red fluorescent antibodies were mixed and fused with mouse cells having membrane antigens for green fluorescent antibodies. Treatment of the resulting composite cells with red- and green-fluorescent-labeled antibodies revealed a rapid mixing of the membrane antigens in the composite membrane. This experiment demonstrated the lateral mobility of membrane proteins.

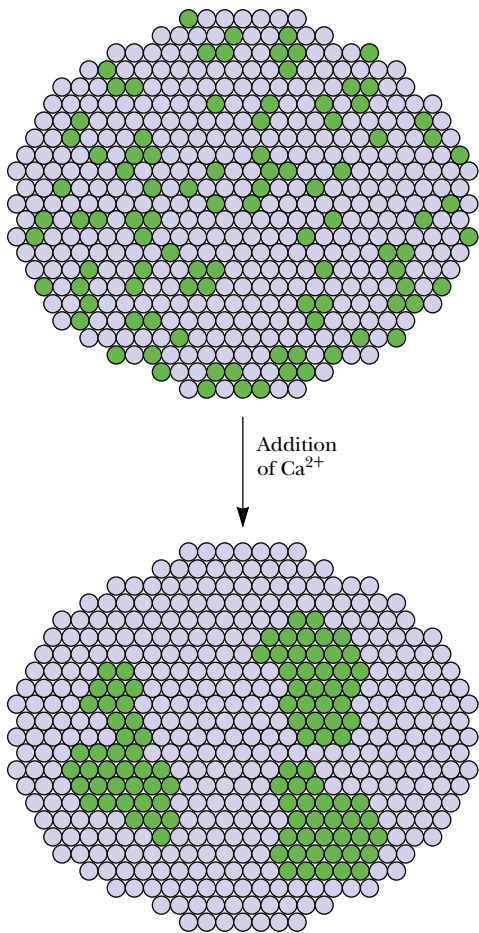


FIGURE 9.8 • An illustration of the concept of lateral phase separations in a membrane. Phase separations of phosphatidylserine (green circles) can be induced by divalent cations such as Ca^{2+} .

and phosphatidylethanolamine (PE) or from PS, PE, and phosphatidylcholine. Ca^{2+} added to these membranes forms complexes with the negatively charged serine carboxyls, causing the PS to cluster and separate from the other lipids. Such metal-induced lipid phase separations have been shown to regulate the activity of membrane-bound enzymes.

There are other ways in which the lateral organization (and asymmetry) of lipids in biological membranes can be altered. For example, cholesterol can intercalate between the phospholipid fatty acid chains, its polar hydroxyl group associated with the polar head groups. In this manner, patches of cholesterol and phospholipids can form in an otherwise homogeneous sea of pure phospholipid. This lateral asymmetry can in turn affect the function of membrane proteins and enzymes. The lateral distribution of lipids in a membrane can also be affected by proteins in the membrane. Certain integral membrane proteins prefer associations with specific lipids. Proteins may select unsaturated lipid chains over saturated chains or may prefer a specific head group over others.

Proteins Exhibit Lateral Membrane Asymmetry

Membrane proteins in many cases are randomly distributed through the plane of the membrane. This was one of the corollaries of the fluid mosaic model of Singer and Nicholson and has been experimentally verified using electron microscopy. Electron micrographs show that integral membrane proteins are often randomly distributed in the membrane, with no apparent long-range order.

However, membrane proteins can also be distributed in nonrandom ways across the surface of a membrane. This can occur for several reasons. Some proteins must interact intimately with certain other proteins, forming multi-subunit complexes that perform specific functions in the membrane. A few integral membrane proteins are known to *self-associate* in the membrane, forming large multimeric clusters. **Bacteriorhodopsin**, a light-driven proton pump protein, forms such clusters, known as “purple patches,” in the membranes of *Halobacterium halobium* (Figure 9.9). The bacteriorhodopsin protein in these purple patches forms highly ordered, two-dimensional crystals.

Transverse Membrane Asymmetry

Membrane asymmetries in the **transverse** direction (from one side of the membrane to the other) can be anticipated when one considers that many properties of a membrane depend upon its two-sided nature. Properties that are a consequence of membrane “sidedness” include membrane transport, which is driven in one direction only, the effects of hormones at the outsides of cells, and the immunological reactions that occur between cells (necessarily involving only the outside surfaces of the cells). One would surmise that the proteins involved in these and other interactions must be arranged asymmetrically in the membrane.

Protein Transverse Asymmetry

Protein transverse asymmetries have been characterized using chemical, enzymatic, and immunological labeling methods. Working with **glycophorin**, the major glycoprotein in the erythrocyte membrane (discussed in Section 9.2), Mark Bretscher was the first to demonstrate the asymmetric arrangement of an integral membrane protein. Treatment of whole erythrocytes with trypsin released the carbohydrate groups of glycophorin (in the form of several small glycopeptides). Because trypsin is much too large to penetrate the erythrocyte membrane, the N-terminus of glycophorin, which contains the carbohydrate

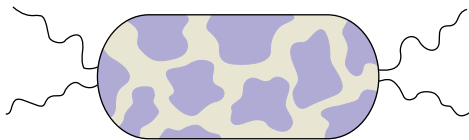


FIGURE 9.9 • The purple patches of *Halobacterium halobium*.

moieties, must be exposed to the outside surface of the membrane. Bretscher showed that [³⁵S]-formylmethionylsulfone methyl phosphate could label the C-terminus of glycophorin with ³⁵S in erythrocyte membrane fragments but not in intact erythrocytes. This clearly demonstrated that the C-terminus of glycophorin is uniformly exposed to the interior surface of the erythrocyte membrane. Since that time, many integral membrane proteins have been shown to be oriented uniformly in their respective membranes.

Lipid Transverse Asymmetry

Phospholipids are also distributed asymmetrically across many membranes. In the erythrocyte, phosphatidylcholine (PC) comprises about 30% of the total phospholipid in the membrane. Of this amount, 76% is found in the outer monolayer and 24% is found in the inner monolayer. Since this early observation, the lipids of many membranes have been found to be asymmetrically distributed between the inner and outer monolayers. Figure 9.10 shows the asymmetric distribution of phospholipids observed in the human erythrocyte membrane. Asymmetric lipid distributions are important to cells in several ways. The carbohydrate groups of glycolipids (and of glycoproteins) always face the outside surface of plasma membranes where they participate in cell recognition phenomena. Asymmetric lipid distributions may also be important to various integral membrane proteins, which may prefer particular lipid classes in the inner and outer monolayers. The total charge on the inner and outer surfaces of a membrane depends on the distribution of lipids. The resulting charge differences affect the membrane potential, which in turn is known to modulate the activity of certain ion channels and other membrane proteins.

How are transverse lipid asymmetries created and maintained in cell membranes? From a thermodynamic perspective, these asymmetries could only occur by virtue of asymmetric syntheses of the bilayer itself or by energy-dependent asymmetric transport mechanisms. Without at least one of these, lipids of all kinds would eventually distribute equally between the two monolayers of a membrane. In eukaryotic cells, phospholipids, glycolipids, and cholesterol are synthesized by enzymes located in (or on the surface of) the endoplasmic reticulum (ER) and the Golgi system (discussed in Chapter 25). Most if not all of these biosynthetic processes are asymmetrically arranged across the membranes of the ER and Golgi. There is also a separate and continuous flow of phospholipids, glycolipids, and cholesterol from the ER and Golgi to other membranes in the cell, including the plasma membrane. This flow is mediated by specific **lipid transfer proteins**. Most cells appear to contain such proteins.

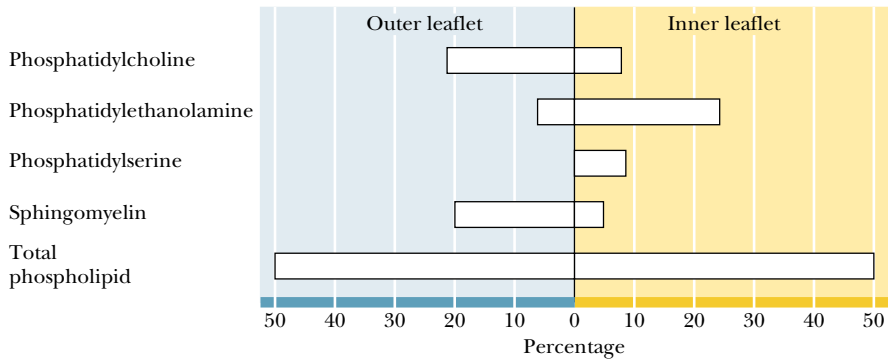


FIGURE 9.10 • Phospholipids are arranged asymmetrically in most membranes, including the human erythrocyte membrane, as shown here. Values are mole percentages. (After Rothman and Lenard, 1977. *Science* **194**:1744.)

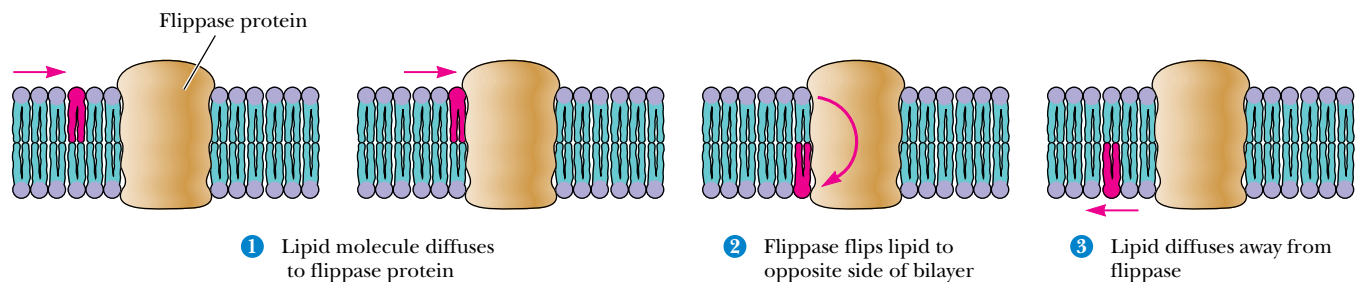


FIGURE 9.11 • Phospholipids can be “flipped” across a bilayer membrane by the action of flippase proteins. When, by normal diffusion through the bilayer, the lipid encounters a flippase, it can be moved quickly to the other face of the bilayer.

Flippases: Proteins That Flip Lipids Across the Membrane

Proteins that can “flip” phospholipids from one side of a bilayer to the other have also been identified in several tissues (Figure 9.11). Called **flippases**, these proteins reduce the half-time for phospholipid movement across a membrane from 10 days or more to a few minutes or less. Some of these systems may operate passively, with no required input of energy, but passive transport alone cannot establish or maintain asymmetric transverse lipid distributions. However, rapid phospholipid movement from one monolayer to the other occurs in an *ATP-dependent* manner in erythrocytes. Energy-dependent lipid flippase activity may be responsible for the creation and maintenance of transverse lipid asymmetries.

Membrane Phase Transitions

Lipids in bilayers undergo radical changes in physical state over characteristic narrow temperature ranges. These changes are in fact true **phase transitions**, and the temperatures at which these changes take place are referred to as **transition temperatures** or **melting temperatures** (T_m). These phase transitions involve substantial changes in the organization and motion of the fatty acyl chains within the bilayer. The bilayer below the phase transition exists in a closely packed gel state, with the fatty acyl chains relatively immobilized in a tightly packed array (Figure 9.12). In this state, the anti conformation is adopted by all the carbon–carbon bonds in the lipid chains. This leaves the lipid chains in their fully extended conformation. As a result, the surface area per lipid is minimal and the bilayer thickness is maximal. Above the transition temperature, a liquid crystalline state exists in which the mobility of fatty acyl chains is intermediate between solid and liquid alkane. In this more fluid, liquid crystalline state, the carbon–carbon bonds of the lipid chains more readily adopt gauche conformations (Figure 9.13). As a result, the surface area per lipid increases and the bilayer thickness decreases by 10 to 15%.

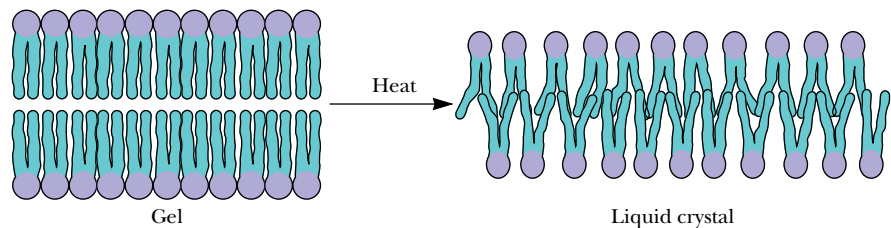


FIGURE 9.12 • An illustration of the gel-to-liquid crystalline phase transition, which occurs when a membrane is warmed through the transition temperature, T_m . Notice that the surface area must increase and the thickness must decrease as the membrane goes through a phase transition. The mobility of the lipid chains increases dramatically.

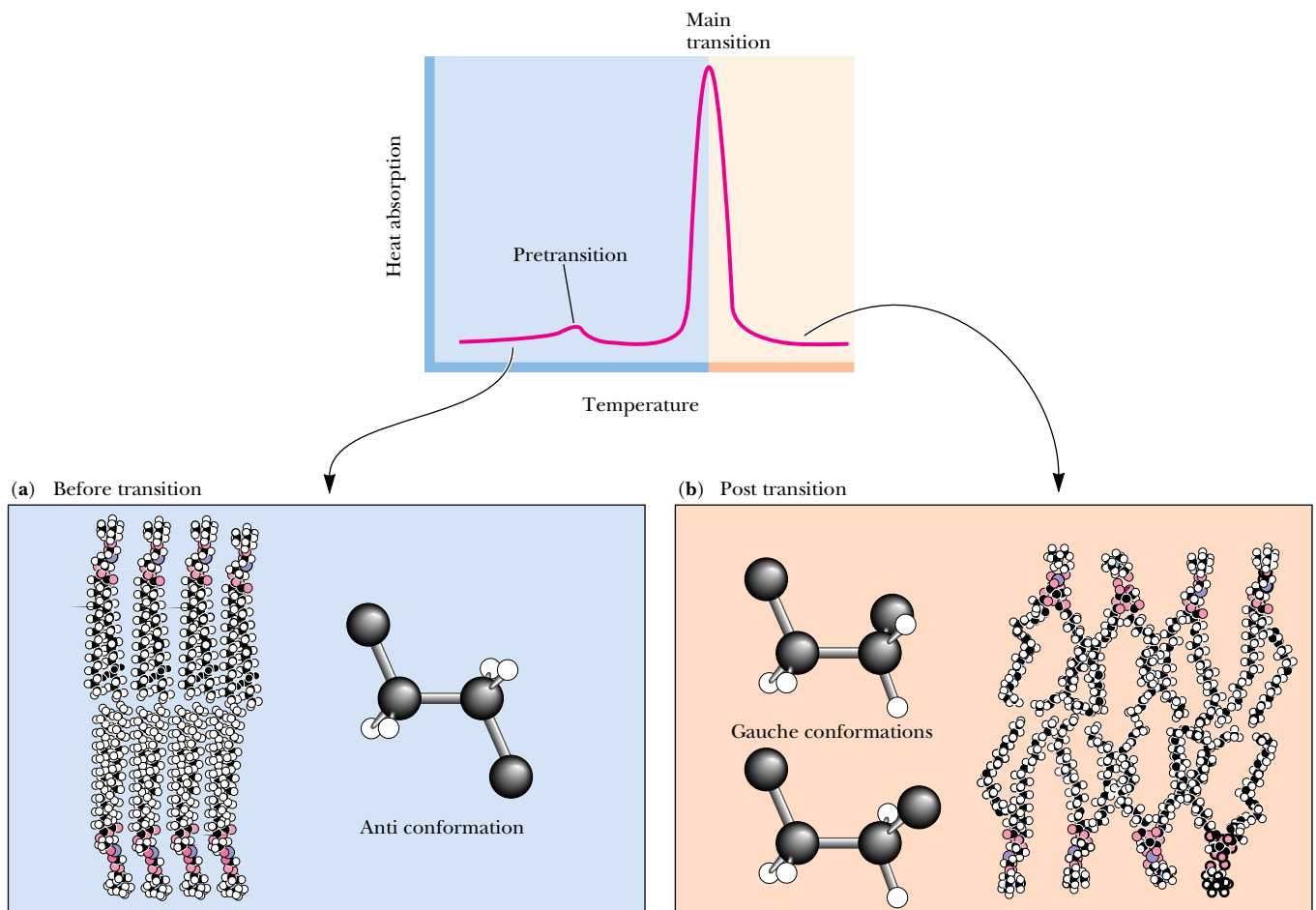


FIGURE 9.13 • Membrane lipid phase transitions can be detected and characterized by measuring the rate of absorption of heat by a membrane sample in a calorimeter (see Chapter 3 for a detailed discussion of calorimetry). Pure, homogeneous bilayers (containing only a single lipid component) give sharp calorimetric peaks. Egg PC contains a variety of fatty acid chains and thus yields a broad calorimetric peak. Below the phase transition, lipid chains primarily adopt the anti conformation. Above the phase transition, lipid chains have absorbed a substantial amount of heat. This is reflected in the adoption of higher-energy conformations, including the gauche conformations shown.

The sharpness of the transition in pure lipid preparations shows that the phase change is a cooperative behavior. This is to say that the behavior of one or a few molecules affects the behavior of many other molecules in the vicinity. The sharpness of the transition then reflects the number of molecules that are acting in concert. Sharp transitions involve large numbers of molecules all “melting” together.

Phase transitions have been characterized in a number of different pure and mixed lipid systems. Table 9.1 shows a comparison of the transition temperatures observed for several different phosphatidylcholines with different fatty acyl chain compositions. General characteristics of bilayer phase transitions include the following:

1. The transitions are always endothermic; heat is absorbed as the temperature increases through the transition (Figure 9.13).
2. Particular phospholipids display characteristic transition temperatures (T_m). As shown in Table 9.1, T_m increases with chain length, decreases with unsaturation, and depends on the nature of the polar head group.
3. For pure phospholipid bilayers, the transition occurs over a narrow temperature range. The phase transition for dimyristoyl lecithin has a peak width of about 0.2°C .
4. Native biological membranes also display characteristic phase transitions, but these are broad and strongly dependent on the lipid and protein composition of the membrane.

Table 9.1

Phase Transition Temperatures for Phospholipids in Water

Phospholipid	Transition Temperature (<i>T_m</i>), °C
Dipalmitoyl phosphatidic acid (Di 16:0 PA)	67
Dipalmitoyl phosphatidylethanolamine (Di 16:0 PE)	63.8
Dipalmitoyl phosphatidylcholine (Di 16:0 PC)	41.4
Dipalmitoyl phosphatidylglycerol (Di 16:0 PG)	41.0
Dilauroyl phosphatidylcholine (Di 14:0 PC)	23.6
Distearoyl phosphatidylcholine (Di 18:0 PC)	58
Di oleoyl phosphatidylcholine (Di 18:1 PC)	−22
1-Stearoyl-2-oleoyl-phosphatidylcholine (1-18:0, 2-18:1 PC)	3
Egg phosphatidylcholine (Egg PC)	−15

Adapted from Jain, M., and Wagner, R. C., 1980. *Introduction to Biological Membranes*. New York: John Wiley and Sons; Martonosi, A., ed., 1982. *Membranes and Transport*, Vol. 1. New York: Plenum Press.

- 5. With certain lipid bilayers, a change of physical state referred to as a *pre-transition* occurs 5° to 15°C below the phase transition itself. These pre-transitions involve a tilting of the hydrocarbon chains.
- 6. A volume change is usually associated with phase transitions in lipid bilayers.
- 7. Bilayer phase transitions are sensitive to the presence of solutes that interact with lipids, including multivalent cations, lipid-soluble agents, peptides, and proteins.

Cells adjust the lipid composition of their membranes to maintain proper fluidity as environmental conditions change.

9.2 • Structure of Membrane Proteins

The lipid bilayer constitutes the fundamental structural unit of all biological membranes. Proteins, in contrast, carry out essentially all of the active functions of membranes, including transport activities, receptor functions, and other related processes. As suggested by Singer and Nicolson, most membrane proteins can be classified as peripheral or integral. The **peripheral proteins** are globular proteins that interact with the membrane mainly through electrostatic and hydrogen-bonding interactions with integral proteins. Although peripheral proteins are not discussed further here, many proteins of this class are described in the context of other discussions throughout this textbook. **Integral proteins** are those that are strongly associated with the lipid bilayer, with a portion of the protein embedded in, or extending all the way across, the lipid bilayer. Another class of proteins not anticipated by Singer and Nicolson, the **lipid-anchored proteins**, are important in a variety of functions in different cells and tissues. These proteins associate with membranes by means of a variety of covalently linked lipid anchors.

Integral Membrane Proteins

Despite the diversity of integral membrane proteins, most fall into two general classes. One of these includes proteins attached or anchored to the membrane by only a small hydrophobic segment, such that most of the protein extends out into the water solvent on one or both sides of the membrane. The other class includes those proteins that are more or less globular in shape and more totally embedded in the membrane, exposing only a small surface to the water solvent outside the membrane. In general, those structures of integral membrane protein within the nonpolar core of the lipid bilayer are dominated by α -helices or β -sheets because these secondary structures neutralize the highly polar N—H and C=O functions of the peptide backbone through H-bond formation.

A Protein with a Single Transmembrane Segment

In the case of the proteins that are anchored by a small hydrophobic polypeptide segment, that segment often takes the form of a single α -helix. One of the best examples of a membrane protein with such an α -helical structure is **glycophorin**. Most of glycophorin’s mass is oriented on the outside surface of the cell, exposed to the aqueous milieu (Figure 9.14). A variety of hydrophilic oligosaccharide units are attached to this extracellular domain. These oligosaccharide groups constitute the ABO and MN blood group antigenic specificity.

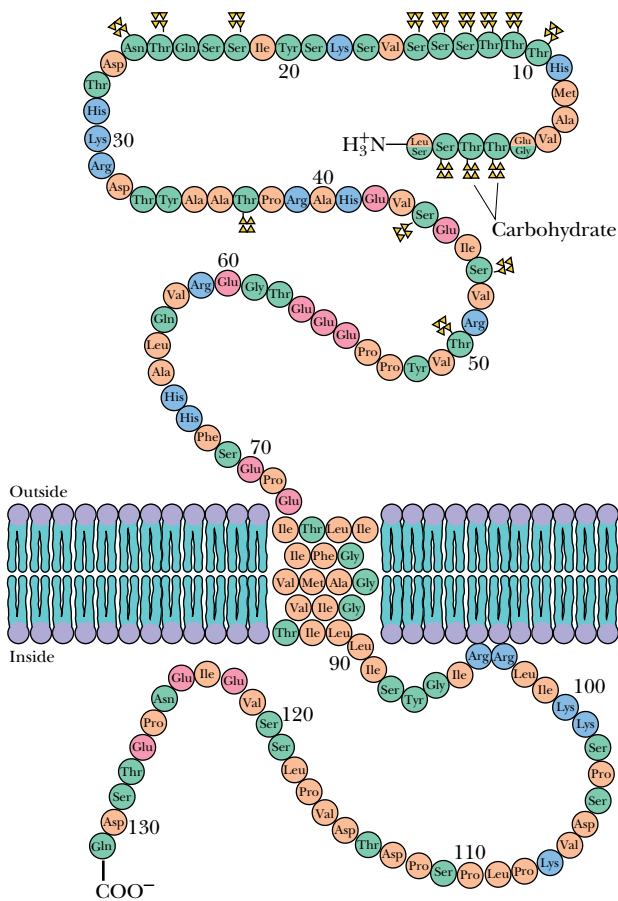


FIGURE 9.14 • Glycophorin A spans the membrane of the human erythrocyte via a single α -helical transmembrane segment. The C-terminus of the peptide, whose sequence is shown here, faces the cytosol of the erythrocyte; the N-terminal domain is extracellular. Points of attachment of carbohydrate groups are indicated.

ties of the red cell. This extracellular portion of the protein also serves as the receptor for the influenza virus. Glycophorin has a total molecular weight of about 31,000 and is approximately 40% protein and 60% carbohydrate. The glycophorin primary structure consists of a segment of 19 hydrophobic amino acid residues with a short hydrophilic sequence on one end and a longer hydrophilic sequence on the other end. The 19-residue sequence is just the right length to span the cell membrane if it is coiled in the shape of an α -helix. The large hydrophilic sequence includes the amino terminal residue of the polypeptide chain.

Numerous other membrane proteins are also attached to the membrane by means of a single hydrophobic α -helix, with hydrophilic segments extending into either the cytoplasm or the extracellular space. These proteins often function as receptors for extracellular molecules or as recognition sites that allow the immune system to recognize and distinguish the cells of the host organism from invading foreign cells or viruses. The proteins that represent the *major transplantation antigens H2* in mice and *human leukocyte associated (HLA) proteins* in humans are members of this class. Other such proteins include the *surface immunoglobulin receptors* on B lymphocytes and the *spike proteins* of many membrane viruses. The function of many of these proteins depends primarily on their extracellular domain, and thus the segment facing the intracellular surface is often a shorter one.

Bacteriorhodopsin: A 7-Transmembrane Segment Protein

Membrane proteins that take on a more globular shape, instead of the rodlike structure previously described, are often involved with transport activities and other functions requiring a substantial portion of the peptide to be embedded in the membrane. These proteins may consist of numerous hydrophobic α -helical segments joined by hinge regions so that the protein winds in a zig-zag pattern back and forth across the membrane. A well-characterized example of such a protein is **bacteriorhodopsin**, which clusters in purple patches in the membrane of the bacterium *Halobacterium halobium*. The name *Halobacterium* refers to the fact that this bacterium thrives in solutions having high concentrations of sodium chloride, such as the salt beds of San Francisco Bay. *Halobacterium* carries out a light-driven proton transport by means of bacteriorhodopsin, named in reference to its spectral similarities to rhodopsin in the rod outer segments of the mammalian retina. When this organism is deprived of oxygen for oxidative metabolism, it switches to the capture of energy from sunlight, using this energy to pump protons out of the cell. The proton gradient generated by such light-driven proton pumping represents potential energy, which is exploited elsewhere in the membrane to synthesize ATP.

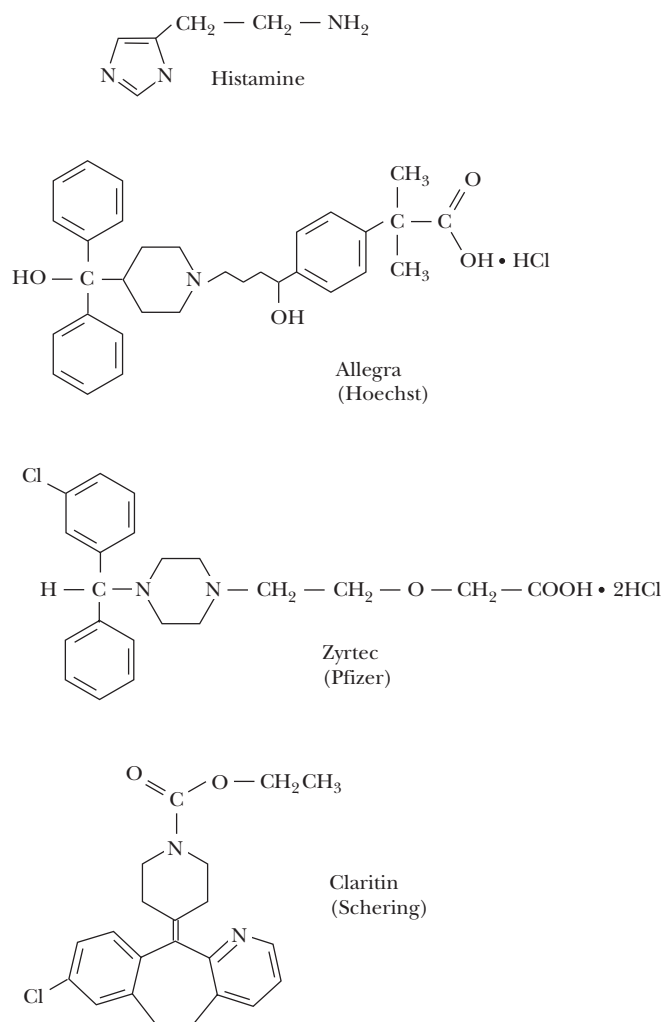
Bacteriorhodopsin clusters in hexagonal arrays (Figure 9.15) in the purple membrane patches of *Halobacterium*, and it was this orderly, repeating arrangement of proteins in the membrane that enabled Nigel Unwin and Richard Henderson in 1975 to determine the bacteriorhodopsin structure. The polypeptide chain crosses the membrane seven times, in seven α -helical segments, with very little of the protein exposed to the aqueous milieu. The bacteriorhodopsin structure has become a model of globular membrane protein structure. Many other integral membrane proteins contain numerous hydrophobic sequences that, like those of bacteriorhodopsin, could form α -helical transmembrane segments. For example, the amino acid sequence of the sodium–potassium transport ATPase contains ten hydrophobic segments of length sufficient to span the plasma membrane. By analogy with bacteriorhodopsin, one would expect that these segments form a globular hydrophobic core that anchors the ATPase in the membrane. The helical segments may also account for the transport properties of the enzyme itself.

Treating Allergies at the Cell Membrane

Allergies represent overreactions of the immune system caused by exposure to foreign substances referred to as allergens. The inhalation of allergens, such as pollen, pet dander, and dust, can cause a variety of allergic responses, including itchy eyes, a runny nose, shortness of breath, and wheezing. Allergies can also be caused by food, drugs, dyes, and other chemicals.

The visible symptoms of such an allergic response are caused by the release of **histamine** (see figure) by mast cells, a type of cell found in loose connective tissue. Histamine dilates blood vessels, increases the permeability of capillaries (allowing antibodies to pass from the capillaries to surrounding tissue), and constricts bronchial air passages. Histamine acts by binding to specialized membrane proteins called **histamine H1 receptors**. These integral membrane proteins possess seven transmembrane α -helical segments, with an extracellular amino terminus and a cytoplasmic carboxy terminus. When histamine binds to the extracellular domain of an H1 receptor, the intracellular domain undergoes a conformation change that stimulates a GTP-binding protein, which in turn activates the allergic response in the affected cell.

A variety of highly effective **antihistamine** drugs are available for the treatment of allergy symptoms. These drugs share the property of binding tightly to histamine H1 receptors, without eliciting the same effects as histamine itself. They are referred to as **histamine H1 receptor antagonists** because they prevent the binding of histamine to the receptors. The structures of Allegra (made by Hoechst Marion Roussel, Inc.), Claritin (by Schering-Plough Corp.), and Zyrtec (by Pfizer) are all shown at right.



The structures of histamine and three antihistamine drugs. ►

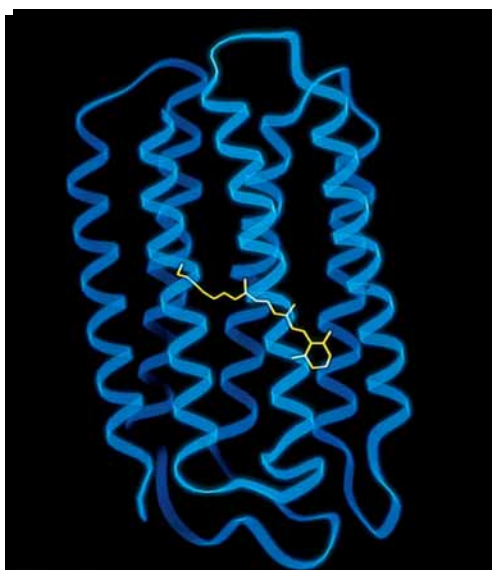
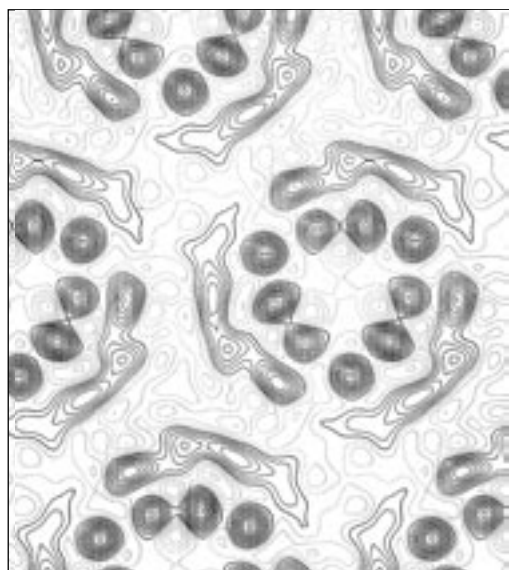


FIGURE 9.15 • An electron density profile illustrating the three centers of threefold symmetry in arrays of bacteriorhodopsin in the purple membrane of *Halobacterium halobium*, together with a computer-generated model showing the seven α -helical transmembrane segments in bacteriorhodopsin. (Electron density map from Stoeknius, W., 1980. Purple membrane of halobacteria: A new light-energy converter. Accounts of Chemical Research 13:337–344. Model on right from Henderson, R., 1990. Model for the structure of bacteriorhodopsin based on high-resolution electron cryo-microscopy. Journal of Molecular Biology 213:899–929.)

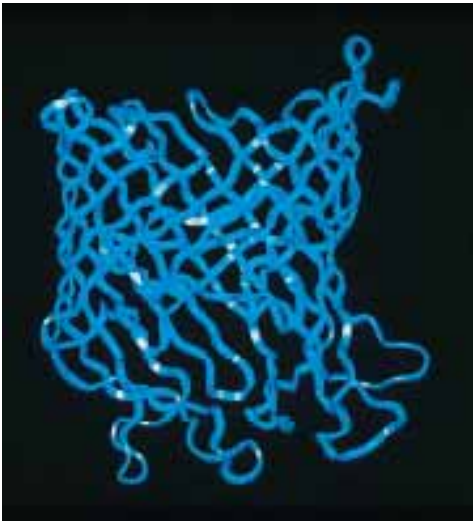


FIGURE 9.16 • The three-dimensional structure of maltoporin from *E. coli*.

Porins—A β -Sheet Motif for Membrane Proteins

The β -sheet is another structural motif that provides extensive hydrogen bonding for transmembrane peptide segments. **Porin** proteins found in the outer membranes (OM) of Gram-negative bacteria such as *Escherichia coli*, and also in the outer mitochondrial membranes of eukaryotic cells, span their respective membranes with large β -sheets. A good example is **maltoporin**, also known as **LamB protein** or **lambda receptor**, which participates in the entry of maltose and maltodextrins into *E. coli*. Maltoporin is active as a trimer. The 421-residue monomer is an aesthetically pleasing 18-strand β -barrel (Figure 9.16). The β -strands are connected to their nearest neighbors either by long loops or by β -turns (Figure 9.17). The long loops are found at the end of the barrel that is exposed to the cell exterior, whereas the turns are located on the intracellular face of the barrel. Three of the loops fold into the center of the barrel.

The amino acid compositions and sequences of the β -strands in porin proteins are novel. Polar and nonpolar residues alternate along the β -strands, with polar residues facing the central pore or cavity of the barrel and nonpolar residues facing out from the barrel where they can interact with the hydrophobic lipid milieu of the membrane. The smallest diameter of the porin channel is about 5 Å. Thus, a maltodextrin polymer (composed of two or more glucose units) must pass through the porin in an extended conformation (like a spaghetti strand).

Lipid-Anchored Membrane Proteins

Certain proteins are found to be covalently linked to lipid molecules. For many of these proteins, covalent attachment of lipid is required for association with a membrane. The lipid moieties can insert into the membrane bilayer, effectively **anchoring** their linked proteins to the membrane. Some proteins with covalently linked lipid normally behave as soluble proteins; others are integral

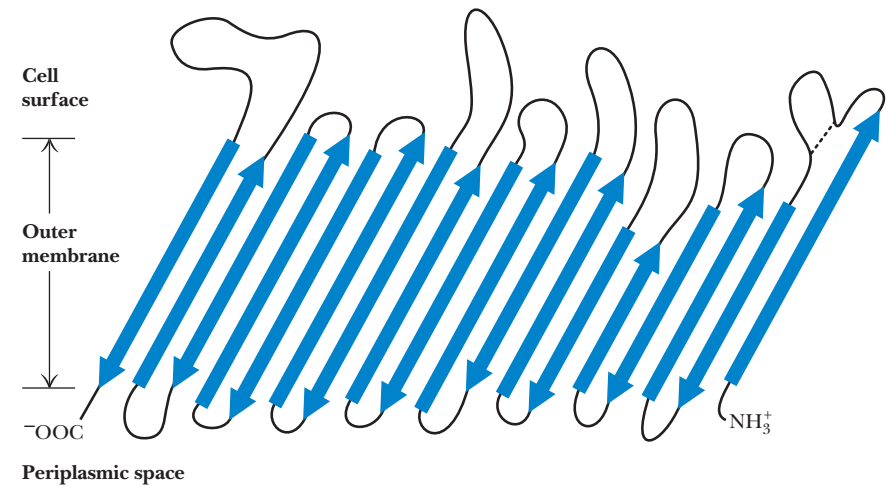


FIGURE 9.17 • The arrangement of the peptide chain in maltoporin from *E. coli*.

A D E E P E R L O O K

Exterminator Proteins—Biological Pest Control at the Membrane

Control of biological pests, including mosquitoes, houseflies, gnats, and tree-consuming predators like the eastern tent caterpillar, is frequently achieved through the use of microbial membrane proteins. For example, several varieties of *Bacillus thuringiensis* produce proteins that bind to cell membranes in the digestive systems of insects that consume them, creating transmembrane ion channels. Leakage of Na^+ , K^+ , and H^+ ions through these membranes in the insect gut destroys crucial ion gradients and interferes with digestion of food. Insects that ingest these toxins eventually die of starvation. *B. thuringiensis* toxins account for more than 90% of sales of biological pest control agents.

B. thuringiensis is a common Gram-positive, spore-forming soil bacterium that produces **inclusion bodies**, microcrystalline clusters of many different proteins. These crystalline proteins, called **δ -endotoxins**, are the ion channel toxins that are sold commercially for pest control. Most such endotoxins are **protoxins**, which are inactive until cleaved to smaller, active proteins by proteases in the gut of a susceptible insect. One such crystalline protoxin,

lethal to mosquitoes, is a 27 kD protein, which is cleaved to form the active 25 kD toxin in the mosquito. This toxin has no effect on membranes at neutral pH, but at pH 9.5 (the pH of the mosquito gut) the toxin forms cation channels in the gut membranes.

This 25 kD protein is not toxic to tent caterpillars, but a larger, 130 kD protein in the *B. thuringiensis* inclusion bodies is cleaved by a caterpillar gut protease to produce a 55 kD toxin that is active in the caterpillar. Remarkably, the strain of *B. thuringiensis* known as *azawai* produces a protoxin with dual specificity: in the caterpillar gut, this 130 kD protein is cleaved to form a 55 kD toxin active in the caterpillar. However, when the same 130 kD protoxin is consumed by mosquitoes or houseflies, it is cleaved to form a 53 kD protein (15 amino acid residues shorter than the caterpillar toxin) that is toxic to these latter organisms. Understanding the molecular basis of the toxicity and specificity of these proteins and the means by which they interact with membranes to form lethal ion channels is a fascinating biochemical challenge with far-reaching commercial implications.

membrane proteins and remain membrane-associated even when the lipid is removed. Covalently bound lipid in these latter proteins can play a role distinct from membrane anchoring. In many cases, attachment to the membrane via the lipid anchor serves to modulate the activity of the protein.

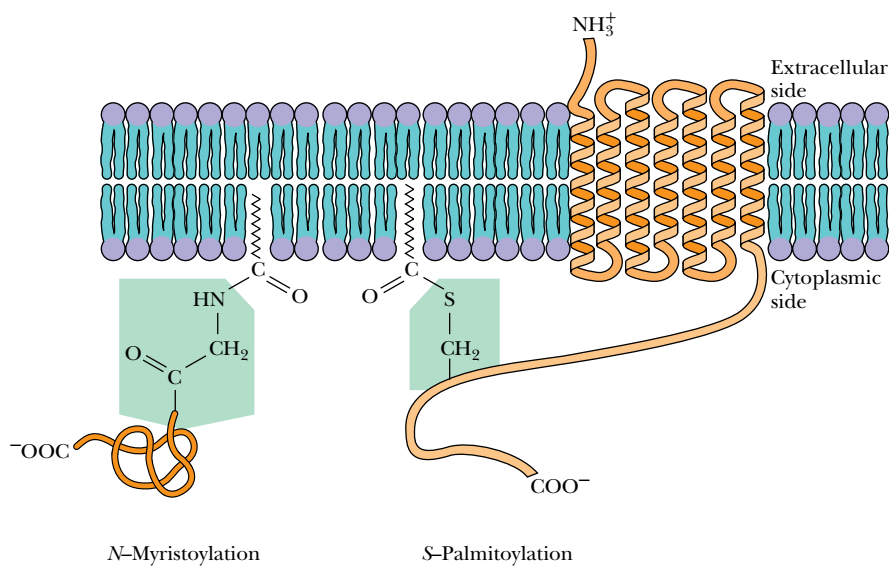
Another interesting facet of lipid anchors is that they are transient. Lipid anchors can be reversibly attached to and detached from proteins. This provides a “switching device” for altering the affinity of a protein for the membrane. Reversible lipid anchoring is one factor in the control of **signal transduction pathways** in eukaryotic cells (Chapter 34).

Four different types of lipid-anchoring motifs have been found to date. These are **amide-linked myristoyl** anchors, **thioester-linked fatty acyl** anchors, **thioether-linked prenyl** anchors, and **amide-linked glycosyl phosphatidylinositol** anchors. Each of these anchoring motifs is used by a variety of membrane proteins, but each nonetheless exhibits a characteristic pattern of structural requirements.

Amide-Linked Myristoyl Anchors

Myristic acid may be linked via an amide bond to the α -amino group of the N-terminal glycine residue of selected proteins (Figure 9.18). The reaction is referred to as **N-myristoylation** and is catalyzed by *myristoyl-CoA:protein N-myristoyltransferase*, known simply as **NMT**. N-Myristoyl-anchored proteins include the catalytic subunit of *cAMP-dependent protein kinase*, the *pp60^{src}* *tyrosine kinase*, the phosphatase known as *calcineurin B*, the α -subunit of *G proteins* (involved in GTP-dependent transmembrane signaling events), and the *gag proteins* of certain retroviruses, including the HIV-1 virus that causes AIDS.

FIGURE 9.18 • Certain proteins are anchored to biological membranes by lipid anchors. Particularly common are the *N*-myristoyl- and *S*-palmitoyl-anchoring motifs shown here. *N*-Myristoylation always occurs at an N-terminal glycine residue, whereas thioester linkages occur at cysteine residues within the polypeptide chain. G-protein-coupled receptors, with seven transmembrane segments, may contain one (and sometimes two) palmitoyl anchors in thioester linkage to cysteine residues in the C-terminal segment of the protein.



Thioester-Linked Fatty Acyl Anchors

A variety of cellular and viral proteins contain fatty acids covalently bound via ester linkages to the side chains of cysteine and sometimes to serine or threonine residues within a polypeptide chain (Figure 9.18). This type of fatty acyl chain linkage has a broader fatty acid specificity than *N*-myristoylation. Myristate, palmitate, stearate, and oleate can all be esterified in this way, with the C₁₆ and C₁₈ chain lengths being most commonly found. Proteins anchored to membranes via fatty acyl thioesters include *G-protein-coupled receptors*, the *surface glycoproteins* of several viruses, and the *transferrin receptor* protein.

Thioether-Linked Prenyl Anchors

As noted in Chapter 7, polyprenyl (or simply prenyl) groups are long-chain polyisoprenoid groups derived from isoprene units. Prenylation of proteins destined for membrane anchoring can involve either **farnesyl** or **geranylgeranyl** groups (Figure 9.19). The addition of a prenyl group typically occurs at the cysteine residue of a carboxy-terminal CAAX sequence of the target protein, where C is cysteine, A is any aliphatic residue, and X can be any amino acid. As shown in Figure 9.19, the result is a thioether-linked farnesyl or geranylgeranyl group. Once the prenylation reaction has occurred, a specific protease cleaves the three carboxy-terminal residues, and the carboxyl group of the now terminal Cys is methylated to produce an ester. All of these modifications appear to be important for subsequent activity of the prenyl-anchored protein. Proteins anchored to membranes via prenyl groups include *yeast mating factors*, the *p21^{ras} protein* (the protein product of the *ras* oncogene; see Chapter 34), and the *nuclear lamins*, structural components of the lamina of the inner nuclear membrane.

Glycosyl Phosphatidylinositol Anchors

Glycosyl phosphatidylinositol, or **GPI**, groups are structurally more elaborate membrane anchors than fatty acyl or prenyl groups. GPI groups modify the carboxy-terminal amino acid of a target protein via an ethanolamine residue linked to an oligosaccharide, which is linked in turn to the inositol moiety of a phosphatidylinositol (Figure 9.20). The oligosaccharide typically consists of

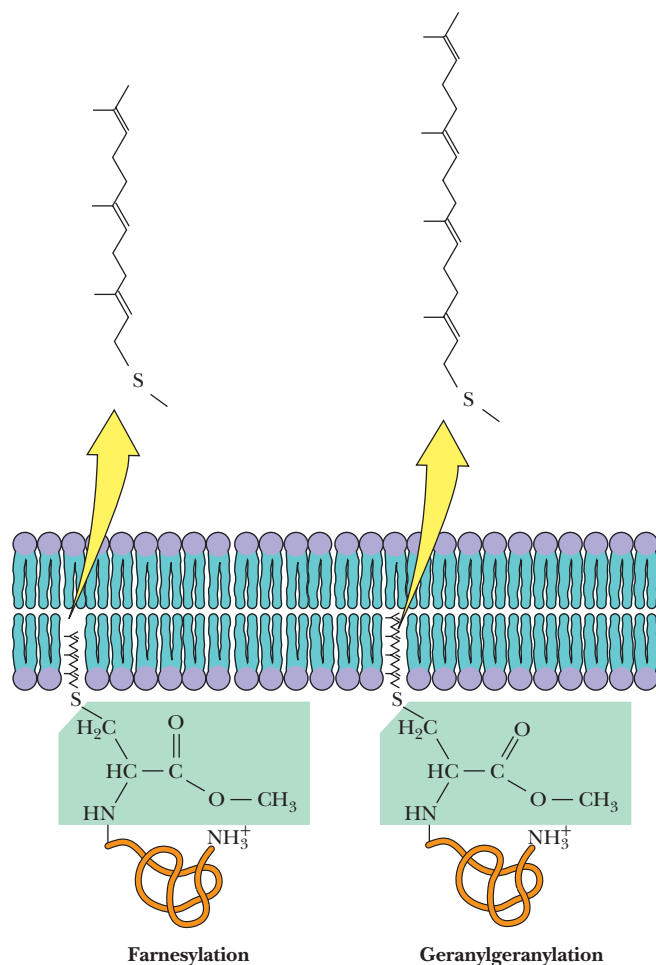


FIGURE 9.19 • Proteins containing the C-terminal sequence CAAX can undergo prenylation reactions that place thioether-linked farnesyl or geranylgeranyl groups at the cysteine side chain. Prenylation is accompanied by removal of the AAX peptide and methylation of the carboxyl group of the cysteine residue, which has become the C-terminal residue.

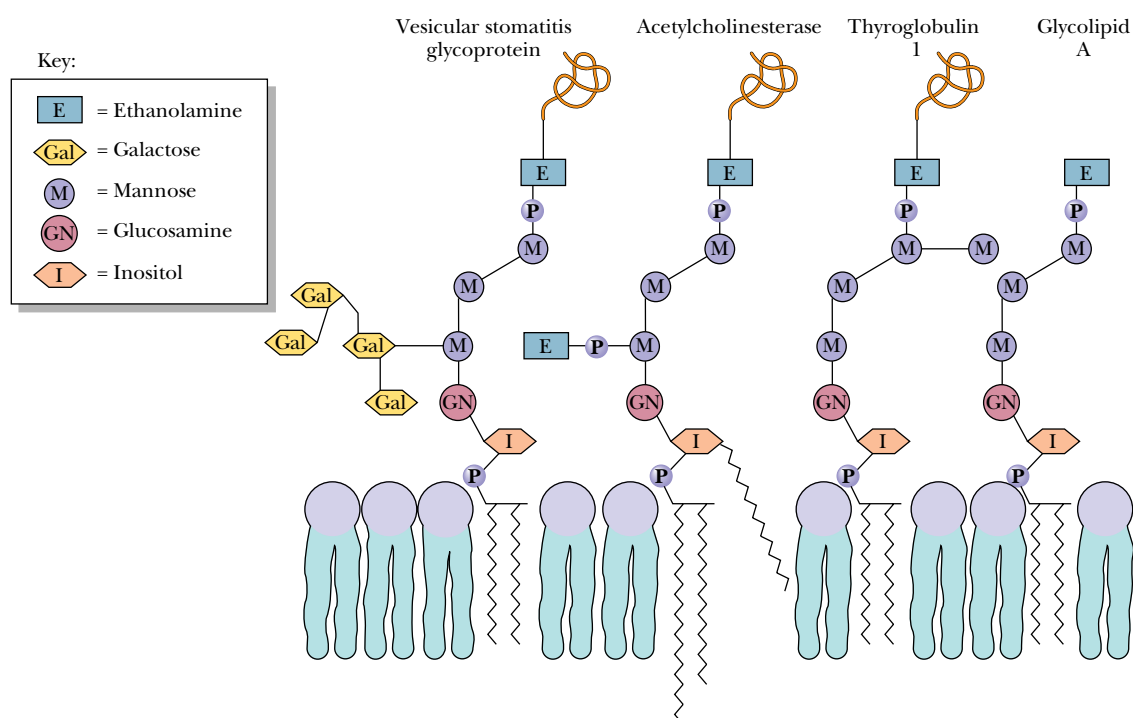


FIGURE 9.20 • The glycosyl phosphatidylinositol (GPI) moiety is an elaborate lipid-anchoring group. Note the core of three mannose residues and a glucosamine. Additional modifications may include fatty acids at the inositol and glycerol —OH groups.

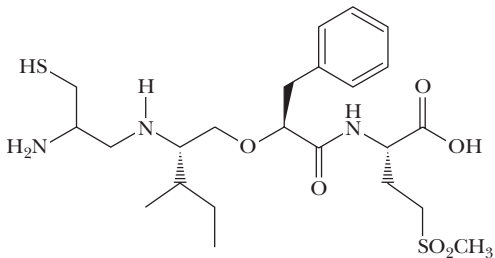
A Prenyl Protein Protease Is a New Chemotherapy Target

The protein called p21^{ras} or simply Ras is a small GTP-binding protein involved in cell signaling pathways that regulate growth and cell division. Mutant forms of Ras cause uncontrolled cell growth, and Ras mutations are involved in one third of all human cancers. Because the signaling activity of Ras is dependent on prenylation, the prenylation reaction itself, as well as the proteolysis of the -AAX motif and the methylation of the prenylated Cys residue, have been considered targets for development of new chemotherapy strategies.

Farnesyl transferase from rat cells is a heterodimer consisting of a 48 kD α -subunit and a 46 kD β -subunit. In the structure shown here, helices 2 to 15 of the α -subunit are folded into seven short coiled coils that together form a crescent-shaped envelope partially surrounding the β -subunit. Twelve helices of the β -subunit form a novel barrel motif that creates the active site of the enzyme. Farnesyl transferase inhibitors, one of which is shown here, are potent suppressors of tumor growth in mice, but their value in humans has not been established.



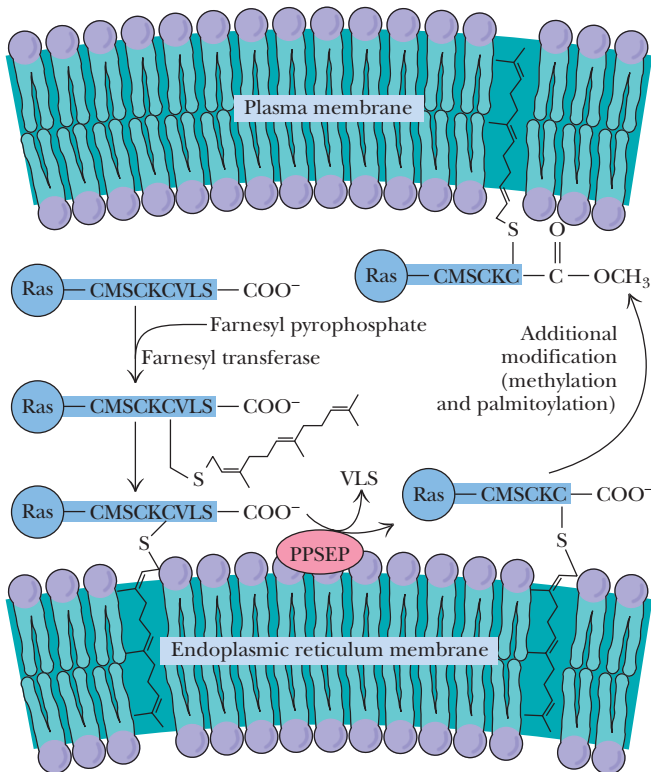
The structure of the farnesyl transferase heterodimer. A novel barrel structure is formed from 12 helical segments in the β -subunit (purple). The α -subunit (yellow) consists largely of seven successive pairs of α -helices that form a series of right-handed antiparallel coiled coils running along the bottom of the structure. These “helical hairpins” are arranged in a double-layered, right-handed super-helix resulting in a crescent-shaped subunit that envelopes part of the subunit.



2(S)-{[(S)-[2(R)-amino-3-mercapto]propylamino-3(S)-methyl]pentyl}oxy-3-phenylpropionyl-methioninesulfone methyl ester

The structure of I-739,749, a farnesyl transferase inhibitor that is a potent tumor growth suppressor.

Mutations that inhibit prenyl transferases cause defective growth or death of cells, raising questions about the usefulness of prenyl transferase inhibitors in chemotherapy. However, Victor Boyartchuk and his colleagues at the University of California, Berkeley, and Acacia Biosciences have shown that the protease that cleaves the -AAX motif from Ras following the prenylation reaction may be a better chemotherapeutic target. They have identified two genes for the prenyl protein protease in the yeast *Saccharomyces cerevisiae* and have shown that deletion of these genes results in loss of proteolytic processing of prenylated proteins, including Ras. Interestingly, normal yeast cells are unaffected by this gene deletion. However, in yeast cells that carry mutant forms of Ras and that display aberrant growth behaviors, deletion of the protease gene restores normal growth patterns. If these remarkable results translate from yeast to human tumor cells, inhibitors of CAAX proteases may be more valuable chemotherapeutic agents than prenyl transferase inhibitors.



The farnesylation and subsequent processing of the Ras protein. Following farnesylation by the FTase, the carboxy-terminal VLS peptide is removed by a prenyl protein-specific endoprotease (PPSEP) in the ER, and then a prenylprotein-specific methyltransferase (PPSMT) donates a methyl group from S-adenosylmethionine (SAM) to the carboxy-terminal S-farnesylated cysteine. Finally, palmitates are added to cysteine residues near the C-terminus of the protein.

a conserved tetrasaccharide core of three mannose residues and a glucosamine, which can be altered by modifications of the mannose residues or addition of galactosyl side chains of various sizes, extra phosphoethanolamines, or additional *N*-acetylgalactose or mannosyl residues (Figure 9.20). The inositol moiety can also be modified by an additional fatty acid, and a variety of fatty acyl groups are found linked to the glycerol group. GPI groups anchor a wide variety of *surface antigens*, *adhesion molecules*, and *cell surface hydrolases* to plasma membranes in various eukaryotic organisms. GPI anchors have not yet been observed in prokaryotic organisms or plants.

9.3 • Membranes and Cell-Surface Polysaccharides

Bacterial Cell Walls

Some of nature's most interesting polysaccharide structures are found in *bacterial cell walls*. Given the strength and rigidity provided by polysaccharide structures, it is not surprising that bacteria use such structures to provide protection for their cellular contents. Bacteria normally exhibit high internal osmotic pressures and frequently encounter variable, often hypotonic exterior conditions. The rigid cell walls synthesized by bacteria maintain cell shape and size and prevent swelling or shrinkage that would inevitably accompany variations in solution osmotic strength.

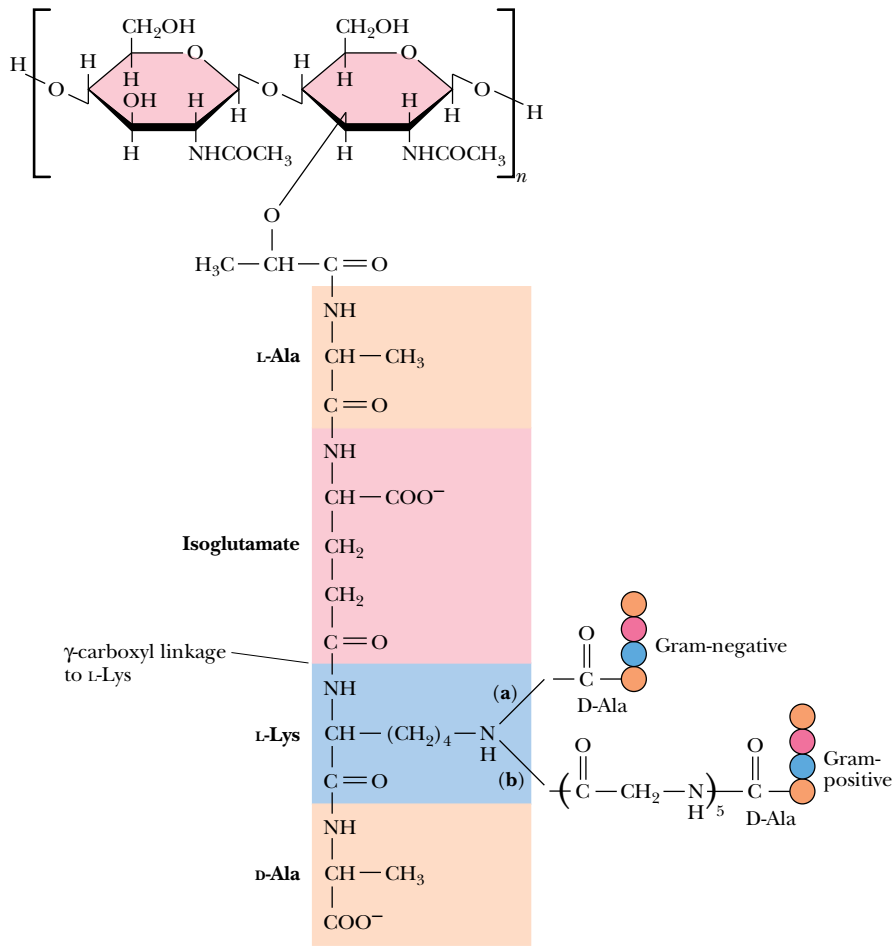
Peptidoglycan

Bacteria are conveniently classified as either **Gram-positive** or **Gram-negative** depending on their response to the so-called *Gram stain*. Despite substantial differences in the various structures surrounding these two types of cells, nearly all bacterial cell walls have a strong, protective peptide–polysaccharide layer called **peptidoglycan**. Gram-positive bacteria have a thick (approximately 25 nm) cell wall consisting of multiple layers of peptidoglycan. This thick cell wall surrounds the bacterial plasma membrane. Gram-negative bacteria, in contrast, have a much thinner (2 to 3 nm) cell wall consisting of a single layer of peptidoglycan sandwiched between the inner and outer lipid bilayer membranes. In either case, peptidoglycan, sometimes called **murein** (from the Latin *murus* for “wall”), is a continuous cross-linked structure—in essence, a single molecule—built around the cell. The structure is shown in Figure 9.21. The backbone is a $\beta(1\rightarrow4)$ linked polymer of alternating *N*-acetylglucosamine and *N*-acetylmuramic acid units. This part of the structure is similar to chitin, but it is joined to a tetrapeptide, usually L-Ala·D-Glu·L-Lys·D-Ala, in which the L-lysine is linked to the γ -COOH of D-glutamate. The peptide is linked to the *N*-acetylmuramic acid units via its D-lactate moiety. The ϵ -amino group of lysine in this peptide is linked to the —COOH of D-alanine of an adjacent tetrapeptide. In Gram-negative cell walls, the lysine ϵ -amino group forms a *direct amide bond* with this D-alanine carboxyl (Figure 9.22). In Gram-positive cell walls, a **pentaglycine chain** bridges the lysine ϵ -amino group and the D-Ala carboxyl group.

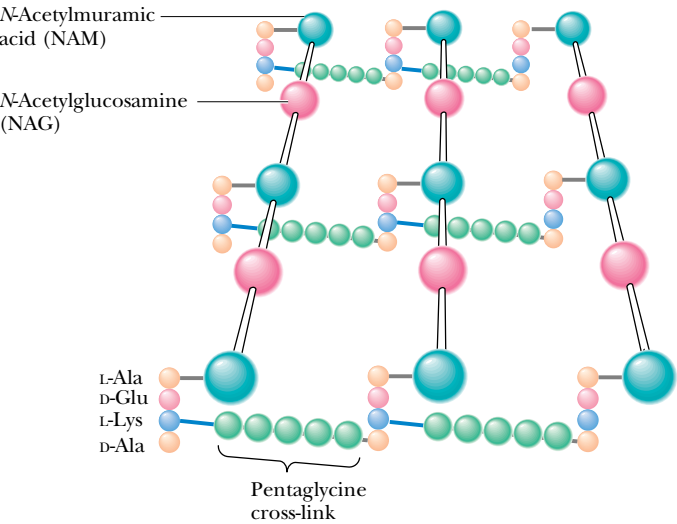
Cell Walls of Gram-Negative Bacteria

In Gram-negative bacteria, the peptidoglycan wall is the rigid framework around which is built an elaborate membrane structure (Figure 9.23). The peptidoglycan layer encloses the *periplasmic space* and is attached to the outer membrane via a group of **hydrophobic proteins**. These proteins, each having 57 amino acid residues, are attached through amide linkages from the side chains of C-terminal lysines of the proteins to diaminopimelic acid groups on the

FIGURE 9.21 • The structure of peptidoglycan. The tetrapeptides linking adjacent backbone chains contain an unusual γ -carboxyl linkage.



(a) Gram-positive cell wall



(b) Gram-negative cell wall

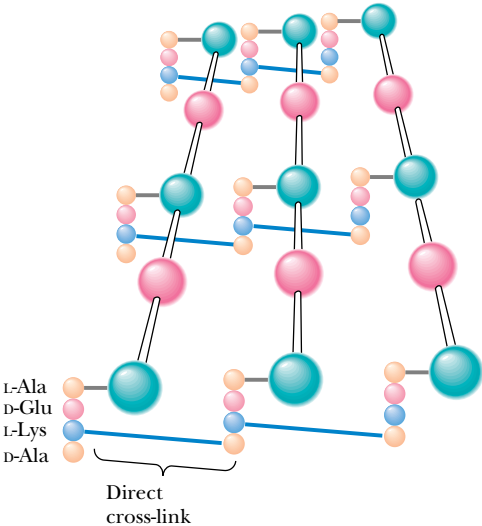


FIGURE 9.22 • (a) The cross-link in Gram-positive cell walls is a pentaglycine bridge. (b) In Gram-negative cell walls, the linkage between the tetrapeptides of adjacent carbohydrate chains in peptidoglycan involves a direct amide bond between the lysine side chain of one tetrapeptide and D-alanine of the other.

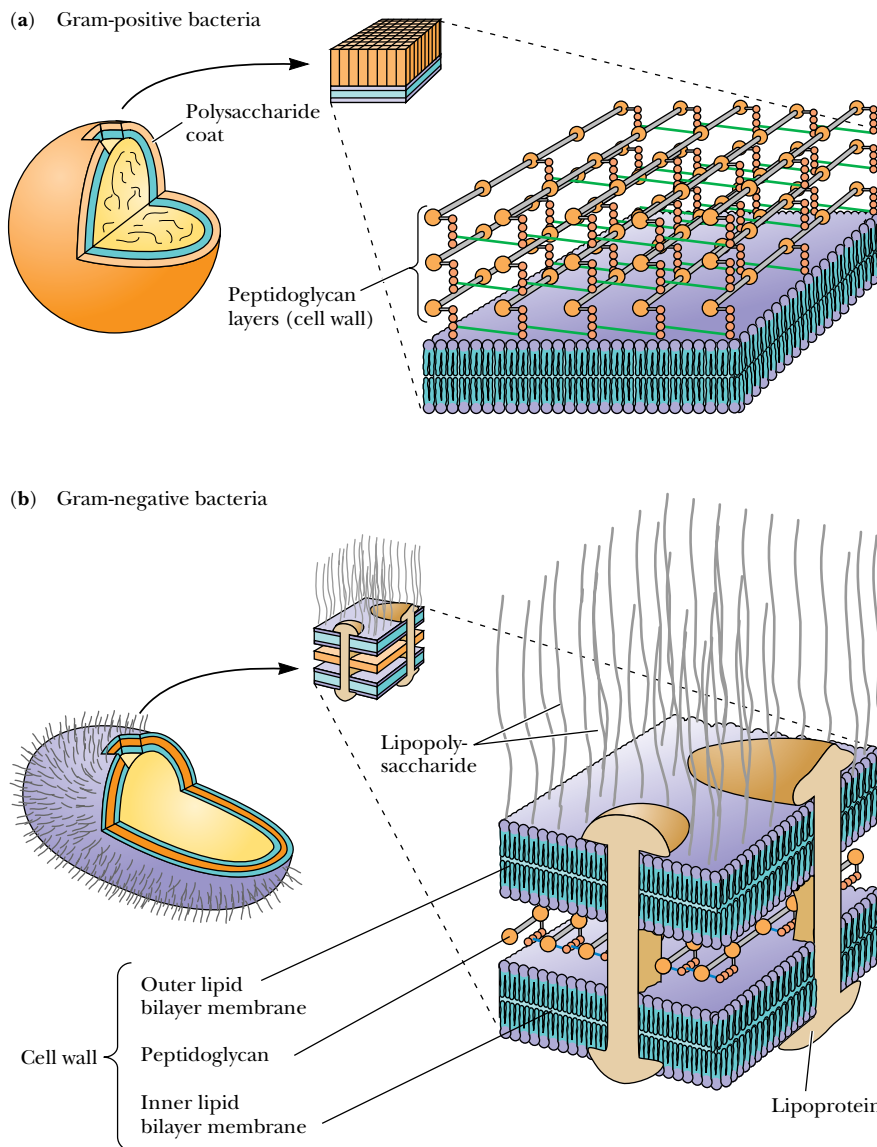
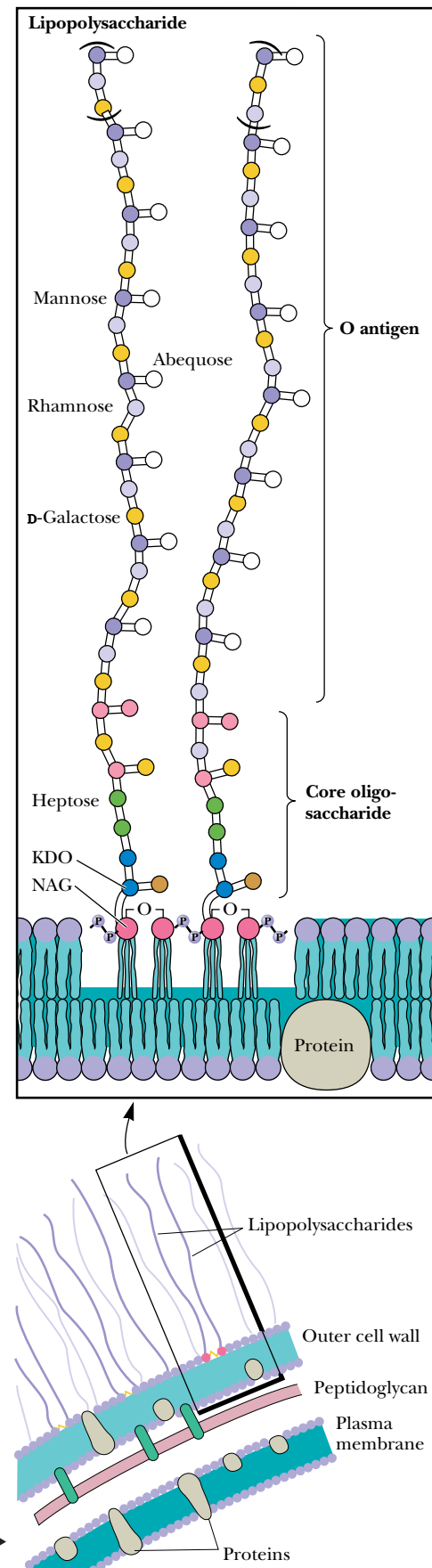


FIGURE 9.23 • The structures of the cell wall and membrane(s) in Gram-positive and Gram-negative bacteria. The Gram-positive cell wall is thicker than that in Gram-negative bacteria, compensating for the absence of a second (outer) bilayer membrane.

peptidoglycan. Diaminopimelic acid replaces one of the D-alanine residues in about 10% of the peptides of the peptidoglycan. On the other end of the hydrophobic protein, the N-terminal residue, a serine, makes a covalent bond to a lipid that is part of the outer membrane.

As shown in Figure 9.24, the outer membrane of Gram-negative bacteria is coated with a highly complex **lipopolysaccharide**, which consists of a lipid group (anchored in the outer membrane) joined to a polysaccharide made up of long chains with many different and characteristic repeating structures

FIGURE 9.24 • Lipopolysaccharide (LPS) coats the outer membrane of Gram-negative bacteria. The lipid portion of the LPS is embedded in the outer membrane and is linked to a complex polysaccharide.



(Figure 9.24). These many different unique units determine the antigenicity of the bacteria; that is, animal immune systems recognize them as foreign substances and raise antibodies against them. As a group, these **antigenic determinants** are called the **O antigens**, and there are thousands of different ones. The *Salmonella* bacteria alone have well over a thousand known O antigens that have been organized into 17 different groups. The great variation in these O antigen structures apparently plays a role in the recognition of one type of cell by another and in evasion of the host immune system.

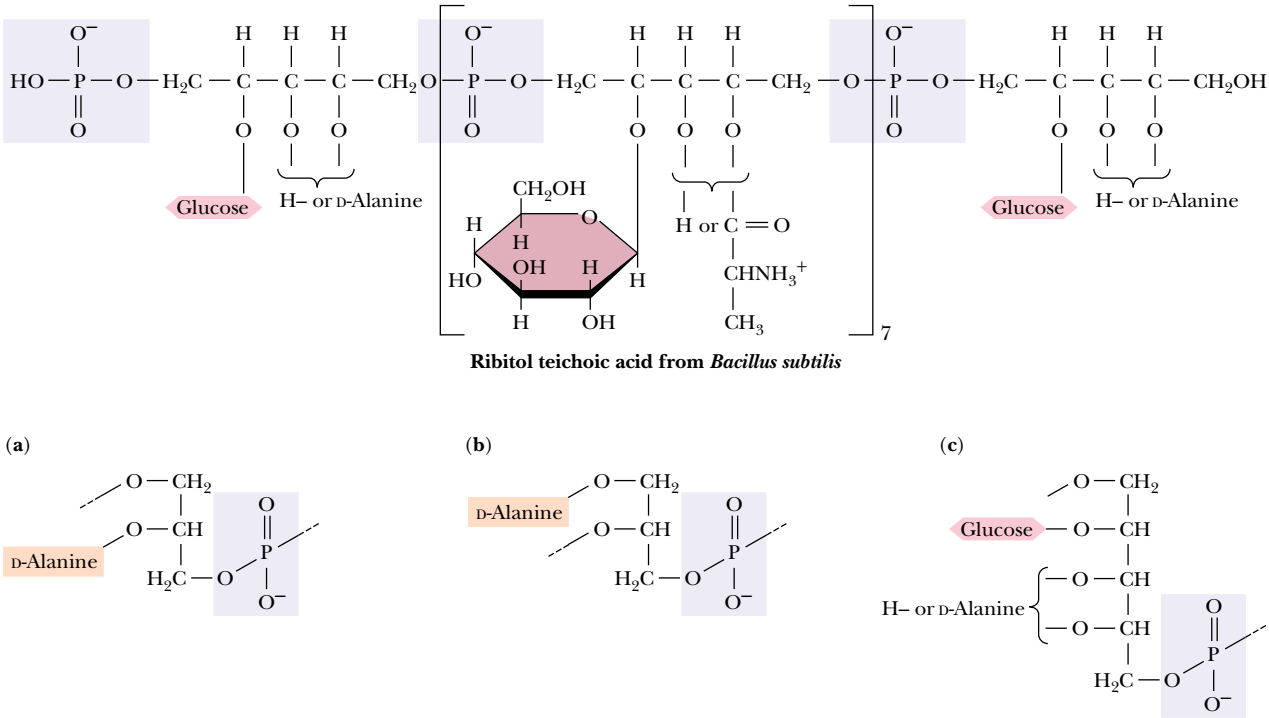
Cell Walls of Gram-Positive Bacteria

In Gram-positive bacteria, the cell exterior is less complex than for Gram-negative cells. Having no outer membrane, Gram-positive cells compensate with a thicker wall. Covalently attached to the peptidoglycan layer are **teichoic acids**, which often account for 50% of the dry weight of the cell wall (Figure 9.25). The teichoic acids are polymers of *ribitol phosphate* or *glycerol phosphate* linked by phosphodiester bonds. In these heteropolysaccharides, the free hydroxyl groups of the ribitol or glycerol are often substituted by glycosidically linked monosaccharides (often glucose or *N*-acetylglucosamine) or disaccharides. *D*-Alanine is sometimes found in ester linkage to the saccharides. Teichoic acids are not confined to the cell wall itself, and they may be present in the inner membranes of these bacteria. Many teichoic acids are antigenic, and they also serve as the receptors for bacteriophages in some cases.

Cell Surface Polysaccharides

Compared to bacterial cells, which are identical within a given cell type (except for O antigen variations), animal cells display a wondrous diversity of structure, constitution, and function. Although each animal cell contains, in its genetic material, the instructions to replicate the entire organism, each differentiated

FIGURE 9.25 • Teichoic acids are covalently linked to the peptidoglycan of Gram-positive bacteria. These polymers of (a, b) glycerol phosphate or (c) ribitol phosphate are linked by phosphodiester bonds.



animal cell carefully controls its composition and behavior within the organism. A great part of each cell’s uniqueness begins at the cell surface. This surface uniqueness is critical to each animal cell because cells spend their entire life span in intimate contact with other cells and must therefore communicate with one another. That cells are able to pass information among themselves is evidenced by numerous experiments. For example, heart *myocytes*, when grown in culture (in glass dishes) establish *synchrony* when they make contact, so that they “beat” or contract in unison. If they are removed from the culture and separated, they lose their synchronous behavior, but if allowed to reestablish cell-to-cell contact, they spontaneously restore their synchronous contractions.

HUMAN BIOCHEMISTRY

Selectins, Rolling Leukocytes, and the Inflammatory Response

Human bodies are constantly exposed to a plethora of bacteria, viruses, and other inflammatory substances. To combat these infectious and toxic agents, the body has developed a carefully regulated inflammatory response system. Part of that response is the orderly migration of leukocytes to sites of inflammation. Leukocytes literally roll along the vascular wall and into the tissue site of inflammation. This rolling movement is mediated by reversible adhesive interactions between the leukocytes and the vascular surface.

These interactions involve adhesion proteins called **selectins**, which are found both on the rolling leukocytes and on the endothelial cells of the vascular walls. Selectins have a characteristic domain structure, consisting of an N-terminal extracellular lectin domain, a single epidermal growth factor (EGF) domain, a series of two to nine short consensus repeat (SCR) domains, a single transmembrane segment, and a short cytoplasmic domain. Lectin domains, first characterized in plants, bind carbohydrates

with high affinity and specificity. Selectins of three types are known—**E-selectins**, **L-selectins**, and **P-selectins**. L-selectin is found on the surfaces of leukocytes, including neutrophils and lymphocytes, and binds to carbohydrate ligands on endothelial cells. The presence of L-selectin is a necessary component of leukocyte rolling. P-selectin and E-selectin are located on the vascular endothelium and bind with carbohydrate ligands on leukocytes. Typical neutrophil cells possess 10,000 to 20,000 P-selectin binding sites. Selectins are expressed on the surfaces of their respective cells by exposure to inflammatory signal molecules, such as histamine, hydrogen peroxide, and bacterial endotoxins. P-selectins, for example, are stored in intracellular granules and are transported to the cell membrane within seconds to minutes of exposure to a triggering agent.

Substantial evidence supports the hypothesis that selectin–carbohydrate ligand interactions modulate the rolling of leukocytes along the vascular wall. Studies with L-selectin–deficient and P-selectin–deficient leukocytes show that L-selectins mediate weaker adherence of the leukocyte to the vascular wall and promote faster rolling along the wall. P-selectins conversely promote stronger adherence and slower rolling. Thus, leukocyte rolling velocity in the inflammatory response could be modulated by variable exposure of P-selectins and L-selectins at the surfaces of endothelial cells and leukocytes, respectively.

A diagram showing the interactions of selectins with their receptors.

The selectin family of adhesion proteins.

Kidney cells grown in culture with liver cells seek out and make contact with other kidney cells and avoid contact with liver cells. Cells grown in culture grow freely until they make contact with one another, at which point growth stops, a phenomenon well known as **contact inhibition**. One important characteristic of cancerous cells is the loss of contact inhibition.

As these and many other related phenomena show, it is clear that molecular structures on one cell are recognizing and responding to molecules on the adjacent cell or to molecules in the **extracellular matrix**, the complex “soup” of connective proteins and other molecules that exists outside of and among cells. Many of these interactions involve *glycoproteins* on the cell surface and *proteoglycans* in the extracellular matrix. The “information” held in these special carbohydrate-containing molecules is not encoded directly in the genes (as with proteins), but is determined instead by expression of the appropriate enzymes that assemble carbohydrate units in a characteristic way on these molecules. Also, by virtue of the several hydroxyl linkages that can be formed with each carbohydrate monomer, these structures can be more information-rich than proteins and nucleic acids, which can form only linear polymers. A few of these glycoproteins and their unique properties are described in the following sections.

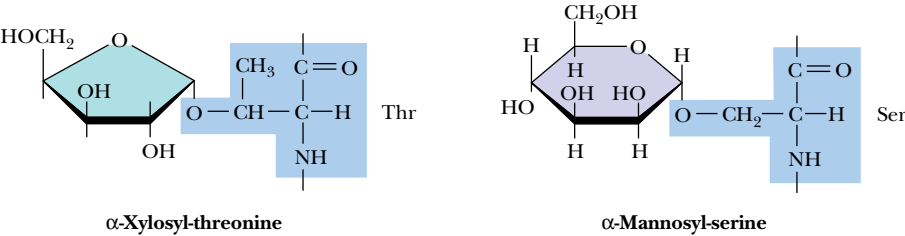
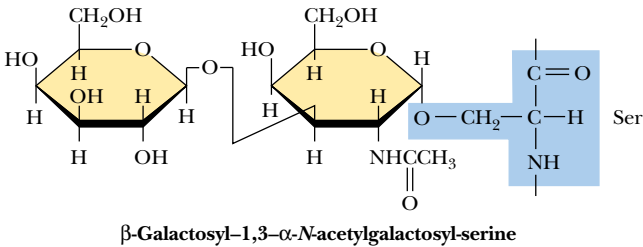
9.4 • Glycoproteins

Many proteins found in nature are **glycoproteins** because they contain covalently linked oligo- and polysaccharide groups. The list of known glycoproteins includes structural proteins, enzymes, membrane receptors, transport proteins, and immunoglobulins, among others. In most cases, the precise function of the bound carbohydrate moiety is not understood.

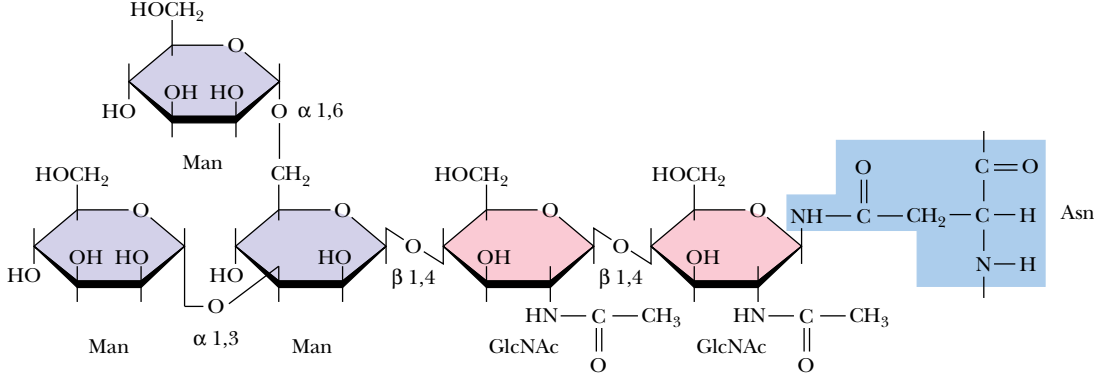
Carbohydrate groups may be linked to polypeptide chains via the hydroxyl groups of serine, threonine, or hydroxylysine residues (in **O-linked saccharides**) (Figure 9.26a) or via the amide nitrogen of an asparagine residue (in **N-linked saccharides**) (Figure 9.26b). The carbohydrate residue linked to the protein in O-linked saccharides is usually an *N*-acetylgalactosamine, but mannose, galactose, and xylose residues linked to protein hydroxyls are also found (Figure 9.26a). Oligosaccharides O-linked to glycophorin (see Figure 9.14) involve *N*-acetylgalactosamine linkages and are rich in sialic acid residues (Figure 9.14). N-linked saccharides always have a unique core structure composed of two *N*-acetylglucosamine residues linked to a branched mannose triad (Figure 9.26b, c). Many other sugar units may be linked to each of the mannose residues of this branched core.

O-Linked saccharides are often found in cell surface glycoproteins and in **mucins**, the large glycoproteins that coat and protect mucous membranes in the respiratory and gastrointestinal tracts in the body. Certain viral glycoproteins also contain O-linked sugars. O-Linked saccharides in glycoproteins are often found clustered in richly glycosylated domains of the polypeptide chain. Physical studies on mucins show that they adopt rigid, extended structures so that an individual mucin molecule ($M_r = 10^7$) may extend over a distance of 150 to 200 nm in solution. Inherent steric interactions between the sugar residues and the protein residues in these cluster regions cause the peptide core to fold into an extended and relatively rigid conformation. This interesting effect may be related to the function of O-linked saccharides in glycoproteins. It allows aggregates of mucin molecules to form extensive, intertwined networks, even at low concentrations. These viscous networks protect the mucosal surface of the respiratory and gastrointestinal tracts from harmful environmental agents.

(a) O-linked saccharides



(b) Core oligosaccharides in N-linked glycoproteins



(c) N-linked glycoproteins

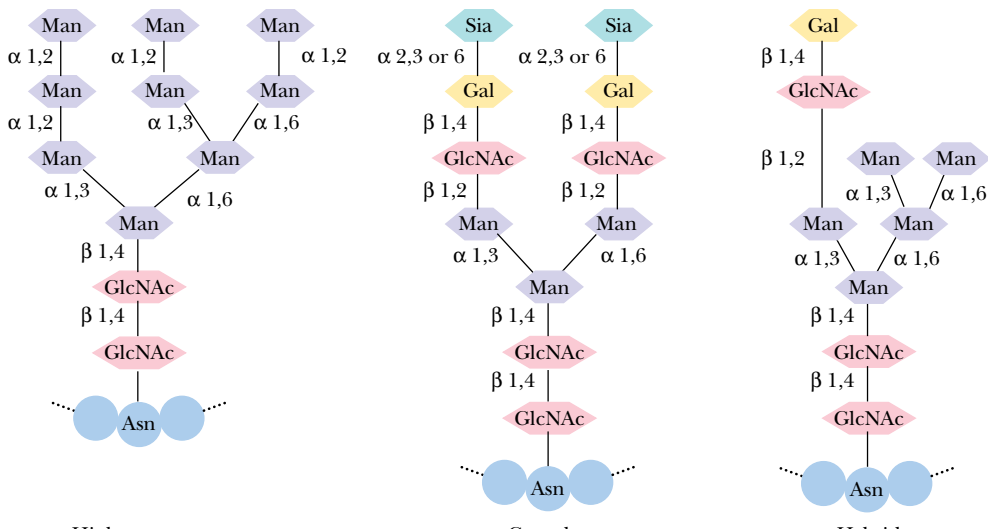


FIGURE 9.26 • The carbohydrate moieties of glycoproteins may be linked to the protein via (a) serine or threonine residues (in the O-linked saccharides) or (b) asparagine residues (in the N-linked saccharides). (c) N-Linked glycoproteins are of three types: high mannose, complex, and hybrid, the latter of which combines structures found in the high mannose and complex saccharides.

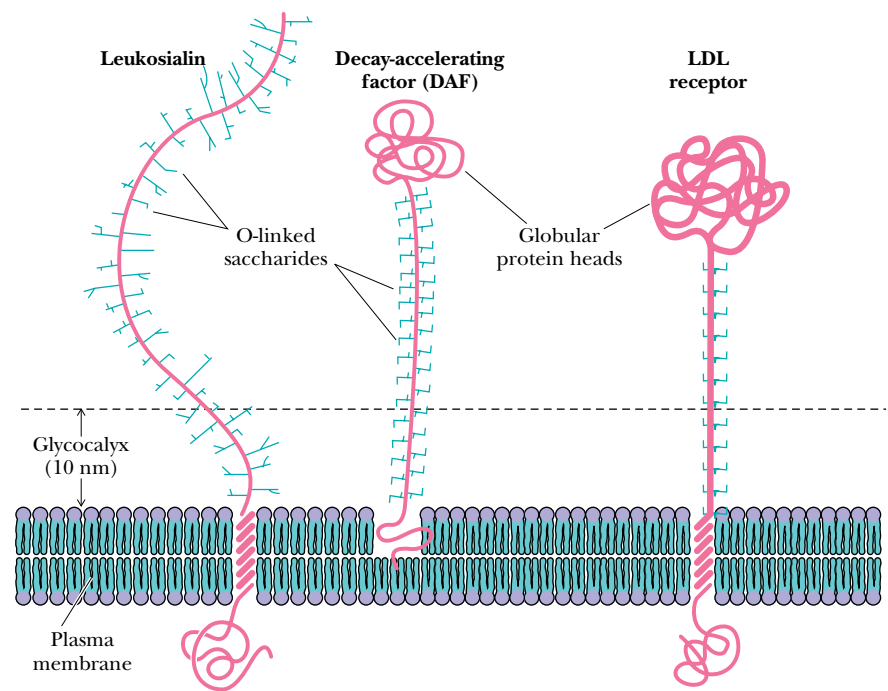


FIGURE 9.27 • The O-linked saccharides of glycoproteins appear in many cases to adopt extended conformations that serve to extend the functional domains of these proteins above the membrane surface. (Adapted from Jentoft, N., 1990, Trends in Biochemical Sciences 15:291–294.)

There appear to be two structural motifs for membrane glycoproteins containing O-linked saccharides. Certain glycoproteins, such as **leukosialin**, are O-glycosylated throughout much or most of their extracellular domain (Figure 9.27). Leukosialin, like mucin, adopts a highly extended conformation, allowing it to project great distances above the membrane surface, perhaps protecting the cell from unwanted interactions with macromolecules or other cells. The second structural motif is exemplified by the **low density lipoprotein (LDL) receptor** and by **decay accelerating factor (DAF)**. These proteins contain a highly O-glycosylated stem region that separates the transmembrane domain from the globular, functional extracellular domain. The O-glycosylated stem serves to raise the functional domain of the protein far enough above the membrane surface to make it accessible to the extracellular macromolecules with which it interacts.

Antifreeze Glycoproteins

A unique family of O-linked glycoproteins permits fish to live in the icy seawater of the Arctic and Antarctic regions where water temperature may reach as low as -1.9°C . **Antifreeze glycoproteins (AFGPs)** are found in the blood of nearly all Antarctic fish and at least five Arctic fish. These glycoproteins have the peptide structure



where n can be 4, 5, 6, 12, 17, 28, 35, 45, or 50. Each of the threonine residues is glycosylated with the disaccharide β -galactosyl-(1 $\rightarrow$ 3)- α -N-acetylgalactos-

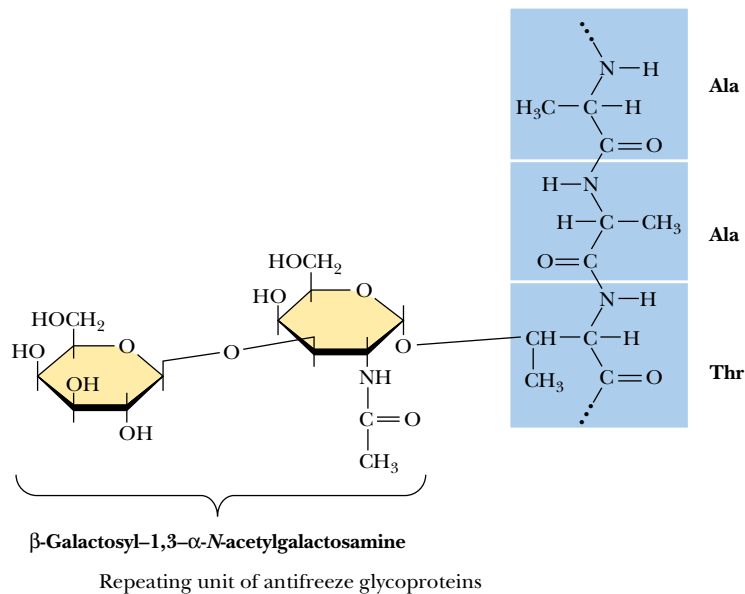


FIGURE 9.28 • The structure of the repeating unit of antifreeze glycoproteins, a disaccharide consisting of β -galactosyl-(1 $\rightarrow$ 3)- α -N-acetylgalactosamine in glycosidic linkage to a threonine residue.

amine (Figure 9.28). This glycoprotein adopts a **flexible rod** conformation with regions of threefold left-handed helix. The evidence suggests that antifreeze glycoproteins may inhibit the formation of ice in the fish by binding specifically to the growth sites of ice crystals, inhibiting further growth of the crystals.

N-Linked Oligosaccharides

N-Linked oligosaccharides are found in many different proteins, including immunoglobulins G and M, ribonuclease B, ovalbumin, and peptide hormones (Figure 9.29). Many different functions are known or suspected for *N*-glycosylation of proteins. Glycosylation can affect the physical and chemical properties of proteins, altering solubility, mass, and electrical charge. Carbohydrate moieties have been shown to stabilize protein conformations and protect proteins against proteolysis. Eukaryotic organisms use posttranslational additions of N-linked oligosaccharides to direct selected proteins to various intracellular organelles.

Oligosaccharide Cleavage as a Timing Device for Protein Degradation

The slow cleavage of monosaccharide residues from N-linked glycoproteins circulating in the blood targets these proteins for degradation by the organism. The liver contains specific receptor proteins that recognize and bind glycoproteins that are ready to be degraded and recycled. Newly synthesized serum glycoproteins contain N-linked **triantennary** (three-chain) oligosaccharides having structures similar to those in Figure 9.30, in which sialic acid residues cap galactose residues. As these glycoproteins circulate, enzymes on the blood vessel walls cleave off the sialic acid groups, exposing the galactose residues. In

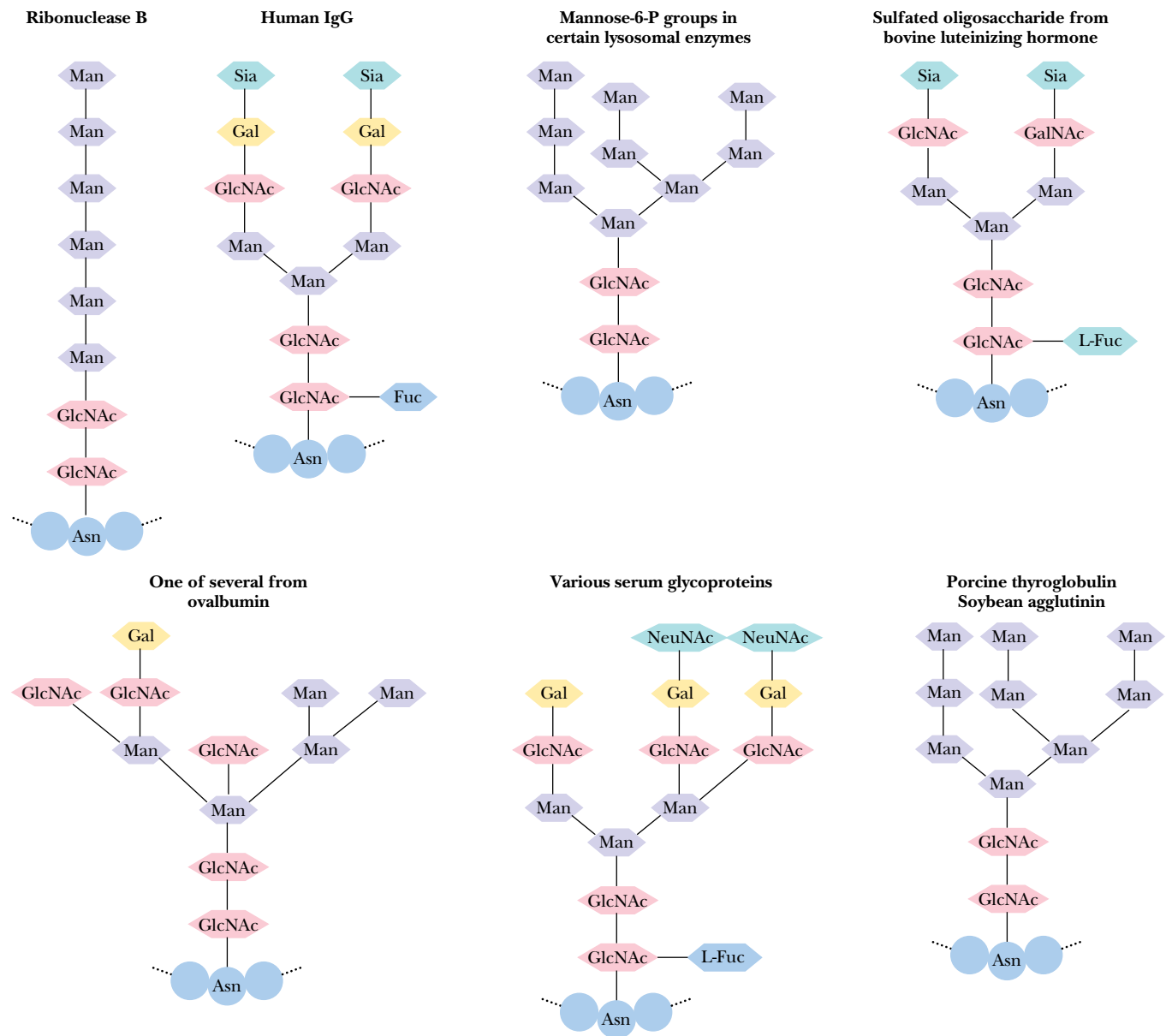


FIGURE 9.29 • Some of the oligosaccharides found in N-linked glycoproteins.

the liver, the **asialoglycoprotein receptor** binds the exposed galactose residues of these glycoproteins with very high affinity ($K_D = 10^{-9}$ to 10^{-8} M). The complex of receptor and glycoprotein is then taken into the cell by **endocytosis**, and the glycoprotein is degraded in cellular lysosomes. Highest affinity binding of glycoprotein to the asialoglycoprotein receptor requires three free galactose residues. Oligosaccharides with only one or two exposed galactose residues bind less tightly. This is an elegant way for the body to keep track of how long glycoproteins have been in circulation. Over a period of time, anywhere from a few hours to weeks, the sialic acid groups are cleaved one by one. The longer the glycoprotein circulates and the more sialic acid residues are removed, the more galactose residues become exposed so that the glycoprotein is eventually bound to the liver receptor.

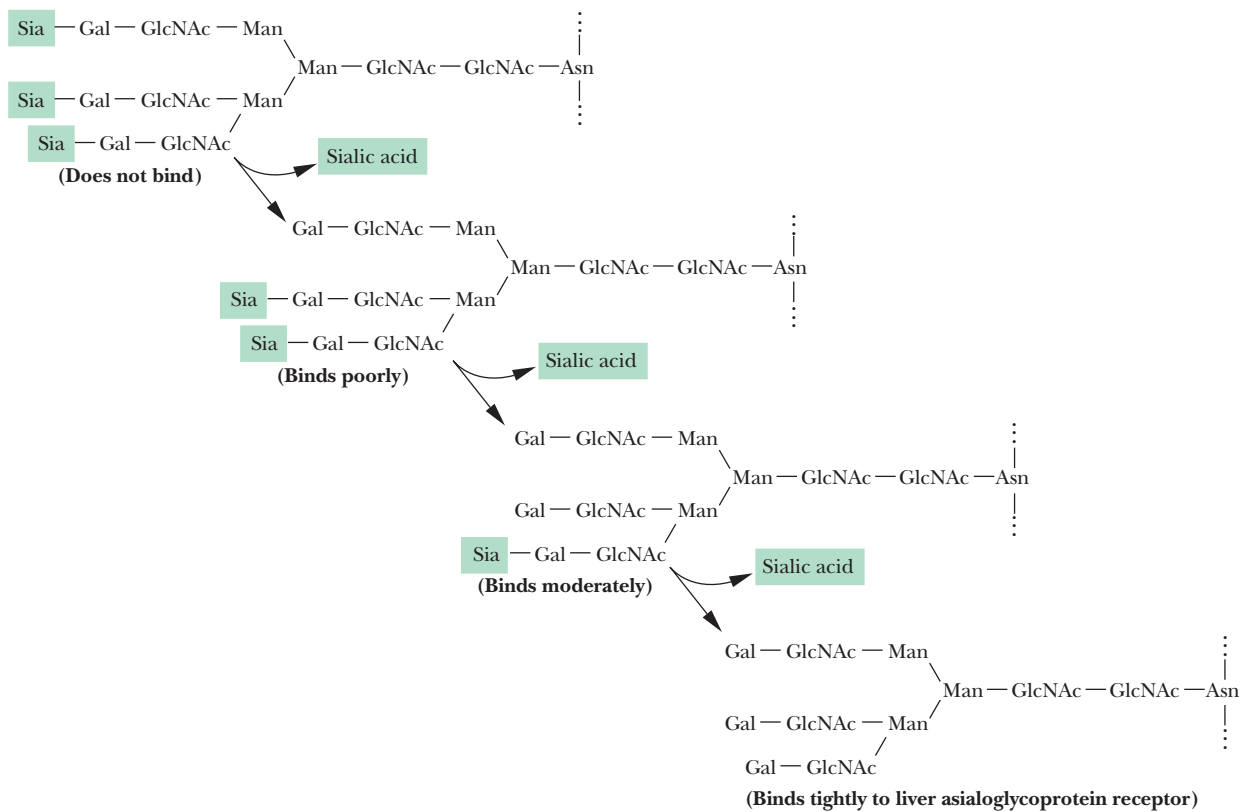


FIGURE 9.30 • Progressive cleavage of sialic acid residues exposes galactose residues. Binding to the asialoglycoprotein receptor in the liver becomes progressively more likely as more Gal residues are exposed.

9.5 • Proteoglycans

Proteoglycans are a family of glycoproteins whose carbohydrate moieties are predominantly **glycosaminoglycans**. The structures of only a few proteoglycans are known, and even these few display considerable diversity (Figure 9.31). They range in size from **serglycin**, having 104 amino acid residues (10.2 kD) to **versican**, having 2409 residues (265 kD). Each of these proteoglycans contains one or two types of covalently linked glycosaminoglycans (Table 9.2). In the known proteoglycans, the glycosaminoglycan units are O-linked to serine residues of Ser-Gly dipeptide sequences. Serglycin is named for a unique central domain of 49 amino acids composed of alternating serine and glycine residues. The **cartilage matrix proteoglycan** contains 117 Ser-Gly pairs to which chondroitin sulfates attach. **Decorin**, a small proteoglycan secreted by fibroblasts and found in the extracellular matrix of connective tissues, contains only three Ser-Gly pairs, only one of which is normally glycosylated. In addition to glycosaminoglycan units, proteoglycans may also contain other N-linked and O-linked oligosaccharide groups.

Functions of Proteoglycans

Proteoglycans may be *soluble* and located in the extracellular matrix, as is the case for serglycin, versican, and the cartilage matrix proteoglycan, or they may be *integral transmembrane proteins*, such as **syndecan**. Both types of proteoglycan

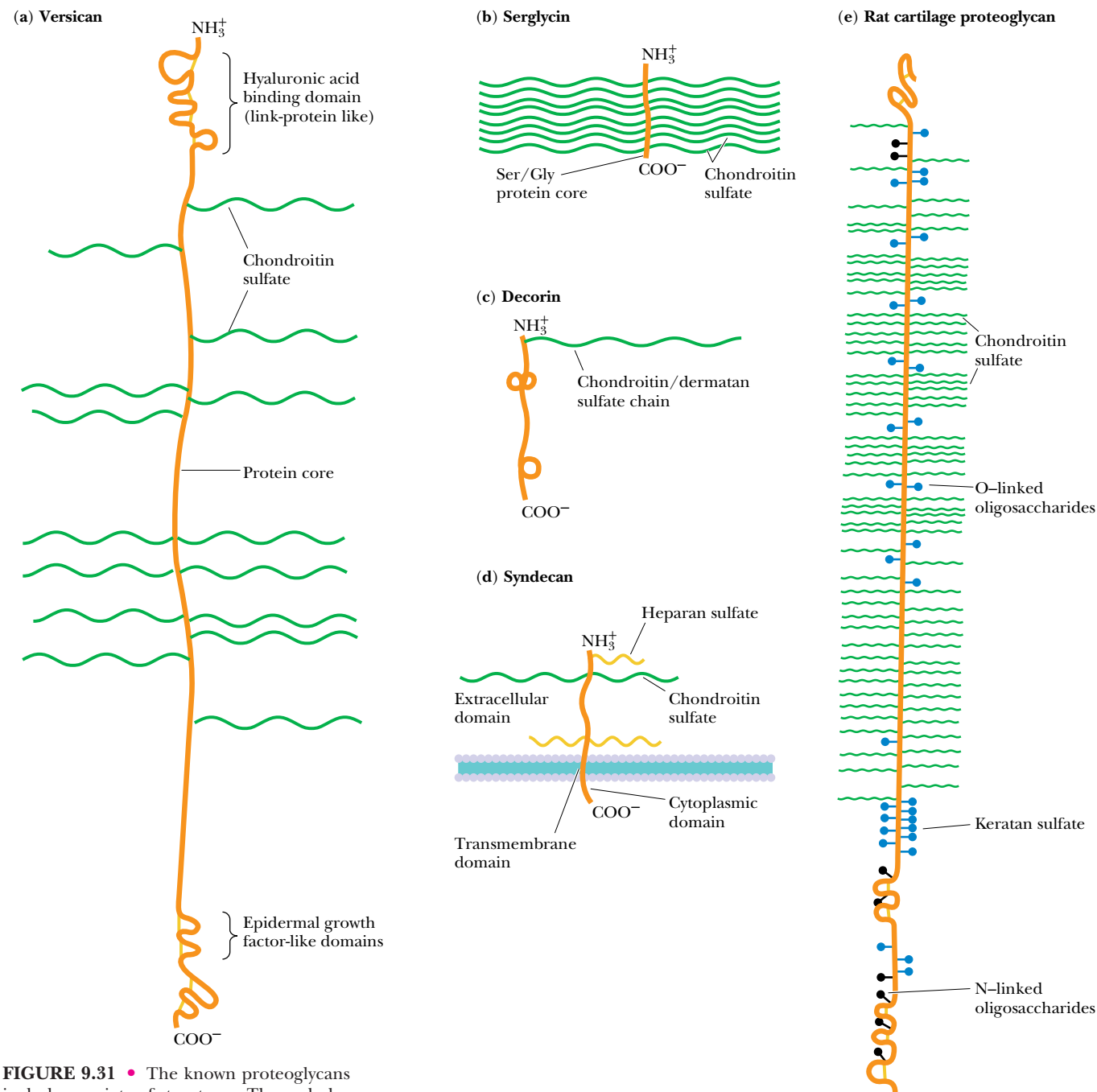


FIGURE 9.31 • The known proteoglycans include a variety of structures. The carbohydrate groups of proteoglycans are predominantly glycosaminoglycans O-linked to serine residues. Proteoglycans include both soluble proteins and integral transmembrane proteins.

appear to function by interacting with a variety of other molecules through their glycosaminoglycan components and through specific receptor domains in the polypeptide itself. For example, syndecan (from the Greek *syndein* meaning “to bind together”) is a transmembrane proteoglycan that associates intracellularly with the actin cytoskeleton (Chapter 17). Outside the cell, it interacts with **fibronectin**, an extracellular protein that binds to several cell surface proteins and to components of the extracellular matrix. The ability of syndecan to participate in multiple interactions with these target molecules allows them to act as a sort of “glue” in the extracellular space, linking components of the extracellular matrix, facilitating the binding of cells to the matrix, and mediating the binding of growth factors and other soluble molecules to the matrix and to cell surfaces (Figure 9.32).

Table 9.2
Some Proteoglycans of Known Sequence

Proteoglycan	Glycosaminoglycan	Protein M _r	Number of Amino Acid Residues
<i>Secreted or extracellular matrix proteoglycans</i>			
Large aggregating cartilage proteoglycans	CS/KS*	220,952	2124
Versican	CS/DS	265,048	2409
Decorin	CS/DS	38,000	329
<i>Intracellular granule proteoglycan</i>			
Serglycin (PG19)	CS/DS	10,190	104
<i>Membrane-intercalated proteoglycans</i>			
Syndecan	HS/CS	38,868	311

*CS, chondroitin sulfate; DS, dermatan sulfate; HS, heparan sulfate (an analog of heparin); KS, keratan sulfate. These glycosaminoglycans are polymers consisting of the repeating disaccharides: glucuronic acid *N*-acetylgalactosamine (CS), iduronic acid *N*-acetylgalactosamine (DS), iduronic acid *N*-acetylglucosamine (HS and heparin), and galactose *N*-acetylglucosamine (KS). DS, HS, and heparin also contain some disaccharide units in which the uronic acid is glucuronic acid instead of iduronic acid. These glycosaminoglycans and CS are generally bound to the hydroxyl group of a serine residue to give the sequence (disaccharide) *n*GlcUA-Gal-Gal-Xyl-*O* Ser. Keratan sulfate has a different linkage region and can be either O- or N-linked. The sugars in the repeating disaccharide unit are sulfated to various degrees. By comparison, hyaluronic acid is a polymer of glucuronic acid and glucosamine that is not sulfated and does not attach covalently to a protein core.

Adapted from Ruoslahti, E., 1989. *Journal of Biological Chemistry* **264**:13369–13372.

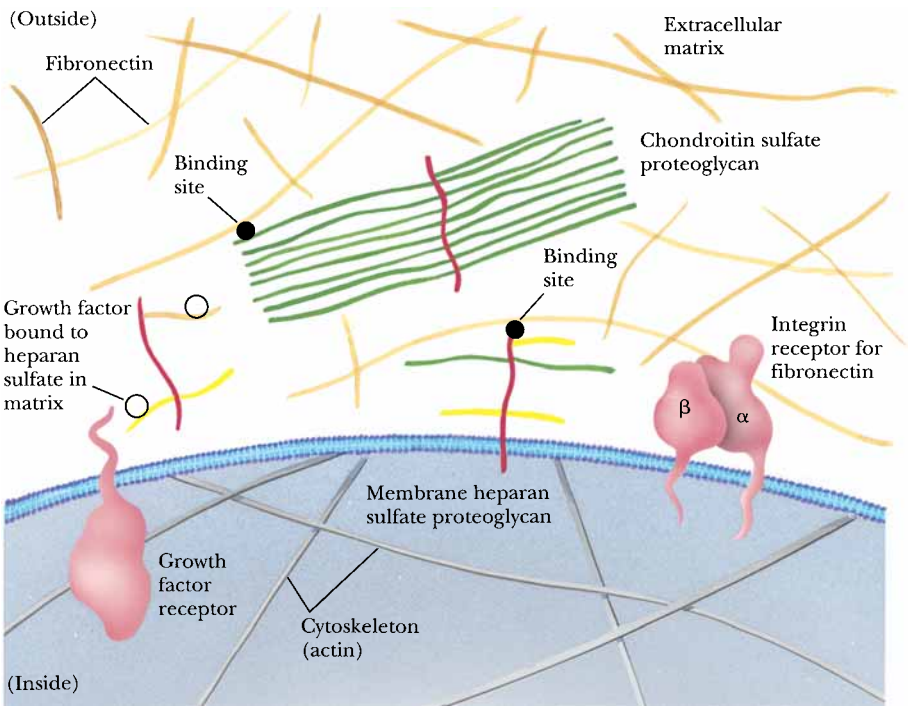
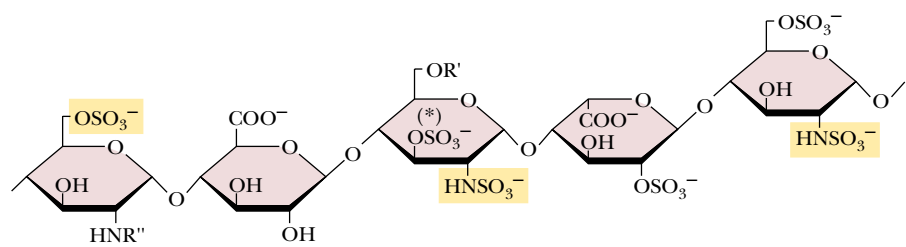


FIGURE 9.32 • Proteoglycans serve a variety of functions on the cytoplasmic and extracellular surfaces of the plasma membrane. Many of these functions appear to involve the binding of specific proteins to the glycosaminoglycan groups.

FIGURE 9.33 • A portion of the structure of heparin, a carbohydrate having anticoagulant properties. It is used by blood banks to prevent the clotting of blood during donation and storage and also by physicians to prevent the formation of life-threatening blood clots in patients recovering from serious injury or surgery. This sulfated pentasaccharide sequence in heparin binds with high affinity to antithrombin III, accounting for this anticoagulant activity. The 3-O-sulfate marked by an asterisk is essential for high-affinity binding of heparin to antithrombin III.



Many of the functions of proteoglycans involve the binding of specific proteins to the glycosaminoglycan groups of the proteoglycan. The glycosaminoglycan binding sites on these specific proteins contain multiple basic amino acid residues. The amino acid sequences BBXB and BBBXXB (where B is a basic amino acid and X is any amino acid) recur repeatedly in these binding domains. Basic amino acids such as lysine and arginine provide charge neutralization for the negative charges of glycosaminoglycan residues, and in many cases, the binding of extracellular matrix proteins to glycosaminoglycans is primarily charge-dependent. For example, more highly sulfated glycosaminoglycans bind more tightly to fibronectin. Certain protein–glycosaminoglycan interactions, however, require a specific carbohydrate sequence. A particular pentasaccharide sequence in heparin, for example, binds tightly to antithrombin III (Figure 9.33), accounting for the anticoagulant properties of heparin. Other glycosaminoglycans interact much more weakly.

Proteoglycans May Modulate Cell Growth Processes

Several lines of evidence raise the possibility of modulation or regulation of cell growth processes by proteoglycans. First, heparin and heparan sulfate are known to inhibit cell proliferation in a process involving internalization of the glycosaminoglycan moiety and its migration to the cell nucleus. Second, **fibroblast growth factor** binds tightly to heparin and other glycosaminoglycans, and the heparin–growth factor complex protects the growth factor from degradative enzymes, thus enhancing its activity. There is evidence that binding of fibroblast growth factors by proteoglycans and glycosaminoglycans in the extracellular matrix creates a reservoir of growth factors for cells to use. Third, **transforming growth factor β** has been shown to stimulate the synthesis and secretion of proteoglycans in certain cells. Fourth, several proteoglycan core proteins, including versican and **lymphocyte homing receptor**, have domains similar in sequence to **epidermal growth factor** and **complement regulatory factor**. These growth factor domains may interact specifically with growth factor receptors in the cell membrane in processes that are not yet understood.

Proteoglycans Make Cartilage Flexible and Resilient

Cartilage matrix proteoglycan is responsible for the flexibility and resilience of cartilage tissue in the body. In cartilage, long filaments of hyaluronic acid are studded or coated with proteoglycan molecules, as shown in Figure 9.34. The hyaluronate chains can be as long as 4 μm and can coordinate 100 or more proteoglycan units. Cartilage proteoglycan possesses a **hyaluronic acid binding domain** on the NH_2 -terminal portion of the polypeptide, which binds to hyaluronate with the assistance of a **link protein**. The proteoglycan–hyaluronate aggregates can have molecular weights of 2 million or more.

The proteoglycan–hyaluronate aggregates are highly hydrated by virtue of strong interactions between water molecules and the polyanionic complex.

When cartilage is compressed (such as when joints absorb the impact of walking or running), water is briefly squeezed out of the cartilage tissue and then reabsorbed when the stress is diminished. This reversible hydration gives cartilage its flexible, shock-absorbing qualities and cushions the joints during physical activities that might otherwise injure the involved tissues.

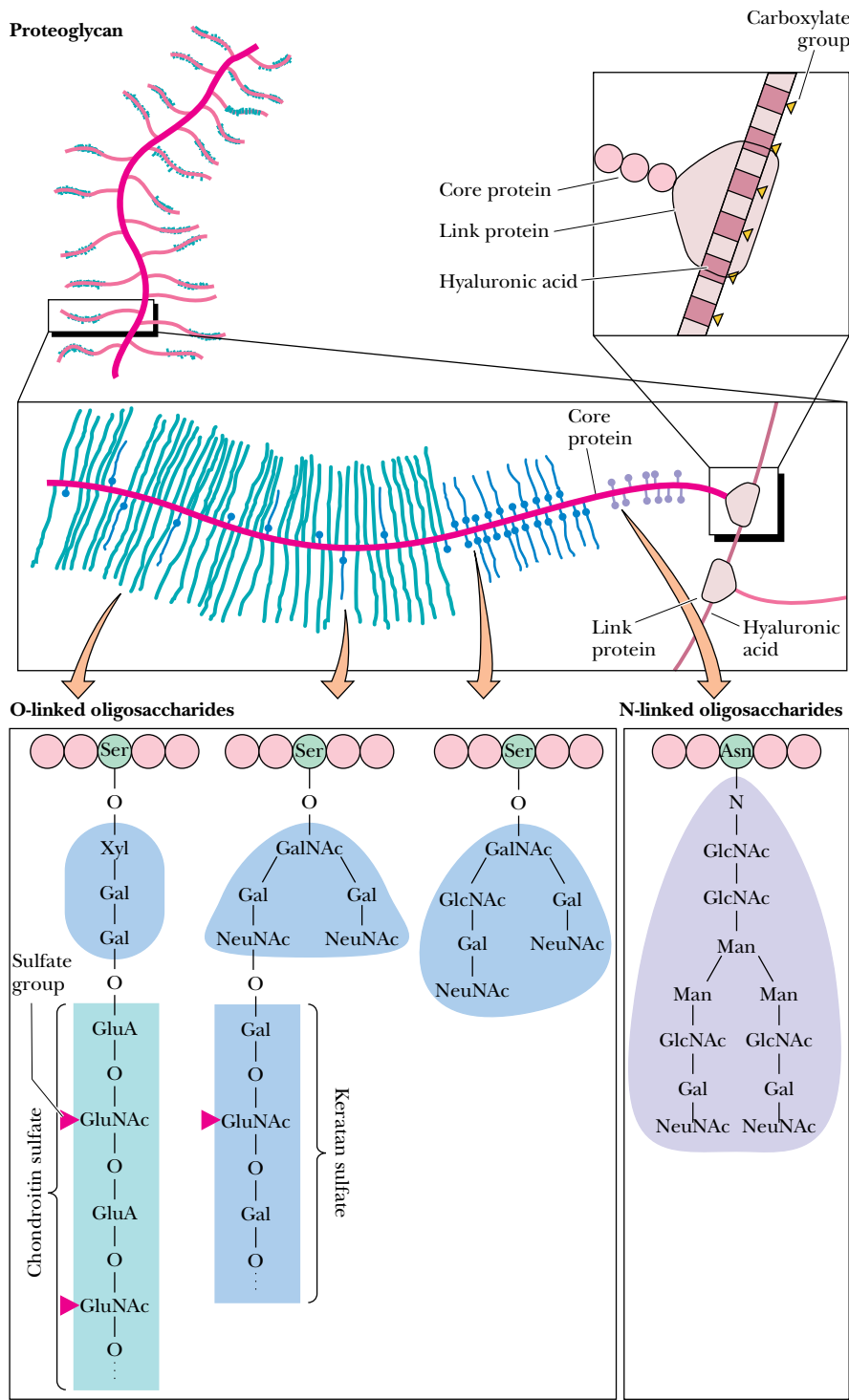


FIGURE 9.34 • Hyaluronate (see Figure 7.33) forms the backbone of proteoglycan structures, such as those found in cartilage. The proteoglycan subunits consist of a core protein containing numerous O-linked and N-linked glycosaminoglycans. In cartilage, these highly hydrated proteoglycan structures are enmeshed in a network of collagen fibers. Release (and subsequent reabsorption) of water by these structures during compression accounts for the shock-absorbing qualities of cartilaginous tissue.

PROBLEMS

1. In Problem 1(b) in chapter 8 (page 257) you were asked to draw all the phosphatidylserine isomers that can be formed from palmitic and linolenic acids. Which of the PS isomers are not likely to be found in biological membranes?
2. The purple patches of the *Halobacterium halobium* membrane, which contain the protein bacteriorhodopsin, are approximately 75% protein and 25% lipid. If the protein molecular weight is 26,000 and an average phospholipid has a molecular weight of 800, calculate the phospholipid to protein mole ratio.
3. Sucrose gradients for separation of membrane proteins must be able to separate proteins and protein–lipid complexes having a wide range of densities, typically 1.00 to 1.35 g/mL.
- a. Consult reference books (such as the *CRC Handbook of Biochemistry*) and plot the density of sucrose solutions versus percent sucrose by weight (g sucrose per 100 g solution), and versus percent by volume (g sucrose per 100 mL solution). Why is one plot linear and the other plot curved?
- b. What would be a suitable range of sucrose concentrations for separation of three membrane-derived protein–lipid complexes with densities of 1.03, 1.07, and 1.08 g/mL?
4. Phospholipid lateral motion in membranes is characterized by

- a diffusion coefficient of about 1×10^{-8} cm²/sec. The distance traveled in two dimensions (across the membrane) in a given time is $r = (4Dt)^{1/2}$, where r is the distance traveled in centimeters, D is the diffusion coefficient, and t is the time during which diffusion occurs. Calculate the distance traveled by a phospholipid across a bilayer in 10 msec (milliseconds).
5. Protein lateral motion is much slower than that of lipids because proteins are larger than lipids. Also, some membrane proteins can diffuse freely through the membrane, whereas others are bound or anchored to other protein structures in the membrane. The diffusion constant for the membrane protein fibronectin is approximately 0.7×10^{-12} cm²/sec, whereas that for rhodopsin is about 3×10^{-9} cm²/sec.
- a. Calculate the distance traversed by each of these proteins in 10 msec.
- b. What could you surmise about the interactions of these proteins with other membrane components?
6. Discuss the effects on the lipid phase transition of pure dimyristoyl phosphatidylcholine vesicles of added (a) divalent cations, (b) cholesterol, (c) distearoyl phosphatidylserine, (d) dioleoyl phosphatidylcholine, and (e) integral membrane proteins.

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It takes a membrane to make sense out of disorder in biology. You have to be able to catch energy and hold it, storing precisely the needed amount and releasing it in measured shares. A cell does this, and so do the organelles inside. . . . To stay alive, you have to be able to hold out against equilibrium, maintain imbalance, bank against entropy, and you can only transact this business with membranes in our kind of world.

LEWIS THOMAS, “The World’s Biggest Membrane,” *The Lives of a Cell* (1974)

OUTLINE

- 10.1 • Passive Diffusion
- 10.2 • Facilitated Diffusion
- 10.3 • Active Transport Systems
- 10.4 • Transport Processes Driven by ATP
- 10.5 • Transport Processes Driven by Light
- 10.6 • Transport Processes Driven by Ion Gradients
- 10.7 • Group Translocation
- 10.8 • Specialized Membrane Pores
- 10.9 • Ionophore Antibiotics

Chapter 10

Membrane Transport



“Drawbridge at Arles with a Group of Washerwomen” (1888) by Vincent van Gogh (Rijksmuseum Kroller-Muller; photo by Erich Lessing/Art Resource)

Transport processes are vitally important to all life forms because all cells must exchange materials with their environment. Cells must obviously have ways to bring nutrient molecules into the cell and ways to send waste products and toxic substances out. Also, inorganic electrolytes must be able to pass in and out of cells and across organelle membranes. All cells maintain **concentration gradients** of various metabolites across their plasma membranes and also across the membranes of intracellular organelles. By their very nature, cells maintain a very large amount of potential energy in the form of such concentration gradients. Sodium and potassium ion gradients across the plasma membrane mediate the transmission of nerve impulses and the normal functions of the brain, heart, kidneys, and liver, among other organs. Storage and release of calcium from cellular compartments controls muscle contraction, and also the response of many cells to hormonal signals. High acid concentrations in the stomach are required for the digestion of food. Extremely high hydrogen ion gradients

are maintained across the plasma membranes of the mucosal cells lining the stomach in order to maintain high acid levels in the stomach yet protect the cells that constitute the stomach walls from the deleterious effects of such acid.

In this chapter, we shall consider the molecules and mechanisms that mediate these transport activities. In nearly every case, the molecule or ion transported is water-soluble, yet moves across the hydrophobic, impermeable lipid membrane at a rate high enough to serve the metabolic and physiologic needs of the cell. This perplexing problem is solved in each case by a specific transport protein. The transported species either diffuses through a channel-forming protein or is carried by a carrier protein. Transport proteins are all classed as **integral membrane proteins** (Chapter 9), ranging in size from small peptides to large, multisubunit protein complexes.

Some transport proteins merely provide a path for the transported species, whereas others couple an enzymatic reaction with the transport event. In all cases, transport behavior depends on the interactions of the transport protein not only with solvent water but with the lipid milieu of the membrane as well. The dynamic and asymmetric nature of the membrane and its components (Chapter 9) plays an important part in the function of these transport systems.

From a thermodynamic and kinetic perspective, there are only three types of membrane transport processes: *passive diffusion*, *facilitated diffusion*, and *active transport*. To be thoroughly appreciated, membrane transport phenomena must be considered in terms of thermodynamics. Some of the important kinetic considerations also will be discussed.

10.1 • Passive Diffusion

Passive diffusion is the simplest transport process. In passive diffusion, the transported species moves across the membrane in the thermodynamically favored direction without the help of any specific transport system/molecule. For an uncharged molecule, passive diffusion is an entropic process, in which movement of molecules across the membrane proceeds until the concentration of the substance on both sides of the membrane is the same. For an uncharged molecule, the free energy difference between side 1 and side 2 of a membrane (Figure 10.1) is given by

$$\Delta G = G_2 - G_1 = RT \ln \frac{[C_2]}{[C_1]} \quad (10.1)$$

The difference in concentrations, $[C_2] - [C_1]$, is termed the **concentration gradient**, and ΔG here is the **chemical potential difference**.

Passive Diffusion of a Charged Species

For a charged species, the situation is slightly more complicated. In this case, the movement of a molecule across a membrane depends on its **electrochemical potential**. This is given by

$$\Delta G = G_2 - G_1 = RT \ln \frac{[C_2]}{[C_1]} + Z\mathcal{F}\Delta\psi \quad (10.2)$$

where Z is the **charge** on the transported species, $\mathcal{F}$ is **Faraday's constant** (the charge on 1 mole of electrons = 96,485 coulombs/mol = 96,485 joules/volt • mol, because 1 volt = 1 joule/coulomb), and $\Delta\psi$ is the electric potential difference (that is, voltage difference) across the membrane. The second term in the expression thus accounts for the movement of a charge across a potential

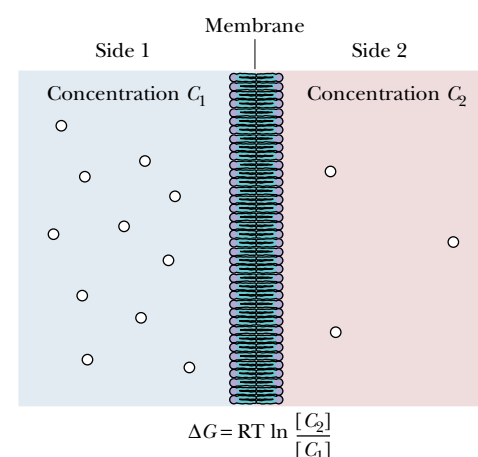


FIGURE 10.1 • Passive diffusion of an uncharged species across a membrane depends only on the concentrations (C_1 and C_2) on the two sides of the membrane.

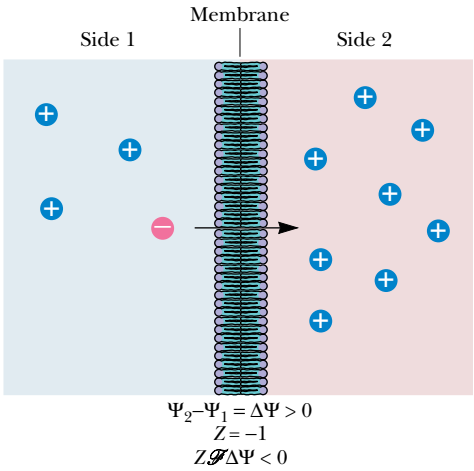


FIGURE 10.2 • The passive diffusion of a charged species across a membrane depends upon the concentration and also on the charge of the particle, Z , and the electrical potential difference across the membrane, $\Delta\psi$.

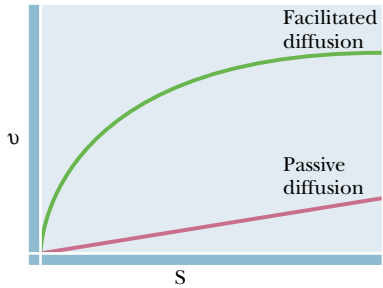


FIGURE 10.3 • Passive diffusion and facilitated diffusion may be distinguished graphically. The plots for facilitated diffusion are similar to plots of enzyme-catalyzed processes (Chapter 14) and they display saturation behavior.

difference. Note that the effect of this second term on ΔG depends on the magnitude and the sign of both Z and $\Delta\psi$. For example, as shown in Figure 10.2, if side 2 has a higher potential than side 1 (so that $\Delta\psi$ is positive), for a negatively charged ion the term $Z\mathcal{F}\Delta\psi$ makes a negative contribution to ΔG .

In other words, the negative charge is spontaneously attracted to the more positive potential—and ΔG is negative. In any case, if the sum of the two terms on the right side of Equation 10.2 is a negative number, transport of the ion in question from side 1 to side 2 would occur spontaneously. The driving force for passive transport is the ΔG term for the transported species itself.

10.2 • Facilitated Diffusion

The transport of many substances across simple lipid bilayer membranes via passive diffusion is far too slow to sustain life processes. On the other hand, the transport rates for many ions and small molecules across actual biological membranes is much higher than anticipated from passive diffusion alone. This difference is due to specific proteins in the membrane that **facilitate** transport of these species across the membrane. Similar proteins capable of effecting **facilitated diffusion** of a variety of solutes are present in essentially all natural membranes. Such proteins have two features in common: (a) they facilitate net movement of solutes only in the thermodynamically favored direction (that is, $\Delta G < 0$), and (b) they display a measurable affinity and specificity for the transported solute. Consequently, facilitated diffusion rates display **saturation behavior** similar to that observed with substrate binding by enzymes (Chapter 14). Such behavior provides a simple means for distinguishing between passive diffusion and facilitated diffusion experimentally. The dependence of transport rate on solute concentration takes the form of a rectangular hyperbola (Figure 10.3), so that the transport rate approaches a limiting value, $V_{\max}$, at very high solute concentration. Figure 10.3 also shows the graphical behavior exhibited by simple passive diffusion. Because passive diffusion does not involve formation of a specific solute:protein complex, the plot of rate versus concentration is linear, not hyperbolic.

Glucose Transport in Erythrocytes Occurs by Facilitated Diffusion

Many transport processes in a variety of cells occur by facilitated diffusion. Table 10.1 lists just a few of these. The **glucose transporter** of erythrocytes illustrates many of the important features of facilitated transport systems. Although glucose transport operates variously by passive diffusion, facilitated diffusion, or active transport mechanisms, depending on the particular cell, the **glucose transport system** of erythrocytes (red blood cells) operates exclusively by facilitated diffusion. The erythrocyte glucose transporter has a molecular mass of approximately 55 kD and is found on SDS polyacrylamide electrophoresis gels (Figure 10.4) as **band 4.5**. Typical erythrocytes contain around 500,000 copies of this protein. The active form of this transport protein in the erythrocyte membrane is a trimer. Hydropathy analysis of the amino acid sequence of the erythrocyte glucose transporter has provided a model for the structure of the protein (Figure 10.5). In this model, the protein spans the membrane 12 times, with both the N- and C-termini located on the cytoplasmic side. Transmembrane segments M7, M8, and M11 comprise a hydrophilic transmembrane channel, with segments M9 and M10 forming a relatively hydrophobic pocket adjacent to the glucose-binding site. Cytochalasin B, a fungal metabolite (Figure 10.6), is a competitive inhibitor of glucose transport. The mechanism of glucose transport is not well understood. An **alternating conformation model**, in

Table 10.1

Facilitated Transport Systems			
Permeant	Cell Type	K_m (mM)	$V_{\max}$ (mM/min)
D-Glucose	Erythrocyte	4–10	100–500
Chloride	Erythrocyte	25–30	
cAMP	Erythrocyte	0.0047	0.028
Phosphate	Erythrocyte	80	2.8
D-Glucose	Adipocytes	20	
D-Glucose	Yeast	5	
Sugars and amino acids	Tumor cells	0.5–4	2–6
D-Glucose	Rat liver	30	
D-Glucose	<i>Neurospora crassa</i>	8.3	46
Choline	Synaptosomes	0.083	
L-Valine	<i>Arthrobotrys conoides</i>	0.15–0.75	

Source: Adapted from Jain, M., and Wagner, R., 1980. *Introduction to Biological Membranes*. New York: Wiley.

which the glucose-binding site is alternately exposed to the cytoplasmic and extracellular surfaces of the membrane, has been proposed but remains controversial. Many other glucose transport proteins with sequences that are homologous to the erythrocyte glucose transporter have been identified in muscle, liver, and most other animal tissues. The reduced ability of insulin to stimulate glucose transport in diabetic patients is due to reduced expression of some, but not all, of these glucose transport proteins.

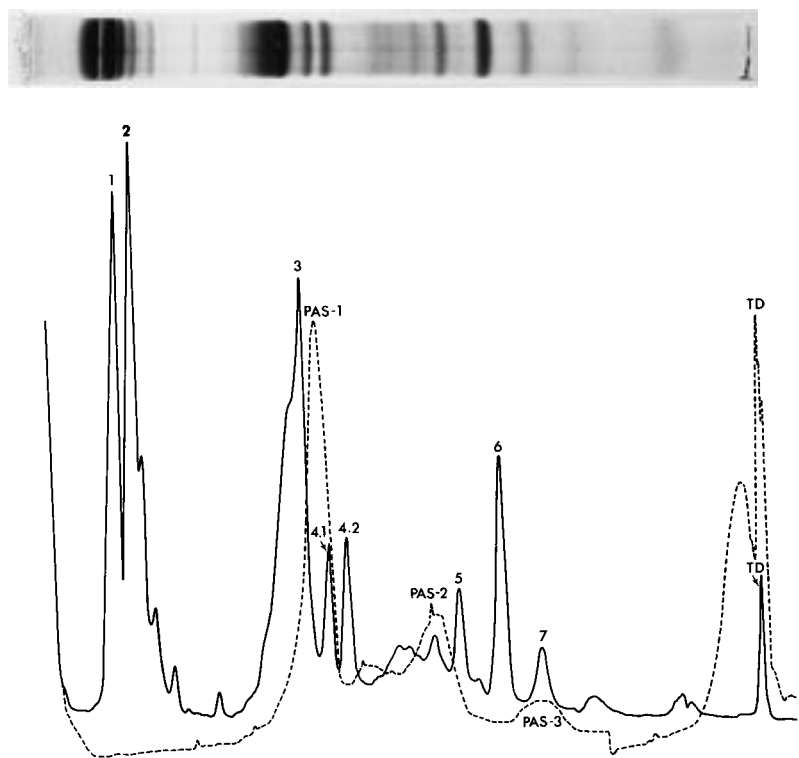
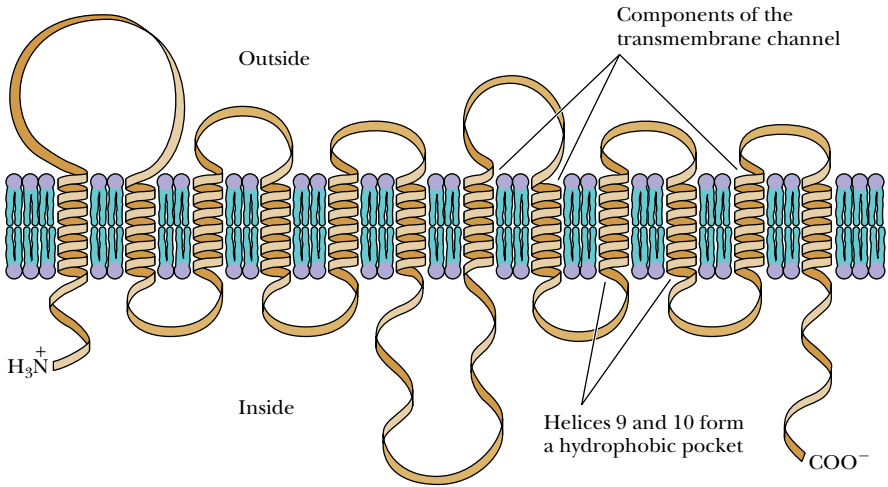


FIGURE 10.4 • SDS-gel electrophoresis of erythrocyte membrane proteins (*top*) and a densitometer tracing of the same gel (*bottom*). The region of the gel between band 4.2 and band 5 is referred to as zone 4.5 or “band 4.5.” The bands are numbered from the top of the gel (high molecular weights) to the bottom (low molecular weights). Band 3 is the anion-transporting protein and band 4.5 is the glucose transporter. The dashed line shows the staining of the gel by periodic acid–Schiff’s reagent (PAS), which stains carbohydrates. Three “PAS bands” (PAS-1, PAS-2, PAS-3) indicate the positions of glycoproteins in the gel. (Photo courtesy of Theodore Steck, University of Chicago)

FIGURE 10.5 • A model for the arrangement of the glucose transport protein in the erythrocyte membrane. Hydropathy analysis is consistent with 12 transmembrane helical segments.



The Anion Transporter of Erythrocytes Also Operates by Facilitated Diffusion

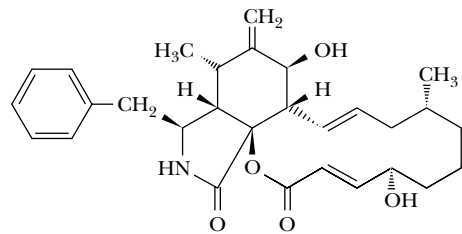
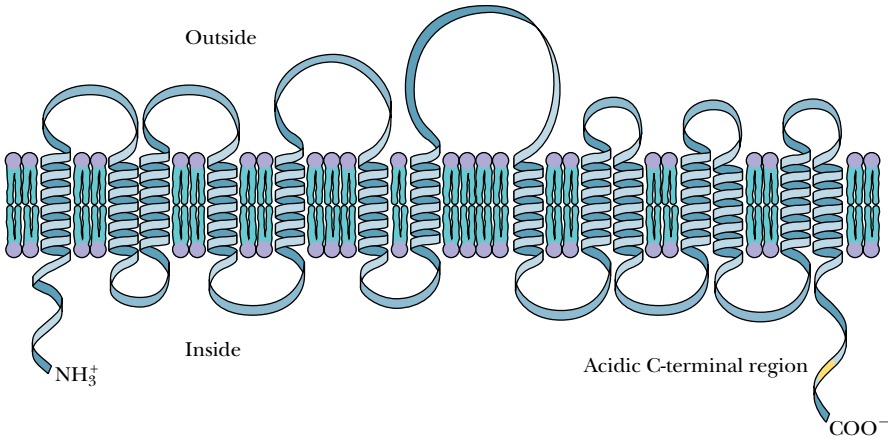


FIGURE 10.6 • The structure of cytochalasin B.

The **anion transport system** is another facilitated diffusion system of the erythrocyte membrane. Chloride and bicarbonate (HCO_3^-) ions are exchanged across the red cell membrane by a 95-kD transmembrane protein. This protein is abundant in the red cell membrane and is represented by **band 3** on SDS electrophoresis gels (Figure 10.4). The gene for the human erythrocyte anion transporter has been sequenced and hydropathy analysis has yielded a model for the arrangement of the protein in the red cell membrane (Figure 10.7). The model has 14 transmembrane segments, and the sequence includes 3 regions: a hydrophilic, cytoplasmic domain (residues 1 through 403) that interacts with numerous cytoplasmic and membrane proteins; a hydrophobic domain (residues 404 through 882) that comprises the anion transporting channel; and an acidic, C-terminal domain (residues 883 through 911). This transport system facilitates a one-for-one exchange of chloride and bicarbonate, so that the net transport process is electrically neutral. The net direction of anion flow through this protein depends on the sum of the chloride and bicarbonate concentration gradients. Typically, carbon dioxide is collected by red cells in respiring tissues (by means of $\text{Cl}^- \rightleftharpoons \text{HCO}_3^-$ exchange) and is then carried in the blood to the lungs, where bicarbonate diffuses out of the erythrocytes in exchange for Cl^- ions.

FIGURE 10.7 • A model for the arrangement of the anion transport protein in the membrane, based on hydropathy analysis.



10.3 • Active Transport Systems

Passive and facilitated diffusion systems are relatively simple, in the sense that the transported species flow downhill energetically, that is, from high concentration to low concentration. However, other transport processes in biological systems must be *driven* in an energetic sense. In these cases, the transported species moves from low concentration to high concentration, and thus the transport requires *energy input*. As such, it is considered an **active transport system**. The most common energy input is **ATP hydrolysis** (see Chapter 3), with hydrolysis being *tightly coupled* to the transport event. Other energy sources also drive active transport processes, including *light energy* and the *energy stored in ion gradients* (concentration differences of an ion (or solute) across a membrane represent an energized state (see Chapter 21)). The original ion gradient is said to arise from a **primary active transport** process, and the transport that depends on the ion gradient for its energy input is referred to as a **secondary active transport** process (see discussion of amino acid and sugar transport, Section 10.6). When transport results in a net movement of electric charge across the membrane, it is referred to as an **electrogenic transport** process. If no net movement of charge occurs during transport, the process is electrically neutral.

All Active Transport Systems Are Energy-Coupling Devices

Hydrolysis of ATP is essentially a chemical process, whereas movement of species across a membrane is a mechanical process (that is, movement). An active transport process that depends on ATP hydrolysis thus couples chemical free energy to mechanical (translational) free energy. The bacteriorhodopsin protein in *Halobacterium halobium* couples light energy and mechanical energy. Oxidative phosphorylation (Chapter 21) involves coupling between electron transport, proton translocation, and the capture of chemical energy in the form of ATP synthesis. Similarly, the overall process of photosynthesis (Chapter 22) amounts to a coupling between captured light energy, proton translocation, and chemical energy stored in ATP.

10.4 • Transport Processes Driven by ATP

Monovalent Cation Transport: Na^+, K^+ -ATPase

All animal cells actively extrude Na^+ ions and accumulate K^+ ions. These two transport processes are driven by **Na^+, K^+ -ATPase**, also known as the **sodium pump**, an integral protein of the plasma membrane. Most animal cells maintain cytosolic concentrations of Na^+ and K^+ of 10 mM and 100 mM, respectively. The extracellular milieu typically contains about 100 to 140 mM Na^+ and 5 to 10 mM K^+ . Potassium is required within the cell to activate a variety of processes, whereas high intracellular sodium concentrations are inhibitory. The transmembrane gradients of Na^+ and K^+ and the attendant gradients of Cl^- and other ions provide the means by which neurons communicate (see Chapter 34). They also serve to regulate cellular volume and shape. Animal cells also depend upon these Na^+ and K^+ gradients to drive transport processes involving amino acids, sugars, nucleotides, and other substances. In fact, maintenance of these Na^+ and K^+ gradients consumes large amounts of energy in animal cells—20 to 40% of total metabolic energy in many cases and up to 70% in neural tissue.

The Na^+ - and K^+ -dependent ATPase comprises two subunits, an α -subunit of 1016 residues (120 kD) and a 35-kD β -subunit. The sodium pump actively

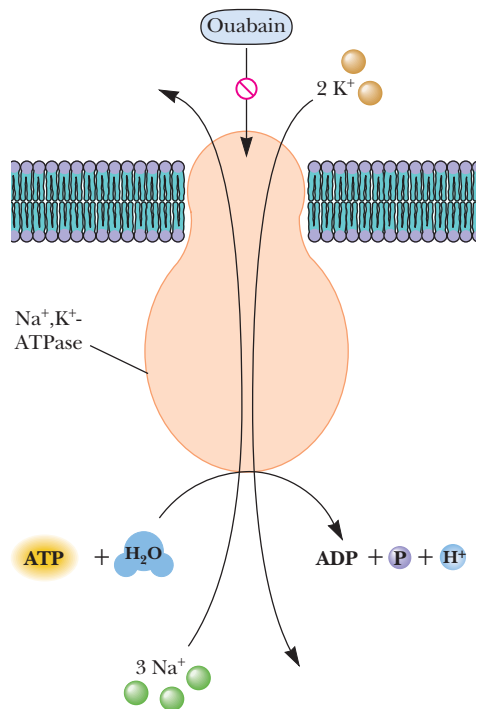
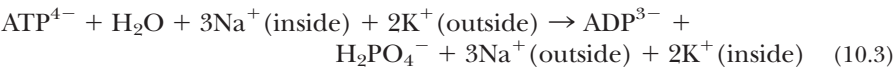


FIGURE 10.8 • A schematic diagram of the Na⁺,K⁺-ATPase in mammalian plasma membrane. ATP hydrolysis occurs on the cytoplasmic side of the membrane, Na⁺ ions are transported out of the cell, and K⁺ ions are transported in. The transport stoichiometry is 3 Na⁺ out and 2 K⁺ in per ATP hydrolyzed. The specific inhibitor ouabain (Figure 7.12) and other cardiac glycosides inhibit Na⁺,K⁺-ATPase by binding on the extracellular surface of the pump protein.

pumps three Na⁺ ions out of the cell and two K⁺ ions into the cell per ATP hydrolyzed:

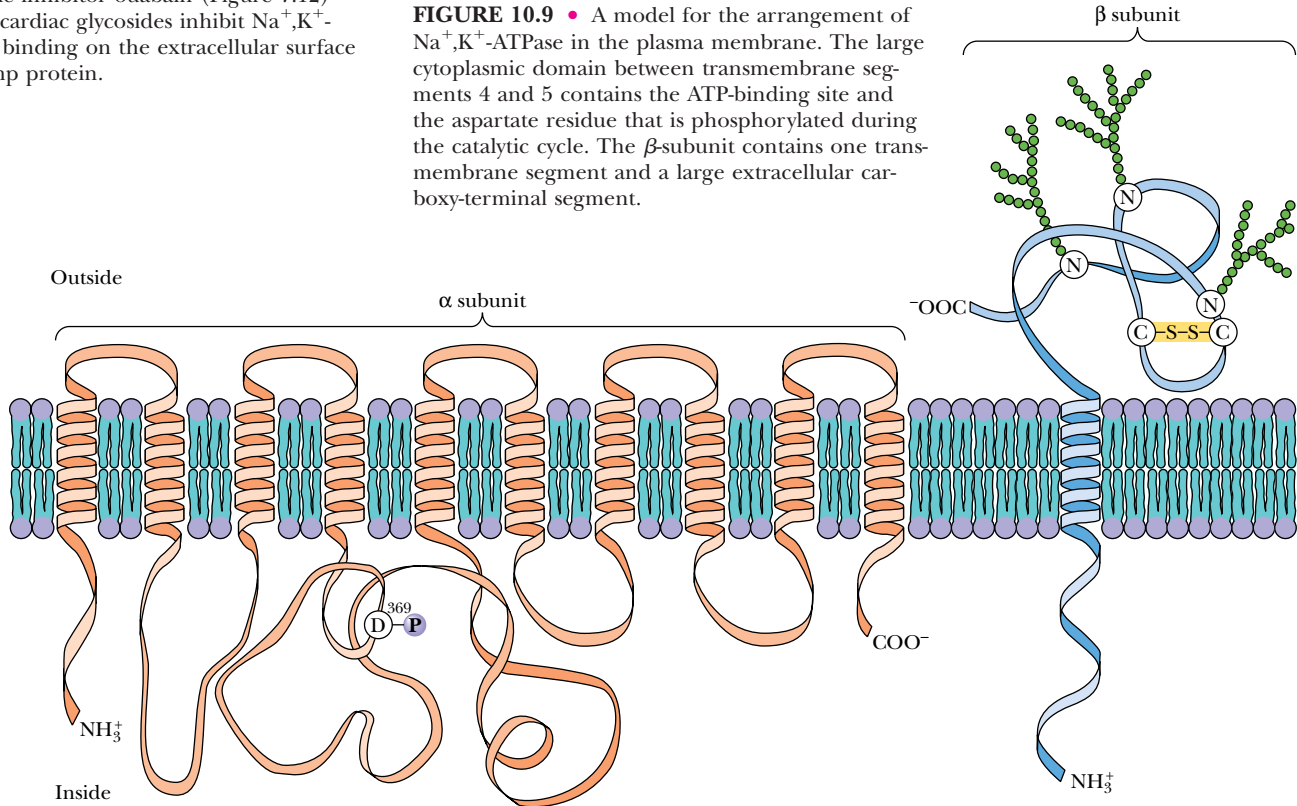


ATP hydrolysis occurs on the cytoplasmic side of the membrane (Figure 10.8), and the net movement of one positive charge outward per cycle makes the sodium pump electrogenic in nature.

Hydropathy analysis of the amino acid sequences of the α- and β-subunits and chemical modification studies have led to a model for the arrangement of the ATPase in the plasma membrane (Figure 10.9). The model describes 10 transmembrane α-helices in the α-subunit, with two large cytoplasmic domains. The larger of these, between transmembrane segments 4 and 5, has been implicated as the ATP-binding domain. The enzyme is covalently phosphorylated at an aspartate residue on the α-subunit in the course of ATP hydrolysis. The covalent E-P intermediate was trapped and identified using tritiated sodium borohydride (Figure 10.10).

A minimal mechanism for Na⁺,K⁺-ATPase postulates that the enzyme cycles between two principal conformations, denoted E₁ and E₂ (Figure 10.11). E₁ has a high affinity for Na⁺ and ATP and is rapidly phosphorylated in the presence of Mg²⁺ to form E₁-P, a state which contains *three occluded Na⁺ ions* (occluded in the sense that they are tightly bound and not easily dissociated from the enzyme in this conformation). A conformation change yields E₂-P, a form of the enzyme with relatively low affinity for Na⁺, but a high affinity for K⁺. This state presumably releases 3 Na⁺ ions and binds 2 K⁺ ions on the outside of the cell. Dephosphorylation leaves E₂K₂, a form of the enzyme with *two*

FIGURE 10.9 • A model for the arrangement of Na⁺,K⁺-ATPase in the plasma membrane. The large cytoplasmic domain between transmembrane segments 4 and 5 contains the ATP-binding site and the aspartate residue that is phosphorylated during the catalytic cycle. The β-subunit contains one transmembrane segment and a large extracellular carboxy-terminal segment.



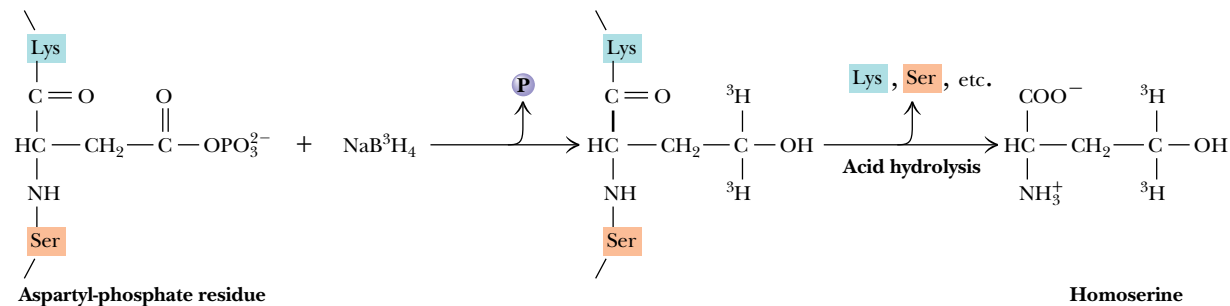


FIGURE 10.10 • The reaction of tritiated sodium borohydride with the aspartyl phosphate at the active site of Na^+, K^+ -ATPase. Acid hydrolysis of the enzyme following phosphorylation and sodium borohydride treatment yields a tripeptide containing serine, homoserine (derived from the aspartyl-phosphate), and lysine as shown. The site of phosphorylation is Asp³⁶⁹ in the large cytoplasmic domain of the ATPase.

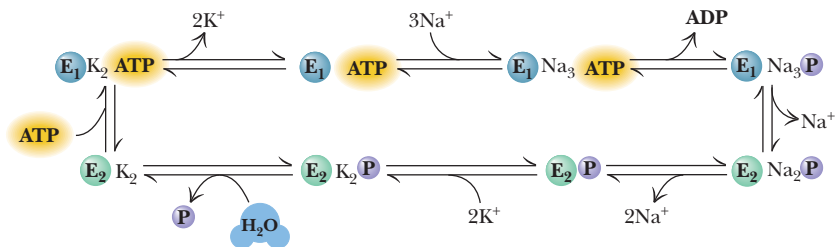


FIGURE 10.11 • A mechanism for Na^+, K^+ -ATPase. The model assumes two principal conformations, E_1 and E_2 . Binding of Na^+ ions to E_1 is followed by phosphorylation and release of ADP. Na^+ ions are transported and released and K^+ ions are bound before dephosphorylation of the enzyme. Transport and release of K^+ ions complete the cycle.

occluded K^+ ions. A conformation change, which appears to be accelerated by the binding of ATP (with a relatively low affinity), releases the bound K^+ inside the cell and returns the enzyme to the E_1 : ATP state. Enzyme forms with occluded cations represent states of the enzyme with cations bound in the transport channel. The alternation between high and low affinities for Na^+ , K^+ , and ATP serves to tightly couple the hydrolysis of ATP and ion binding and transport.

Na^+, K^+ -ATPase Is Inhibited by Cardiac Glycosides

Plant and animal steroids such as *ouabain* (Figure 10.12) specifically inhibit Na^+, K^+ -ATPase and ion transport. These substances are traditionally referred to as **cardiac glycosides** or **cardiotonic steroids**, both names derived from the potent effects of these molecules on the heart. These molecules all possess a *cis*-configuration of the C-D ring junction, an unsaturated lactone ring (5- or 6-membered) in the β -configuration at C-17, and a β -OH at C-14. There may be one or more sugar residues at C-3. The sugar(s) are not required for inhibition, but do contribute to water solubility of the molecule. Cardiac glycosides bind exclusively to the extracellular surface of Na^+, K^+ -ATPase when it is in the E_2 -P state, forming a very stable E_2 -P (cardiac glycoside) complex.

Medical researchers studying high blood pressure have consistently found that people with hypertension have high blood levels of some sort of Na^+, K^+ -

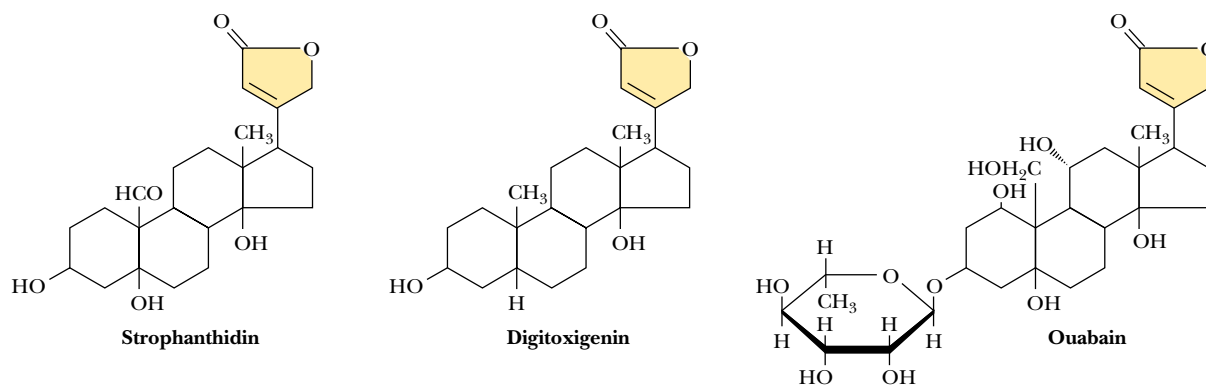


FIGURE 10.12 • The structures of several cardiac glycosides. The lactone rings are yellow.

ATPase inhibitor. In such patients, inhibition of the sodium pump in the cells lining the blood vessel wall results in accumulation of sodium and calcium in these cells and the narrowing of the vessels to create hypertension. An 8-year study aimed at the isolation and identification of the agent responsible for these effects by researchers at the University of Maryland Medical School and the Upjohn Laboratories in Michigan recently yielded a surprising result. Mass spectrometric analysis of compounds isolated from many hundreds of gallons of blood plasma has revealed that the hypertensive agent is ouabain itself or a closely related molecule!

Calcium Transport: Ca^{2+} -ATPase

Calcium, an ion acting as a cellular signal in virtually all cells (see Chapter 34), plays a special role in muscles. It is the signal that stimulates muscles to contract (Chapter 17). In the resting state, the levels of Ca^{2+} near the muscle fibers are very low (approximately $0.1 \mu\text{M}$), and nearly all of the calcium ion in muscles is sequestered inside a complex network of vesicles called the **sarcoplasmic reticulum**, or **SR** (see Figure 17.2). Nerve impulses induce the sarcoplasmic reticulum membrane to quickly release large amounts of Ca^{2+} , with cytosolic levels rising to approximately $10 \mu\text{M}$. At these levels, Ca^{2+} stimulates contraction. Relaxation of the muscle requires that cytosolic Ca^{2+} levels be reduced to their resting levels. This is accomplished by an ATP-driven Ca^{2+} transport protein known as the **Ca^{2+} -ATPase**. This enzyme is the most abundant protein in the SR membrane, accounting for 70 to 80% of the SR protein. Ca^{2+} -ATPase bears many similarities to the Na^+, K^+ -ATPase. It has an α -subunit of the same approximate size, it forms a covalent E-P intermediate during ATP hydrolysis, and its mechanism of ATP hydrolysis and ion transport is similar in many ways to that of the sodium pump.

The amino acid sequence of the α -subunit is homologous with the sodium pump α -subunit, particularly around the phosphorylation site and the ATP-binding site (Figure 10.13). Ten transmembrane helical segments are predicted from hydropathy analysis, as well as a “stalk” consisting of five helical segments (Figure 10.14). This stalk lies between the membrane surface and the globular cytoplasmic domain containing the nucleotide-binding domain and the site of phosphorylation. The E-P formed by SR Ca^{2+} -ATPase is an aspartyl phosphate like that of Na^+, K^+ -ATPase, in this case Asp residue 351.

(a) Phosphorylation domain

	Res. no.																				
Na ⁺ , K ⁺ -ATPase, α	363	T	S	T	I	C	S	D	K	T	G	T	L	T	Q	N	R	M			
H ⁺ , K ⁺ -ATPase	379	T	S	V	I	C	S	D	K	T	G	T	L	T	Q	N	R	M			
Ca ²⁺ -ATPase, SR	345	T	S	V	I	C	S	D	K	T	G	T	L	T	T	N	Q	M			
H ⁺ -ATPase, yeast	372	V	E	I	L	C	S	D	K	T	G	T	L	T	K	N	K	L			
K ⁺ -ATPase, <i>Streptococcus faecalis</i>	273	L	D	V	I	M	L	D	K	T	G	T	L	T	Q	G	K	F			
F ₁ ATPase, <i>E. coli</i>	280	Q	E	R	I	T	S	T	K	T	G	S	I	T	S	V	Q	A			
F ₁ ATPase, bovine	293	Q	E	R	I	T	T	T	K	K	G	S	I	T	S	V	Q	A			

(b) Segment-binding FITC, an ATP-site probe for ATPases

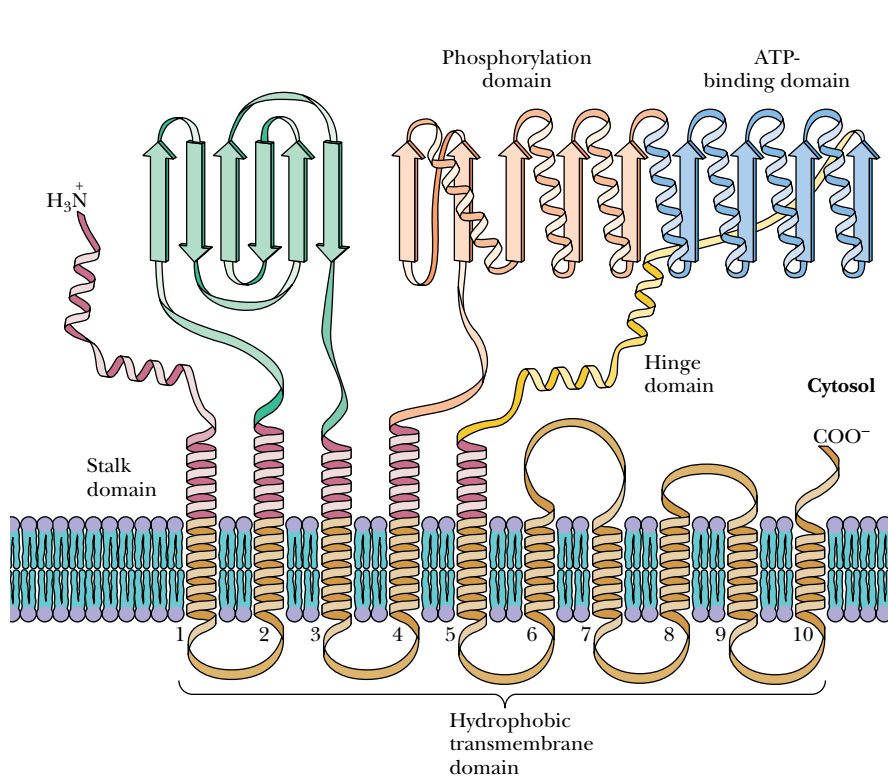
Na ⁺ , K ⁺ -ATPase, α	496	P	Q	H	L	L	V	M	K	G	A	P	E	R	I	L	D	R	C	S	S
H ⁺ , K ⁺ -ATPase	510	P	R	H	L	L	V	M	K	G	A	P	E	R	V	L	E	R	C	S	S
Ca ²⁺ -ATPase, SR	508	V	G	N	K	M	F	V	K	G	A	P	E	G	V	I	D	R	C	N	Y
Ca ²⁺ -ATPase, plasma membrane						M	Y	S	K	G	A	S	E	I	I	L	R				
H ⁺ -ATPase, yeast	467	G	E	R	I	V	C	V	K	G	A	P	L	S	A	L	K	T	V	E	E
H ⁺ -ATPase, <i>Neurospora</i>	467	G	E	R	I	T	C	V	K	G	A	P	L	F	V	L	K	T	V	E	E

(c) ATP-binding region

Na ⁺ , K ⁺ -ATPase, α	543	L	G	E	R	V	—	L	G	F	C	H	L	F	L	P	D	E	Q	F	P
H ⁺ , K ⁺ -ATPase	613	L	K	C	R	T	—	A	G	I	R	V	I	M	V	T	G	D	H	P	I
Ca ²⁺ -ATPase, SR	611	Q	L	C	R	D	—	A	G	I	R	V	I	M	I	T	G	D	N	K	G
H ⁺ -ATPase, <i>Neurospora</i>	544	C	E	A	K	T	—	L	G	L	S	I	K	M	L	T	G	D	A	V	G
H ⁺ -ATPase, yeast	545	S	E	A	R	H	—	L	G	L	R	V	K	M	L	T	G	D	A	V	G
F ₁ ATPase, bovine	243	E	Y	F	R	D	Q	E	G	Q	D	V	L	L	F	I	D	N	I	F	R
F ₁ ATPase, <i>E. coli</i>	267	E	Y	F	R	D	—	R	G	E	D	A	L	I	I	Y	D	D	L	S	K
ATP-ADP exchange protein	277	V	L	—	R	G	N	G	G	A	F	V	L	V	L	Y	D	E	I	K	K
Adenylate kinase	104	E	F	E	R	K	—	I	G	Q	P	T	L	L	L	Y	V	D	A	G	P
Phosphofructokinase	87	E	Q	L	K	K	—	H	G	I	Q	G	L	V	V	I	G	G	D	G	S

(d) Segment-binding active-site probes in Na⁺, K⁺-ATPase

Na ⁺ , K ⁺ -ATPase, α	701	Q	G	A	I	V	A	V	T	G	D	G	V	N	D	S	P	A	L	K	K
H ⁺ , K ⁺ -ATPase	717	L	G	A	I	V	A	V	T	G	D	G	V	N	D	S	P	A	L	K	K
Ca ²⁺ -ATPase, SR	694	Y	D	E	I	T	A	M	T	G	D	G	V	N	D	A	P	A	L	K	K
H ⁺ -ATPase, <i>Neurospora</i>	625	R	G	Y	L	V	A	M	T	G	D	G	V	N	D	A	P	S	L	K	K
H ⁺ -ATPase, yeast	625	R	G	Y	L	V	A	M	T	G	D	G	V	N	D	A	P	S	L	K	K
K ⁺ -ATPase, <i>Streptococcus faecalis</i>	467	Q	G	K	K	V	I	M	V	G	D	G	I	N	D	A	P	S	L	A	R



▲ **FIGURE 10.13** • Some of the sequence homologies in the nucleotide binding and phosphorylation domains of Na⁺,K⁺-ATPase, Ca²⁺-ATPase, and gastric H⁺,K⁺-ATPase. (Adapted from Jørgensen, P. L., and Andersen, J. P., 1988. Structural basis for E₁ - E₂ conformational transitions in Na⁺,K⁺-pump and Ca²⁺-pump proteins. Journal of Membrane Biology 103:95–120)

◀ **FIGURE 10.14** • The arrangement of Ca²⁺-ATPase in the sarcoplasmic reticulum membrane. Ten transmembrane segments are postulated on the basis of hydropathy analysis.

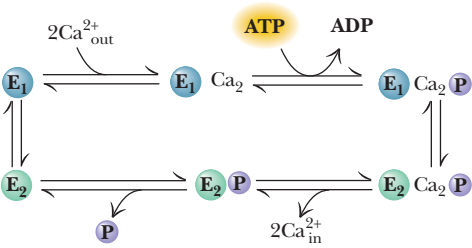


FIGURE 10.15 • A mechanism for Ca^{2+} -ATPase from sarcoplasmic reticulum. Note the similarity to the mechanism of Na^+, K^+ -ATPase (see also Figure 10.11). (“Out” here represents the cytosol; “in” represents the lumen of the SR.)

Two Ca^{2+} ions are transported into the SR per ATP hydrolyzed by this enzyme, and the mechanism (Figure 10.15) appears to involve two major conformations, E_1 and E_2 , just as the Na^+, K^+ -ATPase mechanism does. Calcium ions are strongly occluded in the $\text{E}_1\text{-Ca}_2\text{-P}$ state, and these occluded ions do not dissociate from the enzyme until the enzyme converts to the $\text{E}_2\text{-Ca}_2\text{-P}$ state, which has a very low affinity for Ca^{2+} . In the $\text{E}_1\text{-Ca}_2\text{-P}$ state, the transported Ca^{2+} ions are bound in the transport channel.

HUMAN BIOCHEMISTRY

Cardiac Glycosides: Potent Drugs from Ancient Times

The cardiac glycosides have a long and colorful history. Many species of plants producing these agents grow in tropical regions and have been used by natives in South America and Africa to prepare poisoned arrows used in fighting and hunting. Zulus in South Africa, for example, have used spears tipped with cardiac glycoside poisons. The sea onion, found commonly in southern Europe and northern Africa, was used by the Romans and the Egyptians as a cardiac stimulant, diuretic, and expectorant. The Chinese have long used a medicine made from the skins of certain toads for similar purposes. Cardiac glycosides are also found in several species of domestic plants, including the foxglove, lily of the valley, oleander (figure), and milkweed plant. Monarch butterflies acquire these compounds by feeding on milkweed and then storing the cardiac glycosides in their exoskeletons. Cardiac glycosides deter predation of monarch butterflies by birds, which learn by experience not to feed on monarchs. Viceroy butterflies mimic monarchs in overall appearance. Although viceroys con-

tain no cardiac glycosides and are edible, they are avoided by birds that mistake them for monarchs.

In 1785, the physician and botanist William Withering described the medicinal uses for agents derived from the foxglove plant. In modern times, **digitalis** (a preparation from dried leaves of the foxglove, *Digitalis purpurea*) and other purified cardiotonic steroids have been used to increase the contractile force of heart muscle, to slow the rate of beating, and to restore normal function in hearts undergoing fibrillation (a condition in which heart valves do not open and close rhythmically, but rather remain partially open, fluttering in an irregular and ineffective way). Inhibition of the cardiac sodium pump increases the intracellular Na^+ concentration, leading to stimulation of the $\text{Na}^+\text{-Ca}^{2+}$ exchanger, which extrudes sodium in exchange for inward movement of calcium. Increased intracellular Ca^{2+} stimulates muscle contraction. Careful use of digitalis drugs has substantial therapeutic benefit for heart patients.



(a) Oleander



(b) Monarch butterfly



(c) Viceroy

(a) Cardiac glycoside inhibitors of Na^+, K^+ -ATPase are produced by many plants, including foxglove, lily of the valley, milkweed, and oleander (shown here). (b) The monarch butterfly, which concentrates cardiac glycosides in its exoskeleton, is shunned by predatory birds. (c) Predators also avoid the viceroy, even though it contains no cardiac glycosides, because it is similar in appearance to the monarch.

The Gastric H⁺,K⁺-ATPase

Production of protons is a fundamental activity of cellular metabolism, and proton production plays a special role in the stomach. The highly acidic environment of the stomach is essential for the digestion of food in all animals. The pH of the stomach fluid is normally 0.8 to 1. The pH of the parietal cells of the gastric mucosa in mammals is approximately 7.4. This represents a **pH gradient** across the mucosal cell membrane of 6.6, the largest known transmembrane gradient in eukaryotic cells. This enormous gradient must be maintained constantly so that food can be digested in the stomach without damage to the cells and organs adjacent to the stomach. The gradient of H⁺ is maintained by an **H⁺,K⁺-ATPase**, which uses the energy of hydrolysis of ATP to pump H⁺ out of the mucosal cells and into the stomach interior in exchange for K⁺ ions. This transport is electrically neutral, and the K⁺ that is transported into the mucosal cell is subsequently pumped back out of the cell together with Cl⁻ in a second electroneutral process (Figure 10.16). Thus, the net transport effected by these two systems is the movement of HCl into the interior of the stomach. (Only a small amount of K⁺ is needed because it is recycled.) The H⁺,K⁺-ATPase bears many similarities to the plasma membrane Na⁺,K⁺-ATPase and the SR Ca²⁺-ATPase described above. It has a similar molecular weight, forms an E-P intermediate, and many parts of its peptide sequence are homologous with the Na⁺,K⁺-ATPase and Ca²⁺-ATPase (Figure 10.13).

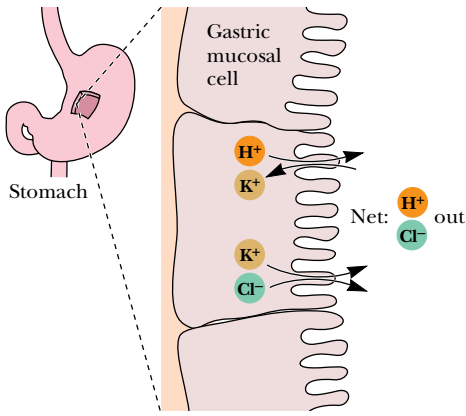


FIGURE 10.16 • The H⁺,K⁺-ATPase of gastric mucosal cells mediates proton transport into the stomach. Potassium ions are recycled by means of an associated K⁺/Cl⁻ cotransport system. The action of these two pumps results in net transport of H⁺ and Cl⁻ into the stomach.

Bone Remodeling by Osteoclast Proton Pumps

Other proton-translocating ATPases exist in eukaryotic and prokaryotic systems. **Vacuolar ATPases** are found in vacuoles, lysosomes, endosomes, Golgi, chromaffin granules, and coated vesicles. Various H⁺-transporting ATPases occur in yeast and bacteria as well. H⁺-transporting ATPases found in **osteoclasts** (multinucleate cells that break down bone during normal bone remodeling) provide a source of circulating calcium for soft tissues such as nerves and muscles. About 5% of bone mass in the human body undergoes remodeling at any given time. Once growth is complete, the body balances formation of new bone tissue by cells called **osteoblasts** with resorption of existing bone matrix by osteoclasts. Osteoclasts possess proton pumps—similar to vacuolar ATPases—on the portion of the plasma membrane that attaches to the bone. This region of the osteoclast membrane is called the **ruffled border**. The osteoclast attaches to the bone in the manner of a cup turned upside down on a saucer (Figure 10.17), leaving an extracellular space between the bone surface and the cell. The H⁺-ATPases in the ruffled border pump protons into this space, creating an acidic solution that dissolves the bone mineral matrix. Bone mineral is primarily an inorganic mixture of calcium carbonate and hydroxyapatite (calcium phosphate). In this case, transport of protons out of the osteoclasts lowers the pH of the extracellular space near the bone to about 4, solubilizing the hydroxyapatite.

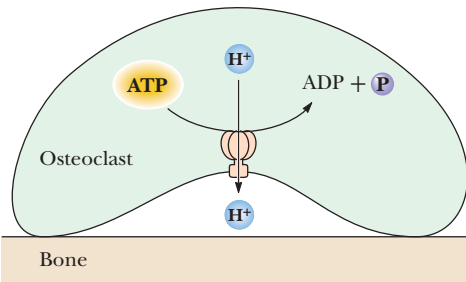


FIGURE 10.17 • Proton pumps cluster on the ruffled border of osteoclast cells and function to pump protons into the space between the cell membrane and the bone surface. High proton concentration in this space dissolves the mineral matrix of the bone.

ATPases That Transport Peptides and Drugs

Species other than protons and inorganic ions are also transported across certain membranes by specialized ATPases. Yeast (*Saccharomyces cerevisiae*) has one such system. Yeasts exist in two haploid mating types, designated **a** and **α**. Each mating type produces a mating factor (**a-factor** or **α-factor**, respectively) and responds to the mating factor of the opposite type. The α-factor is a peptide that is inserted into the ER during translation on the ribosome. α-Factor is glycosylated in the ER and then secreted from the cell. On the other hand, the

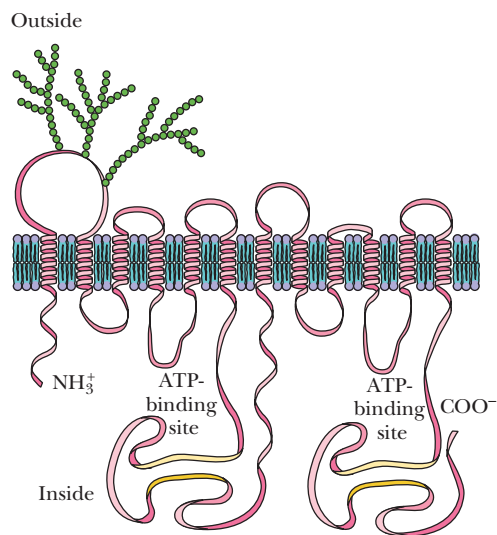
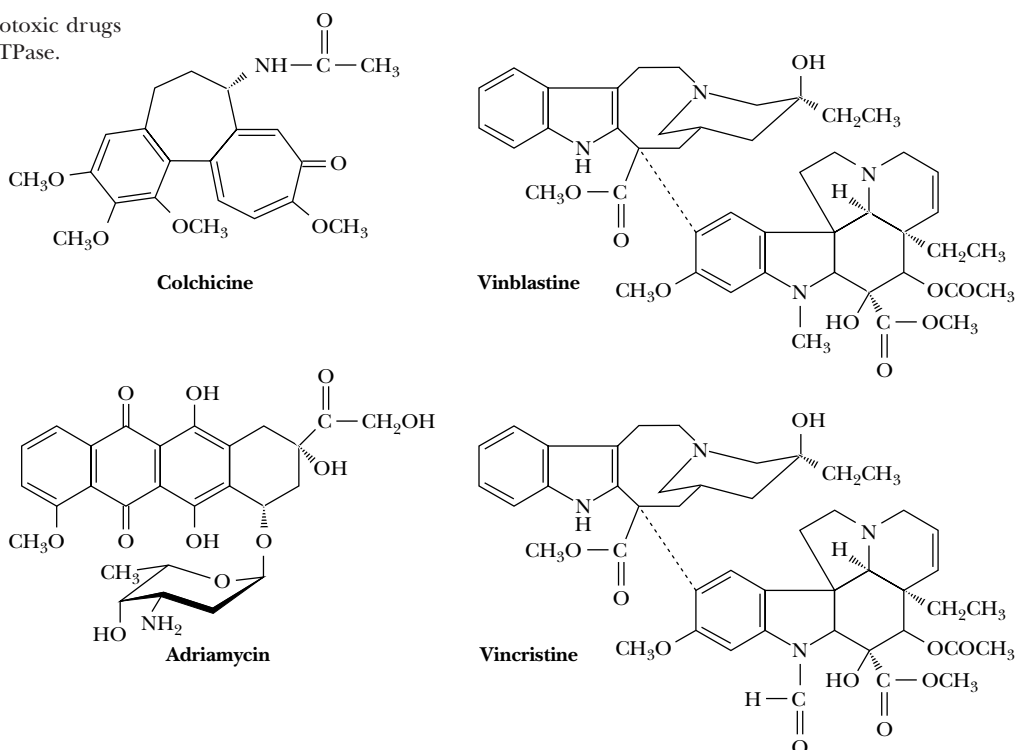


FIGURE 10.18 • A model for the structure of the a-factor transport protein in the yeast plasma membrane. Gene duplication has yielded a protein with two identical halves, each half containing six transmembrane helical segments and an ATP-binding site. Like the yeast a-factor transporter, the multidrug transporter is postulated to have 12 transmembrane helices and 2 ATP-binding sites.

a-factor is a 12-amino acid peptide made from a short precursor. Export of this peptide from the cell is carried out by a 1290-residue protein, which consists of two identical halves joined together—a tandem duplication. Each half contains six putative transmembrane segments arranged in pairs, and a conserved hydrophilic cytoplasmic domain containing a consensus sequence for an ATP-binding site (Figure 10.18). This protein uses the energy of ATP hydrolysis to export the 12-residue a-factor from the cell. In yeast cells that produce mutant forms of the a-factor ATPase, a-factor is not excreted and accumulates to high levels inside the cell.

Proteins very similar to the yeast a-factor transporter have been identified in a variety of prokaryotic and eukaryotic cells, and one of these appears to be responsible for the acquisition of **drug resistance** in many human malignancies. Clinical treatment of human cancer often involves chemotherapy, the treatment with one or more drugs that selectively inhibit the growth and proliferation of tumorous tissue. However, the efficacy of a given chemotherapeutic drug often decreases with time, owing to an acquired resistance. Even worse, the acquired resistance to a single drug usually results in a simultaneous resistance to a wide spectrum of drugs with little structural or even functional similarity to the original drug, a phenomenon referred to as **multidrug resistance**, or **MDR**. This perplexing problem has been traced to the induced expression of a 170-kD plasma membrane glycoprotein known as the **P-glycoprotein** or the **MDR ATPase**. Like the yeast a-factor transporter, MDR ATPase is a tandem repeat, each half consisting of a hydrophobic sequence with six transmembrane segments followed by a hydrophilic, cytoplasmic sequence containing a consensus ATP-binding site (Figure 10.18). The protein uses the energy of ATP hydrolysis to actively transport a wide variety of drugs (Figure 10.19) out of the cell. Ironically, it is probably part of a sophisticated protection system for the cell and the organism. Organic molecules of various types and structures that might diffuse across the plasma membrane are apparently recognized by this

FIGURE 10.19 • Some of the cytotoxic drugs that are transported by the MDR ATPase.



protein and actively extruded from the cell. Despite the cancer-fighting nature of chemotherapeutic agents, the MDR ATPase recognizes these agents as cellular intruders and rapidly removes them. It is not yet understood how this large protein can recognize, bind, and transport such a broad group of diverse molecules, but it is known that the yeast a-factor ATPase and the MDR ATPase are just two members of a **superfamily** of transport proteins, many of whose functions are not yet understood.

10.5 • Transport Processes Driven by Light

As noted previously, certain biological transport processes are driven by light energy rather than by ATP. Two well-characterized systems are **bacteriorhodopsin**, the light-driven H^+ -pump, and **halorhodopsin**, the light-driven Cl^- -pump, of *Halobacterium halobium*, an archaebacterium that thrives in high-salt media. *H. halobium* grows optimally at an NaCl concentration of 4.3 M. It was extensively characterized by Walther Stoeckenius, who found it growing prolifically in the salt pools near San Francisco Bay, where salt is commercially extracted from seawater. *H. halobium* carries out normal respiration if oxygen and metabolic energy sources are plentiful. However, when these substrates are lacking, *H. halobium* survives by using bacteriorhodopsin and halorhodopsin to capture light energy. In oxygen- and nutrient-deficient conditions, **purple patches** appear on the surface of *H. halobium* (Figure 10.20). These purple patches of membrane are 75% protein, the only protein being **bacteriorhodopsin (bR)**. The purple color arises from a retinal molecule that is covalently bound in a Schiff base linkage with an ϵ - NH_2 group of Lys²¹⁶ on each bacteriorhodopsin protein (Figure 10.21). Bacteriorhodopsin is a 26-kD transmembrane protein that packs so densely in the membrane that it naturally forms a two-dimensional crystal in the plane of the membrane. The structure of bR has been elucidated by image enhancement analysis of electron microscopic data, which reveals seven transmembrane helical protein segments. The retinal moiety lies parallel to the membrane plane, about 1 nm below the membrane's outer surface (Figure 9.15).

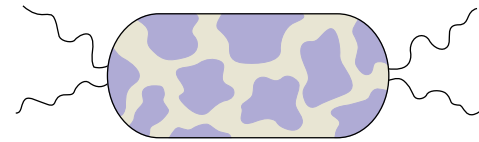


FIGURE 10.20 • A schematic drawing of *Halobacterium halobium*. The purple patches contain bacteriorhodopsin (bR).

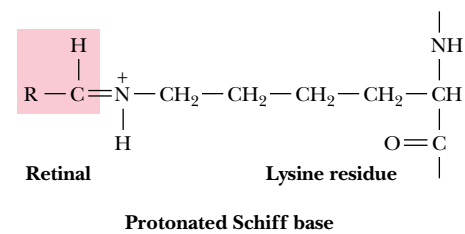


FIGURE 10.21 • The Schiff base linkage between the retinal chromophore and Lys²¹⁶.

A Model for Light-Driven Proton Transport

The mechanism of the light-driven transport of protons by bacteriorhodopsin is complex, but a partial model has emerged (Figure 10.22). A series of intermediate states, named for the wavelengths (in nm) of their absorption spectra, has been identified. Absorption of a photon of light by the bR₅₆₈ form (in which the Schiff base at Lys²¹⁶ is protonated) converts the retinal from the all-*trans* configuration to the 13-*cis* isomer. Passage through several different intermediate states results in outward transport of 2 H⁺ ions per photon absorbed, and the return of the bound retinal to the all-*trans* configuration. It appears that the transported protons are in fact protons from the protonated Schiff base. The proton gradient thus established represents chemical energy that can be used by *H. halobium* to drive ATP synthesis and the movement of molecules across the cell membrane (see Chapter 21).

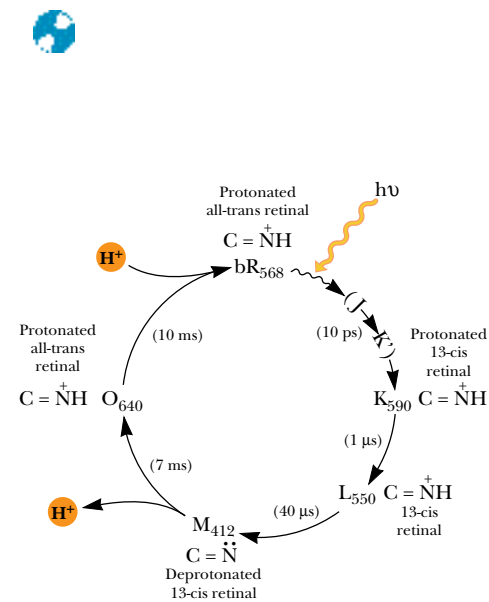


FIGURE 10.22 • The reaction cycle of bacteriorhodopsin. The intermediate states are indicated by letters, with subscripts to indicate the absorption maxima of the states. Also indicated for each state is the configuration of the retinal chromophore (all-*trans* or 13-*cis*) and the protonation state of the Schiff base (C=N: or C=N⁺H).

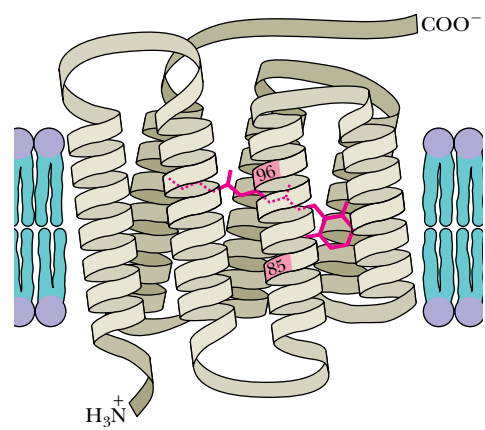


FIGURE 10.23 • The folding of halorhodopsin with the transmembrane segments indicated. The only lysine residue in the protein is Lys²⁴², to which the retinal chromophore is covalently linked.

Light-Driven Chloride Transport in *H. halobium*

Anion transport, on the other hand, is handled by a second light-driven ion pump in the *H. halobium* membrane. The inward transport of Cl[−] ion is mediated by *halorhodopsin*, a 27-kD protein whose primary structure and arrangement in the membrane (Figure 10.23) is very similar to that of bacteriorhodopsin. Although halorhodopsin does not exist naturally as a tightly packed two-dimensional crystal in the membrane, it does have a retinal chromophore, bound covalently at Lys²⁴², the only lysine in the protein. The transmembrane portion of halorhodopsin is 36% homologous with bacteriorhodopsin. The conserved residues are concentrated in the central core formed in both proteins by the seven transmembrane helices (Figure 10.24). Like bacteriorhodopsin, halorhodopsin undergoes a cycle of light-driven con-

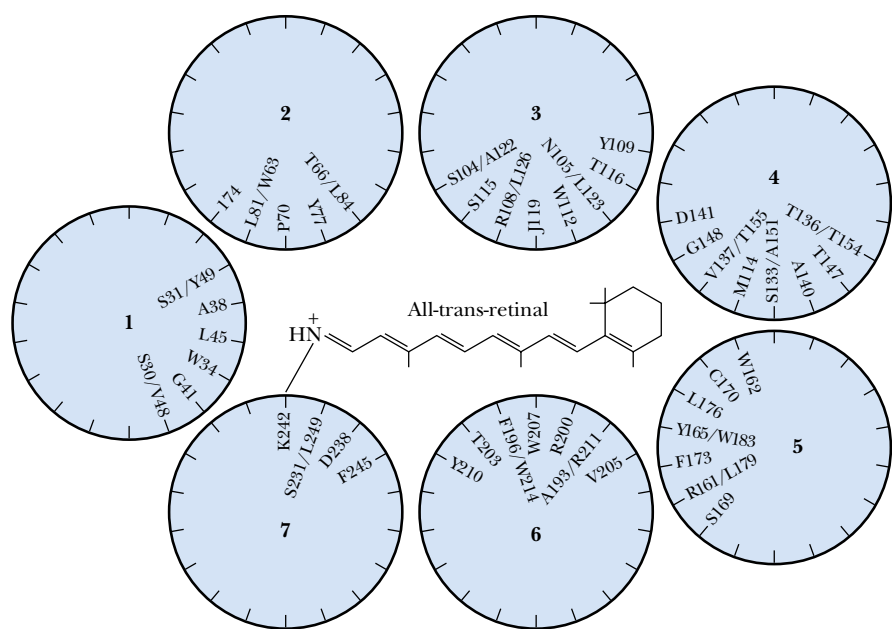


FIGURE 10.24 • A helical wheel model of halorhodopsin. The amino acids facing the polar, hydrophilic core of the protein are shown. Of these 60 residues, 36 are conserved between halorhodopsin and bacteriorhodopsin. (Adapted from Oesterhelt, D., and Tittor, J., 1989. Trends in Biochemical Sciences 14:57–61.)

formational changes (Figure 10.25), but no deprotonation of the Schiff base occurs during the halorhodopsin photocycle. Given the striking similarity of structures for these two proteins, it is intriguing to ask why bacteriorhodopsin pumps H^+ but not Cl^- and why halorhodopsin pumps Cl^- but not H^+ . The first question may be answered by the work of H. G. Khorana and his coworkers, who replaced Asp^{85} and Asp^{96} in bacteriorhodopsin with asparagine and found that either substitution caused a drastic reduction in H^+ transport. Dieter Oesterhelt and coworkers have shown that Asp^{85} and Asp^{96} are important in the deprotonation and reprotonation, respectively, of the Schiff base in bacteriorhodopsin. The absence of these two crucial residues in halorhodopsin may explain why the latter protein can't reversibly deprotonate the Schiff base and why halorhodopsin doesn't pump protons.

10.6 • Transport Processes Driven by Ion Gradients

Amino Acid and Sugar Transport

The gradients of H^+ , Na^+ , and other cations and anions established by ATPases and other energy sources can be used for **secondary active transport** of various substrates. The best-understood systems use Na^+ or H^+ gradients to transport amino acids and sugars in certain cells. Many of these systems operate as **symports**, with the ion and the transported amino acid or sugar moving in the same direction (that is, into the cell). In **antiport** processes, the ion and the other transported species move in opposite directions. (For example, the anion transporter of erythrocytes is an antiport.) **Proton symport** proteins are used by *E. coli* and other bacteria to accumulate lactose, arabinose, ribose, and a variety of amino acids. *E. coli* also possesses Na^+ -symport systems for melibiose as well as for glutamate and other amino acids.

Table 10.2 lists several systems that transport amino acids into mammalian cells. The accumulation of neutral amino acids in the liver by System A rep-

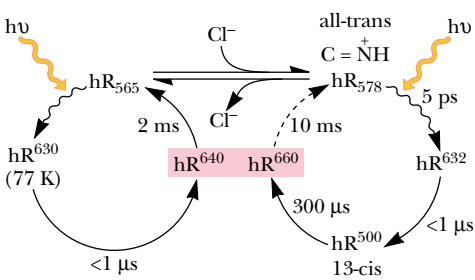


FIGURE 10.25 • The photocycle of light-adapted halorhodopsin (hR), shown in the presence and absence of chloride. The superscripts indicate the maxima of the difference spectra between hR and the intermediates.

Table 10.2

Some Mammalian Amino Acid Transport Systems			
System Designation	Ion Dependence	Amino Acids Transported	Cellular Source
A	Na^+	Neutral amino acids	Ehrlich ascites cells Chinese hamster ovary cells Hepatocytes
ASC	Na^+	Neutral amino acids	
L	Na^+ -independent	Branched-chain and aromatic amino acids	
N	Na^+	Nitrogen-containing side chains (Gln, Asn, His, etc.)	
γ^+	Na^+ -independent	Cationic amino acids	Hepatocytes Chinese hamster ovary cells
x_{AG}^-	Na^+	Aspartate and glutamate	
P	Na^+	Proline	

Adapted from: Collarini, E. J., and Oxender, D. L., 1987. Mechanisms of transport of amino acids across membranes. *Annual Review of Nutrition* 7:75–90.

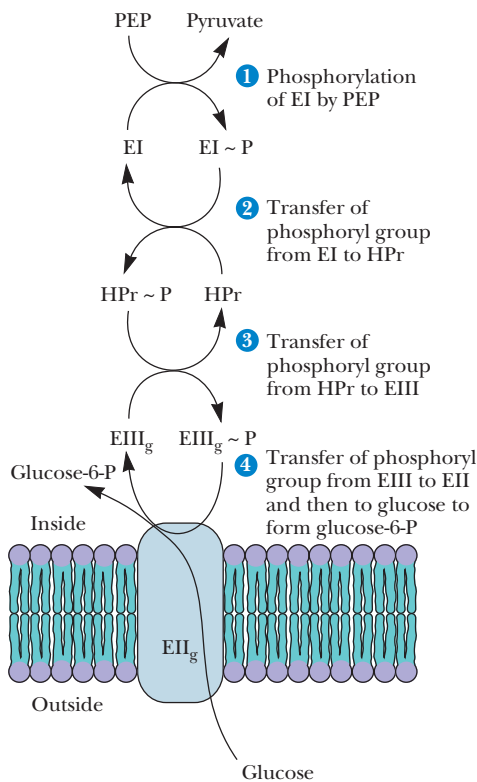
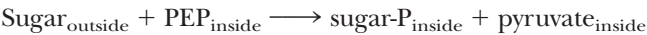


FIGURE 10.26 • Glucose transport in *E. coli* is mediated by the PEP-dependent phosphotransferase system. Enzyme I is phosphorylated in the first step by PEP. Successive phosphoryl transfers to HPr and Enzyme III in Steps 2 and 3 are followed by transport and phosphorylation of glucose. Enzyme II is the sugar transport channel.

resents an important metabolic process. Thus, plasma membrane transport of alanine is the rate-limiting step in hepatic alanine metabolism. This system is normally expressed at low levels in the liver, but substrate deprivation and hormonal activation both stimulate System A expression.

10.7 • Group Translocation

Certain bacteria possess a novel and versatile system for the inward transport of certain sugars. In this process, the sugar becomes phosphorylated during its transport across the membrane; that is, transport and phosphorylation are tightly coupled. This type of process, in which a chemical modification accompanies transport, has been denoted **group translocation**. Several such systems are known, but the best understood is the **phosphoenolpyruvate:glucose phosphotransferase system**, or simply the **phosphotransferase system** (or **PTS**), discovered by Saul Roseman of Johns Hopkins University in 1964. The advantage of this system lies in the fact that the sugars, once phosphorylated, are trapped in the cell. Membranes are permeable to simple sugars but impermeable to sugar phosphates, which are negatively charged. The overall reaction for the phosphotransferase is:



The subscripts illustrate an important point: the phosphoryl transfer occurs entirely on the inside surface of the bacterial membrane.

Several unique features distinguish the phosphotransferase. First, phosphoenolpyruvate is both the phosphoryl donor and the energy source for sugar transport. Second, four different proteins are required for this transport. Two of these proteins (**Enzyme I** and **HPr**) are general and are required for the phosphorylation of all PTS-transported sugars. The other two proteins (**Enzyme II** and **Enzyme III**) are specific for the particular sugar to be transported.

The first step in the phosphotransferase reaction (Figure 10.26) is the phosphorylation of Enzyme I by PEP to form a reactive phosphohistidine intermediate (Figure 10.27). This is followed by phosphoryl transfer to a histidyl residue of HPr, followed by phosphorylation of Enzyme III. At the same time, the sugar to be transported is bound on the outside surface of the cell by Enzyme II, which constitutes the sugar transport channel. As the sugar is moved to the inside surface of the membrane, the phosphoryl group is transferred from Enzyme III to the sugar, forming the desired sugar phosphate, which is

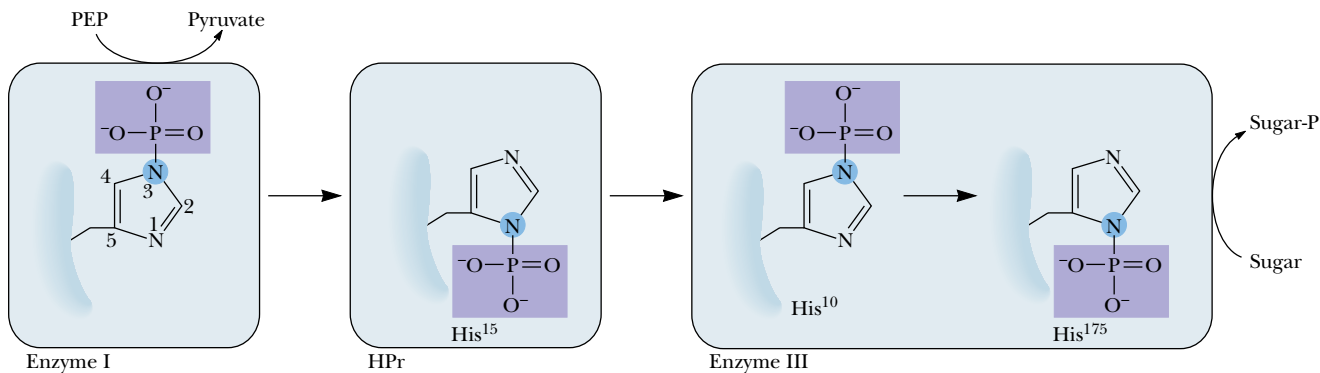


FIGURE 10.27 • The path of the phosphoryl group through the PTS mechanism. Reactive phosphohistidine intermediates of Enzyme I, HPr, and Enzyme III transfer phosphoryl groups from PEP to the transported sugar.

released into the cytoplasm. (In some cases, for example the *E. coli* mannitol system, no Enzyme III has been identified. In these cases, the C-terminal end of the relevant Enzyme II, which resembles an Enzyme III-type sequence, substitutes for Enzyme III.)

10.8 • Specialized Membrane Pores

Porins in Gram-Negative Bacterial Membranes

The membrane transport systems described previously (and many others like them) are relatively specific and function to transport either a single substrate or a very limited range of substrates under normal conditions. At the same time, several rather nonspecific systems also carry out transport processes. One such class of nonspecific transport proteins is found in the outer membranes of Gram-negative bacteria and mitochondria. Low-molecular-weight nutrients and certain other molecules, such as some antibiotics, cross this outer membrane, but larger molecules such as proteins cannot. The ability of the outer membrane to act as a molecular sieve is due to proteins called **porins** (Chapter 9). Alternatively, these molecules have been referred to as **peptidoglycan-associated proteins** or simply **matrix proteins**. **General porins** form nonspecific pores across the outer membrane and sort molecules according to molecular size, whereas **specific porins** contain binding sites for particular substrates. Porins from several organisms have been isolated and characterized (Table 10.3). Molecular masses of the porins generally range from 30 kD to 50 kD. Most (but not all) porins are arranged in the outer membrane as trimers of identical subunits. The molecular exclusion limits clearly depend on the size of the pore formed by the porin molecule. The pores formed by *E. coli* and *S. typhimurium* porins are relatively small, but porin F from *Pseudomonas aeruginosa* creates a much larger pore, with an exclusion limit of approximately 6 kD. Specific porins *LamB* and *Tsx* of *E. coli* and porins *P* and *DI* of *P. aeruginosa* possess specific binding sites for maltose and related oligosaccharides (Table 10.4), nucleosides, anions, and glucose, respectively.

Table 10.3		
Properties of Some General Porins		
Porin and Bacterial Source	Pore Diameter (nm)	M _r Exclusion Limit
<i>E. coli</i>		
OmpF	1.2	
OmpC	1.1	600
PhoE	1.2	
<i>S. typhimurium</i>		
M _r 38,000	1.4	
M _r 39,000	1.4	700
M _r 40,000	1.4	
<i>P. aeruginosa</i>		
F	2.2	6000

Source: Adapted from Benz, R., 1984. Structure and selectivity of porin channels. *Current Topics in Membrane Transport* **21**:199–219; and Benz, R., 1988. Structure and function of porins from Gram-negative bacteria. *Annual Review of Microbiology* **42**:359–393.

Table 10.4

Binding and Permeation Properties of the LamB Channel for Different Sugars

Sugar	K_s (mM)*	P (s ⁻¹)†
Maltose	10	100
Maltotriose	0.40	66
Maltoheptaose	0.067	2.5
Lactose	56	9
Sucrose	15	2.5
D-Glucose	110	290
L-Glucose	46	—
D-Galactose	42	225
D-Fructose	600	135
D-Mannose	160	160
Stachyose	50	<1

*Half-saturation constant (concentration for 50% saturation of the transport protein).

†Rate of permeation relative to that of maltose. Data adjusted to 100 s⁻¹ for maltose. The LamB-containing liposomes were added to buffer solutions containing 40 mM of the corresponding test sugars.

Source: Adapted from Benz, R. 1988. Structure and function of porins from Gram-negative bacteria. *Annual Review of Microbiology* 42:359–393.

Porins show high degrees of sequence homology and similarity. The most intriguing feature of porin secondary and tertiary structure is this: In contrast to nearly all other membrane proteins that adopt α -helical structures in the transmembrane segments, porins show little or no evidence of α -helical domains and segments. Instead, the porins and other outer membrane proteins adopt β -sheet structures for their membrane-spanning segments. Models of membrane insertion, which involve β -strands arranged perpendicular to the membrane plane, have been proposed for several porins (Figure 10.28). The crystal structure of the porin from *Rhodobacter capsulatus* shows a trimer in which each monomer forms a pore (Figure 10.29). The monomer pore consists of a 16-stranded β -barrel that traverses the membrane as a tube. The tube is nar-

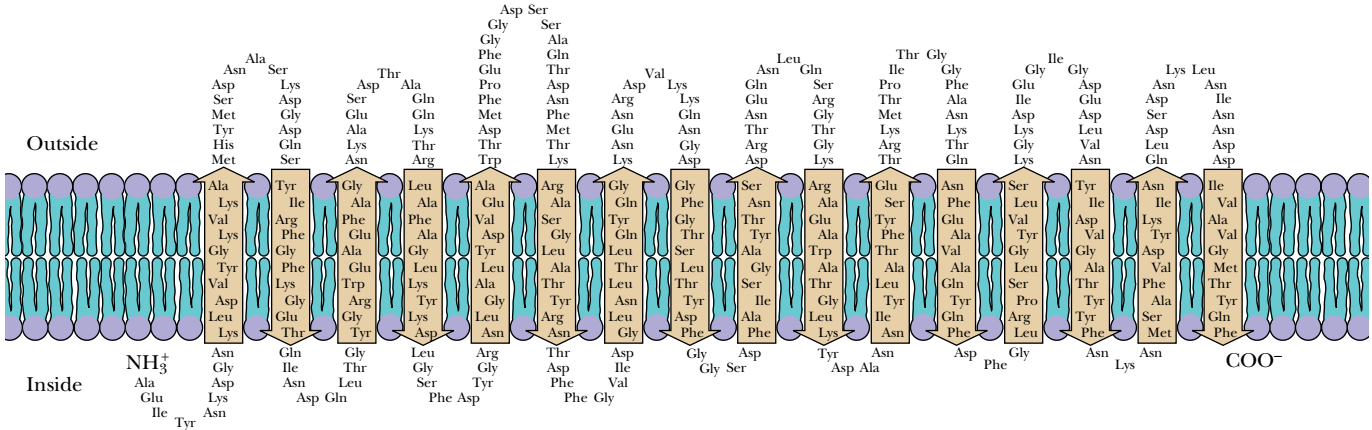


FIGURE 10.28 • A model for the arrangement of the porin PhoE in the outer membrane of *E. coli*. The transmembrane segments are strands of β -sheet.

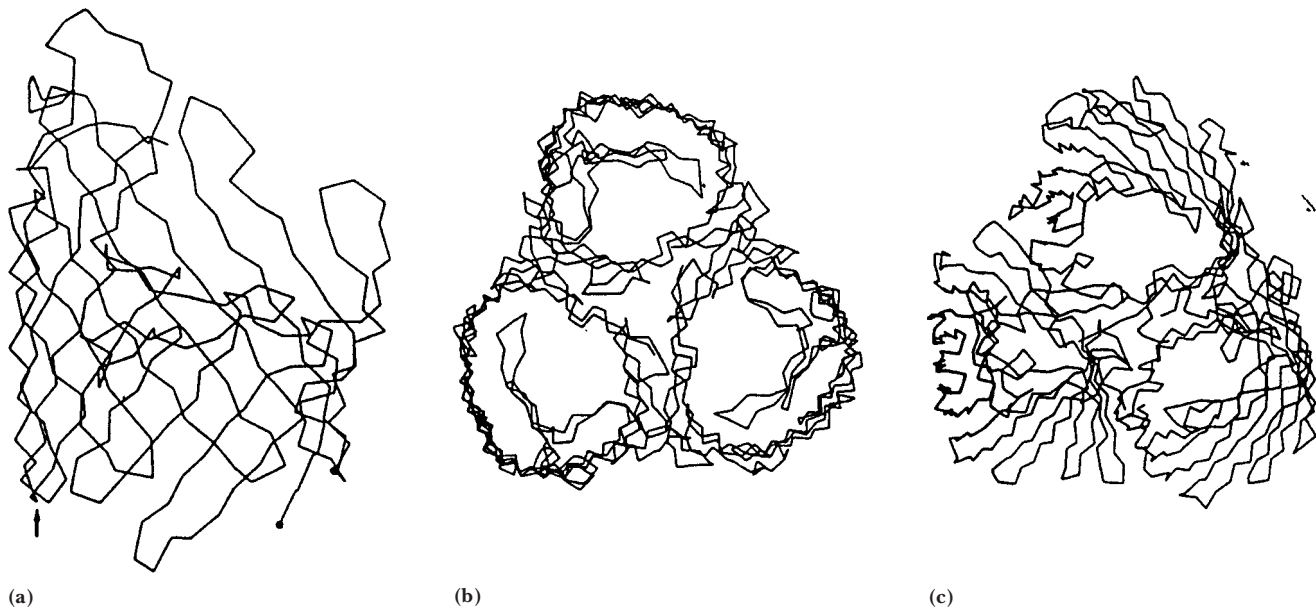


FIGURE 10.29 • Three-dimensional reconstruction of porin from *Rhodobacter capsulatus*. Drawings of (a) side view of porin monomer showing β -sheet structure. (b) Top view and (c) nearly top view of porin trimer.

rowed near the center by peptide chain segments protruding from the inner wall of the barrel. These chain segments form an “eyelet” about 1 nm long and 0.6 to 1 nm across. The eyelet is postulated to determine the exclusion limit for particles diffusing through the pore.

Porins and the other outer membrane proteins of Gram-negative bacteria appear to be the only known membrane proteins that have chosen the β -strand over the α -helix. Why might this be? Among other reasons, there is an advantage of genetic economy in the use of β -strands to traverse the membrane instead of α -helices. An α -helix requires 21 to 25 amino acid residues to span a typical biological membrane; a β -strand can cross the same membrane with 9 to 11 residues. Therefore, a given amount of genetic information could encode a larger number of membrane-spanning segments using a β -strand motif instead of α -helical arrays. Further, β -strands can present alternating hydrophobic and hydrophilic R groups along their length, with hydrophobic R groups facing the lipid bilayer and hydrophilic R groups facing the water-filled channel (Chapter 9).

The Pore-Forming Toxins

Many organisms produce lethal molecules known as **pore-forming toxins**, which insert themselves in a host cell’s plasma membrane to form a channel or pore. Pores formed by such toxins can kill the host cell by collapsing ion gradients or by facilitating the entry of toxic agents into the cell. Produced by a variety of organisms and directed toward a similarly diverse range of target cells, these toxins nonetheless share certain features in common. The structures of these remarkable toxins have provided valuable insights into the mechanisms of their membrane insertion and also into the architecture of membrane proteins.

Colicins are pore-forming proteins, produced by certain strains of *E. coli*, that kill or inhibit the growth of other, competing bacteria and even other strains of *E. coli* (a process known as *allelopathy*). Channel-forming colicins are released as soluble monomers. Upon encountering a host cell, the colicin molecule traverses the bacterial outer membrane and periplasm, then inserts itself

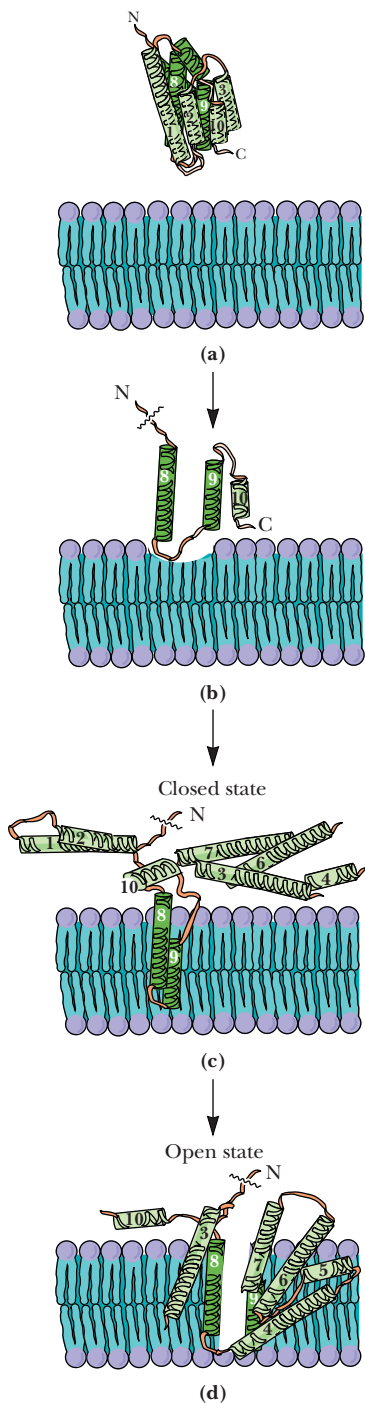


FIGURE 10.31 • The umbrella model of membrane channel protein insertion. Hydrophobic helices insert directly into the core of the membrane, with amphipathic helices arrayed on the surface like an open umbrella. A trigger signal (low pH or a voltage gradient) draws some of the amphipathic helices into and across the membrane, causing the pore to open.

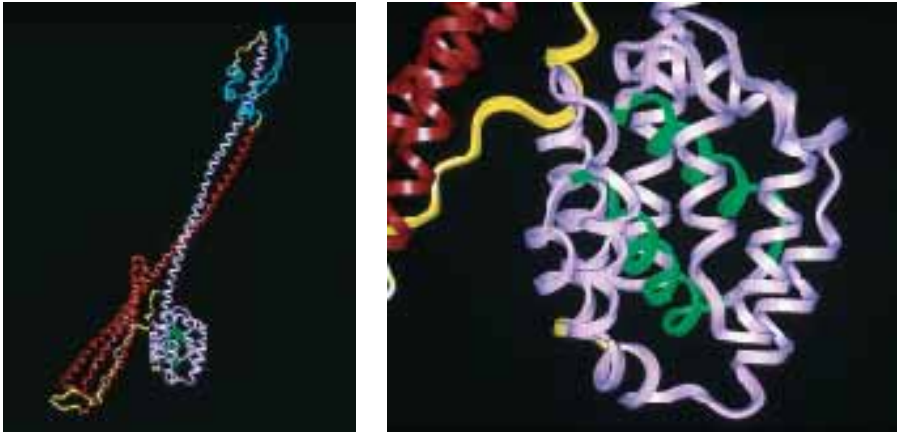


FIGURE 10.30 • The structure of colicin Ia. Colicin Ia, with a total length of 210 Å, spans the periplasmic space of a Gram-negative bacterium host, with the R (receptor-binding) domain (blue) anchored to proteins in the outer membrane and the C domain (violet) forming a channel in the inner membrane. The T (translocation) domain is shown in red. The image on the right shows details of the C domain, including helices 8 and 9 (green), which are highly hydrophobic.

into the inner (plasma) membrane. The channel thus formed is monomeric and a single colicin molecule can kill a host cell. The structure of colicin Ia, a 626-residue protein, is shown in Figure 10.30. It consists of three domains, termed the **T (translocation) domain**, the **R (receptor-binding) domain**, and the **C (channel-forming) domain**. The T domain mediates translocation across the outer membrane, the R domain binds to an outer-membrane receptor, and the C-domain creates a voltage-gated channel across the inner membrane. The T, R, and C domains are separated by long (160 Å) α-helical segments. The peptide is folded at the R domain, so that the C and T domains are juxtaposed and the two long helices form an underwound antiparallel coiled coil. The protein is unusually elongated—210 Å from end to end—with the T and C domains at one end and R at the other. This unusual design permits colicin Ia to span the periplasmic space (which has an average width of 150 Å) and insert in the inner membrane.

The nature of the channel-forming domain provides clues to the process of channel formation in the inner membrane. The C domain consists of a 10-helix bundle, with helices 8 and 9 forming an unusually hydrophobic hairpin structure. The other eight helices are amphipathic and serve to stabilize hydrophobic helices 8 and 9 in solution. When this domain inserts in the inner membrane, helices 8 and 9 inject themselves into the hydrophobic membrane core, leaving the other helices behind on the membrane surface (Figure 10.31). Application of a transmembrane potential (voltage) then triggers the amphipathic helices to insert into the membrane, with their hydrophobic faces facing the hydrophobic bilayer and their polar faces forming the channel surface. This model is hypothetical, but it is supported by studies showing that channel opening involves dramatic structural changes and that helices 2 to 5 move across the membrane during channel opening.

Interestingly, certain other pore-forming toxins possess helix-bundle motifs that may participate in channel formation, in a manner similar to that proposed for colicin Ia. For example, the **δ-endotoxin** produced by *Bacillus thuringiensis* is toxic to Coleoptera insects (beetles) and is composed of three domains, including a seven-helix bundle, a three-sheet domain, and a β-sandwich. In the seven-helix bundle, helix 5 is highly hydrophobic, and the other six helices are amphipathic. In solution (Figure 10.32), the six amphipathic

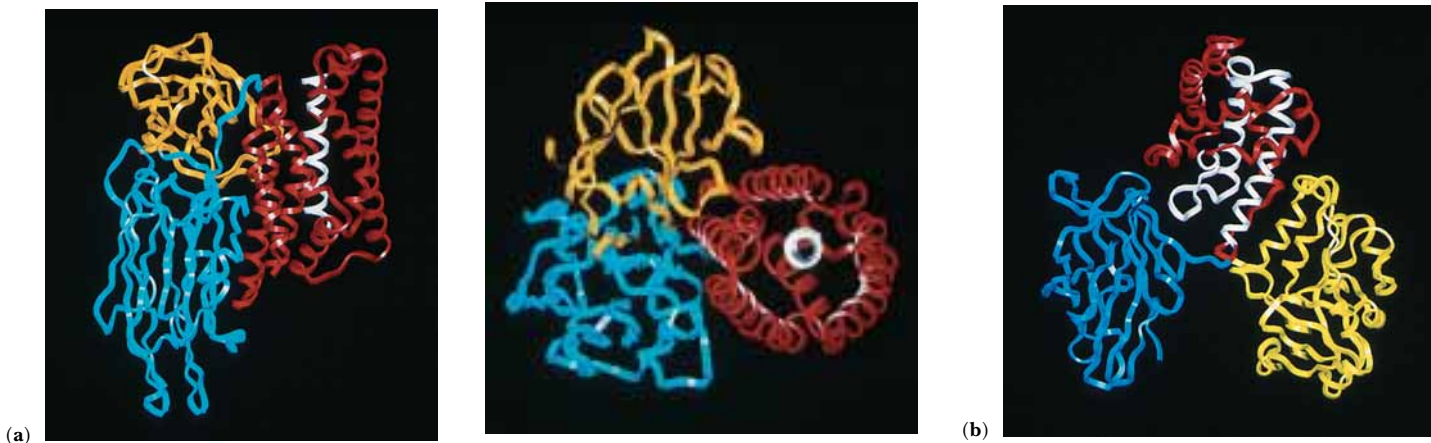


FIGURE 10.32 • The structures of (a) δ -endotoxin (two views) from *Bacillus thuringiensis* and (b) diphtheria toxin from *Corynebacterium diphtheriae*. Each of these toxins possesses a bundle of α -helices which is presumed to form the transmembrane channel when the toxin is inserted across the host membrane. In δ -endotoxin, helix 5 (white) is surrounded by 6 helices (red) in a 7-helix bundle. In diphtheria toxin, three hydrophobic helices (white) lie at the center of the transmembrane domain (red).

helices surround helix 5, with their nonpolar faces apposed to helix 5 and their polar faces directed to the solvent. Membrane insertion and channel formation may involve initial insertion of helix 5, as in Figure 10.31, followed by insertion of the amphipathic helices, so that their nonpolar faces contact the bilayer lipids and their polar faces line the channel.

There are a number of other toxins for which the helical channel model is inappropriate. These include **α -hemolysin** from *Staphylococcus aureus*, **aerolysin** from *Aeromonas hydrophila*, and the **anthrax toxin protective antigen** from *Bacillus anthracis*. The membrane-spanning domains of these proteins do not possess long stretches of hydrophobic residues that could form α -helical transmembrane segments. They do, however, contain substantial peptide segments of alternating hydrophobic and polar residues. Like the porins, such segments can adopt β -strand structures, such that one side of the β -strand is hydrophobic and the other side is polar. Oligomeric association of several such segments can produce a β -barrel motif, with the inside of the barrel lined with polar residues and the outside of the barrel coated with hydrophobic residues—a motif that can be accommodated readily in a bilayer membrane, creating a polar transmembrane channel.

α -Hemolysin, a 33.2-kD monomer protein, forms a mushroom-shaped heptameric pore, 100 Å in length, with a diameter that ranges from 14 Å to 46 Å (Figure 10.33). In this structure, each monomer contributes two β -strands 65 Å long, which are connected by a hairpin turn. The interior of the 14-stranded β -barrel structure is hydrophilic and the hydrophobic outer surface of the barrel is 28 Å wide. Pores formed by α -hemolysin in human erythrocytes, platelets, and lymphocytes allow rapid Ca^{2+} influx into these cells with toxic consequences.

Aeromonas hydrophila is a bacterium that causes diarrheal diseases and deep wound infections. These complications arise due to pore formation in sensitive cells by the protein toxin aerolysin. Proteolytic processing of the 52-kD precursor **proaerolysin** (Figure 10.34) produces the toxic form of the protein, aerolysin. Like α -hemolysin, aerolysin monomers associate to form a heptameric transmembrane pore. Michael Parker and coworkers have proposed

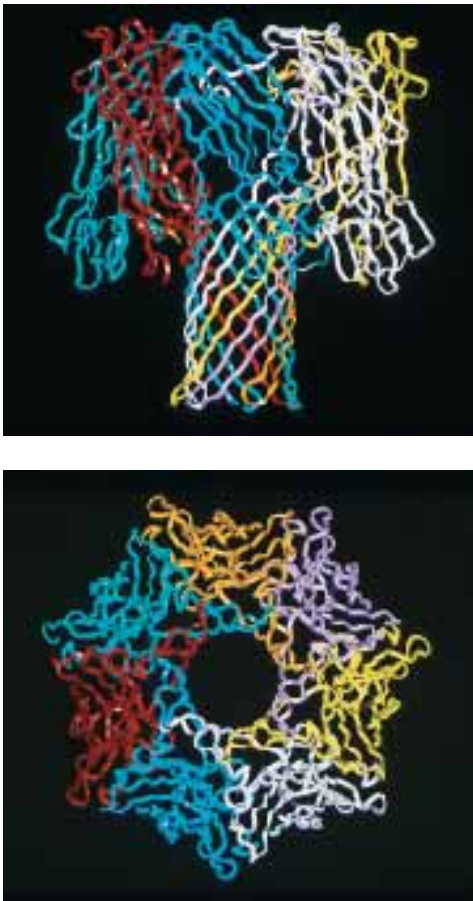


FIGURE 10.33 • The structure of the heptameric channel formed by α -hemolysin. Each of the seven subunits contributes a β -sheet hairpin to the transmembrane channel.



FIGURE 10.34 • The structure of proaerolysin, produced by *Aeromonas hydrophila*. Proteolysis of this precursor yields the active form, aerolysin, which is responsible for the pathogenic effects of the bacterium in deep wound infections and diarrheal diseases. Like hemolysin, aerolysin monomers associate to form heptameric membrane pores. The three β -strands that contribute to the formation of the heptameric pore are shown in red. The N-terminal domain (residues 1–80, yellow) is a small lobe that protrudes from the rest of the protein.

Alamethicin I¹:
Ac-Aib-Pro-Aib-Ala-Aib-Ala-Gln-Aib-Val-Aib-Gly-Leu-Aib-Pro-Val-Aib-Aib-Glu-Gln-Phol*

Cecropin A:
Lys-Trp-Lys-Leu-Phe-Lys-Lys-Ile-Glu-Lys-Val-Gly-Gln-Asn-Ile-Arg-Asp-Gly-Ile-Ile-Lys-Ala-Gly-Pro-Ala-Val-Ala-Val-Val-Gly-Gln-Ala-Thr-Gln-Ile-Ala-Lys-NH₂

Melittin:
Gly-Ile-Gly-Ala-Val-Leu-Lys-Val-Leu-Thr-Thr-Gly-Leu-Pro-Ala-Leu-Ile-Ser-Trp-Ile-Lys-Arg-Lys-Arg-Gln-Gln-NH₂

Magainin 2 amide:
Gly-Ile-Gly-Lys-Phe-Leu-His-Ser-Ala-Lys-Lys-Phe-Gly-Lys-Ala-Phe-Val-Gly-Glu-Ile-Met-Asn-Ser-NH₂

FIGURE 10.35 • The amino acid sequences of several amphipathic peptide antibiotics. α -Helices formed from these peptides cluster polar residues on one face of the helix, with nonpolar residues at other positions.



that each monomer in this aggregate contributes three β -strands to the β -barrel pore. Each of these β -strands (residues 277 to 287, 290 to 302, and 410 to 422) consists of alternating hydrophobic and polar residues, so that the pore once again places polar residues toward the water-filled channel and nonpolar residues facing the lipid bilayer.

Whether crossing the membrane with aggregates of amphipathic α -helices or β -barrels, these pore-forming toxins represent Nature’s accommodation to a structural challenge facing all protein-based transmembrane channels: the need to provide hydrogen-bonding partners for the polypeptide backbone N—H and C=O groups in an environment (the bilayer interior) that lacks hydrogen-bond donors or acceptors. The solution to this problem is found, of course, in the extensive hydrogen-bonding possibilities of α -helices and β -sheets.

Amphipathic Helices Form Transmembrane Ion Channels

Recently, a variety of natural peptides that form transmembrane channels have been identified and characterized. Melittin (Figure 10.35) is a bee venom toxin peptide of 26 residues. The cecropins are peptides induced in *Hyalophora cecropia* (Figure 10.36) and other related silkworms when challenged by bacterial infections. These peptides are thought to form α -helical aggregates in mem-

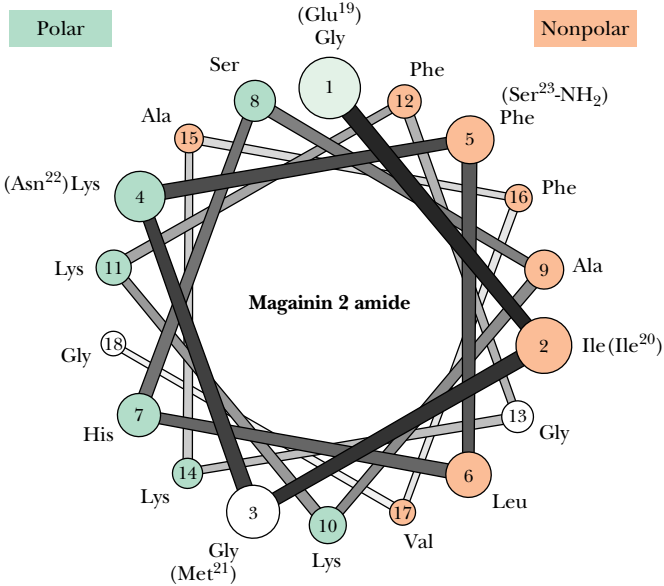




FIGURE 10.36 • Adult (*left*) and caterpillar (*right*) stages of the cecropia moth, *Hyalophora cecropia*. (*left*, Greg Neise/Visuals Unlimited; *right*, Patti Murray/Animals, Animals)

branes, creating an ion channel in the center of the aggregate. The unifying feature of these helices is their **amphipathic** character, with polar residues clustered on one face of the helix and nonpolar residues elsewhere. In the membrane, the polar residues face the ion channel, leaving the nonpolar residues elsewhere on the helix to interact with the hydrophobic interior of the lipid bilayer.

A D E E P E R L O O K

Melittin—How to Sting Like a Bee

The stings of many stinging insects, like wasps, hornets, and bumblebees, cause a pain that, although mild at first, increases in intensity over 2 to 30 minutes, with a following period of swelling that may last for several days. The sting of the honey bee (*Apis mellifera*), on the other hand, elicits a sharp, stabbing pain within 10 seconds. This pain may last for several minutes and is followed by several hours of swelling and itching. The immediate, intense pain is caused by **melittin**, a 26-residue peptide that constitutes about half of the 50 μg (dry weight) of material injected during the “sting” (in a total volume of only 0.5 μL). How does this simple peptide cause the intense pain that accompanies a bee sting?

The pain appears to arise from the formation of melittin pores in the membranes of **nociceptors**, free nerve endings that detect harmful (“noxious”—thus the name) stimuli of violent mechanical stress, high temperatures, and irritant chemicals. The creation of pores by melittin depends on the nociceptor membrane potential. Melittin in water solution is tetrameric. However, melittin interacting with membranes in the absence of a membrane potential is monomeric and shows no evidence of oligomer

formation. When an electrical potential (voltage) is applied across the membrane, melittin tetramers form and the membrane becomes permeable to anions such as chloride. Nociceptor membranes maintain a resting potential of 70 mV (negative inside). When melittin binds to the nociceptor membrane, the flow of chloride ions out of the cell diminishes the transmembrane potential, stimulating the nerve and triggering a pain response and also inducing melittin tetramers to dissociate. When the membrane potential is re-established, melittin tetramers reform and the cycle is repeated over and over, causing a prolonged and painful stimulation of the nociceptors. The pain of the sting eventually lessens, perhaps due to the molecules of melittin diffusing apart, so that tetramers can no longer form.

Although the honey bee’s sting is unpleasant, this tiny creature is crucial to the world’s agricultural economy. Honey bees produce more than \$100 million worth of honey each year, and, more importantly, the pollination of numerous plants by honey bees is responsible for the production of \$20 billion worth of crops in the United States alone.

Gap Junctions in Mammalian Cell Membranes

When cells lie adjacent to each other in animal tissues, they are often connected by **gap junction** structures, which permit the passive flow of small molecules from one cell to the other. Such junctions essentially connect the cells metabolically, providing a means of chemical transfer and communication. In certain tissues, such as heart muscle that is not innervated, gap junctions permit very large numbers of cells to act synchronously. Gap junctions also provide a means for transport of nutrients to cells disconnected from the circulatory system, such as the lens cells of the eye.

Gap junctions are formed from hexameric arrays of a single 32-kD protein. Each subunit of the array is cylindrical, with a length of 7.5 nm and a diameter of 2.5 nm. The subunits of the hexameric array are normally tilted with respect to the sixfold axis running down the center of the hexamer (Figure 10.37). In this conformation, a central pore having a diameter of about 1.8 to 2.0 nm is created, and small molecules (up to masses of 1 kD to 1.2 kD) can pass through unimpeded. Proteins, nucleic acids, and other large structures cannot. A complete gap junction is formed from two such hexameric arrays, one from each cell. A twisting, sliding movement of the subunits narrows the channel and closes the gap junction. This closure is a cooperative process, and a localized conformation change at the cytoplasmic end assists in the closing of the channels. Because the closing of the gap junction does not appear to involve massive conformational changes in the individual subunits, the free energy change for closure is small.

Although gap junctions allow cells to communicate metabolically under normal conditions, the ability to close gap junctions provides the tissue with an important intercellular regulation mechanism. In addition, gap junctions provide a means to protect adjacent cells if one or more cells are damaged or

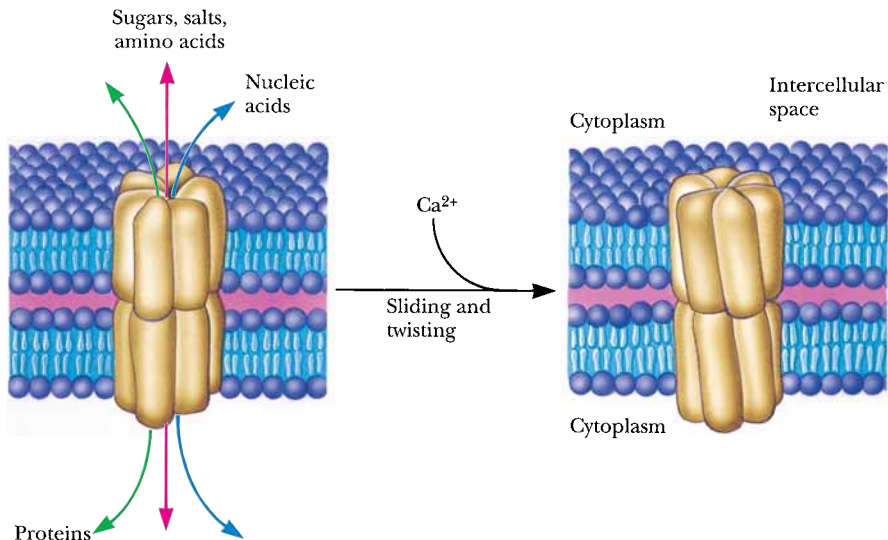


FIGURE 10.37 • Gap junctions consist of hexameric arrays of cylindrical protein subunits in the plasma membrane. The subunit cylinders are tilted with respect to the axis running through the center of the gap junction. A gap junction between cells is formed when two hexameric arrays of subunits in separate cells contact each other and form a pore through which cellular contents may pass. Gap junctions close by means of a twisting, sliding motion in which the subunits decrease their tilt with respect to the central axis. Closure of the gap junction is Ca^{2+} -dependent.

stressed. To these ends, gap junctions are sensitive to membrane potentials, hormonal signals, pH changes, and intracellular calcium levels. Dramatic changes in pH or Ca^{2+} concentration in a cell may be a sign of cellular damage or death. In order to protect neighboring cells from the propagation of such effects, gap junctions close in response to decreased pH or prolonged increases in intracellular Ca^{2+} . Under normal conditions of intracellular Ca^{2+} levels ($<10^{-7} \text{ M}$), gap junctions are open and intercellular communication is maintained. When calcium levels rise to 10^{-5} M or higher, the junctions, sensing danger, rapidly close.

10.9 • Ionophore Antibiotics

All of the transport systems examined thus far are relatively large proteins. Several small molecule toxins produced by microorganisms facilitate ion transport across membranes. Due to their relative simplicity, these molecules, the **ionophore antibiotics**, represent paradigms of the **mobile carrier** and **pore** or **channel** models for membrane transport. Mobile carriers are molecules that form complexes with particular ions and diffuse freely across a lipid membrane (Figure 10.38). Pores or channels, on the other hand, adopt a fixed orientation in a membrane, creating a hole that permits the transmembrane movement of ions. These pores or channels may be formed from monomeric or (more often) multimeric structures in the membrane.

Carriers and channels may be distinguished on the basis of their temperature dependence. Channels are comparatively insensitive to membrane phase transitions and show only a slight dependence of transport rate on temperature. Mobile carriers, on the other hand, function efficiently above a membrane phase transition, but only poorly below it. Consequently, mobile carrier systems often show dramatic increases in transport rate as the system is heated through its phase transition. Figure 10.39 displays the structures of several of these interesting molecules. As might be anticipated from the variety of structures represented here, these molecules associate with membranes and facilitate transport by different means.

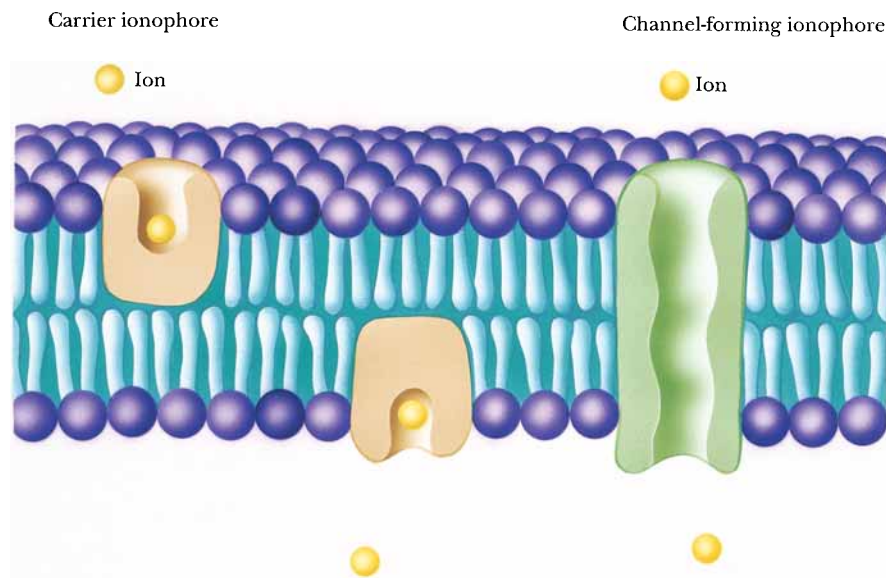


FIGURE 10.38 • Schematic drawings of mobile carrier and channel ionophores. Carrier ionophores must move from one side of the membrane to the other, acquiring the transported species on one side and releasing it on the other side. Channel ionophores span the entire membrane.

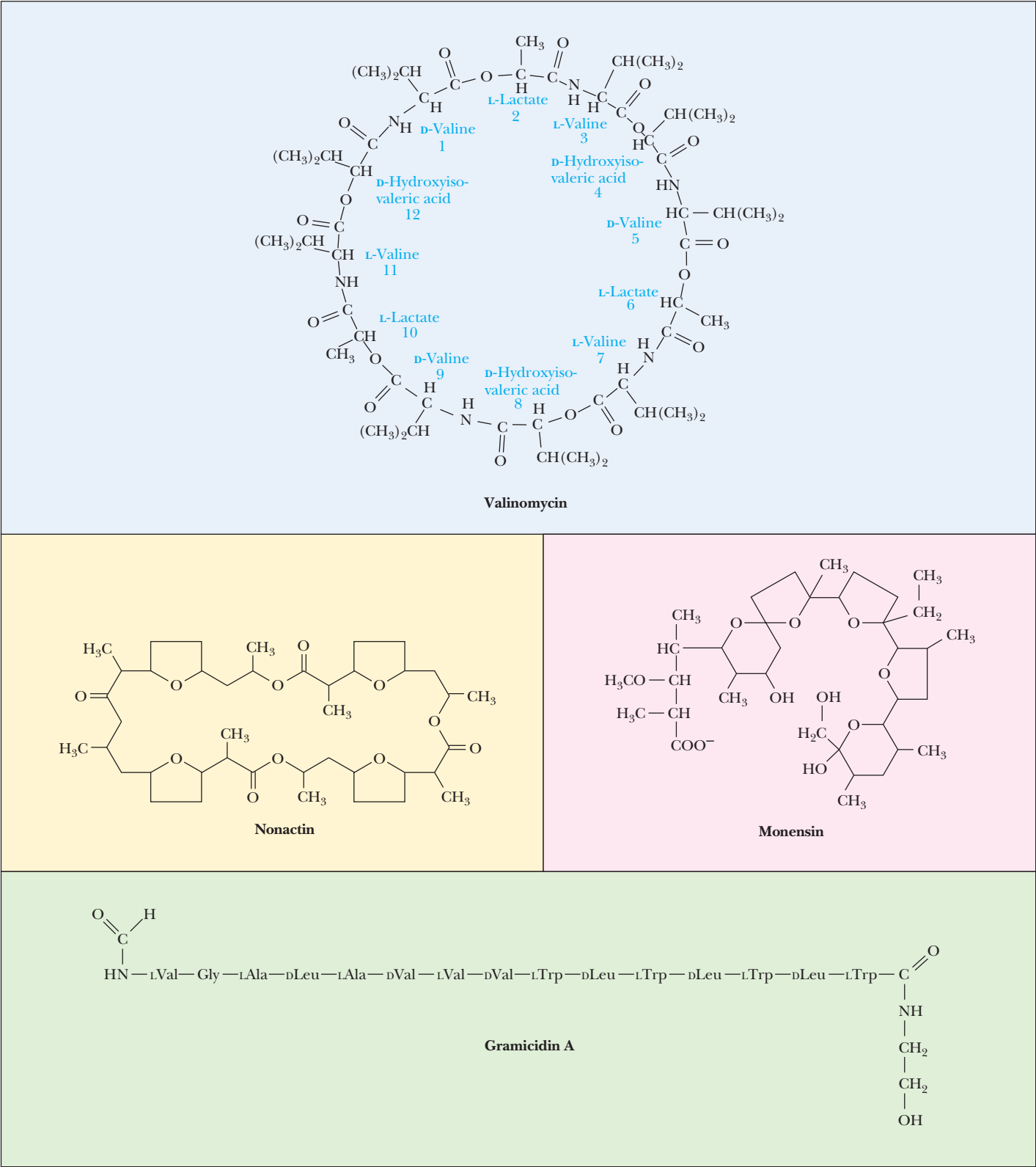


FIGURE 10.39 • Structures of several ionophore antibiotics. Valinomycin consists of three repeats of a four-unit sequence. Because it contains both peptide and ester bonds, it is referred to as a depsipeptide.

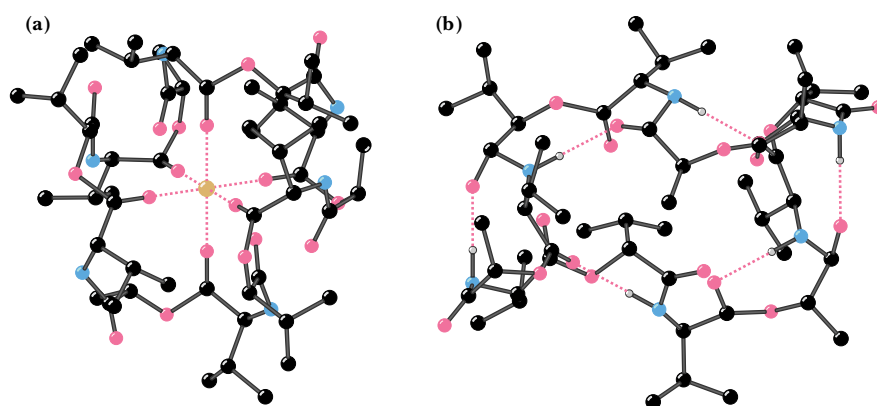


FIGURE 10.40 • The structures of (a) the valinomycin- K^+ complex and (b) uncomplexed valinomycin.

Valinomycin Is a Mobile Carrier Ionophore

Valinomycin (isolated from *Streptomyces fulvissimus*) is a cyclic structure containing 12 units made from four different residues. Two are amino acids (L-valine and D-valine); the other two residues, L-lactate and D-hydroxyisovalerate, contribute ester linkages. Valinomycin is a **depsipeptide**, that is, a molecule with both peptide and ester bonds. (Considering the 12 units in the structure, valinomycin is called a **dodecadepsipeptide**.) Valinomycin consists of the 4-unit sequence (D-valine, L-lactate, L-valine, D-hydroxyisovalerate), repeated three times to form the cyclic structure in Figure 10.39. The structures of uncomplexed valinomycin and the K^+ -valinomycin complex have been studied by X-ray crystallography (Figure 10.40). The structure places K^+ at the center of the valinomycin ring, coordinated with the carbonyl oxygens of the 6 valines. The polar groups of the valinomycin structure are positioned toward the center of the ring, whereas the nonpolar groups (the methyl and isopropyl side chains) are directed outward from the ring. The hydrophobic exterior of valinomycin interacts favorably with low dielectric solvents and with the hydrophobic interiors of lipid bilayers. Moreover, the central carbonyl groups completely surround the K^+ ion, shielding it from contact with nonpolar solvents or the hydrophobic membrane interior. As a result, the K^+ -valinomycin complex freely diffuses across biological membranes and effects rapid, passive K^+ transport (up to 10,000 K^+ /sec) in the presence of K^+ gradients.

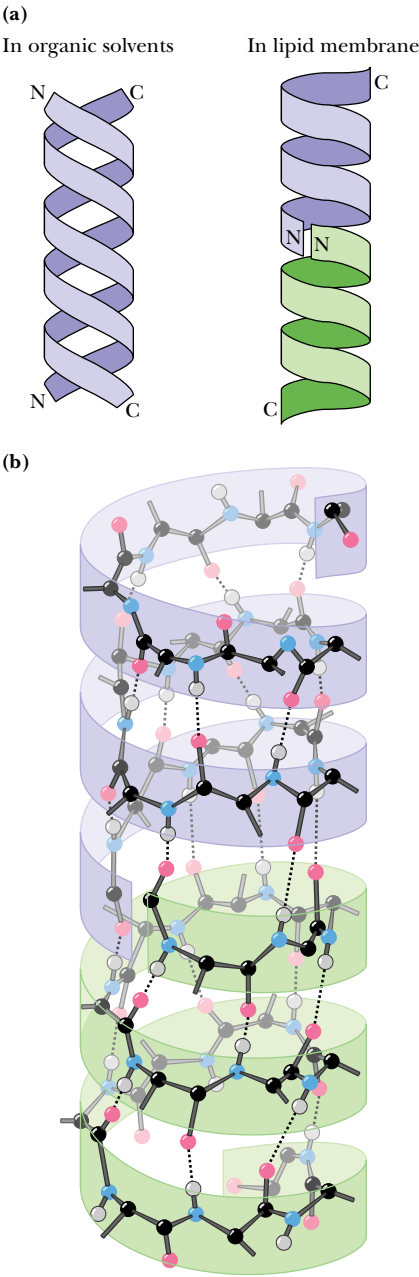
Valinomycin displays a striking selectivity with respect to monovalent cation binding. It binds K^+ and Rb^+ tightly, but shows about a thousandfold lower affinity for Na^+ and Li^+ . The smaller ionic radii of Na^+ and Li^+ (compared to K^+ and Rb^+) may be responsible in part for the observed differences. However, another important difference between Na^+ and K^+ is shown in Table 10.5. The **free energy of hydration** for an ion is the stabilization achieved by hydrating that ion. The process of dehydration, a prerequisite to forming the ion-valinomycin complex, requires energy input. As shown in Table 10.5, considerably more energy is required to desolvate a Na^+ ion than to desolvate a K^+ ion. It is thus easier to form the K^+ -valinomycin complex than to form the corresponding Na^+ complex.

Other mobile carrier ionophores include *monensin* and *nonactin* (Figure 10.39). The unifying feature in all these structures is an inward orientation of polar groups (to coordinate the central ion) and outward orientation of non-

Table 10.5

Properties of Alkali Cations

Ion	Atomic Number	Ionic Radius (nm)	Hydration Free Energy, ΔG (kJ/mol)
Li ⁺	3	0.06	−410
Na ⁺	11	0.095	−300
K ⁺	19	0.133	−230
Rb ⁺	37	0.148	−210
Cs ⁺	55	0.169	−200



polar residues (making these complexes freely soluble in the hydrophobic membrane interior).

Gramicidin Is a Channel-Forming Ionophore

In contrast to valinomycin, all protein-derived membrane transport systems appear to function as channels, not mobile carriers. All of the proteins discussed in this chapter use multiple transmembrane segments to create channels in the membrane, through which species are transported. For this reason, it may be more relevant to consider the **pore** or **channel** ionophores. **Gramicidin** from *Bacillus brevis* (Figure 10.41) is a linear peptide of 15 residues and is a prototypical channel ionophore. Gramicidin contains alternating L- and D-residues, a formyl group at the N-terminus, and an ethanolamine at the C-terminus. The predominance of hydrophobic residues in the gramicidin structure facilitates its incorporation into lipid bilayers and membranes. Once incorporated in lipid bilayers, it permits the rapid diffusion of many different cations. Gramicidin possesses considerably less ionic specificity than does valinomycin, but permits higher transport rates. A single gramicidin channel can transport as many as 10 million K⁺ ions per second. Protons and all alkali cations can diffuse through gramicidin channels, but divalent cations such as Ca²⁺ block the channel.

Gramicidin forms two different helical structures. A double helical structure predominates in organic solvents (Figure 10.41), whereas a helical dimer is formed in lipid membranes. (An α -helix cannot be formed by gramicidin, because it has both D- and L-amino acid residues.) The helical dimer is a head-to-head or amino terminus-to-amino terminus (N-to-N) dimer oriented perpendicular to the membrane surface, with the formyl groups at the bilayer center and the ethanolamine moieties at the membrane surface. The helix is unusual, with 6.3 residues per turn and a central hole approximately 0.4 nm in diameter. The hydrogen-bonding pattern in this structure, in which N—H groups alternately point up and down the axis of the helix to hydrogen-bond with carbonyl groups, is reminiscent of a β -sheet. For this reason this structure has often been referred to as a β -helix.

FIGURE 10.41 • (a) Gramicidin forms a double helix in organic solvents; a helical dimer is the preferred structure in lipid bilayers. The structure is a head-to-head, left-handed helix, with the carboxy-termini of the two monomers at the ends of the structure. (b) The hydrogen-bonding pattern resembles that of a parallel β -sheet.

PROBLEMS

- 1. Calculate the free energy difference at 25°C due to a galactose gradient across a membrane, if the concentration on side 1 is 2 mM and the concentration on side 2 is 10 mM.
- 2. Consider a phospholipid vesicle containing 10 mM Na⁺ ions. The vesicle is bathed in a solution that contains 52 mM Na⁺ ions, and the electrical potential difference across the vesicle membrane $\Delta\psi = \psi_{\text{outside}} - \psi_{\text{inside}} = -30$ mV. What is the electrochemical potential at 25°C for Na⁺ ions?
- 3. Transport of histidine across a cell membrane was measured at several histidine concentrations:

[Histidine], μM	Transport, $\mu\text{mol/min}$
2.5	42.5
7	119
16	272
31	527
72	1220

Does this transport operate by passive diffusion or by facilitated diffusion?

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- 4. Fructose is present outside a cell at 1 μM concentration. An active transport system in the plasma membrane transports fructose into this cell, using the free energy of ATP hydrolysis to drive fructose uptake. Assume that one fructose is transported per ATP hydrolyzed, that ATP is hydrolyzed on the intracellular surface of the membrane, and that the concentrations of ATP, ADP, and P_i are 3 mM, 1 mM, and 0.5 mM, respectively. T = 298 K. What is the highest intracellular concentration of fructose that this transport system can generate? (*Hint*: Refer to Chapter 3 to recall the effects of concentration on free energy of ATP hydrolysis.)
- 5. The rate of K⁺ transport across bilayer membranes reconstituted from dipalmitoylphosphatidylcholine (DPPC) and nigericin is approximately the same as that observed across membranes reconstituted from DPPC and *cecropin a* at 35°C. Would you expect the transport rates across these two membranes also to be similar at 50°C? Explain.
- 6. In this chapter, we have examined coupled transport systems that rely on ATP hydrolysis, on primary gradients of Na⁺ or H⁺, and on phosphotransferase systems. Suppose you have just discovered an unusual strain of bacteria that transports rhamnose across its plasma membrane. Suggest experiments that would test whether it was linked to any of these other transport systems.

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Chapter 11

Nucleotides and Nucleic Acids



Francis Crick and James Watson point out features of their model for the structure of DNA. (© A. Barrington Brown/Science Source/Photo Researchers, Inc.)

Nucleotides and **nucleic acids** are biological molecules that possess heterocyclic nitrogenous bases as principal components of their structure. The biochemical roles of nucleotides are numerous; they participate as essential intermediates in virtually all aspects of cellular metabolism. Serving an even more central biological purpose are the nucleic acids, the elements of heredity and the agents of genetic information transfer. Just as proteins are linear polymers of amino acids, nucleic acids are linear polymers of nucleotides. Like the letters in this sentence, the orderly sequence of nucleotide residues in a nucleic acid can encode information. The two basic kinds of nucleic acids are **deoxyribonucleic acid (DNA)** and **ribonucleic acid (RNA)**. Complete hydrolysis of nucleic acids liberates nitrogenous bases, a five-carbon sugar, and phosphoric acid in equal amounts. The five-carbon sugar in DNA is 2-deoxyribose; in RNA,

We have discovered the secret of life!

Proclamation by Francis H. C. Crick to patrons of The Eagle, a pub in Cambridge, England (1953)

OUTLINE

- 11.1 • Nitrogenous Bases
- 11.2 • The Pentoses of Nucleotides and Nucleic Acids
- 11.3 • Nucleosides Are Formed by Joining a Nitrogenous Base to a Sugar
- 11.4 • Nucleotides Are Nucleoside Phosphates
- 11.5 • Nucleic Acids Are Polynucleotides
- 11.6 • Classes of Nucleic Acids
- 11.7 • Hydrolysis of Nucleic Acids

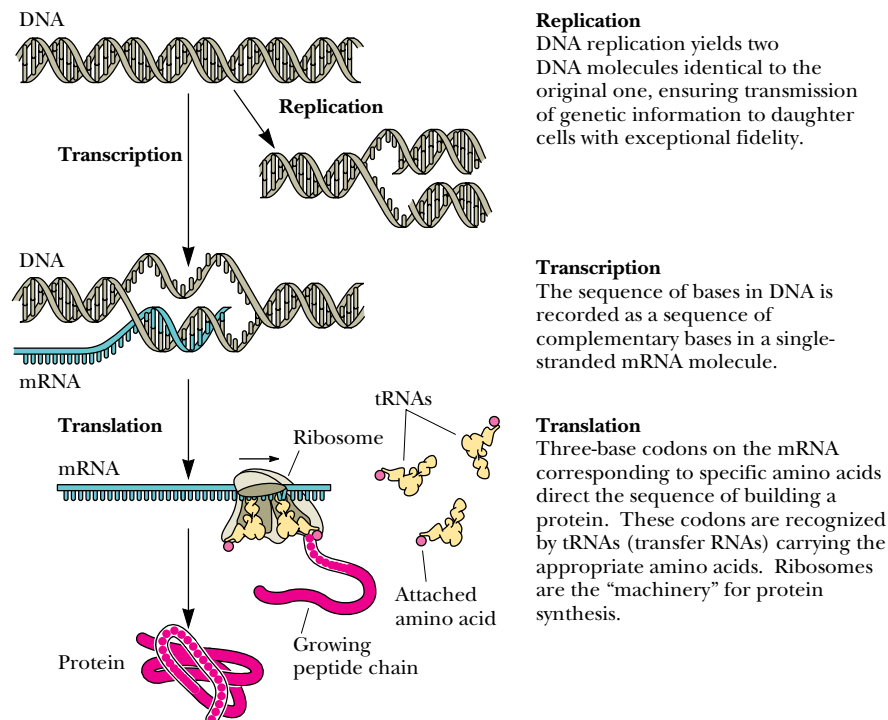


FIGURE 11.1 • The fundamental process of information transfer in cells. Information encoded in the nucleotide sequence of DNA is transcribed through synthesis of an RNA molecule whose sequence is dictated by the DNA sequence. As the sequence of this RNA is read (as groups of three consecutive nucleotides) by the protein synthesis machinery, it is translated into the sequence of amino acids in a protein. This information transfer system is encapsulated in the dogma: DNA → RNA → protein.

it is ribose. (See Chapter 7 for a detailed discussion of sugars and other carbohydrates.) DNA is the repository of genetic information in cells, while RNA serves in the transcription and translation of this information (Figure 11.1). An interesting exception to this rule is that some viruses have their genetic information stored as RNA.

This chapter describes the chemistry of nucleotides and the major classes of nucleic acids. Chapter 12 presents methods for determination of nucleic acid primary structure (nucleic acid sequencing) and describes the higher orders of nucleic acid structure. Chapter 13 introduces the *molecular biology of recombinant DNA*: the construction and uses of novel DNA molecules assembled by combining segments from other DNA molecules.

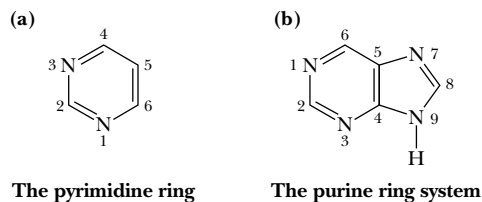


FIGURE 11.2 • (a) The pyrimidine ring system; by convention, atoms are numbered as indicated. (b) The purine ring system, atoms numbered as shown.

11.1 • Nitrogenous Bases

The bases of nucleotides and nucleic acids are derivatives of either **pyrimidine** or **purine**. Pyrimidines are six-membered heterocyclic aromatic rings containing two nitrogen atoms (Figure 11.2a). The atoms are numbered in a clockwise fashion, as shown in the figure. The purine ring structure is represented by the combination of a pyrimidine ring with a five-membered imidazole ring to yield a fused ring system (Figure 11.2b). The nine atoms in this system are numbered according to the convention shown.

The pyrimidine ring system is planar, while the purine system deviates somewhat from planarity in having a slight pucker between its imidazole and pyrimidine portions. Both are relatively insoluble in water, as might be expected from their pronounced aromatic character.

Common Pyrimidines and Purines

The common naturally occurring pyrimidines are **cytosine**, **uracil**, and **thymine** (5-methyluracil) (Figure 11.3). Cytosine and thymine are the pyrimidines typically found in DNA, whereas cytosine and uracil are common in RNA. To view this generality another way, the uracil component of DNA occurs as the 5-methyl variety, thymine. Various pyrimidine derivatives, such as dihydrouracil, are present as minor constituents in certain RNA molecules.

Adenine (6-amino purine) and **guanine** (2-amino-6-oxy purine), the two common purines, are found in both DNA and RNA (Figure 11.4). Other naturally occurring purine derivatives include **hypoxanthine**, **xanthine**, and **uric acid** (Figure 11.5). Hypoxanthine and xanthine are found only rarely as constituents of nucleic acids. Uric acid, the most oxidized state for a purine derivative, is never found in nucleic acids.

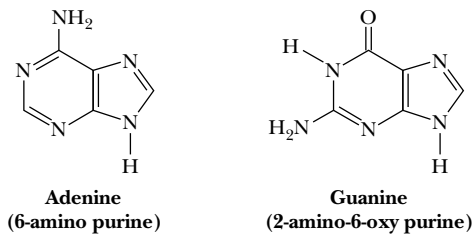


FIGURE 11.4 • The common purine bases—adenine and guanine—in the tautomeric forms predominant at pH 7.

Properties of Pyrimidines and Purines

The aromaticity of the pyrimidine and purine ring systems and the electron-rich nature of their —OH and —NH₂ substituents endow them with the capacity to undergo **keto–enol tautomeric shifts**. That is, pyrimidines and purines exist as tautomeric pairs, as shown in Figure 11.6 for uracil. The keto tautomer is called a **lactam**, whereas the enol form is a **lactim**. The lactam form vastly predominates at neutral pH. In other words, p*K*_a values for ring nitrogen atoms 1 and 3 in uracil are greater than 8 (the p*K*_a value for N-3 is 9.5) (Table 11.1).

Table 11.1			
Proton Dissociation Constants (p <i>K</i> _a Values) for Nucleotides			
Nucleotide	p <i>K</i> _a Base-N	p <i>K</i> ₁ Phosphate	p <i>K</i> ₂ Phosphate
5'-AMP	3.8 (N-1)	0.9	6.1
5'-GMP	9.4 (N-1) 2.4 (N-7)	0.7	6.1
5'-CMP	4.5 (N-3)	0.8	6.3
5'-UMP	9.5 (N-3)	1.0	6.4

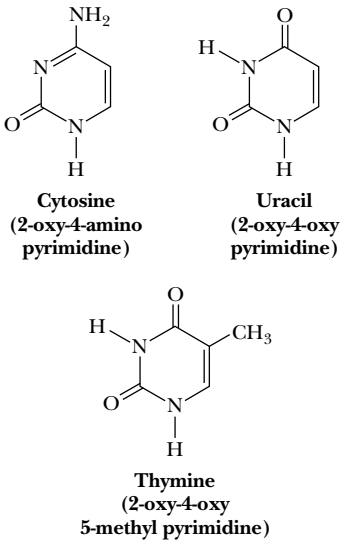


FIGURE 11.3 • The common pyrimidine bases—cytosine, uracil, and thymine—in the tautomeric forms predominant at pH 7.

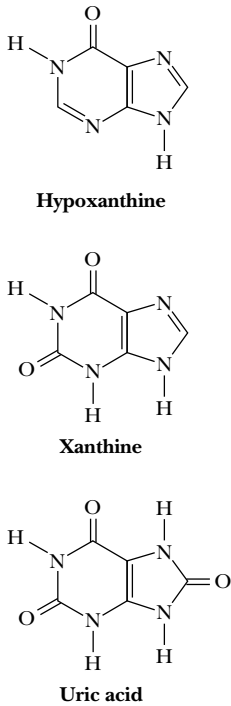


FIGURE 11.5 • Other naturally occurring purine derivatives—hypoxanthine, xanthine, and uric acid.

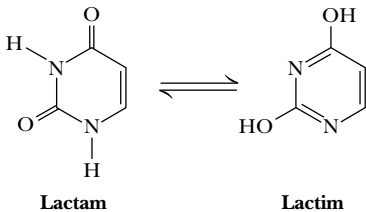
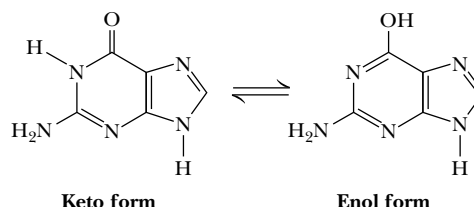


FIGURE 11.6 • The keto/enol tautomerism of uracil.

FIGURE 11.7 • The tautomerism of the purine, guanine.

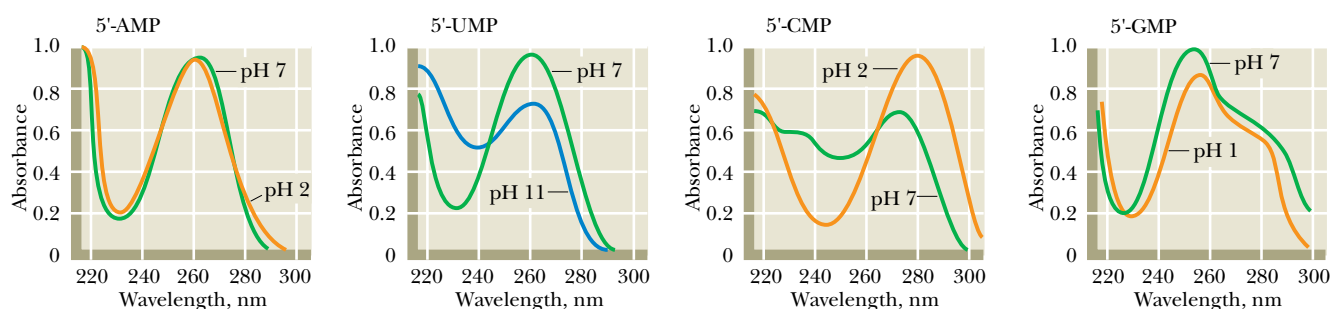
In contrast, as might be expected from the form of cytosine that predominates at pH 7, the pK_a value for N-3 in this pyrimidine is 4.5. Similarly, tautomeric forms can be represented for purines, as given for guanine in Figure 11.7.

Here, the pK_a value is 9.4 for N-1 and less than 5 for N-3. These pK_a values specify whether hydrogen atoms are associated with the various ring nitrogens at neutral pH. As such, they are important in determining whether these nitrogens serve as H-bond donors or acceptors. Hydrogen bonding between purine and pyrimidine bases is fundamental to the biological functions of nucleic acids, as in the formation of the double helix structure of DNA (see Section 11.6). The important functional groups participating in H-bond formation are the amino groups of cytosine, adenine, and guanine; the ring nitrogens at position 3 of pyrimidines and position 1 of purines; and the strongly electronegative oxygen atoms attached at position 4 of uracil and thymine, position 2 of cytosine, and position 6 of guanine (see Figure 11.21).

Another property of pyrimidines and purines is their strong absorbance of ultraviolet (UV) light, which is also a consequence of the aromaticity of their heterocyclic ring structures. Figure 11.8 shows characteristic absorption spectra of several of the common bases of nucleic acids—adenine, uracil, cytosine, and guanine—in their nucleotide forms: AMP, UMP, CMP, and GMP (see Section 11.4). This property is particularly useful in quantitative and qualitative analysis of nucleotides and nucleic acids.

11.2 • The Pentoses of Nucleotides and Nucleic Acids

Five-carbon sugars are called **pentoses** (see Chapter 7). RNA contains the pentose D-ribose, while 2-deoxy-D-ribose is found in DNA. In both instances, the pentose is in the five-membered ring form known as *furanose*: D-ribofuranose for RNA and 2-deoxy-D-ribofuranose for DNA (Figure 11.9). When these ribofuranoses are found in nucleotides, their atoms are numbered as 1', 2', 3', and so on to distinguish them from the ring atoms of the nitrogenous bases. As we shall see, the seemingly minor difference of a hydroxyl group at the 2'-position has far-reaching effects on the secondary structures available to RNA and DNA, as well as their relative susceptibilities to chemical and enzymatic hydrolysis.

**FIGURE 11.8** • The UV absorption spectra of the common ribonucleotides.

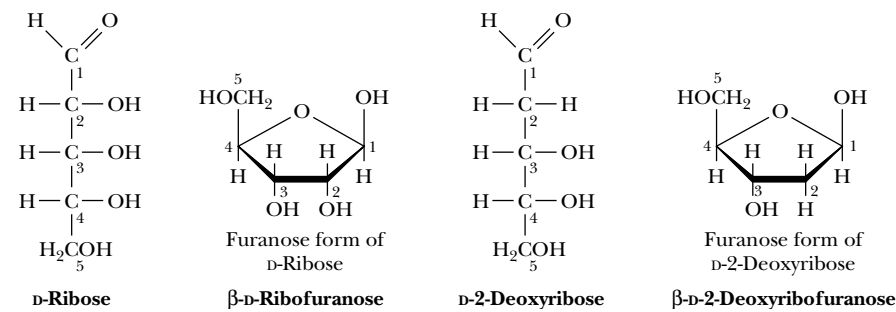
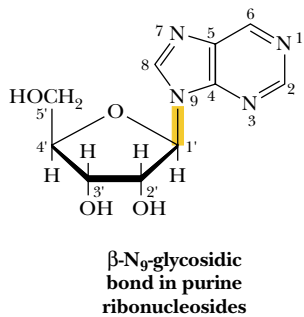
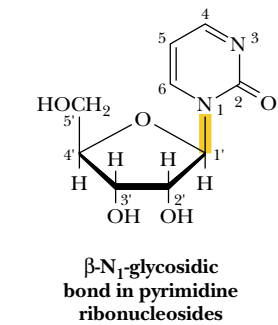


FIGURE 11.9 • Furanose structures—ribose and deoxyribose.



11.3 • Nucleosides Are Formed by Joining a Nitrogenous Base to a Sugar

Nucleosides are compounds formed when a base is linked to a sugar via a **glycosidic bond** (Figure 11.10). Glycosidic bonds by definition involve the carbonyl carbon atom of the sugar, which in cyclic structures is joined to the ring O atom (see Chapter 7). Such carbon atoms are called **anomeric**. In nucleosides, the bond is an *N*-glycoside because it connects the anomeric C-1' to N-1 of a pyrimidine or to N-9 of a purine. Glycosidic bonds can be either α or β , depending on their orientation relative to the anomeric C atom. Glycosidic bonds in nucleosides and nucleotides are always of the β -configuration, as represented in Figure 11.10. Nucleosides are named by adding the ending *-idine* to the root name of a pyrimidine or *-osine* to the root name of a purine. The common nucleosides are thus **cytidine**, **uridine**, **thymidine**, **adenosine**, and **guanosine**. The structures shown in Figure 11.11 are *ribonucleosides*. *Deoxyribonucleosides*, in contrast, lack a 2'-OH group on the pentose. The nucleoside formed by hypoxanthine and ribose is **inosine**.

FIGURE 11.10 • β -Glycosidic bonds link nitrogenous bases and sugars to form nucleosides.

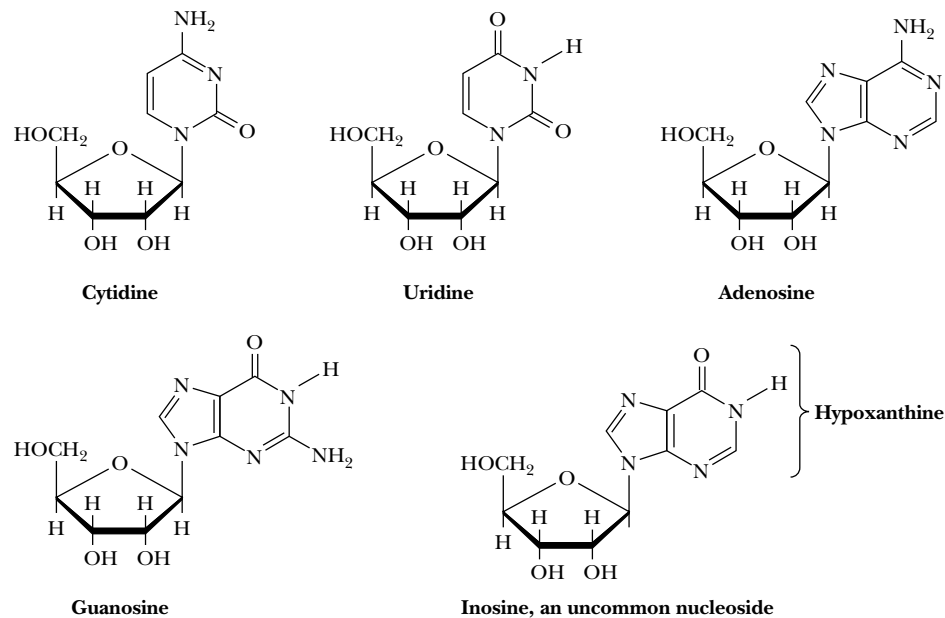


FIGURE 11.11 • The common ribonucleosides—cytidine, uridine, adenosine, and guanosine. Also, inosine drawn in anti conformation.

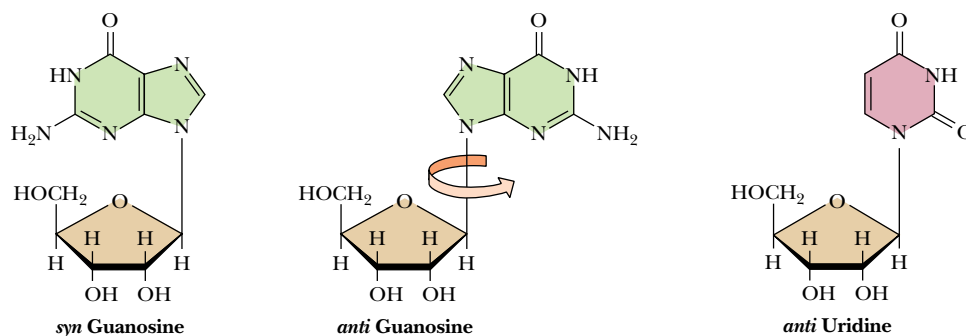


FIGURE 11.12 • Rotation around the glycosidic bond is sterically hindered; syn versus anti conformations in nucleosides are shown.

Nucleoside Conformation

In nucleosides, rotation of the base about the glycosidic bond is sterically hindered, principally by the hydrogen atom on the C-2' carbon of the furanose. (This hindrance is most easily seen and appreciated by manipulating accurate molecular models of these structures.) Consequently, nucleosides and nucleotides (see next section) exist in either of two conformations, designated *syn* and *anti* (Figure 11.12). For pyrimidines in the *syn* conformation, the oxygen substituent at position C-2 lies immediately above the furanose ring; in the *anti* conformation, this steric interference is avoided. Consequently, pyrimidine nucleosides favor the *anti* conformation. Purine nucleosides can adopt either the *syn* or *anti* conformation. In either conformation, the roughly planar furanose and base rings are not coplanar but lie at approximately right angles to one another.

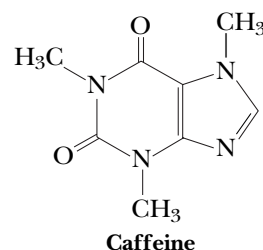
HUMAN BIOCHEMISTRY

Adenosine: A Nucleoside with Physiological Activity

For the most part, nucleosides have no biological role other than to serve as component parts of nucleotides. Adenosine is an exception. In mammals, adenosine functions as an **autocoid**, or “local hormone.” This nucleoside circulates in the bloodstream, acting locally on specific cells to influence such diverse physiological phenomena as blood vessel dilation, smooth muscle contraction, neuronal discharge, neurotransmitter release, and metabolism of fat. For example, when muscles work hard, they release adenosine, causing the surrounding blood vessels to dilate, which in turn increases the flow of blood and its delivery of O₂ and nutrients to the muscles. In a different autocoid role, adenosine acts in regulating heartbeat. The natural rhythm of the heart is controlled by a pacemaker, the sinoatrial node, that cyclically sends a wave of electrical excitation to the heart muscles. By blocking the flow of electrical current, adenosine slows the heart rate. *Supraventricular tachycardia* is a heart condition characterized by a rapid heartbeat. Intravenous injection of adenosine causes a momentary interruption of the rapid cycle of contraction and

restores a normal heart rate. Adenosine is licensed and marketed as *Adenocard*TM to treat supraventricular tachycardia.

In addition, adenosine is implicated in sleep regulation. During periods of extended wakefulness, extracellular adenosine levels rise as a result of metabolic activity in the brain, and this increase promotes sleepiness. During sleep, adenosine levels fall. Caffeine promotes wakefulness by blocking the interaction of extracellular adenosine with its neuronal receptors.*



*Porrka-Heiskanen, T., et al., 1997. Adenosine: A mediator of the sleep-inducing effects of prolonged wakefulness. *Science* **276**:1265–1268.

Nucleosides Are More Water-Soluble Than Free Bases

Nucleosides are much more water-soluble than the free bases because of the hydrophilicity of the sugar moiety. Like glycosides (see Chapter 7), nucleosides are relatively stable in alkali. Pyrimidine nucleosides are also resistant to acid hydrolysis, but purine nucleosides are easily hydrolyzed in acid to yield the free base and pentose.

11.4 • Nucleotides Are Nucleoside Phosphates

A **nucleotide** results when phosphoric acid is esterified to a sugar —OH group of a nucleoside. The nucleoside ribose ring has three —OH groups available for esterification, at C-2', C-3', and C-5' (although 2'-deoxyribose has only two). The vast majority of monomeric nucleotides in the cell are **ribonucleotides** having 5'-phosphate groups. Figure 11.13 shows the structures of the common four *ribonucleotides*, whose formal names are **adenosine 5'-monophosphate**, **guanosine 5'-monophosphate**, **cytidine 5'-monophosphate**, and **uridine 5'-monophosphate**. These compounds are more often referred to by their abbreviations: **5'-AMP**, **5'-GMP**, **5'-CMP**, and **5'-UMP**, or even more simply as **AMP**, **GMP**, **CMP**, and **UMP**. Nucleoside 3'-phosphates and nucleoside 2'-phosphates (3'-NMP and 2'-NMP, where N is a generic designation for “nucleoside”) do not occur naturally, but are biochemically important as products of polynucleotide or nucleic acid hydrolysis. Because the pK_a value for the first dissociation of a proton from the phosphoric acid moiety is 1.0 or less (Table 11.1), the nucleotides have acidic properties. This acidity is implicit in the other names by which these substances are known—**adenylic acid**, **guanylic acid**,

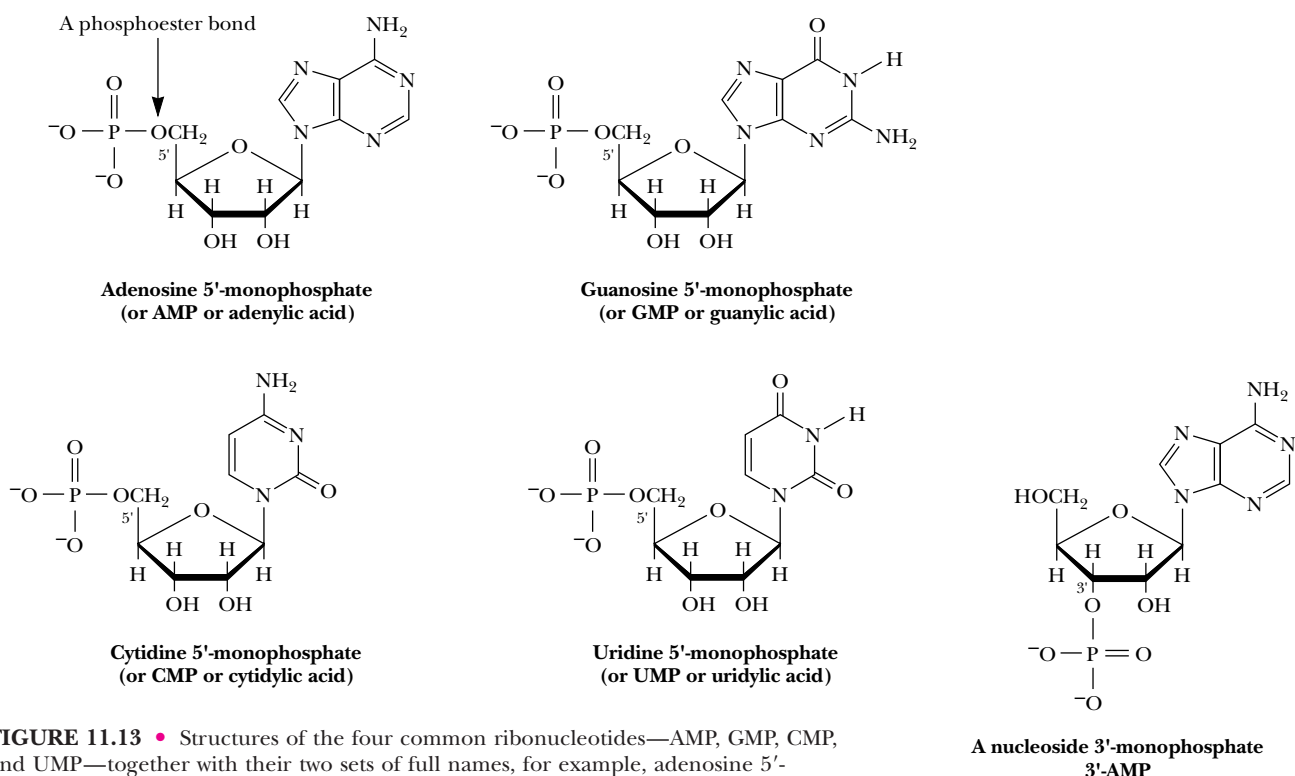


FIGURE 11.13 • Structures of the four common ribonucleotides—AMP, GMP, CMP, and UMP—together with their two sets of full names, for example, adenosine 5'-monophosphate and adenylic acid. Also shown is the nucleoside 3'-AMP.

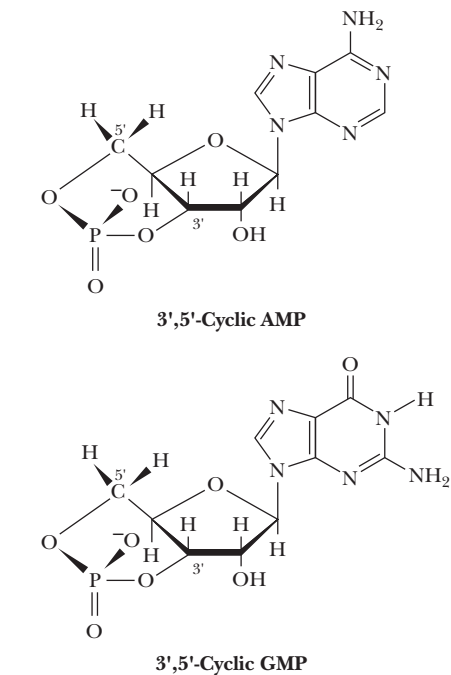


FIGURE 11.14 • Structures of the cyclic nucleotides cAMP and cGMP.

cytidylic acid, and **uridylic acid**. The pK_a value for the second dissociation, pK_2 , is about 6.0, so at neutral pH or above, the net charge on a nucleoside monophosphate is -2 . Nucleic acids, which are polymers of nucleoside monophosphates, derive their name from the acidity of these phosphate groups.

Cyclic Nucleotides

Nucleoside monophosphates in which the phosphoric acid is esterified to *two* of the available ribose hydroxyl groups (Figure 11.14) are found in all cells. Forming two such ester linkages with one phosphate results in a cyclic structure. **3',5'-cyclic AMP**, often abbreviated **cAMP**, and its guanine analog **3',5'-cyclic GMP**, or **cGMP**, are important regulators of cellular metabolism (see Part III: Metabolism and Its Regulation).

Nucleoside Diphosphates and Triphosphates

Additional phosphate groups can be linked to the phosphoryl group of a nucleotide through the formation of phosphoric anhydride linkages, as shown in Figure 11.15. Addition of a second phosphate to AMP creates **adenosine 5'-diphosphate**, or **ADP**, and adding a third yields **adenosine 5'-triphosphate**, or **ATP**. The respective phosphate groups are designated by the Greek letters α , β , and γ , starting with the α -phosphate as the one linked directly to the pentose. The abbreviations **GTP**, **CTP**, and **UTP** represent the other corresponding nucleoside 5'-triphosphates. Like the nucleoside 5'-monophosphates, the nucleoside 5'-diphosphates and 5'-triphosphates all occur in the free state in the cell, as do their deoxyribonucleoside phosphate counterparts, represented as dAMP, dADP, and dATP; dGMP, dGDP, and dGTP; dCMP, dCDP, and dCTP; dUMP, dUDP, and dUTP; and dTMP, dTDP, and dTTP.

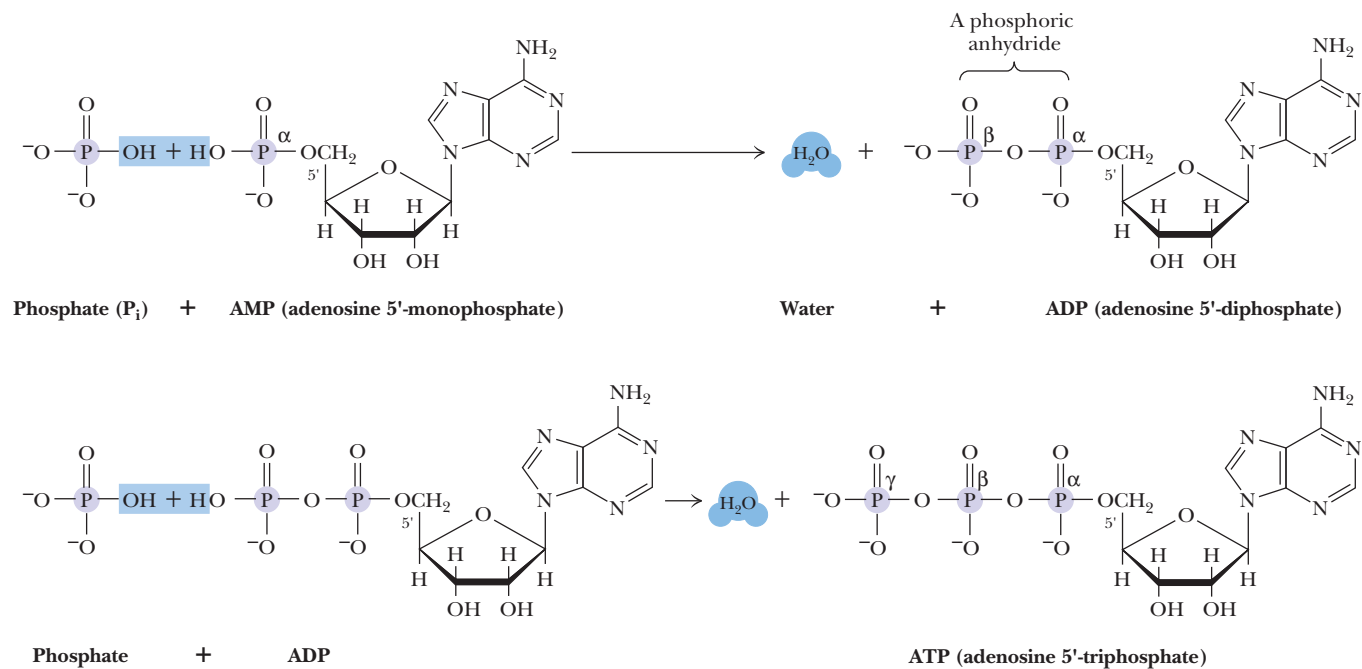


FIGURE 11.15 • Formation of ADP and ATP by the successive addition of phosphate groups via phosphoric anhydride linkages. Note the removal of equivalents of H_2O in these dehydration synthesis reactions.

directly involved in this chemistry. However, a “division of labor” directs ATP to serve as the primary nucleotide in central pathways of energy metabolism, while GTP, for example, is used to drive protein synthesis. Thus, the various nucleotides are channeled in appropriate metabolic directions through specific recognition of the base of the nucleotide. That is, the bases of nucleotides serve solely as *information symbols* aloof from the covalent bond chemistry that goes on. This role as information symbols extends to nucleotide polymers, the nucleic acids, where the bases serve as the information symbols for the code of genetic information.

11.5 • Nucleic Acids Are Polynucleotides

Nucleic acids are linear polymers of nucleotides linked 3' to 5' by **phosphodiester bridges** (Figure 11.17). They are formed as 5'-nucleoside monophosphates are successively added to the 3'-OH group of the preceding nucleotide, a process that gives the polymer a directional sense. Polymers of ribonucleotides are named **ribonucleic acid**, or **RNA**. Deoxyribonucleotide polymers are called **deoxyribonucleic acid**, or **DNA**. Because C-1' and C-4' in deoxyribonucleotides are involved in furanose ring formation and because there is no 2'-OH, only

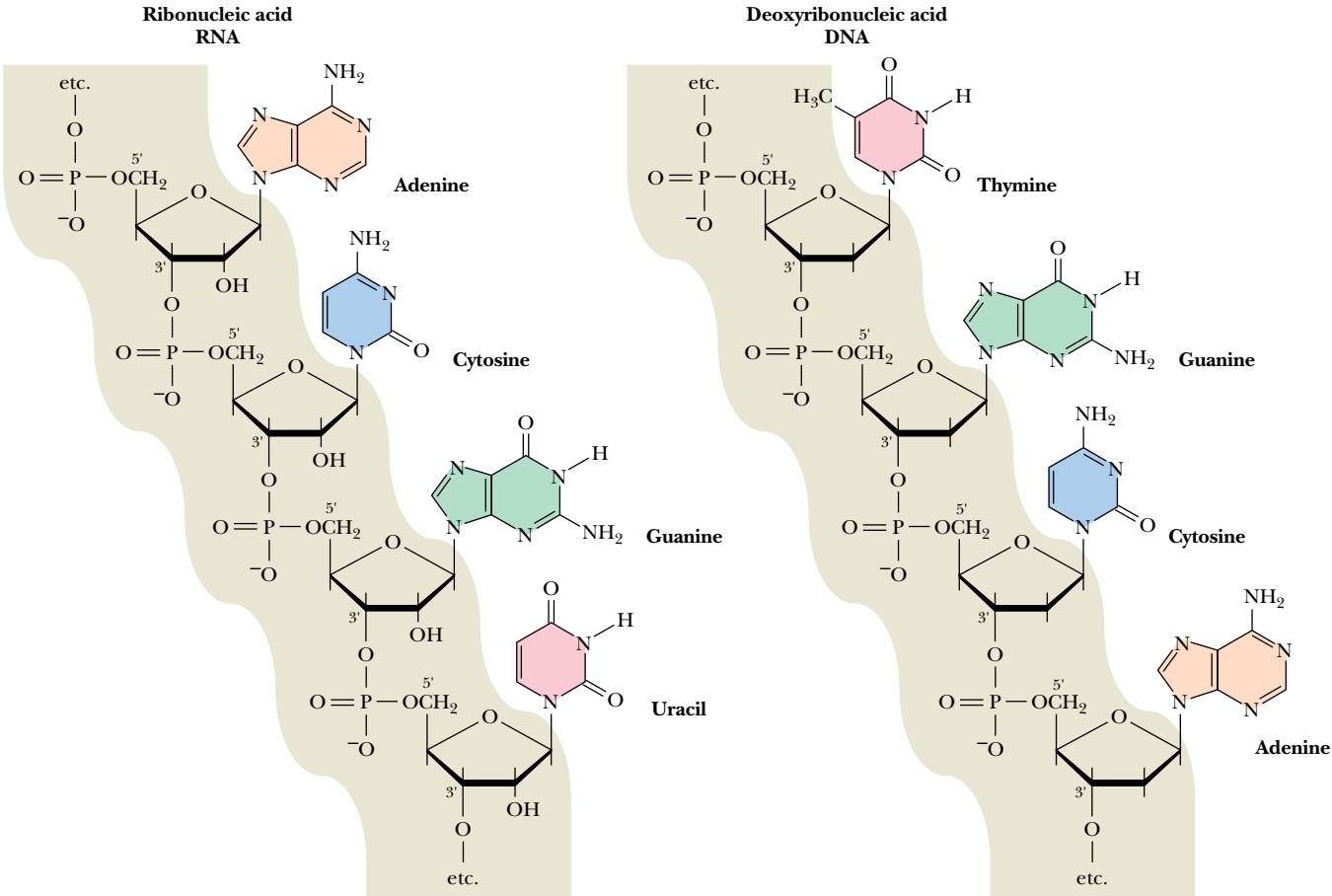


FIGURE 11.17 • 3'-5' phosphodiester bridges link nucleotides together to form polynucleotide chains.

the 3'- and 5'-hydroxyl groups are available for internucleotide phosphodiester bonds. In the case of DNA, a polynucleotide chain may contain hundreds of millions of nucleotide units. Any structural representation of such molecules would be cumbersome at best, even for a short oligonucleotide stretch.

Shorthand Notations for Polynucleotide Structures

Several conventions have been adopted to convey the sense of polynucleotide structures. A repetitious uniformity exists in the covalent backbone of polynucleotides, in which the chain can be visualized as running from 5' to 3' along the atoms of one furanose and thence across the phosphodiester bridge to the furanose of the next nucleotide in line. Thus, this backbone can be portrayed by the symbol of a vertical line representing the furanose and a slash representing the phosphodiester link, as shown in Figure 11.18. The diagonal slash runs from the middle of a furanose line to the bottom of an adjacent one to indicate the 3'- (middle) to 5'- (bottom) carbons of neighboring furanoses joined by the phosphodiester bridge. The base attached to each furanose is indicated above it by a one-letter designation: A, C, G, or U (or T). *The convention in all notations of nucleic acid structure is to read the polynucleotide chain from the 5'-end of the polymer to the 3'-end.* Note that this reading direction actually passes through each phosphodiester from 3' to 5'.

Base Sequence

The only significant variation that commonly occurs in the chemical structure of nucleic acids is the nature of the base at each nucleotide position. These bases are not part of the sugar-phosphate backbone but instead serve as distinctive side chains, much like the R groups of amino acids along a polypeptide backbone. They give the polymer its unique identity. A simple notation of these structures is merely to list the order of bases in the polynucleotide using single capital letters—A, G, C, and U (or T). Occasionally, a lowercase “p” is written between each successive base to indicate the phosphodiester bridge, as in GpApCpGpUpA. A “p” preceding the sequence indicates that the nucleic acid carries a PO_4 on its 5'-end, as in pGpApCpGpUpA; a “p” terminating the sequence connotes the presence of a phosphate on the 3'-OH end, as in GpApCpGpUpAp.

A more common method of representing nucleotide sequences is to omit the “p” and write only the order of bases, such as GACGUA. This notation assumes the presence of the phosphodiester joining adjacent nucleotides. The presence of 3'- or 5'-phosphate termini, however, must still be specified, as in GACGUAp for a 3'- PO_4 terminus. To distinguish between RNA and DNA sequences, DNA sequences are typically preceded by a lowercase “d” to denote deoxy, as in d-GACGTA. From a simple string of letters such as this, any biochemistry student should be able to draw the unique chemical structure for a pentanucleotide, even though it may contain over 200 atoms.

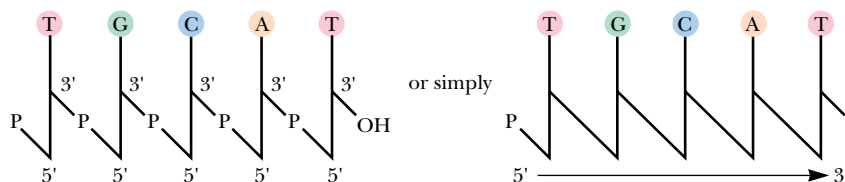


FIGURE 11.18 • Furanoses are represented by lines; phosphodiester bonds are represented by diagonal slashes in this shorthand notation for nucleic acid structures.

11.6 • Classes of Nucleic Acids

The two major classes of nucleic acids are DNA and RNA. DNA has only one biological role, but it is the more central one. The information to make all the functional macromolecules of the cell (even DNA itself) is preserved in DNA and accessed through transcription of the information into RNA copies. Coincident with its singular purpose, there is only a single DNA molecule (or “chromosome”) in simple life forms such as viruses or bacteria. Such DNA molecules must be quite large in order to embrace enough information for making the macromolecules necessary to maintain a living cell. The *Escherichia coli* chromosome has a molecular mass of 2.9×10^9 D and contains over 9 million nucleotides. Eukaryotic cells have many chromosomes, and DNA is found principally in two copies in the diploid chromosomes of the nucleus, but it also occurs in mitochondria and in chloroplasts, where it encodes some of the proteins and RNAs unique to these organelles.

In contrast, RNA occurs in multiple copies and various forms (Table 11.2). Cells contain up to eight times as much RNA as DNA. RNA has a number of important biological functions, and on this basis, RNA molecules are categorized into several major types: **messenger RNA**, **ribosomal RNA**, and **transfer RNA**. Eukaryotic cells contain an additional type, **small nuclear RNA (snRNA)**. With these basic definitions in mind, let’s now briefly consider the chemical and structural nature of DNA and the various RNAs. Chapter 12 elaborates on methods to determine the primary structure of nucleic acids by sequencing methods and discusses the secondary and tertiary structures of DNA and RNA. Part IV, Information Transfer, includes a detailed treatment of the dynamic role of nucleic acids in the molecular biology of the cell.

DNA

The DNA isolated from different cells and viruses characteristically consists of two polynucleotide strands wound together to form a long, slender, helical molecule, the **DNA double helix**. The strands run in opposite directions; that is, they are *antiparallel* and are held together in the double helical structure through *interchain hydrogen bonds* (Figure 11.19). These H bonds pair the bases of nucleotides in one chain to complementary bases in the other, a phenomenon called **base pairing**.

Table 11.2

Various Kinds of RNA Found in an <i>E. coli</i> Cell				
Type	Sedimentation Coefficient	Molecular Weight	Number of Nucleotide Residues	Percentage of Total Cell RNA
mRNA	6–25	25,000–1,000,000	75–3,000	~2
tRNA	~4	23,000–30,000	73–94	16
rRNA	5	35,000	120	82
	16	550,000	1542	
	23	1,100,000	2904	

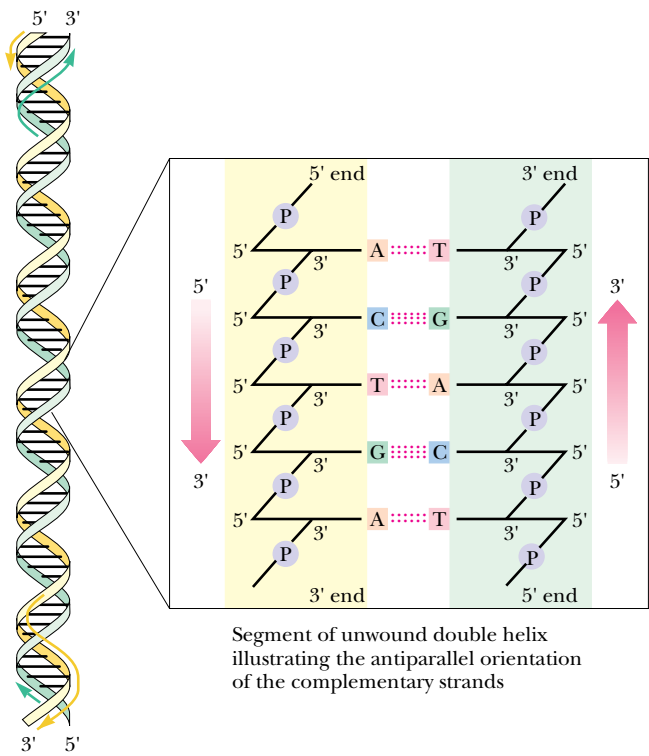


FIGURE 11.19 • The antiparallel nature of the DNA double helix.

Chargaff's Rules

A clue to the chemical basis of base pairing in DNA came from the analysis of the base composition of various DNAs by Erwin Chargaff in the late 1940s. His data showed that the four bases commonly found in DNA (A, C, G, and T) do not occur in equimolar amounts and that the relative amounts of each vary from species to species (Table 11.3). Nevertheless, Chargaff noted that certain pairs of bases, namely, adenine and thymine, and guanine and cytosine, are

Table 11.3
Molar Ratios Leading to the Formulation of Chargaff's Rules

Source	Adenine to Guanine	Thymine to Cytosine	Adenine to Thymine	Guanine to Cytosine	Purines to Pyrimidines
Ox	1.29	1.43	1.04	1.00	1.1
Human	1.56	1.75	1.00	1.00	1.0
Hen	1.45	1.29	1.06	0.91	0.99
Salmon	1.43	1.43	1.02	1.02	1.02
Wheat	1.22	1.18	1.00	0.97	0.99
Yeast	1.67	1.92	1.03	1.20	1.0
<i>Hemophilus influenzae</i>	1.74	1.54	1.07	0.91	1.0
<i>E. coli</i> K-12	1.05	0.95	1.09	0.99	1.0
Avian tubercle bacillus	0.4	0.4	1.09	1.08	1.1
<i>Serratia marcescens</i>	0.7	0.7	0.95	0.86	0.9
<i>Bacillus schatz</i>	0.7	0.6	1.12	0.89	1.0

Source: After Chargaff, E., 1951. *Federation Proceedings* **10**:654–659.

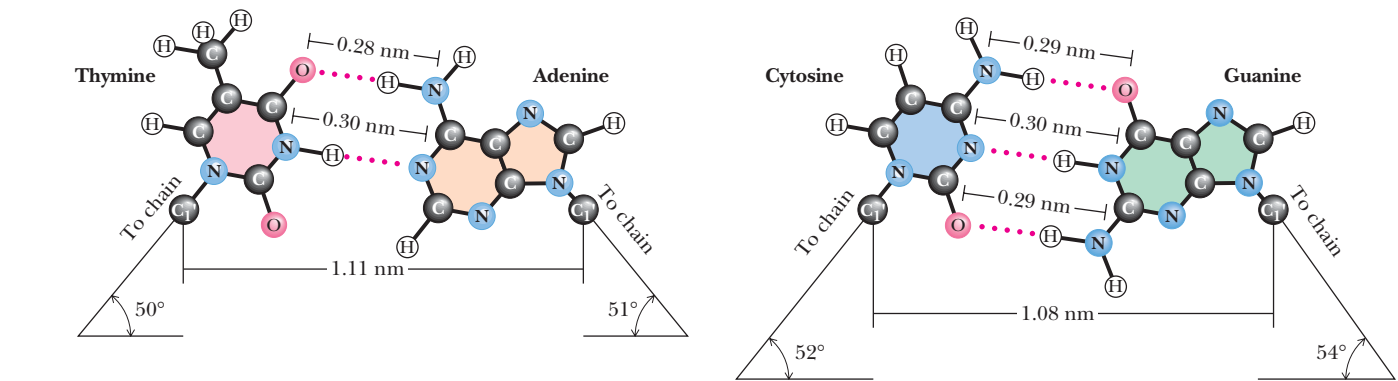


FIGURE 11.20 • The Watson–Crick base pairs A : T and G : C.

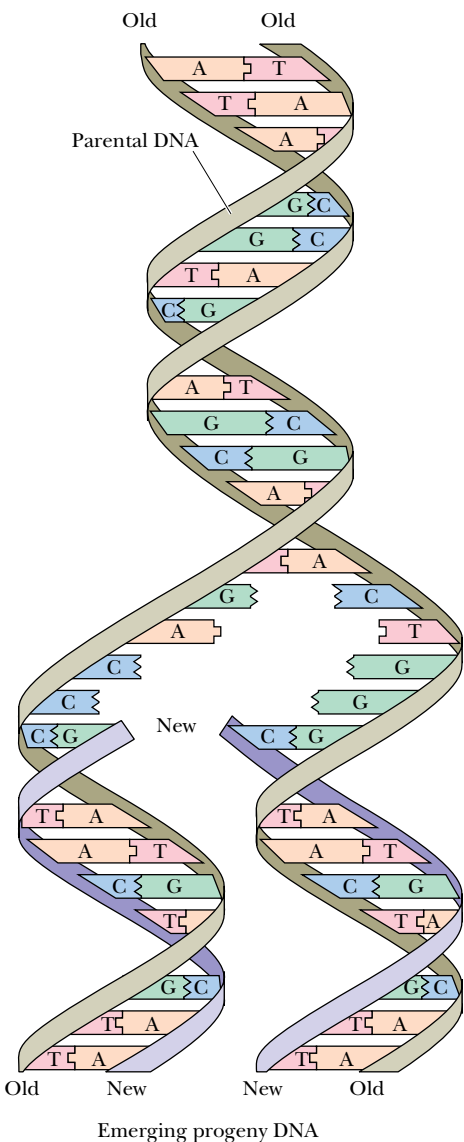


FIGURE 11.21 • Replication of DNA gives identical progeny molecules because base pairing is the mechanism determining the nucleotide sequence synthesized within each of the new strands during replication.

always found in a 1:1 ratio and that the number of pyrimidine residues always equals the number of purine residues. These findings are known as *Chargaff’s rules*: $[A] = [T]$; $[C] = [G]$; $[\text{pyrimidines}] = [\text{purines}]$.

Watson and Crick’s Double Helix

James Watson and Francis Crick, working in the Cavendish Laboratory at Cambridge University in 1953, took advantage of Chargaff’s results and the data obtained by Rosalind Franklin and Maurice Wilkins in X-ray diffraction studies on the structure of DNA to conclude that DNA was a *complementary double helix*. Two strands of deoxyribonucleic acid (sometimes referred to as the *Watson strand* and the *Crick strand*) are held together by hydrogen bonds formed between unique base pairs, always consisting of a purine in one strand and a pyrimidine in the other. Base pairing is very specific: if the purine is adenine, the pyrimidine must be thymine. Similarly, guanine pairs only with cytosine (Figure 11.20). Thus, if an A occurs in one strand of the helix, T must occupy the complementary position in the opposing strand. Likewise, a G in one dictates a C in the other. Because exceptions to this exclusive pairing of A only with T and G only with C are rare, these pairs are taken as the standard or accepted law, and the A:T and G:C base pairs are often referred to as canonical. As Watson recognized from testing various combinations of bases using structurally accurate models, the A:T pair and the G:C pair form spatially equivalent units (Figure 11.20). The backbone-to-backbone distance of an A:T pair is 1.11 nm, virtually identical to the 1.08 nm chain separation in G:C base pairs.

The DNA molecule not only conforms to Chargaff’s rules but also has a profound property relating to heredity: *The sequence of bases in one strand has a complementary relationship to the sequence of bases in the other strand*. That is, the information contained in the sequence of one strand is conserved in the sequence of the other. Therefore, separation of the two strands and faithful replication of each, through a process in which base pairing specifies the nucleotide sequence in the newly synthesized strand, leads to two progeny molecules identical in every respect to the parental double helix (Figure 11.21). Elucidation of the double helical structure of DNA represented one of the most significant events in the history of science. This discovery more than any other marked the beginning of molecular biology. Indeed, upon solving the structure of DNA, Crick proclaimed in *The Eagle*, a pub just across from the Cavendish lab, “We have discovered the secret of life!”

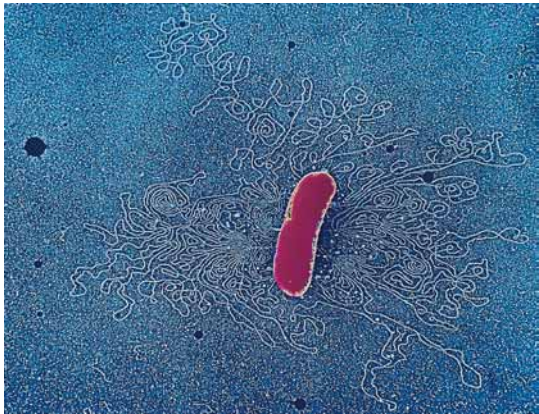


FIGURE 11.22 • If the cell walls of bacteria such as *Escherichia coli* are partially digested and the cells are then osmotically shocked by dilution with water, the contents of the cells are extruded to the exterior. In electron micrographs, the most obvious extruded component is the bacterial chromosome, shown here surrounding the cell. (Dr. Gopal Murti/CNRI/Phototake NYC)

Size of DNA Molecules

Because of the double helical nature of DNA molecules, their size can be represented in terms of the numbers of nucleotide base pairs they contain. For example, the *E. coli* chromosome consists of 4.64×10^6 base pairs (abbreviated bp) or 4.64×10^3 kilobase pairs (kbp). DNA is a threadlike molecule. The diameter of the DNA double helix is only 2 nm, but the length of the DNA molecule forming the *E. coli* chromosome is over 1.6×10^6 nm (1.6 mm). Because the long dimension of an *E. coli* cell is only 2000 nm (0.002 mm), its chromosome must be highly folded. Because of their long, threadlike nature, DNA molecules are easily sheared into shorter fragments during isolation procedures, and it is difficult to obtain intact chromosomes even from the simple cells of prokaryotes.

DNA in the Form of Chromosomes

DNA occurs in various forms in different cells. The single chromosome of prokaryotic cells (Figure 11.22) is typically a circular DNA molecule. Relatively little protein is associated with prokaryotic chromosomes. In contrast, the DNA molecules of eukaryotic cells, each of which defines a chromosome, are linear and richly adorned with proteins. A class of arginine- and lysine-rich basic proteins called **histones** interact ionically with the anionic phosphate groups in the DNA backbone to form **nucleosomes**, structures in which the DNA double helix is wound around a protein “core” composed of pairs of four different histone polypeptides (Figure 11.23; see also Section 12.5 in Chapter 12). Chromosomes also contain a varying mixture of other proteins, so-called **non-histone chromosomal proteins**, many of which are involved in regulating which genes in DNA are transcribed at any given moment. The amount of DNA in a diploid mammalian cell is typically more than 1000 times that found in an *E. coli* cell. Some higher plant cells contain more than 50,000 times as much.

RNA

Messenger RNA

Messenger RNA (**mRNA**) serves to carry the information or “message” that is encoded in genes to the sites of protein synthesis in the cell, where this information is translated into a polypeptide sequence. Because mRNA molecules are transcribed copies of the protein-coding genetic units that comprise most of DNA, mRNA is said to be “the DNA-like RNA.”

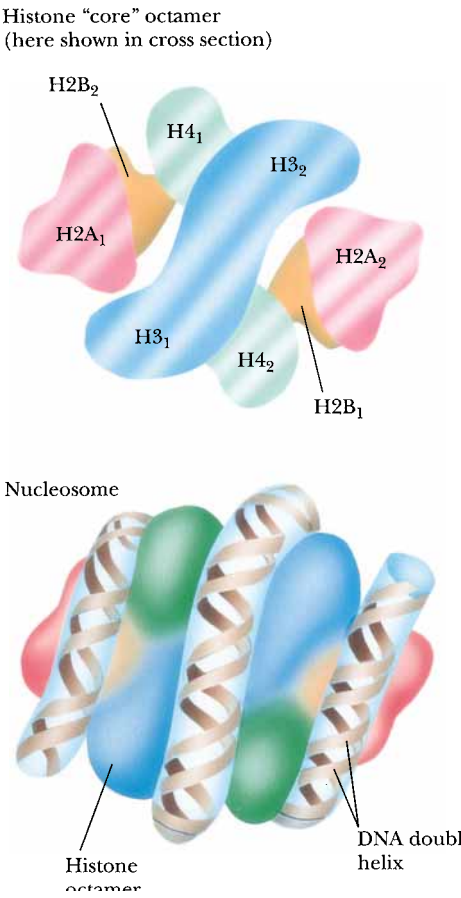


FIGURE 11.23 • A diagram of the histone octamer. Nucleosomes consist of two turns of DNA supercoiled about a histone “core” octamer.

Messenger RNA is synthesized during **transcription**, an enzymatic process in which an RNA copy is made of the sequence of bases along one strand of DNA. This mRNA then directs the synthesis of a polypeptide chain as the information that is contained within its nucleotide sequence is translated into an amino acid sequence by the protein-synthesizing machinery of the ribosomes. Ribosomal RNA and tRNA molecules are also synthesized by transcription of DNA sequences, but unlike mRNA molecules, these RNAs are not subsequently translated to form proteins. Only the genetic units of DNA sequence that encode proteins are transcribed into mRNA molecules. In prokaryotes, a single mRNA may contain the information for the synthesis of several polypeptide chains within its nucleotide sequence (Figure 11.24). In contrast, eukaryotic mRNAs encode only one polypeptide, but are more complex in that they are synthesized in the nucleus in the form of much larger precursor molecules called **heterogeneous nuclear RNA**, or **hnRNA**. hnRNA molecules contain stretches of nucleotide sequence that have no protein-coding capacity. These

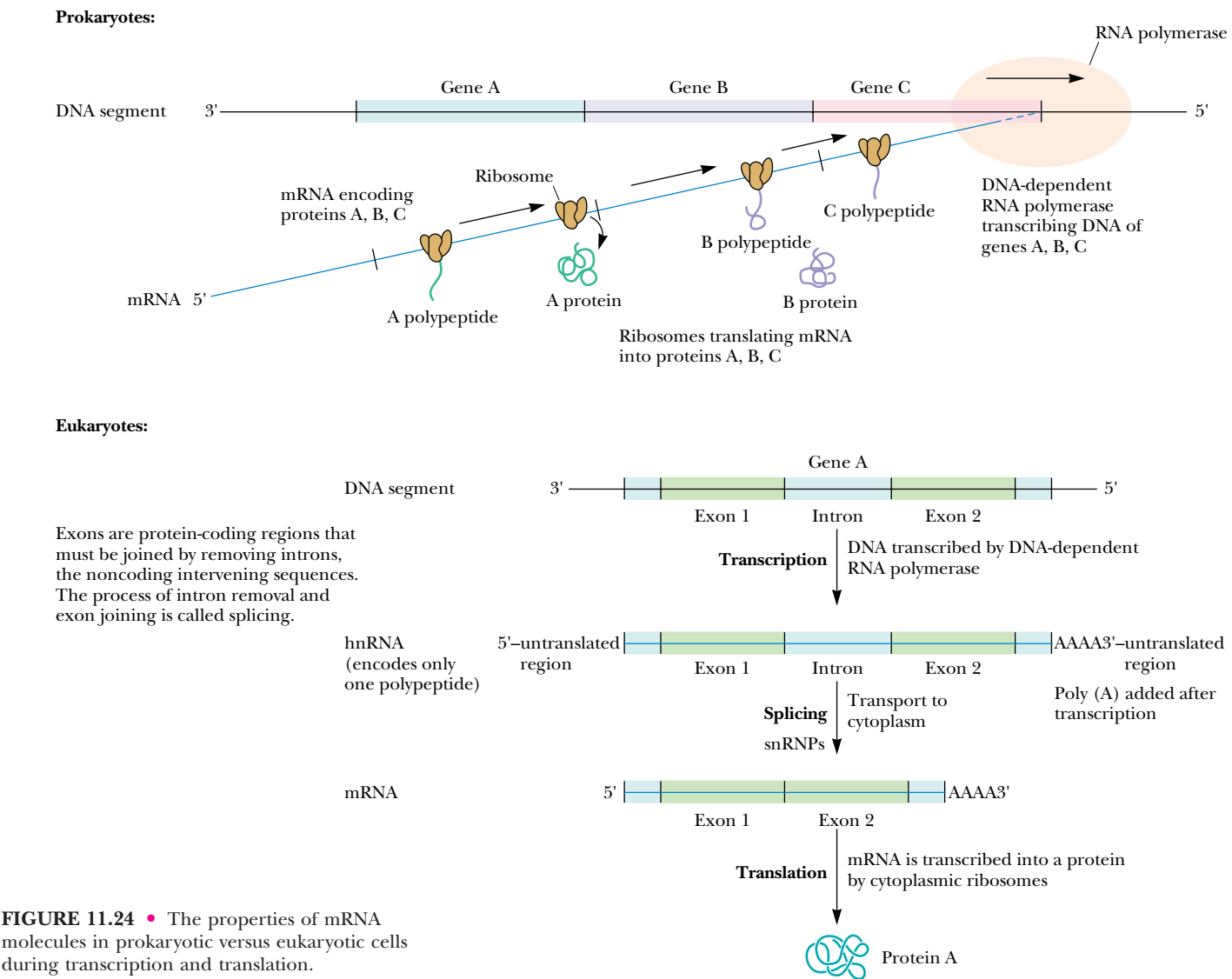


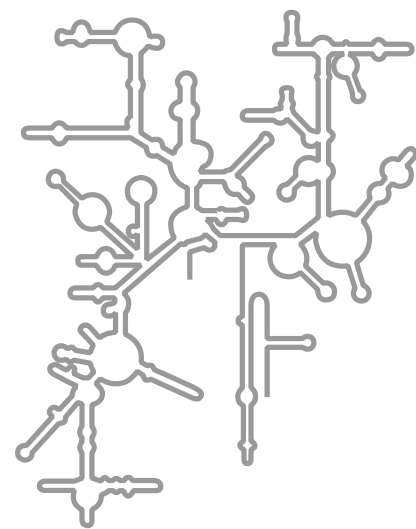
FIGURE 11.24 • The properties of mRNA molecules in prokaryotic versus eukaryotic cells during transcription and translation.

noncoding regions are called **intervening sequences** or **introns** because they intervene between coding regions, which are called **exons**. Introns interrupt the continuity of the information specifying the amino acid sequence of a protein and must be spliced out before the message can be translated. In addition, eukaryotic hnRNA and mRNA molecules have a run of 100 to 200 adenylic acid residues attached at their 3'-ends, so-called **poly(A) tails**. This polyadenylation occurs after transcription has been completed and is believed to contribute to mRNA stability. The properties of messenger RNA molecules as they move through transcription and translation in prokaryotic versus eukaryotic cells are summarized in Figure 11.24.

Ribosomal RNA

Ribosomes, the supramolecular assemblies where protein synthesis occurs, are about 65% RNA of the ribosomal RNA type. Ribosomal RNA (**rRNA**) molecules fold into characteristic secondary structures as a consequence of intramolecular hydrogen bond interactions (marginal figure). The different species of rRNA are generally referred to according to their **sedimentation coefficients**¹ (see the Appendix to Chapter 5), which are a rough measure of their relative size (Table 11.2 and Figure 11.25).

Ribosomes are composed of two subunits of different sizes that dissociate from each other if the Mg²⁺ concentration is below 10⁻³ M. Each subunit is



Ribosomal RNA has a complex secondary structure due to many intrastrand hydrogen bonds.

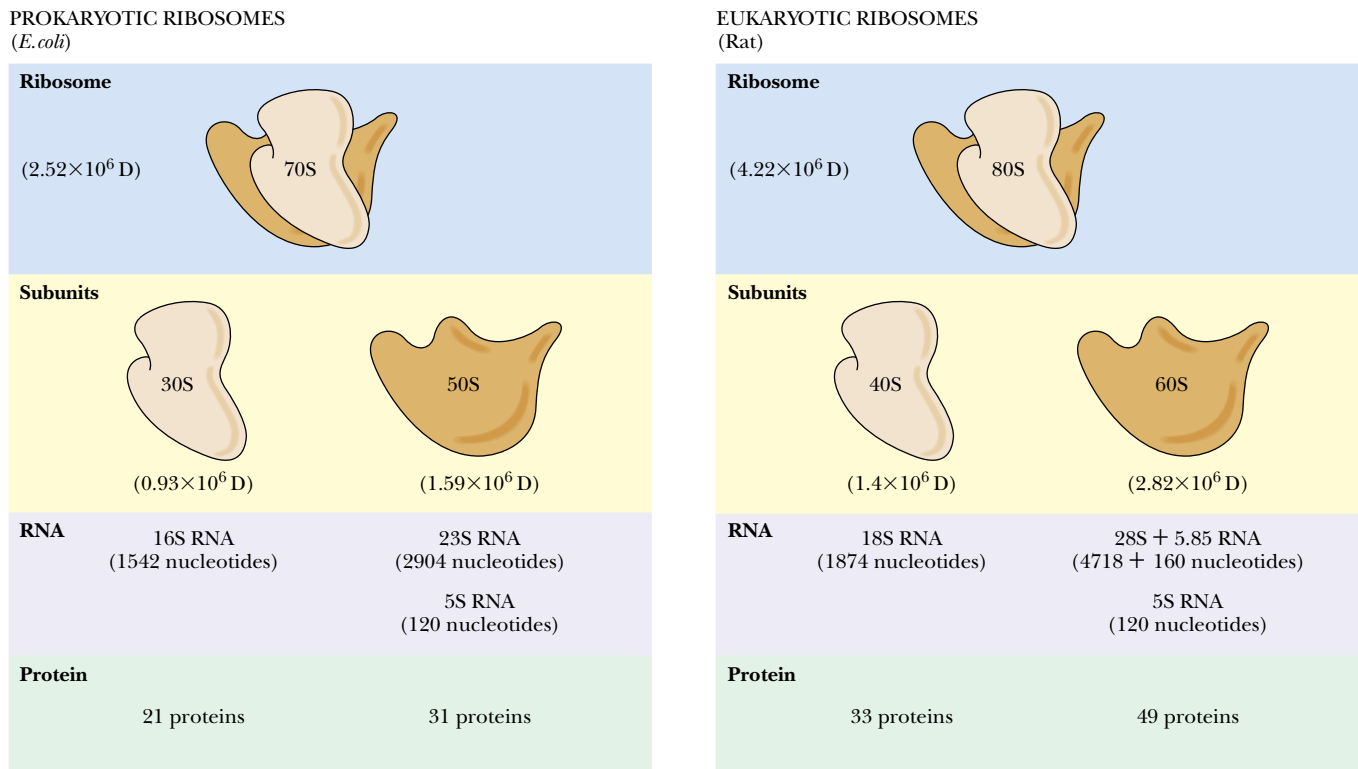


FIGURE 11.25 • The organization and composition of prokaryotic and eukaryotic ribosomes.

¹Sedimentation coefficients are a measure of the velocity with which a particle sediments in a centrifugal force field. Sedimentation coefficients are typically expressed in **Svedbergs** (symbolized S), named to honor The Svedberg, developer of the ultracentrifuge. One S equals 10⁻¹³ sec.

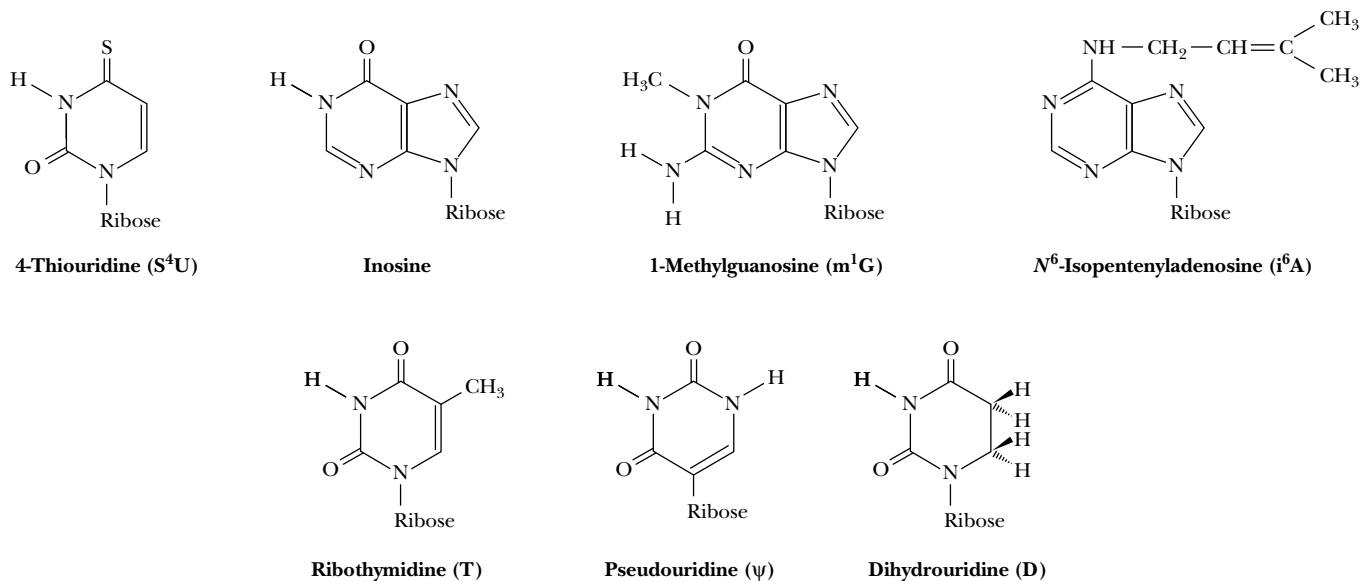
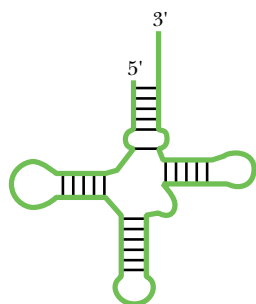


FIGURE 11.26 • Unusual bases of RNA—pseudouridine, ribothymidylic acid, and various methylated bases.

a supramolecular assembly of proteins and RNA and has a total mass of 10^6 daltons or more. *E. coli* ribosomal subunits have sedimentation coefficients of 30S (the small subunit) and 50S (the large subunit). Eukaryotic ribosomes are somewhat larger than prokaryotic ribosomes, consisting of 40S and 60S subunits. The properties of ribosomes and their rRNAs are summarized in Figure 11.25. The 30S subunit of *E. coli* contains a single RNA chain of 1542 nucleotides. This small subunit rRNA itself has a sedimentation coefficient of 16S. The large *E. coli* subunit has two rRNA molecules, a 23S (2904 nucleotides) and a 5S (120 nucleotides). The ribosomes of a typical eukaryote, the rat, have rRNA molecules of 18S (1874 nucleotides) and 28S (4718 bases), 5.8S (160 bases), and 5S (120 bases). The 18S rRNA is in the 40S subunit and the latter three are all part of the 60S subunit.

Ribosomal RNAs characteristically contain a number of specially modified nucleotides, including **pseudouridine** residues, **ribothymidylic acid**, and **methylated bases** (Figure 11.26). The central role of ribosomes in the biosynthesis of proteins is treated in detail in Chapter 33. Here we briefly note the significant point that genetic information in the nucleotide sequence of an mRNA is translated into the amino acid sequence of a polypeptide chain by ribosomes.



Transfer RNA also has a complex secondary structure due to many intrastrand hydrogen bonds.

Transfer RNA

Transfer RNA (**tRNA**) serves as a carrier of amino acid residues for protein synthesis. Transfer RNA molecules also fold into a characteristic secondary structure (marginal figure). The amino acid is attached as an aminoacyl ester to the 3'-terminus of the tRNA. Aminoacyl-tRNAs are the substrates for protein biosynthesis. The tRNAs are the smallest RNAs (size range—23 to 30 kD) and contain 73 to 94 residues, a substantial number of which are methylated or otherwise unusually modified. Transfer RNA derives its name from its role as the carrier of amino acids during the process of protein synthesis (see Chapters 32 and 33). Each of the 20 amino acids of proteins has at least one unique tRNA species dedicated to chauffeuring its delivery to ribosomes for insertion into growing polypeptide chains, and some amino acids are served by several tRNAs. For example, five different tRNAs act in the transfer of leucine into

proteins. In eukaryotes, there are even discrete sets of tRNA molecules for each site of protein synthesis—the cytoplasm, the mitochondrion, and, in plant cells, the chloroplast. All tRNA molecules possess a 3'-terminal nucleotide sequence that reads **-CCA**, and the amino acid is carried to the ribosome attached as an acyl ester to the free 3'-OH of the terminal A residue. These **aminoacyl-tRNAs** are the substrates of protein synthesis, the amino acid being transferred to the carboxyl end of a growing polypeptide. The peptide bond-forming reaction is a catalytic process intrinsic to ribosomes.

Small Nuclear RNAs

Small nuclear RNAs, or **snRNAs**, are a class of RNA molecules found in eukaryotic cells, principally in the nucleus. They are neither tRNA nor small rRNA molecules, although they are similar in size to these species. They contain from 100 to about 200 nucleotides, some of which, like tRNA and rRNA, are methylated or otherwise modified. No snRNA exists as naked RNA. Instead, snRNA is found in stable complexes with specific proteins forming **small nuclear ribonucleoprotein particles**, or **snRNPs**, which are about 10S in size. Their occurrence in eukaryotes, their location in the nucleus, and their relative abundance (1 to 10% of the number of ribosomes) are significant clues to their biological purpose: snRNPs are important in the processing of eukaryotic gene transcripts (hnRNA) into mature messenger RNA for export from the nucleus to the cytoplasm (Figure 11.24).

Significance of Chemical Differences Between DNA and RNA

Two fundamental chemical differences distinguish DNA from RNA:

1. DNA contains 2-deoxyribose instead of ribose.
2. DNA contains thymine instead of uracil.

What are the consequences of these differences and do they hold any significance in common? An argument can be made that, because of these differences, DNA is a more stable polymeric form than RNA. The greater stability of DNA over RNA is consistent with the respective roles these macromolecules have assumed in heredity and information transfer.

Consider first why DNA contains thymine instead of uracil. The key observation is that *cytosine deaminates to form uracil* at a finite rate *in vivo* (Figure 11.27). Because C in one DNA strand pairs with G in the other strand, whereas U would pair with A, conversion of a C to a U could potentially result in a heritable change of a CG pair to a UA pair. Such a change in nucleotide sequence would constitute a *mutation* in the DNA. To prevent this reaction from leading to changes in nucleotide sequence, a cellular repair mechanism “proofreads” DNA, and when a U arising from C deamination is encountered, it is treated as inappropriate and is replaced by a C. If DNA normally contained U rather than T, this repair system could not readily distinguish U formed by C deamination from U correctly paired with A. However, the U in DNA is “5-methyl-U” or, as it is conventionally known, thymine (Figure 11.28). That is, the 5-methyl group on T labels it as if to say “this U belongs; do not replace it.”

The ribose 2'-OH group of RNA is absent in DNA. Consequently, the ubiquitous 3'-O of polynucleotide backbones lacks a vicinal hydroxyl neighbor in DNA. This difference leads to a greater resistance of DNA to alkaline hydrolysis, examined in detail in the following section. To view it another way, RNA is less stable than DNA because its vicinal 2'-OH group makes the 3'-phosphodiester bond susceptible to nucleophilic cleavage (Figure 11.29). For just this reason, it is selectively advantageous for the heritable form of genetic information to be DNA rather than RNA.

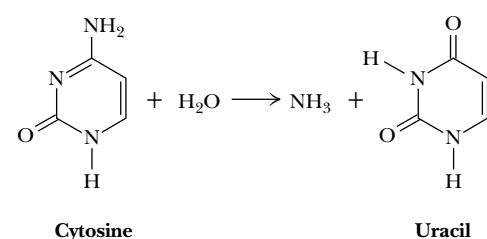


FIGURE 11.27 • Deamination of cytosine forms uracil.

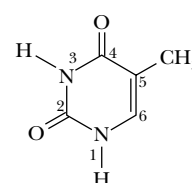
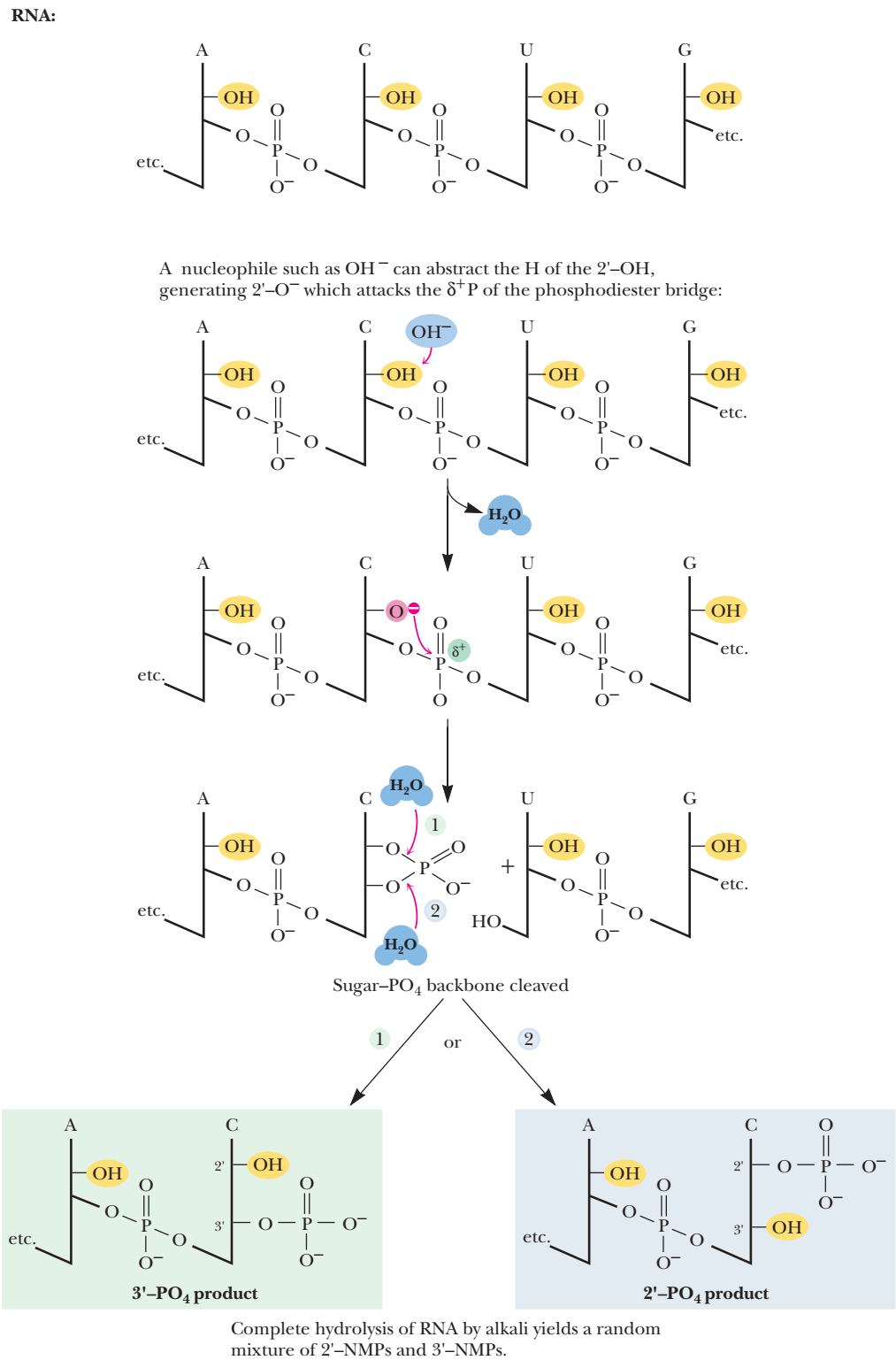
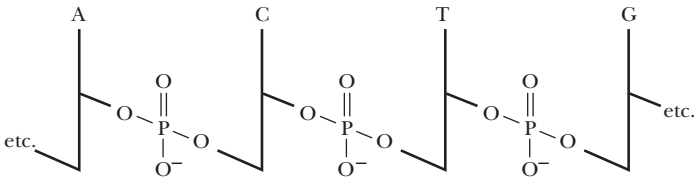


FIGURE 11.28 • The 5-methyl group on thymine labels it as a special kind of uracil.

FIGURE 11.29 • The vicinal —OH groups of RNA are susceptible to nucleophilic attack leading to hydrolysis of the phosphodiester bond and fracture of the polynucleotide chain; DNA lacks a 2'-OH vicinal to its 3'-O-phosphodiester backbone. Alkaline hydrolysis of RNA results in the formation of a mixture of 2'- and 3'-nucleoside monophosphates.



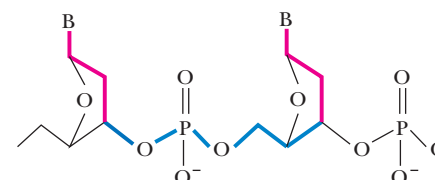
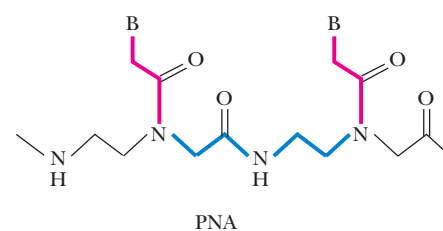
DNA: no 2'-OH; resistant to OH^- :



A DEEPER LOOK

Peptide Nucleic Acids (PNAs) Are Synthetic Mimics of DNA and RNA

Synthetic chemists have invented analogs of DNA (and RNA) in which the sugar–phosphate backbone is replaced by a peptide backbone, creating a polymer appropriately termed a *peptide nucleic acid*, or *PNA*. The PNA peptide backbone was designed so that the space between successive bases was the same as in natural DNA (see figure). PNA consists of repeating units of *N*-(2-aminoethyl)-glycine residues linked by peptide bonds; the bases are attached to this backbone through methylene carbonyl linkages. This chemistry provides six bonds along the backbone between bases and three bonds between the backbone and each base, just like natural DNA. PNA oligomers interact with DNA (and RNA) through specific base-pairing interactions, just as would be expected for a pair of complementary oligonucleotides. PNAs are resistant to nucleases and also are poor substrates for proteases. PNAs thus show great promise as specific diagnostic probes for unique DNA or RNA nucleotide sequences. PNAs also have potential application as antisense drugs (see problem 5 in the end-of-chapter problems).



Note the repeating six bonds (in bold) between base attachments and the three-bond linker between base (B) and backbone.

Buchardt, O., et al., 1993. Peptide nucleic acids and their potential applications in biotechnology. *Trends in Biotechnology* 11:384–386.

11.7 • Hydrolysis of Nucleic Acids

Most reactions of nucleic acid hydrolysis break bonds in the polynucleotide backbone. Such reactions are important because they can be used to manipulate these polymeric molecules. For example, hydrolysis of polynucleotides generates smaller fragments whose nucleotide sequence can be more easily determined.

Hydrolysis by Acid or Base

RNA is relatively resistant to the effects of dilute acid, but gentle treatment of DNA with 1 mM HCl leads to hydrolysis of purine glycosidic bonds and the loss of purine bases from the DNA. The glycosidic bonds between pyrimidine bases and 2'-deoxyribose are not affected, and, in this case, the polynucleotide's sugar–phosphate backbone remains intact. The purine-free polynucleotide product is called **apurinic acid**.

DNA is not susceptible to alkaline hydrolysis. On the other hand, RNA is alkali labile and is readily hydrolyzed by dilute sodium hydroxide. Cleavage is random in RNA, and the ultimate products are a mixture of nucleoside 2'- and 3'-monophosphates. These products provide a clue to the reaction mechanism (Figure 11.29). Abstraction of the 2'-OH hydrogen by hydroxyl anion leaves a 2'-O[−] that carries out a nucleophilic attack on the δ^+ phosphorus atom of the phosphate moiety, resulting in cleavage of the 5'-phosphodiester bond and formation of a cyclic 2',3'-phosphate. This cyclic 2',3'-phosphodiester is unstable and decomposes randomly to either a 2'- or 3'-phosphate ester. DNA has no 2'-OH; therefore DNA is alkali stable.

Enzymatic Hydrolysis

Enzymes that hydrolyze nucleic acids are called **nucleases**. Virtually all cells contain various nucleases that serve important housekeeping roles in the normal course of nucleic acid metabolism. Organs that provide digestive fluids, such as the pancreas, are rich in nucleases and secrete substantial amounts to hydrolyze ingested nucleic acids. Fungi and snake venom are often good sources of nucleases. As a class, nucleases are **phosphodiesterases** because the reaction that they catalyze is the cleavage of phosphodiester bonds by H₂O. Because each internal phosphate in a polynucleotide backbone is involved in two phosphoester linkages, cleavage can potentially occur on either side of the phosphorus (Figure 11.30). Convention labels the 3'-side as *a* and the 5'-side as *b*. Cleavage on the *a* side leaves the phosphate attached to the 5'-position of the adjacent nucleotide, while *b*-side hydrolysis yields 3'-phosphate products. Enzymes or reactions that hydrolyze nucleic acids are characterized as acting at either *a* or *b*. A second convention denotes whether the nucleic acid chain was cleaved at some internal location, *endo*, or whether a terminal nucleotide residue was hydrolytically removed, *exo*. Note that *exo a* cleavage characteristically occurs at the 3'-end of the polymer, whereas *exo b* cleavage involves attack at the 5'-terminus (Figure 11.31).

Nuclease Specificity

Like most enzymes (see Chapter 14), nucleases exhibit selectivity or *specificity* for the nature of the substance on which they act. That is, some nucleases act only on DNA (**DNases**), while others are specific for RNA (the **RNases**). Still

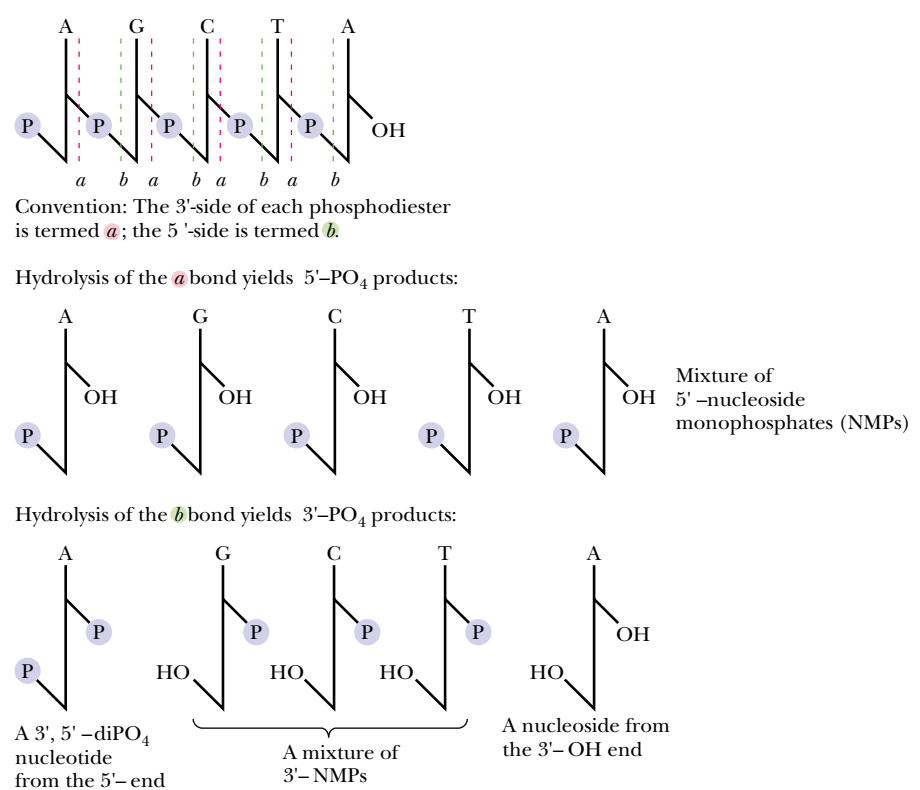


FIGURE 11.30 • Cleavage in polynucleotide chains: *a* cleavage yields 5'-phosphate products, whereas *b* cleavage gives 3'-phosphate products.

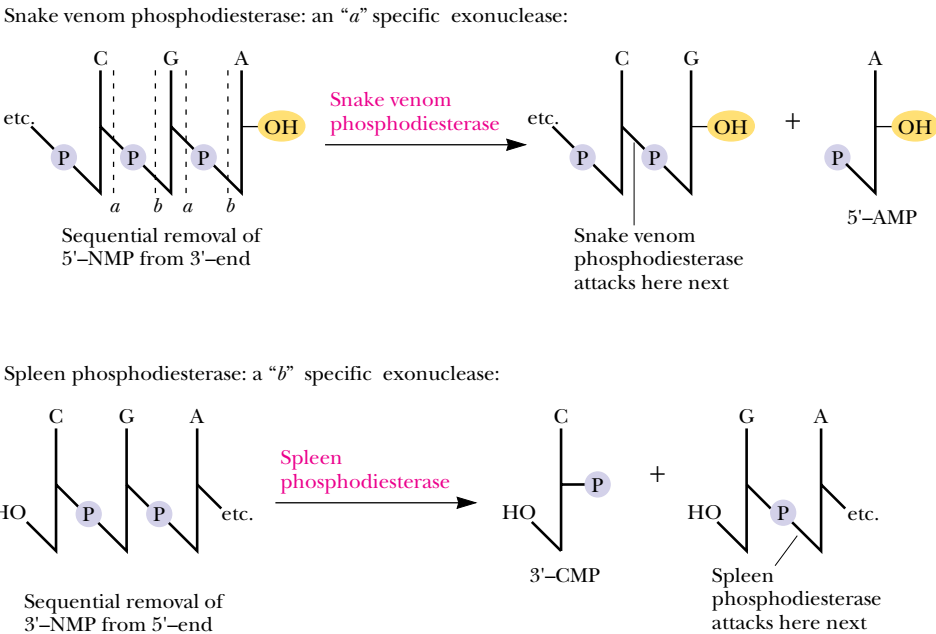


FIGURE 11.31 • Snake venom phosphodiesterase and spleen phosphodiesterase are exonucleases that degrade polynucleotides from opposite ends.

others are nonspecific and are referred to simply as **nucleases**, as in *nuclease S1* (see Table 11.4). Nucleases may also show specificity for only single-stranded nucleic acids or may only act on double helices. Single-stranded nucleic acids are abbreviated by an *ss* prefix, as in *ssRNA*; the prefix *ds* denotes double-stranded. Nucleases may also display a decided preference for acting only at

Table 11.4 Specificity of Various Nucleases			
Enzyme	DNA, RNA, or Both	a or b	Specificity
Exonucleases			
Snake venom phosphodiesterase	Both	a	Starts at 3'-end, 5'-NMP products
Spleen phosphodiesterase	Both	b	Starts at 5'-end, 3'-NMP products
Endonucleases			
RNase A (pancreas)	RNA	b	Where 3'-PO ₄ is to pyrimidine; oligos with pyrimidine 3'-PO ₄ ends
<i>Bacillus subtilis</i> RNase	RNA	b	Where 3'-PO ₄ is to purine; oligos with purine 3'-PO ₄ ends
RNase T ₁	RNA	b	Where 3'-PO ₄ is to guanine
RNase T ₂	RNA	b	Where 3'-PO ₄ is to adenine
DNase I (pancreas)	DNA	a	Preferably between Py and Pu; nicks dsDNA, creating 3'-OH ends
DNase II (spleen, thymus, <i>Staphylococcus aureus</i>)	DNA	b	Oligo products
Nuclease S1	Both	a	Cleaves single-stranded but not double-stranded nucleic acids

Pancreatic RNase is an enzyme specific for *b* cleavage where a pyrimidine base lies to the 3'-side of the phosphodiester; it acts endo. The products are oligonucleotides with pyrimidine-3'-PO₄ ends:

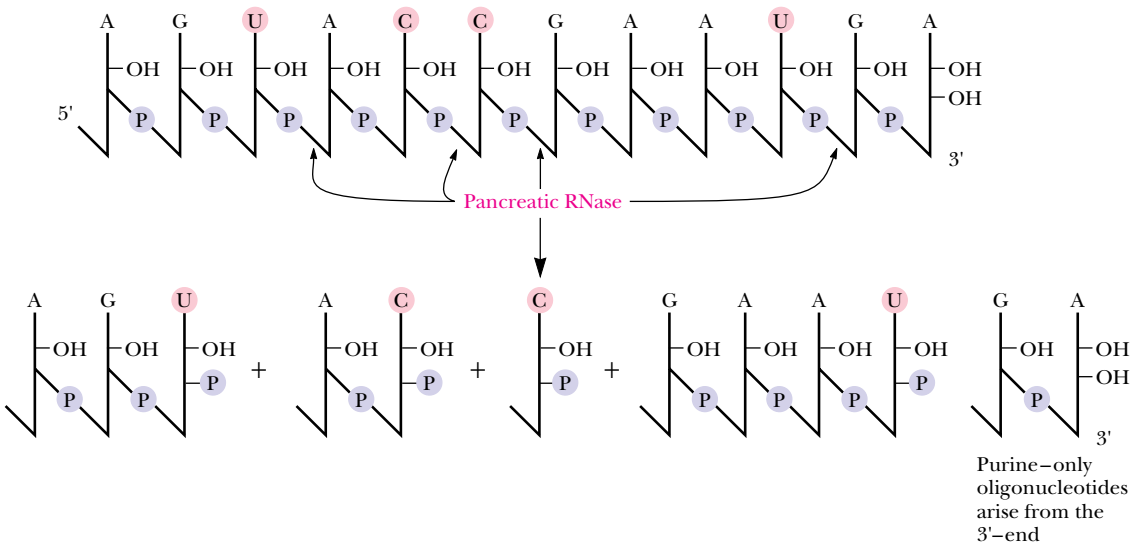


FIGURE 11.32 • An example of nuclease specificity: The specificity of RNA hydrolysis by bovine pancreatic RNase. This RNase cleaves *b* at 3'-pyrimidines, yielding oligonucleotides with pyrimidine 3'-PO₄ ends.

certain bases in a polynucleotide (Figure 11.32), or, as we shall see for *restriction endonucleases*, some nucleases will act only at a particular nucleotide sequence four to eight nucleotides in length. Table 11.4 lists the various permutations in specificity displayed by these nucleases and gives prominent examples of each. To the molecular biologist, nucleases are the surgical tools for the dissection and manipulation of nucleic acids in the laboratory.

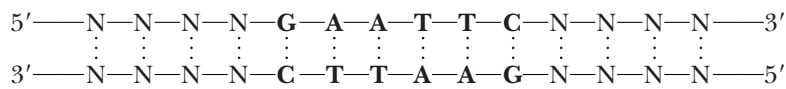
Exonucleases degrade nucleic acids by sequentially removing nucleotides from their ends. Two in common use are *snake venom phosphodiesterase* and *bovine spleen phosphodiesterase* (Figure 11.31). Because they act on either DNA or RNA, they are referred to by the generic name *phosphodiesterase*. These two enzymes have complementary specificities. Snake venom phosphodiesterase acts by *a* cleavage and starts at the free 3'-OH end of a polynucleotide chain, liberating nucleoside 5'-monophosphates. In contrast, the bovine spleen enzyme starts at the 5'-end of a nucleic acid, cleaving *b* and releasing 3'-NMPs.

Restriction Enzymes

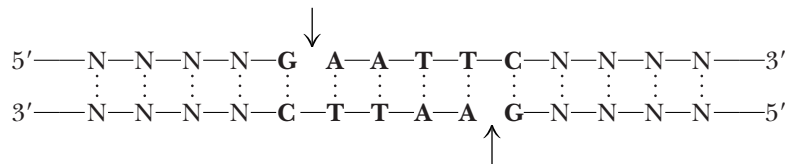
Restriction endonucleases are enzymes, isolated chiefly from bacteria, that have the ability to cleave double-stranded DNA. The term *restriction* comes from the capacity of prokaryotes to defend against or “restrict” the possibility of takeover by foreign DNA that might gain entry into their cells. Prokaryotes degrade foreign DNA by using their unique restriction enzymes to chop it into relatively large but noninfective fragments. Restriction enzymes are classified into three types, I, II, or III. Types I and III require ATP to hydrolyze DNA and can also catalyze chemical modification of DNA through addition of methyl groups to specific bases. Type I restriction endonucleases cleave DNA randomly, while type III recognize specific nucleotide sequences within dsDNA and cut the DNA at or near these sites.

Type II Restriction Endonucleases

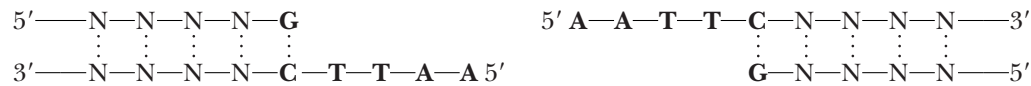
Type II restriction enzymes have received widespread application in the cloning and sequencing of DNA molecules. Their hydrolytic activity is not ATP-dependent, and they do not modify DNA by methylation or other means. Most importantly, they cut DNA within or near particular nucleotide sequences that they specifically recognize. These recognition sequences are typically four or six nucleotides in length and have a twofold axis of symmetry. For example, *E. coli* has a restriction enzyme, *EcoRI*, that recognizes the hexanucleotide sequence GAATTC:



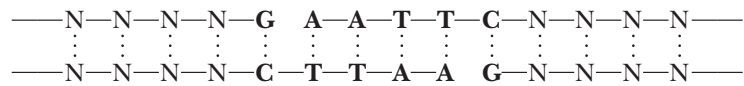
Note the twofold symmetry: the sequence read 5'→3' is the same in both strands.
When *EcoRI* encounters this sequence in dsDNA, it causes a staggered, double-stranded break by hydrolyzing each chain between the G and A residues:



Staggered cleavage results in fragments with protruding single-stranded 5'-ends:



Because the protruding termini of *EcoRI* fragments have complementary base sequences, they can form base pairs with one another.



Therefore, DNA restriction fragments having such “sticky” ends can be joined together to create new combinations of DNA sequence. If the fragments are derived from DNA molecules of different origin, novel recombinant forms of DNA are created.

EcoRI leaves staggered 5'-termini. Other restriction enzymes, such as *PstI*, which recognizes the sequence 5'-CTGCAG-3' and cleaves between A and G, produce cohesive staggered 3'-ends. Still others, such as *BalI*, act at the center of the twofold symmetry axis of their recognition site and generate blunt ends that are noncohesive. *BalI* recognizes 5'-TGGCCA-3' and cuts between G and C.

Table 11.5

Restriction Endonucleases

About 1000 restriction enzymes have been characterized. They are named by italicized three-letter codes, the first a capital letter denoting the genus of the organism of origin, while the next two letters are an abbreviation of the particular species. Because prokaryotes often contain more than one restriction enzyme, the various representatives are assigned letter and number codes as they are identified. Thus, *EcoRI* is the initial restriction endonuclease isolated from *Escherichia coli*, strain R. With one exception (*NciI*), all known type II restriction endonucleases generate fragments with 5'-PO₄ and 3'-OH ends.

Enzyme	Common Isoschizomers	Recognition Sequence	Compatible Cohesive Ends
<i>AluI</i>	<i>AtuI</i> , <i>EcoRII</i>	AG↓CT	Blunt
<i>ApyI</i>		CC↓(A)GG	
<i>AsuII</i>		TT↓CGAA	<i>ClaI</i> , <i>HpaII</i> , <i>TaqI</i>
<i>AvaI</i>		G↓PyCGPuG	<i>SalI</i> , <i>XhoI</i> , <i>XmaI</i>
<i>AvrII</i>		C↓CTAGG	
<i>BalI</i>	<i>AtuI</i> , <i>ApyI</i>	TGG↓CCA	Blunt
<i>BamHI</i>		G↓GATCC	<i>BclI</i> , <i>BglII</i> , <i>MboI</i> , <i>Sau3A</i> , <i>XhoII</i>
<i>BclI</i>		T↓GATCA	<i>BamHI</i> , <i>BglII</i> , <i>MboI</i> , <i>Sau3A</i> , <i>XhoII</i>
<i>BglII</i>		A↓GATCT	<i>BamHI</i> , <i>BclI</i> , <i>MboI</i> , <i>Sau3A</i> , <i>XhoII</i>
<i>BstEII</i>		G↓GTNACC	
<i>BstXI</i>		CCANNNNN↓NTGG	
<i>ClaI</i>		AT↓CGAT	<i>AccI</i> , <i>AcyI</i> , <i>AsyII</i> , <i>HpaII</i> , <i>TaqI</i>
<i>DdeI</i>		C↓TNAG	
<i>EcoRI</i>		G↓AATTC	
<i>EcoRII</i>		↓CC (A)GG	
<i>FnuDII</i>	<i>ThaI</i>	CG↓CG	Blunt
<i>HaeI</i>		(A)GG↓CC(A)	Blunt
<i>HaeII</i>		PuGCGC↓Py	
<i>HaeIII</i>		GG↓CC	Blunt
<i>HincII</i>		GTPy↓PuAC	Blunt
<i>HindIII</i>	<i>Sau3A</i>	A↓AGCTT	
<i>HpaI</i>		GTT↓AAC	Blunt
<i>HpaII</i>		C↓CGG	<i>AccI</i> , <i>AcyI</i> , <i>AsuII</i> , <i>ClaI</i> , <i>TaqI</i>
<i>KpnI</i>		GGTAC↓C	<i>BamHI</i> , <i>BclI</i> , <i>BglII</i> , <i>XhoI</i>
<i>MboI</i>		↓GATC	
<i>MspI</i>		C↓CGG	
<i>MstI</i>		TGC↓GCA	Blunt
<i>NotI</i>		GC↓GGCCGC	
<i>PstI</i>		CTGCA↓G	
<i>SacI</i>		GAGCT↓C	
<i>SalI</i>	<i>SstI</i>	G↓TCGAC	<i>AvaI</i> , <i>XhoI</i>
<i>Sau3A</i>		↓GATC	<i>BamHI</i> , <i>BclI</i> , <i>BglII</i> , <i>MboI</i> , <i>XhoII</i>
<i>SfiI</i>		GGCCNNNN↓NGGCC	
<i>SmaI</i>		CCC↓GGG	Blunt
<i>SphI</i>		GCATG↓C	
<i>SstI</i>	<i>SacI</i>	GAGCT↓C	
<i>TaqI</i>		T↓CGA	<i>AccI</i> , <i>AcyI</i> , <i>AsuII</i> , <i>ClaI</i> , <i>HpaII</i>
<i>XbaI</i>		T↓CTAGA	
<i>XhoI</i>		C↓TCGAG	<i>AvaI</i> , <i>SalI</i>
<i>XhoII</i>		(A)↓GATC(C)	<i>BamHI</i> , <i>BclI</i> , <i>BglII</i> , <i>MboI</i> , <i>Sau3A</i>
<i>XmaI</i>	<i>SmaI</i>	C↓CCGGG	<i>AvaI</i>

Table 11.5 lists many of the commonly used restriction endonucleases and their recognition sites. Because these sites all have twofold symmetry, only the sequence on one strand needs to be designated.

ISOSCHIZOMERS. Different restriction enzymes sometimes recognize and cleave within identical target sequences. For example, *MboI* and *Sau3A* recognize the same tetranucleotide run: 5'-GATC-3'. Both cleave the DNA strands at the same position, namely, on the 5'-side of the G. Such enzymes are called **isoschizomers**, meaning that they cut at the same site. The enzyme *BamHI* is an isoschizomer of *MboI* and *Sau3A* except that it has greater specificity because it acts only at hexanucleotide sequences reading GGATCC. *BamHI* cuts between the two G's, leaving cohesive 5'-ends that can match up with *MboI* or *Sau3A* fragments.

schizo • from the Greek *schizein*, to split

RESTRICTION FRAGMENT SIZE. Assuming random distribution and equimolar proportions for the four nucleotides in DNA, a particular tetranucleotide sequence should occur every 4⁴ nucleotides, or every 256 bases. Therefore, the fragments generated by a restriction enzyme that acts at a four-nucleotide sequence should average about 250 bp in length. “Six-cutters,” enzymes such as *EcoRI* or *BamHI*, will find their unique hexanucleotide sequences on the average once in every 4096 (4⁶) bp of length. Because the genetic code is a triplet code with three bases of DNA specifying one amino acid in a polypeptide sequence, and because polypeptides typically contain at most 1000 amino acid residues, the fragments generated by six-cutters are approximately the size of prokaryotic genes. This property makes these enzymes useful in the construction and cloning of genetically useful recombinant DNA molecules. For the isolation of even larger nucleotide sequences, such as those of genes encoding large polypeptides (or those of eukaryotic genes that are disrupted by large introns), partial or limited digestion of DNA by restriction enzymes can be employed. However, restriction endonucleases that cut only at specific nucleotide sequences 8 or even 13 nucleotides in length are also available, such as *NotI* and *SfiI*.

Restriction Mapping

The application of these sequence-specific nucleases to problems in molecular biology is considered in detail in Chapter 13, but one prominent application is described here. Because restriction endonucleases cut dsDNA at unique sites to generate large fragments, they provide a means for mapping DNA molecules that are many kilobase pairs in length. Restriction digestion of a DNA molecule is in many ways analogous to proteolytic digestion of a protein by an enzyme such as trypsin (see Chapter 5): the restriction endonuclease acts only at its specific sites so that a discrete set of nucleic acid fragments is generated. This action is analogous to trypsin cleavage only at Arg and Lys residues to yield a particular set of tryptic peptides from a given protein. The restriction fragments represent a unique collection of different-sized DNA pieces. Fortunately, this complex mixture can be resolved by *electrophoresis* (see the Appendix to Chapter 5). Electrophoresis of DNA molecules on gels of restricted pore size (as formed in agarose or polyacrylamide media) separates them according to size, the largest being retarded in their migration through the gel pores while the smallest move relatively unhindered. Figure 11.33 shows a hypothetical electrophoretogram obtained for a DNA molecule treated with two different restriction nucleases, alone and in combination. Just as cleavage of a protein with different proteases to generate overlapping fragments allows an ordering of the peptides, restriction fragments can be ordered or “mapped” according to their sizes, as deduced from the patterns depicted in Figure 11.33.

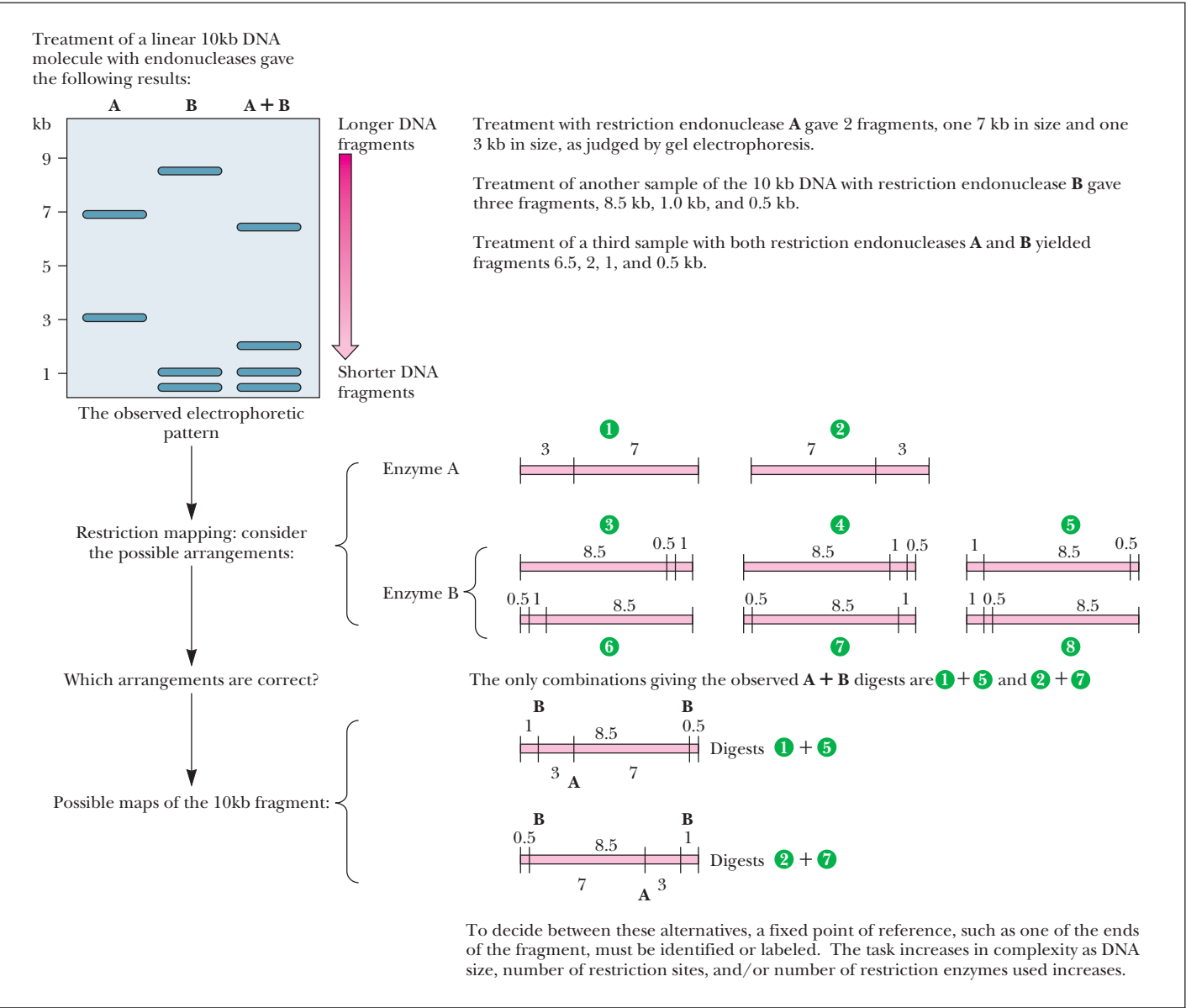


FIGURE 11.33 • Restriction mapping of a DNA molecule as determined by an analysis of the electrophoretic pattern obtained for different restriction endonuclease digests. (Keep in mind that a dsDNA molecule has a unique nucleotide sequence and therefore a definite polarity; thus, fragments from one end are distinctly different from fragments derived from the other end.)

PROBLEMS

1. Draw the chemical structure of pACG.
2. Chargaff's results (Table 11.3) yielded a molar ratio of 1.56 for A to G in human DNA, 1.75 for T to C, 1.00 for A to T, and 1.00 for G to C. Given these values, what are the mole fractions of A, C, G, and T in human DNA?
3. Adhering to the convention of writing nucleotide sequences in the 5' → 3' direction, what is the nucleotide sequence of the DNA strand that is complementary to d-ATCGCAACTGTCACTA?
4. Messenger RNAs are synthesized by RNA polymerases that read along a DNA template strand in the 3' → 5' direction, polymerizing ribonucleotides in the 5' → 3' direction (see Figure 11.24). Give the nucleotide sequence (5' → 3') of the DNA template strand from which the following mRNA segment was transcribed: 5'-UAGUGACAGUUGCGAU-3'.
5. The DNA strand that is complementary to the template strand copied by RNA polymerase during transcription has a nucleotide

sequence identical to that of the RNA being synthesized (except T residues are found in the DNA strand at sites where U residues occur in the RNA). An RNA transcribed from this nontemplate DNA strand would be complementary to the mRNA synthesized by RNA polymerase. Such an RNA is called antisense RNA. A promising strategy to thwart the deleterious effects of genes activated in disease states (such as cancer) is to generate antisense RNAs in affected cells. These antisense RNAs would form double-stranded hybrids with mRNAs transcribed from the activated genes and prevent their translation into protein. Suppose transcription of a cancer-activated gene yielded an mRNA whose sequence included the segment 5'-

UACGGUCUAAGCUGA. What is the corresponding nucleotide sequence (5' → 3') of the template strand in a DNA duplex that might be introduced into these cells so that an antisense RNA could be transcribed from it?

6. A 10-kb DNA fragment digested with restriction endonuclease *Eco*RI yielded fragments 4 kb and 6 kb in size. When digested with *Bam*HI, fragments 1, 3.5, and 5.5 kb were generated. Concomitant digestion with both *Eco*RI and *Bam*HI yielded fragments 0.5, 1, 3, and 5.5 kb in size. Give a possible restriction map for the original fragment.

FURTHER READING

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The Structure of DNA: “A melody for the eye of the intellect, with not a note wasted.”
Horace Freeland Judson, *The Eighth Day of Creation*

OUTLINE

- 12.1 • The Primary Structure of Nucleic Acids
- 12.2 • The ABZs of DNA Secondary Structure
- 12.3 • Denaturation and Renaturation of DNA
- 12.4 • Supercoils and Cruciforms: Tertiary Structure in DNA
- 12.5 • Chromosome Structure
- 12.6 • Chemical Synthesis of Nucleic Acids
- 12.7 • Secondary and Tertiary Structure of RNA

Chapter 12

Structure of Nucleic Acids



“Scherzo in D&A” (detail) by David E. Rodale (1955–1985)

Chapter 11 presented the structure and chemistry of nucleotides and how these units are joined via phosphodiester bonds to form nucleic acids, the biological polymers for information storage and transmission. In this chapter, we investigate biochemical methods that reveal this information by determining the sequential order of nucleotides in a polynucleotide, the so-called **primary structure** of nucleic acids. Then, we consider the higher orders of structure in the nucleic acids, the secondary and tertiary levels. Although the focus here is primarily on the structural and chemical properties of these macromolecules, it is fruitful to keep in mind the biological roles of these remarkable substances. The sequence of nucleotides in nucleic acids is the embodiment of genetic information (see Part IV). We can anticipate that the cellular mechanisms for accessing this information, as well as reproducing it with high fidelity, will be illuminated by knowledge of the chemical and structural qualities of these polymers.

12.1 • The Primary Structure of Nucleic Acids

As recently as 1975, determining the primary structure of nucleic acids (the nucleotide sequence) was a more formidable problem than amino acid sequencing of proteins, simply because nucleic acids contain only four unique monomeric units whereas proteins have twenty. With only four, there are *apparently* fewer specific sites for selective cleavage, distinctive sequences are more difficult to recognize, and the likelihood of ambiguity is greater. The much greater number of monomeric units in most polynucleotides as compared to polypeptides is a further difficulty. Two important breakthroughs reversed this situation so that now sequencing nucleic acids is substantially easier than sequencing polypeptides. One was the discovery of *restriction endonucleases* that cleave DNA at specific oligonucleotide sites, generating unique fragments of manageable size (see Chapter 11). The second is the power of *polyacrylamide gel electrophoresis* separation methods to resolve nucleic acid fragments that differ from one another in length by just one nucleotide.

Sequencing Nucleic Acids

Two basic protocols for nucleic acid sequencing are in widespread use: the **chain termination** or **dideoxy method** of F. Sanger and the **base-specific chemical cleavage method** developed by A. M. Maxam and W. Gilbert. Because both methods are carried out on nanogram amounts of DNA, very sensitive analytical techniques are used to detect the DNA chains following electrophoretic separation on polyacrylamide gels. Typically, the DNA molecules are labeled with radioactive ^{32}P ,¹ and following electrophoresis, the pattern of their separation is visualized by **autoradiography**. A piece of X-ray film is placed over the gel and the radioactive disintegrations emanating from ^{32}P decay create a pattern on the film that is an accurate image of the resolved oligonucleotides. Recently, sensitive biochemical and chemiluminescent methods have begun to supersede the use of radioisotopes as tracers in these experiments.

Chain Termination or Dideoxy Method

To appreciate the rationale of the chain termination or dideoxy method, we first must briefly examine the biochemistry of DNA replication. DNA is a double-helical molecule. In the course of its replication, the sequence of nucleotides in one strand is copied in a complementary fashion to form a new second strand by the enzyme **DNA polymerase**. Each original strand of the double helix serves as **template** for the biosynthesis that yields two daughter DNA duplexes from the parental double helix (Figure 12.1). DNA polymerase carries out this reaction *in vitro* in the presence of the four deoxynucleotide monomers and copies single-stranded DNA, provided a double-stranded region of DNA is artificially generated by adding a **primer**. This primer is merely an oligonucleotide capable of forming a short stretch of dsDNA by base pairing with the ssDNA (Figure 12.2). The primer must have a free 3'-OH end from which the new polynucleotide chain can grow as the first residue is added in the initial step of the polymerization process. DNA polymerases synthesize new strands by adding successive nucleotides in the 5' → 3' direction.

¹Because its longer half-life and lower energy make it more convenient to handle, ^{35}S is replacing ^{32}P as the radioactive tracer of choice in sequencing by the Sanger method. ^{35}S - α -labeled deoxynucleotide analogs provide the source for incorporating radioactivity into DNA.

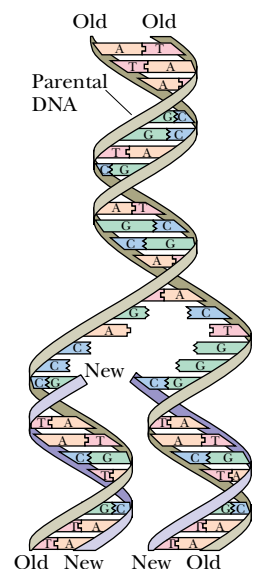
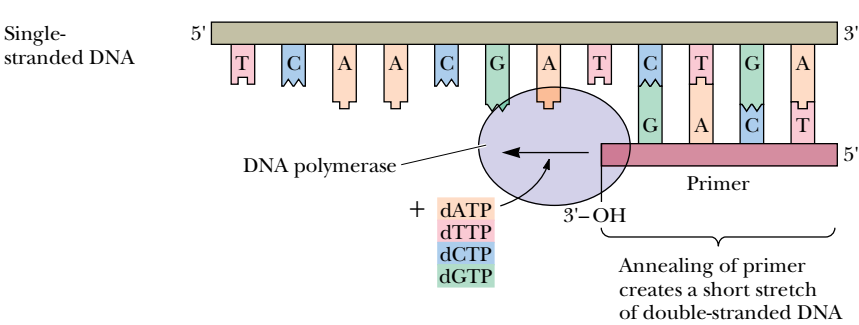


FIGURE 12.1 • DNA replication yields two daughter DNA duplexes identical to the parental DNA molecule. Each original strand of the double helix serves as a template, and the sequence of nucleotides in each of these strands is copied to form a new complementary strand by the enzyme DNA polymerase. By this process, biosynthesis yields two daughter DNA duplexes from the parental double helix.

FIGURE 12.2 • DNA polymerase copies ssDNA *in vitro* in the presence of the four deoxynucleotide monomers, provided a double-stranded region of DNA is artificially generated by adding a primer, an oligonucleotide capable of forming a short stretch of dsDNA by base pairing with the ssDNA. The primer must have a free 3'-OH end from which the new polynucleotide chain can grow as the first residue is added in the initial step of the polymerization process.



Chain Termination Protocol

In the chain termination method of DNA sequencing, a DNA fragment of unknown sequence serves as template in a polymerization reaction using some type of DNA polymerase, usually *Sequenase 2*[®], a genetically engineered version of bacteriophage T7 DNA polymerase that lacks all traces of exonuclease activity that might otherwise degrade the DNA. The primer requirement is met by an appropriate oligonucleotide (this method is also known as the **primed synthesis method** for this reason). Four parallel reactions are run; all four contain the four deoxynucleoside triphosphates dATP, dGTP, dCTP, and dTTP, which are the substrates for DNA polymerase (Figure 12.3 on the facing page). In each of the four reactions, a different 2',3'-**dideoxynucleotide** is included, and it is these dideoxynucleotides that give the method its name.

Because dideoxynucleotides lack 3'-OH groups, these nucleotides cannot serve as acceptors for 5'-nucleotide addition in the polymerization reaction, and thus the chain is terminated where they become incorporated. The concentrations of the four deoxynucleotides and the single dideoxynucleotide in each reaction mixture are adjusted so that the dideoxynucleotide is incorporated infrequently. Therefore, base-specific premature chain termination is only a random, occasional event, and a population of new strands of varying length is synthesized. Four reactions are run, one for each dideoxynucleotide, so that termination, although random, can occur everywhere in the sequence. In each mixture, each newly synthesized strand has a dideoxynucleotide at its 3'-end, and its presence at that position demonstrates that a base of that particular kind was specified by the template. A radioactively labeled dNTP is included in each reaction mixture to provide a tracer for the products of the polymerization process.

Reading Dideoxy Sequencing Gels

The sequencing products are visualized by autoradiography (or similar means) following their separation according to size by polyacrylamide gel electrophoresis (Figure 12.3). Because the smallest fragments migrate fastest upon electrophoresis and because fragments differing by only single nucleotides in length are readily resolved, the autoradiogram of the gel can be read from bottom to top, noting which lane has the next largest band at each step. Thus, the gel in Figure 12.3 is read AGCGTAGC (5' → 3'). Because of the way DNA polymerase acts, this observed sequence is complementary to the corresponding unknown template sequence. Knowing this, the template sequence now can be written GCTACGCT (5' → 3').

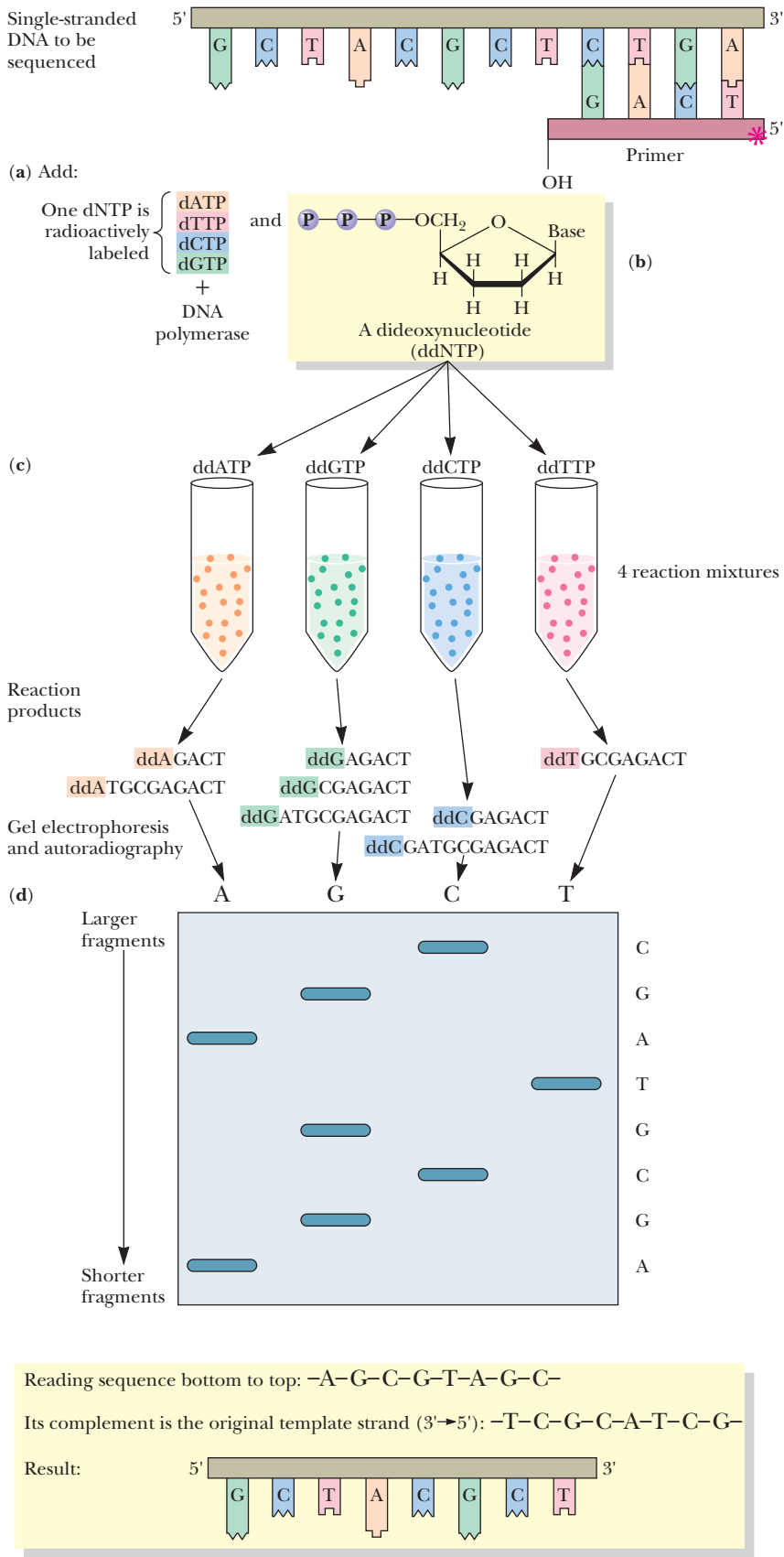


FIGURE 12.3 • The chain termination or dideoxy method of DNA sequencing. (a) DNA polymerase reaction. (b) Structure of dideoxynucleotide. (c) Four reaction mixtures with nucleoside triphosphates plus one dideoxynucleoside triphosphate. (d) Electrophoretogram. Note that the nucleotide sequence as read from the bottom to the top of the gel is the order of nucleotide addition carried out by DNA polymerase.

Base-Specific Chemical Cleavage Method

The base-specific chemical cleavage (or Maxam–Gilbert) method starts with a single-stranded DNA that is labeled at one end with radioactive ^{32}P . (Double-stranded DNA can be used if only one strand is labeled at only one of its ends.) The DNA strand is then randomly cleaved by reactions that specifically fragment its sugar–phosphate backbone only where certain bases have been chemically removed. There is no unique reaction for each of the four bases. However,

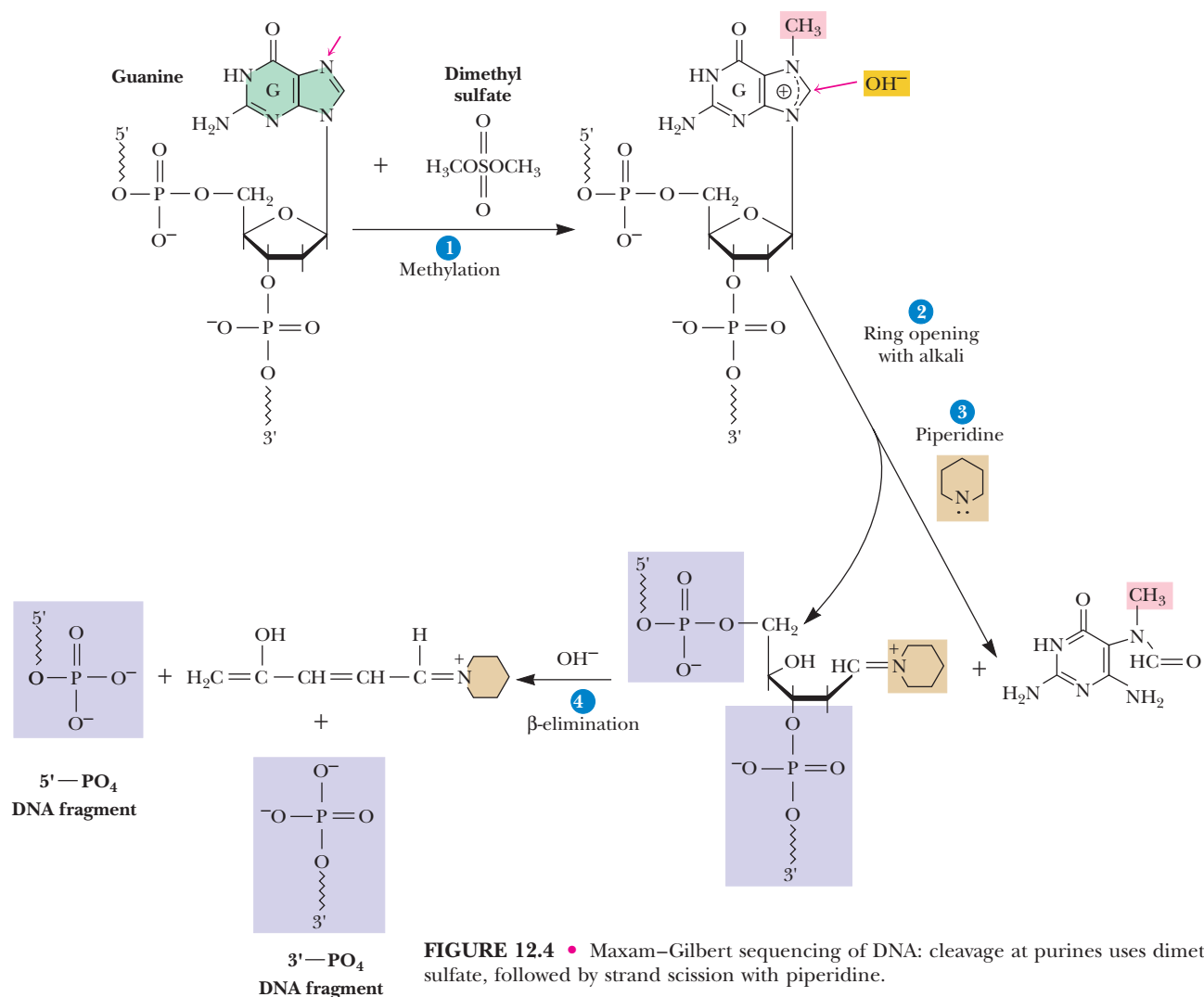


FIGURE 12.4 • Maxam–Gilbert sequencing of DNA: cleavage at purines uses dimethyl sulfate, followed by strand scission with piperidine.

Cleavage at G using dimethyl sulfate, followed by strand scission with piperidine: Under alkaline conditions, **dimethyl sulfate** reacts with guanine to methylate it at the 7-position (1). This substitution leads to instability of the N-9 glycosidic bond, so that in the presence of OH^- and the secondary amine **piperidine** (2), the purine ring is degraded and released. A β -elimination reaction facilitated by piperidine (3) then causes the excision of the naked deoxyribose moiety from the sugar–phosphate backbone, with consequent scission of the DNA strand to yield 5'- and 3'-fragments.

Cleavage at A or G: If the DNA is first treated with acid, dimethyl sulfate methylates adenine at the 3-position as well as guanine at the 7-position (not shown). Subsequent reaction with OH^- and piperidine triggers degradation and displacement of the methylated A or G purine base and strand scission, essentially as indicated here for reaction of dimethyl sulfate with guanine.

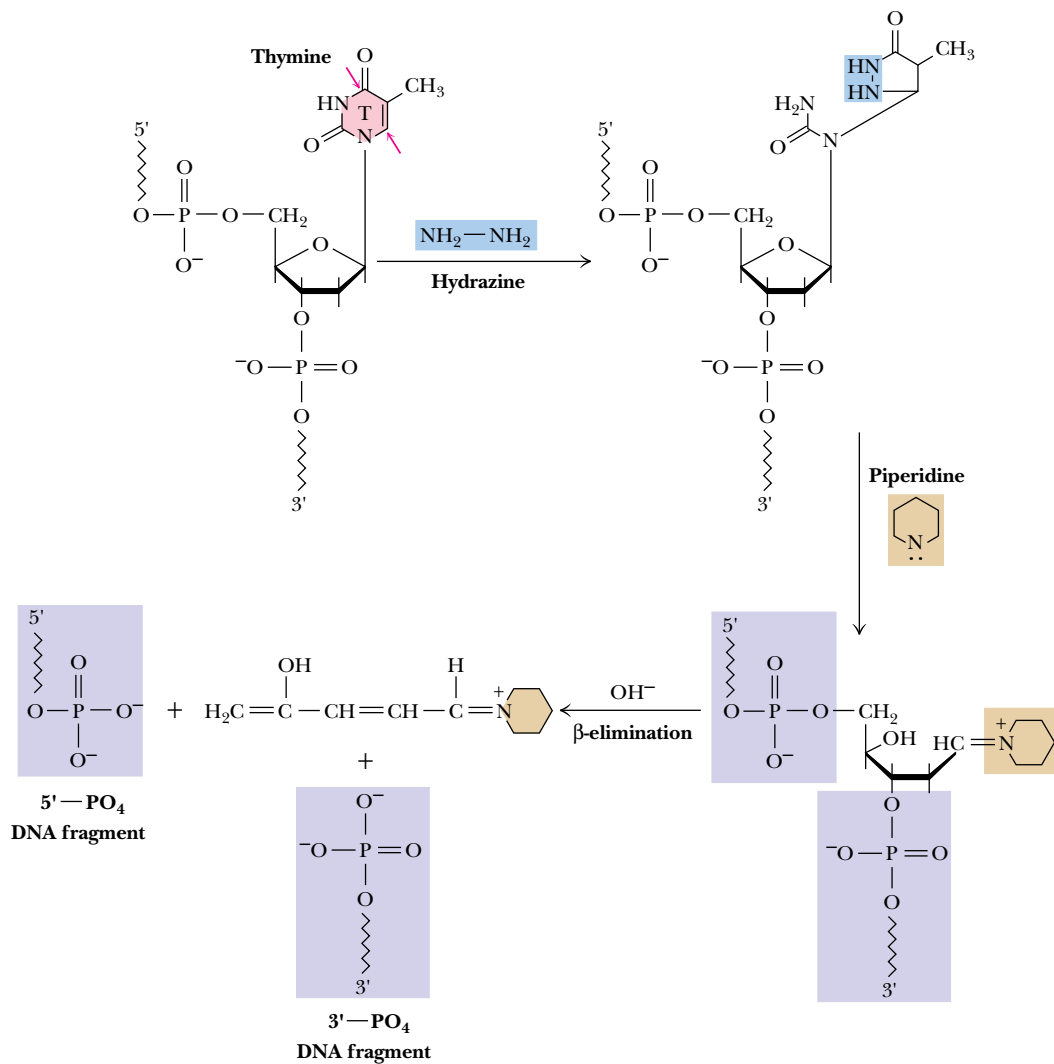
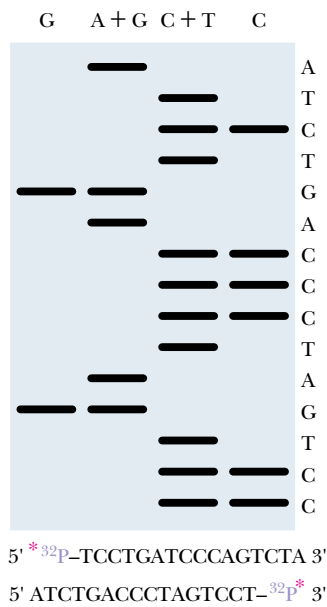


FIGURE 12.5 • Maxam–Gilbert sequencing of DNA: hydrolysis of pyrimidine rings by hydrazine. Hydrazine ($\text{H}_2\text{N--NH}_2$) attacks across the C-4 and C-6 atoms of pyrimidines to open the ring. This degradation subsequently leads to modification of the deoxyribose, rendering it susceptible to β -elimination by piperidine in the presence of hydroxide ion. Shown here is the excision of a T residue. As in Figure 12.4, 5'- and 3'-fragments are produced. The presence of high salt concentrations protects T (but not C) from reaction with hydrazine. In the presence of 2 M NaCl, the reaction shown here occurs only at C residues.

there is a reaction specific to G only and a purine-specific reaction that removes A or G (Figure 12.4). Thus, the difference in these two reactions is a specific indication of where A occurs. Similarly, there is a cleavage reaction specific for the pyrimidines (C+T) (Figure 12.5), which, if run in the presence of 1 or 2 M NaCl, works only with C. Differences in these two are thus attributable to the presence of T in the nucleotide sequence.

Note that the key to Maxam–Gilbert sequencing is to modify a base chemically so that it is removed from its sugar. Then piperidine excises the sugar from its 5'- and 3'-links in a β -elimination reaction. The conditions of chemical cleavage described in Figures 12.4 and 12.5 are generally adjusted so that,



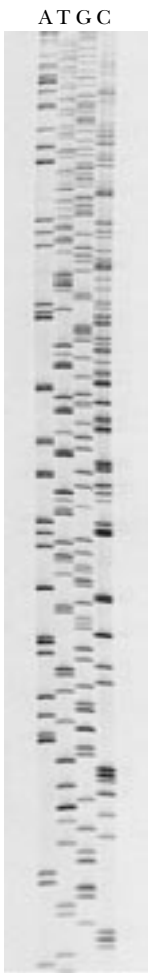
◀ **FIGURE 12.6** • Autoradiogram of a hypothetical electrophoretic pattern obtained for four reaction mixtures, performed as described in Figures 12.4 and 12.5 and run in the four lanes G, A+G, C+T, and C, respectively. Reading this pattern from the bottom up yields the sequence CCTGATCCCAGTCTA. The correct 5' → 3' order is determined by knowing which end of the ssDNA was ³²P-labeled. If the 5'-end was ³²P-labeled, only the 5'-fragments will be evident on the autoradiogram; the 3'-ends will be invisible. Similarly, if the 3'-end was originally labeled, only the 3'-fragments light up the autoradiogram. Assuming that the 5'-end was labeled, the sequence would be CCTGATCCCAGTCTA. If it were the 3'-end, the sequence read in the 5' → 3' convention would be ATCTGACCC-TAGTCC. An interesting feature of the Maxam–Gilbert sequencing procedure is that the base that is “read” in the ladder is actually not present in the oligonucleotide that identifies it. Thus, an unidentified base bears the label at the end of the smallest fragment; this unidentified base is the one that preceded the first identified base. For example, an oligonucleotide of either



or



would yield the same pattern in the autoradiogram. (Indication here of T as the end-labeled nucleotide is arbitrary.)



on average, only a single scission occurs per DNA molecule. However, because a very large number of DNA molecules exist in each reaction mixture, the products are a random collection of different-sized fragments wherein the occurrence of any base is represented by its unique pair of 5'- and 3'-cleavage products. These products form a complete set, the members of which differ in length by only one nucleotide, and they can be resolved by gel electrophoresis into a “ladder,” which can be visualized by autoradiography of the gel if the DNA fragments are radioactively labeled (Figure 12.6).

In principle, the Maxam–Gilbert method can provide the total sequence of a dsDNA molecule just by determining the purine positions on one strand and then the purines on the complementary strand. Complementary base-pairing rules then reveal the pyrimidines along each strand, T complementary to where A is, C complementary to where G occurs. (The analogous approach of locating the pyrimidines on each strand would also provide sufficient information to write the total sequence.)

With current technology, it is possible to read the order of as many as 400 bases from the autoradiogram of a sequencing gel (Figure 12.7). The actual chemical or enzymatic reactions, electrophoresis, and autoradiography are now routine, and a skilled technician can sequence about 1 kbp per week using these manual techniques. The major effort in DNA sequencing is in the isolation and preparation of fragments of interest, such as cloned genes.

Automated DNA Sequencing

In recent years, automated DNA sequencing machines capable of identifying about 10⁴ bases per day have become commercially available. One clever innovation has been the use of fluorescent dyes of different colors to uniquely label the primer DNA introduced into the four sequencing reactions; for example, red for the A reaction, blue for T, green for G, and yellow for C. Then, all four reaction mixtures can be combined and run together on one electrophoretic

◀ **FIGURE 12.7** • A photograph of the autoradiogram from an actual sequencing gel. A portion of the DNA sequence of *nit-6*, the *Neurospora* gene encoding the enzyme nitrite reductase. (James D. Colandene, University of Virginia)

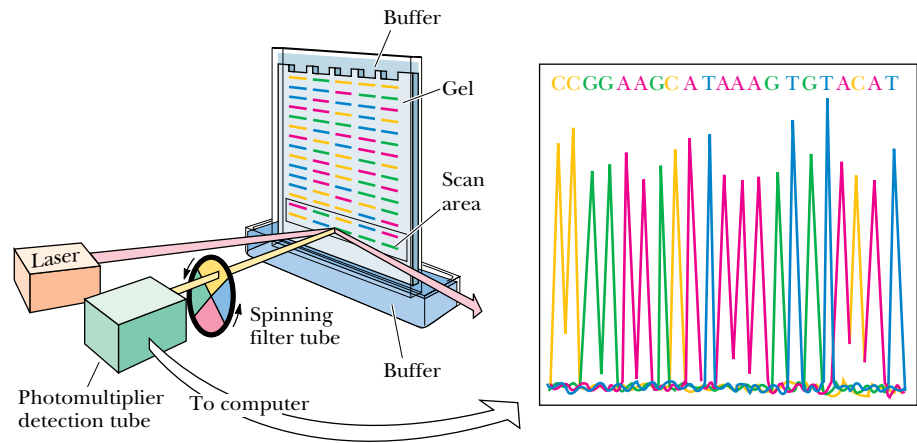


FIGURE 12.8 • Schematic diagram of the methodology used in fluorescent labeling and automated sequencing of DNA. Four reactions are set up, one for each base, and the primer in each is end-labeled with one of four different fluorescent dyes; the dyes serve to color-code the base-specific sequencing protocol (a unique dye is used in each dideoxynucleotide reaction). The four reaction mixtures are then combined and run in one lane. Thus, each lane in the gel represents a different sequencing experiment. As the differently sized fragments pass down the gel, a laser beam excites the dye in the scan area. The emitted energy passes through a rotating color filter and is detected by a fluorometer. The color of the emitted light identifies the final base in the fragment. (*Applied Biosystems, Inc., Foster City, CA*)

gel slab. As the oligonucleotides are separated and pass to the bottom of the gel, each is illuminated by a low-power argon laser beam that causes the dye attached to the primer to fluoresce. The color of the fluorescence is detected automatically, revealing the identity of the primer, and hence the base, immediately (Figure 12.8). The development of such automation has opened the possibility for sequencing the entire human genome, some 2.9 billion bp. Even so, if 100 automated machines operating at peak efficiency were dedicated to the task, it would still take at least 8 years to complete!

12.2 • The ABZs of DNA Secondary Structure

Double-stranded DNA molecules assume one of three secondary structures, termed A, B, and Z. Fundamentally, double-stranded DNA is a regular two-chain structure with hydrogen bonds formed between opposing bases on the two chains (see Chapter 11). Such H-bonding is possible only when the two chains are antiparallel. The polar sugar–phosphate backbones of the two chains are on the outside. The bases are stacked on the inside of the structure; these heterocyclic bases, as a consequence of their π -electron clouds, are hydrophobic on their flat sides. One purely hypothetical conformational possibility for a two-stranded arrangement would be a ladderlike structure (Figure 12.9) in which the base pairs are fixed at 0.6 nm apart because this is the distance between adjacent sugars in the DNA backbone. Because H₂O molecules would be accessible to the spaces between the hydrophobic surfaces of the bases, this conformation is energetically unfavorable. This ladderlike structure converts to a helix when given a simple right-handed twist. Helical twisting brings the base-pair rungs of the ladder closer together, stacking them 0.34 nm apart, without affecting the sugar–sugar distance of 0.6 nm. Because this helix repeats itself approximately every 10 bp, its **pitch** is 3.4 nm. This is the major conformation of DNA in solution and it is called **B-DNA**.

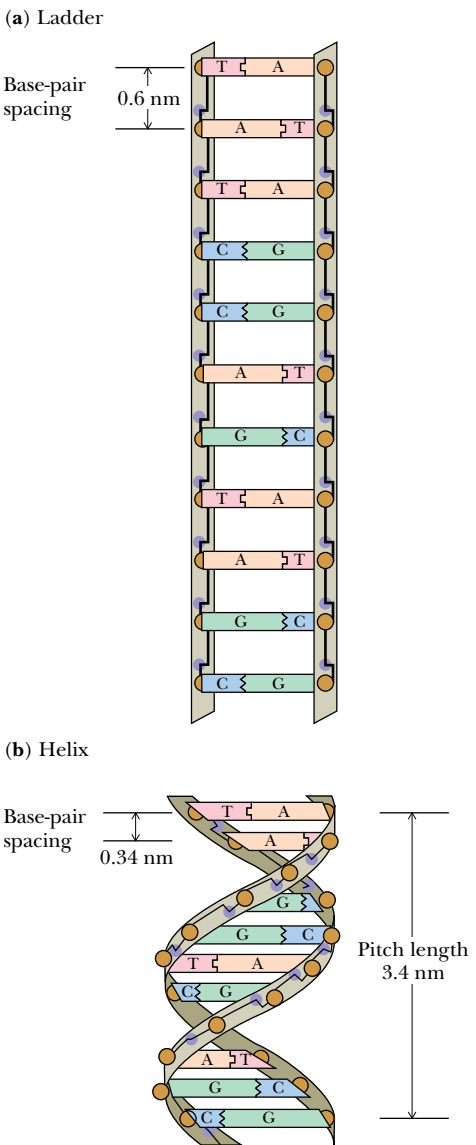


FIGURE 12.9 • (a) Double-stranded DNA as an imaginary ladderlike structure. (b) A simple right-handed twist converts the ladder to a helix.

Structural Equivalence of Watson–Crick Base Pairs

As indicated in Chapter 11, the base pairing in DNA is very specific: the purine adenine pairs with the pyrimidine thymine; the purine guanine pairs with the pyrimidine cytosine. Further, the A:T pair and G:C pair have virtually identical dimensions (Figure 12.10). Watson and Crick realized that units of such similarity could serve as spatially invariant substructures to build a polymer whose exterior dimensions would be uniform along its length, regardless of the sequence of bases.

The DNA Double Helix Is a Stable Structure

Several factors account for the stability of the double-helical structure of DNA. First, both internal and external hydrogen bonds stabilize the double helix. The two strands of DNA are held together by H-bonds that form between the complementary purines and pyrimidines, two in an A:T pair and three in a G:C pair (Figure 12.10), while polar atoms in the sugar–phosphate backbone form external H bonds with surrounding water molecules. Second, the negatively charged phosphate groups are all situated on the exterior surface of the helix in such a way that they have minimal effect on one another and are free to interact electrostatically with cations in solution such as Mg^{2+} . Third, the core of the helix consists of the base pairs, which, in addition to being H-bonded, stack together through hydrophobic interactions and van der Waals forces that contribute significantly to the overall stabilizing energy.

A stereochemical consequence of the way A:T and G:C base pairs form is that the sugars of the respective nucleotides have opposite orientations, and thus the sugar–phosphate backbones of the two chains run in opposite or

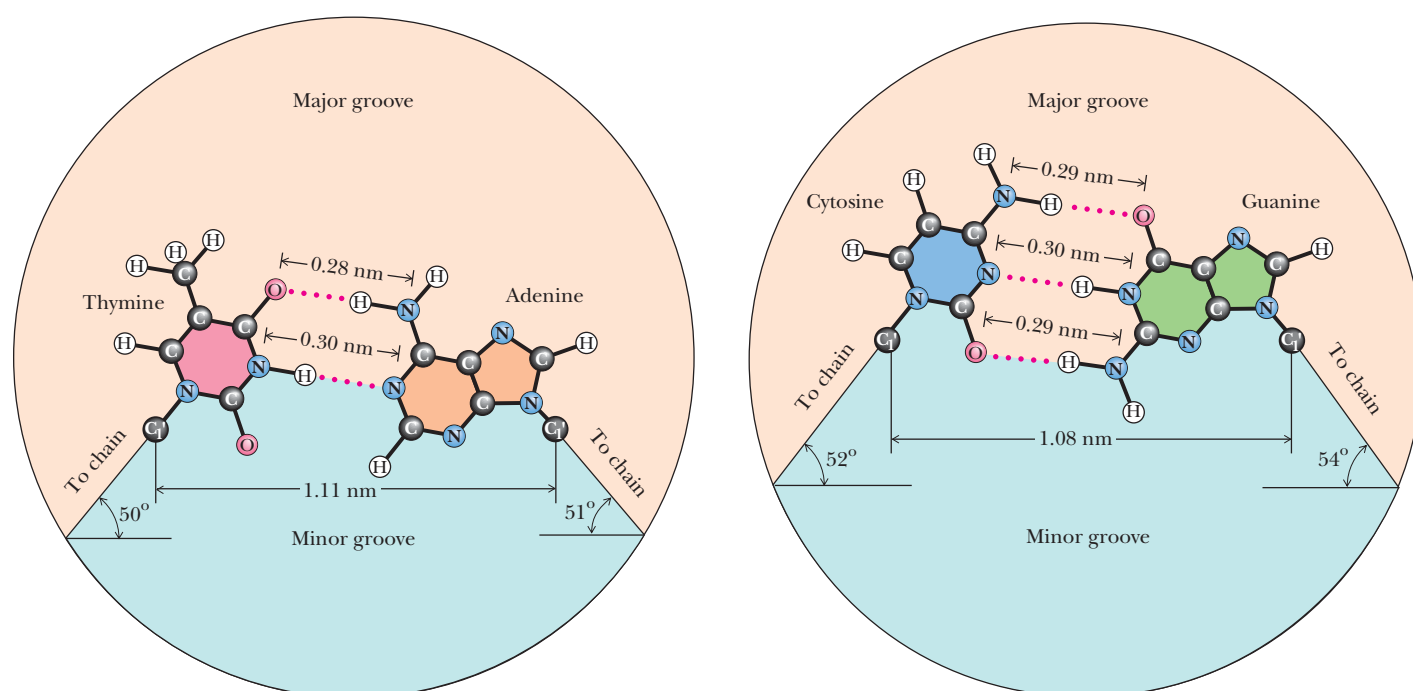


FIGURE 12.10 • Watson–Crick A:T and G:C base pairs. All H bonds in both base pairs are straight, with each H atom pointing directly at its acceptor N or O atom. Linear H bonds are the strongest. The mandatory binding of larger purines with smaller pyrimidines leads to base pairs that have virtually identical dimensions, allowing the two sugar–phosphate backbones to adopt identical helical conformations.

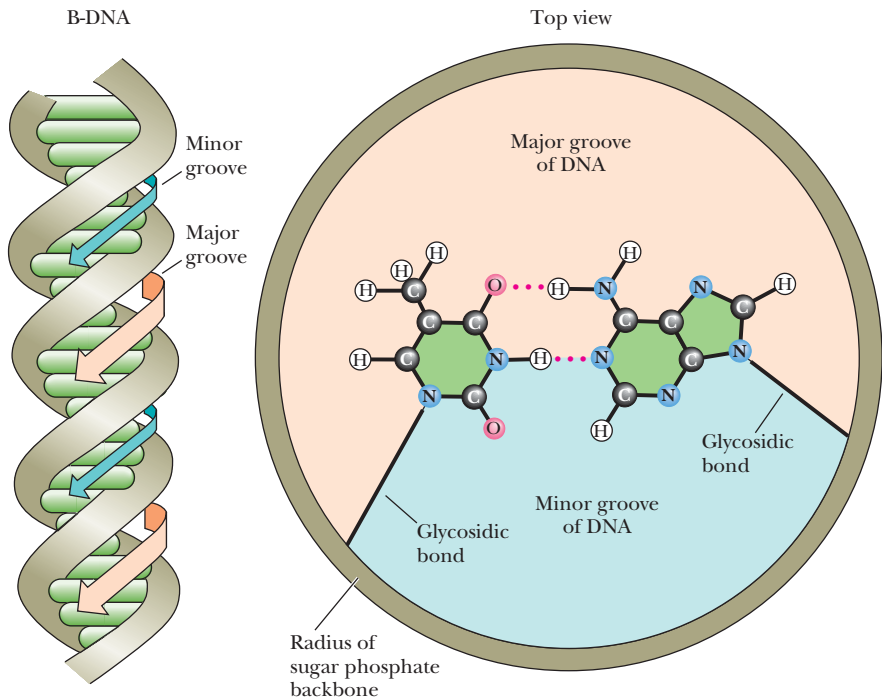


FIGURE 12.11 • The bases in a base pair are not directly across the helix axis from one another along some diameter but rather are slightly displaced. This displacement, and the relative orientation of the glycosidic bonds linking the bases to the sugar–phosphate backbone, leads to differently sized grooves in the cylindrical column created by the double helix, the major groove and the minor groove, each coursing along its length.

“antiparallel” directions. Furthermore, the two glycosidic bonds holding the bases in each base pair are not directly across the helix from each other, defining a common diameter (Figure 12.11). Consequently, the sugar–phosphate backbones of the helix are not equally spaced along the helix axis, and the grooves between them are not the same size. Instead, the intertwined chains create a **major groove** and a **minor groove** (Figure 12.11). The edges of the base pairs have a specific relationship to these grooves. The “top” edges of the base pairs (“top” as defined by placing the glycosidic bond at the bottom, as in Figure 12.10) are exposed along the interior surface or “floor” of the major groove; the base-pair edges nearest to the glycosidic bond form the interior surface of the minor groove. Some proteins that bind to DNA can actually recognize specific nucleotide sequences by “reading” the pattern of H-bonding possibilities presented by the edges of the bases in these grooves. Such DNA–protein interactions provide one step toward understanding how cells regulate the expression of genetic information encoded in DNA (see Chapter 32).

Conformational Variation in Double-Helical Structures

In solution, DNA ordinarily assumes the structure we have been discussing: B-DNA. However, nucleic acids also occur naturally in other double-helical forms. The base-pairing arrangement remains the same, but the sugar–phosphate groupings that constitute the backbone are inherently flexible and can adopt different conformations. One conformational variation is **propeller twist** (Figure 12.12). Propeller twist allows greater overlap between successive bases along a strand of DNA and diminishes the area of contact between bases and solvent water.

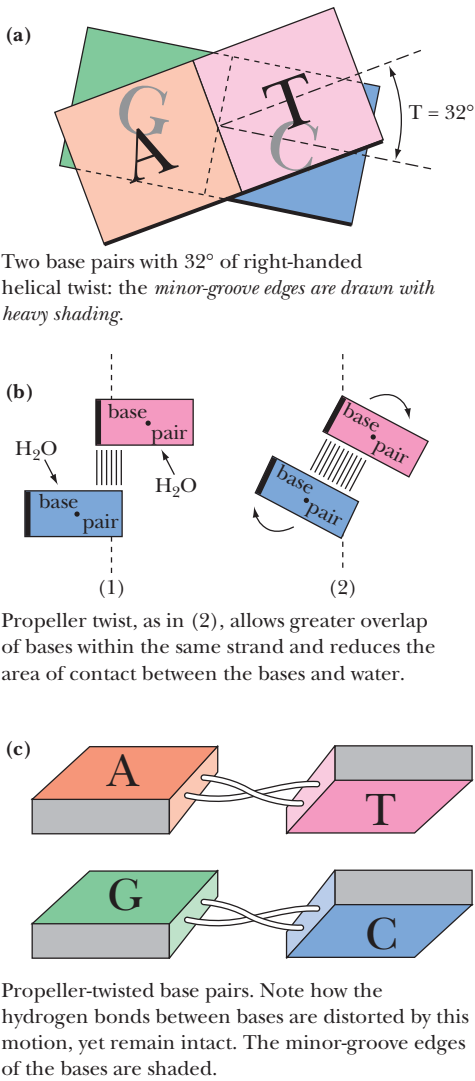
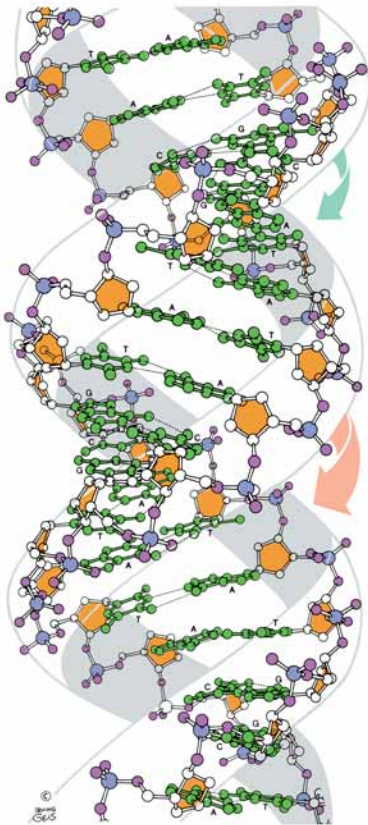
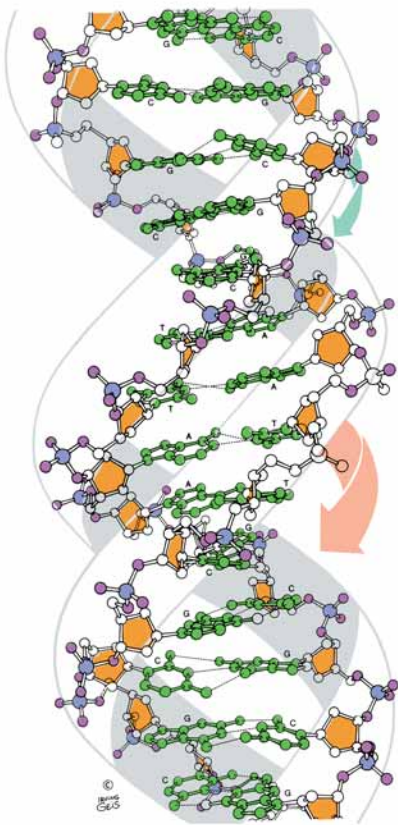


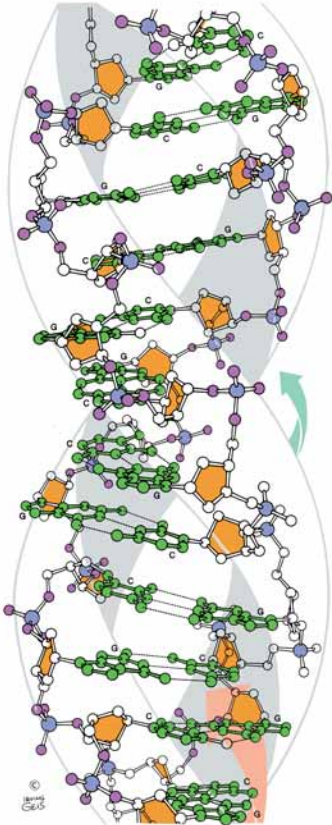
FIGURE 12.12 • Helical twist and propeller twist in DNA. (a) Successive base pairs in B-DNA show a rotation with respect to each other (so-called helical twist) of 36° or so, as viewed down the cylindrical axis of the DNA. (b) Rotation in a different dimension—**propeller twist**—allows the hydrophobic surfaces of bases to overlap better. The view here is edge-on to two successive bases in one DNA strand (as if the two bases on the right-hand strand of DNA in (a) were viewed from the right-hand margin of the page; dots represent end-on views down the glycosidic bonds). Clockwise rotation (as shown here) has a positive sign. (c) The two bases on the left-hand strand of DNA in (a) also show positive propeller twist (a clockwise rotation of the two bases in (a) as viewed from the left-hand margin of the paper). (Adapted from Figure 3.4 in Callandine, C. R., and Drew, H. R., 1992. Understanding DNA: The Molecule and How It Works. London: Academic Press.)



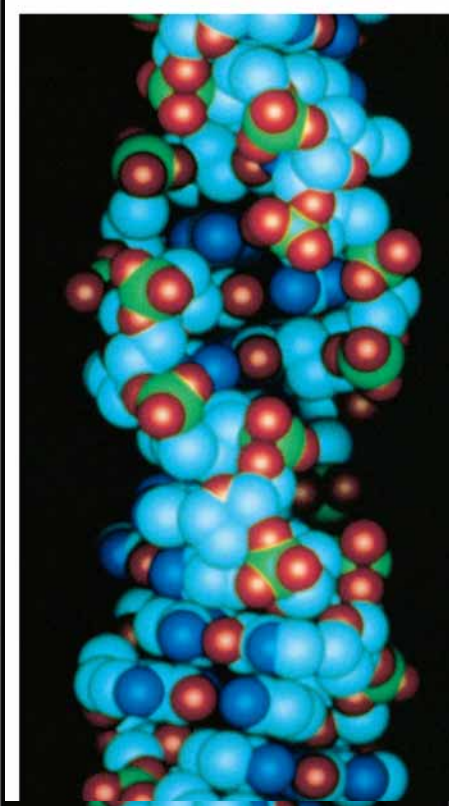
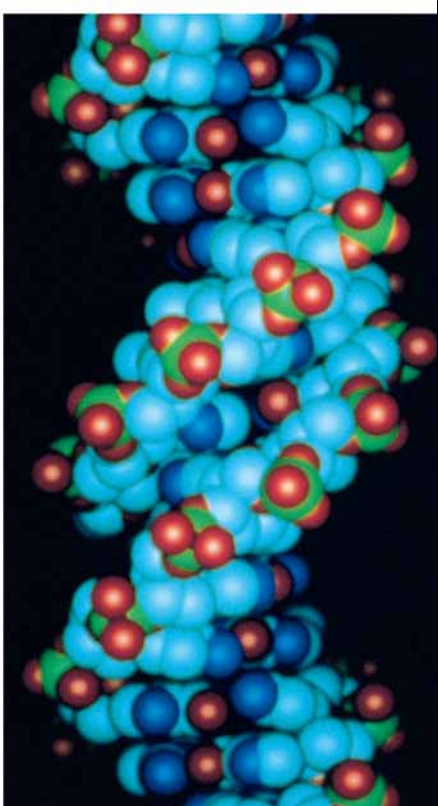
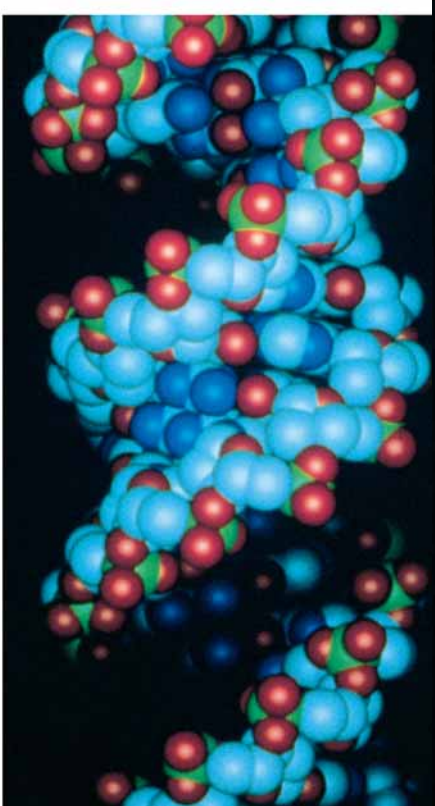
A
DNA



B
DNA



Z
DNA



Alternative Form of Right-Handed DNA

An alternative form of the right-handed double helix is **A-DNA**. A-DNA molecules differ in a number of ways from B-DNA. The pitch, or distance required to complete one helical turn, is different. In B-DNA, it is 3.4 nm, whereas in A-DNA it is 2.46 nm. One turn in A-DNA requires 11 bp to complete. Depending on local sequence, 10 to 10.6 bp define one helical turn in B-form DNA. In A-DNA, the base pairs are no longer nearly perpendicular to the helix axis but instead are tilted 19° with respect to this axis. Successive base pairs occur every 0.23 nm along the axis, as opposed to 0.332 nm in B-DNA. The B-form of DNA is thus longer and thinner than the short, squat A-form, which has its base pairs displaced around, rather than centered on, the helix axis. Figure 12.13 shows the relevant structural characteristics of the A- and B-forms of DNA. (Z-DNA, another form of DNA to be discussed shortly, is also depicted in Figure 12.13.) A comparison of the structural properties of A-, B-, and Z-DNA is summarized in Table 12.1.

Although relatively dehydrated DNA fibers can be shown to adopt the A-conformation under physiological conditions, it is unclear whether DNA ever assumes this form *in vivo*. However, double-helical DNA:RNA hybrids probably have an A-like conformation. The 2'-OH in RNA sterically prevents dou-

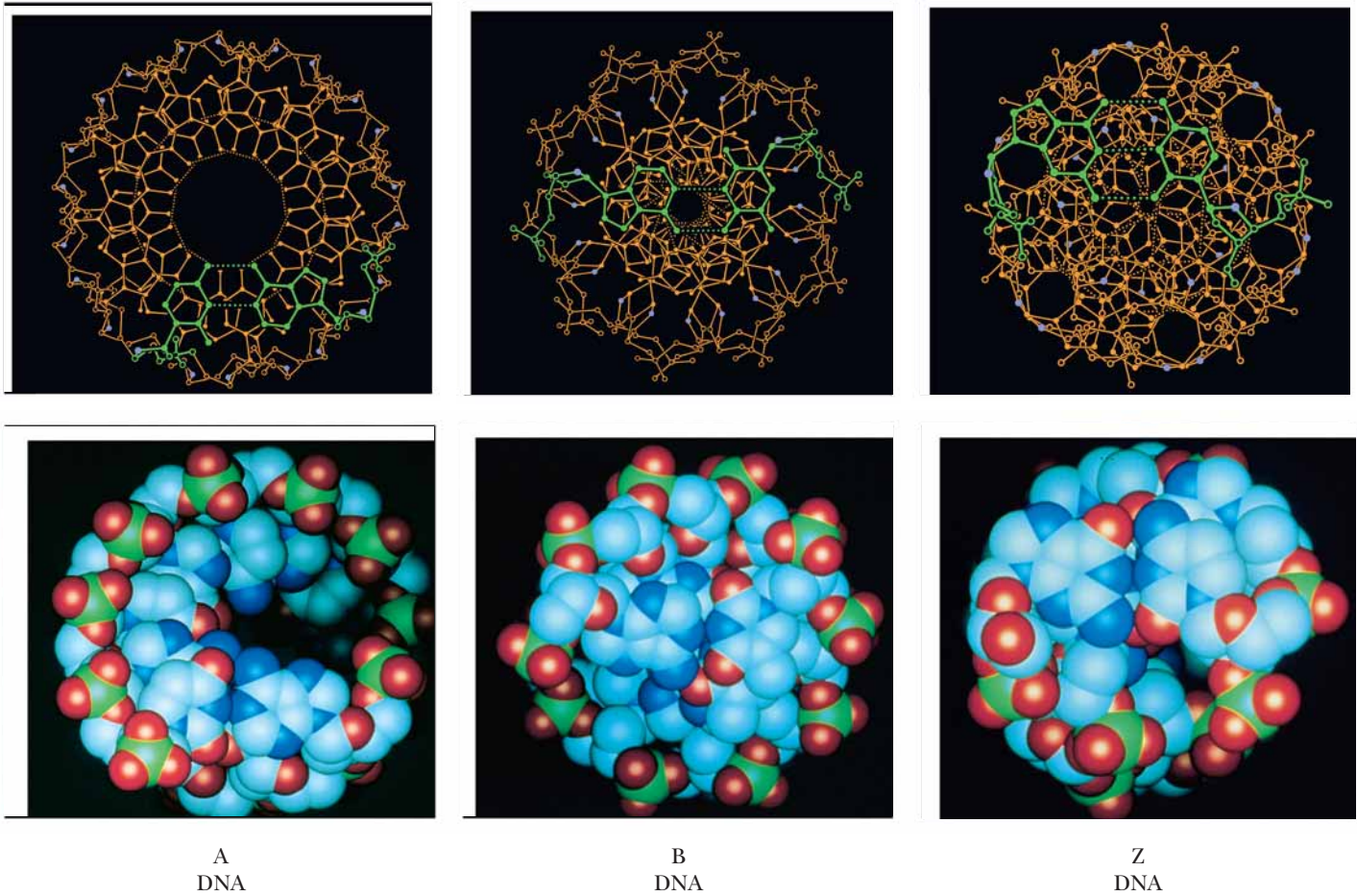


FIGURE 12.13 • (here and on the facing page) Comparison of the A-, B-, and Z-forms of the DNA double helix. The distance required to complete one helical turn is shorter in A-DNA than it is in B-DNA. The alternating pyrimidine–purine sequence of Z-DNA is the key to the “left-handedness” of this helix. (Robert Stodola, Fox Chase Cancer Research Center, and Irving Geis)

Table 12.1
Comparison of the Structural Properties of A-, B-, and Z-DNA

	Double Helix Type		
	A	B	Z
Overall proportions	Short and broad	Longer and thinner	Elongated and slim
Rise per base pair	2.3 Å	3.32 Å ± 0.19 Å	3.8 Å
Helix packing diameter	25.5 Å	23.7 Å	18.4 Å
Helix rotation sense	Right-handed	Right-handed	Left-handed
Base pairs per helix repeat	1	1	2
Base pairs per turn of helix	~11	~10	12
Mean rotation per base pair	33.6°	35.9° ± 4.2°	−60°/2
Pitch per turn of helix	24.6 Å	33.2 Å	45.6 Å
Base-pair tilt from the perpendicular	+19°	−1.2° ± 4.1°	−9°
Base-pair mean propeller twist	+18°	+16° ± 7°	~0°
Helix axis location	Major groove	Through base pairs	Minor groove
Major groove proportions	Extremely narrow but very deep	Wide and with intermediate depth	Flattened out on helix surface
Minor groove proportions	Very broad but shallow	Narrow and with intermediate depth	Extremely narrow but very deep
Glycosyl bond conformation	anti	anti	anti at C, syn at G

Adapted from Dickerson, R. L., et al., 1982. *Cold Spring Harbor Symposium on Quantitative Biology* 47:14.

ble-helical regions of RNA chains from adopting the B-form helical arrangement. Importantly, double-stranded regions in RNA chains assume an A-like conformation, with their bases strongly tilted with respect to the helix axis.

Z-DNA: A Left-Handed Double Helix

Z-DNA was first recognized by Alexander Rich and his colleagues at MIT in X-ray analysis of the synthetic deoxynucleotide dCpGpCpGpCpG, which crystallized into an antiparallel double helix of unexpected conformation. The alternating pyrimidine–purine (Py–Pu) sequence of this oligonucleotide is the key to its unusual properties. The *N*-glycosyl bonds of G residues in this alternating copolymer are rotated 180° with respect to their conformation in B-DNA, so that now the purine ring is in the syn rather than the anti conformation (Figure 12.14). The C residues remain in the anti form. Because the G ring is “flipped,” the C ring must also flip to maintain normal Watson–Crick base pairing. However, pyrimidine nucleosides do not readily adopt the syn conformation because it creates steric interference between the pyrimidine C-2 oxy substituent and atoms of the pentose. Because the cytosine ring does not rotate relative to the pentose, the whole C nucleoside (base and sugar) must flip 180° (Figure 12.15). It is topologically possible for the G to go syn and the C nucleoside to undergo rotation by 180° without breaking and re-forming the G:C hydrogen bonds. In other words, the B to Z structural transition can take place without disruption of the bonding relationships among the atoms involved.

Because alternate nucleotides assume different conformations, the repeating unit on a given strand in the Z-helix is the dinucleotide. That is, for any number of bases, *n*, along one strand, *n* – 1 dinucleotides must be considered. For example, a GpCpGpC subset of sequence along one strand is comprised

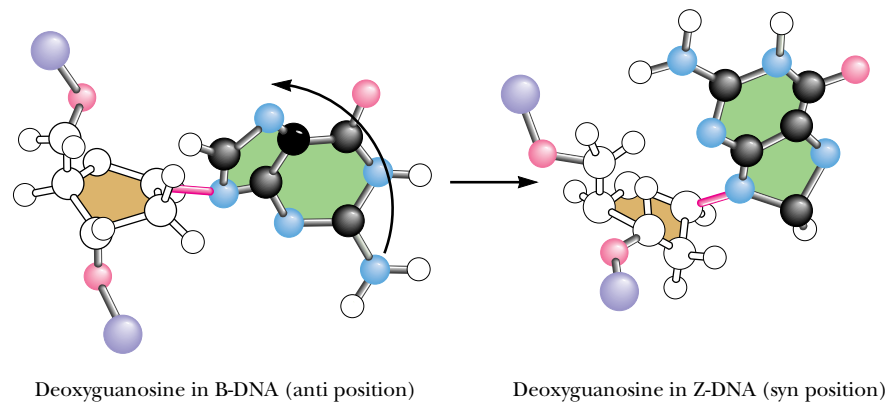


FIGURE 12.14 • Comparison of the deoxyguanosine conformation in B- and Z-DNA. In B-DNA, the C1'–N9 glycosyl bond is always in the anti position (*left*). In contrast, in the left-handed Z-DNA structure, this bond rotates (as shown) to adopt the syn conformation.

of *three* successive dinucleotide units: GpC, CpG, and GpC. (In B-DNA, the nucleotide conformations are essentially uniform and the repeating unit is the mononucleotide.) It follows that the CpG sequence is distinct conformationally from the GpC sequence along the alternating copolymer chains in the Z-double helix. The conformational alterations going from B to Z realign the sugar–phosphate backbone along a zigzag course that has a left-handed orientation (Figure 12.13), thus the designation *Z-DNA*. Note that in any GpCpGp subset, the sugar–phosphates of GpC form the horizontal “zig” while the CpG backbone segment forms the vertical “zag.” The mean rotation angle circumscribed around the helix axis is -15° for a CpG step and -45° for a GpC step (giving -60° for the dinucleotide repeat). The minus sign denotes a left-handed or counterclockwise rotation about the helix axis. Z-DNA is more elongated and slimmer than B-DNA.

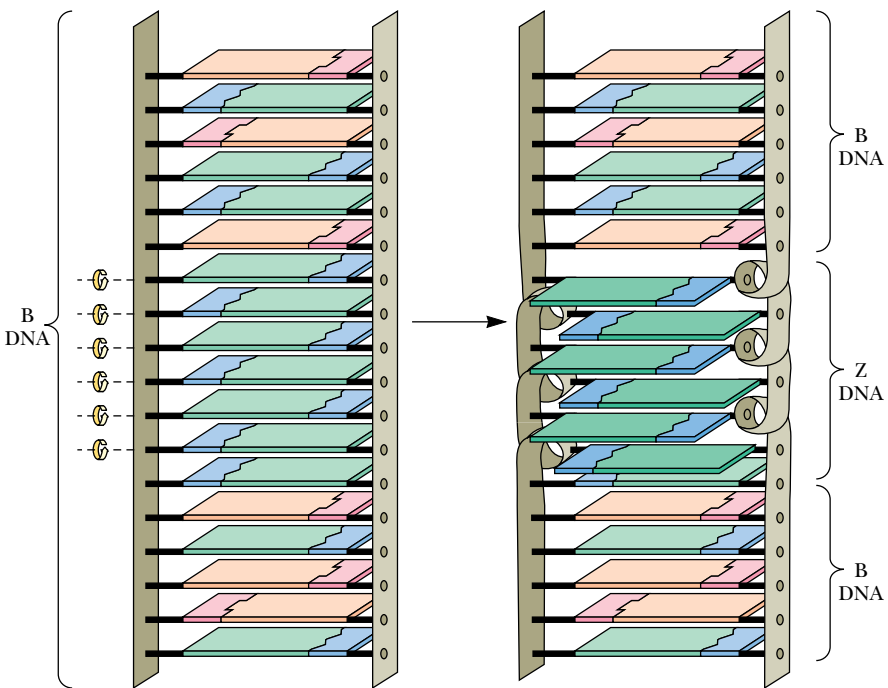


FIGURE 12.15 • The change in topological relationships of base pairs from B- to Z-DNA. A six-base-pair segment of B-DNA is converted to Z-DNA through rotation of the base pairs, as indicated by the curved arrows. The purine rings (green) of the deoxyguanosine nucleosides rotate via an anti to syn change in the conformation of the guanine–deoxyribose glycosidic bond; the pyrimidine rings (blue) are rotated by flipping the entire deoxycytidine nucleoside (base *and* deoxyribose). As a consequence of these conformational changes, the base pairs in the Z-DNA region no longer share π, π stacking interactions with adjacent B-DNA regions.

Cytosine Methylation and Z-DNA

The Z-form can arise in sequences that are not strictly alternating Py–Pu. For example, the hexanucleotide ${}^{\text{m}5}\text{CGAT}{}^{\text{m}5}\text{CG}$, a Py–Pu–Pu–Py–Py–Pu sequence containing two 5-methylcytosines (${}^{\text{m}5}\text{C}$), crystallizes as Z-DNA. Indeed, the *in vivo* methylation of C at the 5-position is believed to favor a B to Z switch because, in B-DNA, these hydrophobic methyl groups would protrude into the aqueous environment of the major groove and destabilize its structure. In Z-DNA, the same methyl groups can form a stabilizing hydrophobic patch. It is likely that the Z-conformation naturally occurs in specific regions of cellular DNA, which otherwise is predominantly in the B-form. Furthermore, because methylation is implicated in gene regulation, the occurrence of Z-DNA may affect the expression of genetic information (see Part IV, Information Transfer).

The Double Helix in Solution

The long-range structure of B-DNA in solution is not a rigid, linear rod. Instead, DNA behaves as a dynamic, flexible molecule. Localized thermal fluctuations temporarily distort and deform DNA structure over short regions. Base and backbone ensembles of atoms undergo elastic motions on a time scale of nanoseconds. To some extent, these effects represent changes in rotational angles of the bonds comprising the polynucleotide backbone. These changes are also influenced by sequence-dependent variations in base-pair stacking. The consequence is that the helix bends gently. When these variations are summed over the great length of a DNA molecule, the net result of these bending motions is that, at any given time, the double helix assumes a roughly spherical shape, as might be expected for a long, semi-rigid rod undergoing apparently random coiling. It is also worth noting that, on close scrutiny, the surface of the double helix is *not* that of a totally featureless, smooth, regular “barber pole” structure. Different base sequences impart their own special signatures to the molecule by subtle influences on such factors as the groove width, the angle between the helix axis and base planes, and the mechanical rigidity. Certain regulatory proteins bind to specific DNA sequences and participate in activating or suppressing expression of the information encoded therein. These proteins bind at unique sites by virtue of their ability to recognize novel structural characteristics imposed on the DNA by the local nucleotide sequence.

Intercalating Agents Distort the Double Helix

Aromatic macrocycles, flat hydrophobic molecules composed of fused, heterocyclic rings, such as **ethidium bromide**, **acridine orange**, and **actinomycin D** (Figure 12.16), can insert between the stacked base pairs of DNA. The bases are forced apart to accommodate these so-called **intercalating agents**, causing an unwinding of the helix to a more ladderlike structure. The deoxyribose–phosphate backbone is almost fully extended as successive base pairs are displaced 0.7 nm from one another, and the rotational angle about the helix axis between adjacent base pairs is reduced from 36° to 10° .

Dynamic Nature of the DNA Double Helix in Solution

Intercalating substances insert with ease into the double helix, indicating that the van der Waals bonds they form with the bases sandwiching them are more favorable than similar bonds between the bases themselves. Furthermore, the fact that these agents slip in suggests that the double helix must temporarily unwind and present gaps for these agents to occupy. That is, the DNA double helix in solution must be represented by a set of metastable alternatives to the standard B-conformation. These alternatives constitute a flickering repertoire of dynamic structures.

intercalate • to insert between others

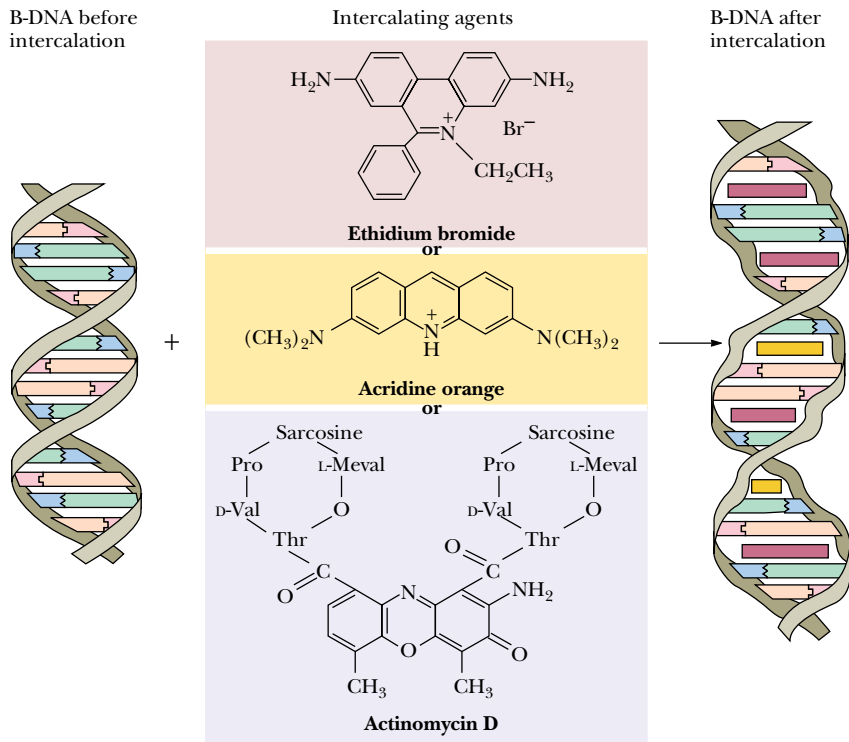
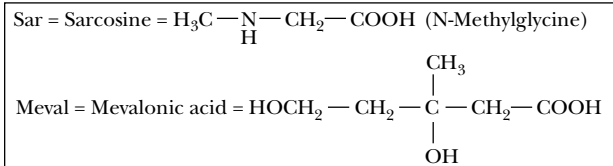


FIGURE 12.16 • The structures of ethidium bromide, acridine orange, and actinomycin D, three intercalating agents, and their effects on DNA structure.

12.3 • Denaturation and Renaturation of DNA

Thermal Denaturation and Hyperchromic Shift

When duplex DNA molecules are subjected to conditions of pH, temperature, or ionic strength that disrupt hydrogen bonds, the strands are no longer held together. That is, the double helix is **denatured** and the strands separate as individual random coils. If temperature is the denaturing agent, the double helix is said to *melt*. The course of this dissociation can be followed spectrophotometrically because the relative absorbance of the DNA solution at 260 nm increases as much as 40% as the bases unstack. This absorbance increase, or **hyperchromic shift**, is due to the fact that the aromatic bases in DNA interact via their π electron clouds when stacked together in the double helix. Because the UV absorbance of the bases is a consequence of π electron transitions, and because the potential for these transitions is diminished when the bases stack, the bases in duplex DNA absorb less 260-nm radiation than expected for their numbers. Unstacking alleviates this suppression of UV absorbance. The rise in absorbance coincides with strand separation, and the midpoint of the absorbance increase is termed the **melting temperature**, T_m .

FIGURE 12.17 • Heat denaturation of DNA from various sources, so-called melting curves. The midpoint of the melting curve is defined as the melting temperature, T_m . (From Marmur, J., 1959. *Nature* **183**:1427–1429.)

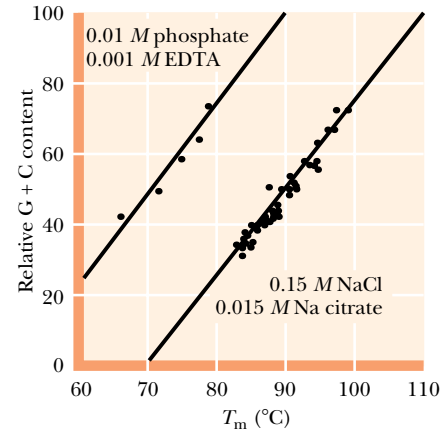
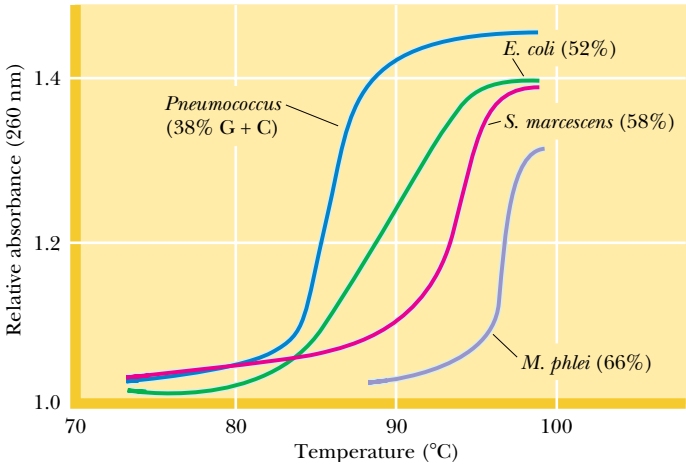


FIGURE 12.18 • The dependence of melting temperature on relative (G + C) content in DNA. Note that T_m increases if ionic strength is raised at constant pH (pH 7); 0.01 M phosphate+0.001 M EDTA versus 0.15 M NaCl/0.015 M Na citrate. In 0.15 M NaCl/0.015 M Na citrate, duplex DNA consisting of 100% A:T pairs melts at less than 70°C, whereas DNA of 100% G:C has a T_m greater than 110°C. (From Marmur, J., and Doty, P., 1962. *Journal of Molecular Biology* **5**:120.)

(Figure 12.17). DNAs differ in their T_m values because they differ in relative G + C content. The higher the G + C content of a DNA, the higher its melting temperature because G:C pairs are held by three H bonds whereas A:T pairs have only two. The dependence of T_m on the G + C content is depicted in Figure 12.18. Also note that T_m is dependent on the ionic strength of the solution; the lower the ionic strength, the lower the melting temperature. At 0.2 M Na^+ , $T_m = 69.3 + 0.41(\% \text{ G} + \text{C})$. Ions suppress the electrostatic repulsion between the negatively charged phosphate groups in the complementary strands of the helix, thereby stabilizing it. (DNA in pure water melts even at room temperature.) At high concentrations of ions, T_m is raised and the transition between helix and coil is sharp.

pH Extremes or Strong H-Bonding Solutes Also Denature DNA Duplexes

At pH values greater than 10, extensive deprotonation of the bases occurs, destroying their hydrogen bonding potential and denaturing the DNA duplex. Similarly, extensive protonation of the bases below pH 2.3 disrupts base pairing. Alkali is the preferred denaturant because, unlike acid, it does not hydrolyze the glycosidic linkages in the sugar–phosphate backbone. Small solutes that readily form H bonds are also DNA denaturants at temperatures below T_m if present in sufficiently high concentrations to compete effectively with the H-bonding between the base pairs. Examples include formamide and urea.

DNA Renaturation

Denatured DNA will **renature** to re-form the duplex structure if the denaturing conditions are removed (that is, if the solution is cooled, the pH is returned to neutrality, or the denaturants are diluted out). Renaturation requires reassociation of the DNA strands into a double helix, a process termed **reannealing**. For this to occur, the strands must realign themselves so that their complementary bases are once again in register and the helix can be zippered up (Figure 12.19). Renaturation is dependent both on DNA concentration and time. Many of the realignments are imperfect, and thus the strands must dissociate again to allow for proper pairings to be formed. The process occurs more quickly if the temperature is warm enough to promote diffusion of the large DNA molecules but not so warm as to cause melting.

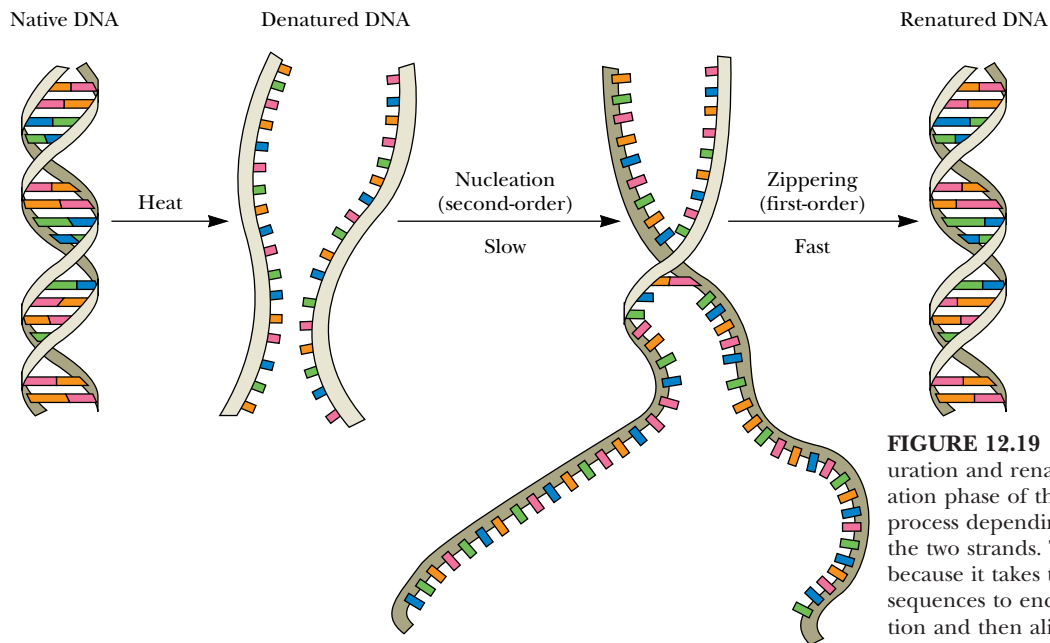


FIGURE 12.19 • Steps in the thermal denaturation and renaturation of DNA. The nucleation phase of the reaction is a second-order process depending on sequence alignment of the two strands. This process takes place slowly because it takes time for complementary sequences to encounter one another in solution and then align themselves in register. Once the sequences are aligned, the strands zipper up quickly.

Renaturation Rate and DNA Sequence Complexity— c_0t Curves

The renaturation rate of DNA is an excellent indicator of the sequence complexity of DNA. For example, bacteriophage T₄ DNA contains about 2×10^5 nucleotide pairs, whereas *Escherichia coli* DNA possesses 4.64×10^6 . *E. coli* DNA is considerably more complex in that it encodes more information. Expressed another way, for any given amount of DNA (in grams), the sequences represented in an *E. coli* sample are more heterogeneous, that is, more dissimilar from one another, than those in an equal weight of phage T₄ DNA. Therefore, it will take the *E. coli* DNA strands longer to find their complementary partners and reanneal. This situation can be analyzed quantitatively.

If c is the concentration of single-stranded DNA at time t , then the second-order rate equation for two complementary strands coming together is given by the rate of decrease in c :

$$-dc/dt = k_2c^2$$

where k_2 is the second-order rate constant. Starting with a concentration, c_0 , of completely denatured DNA at $t = 0$, the amount of single-stranded DNA remaining at some time t is

$$c/c_0 = 1/(1 + k_2c_0t)$$

where the units of c are mol of nucleotide per L and t is in seconds. The time for half of the DNA to renature (when $c/c_0 = 0.5$) is defined as $t = t_{1/2}$. Then,

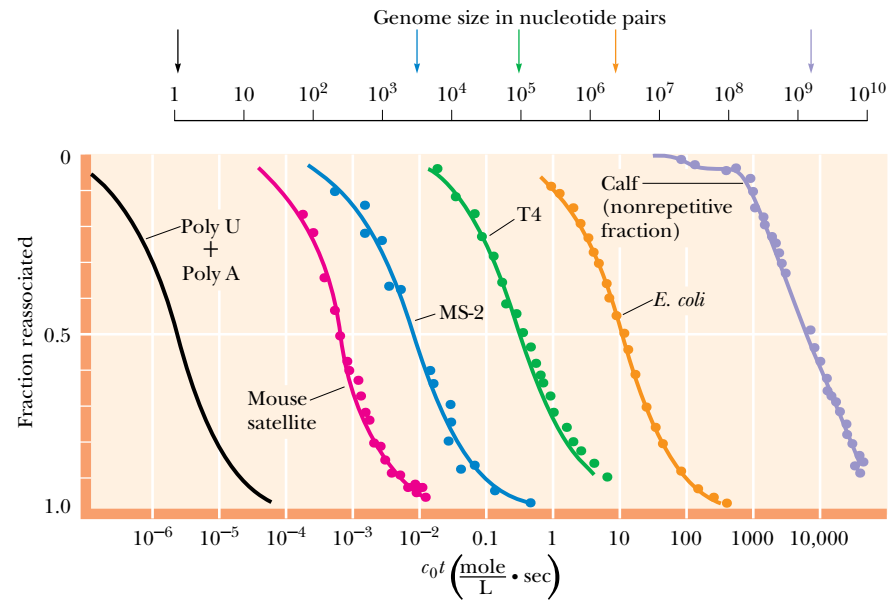
$$0.5 = 1/(1 + k_2c_0t_{1/2}) \quad \text{and thus} \quad 1 + k_2c_0t_{1/2} = 2$$

yielding

$$c_0t_{1/2} = 1/k_2$$

A graph of the fraction of single-stranded DNA reannealed (c/c_0) as a function of c_0t on a semilogarithmic plot is referred to as a c_0t (pronounced “cot”) **curve** (Figure 12.20). The rate of reassociation can be followed spectrophotometrically by the UV absorbance decrease as duplex DNA is formed. Note that

FIGURE 12.20 • These c_0t curves show the rates of reassociation of denatured DNA from various sources and illustrate how the rate of reassociation is inversely proportional to genome complexity. The DNA sources are as follows: poly A+poly U, a synthetic DNA duplex of poly A and poly U polynucleotide chains; mouse satellite DNA, a fraction of mouse DNA in which the same sequence is repeated many thousands of times; MS-2 dsRNA, the double-stranded form of RNA found during replication of MS-2, a simple bacteriophage; T4 DNA, the DNA of a more complex bacteriophage; *E. coli* DNA, bacterial DNA; calf DNA (nonrepetitive fraction), mammalian DNA (calf) from which the highly repetitive DNA fraction (satellite DNA) has been removed. Arrows indicate the genome size (in bp) of the various DNAs. (From Britten, R. J., and Kohne, D. E., 1968. Science **161**:529–540.)



relatively more complex DNAs take longer to renature, as reflected by their greater $c_0t_{1/2}$ values. Poly A and poly U (Figure 12.20) are minimally complex in sequence and anneal rapidly to form a double-stranded A:U polynucleotide. *Mouse satellite DNA* is a highly repetitive subfraction of mouse DNA. Its lack of sequence heterogeneity is seen in its low $c_0t_{1/2}$ value. MS-2 is a small bacteriophage whose genetic material is RNA. Calf thymus DNA is the mammalian representative in Figure 12.20.

Nucleic Acid Hybridization

If DNA from two different species are mixed, denatured, and allowed to cool slowly so that reannealing can occur, artificial **hybrid duplexes** may form, provided the DNA from one species is similar in nucleotide sequence to the DNA of the other. The degree of hybridization is a measure of the sequence similarity or *relatedness* between the two species. Depending on the conditions of the experiment, about 25% of the DNA from a human forms hybrids with mouse DNA, implying that some of the nucleotide sequences (genes) in humans are very similar to those in mice. Mixed RNA:DNA hybrids can be created *in vitro* if single-stranded DNA is allowed to anneal with RNA copies of itself, such as those formed when genes are transcribed into mRNA molecules.

Nucleic acid hybridization is a commonly employed procedure in molecular biology. First, it can reveal evolutionary relationships. Second, it gives researchers the power to identify specific genes selectively against a vast background of irrelevant genetic material. An appropriately labeled oligo- or polynucleotide, referred to as a **probe**, is constructed so that its sequence is complementary to a target gene. The probe specifically base pairs with the target gene, allowing identification and subsequent isolation of the gene. Also, the quantitative expression of genes (in terms of the amount of mRNA synthesized) can be assayed by hybridization experiments.

Buoyant Density of DNA

Not only the melting temperature of DNA but also its density in solution is dependent on relative G:C content. G:C-rich DNA has a significantly higher density than A:T-rich DNA. Furthermore, a linear relationship exists between the buoyant densities of DNA from different sources and their G:C content

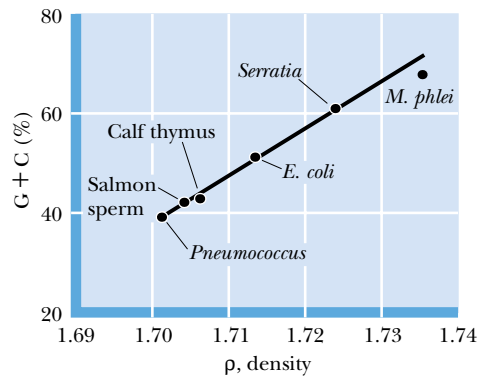


FIGURE 12.21 • The relationship of the densities (in g/mL) of DNAs from various sources and their G:C content. (From Doty, P., 1961. Harvey Lectures 55:103.)

(Figure 12.21). The density of DNA, ρ (in g/mL), as a function of its G:C content is given by the equation $\rho = 1.660 + 0.098(\text{GC})$, where (GC) is the mole fraction of (G + C) in the DNA. Because of its relatively high density, DNA can be purified from cellular material by a form of density gradient centrifugation known as *isopycnic centrifugation* (see Appendix to this chapter).

12.4 • Supercoils and Cruciforms: Tertiary Structure in DNA

The conformations of DNA discussed thus far are variations sharing a common secondary structural theme, the double helix, in which the DNA is assumed to be in a regular, linear form. DNA can also adopt regular structures of higher complexity in several ways. For example, many DNA molecules are circular. Most, if not all, bacterial chromosomes are covalently closed, circular DNA duplexes, as are almost all plasmid DNAs. **Plasmids** are naturally occurring, self-replicating, circular, extrachromosomal DNA molecules found in bacteria; plasmids carry genes specifying novel metabolic capacities advantageous to the host bacterium. Various animal virus DNAs are circular as well.

Supercoils

In duplex DNA, the two strands are wound about each other once every 10 bp, that is, once every turn of the helix. Double-stranded circular DNA (or linear DNA duplexes whose ends are not free to rotate), form **supercoils** if the strands are underwound (*negatively supercoiled*) or overwound (*positively supercoiled*) (Figure 12.22). Underwound duplex DNA has fewer than the natural number of turns, whereas overwound DNA has more. DNA supercoiling is analogous to twisting or untwisting a two-stranded rope so that it is torsionally stressed. Negative supercoiling introduces a torsional stress that favors unwinding of the right-handed B-DNA double helix, while positive supercoiling overwinds such a helix. Both forms of supercoiling compact the DNA so that it sediments faster upon ultracentrifugation or migrates more rapidly in an electrophoretic gel in comparison to **relaxed DNA** (DNA that is not supercoiled).

Linking Number

The basic parameter characterizing supercoiled DNA is the **linking number** (L). This is the number of times the two strands are intertwined, and, provided both strands remain covalently intact, L cannot change. In a relaxed circular

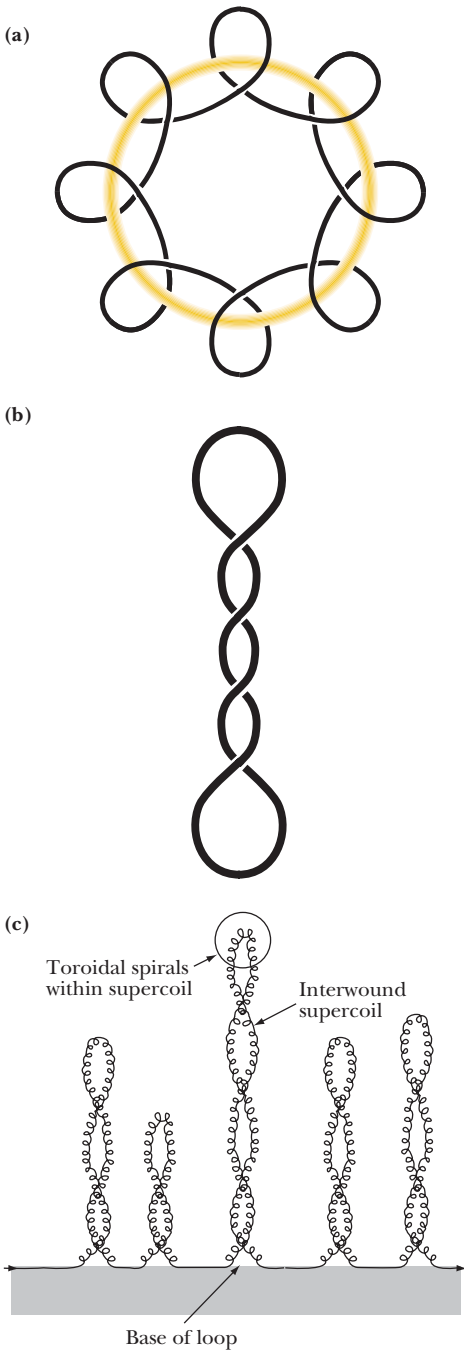


FIGURE 12.22 • Toroidal and interwound varieties of DNA supercoiling. (a) The DNA is coiled in a spiral fashion about an imaginary toroid. (b) The DNA interwinds and wraps about itself. (c) Supercoils in long, linear DNA arranged into loops whose ends are restrained—a model for chromosomal DNA. (Adapted from Figures 6.1 and 6.2 in Callandine, C. R., and Drew, H. R., 1992. Understanding DNA: The Molecule and How It Works. London: Academic Press.)

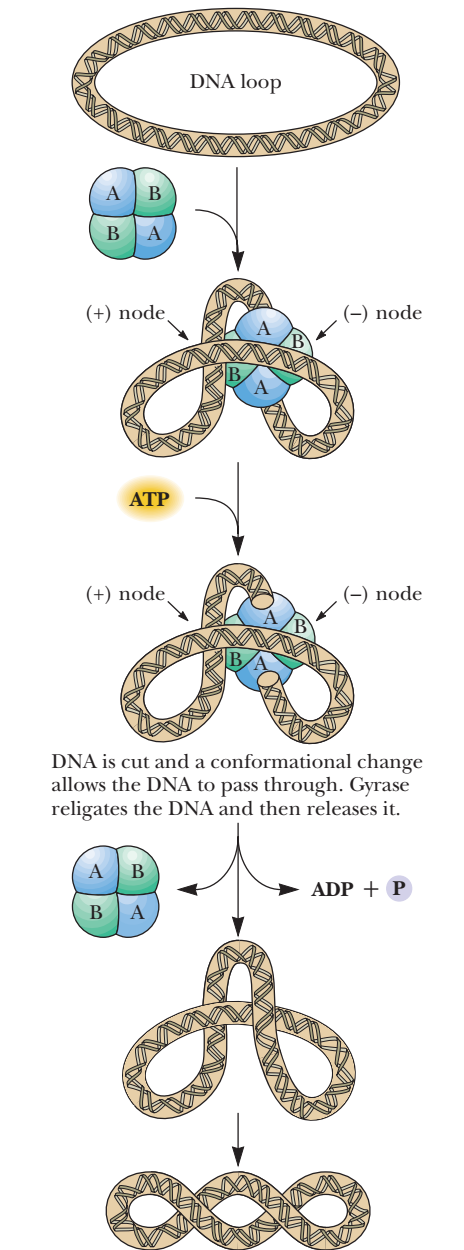


FIGURE 12.24 • A simple model for the action of bacterial DNA gyrase (topoisomerase II). The A-subunits cut the DNA duplex and then hold onto the cut ends. Conformational changes occur in the enzyme that allow a continuous region of the DNA duplex to pass between the cut ends and into an internal cavity of the protein. The cut ends are then re-ligated, and the intact DNA duplex is released from the enzyme. The released intact circular DNA now contains two negative supercoils as a consequence of DNA gyrase action.

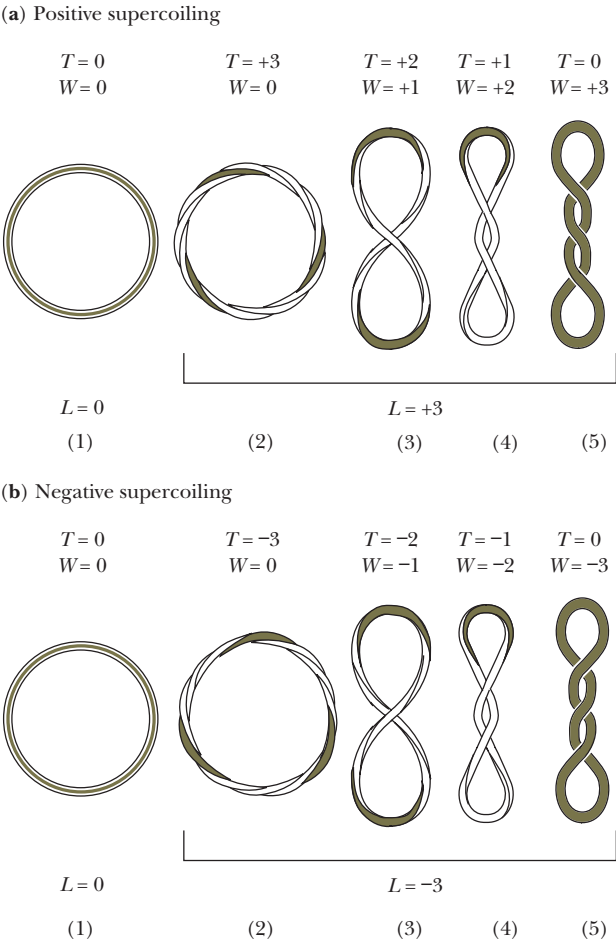


FIGURE 12.23 • Supercoiled DNA topology. (Adapted from Figures 6.5 and 6.6 in Callandine, C. R., and Drew, H. R., 1992. Understanding DNA: The Molecule and How It Works. London: Academic Press.)

DNA duplex of 400 bp, L is 40 (assuming 10 bp per turn in B-DNA). The linking number for relaxed DNA is usually taken as the reference parameter and is written as L_0 . L can be equated to the **twist** (T) and **writhe** (W) of the duplex, where twist is the number of helical turns and writhe is the number of supercoils:

$$L = T + W$$

Figure 12.23a shows the values of T and W for various positively and negatively supercoiled circular DNAs. In any closed, circular DNA duplex that is relaxed, $W = 0$. A relaxed circular DNA of 400 bp has 40 helical turns, $T = L = 40$. This linking number can only be changed by breaking one or both strands of the DNA, winding them tighter or looser, and rejoining the ends. Enzymes capable of carrying out such reactions are called **topoisomerases** because they change the topological state of DNA. Topoisomerase falls into two basic classes, I and II. Topoisomerases of the I type cut one strand of a DNA double helix, pass the other strand through, and then rejoin the cut ends. Topoisomerase II enzymes cut both strands of a dsDNA, pass a region of the DNA duplex between the cut ends, and then rejoin the ends (Figure 12.24). Topoisomerases are important players in DNA replication (see Chapter 30).

DNA Gyrase

The bacterial enzyme **DNA gyrase** is a topoisomerase that introduces negative supercoils into DNA in the manner shown in Figure 12.24. Suppose DNA gyrase puts four negative supercoils into the 400-bp circular duplex, then $W = -4$, T remains the same, and $L = 36$ (Figure 12.25). In actuality, the negative supercoils cause a torsional stress on the molecule so that T tends to decrease; that is, the helix becomes a bit unwound so that base pairs are separated. The extreme would be that T would decrease by 4 and the supercoiling would be removed ($T = 36$, $L = 36$, and $W = 0$). Usually the real situation is a compromise in which the negative value of W is reduced, T decreases slightly, and these changes are distributed over the length of the circular duplex so that no localized unwinding of the helix ensues. Although the parameters T and W are conceptually useful, neither can be measured experimentally at the present time.

Superhelix Density

The difference between the linking number of a DNA and the linking number of its relaxed form is ΔL : $\Delta L = (L - L_0)$. In our example with four negative supercoils, $\Delta L = -4$. The **superhelix density** or **specific linking difference** is defined as $\Delta L/L_0$ and is sometimes termed *sigma*, σ . For our example, $\sigma = -4/40$, or -0.1 . As a ratio, σ is a measure of supercoiling that is independent of length. Its sign reflects whether the supercoiling tends to unwind (*negative* σ) or overwind (*positive* σ) the helix. In other words, the superhelix density states the number of supercoils per 10 bp, which also is the same as the number of supercoils per B-DNA repeat. Circular DNA isolated from natural sources is always found in the underwound, negatively supercoiled state.

Toroidal Supercoiled DNA

Negatively supercoiled DNA can arrange into a toroidal state (Figure 12.26). The toroidal state of negatively supercoiled DNA is stabilized by wrapping around proteins which serve as spools for the DNA “ribbon.” This toroidal con-

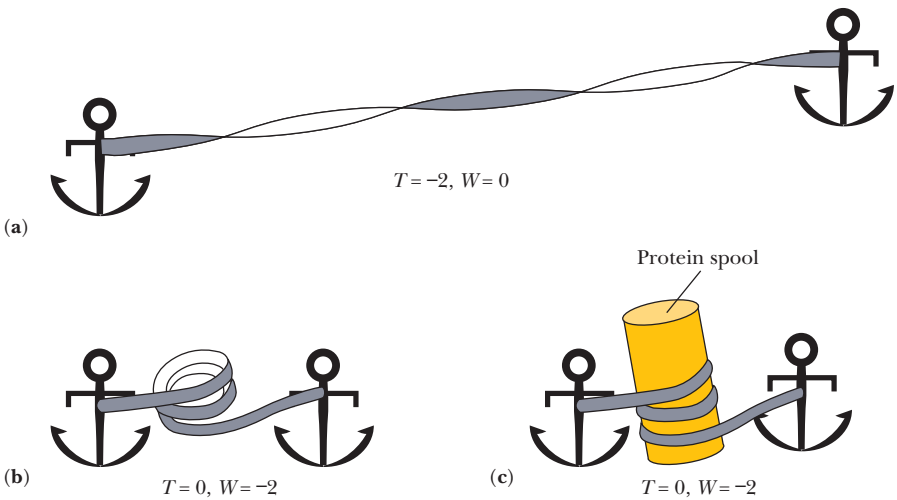


FIGURE 12.26 • Supercoiled DNA in a toroidal form wraps readily around protein “spools.” A twisted segment of linear DNA with two negative supercoils (a) can collapse into a toroidal conformation if its ends are brought closer together (b). Wrapping the DNA toroid around a protein “spool” stabilizes this conformation of supercoiled DNA (c). (Adapted from Figure 6.6 in Callandine, C. R., and Drew, H. R., 1992. Understanding DNA: The Molecule and How It Works. London: Academic Press.)

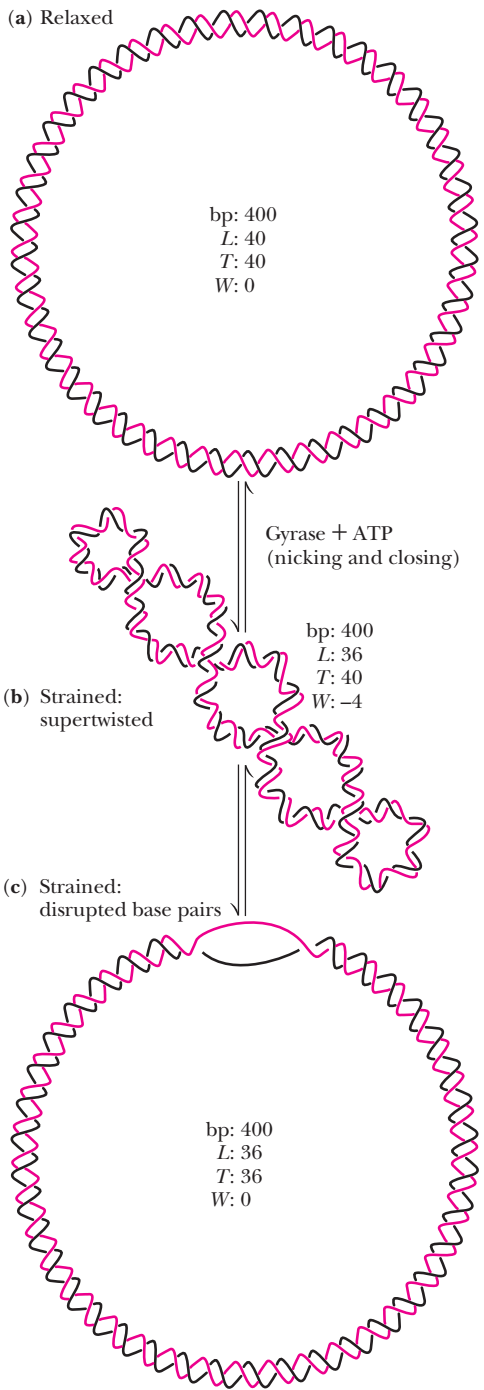


FIGURE 12.25 • A 400-bp circular DNA molecule in different topological states: (a) relaxed, (b) negative supercoils distributed over the entire length, and (c) negative supercoils creating a localized single-stranded region. Negative supercoiling has the potential to cause localized unwinding of the DNA double helix so that single-stranded regions (or bubbles) are created.

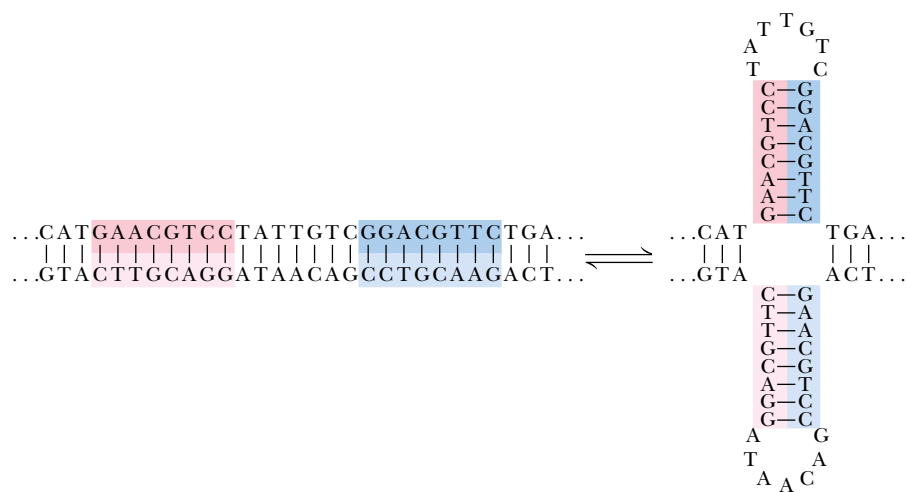


FIGURE 12.27 • The formation of a cruciform structure from a palindromic sequence within DNA. The self-complementary inverted repeats can rearrange to form hydrogen-bonded cruciform loops.

formation of DNA is found in protein:DNA interactions that are the basis of phenomena as diverse as chromosome structure (see Figure 12.31) and gene expression.

Cruciforms

Palindromes are words, phrases, or sentences that are the same when read backward or forward, such as “radar,” “sex at noon taxes,” “Madam, I’m Adam,” and “a man, a plan, a canal, Panama.” DNA sequences that are **inverted repeats**, or palindromes, have the potential to form a tertiary structure known as a **cruciform** (literally meaning “cross-shaped”) if the normal interstrand base pairing is replaced by intrastrand pairing (Figure 12.27). In effect, each DNA strand folds back on itself in a hairpin structure to align the palindrome in base-pairing register. Such cruciforms are never as stable as normal DNA duplexes because an unpaired segment must exist in the loop region. However, negative supercoiling causes a localized disruption of hydrogen bonding between base pairs in DNA and may promote formation of cruciform loops. Cruciform structures have a twofold rotational symmetry about their centers and potentially create distinctive recognition sites for specific DNA-binding proteins.

12.5 • Chromosome Structure

A typical human cell is 20 μm in diameter. Its genetic material consists of 23 pairs of dsDNA molecules in the form of **chromosomes**, the average length of which is 3×10^9 bp/23 or 1.3×10^8 nucleotide pairs. At 0.34 nm/bp in B-DNA, this represents a DNA molecule 5 cm long. Together, these 46 dsDNA molecules amount to more than 2 m of DNA that must be packaged into a nucleus perhaps 5 μm in diameter! Clearly, the DNA must be condensed by a factor of more than 10^5 . This remarkable task is accomplished by neatly wrapping the DNA around protein spools called **nucleosomes** and then packing the nucleosomes to form a helical filament that is arranged in loops associated with the **nuclear matrix**, a skeleton or scaffold of proteins providing a structural framework within the nucleus.

Table 12.2

Properties of Histones

Histone	Ratio of Lysine to Arginine	M_r	Copies per Nucleosome
H1	59/3	21,200	1 (not in bead)
H2A	13/13	14,100	2 (in bead)
H2B	20/8	13,900	2 (in bead)
H3	13/17	15,100	2 (in bead)
H4	11/14	11,400	2 (in bead)

Nucleosomes

The DNA in a eukaryotic cell nucleus during the interphase between cell divisions exists as a nucleoprotein complex called **chromatin**. The proteins of chromatin fall into two classes: **histones** and **nonhistone chromosomal proteins**. Histones are abundant structural proteins, whereas the nonhistone class is represented only by a few copies each of many diverse proteins involved in genetic regulation. The histones are relatively small, positively charged arginine- or lysine-rich proteins that interact via ionic bonds with the negatively charged phosphate groups on the polynucleotide backbone. Five distinct histones are known: **H1**, **H2A**, **H2B**, **H3**, and **H4** (Table 12.2). Pairs of histones H2A, H2B, H3, and H4 aggregate to form an octameric core structure, which is the core of the **nucleosome**, around which the DNA helix is wound (see Figure 11.23).

If chromatin is swelled suddenly in water and prepared for viewing in the electron microscope, the nucleosomes are evident as “beads on a string,” dsDNA being the string (Figure 12.28). The structure of the histone octamer core has been determined by X-ray crystallography without DNA by E. N. Moudrianakis’s laboratory (Figure 12.29) and wrapped with DNA by T. J.

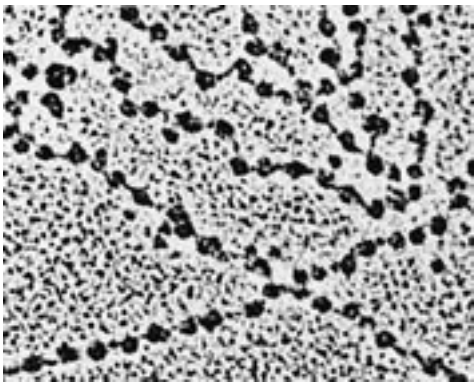


FIGURE 12.28 • Electron micrograph of *Drosophila melanogaster* chromatin after swelling reveals the presence of nucleosomes as “beads on a string.” (Electron micrograph courtesy of Oscar L. Miller, Jr., of the University of Virginia)

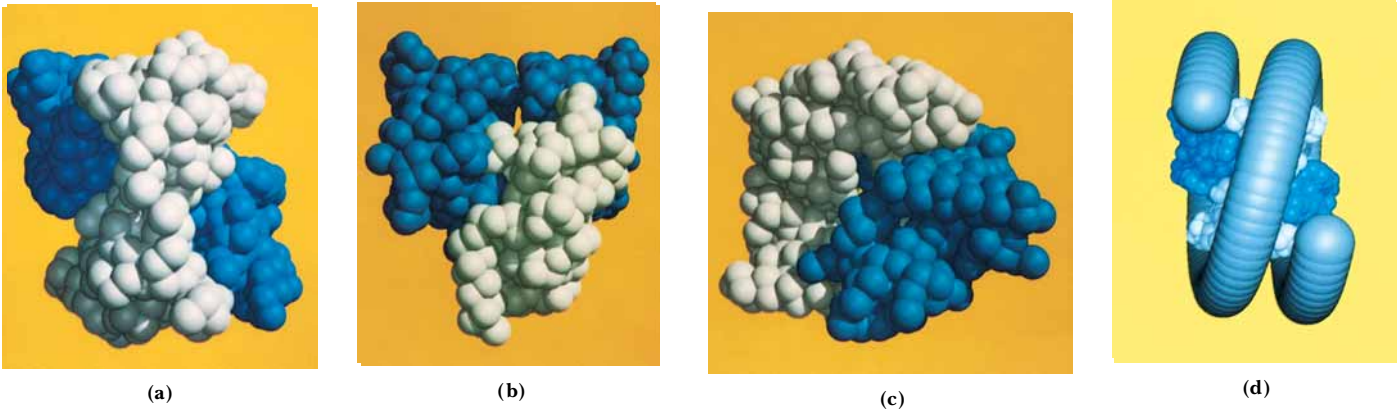


FIGURE 12.29 • Four orthogonal views of the histone octamer as determined by X-ray crystallography: (a) front view; (b) top view; and (c) disk view, that is, as viewed down the long axis of the chromatin fiber. In the (c) perspective, the DNA duplex would wrap around the octamer, with the axis of the DNA supercoil perpendicular to the plane of the picture. (d) Suggested appearance of the nucleosome when wrapped with DNA. (Photographs courtesy of Evangelos N. Moudrianakis of Johns Hopkins University)



FIGURE 12.30 • (a) Deduced structure of the nucleosome core particle wrapped with 1.65 turns of DNA (146 bp). The DNA is shown as a ribbon. (*left*) View down the axis of the nucleosome; (*right*) view perpendicular to the axis. (b) One-half of the nucleosome core particle with 73 bp of DNA, as viewed down the nucleosome axis. Note that the DNA does not wrap in a uniform circle about the histone core, but instead follows a course consisting of a series of somewhat straight segments separated by bends. (*Adapted from Luger, C., et al., 1997. Crystal structure of the nucleosome core particle at 2.8 Å resolution. Nature 389:251–260. Photos courtesy of T. J. Richmond, ETH-Hönggerberg, Zurich, Switzerland.*)

Richmond and collaborators (Figure 12.30). The octamer (Figure 12.29) has surface landmarks that guide the course of the DNA around the octamer; 146 bp of B-DNA in a flat, left-handed superhelical conformation make 1.65 turns around the histone core (Figure 12.30), which itself is a protein superhelix consisting of a spiral array of the four histone dimers. Histone 1, a three-domain protein, serves to seal the ends of the DNA turns to the nucleosome core and to organize the additional 40 to 60 bp of DNA that link consecutive nucleosomes.

Organization of Chromatin and Chromosomes

A higher order of chromatin structure is created when the nucleosomes, in their characteristic beads-on-a-string motif, are wound in the fashion of a *solenoid* having six nucleosomes per turn (Figure 12.31). The resulting 30-nm filament contains about 1200 bp in each of its solenoid turns. Interactions between the respective H1 components of successive nucleosomes stabilize the 30-nm filament. This 30-nm filament then forms long DNA loops of variable length, each containing on average between 60,000 and 150,000 bp. Electron microscopic analysis of human chromosome 4 suggests that 18 such loops are then arranged radially about the circumference of a single turn to form a **miniband unit** of the chromosome. According to this model, approximately 10⁶ of these minibands are arranged along a central axis in each of the chromatids of human chromosome 4 that form at mitosis (Figure 12.31). Despite intensive study, much of the higher-order structure of chromosomes remains a mystery.

solenoid • a coil wound in the form of a helix

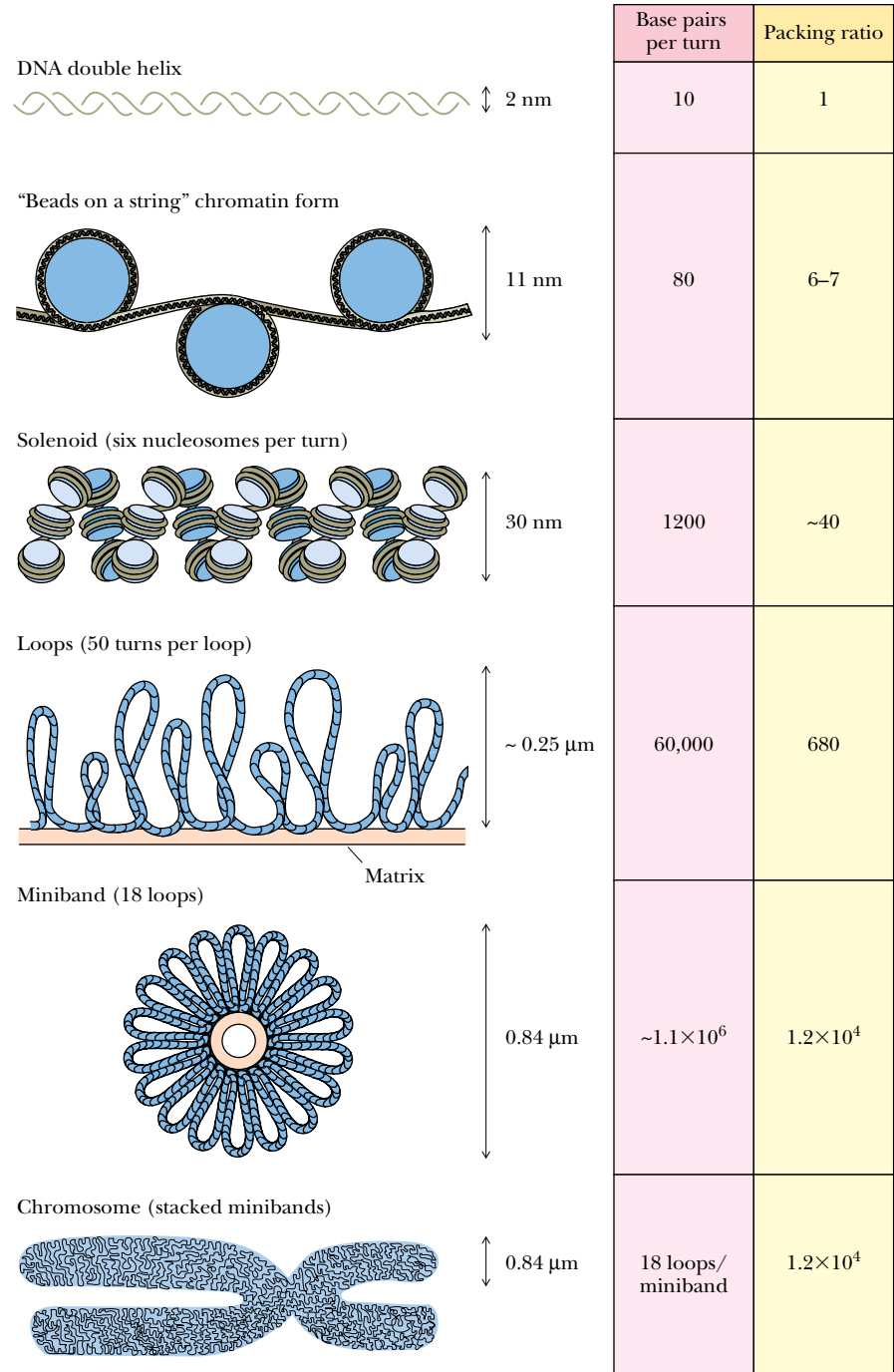


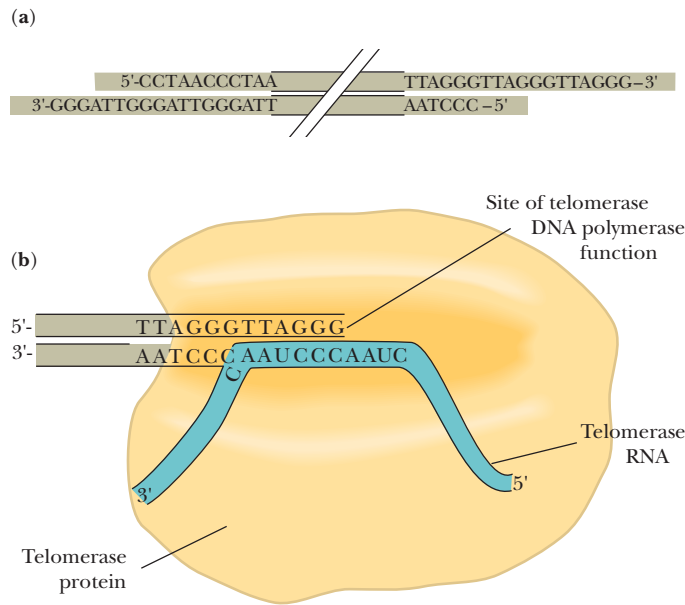
FIGURE 12.31 • A model for chromosome structure, human chromosome 4. The 2-nm DNA helix is wound twice around histone octamers to form 10-nm nucleosomes, each of which contains 160 bp (80 per turn). These nucleosomes are then wound in solenoid fashion with six nucleosomes per turn to form a 30-nm filament. In this model, the 30-nm filament forms long DNA loops, each containing about 60,000 bp, which are attached at their base to the nuclear matrix. Eighteen of these loops are then wound radially around the circumference of a single turn to form a miniband unit of a chromosome. Approximately 10⁶ of these minibands occur in each chromatid of human chromosome 4 at mitosis.

HUMAN BIOCHEMISTRY

Telomeres and Tumors

Eukaryotic chromosomes are linear. The ends of chromosomes have specialized structures known as **telomeres**. The telomeres of virtually all eukaryotic chromosomes consist of short, tandemly repeated nucleotide sequences at the ends of the chromosomal DNA. For example, the telomeres of human germline (sperm and egg) cells contain between 1000 and 1700 copies of the hexameric repeat TTAGGG (see figure). Telomeres are believed to be responsible for maintaining chromosomal integrity by protecting against DNA degradation or rearrangement. Telomeres are added to the ends of chromosomal DNA by an RNA-containing enzyme known as **telomerase** (Chapter 30); telomerase is an unusual DNA polymerase that was discovered in 1985 by Elizabeth Blackburn and Carol Greider of the University of California, San Francisco.

However, most normal somatic cells lack telomerase. Consequently, upon every cycle of cell division when the cell replicates its DNA, about 50-nucleotide portions are lost from the end of each telomere. Thus, over time, the telomeres of somatic cells in animals become shorter and shorter, eventually leading to chromosome instability and cell death. This phenomenon has led some scientists to espouse a “telomere theory of aging” that implicates telomere shortening as the principal factor in cell, tissue, and even organism aging. Interestingly, cancer cells appear “immortal” because they continue to reproduce indefinitely. A survey of 20 different tumor types by Geron Corporation of Menlo Park, California, revealed that all contained telomerase activity.



(a) Telomeres on human chromosomes consist of the hexanucleotide sequence TTAGGG repeated between 1000 and 1700 times. These TTAGGG tandem repeats are attached to the 3'-ends of the DNA strands and are paired with the complementary sequence 3'-AATCCC-5' on the other DNA strand. Thus, a G-rich region is created at the 3'-end of each DNA strand and a C-rich region is created at the 5'-end of each DNA strand. Typically, at each end of the chromosome, the G-rich strand protrudes 12 to 16 nucleotides beyond its complementary C-rich strand. (b) Like other telomerases, human telomerase is a ribonucleoprotein. The ribonucleic acid of human telomerase is an RNA molecule 962 nucleotides long. This RNA serves as the template for the DNA polymerase activity of telomerase. Nucleotides 46 to 56 of this RNA are CUAACCCUAAC and provide the template function for the telomerase-catalyzed addition of TTAGGG units to the 3'-end of a DNA strand.

12.6 • Chemical Synthesis of Nucleic Acids

Laboratory synthesis of oligonucleotide chains of defined sequence presents some of the same problems encountered in chemical synthesis of polypeptides (see Chapter 5). First, functional groups on the monomeric units (in this case, bases) are reactive under conditions of polymerization and therefore must be protected by blocking agents. Second, to generate the desired sequence, a phosphodiester bridge must be formed between the 3'-O of one nucleotide (B) and the 5'-O of the preceding one (A) in a way that precludes the unwanted bridging of the 3'-O of A with the 5'-O of B. Finally, recoveries at each step must be high so that overall yields in the multistep process are acceptable. As in peptide synthesis (see Chapter 5), *solid phase methods* are used to overcome some

of these problems. Commercially available automated instruments, called **DNA synthesizers** or “gene machines,” are capable of carrying out the synthesis of oligonucleotides of 150 bases or more.

Phosphoramidite Chemistry

Phosphoramidite chemistry is currently the accepted method of oligonucleotide synthesis. The general strategy involves the sequential addition of nucleotide units as *nucleoside phosphoramidite* derivatives to a nucleoside covalently attached to the insoluble resin. Excess reagents, starting materials, and side products are removed after each step by filtration. After the desired oligonucleotide has been formed, it is freed of all blocking groups, hydrolyzed from the resin, and purified by gel electrophoresis. The four-step cycle is shown in Figure 12.32. Chemical synthesis takes place in the 3' → 5' direction (the reverse of the biological polymerization direction).

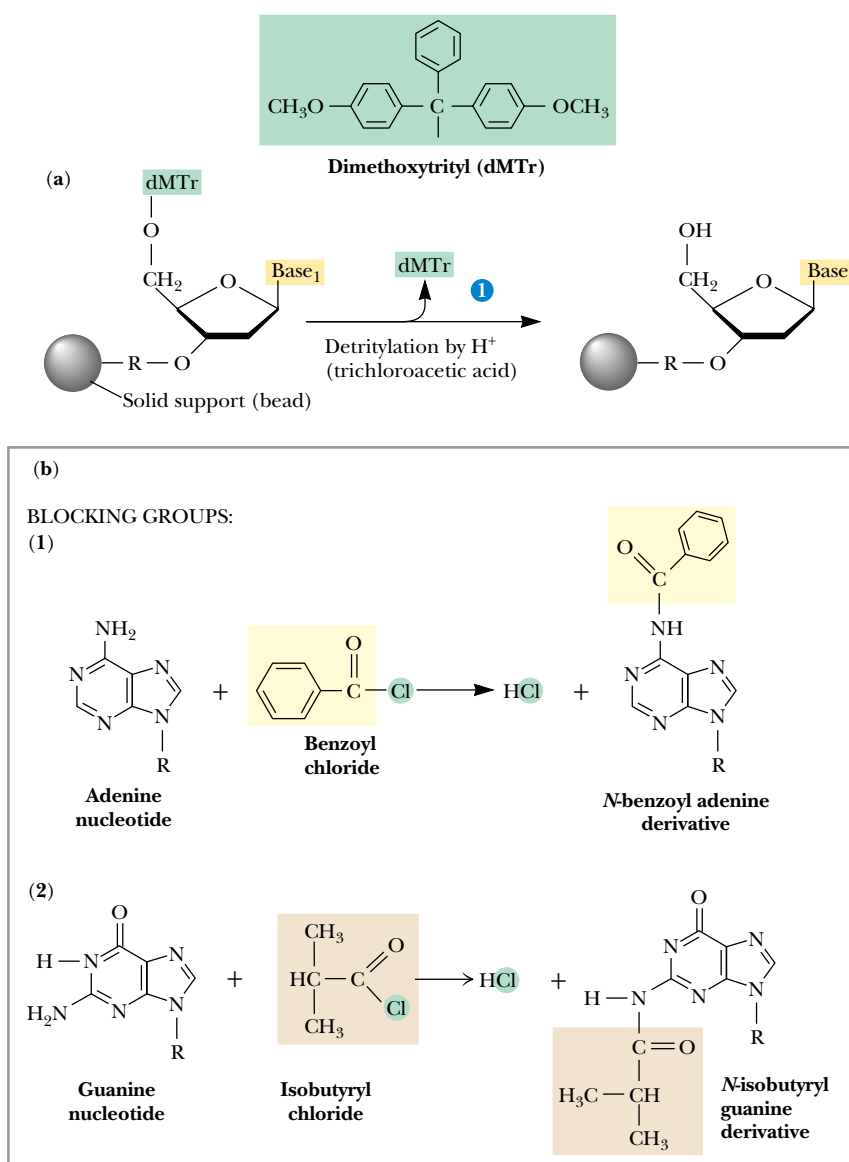


FIGURE 12.32 • Solid phase oligonucleotide synthesis. The four-step cycle starts with the first base in nucleoside form (N-1) attached by its 3'-OH group to an insoluble, inert resin or matrix, typically either controlled pore glass (CPG) or silica beads. Its 5'-OH is blocked with a dimethoxytrityl (DMTr) group (a). If the base has reactive —NH₂ functions, as in A, G, or C, then *N*-benzoyl or *N*-isobutyryl derivatives are used to prevent their reaction (b). In step 1, the DMTr protecting group is removed by trichloroacetic acid treatment. Step 2 is the coupling step: the second base (N-2) is added in the form of a nucleoside phosphoramidite derivative whose 5'-OH bears a DMTr blocking group so it cannot polymerize with itself (c). (Figure continued on following page.)

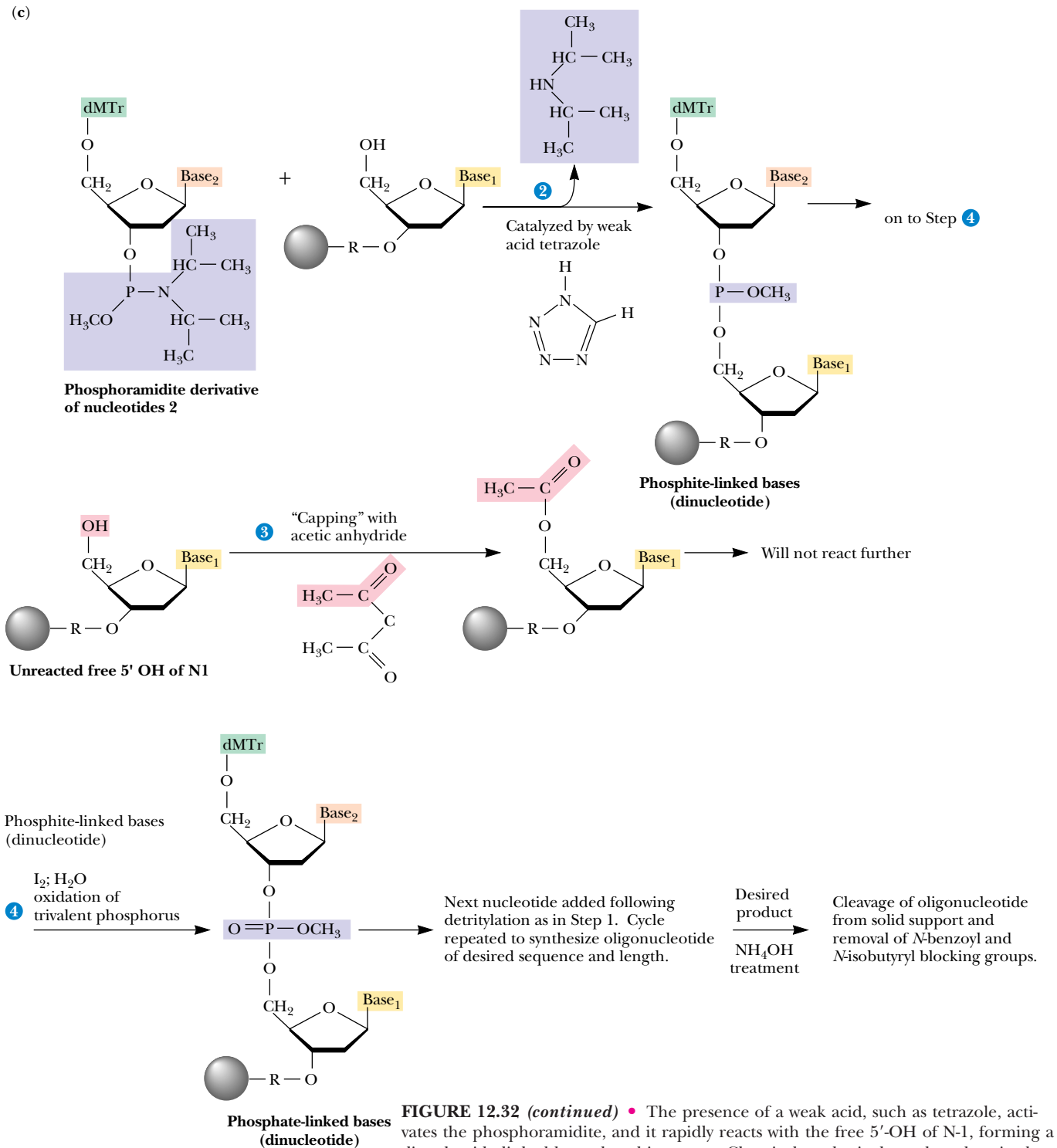


FIGURE 12.32 (continued) • The presence of a weak acid, such as tetrazole, activates the phosphoramidite, and it rapidly reacts with the free 5'-OH of N-1, forming a dinucleotide linked by a phosphite group. Chemical synthesis thus takes place in the 3' → 5' direction. Unreacted free 5'-OHs of N-1 (usually only 2–6% of the total) are blocked from further participation in the polymerization process by acetylation with acetic anhydride in step 3, referred to as *capping*. The phosphite linkage between N-1 and N-2 is highly reactive and, in step 4, it is oxidized by aqueous iodine (I₂) to form the desired more stable phosphate group. This completes the cycle. Subsequent cycles add successive residues to the resin-immobilized chain. When the chain is complete, it is cleaved from the support with NH₄OH, which also removes the *N*-benzoyl and *N*-isobutyryl protecting groups from the amino functions on the A, G, and C residues.

Chemically Synthesized Genes

Table 12.3 lists some of the genes that have been chemically synthesized. Because protein-coding genes are characteristically much larger than the 150-bp practical limit on oligonucleotide synthesis, their synthesis involves joining a series of oligonucleotides to assemble the overall sequence. A prime example of such synthesis is the gene for rhodopsin.

Figure 12.33 illustrates the strategy used in the total synthesis of the gene for bovine rhodopsin. This gene, which is 1057 base pairs long, encodes the 348 amino-acid photoreceptor protein of the vertebrate retina. Theoretically, no gene is beyond the scope of these methods, a fact that opens the door to an incredibly exciting range of possibilities for investigating structure–function relationships in the organization and expression of hereditary material.

Table 12.3

Some Chemically Synthesized Genes	
Gene	Size (bp)
tRNA	126
α-Interferon	542
Secretin	81
γ-Interferon	453
Rhodopsin	1057
Proenkephalin	77
Connective tissue activating peptide III	280
Lysozyme	385
Tissue plasminogen activator	1610
c-Ha-ras	576
RNase T1	324
Cytochrome <i>b</i> ₅	330
Bovine intestinal Ca-binding protein	298
Hirudin	226
RNase A	375

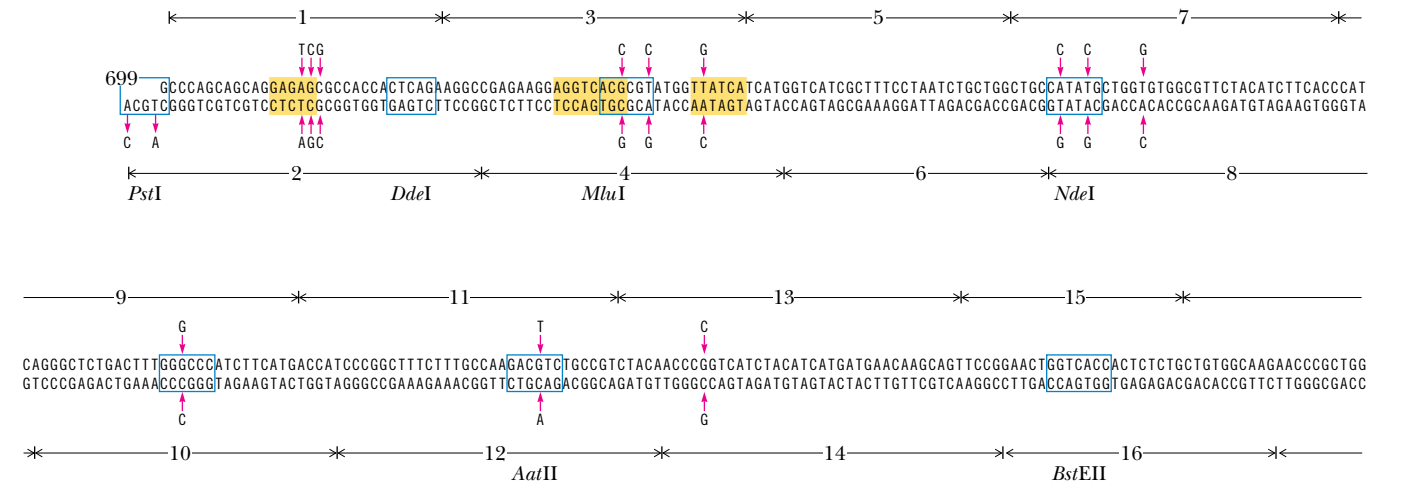


FIGURE 12.33 • Total synthesis of the bovine rhodopsin gene was achieved by joining 72 synthetic oligonucleotides, 36 representing one strand and 36 the complementary strand. These oligonucleotides are overlapping. Once synthesized, the various oligonucleotides, each 15 to 40 nucleotides long, were assembled by annealing and enzymatic ligation into three large fragments, representing nucleotides –5 to 338 (–5 meaning 5 nucleotides before the start of the coding region), 335 to 702, and 699 to 1052. The total gene was then created by joining these fragments. This figure shows only one fragment (fragment PB, comprising nucleotides 699 through 1052), assembled from 20 complementary oligonucleotides whose ends overlap. Odd-numbered oligonucleotides (1, 3, 5, . . .) compose the 5' → 3' strand; even-numbered ones (2, 4, 6, . . .) represent the 3' → 5' strand. (Vertical arrows indicate nucleotides that were changed from the native gene sequence. Restriction sites are shown boxed in blue lines; those removed from the gene through nucleotide substitutions are shown as yellow shaded boxes.) Note the single-stranded overhangs at either end of the 3' → 5' strand. The sequences at these overhangs correspond to restriction endonuclease sites (*Pst*I and *Bam*HI), which facilitate subsequent manipulation of the fragment in gene assembly and cloning.

12.7 • Secondary and Tertiary Structure of RNA

RNA molecules (see Chapter 11) are typically single-stranded. Nevertheless, they are often rich in double-stranded regions that form when complementary sequences within the chain come together and join via **intrastrand hydrogen bonding**. RNA strands cannot fold to form B-DNA type double helices because their 2'-OH groups are a steric hindrance to this conformation. Instead, RNA double helices adopt a conformation similar to the A-form of DNA, having about 11 bp per turn, and the bases strongly tilted from the plane perpendicular to the helix axis (see Figure 12.13). Both tRNA and rRNA have characteristic secondary structures formed in this manner. Secondary structures are presumed to exist in mRNA species as well, although their nature is as yet little understood. (The functions of tRNA, rRNA, and mRNA are discussed in detail in Part IV: Information Transfer.)

Transfer RNA

In tRNA molecules, which contain from 73 to 94 nucleotides in a single chain, a majority of the bases are hydrogen-bonded to one another. Figure 12.34 shows the structure that typifies tRNAs. *Hairpin turns* bring complementary stretches of bases in the chain into contact so that double-helical regions form. Because of the arrangement of the complementary stretches along the chain, the overall pattern of H-bonding can be represented as a *cloverleaf*. Each cloverleaf consists of four H-bonded segments—three loops and the stem where the 3'- and 5'-ends of the molecule meet. These four segments are designated the *acceptor stem*, the *D loop*, the *anticodon loop*, and the *TψC loop*.

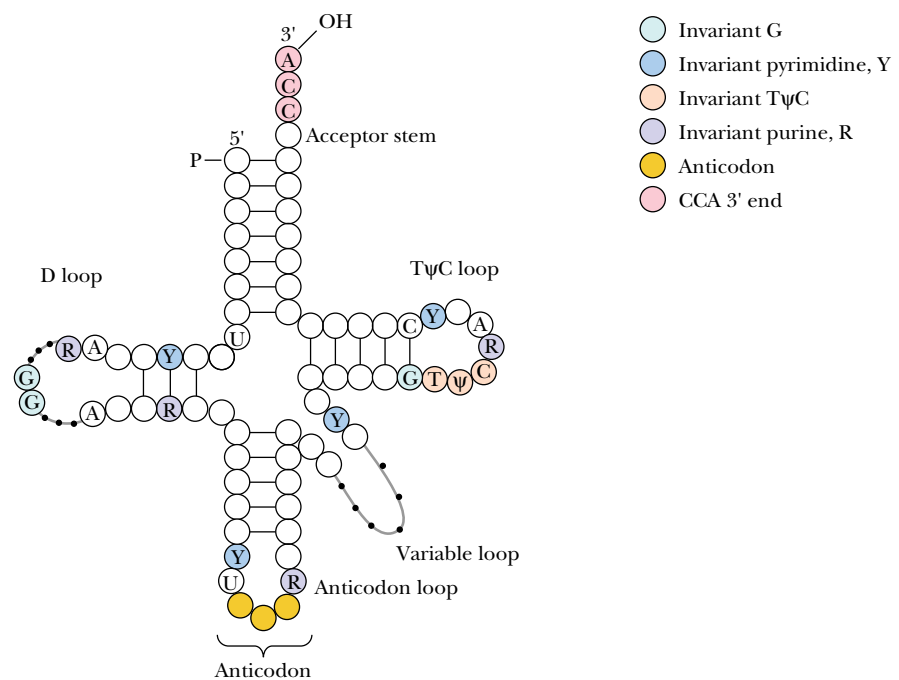


FIGURE 12.34 • A general diagram for the structure of tRNA. The positions of invariant bases as well as bases that seldom vary are shown in color. The numbering system is based on yeast tRNA^{Phe}. R = purine; Y = pyrimidine. Dotted lines denote sites in the D loop and variable loop regions where varying numbers of nucleotides are found in different tRNAs.

tRNA Secondary Structure

The *acceptor stem* is where the amino acid is linked to form the aminoacyl-tRNA derivative, which serves as the amino acid–donating species in protein synthesis; this is the physiological role of tRNA. The amino acid adds to the 3'-OH of the 3'-terminal A nucleotide (Figure 12.35). The 3'-end of tRNA is invariably CCA-3'-OH. This CCA sequence plus a fourth nucleotide extends beyond the double-helical portion of the acceptor stem. The *D loop* is so named because this tRNA loop often contains dihydrouridine, or D, residues. In addition to dihydrouridine, tRNAs characteristically contain a number of unusual bases, including inosine, thiouridine, pseudouridine, and hypermethylated purines (see Figure 11.26). The *anticodon loop* consists of a double-helical segment and seven unpaired bases, three of which are the **anticodon**. (The anticodon is the three-nucleotide unit that recognizes and base pairs with a particular mRNA **codon**, a complementary three-base unit in mRNA which is the genetic information that specifies an amino acid.) Reading 3' → 5', the anticodon is invariably preceded by a purine (often an alkylated one) and followed by a U. Anticodon base pairing to the codon on mRNA allows a particular tRNA species to deliver its amino acid to the protein-synthesizing apparatus. It represents the key event in translating the information in the nucleic acid sequence so that the appropriate amino acid is inserted at the right place in the amino acid sequence of the protein being synthesized. Next along the tRNA sequence in the 5' → 3' direction comes a loop that varies from tRNA to tRNA in the number of residues that it has, the so-called **extra** or **variable loop**. The last loop in the tRNA, reading 5' → 3', is the **TψC loop**, which contains seven unpaired bases including the sequence TψC, where ψ is the symbol for **pseudouridine**. Ribosomes bind tRNAs through recognition of this TψC loop. Almost all of the invariant residues common to tRNAs lie within the non–hydrogen-bonded regions of the cloverleaf structure (Figure 12.34). Figure 12.36 depicts the complete nucleotide sequence and cloverleaf structure of yeast alanine tRNA.

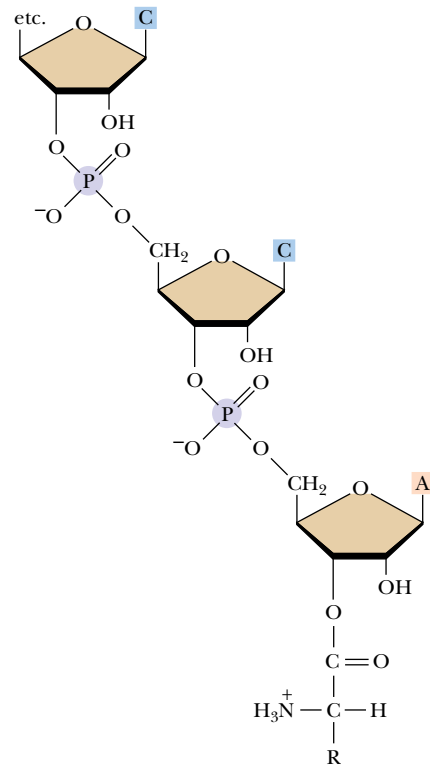


FIGURE 12.35 • Amino acids are linked to the 3'-OH end of tRNA molecules by an ester bond formed between the carboxyl group of the amino acid and the 3'-OH of the terminal ribose of the tRNA.

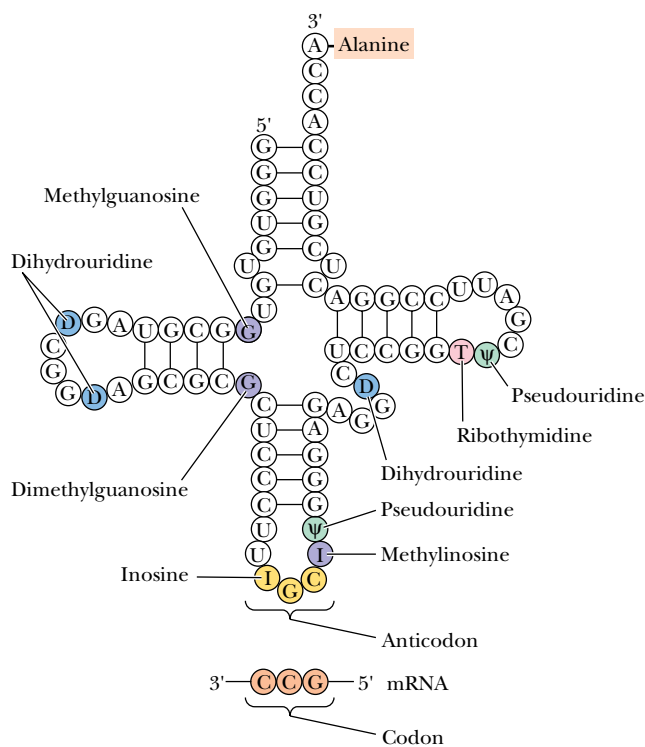


FIGURE 12.36 • The complete nucleotide sequence and cloverleaf structure of yeast alanine tRNA.

tRNA Tertiary Structure

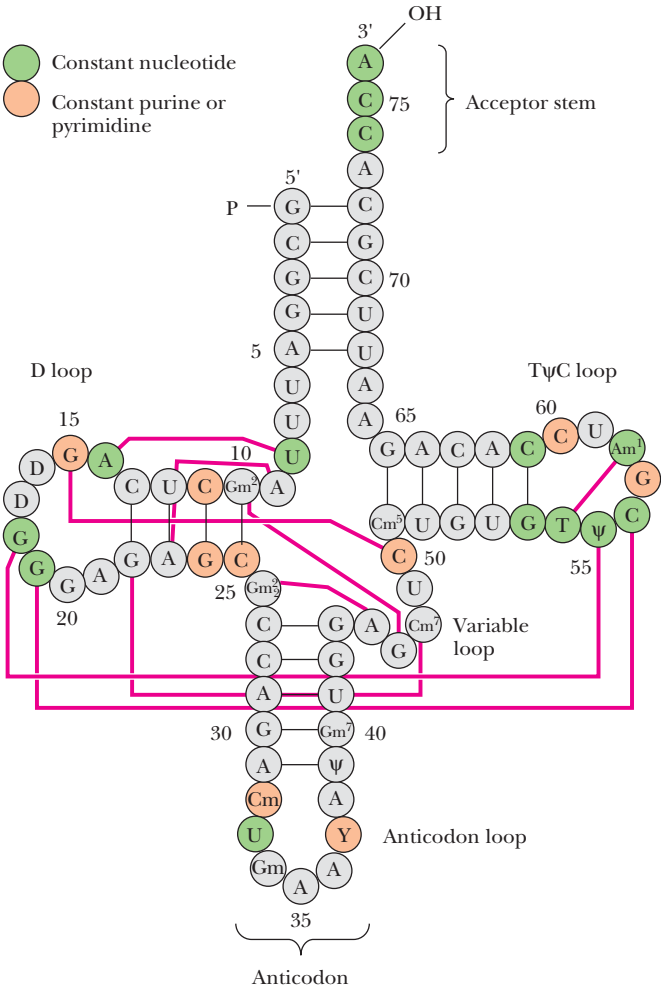
Tertiary structure in tRNA arises from hydrogen-bonding interactions between bases in the D loop with bases in the variable and TψC loops, as shown for yeast phenylalanine tRNA in Figure 12.37. Note that these H bonds involve the invariant nucleotides of tRNAs, thus emphasizing the importance of the tertiary structure they create to the function of tRNAs in general. These H bonds fold the D and TψC arms together and bend the cloverleaf into the stable L-shaped tertiary form (Figure 12.38). Many of these H bonds involve base pairs that are not canonical A:T or G:C pairings (Figure 12.38). The amino acid acceptor stem is at one end of the L, separated by 7 nm or so from the anticodon at the opposite end of the L. The D and TψC loops form the corner of the L. In the L-conformation, the bases are oriented to maximize hydrophobic stacking interactions between their flat faces. Such stacking is a second major factor contributing to L-form stabilization.

Ribosomal RNA

rRNA Secondary Structure

Ribosomes, the protein-synthesizing machinery of cells, are composed of two **subunits**, called **small** and **large**, and ribosomal RNAs are integral components of these subunits (see Table 11.2). A large degree of *intrastrand sequence com-*

FIGURE 12.37 • Tertiary interactions in yeast phenylalanine tRNA. The molecule is presented in the conventional cloverleaf secondary structure generated by intrastrand hydrogen bonding. Solid lines connect bases that are hydrogen-bonded when this cloverleaf pattern is folded into the characteristic tRNA tertiary structure (see also Figure 12.36).



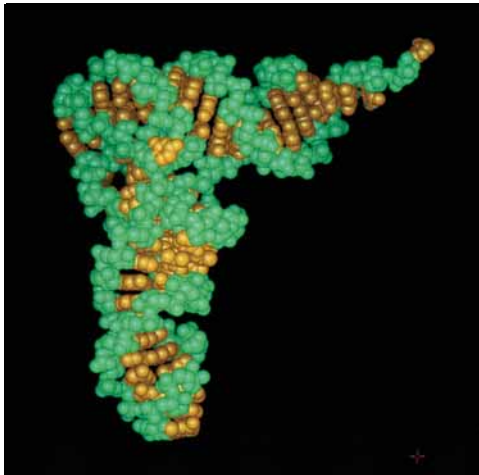
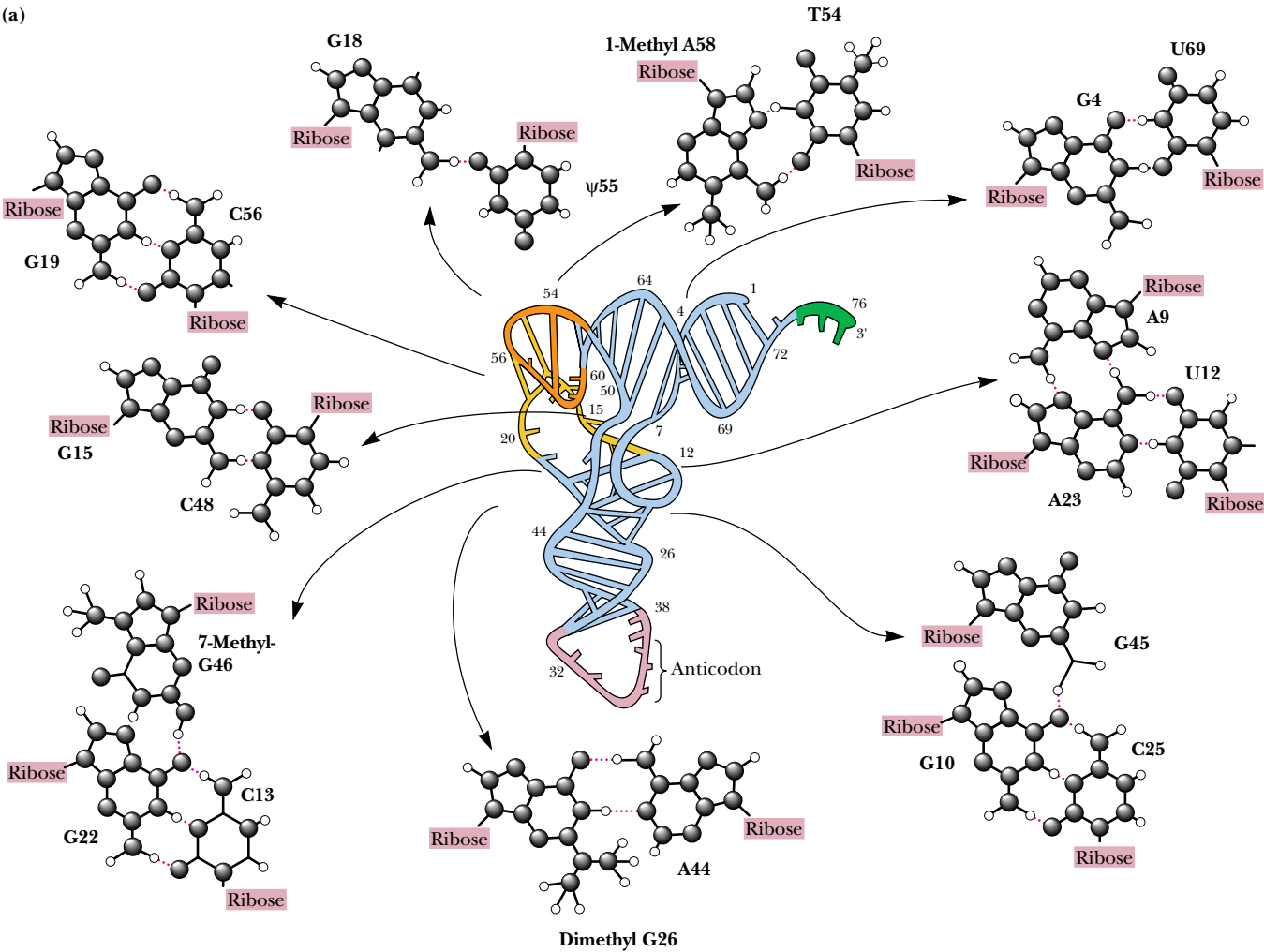
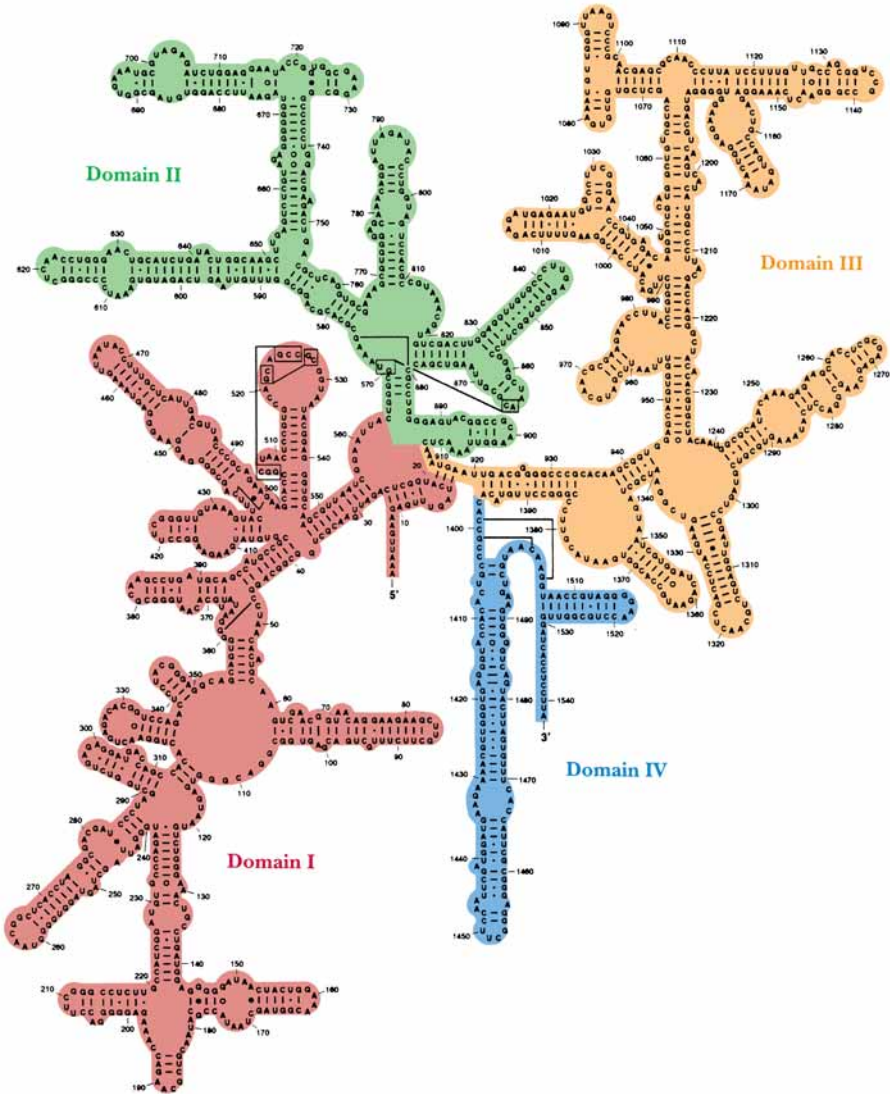


FIGURE 12.38 • (a) The three-dimensional structure of yeast phenylalanine tRNA as deduced from X-ray diffraction studies of its crystals. The tertiary folding is illustrated in the center of the diagram with the ribose–phosphate backbone presented as a continuous ribbon; H bonds are indicated by crossbars. Unpaired bases are shown as short, unconnected rods. The anticodon loop is at the bottom and the -CCA 3'-OH acceptor end is at the top right. The various types of noncanonical hydrogen-bonding interactions observed between bases surround the central molecule. Three of these structures show examples of unusual H-bonded interactions involving three bases; these interactions aid in establishing tRNA tertiary structure. (b) A space-filling model of the molecule. (After Kim, S. H., in Schimmel, P., Söll, D., and Abelson, J. N., eds., 1979. *Transfer RNA: Structure, Properties, and Recognition*. New York: Cold Spring Harbor Laboratory.)

plementarity is found in all rRNA strands, and all assume a highly folded pattern that allows base pairing between these complementary segments. Figure 12.39 shows the secondary structure assigned to the *E. coli* 16S rRNA. This structure is based on alignment of the nucleotide sequence into H-bonding segments. The reliability of these alignments is then tested through a comparative analysis of whether identical secondary structures can be predicted from

FIGURE 12.39 • The proposed secondary structure for *E. coli* 16S rRNA, based on comparative sequence analysis in which the folding pattern is assumed to be conserved across different species. The molecule can be subdivided into four domains—I, II, III, and IV—on the basis of contiguous stretches of the chain that are closed by long-range base-pairing interactions. **I**, the 5'-domain, includes nucleotides 27 through 556. **II**, the central domain, runs from nucleotide 564 to 912. Two domains comprise the 3'-end of the molecule. **III**, the major one, comprises nucleotides 923 to 1391. **IV**, the 3'-terminal domain, covers residues 1392 to 1541.



nucleotide sequences of 16S-like rRNAs from other species. If so, then such structures are apparently conserved. The approach is based on the thesis that, because ribosomal RNA species (regardless of source) serve common roles in protein synthesis, it may be anticipated that they share structural features. The structure is marvelously rich in short, helical segments separated and punctuated by single-stranded loops.

Comparison of rRNAs from Various Species

If a phylogenetic comparison is made of the 16S-like rRNAs from an archaeobacterium (*Halobacterium volcanii*), a eubacterium (*E. coli*), and a eukaryote (the yeast *Saccharomyces cerevisiae*), a striking similarity in secondary structure emerges (Figure 12.40). Remarkably, these secondary structures are similar despite the fact that the nucleotide sequences of these rRNAs themselves exhibit a low degree of similarity. Apparently, evolution is acting at the level of rRNA secondary structure, not rRNA nucleotide sequence. Similar conserved folding patterns are seen for the 23S-like and 5S-like rRNAs that reside in the

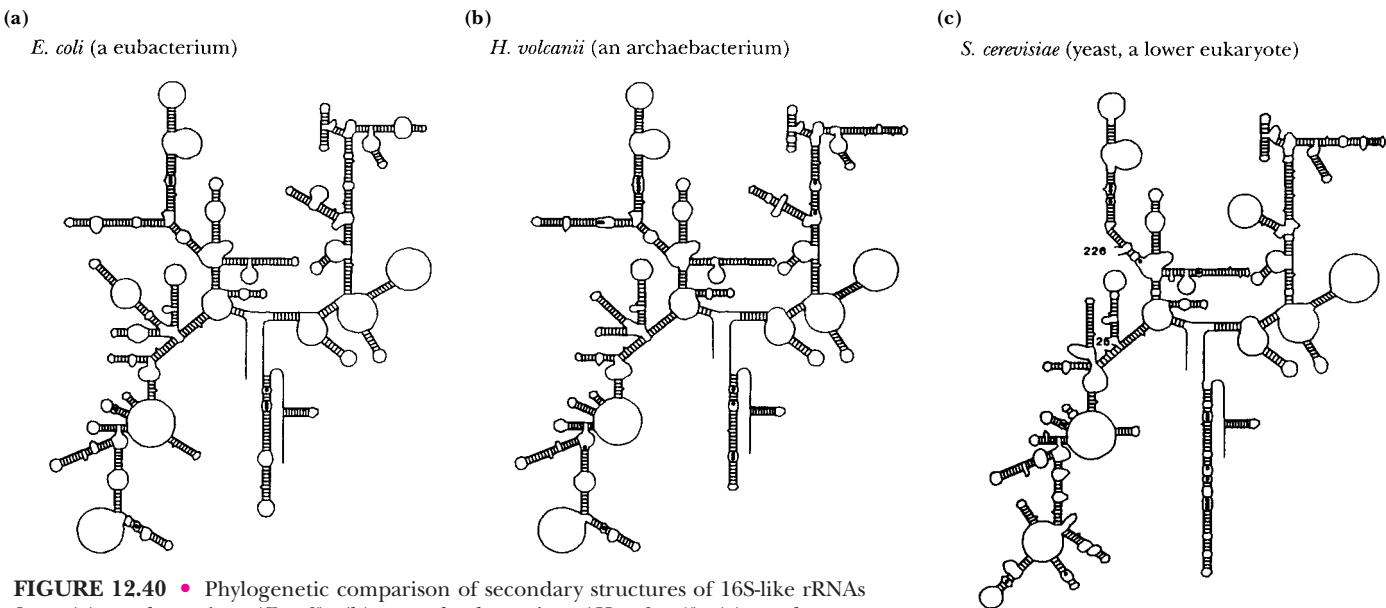


FIGURE 12.40 • Phylogenetic comparison of secondary structures of 16S-like rRNAs from (a) a eubacterium (*E. coli*), (b) an archaeobacterium (*H. volcanii*), (c) a eukaryote (*S. cerevisiae*, a yeast).

large ribosomal subunits of various species. An insightful conclusion may be drawn regarding the persistence of such strong secondary structure conservation despite the millennia that have passed since these organisms diverged: *all ribosomes are constructed to a common design and all function in a similar manner.*

rRNA Tertiary Structure

Despite the unity in secondary structural patterns, little is known about the three-dimensional, or tertiary, structure of rRNAs. Even less is known about the quaternary interactions that occur when ribosomal proteins combine with rRNAs and when the ensuing ribonucleoprotein complexes, the small and large subunits, come together to form the complete ribosome. Furthermore, assignments of functional roles to rRNA molecules are still tentative and approximate. (We return to these topics in Chapter 33.)

PROBLEMS

1. The oligonucleotide d-ATGCCTGACT was subjected to sequencing by (a) Sanger’s dideoxy method and (b) Maxam and Gilbert’s chemical cleavage method, and the products were analyzed by electrophoresis on a polyacrylamide gel. Draw diagrams of the gel-banding patterns obtained for (a) and (b).

2. The result of sequence determination of an oligonucleotide as performed by the Sanger dideoxy chain termination method is displayed at right.

What is the sequence of the original oligonucleotide? A second sample of the oligonucleotide was 3’-end labeled with ³²P and then subjected to the Maxam–Gilbert chemical cleavage sequencing protocol. Draw a diagram depicting the pattern seen on the autoradiogram of the Maxam–Gilbert sequencing gel.

A	C	G	T

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3. X-ray diffraction studies indicate the existence of a novel double-stranded DNA helical conformation in which ΔZ (the rise per base pair) = 0.32 nm and P (the pitch) = 3.36 nm. What are the other parameters of this novel helix: (a) the number of base pairs per turn, (b) $\Delta\phi$ (the mean rotation per base pair), and (c) c (the true repeat)?
4. A 41.5-nm-long duplex DNA molecule in the B-conformation adopts the A-conformation upon dehydration. How long is it now? What is its approximate number of base pairs?
5. If 80% of the base pairs in a duplex DNA molecule (12.5 kbp) are in the B-conformation and 20% are in the Z-conformation, what is the length of the molecule?
6. A “relaxed,” circular, double-stranded DNA molecule (1600 bp) is in a solution where conditions favor 10 bp per turn. What is the value of L_0 for this DNA molecule? Suppose DNA gyrase introduces 12 negative supercoils into this molecule. What are the values of L , W , and T now? What is the superhelical density, σ ?
7. Suppose one double-helical turn of a superhelical DNA molecule changes conformation from B-form to Z-form. What are the changes in L , W , and T ? Why do you suppose the transition of DNA from B-form to Z-form is favored by negative supercoiling?
8. There is one nucleosome for every 200 bp of eukaryotic DNA. How many nucleosomes are in a diploid human cell? Nucleosomes

- can be approximated as disks 11 nm in diameter and 6 nm long. If all the DNA molecules in a diploid human cell are in the B-conformation, what is the sum of their lengths? If this DNA is now arrayed on nucleosomes in the “beads-on-a-string” motif, what is its approximate total length?
9. The characteristic secondary structures of tRNA and rRNA molecules are achieved through intrastrand hydrogen bonding. Even for the small tRNAs, remote regions of the nucleotide sequence interact via H-bonding when the molecule adopts the cloverleaf pattern. Using Figure 12.34 as a guide, draw the primary structure of a tRNA and label the positions of its various self-complementary regions.
10. Using the data in Table 11.3, arrange the DNAs from the following sources in order of increasing T_m : human, salmon, wheat, yeast, *E. coli*.
11. The DNAs from mice and rats have (G+C) contents of 44% and 40%, respectively. Calculate the T_m s for these DNAs in 0.2 M NaCl. If samples of these DNAs were inadvertently mixed, how might they be separated from one another? Describe the procedure and the results (hint: see the Appendix to this chapter).
12. Calculate the density (ρ) of avian tubercle bacillus DNA from the data presented in Table 11.3 and the equation $\rho = 1.660 + 0.098(\text{GC})$, where (GC) is the mole fraction of (G+C) in DNA.

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Appendix to Chapter 12

Isopycnic Centrifugation and Buoyant Density of DNA

Density gradient ultracentrifugation is a variant of the basic technique of ultracentrifugation (discussed in the Appendix to Chapter 5). Density gradient centrifugation can be used to isolate DNA. The densities of DNAs are about the same as concentrated solutions of cesium chloride, CsCl (1.6 to 1.8 g/mL). Centrifugation of CsCl solutions at very high rotational speeds, where the centrifugal force becomes 10^5 times stronger than the force of gravity, causes the formation of a density gradient within the solution. This gradient is the result of a balance that is established between the sedimentation of the salt ions toward the bottom of the tube and their diffusion upward toward regions of lower concentration. If DNA is present in the centrifuged CsCl solution, it moves to a position of equilibrium in the gradient equivalent to its buoyant density (Figure A12.1). For this reason, this technique is also called **isopycnic centrifugation**.

Cesium chloride centrifugation is an excellent means of removing RNA and proteins in the purification of DNA. The density of DNA is typically slightly greater than 1.7 g/cm^3 , while the density of RNA is more than 1.8 g/cm^3 . Proteins have densities less than 1.3 g/cm^3 . In CsCl solutions of appropriate density, the DNA bands near the center of the tube, RNA pellets to the bottom, and the proteins float near the top. Single-stranded DNA is denser than double-helical DNA. The irregular structure of randomly coiled ssDNA allows the atoms to pack together through van der Waals interactions. These interactions compact the molecule into a smaller volume than that occupied by a hydrogen-bonded double helix.

The net movement of solute particles in an ultracentrifuge is the result of two processes: diffusion (from regions of higher concentration to regions of lower concentration) and sedimentation due to centrifugal force (in the direction away from the axis of rotation). In general, diffusion rates for molecules are inversely proportional to their molecular weight—larger molecules diffuse more slowly than smaller ones. On the other hand, sedimentation rates increase with increasing molecular weight. A macromolecular species that has reached its position of equilibrium in isopycnic centrifugation has formed a concentrated band of material.

Essentially three effects are influencing the movement of the molecules in creating this concentration zone: (1) diffusion away to regions of lower concentration; (2) sedimentation of molecules situated at positions of slightly lower solution density in the density gradient; and (3) flotation (buoyancy or “reverse sedimentation”) of molecules that have reached positions of slightly greater solution density in the gradient. The consequence of the physics of these effects is that, at equilibrium, *the width of the concentration band established by the macro-*

isopycnic • same density

molecular species is inversely proportional to the square root of its molecular weight. That is, a population of large molecules will form a concentration band that is narrower than the band formed by a population of small molecules. For example, the band width formed by dsDNA will be less than the band width formed by the same DNA when dissociated into ssDNA.

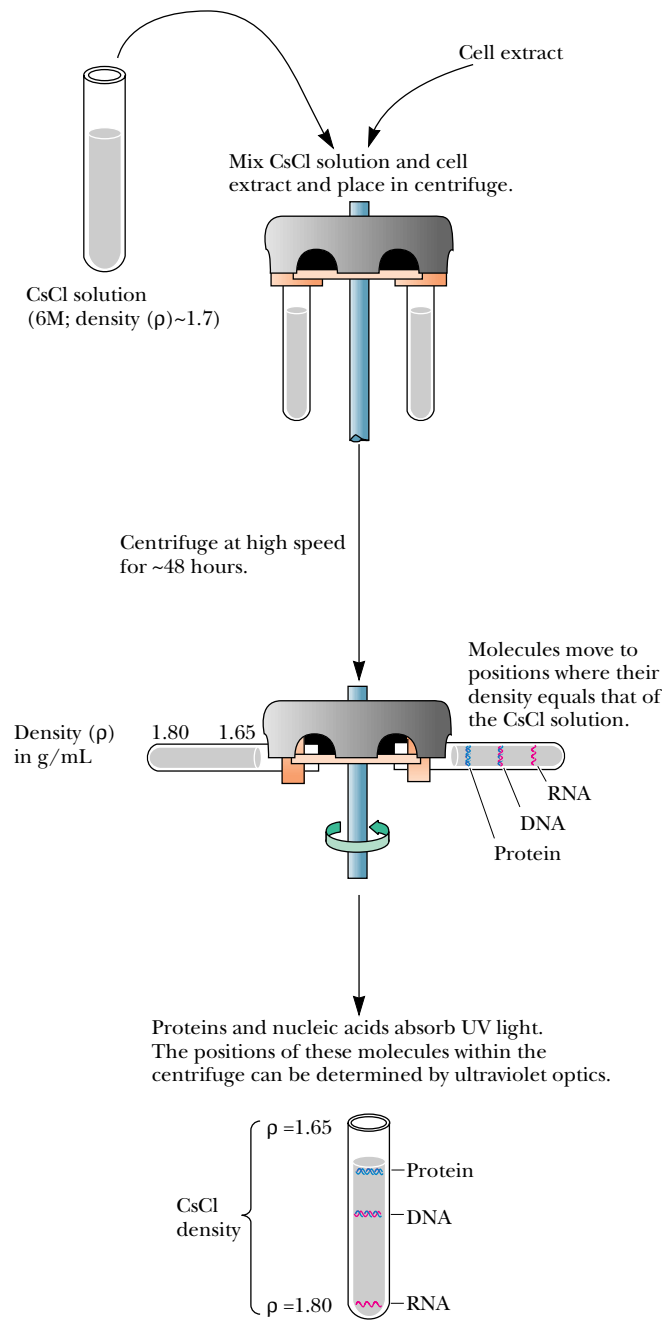


FIGURE A12.1 • Density gradient centrifugation is a common method of separating macromolecules, particularly nucleic acids, in solution. A cell extract is mixed with a solution of CsCl to a final density of about 1.7 g/cm³ and centrifuged at high speed (40,000 rpm, giving relative centrifugal forces of about 200,000 *g*). The biological macromolecules in the extract will move to equilibrium positions in the CsCl gradient that reflect their buoyant densities.

Chapter 13

Recombinant DNA: Cloning and Creation of Chimeric Genes



The Chimera of Arezzo, of Etruscan origin and probably from the 5th century B.C., was found near Arezzo, Italy, in 1553. Chimeric animals existed only in the imagination of the ancients. But the ability to create chimeric DNA molecules is a very real technology that has opened up a whole new field of scientific investigation.
(Scala/Art Resource, Chimera, Museo Archeologico, Florence, Italy)

In the early 1970s, technologies for the laboratory manipulation of nucleic acids emerged. In turn, these technologies led to the construction of DNA molecules composed of nucleotide sequences taken from different sources. The products of these innovations, **recombinant DNA molecules**,¹ opened exciting new avenues of investigation in molecular biology and genetics, and a new field

¹The advent of molecular biology, like that of most scientific disciplines, has generated a jargon all its own. Learning new fields often requires gaining familiarity with a new vocabulary. We will soon see that many words—*vector*, *amplification*, and *insert* are but a few examples—have been bent into new meanings to describe the marvels of this new biology.

. . . how many vain chimeras have you created? . . . Go and take your place with the seekers after gold.

LEONARDO DA VINCI, *The Notebooks* (1508–1518), Volume II, Chapter 25

OUTLINE

- 13.1 • Cloning
- 13.2 • DNA Libraries
- 13.3 • Polymerase Chain Reaction (PCR)
- 13.4 • Recombinant DNA Technology: An Exciting Scientific Frontier

amplification • the production of multiple copies

was born—**recombinant DNA technology**. **Genetic engineering** is the application of this technology to the manipulation of genes. These advances were made possible by methods for **amplification** of any particular DNA segment, regardless of source, within bacterial host cells. Or, in the language of recombinant DNA technology, the **cloning** of virtually any DNA sequence became feasible.

13.1 • Cloning

In classical biology, a *clone* is a population of identical organisms derived from a single parental organism. For example, the members of a colony of bacterial cells that arise from a single cell on a petri plate are clones. Molecular biology has borrowed the term to mean a collection of molecules or cells all identical to an original molecule or cell. So, if the original cell on the petri plate harbored a recombinant DNA molecule in the form of a plasmid, the plasmids within the millions of cells in a bacterial colony represent a clone of the original DNA molecule, and these molecules can be isolated and studied. Furthermore, if the cloned DNA molecule is a gene (or part of a gene), that is, it encodes a functional product, a new avenue to isolating and studying this product has opened. Recombinant DNA methodology offers exciting new vistas in biochemistry.

Plasmids

Plasmids are naturally occurring, circular, extrachromosomal DNA molecules (see Chapter 12). Natural strains of the common colon bacterium *Escherichia coli* isolated from various sources harbor diverse plasmids. Often these plasmids carry genes specifying novel metabolic activities that are advantageous to the host bacterium. These activities range from catabolism of unusual organic substances to metabolic functions that endow the host cells with resistance to antibiotics, heavy metals, or bacteriophages. Plasmids that are able to perpetuate themselves in *E. coli*, the bacterium favored by bacterial geneticists and molecular biologists, have become the darlings of recombinant DNA technology. Because restriction endonuclease digestion of plasmids can generate fragments with overlapping or “sticky” ends, artificial plasmids can be constructed by ligating different fragments together. Such artificial plasmids were among the earliest recombinant DNA molecules. These recombinant molecules can be autonomously replicated, and hence propagated, in suitable bacterial host cells, provided they still possess a site signaling where DNA replication can begin (a so-called **origin of replication** or *ori* sequence).

ligation • the act of joining

Plasmids as Cloning Vectors

The idea arose that “foreign” DNA sequences could be inserted into artificial plasmids and that these foreign sequences would be carried into *E. coli* and propagated as part of the plasmid. That is, these plasmids could serve as **cloning vectors** to carry genes. (The word *vector* is used here in the sense of “a vehicle or carrier.”) Plasmids useful as cloning vectors possess three common features: **a replicator, a selectable marker, and a cloning site** (Figure 13.1). A *replicator* is an origin of replication, or *ori*. The *selectable marker* is typically a gene conferring resistance to an antibiotic. Only those cells containing the cloning vector will grow in the presence of the antibiotic. Therefore, growth on antibiotic-containing media “selects for” plasmid-containing cells. Typically, the *cloning*

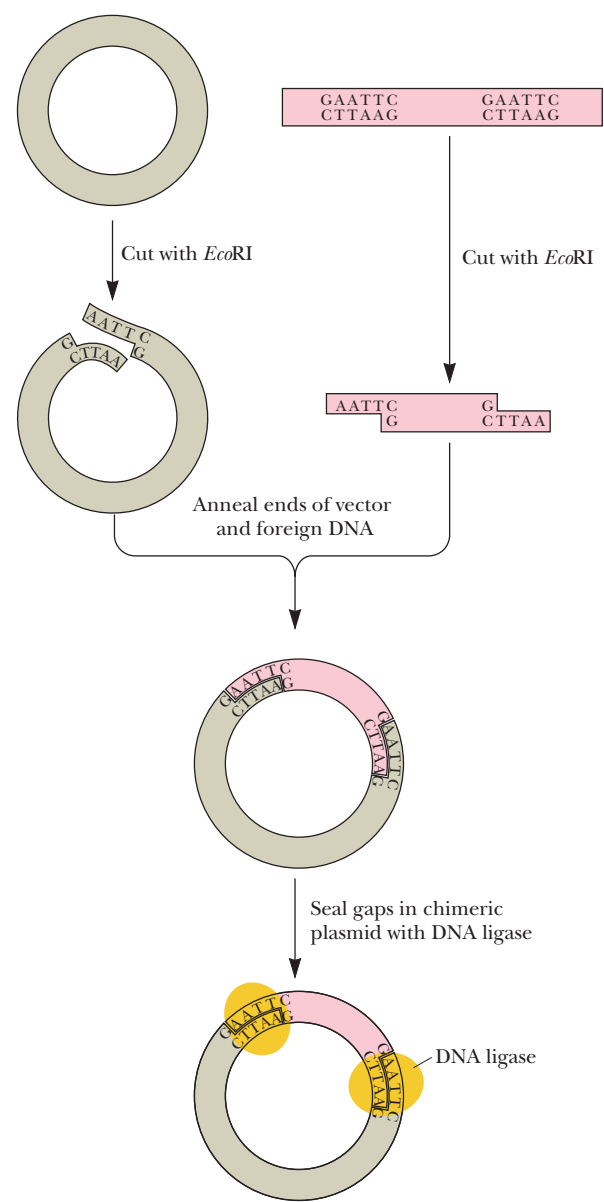
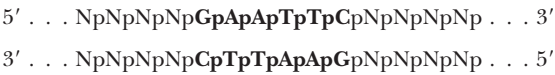
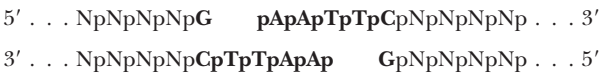


FIGURE 13.3 • Restriction endonuclease *EcoRI* cleaves double-stranded DNA. The recognition site for *EcoRI* is the hexameric sequence GAATTC:



Cleavage occurs at the G residue on each strand so that the DNA is cut in a staggered fashion, leaving 5'-overhanging single-stranded ends (sticky ends):



An *EcoRI* restriction fragment of foreign DNA can be inserted into a plasmid having an *EcoRI* cloning site by (a) cutting the plasmid at this site with *EcoRI*, annealing the linearized plasmid with the *EcoRI* foreign DNA fragment, and (b) sealing the nicks with DNA ligase.

Construction of Chimeric Plasmids

Creation of chimeric plasmids requires joining the ends of the foreign DNA insert to the ends of a linearized plasmid (Figure 13.2). This ligation is facilitated if the ends of the plasmid and the insert have complementary, single-stranded overhangs. Then these ends can base-pair with one another, annealing the two molecules together. One way to generate such ends is to cleave the DNA with restriction enzymes that make staggered cuts; many such restriction endonucleases are available (see Table 11.5). For example, if the sequence to be inserted is an *Eco*RI fragment and the plasmid is cut with *Eco*RI, the single-stranded sticky ends of the two DNAs can anneal (Figure 13.3). The interruptions in the sugar–phosphate backbone of DNA can then be sealed with DNA ligase to yield a covalently closed, circular chimeric plasmid. DNA ligase is an enzyme that covalently links adjacent 3'-OH and 5'-PO₄ groups. An inconvenience of this strategy is that *any* pair of *Eco*RI sticky ends can anneal with each other. So, plasmid molecules can reanneal with themselves, as can the foreign DNA restriction fragments. These DNAs can be eliminated by selection schemes designed to identify only those bacteria containing chimeric plasmids.

Blunt-end ligation is an alternative method for joining different DNAs. This method depends on the ability of **phage T4 DNA ligase** to covalently join the ends of any two DNA molecules (even those lacking 3'- or 5'-overhangs) (Figure 13.4). Some restriction endonucleases cut DNA so that blunt ends are formed (see Table 11.5). Because there is no control over which pair of DNAs are blunt-end ligated by T4 DNA ligase, strategies to identify the desired products must be applied.

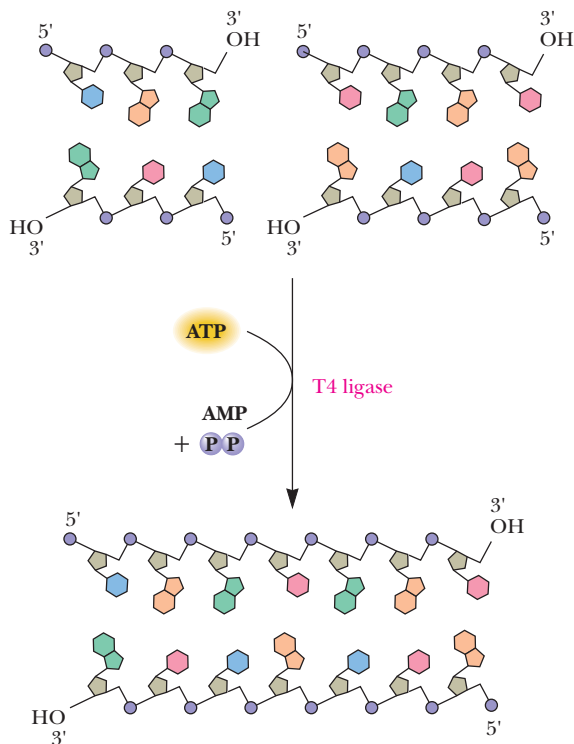


FIGURE 13.4 • Blunt-end ligation using phage T4 DNA ligase, which catalyzes the ATP-dependent ligation of DNA molecules. AMP and PP_i are by-products.

A great number of variations on these basic themes have emerged. For example, short synthetic DNA duplexes whose nucleotide sequence consists of little more than a restriction site can be blunt-end ligated onto any DNA. These short DNAs are known as **linkers**. Cleavage of the ligated DNA with the restriction enzyme then leaves tailor-made sticky ends useful in cloning reactions (Figure 13.5). Similarly, many vectors contain a **polylinker** cloning site, a short region of DNA sequence bearing numerous restriction sites.

Promoters and Directional Cloning

Note that the strategies discussed thus far create hybrids in which the orientation of the DNA insert within the chimera is random. Sometimes it is desirable to insert the DNA in a particular orientation. For example, an experimenter might wish to insert a particular DNA (a gene) in a vector so that its gene product is synthesized. To do this, the DNA must be placed downstream from a **promoter**. A promoter is a nucleotide sequence lying upstream of a

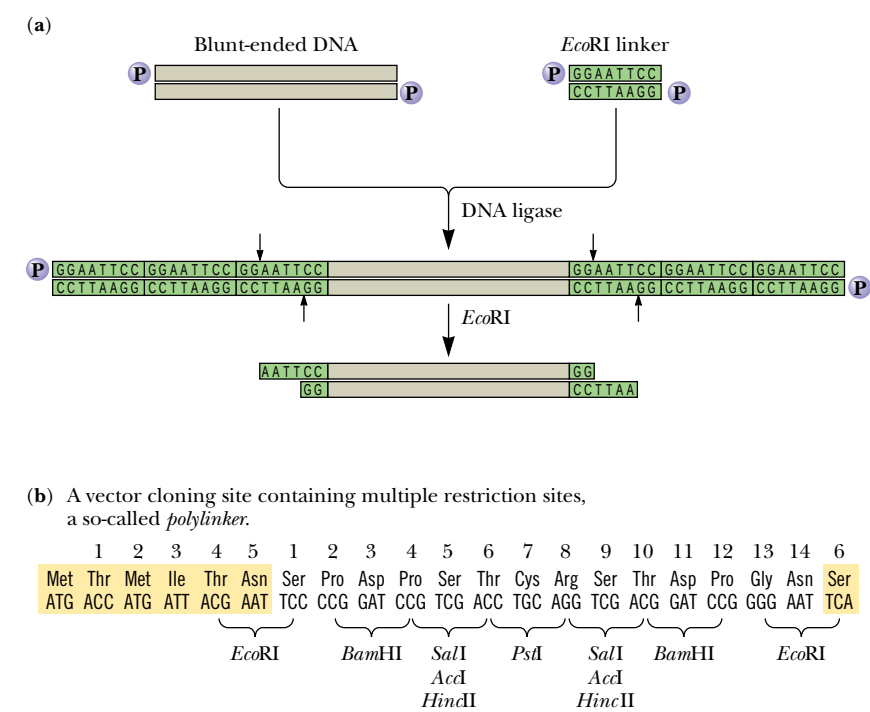


FIGURE 13.5 • (a) The use of linkers to create tailor-made ends on cloning fragments. Synthetic oligonucleotide duplexes whose sequences represent *EcoRI* restriction sites are blunt-end ligated to a DNA molecule using T4 DNA ligase. Note that the ligation reaction can add multiple linkers on each end of the blunt-ended DNA. *EcoRI* digestion removes all but the terminal one, leaving the desired 5'-overhangs. (b) Cloning vectors often have polylinkers consisting of a multiple array of restriction sites at their cloning sites, so restriction fragments generated by a variety of endonucleases can be incorporated into the vector. Note that the polylinker is engineered not only to have multiple restriction sites but also to have an uninterrupted sequence of codons, so this region of the vector has the potential for translation into protein. The sequence shown is the cloning site for the vectors M13mp7 and pUC7; the colored amino acid residues are contiguous with the coding sequence of the *lacZ* gene carried by this vector (see Figure 13.18). (a, Adapted from Figure 3.16.3; b, adapted from Figure 1.14.2, in Ausubel, F. M., et al., 1987, Current Protocols in Molecular Biology. New York: John Wiley & Sons.)

gene that controls expression of the gene. RNA polymerase molecules bind specifically at promoters and initiate transcription of adjacent genes, copying template DNA into RNA products. One way to insert DNA so that it will be properly oriented with respect to the promoter is to create DNA molecules whose ends have different overhangs. Ligation of such molecules into the plasmid vector can only take place in one orientation, to give **directional cloning** (Figure 13.6).

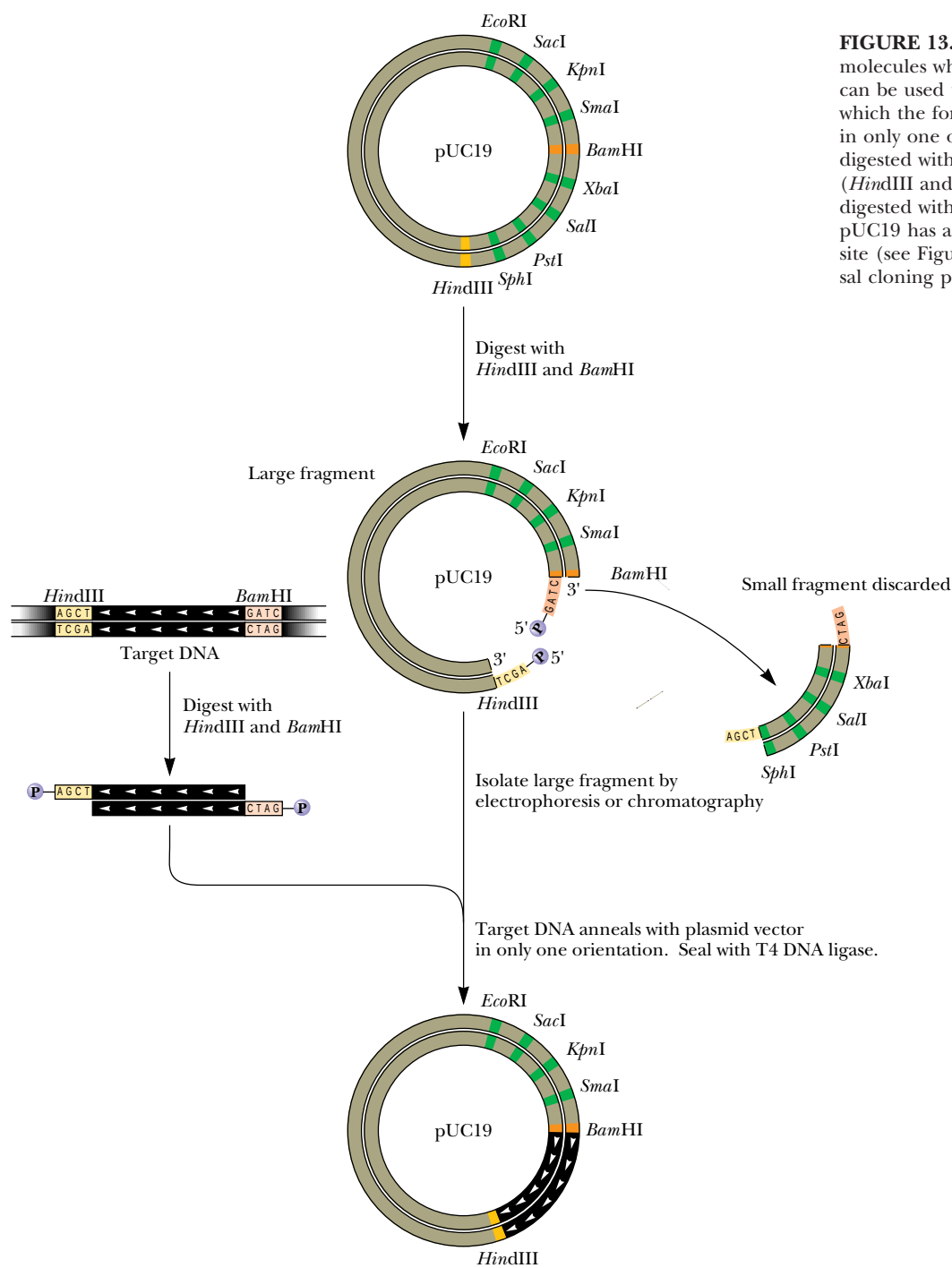


FIGURE 13.6 • Directional cloning. DNA molecules whose ends have different overhangs can be used to form chimeric constructs in which the foreign DNA can enter the plasmid in only one orientation. The foreign DNA is digested with two different restriction enzymes (*HindIII* and *BamHI*), and the plasmid is digested with the same two enzymes. Note that pUC19 has a polylinker or universal cloning site (see Figure 13.5b); pUC stands for universal cloning plasmid.

Biologically Functional Chimeric Plasmids

The first biologically functional chimeric DNA molecules constructed *in vitro* were assembled from parts of different plasmids in 1973 by Stanley Cohen, Annie Chang, Herbert Boyer, and Robert Helling. These plasmids were used to **transform** recipient *E. coli* cells (*transformation* means the uptake and repli-

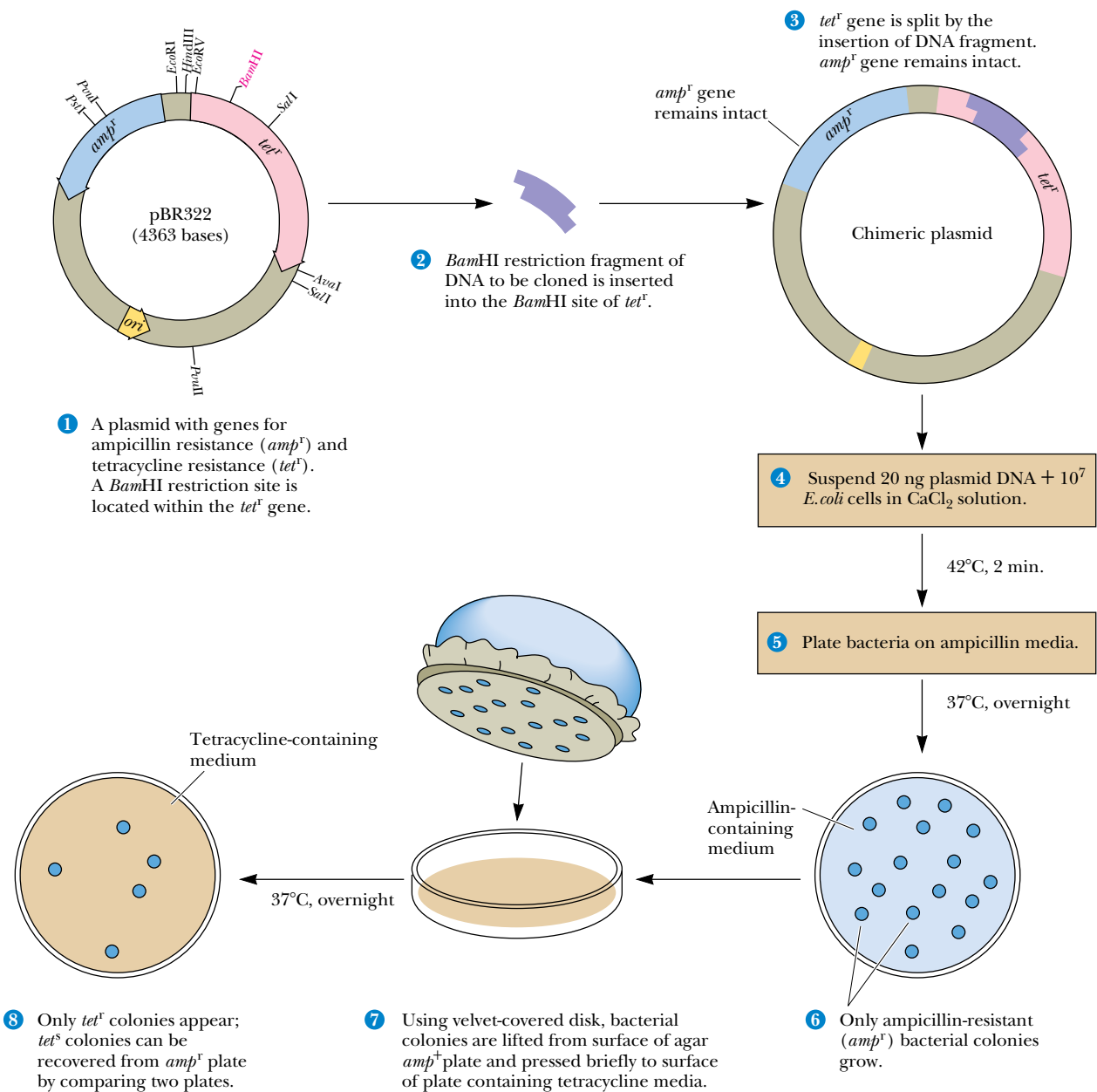


FIGURE 13.7 • A typical bacterial transformation experiment. Here the plasmid pBR322 is the cloning vector. (1) Cleavage of pBR322 with restriction enzyme *Bam*HI, followed by (2) annealing and ligation of inserts generated by *Bam*HI cleavage of some foreign DNA, (3) creates a chimeric plasmid. (4) The chimeric plasmid is then used to transform Ca^{2+} -treated heat-shocked *E. coli* cells, and the bacterial sample is plated on a petri plate. (5) Following incubation of the petri plate overnight at 37°C, (6) colonies of *amp^r* bacteria are evident. (7) Replica plating of these bacteria on plates of tetracycline-containing media (8) reveals which colonies are *tet^r* and which are tetracycline sensitive (*tet^s*). Only the *tet^s* colonies possess plasmids with foreign DNA inserts.

cation of exogenous DNA by a recipient cell; see Chapter 29). The bacterial cells were rendered somewhat permeable to DNA by Ca^{2+} treatment and a brief 42°C heat shock. Although less than 0.1% of the Ca^{2+} -treated bacteria became competent for transformation following such treatment, transformed bacteria could be selected by their resistance to certain antibiotics (Figure 13.7). Consequently, the chimeric plasmids must have been biologically functional in at least two aspects: they replicated stably within their hosts and they expressed the drug resistance markers they carried.

In general, plasmids used as cloning vectors are engineered to be small, 2.5 kbp to about 10 kbp in size, so that the size of the insert DNA can be maximized. These plasmids have only a single origin of replication, so the time necessary for complete replication depends on the size of the plasmid. Under selective pressure in a growing culture of bacteria, overly large plasmids are prone to delete any nonessential “genes,” such as any foreign inserts. Such deletion would thwart the purpose of most cloning experiments. The useful upper limit on cloned inserts in plasmids is about 10 kbp. Many eukaryotic genes exceed this size.

Bacteriophage λ as a Cloning Vector

The genome of bacteriophage λ (lambda) (Figure 13.8) is a 48.5-kbp linear DNA molecule that is packaged into the head of the bacteriophage. The middle one-third of this genome is not essential to phage infection, so λ phage DNA has been engineered so that foreign DNA molecules up to 16 kbp can be inserted into this region for cloning purposes. *In vitro* packaging systems are then used to package the chimeric DNA into phage heads which, when assembled with phage tails, form infective phage particles. Bacteria infected with these recombinant phage produce large numbers of phage progeny before they lyse, and large amounts of recombinant DNA can be easily purified from the lysate.

Cosmids

The DNA incorporated into phage heads by bacteriophage λ packaging systems must satisfy only a few criteria. It must possess a 14-bp sequence known as *cos* (which stands for *cohesive end site*) at each of its ends, and these *cos* sequences must be separated by no fewer than 36 kbp and no more than 51 kbp of DNA. Essentially any DNA satisfying these minimal requirements will be packaged and assembled into an infective phage particle. Other cloning features such as an *ori*, selectable markers, and a polylinker are joined to the *cos* sequence so that the cloned DNA can be propagated and selected in host cells. These features have been achieved by placing *cos* sequences on either side of cloning sites in plasmids to create **cosmid vectors** that are capable of carrying DNA inserts about 40 kbp in size (Figure 13.9). Because cosmids lack essential phage genes, they reproduce in host bacteria as plasmids.

Shuttle Vectors

Shuttle vectors are plasmids capable of propagating and transferring (“shuttling”) genes between two different organisms, one of which is typically a prokaryote (*E. coli*) and the other a eukaryote (for example, yeast). Shuttle vectors must have unique origins of replication for each cell type as well as different markers for selection of transformed host cells harboring the vector (Figure 13.10). Shuttle vectors have the advantage that eukaryotic genes can be cloned in bacterial hosts, yet the expression of these genes can be analyzed in appropriate eukaryotic backgrounds.



FIGURE 13.8 • Electron micrograph of bacteriophage λ . (Robley C. Williams, University of California/BPS)

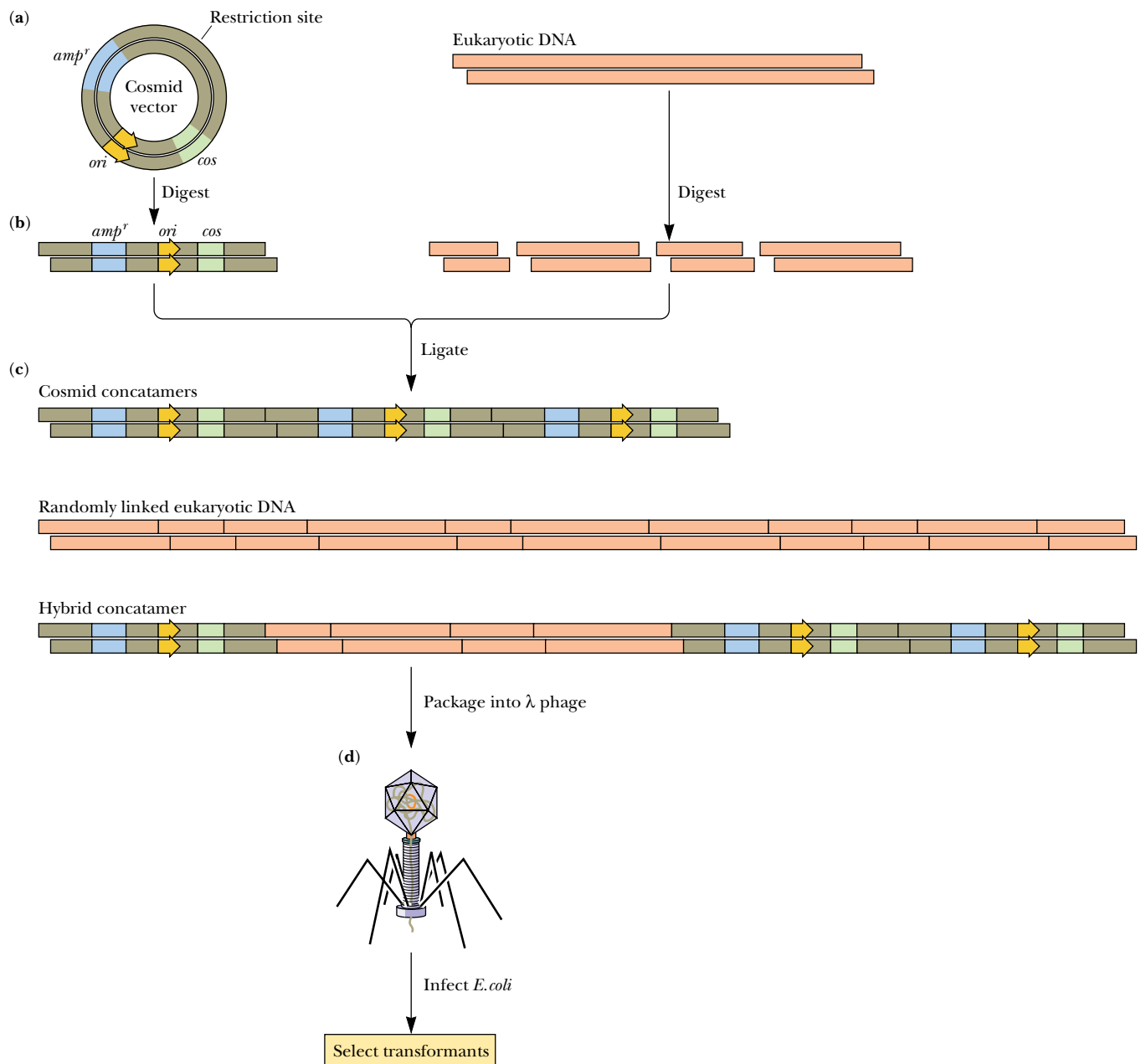
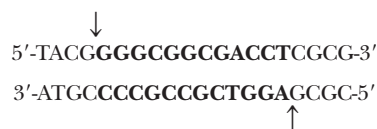


FIGURE 13.9 • Cosmid vectors for cloning large DNA fragments. (a) Cosmid vectors are plasmids that carry a selectable marker such as *amp^r*, an origin of replication (*ori*), a polylinker suitable for insertion of foreign DNA, and (b) a *cos* sequence. Both the plasmid and the foreign DNA to be cloned are cut with a restriction enzyme, and the two DNAs are then ligated together. (c) The ligation reaction leads to the formation of hybrid concatamers, molecules in which plasmid sequences and foreign DNAs are linked in series in no particular order. The bacteriophage λ packaging extract contains the restriction enzyme that recognizes *cos* sequences and cleaves at these sites. (d) DNA molecules of the proper size (36 to 51 kbp) are packaged into phage heads, forming infective phage particles. (e) The *cos* sequence is



Endonuclease cleavage at the sites indicated by arrows leaves 12-bp cohesive ends. (a–d, Adapted from Figure 1.10.7 in Ausubel, F. M., et al., eds., 1987. Current Protocols in Molecular Biology. New York: John Wiley & Sons; e, from Figure 4 in Murialdo, H., 1991. Annual Review of Biochemistry 60:136.)

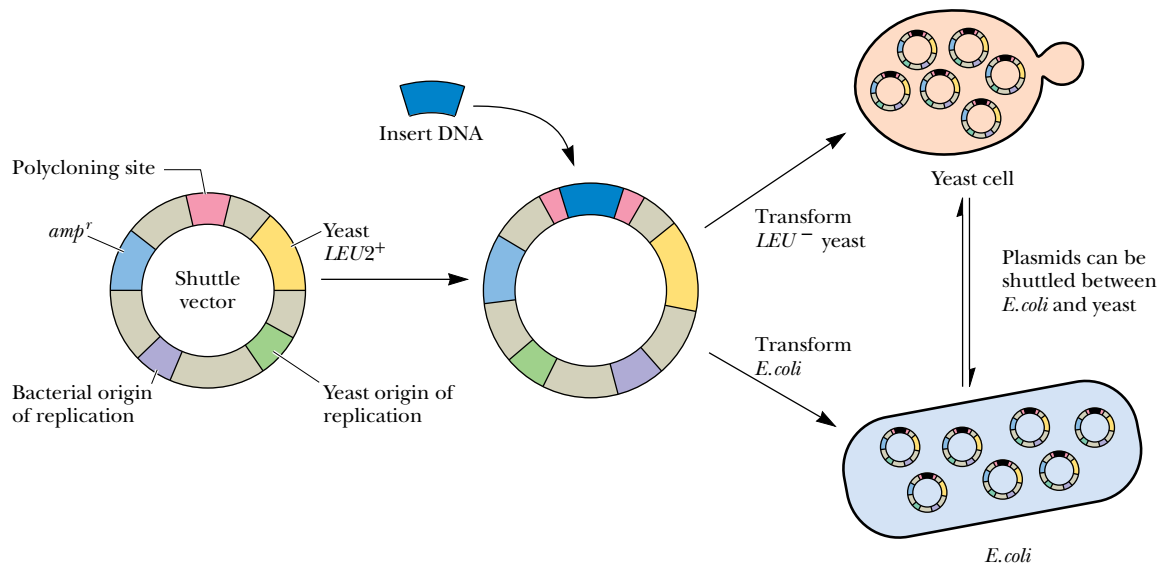


FIGURE 13.10 • A typical shuttle vector. This vector has both yeast and bacterial origins of replication, *amp^r* (ampicillin resistance gene for selection in *E. coli*) and *LEU2⁺*, a gene in the yeast pathway for leucine biosynthesis. The recipient yeast cells are *LEU2⁻* (defective in this gene) and thus require leucine for growth. *LEU2⁻* yeast cells transformed with this shuttle vector can be selected on medium lacking any leucine supplement. (Adapted from Figure 19-5 in Watson J. D., et al., 1987. The Molecular Biology of the Gene. Menlo Park, CA: Benjamin-Cummings.)

Artificial Chromosomes

DNA molecules 2 megabase pairs in length have been successfully propagated in yeast by creating **yeast artificial chromosomes** or **YACs**. Further, such YACs have been transferred into transgenic mice for the analysis of large genes or multigenic DNA sequences *in vivo*, that is, within the living animal. For these large DNAs to be replicated in the yeast cell, YAC constructs must include not only an origin of replication (known in yeast terminology as an *autonomously replicating sequence* or *ARS*) but also a centromere and telomeres. Recall that centromeres provide the site for attachment of the chromosome to the spindle during mitosis and meiosis, and telomeres are nucleotide sequences defining the ends of chromosomes. Telomeres are essential for proper replication of the chromosome.

13.2 • DNA Libraries

A DNA library is a set of cloned fragments that collectively represent the genes of a particular organism. Particular genes can be isolated from DNA libraries, much as books can be obtained from conventional libraries. The secret is knowing where and how to look.

Genomic Libraries

Any particular gene constitutes only a small part of an organism’s genome. For example, if the organism is a mammal whose entire genome encompasses some 10⁶ kbp and the gene is 10 kbp, then the gene represents only 0.001% of the total nuclear DNA. It is impractical to attempt to recover such rare sequences directly from isolated nuclear DNA because of the overwhelming amount of

extraneous DNA sequences. Instead, a **genomic library** is prepared by isolating total DNA from the organism, digesting it into fragments of suitable size, and cloning the fragments into an appropriate vector. This approach is called *shot-gun cloning* because the strategy has no way of targeting a particular gene but instead seeks to clone all the genes of the organism at one time. The intent is that at least one recombinant clone will contain at least part of the gene of interest. Usually, the isolated DNA is only partially digested by the chosen restriction endonuclease so that not every restriction site is cleaved in every DNA molecule. Then, even if the gene of interest contains a susceptible restriction site, some intact genes might still be found in the digest. Genomic libraries have been prepared from hundreds of different species.

Many clones must be created to be confident that the genomic library contains the gene of interest. The probability, P , that some number of clones, N , contains a particular fragment representing a fraction, f , of the genome is

$$P = 1 - (1 - f)^N$$

Thus,

$$N = \ln(1 - P) / \ln(1 - f)$$

For example, if the library consists of 10-kbp fragments of the *E. coli* genome (4640 kbp total), over 2000 individual clones must be screened to have a 99% probability ($P = 0.99$) of finding a particular fragment. Since $f = 10/4640 = 0.0022$ and $P = 0.99$, $N = 2093$. For a 99% probability of finding a particular

CRITICAL DEVELOPMENTS IN BIOCHEMISTRY

Combinatorial Libraries

Specific recognition and binding of other molecules is a defining characteristic of any protein or nucleic acid. Often, target ligands of a particular protein are unknown, or, in other instances, a unique ligand for a known protein may be sought in the hope of blocking the activity of the protein or otherwise perturbing its function. **Combinatorial libraries** are the products of emerging strategies to facilitate the identification and characterization of possible ligands for a protein. These strategies are also applicable to the study of nucleic acids. Unlike genomic libraries, combinatorial libraries consist of synthetic oligomers. Arrays of synthetic oligonucleotides printed as tiny dots on miniature solid supports are known as **DNA chips**. Specifically, combinatorial libraries contain very large numbers of chemically synthesized molecules (such as peptides or oligonucleotides) with randomized sequences or structures. Such libraries are designed and constructed with the hope that one molecule among a vast number will be recognized as a ligand by the protein (or nucleic acid) of interest. If so, perhaps that molecule will be useful in a pharmaceutical application, for instance as a drug to treat a disease involving the protein to which it binds.

An example of this strategy is the preparation of a **synthetic combinatorial library** of hexapeptides. The maximum number of sequence combinations for hexapeptides is 20^6 or 64,000,000. One approach to simplify preparation and screening possibilities for such a library is to specify the first two amino acids in the hexa-

peptide while the next four are randomly chosen. In this approach, 400 libraries (20^2) are synthesized, each of which is unique in terms of the amino acids at positions 1 and 2 but random at the other four positions (as in AAXXXX, ACXXXX, ADXXXX, etc.) so that each of the 400 libraries contains 20^4 or 160,000 different sequence combinations. Screening these libraries with the protein of interest reveals which of the 400 libraries contains a ligand with high affinity. This library is then systematically expanded by specifying the first 3 amino acids (knowing from the chosen 1-of-400 libraries which amino acids are best as the first 2); only 20 synthetic libraries (each containing 20^3 or 8000 hexapeptides) are made here (one for each third-position possibility, the remaining three positions being randomized). Selection for ligand binding, again with the protein of interest, reveals the best of these 20, and this particular library is then varied systematically at the fourth position, creating 20 more libraries (each containing 20^2 or 400 hexapeptides). This cycle of synthesis, screening, and selection is repeated until all six positions in the hexapeptide are optimized to create the best ligand for the protein. A variation on this basic strategy using synthetic oligonucleotides rather than peptides identified a unique 15-mer (sequence GGTGGTGTGGTTGG) with high affinity ($K_D = 2.7$ nM) toward thrombin, a serine protease in the blood coagulation pathway. Thrombin is a major target for the pharmacological prevention of clot formation in coronary thrombosis.

(From Cortese, R., 1996. Combinatorial Libraries: Synthesis, Screening and Application Potential. Berlin: Walter de Gruyter.)

FIGURE 13.11 • Screening a genomic library by colony hybridization (or plaque hybridization). Host bacteria transformed with a plasmid-based genomic library or infected with a bacteriophage-based genomic library are plated on a petri plate and incubated overnight to allow bacterial colonies (or phage plaques) to form. A replica of the bacterial colonies (or plaques) is then obtained by overlaying the plate with a nitrocellulose disc (1). Nitrocellulose strongly binds nucleic acids; single-stranded nucleic acids are bound more tightly than double-stranded nucleic acids. (Nylon membranes with similar nucleic acid– and protein-binding properties are also used.) Once the nitrocellulose disc has taken up an impression of the bacterial colonies (or plaques), it is removed and the petri plate is set aside and saved. The disc is treated with 2 M NaOH, neutralized, and dried (2). NaOH both lyses any bacteria (or phage particles) and dissociates the DNA strands. When the disc is dried, the DNA strands become immobilized on the filter. The dried disc is placed in a sealable plastic bag, and a solution containing heat-denatured (single-stranded), labeled probe is added (3). The bag is incubated to allow annealing of the probe DNA to any target DNA sequences that might be present on the nitrocellulose. The filter is then washed, dried, and placed on a piece of X-ray film to obtain an autoradiograph (4). The position of any spots on the X-ray film reveals where the labeled probe has hybridized with target DNA (5). The location of these spots can be used to recover the genomic clone from the bacteria (or plaques) on the original petri plate.

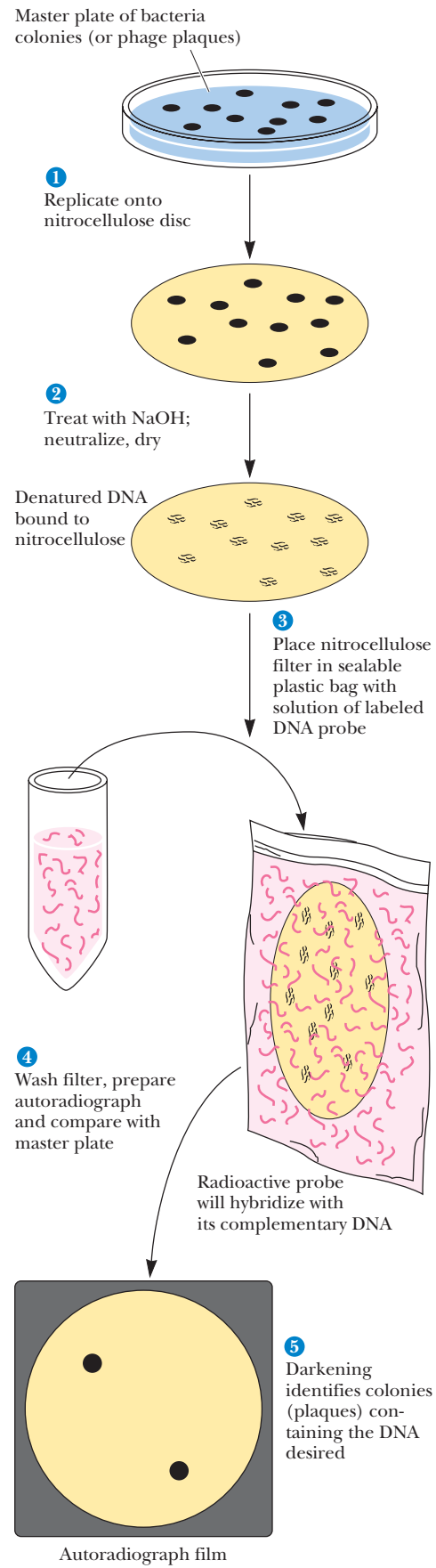
sequence within the 3×10^6 kbp human genome, N would equal almost 1.4 million if the cloned fragments averaged 10 kbp in size. The need for cloning vectors capable of carrying very large DNA inserts becomes obvious from these numbers.

Screening Libraries

A common method of screening plasmid-based genomic libraries is to carry out a **colony hybridization experiment**. The protocol is similar for phage-based libraries except that bacteriophage plaques, not bacterial colonies, are screened. In a typical experiment, host bacteria containing either a plasmid-based or bacteriophage-based library are plated out on a petri dish and allowed to grow overnight to form colonies (or in the case of phage libraries, plaques) (Figure 13.11). A replica of the bacterial colonies (or plaques) is then obtained by overlaying the plate with a nitrocellulose disc. The disc is removed, treated with alkali to dissociate bound DNA duplexes into single-stranded DNA, dried, and placed in a sealed bag with labeled probe (see the box on Southern blotting). If the probe DNA is duplex DNA, it must be denatured by heating at 70°C. The probe and target DNA complementary sequences must be in a single-stranded form if they are to hybridize with one another. Any DNA sequences complementary to probe DNA will be revealed by autoradiography of the nitrocellulose disc. Bacterial colonies (phage plaques) containing clones bearing target DNA are identified on the film and can be recovered from the master plate.

Probes for Southern Hybridization

Clearly, specific probes are essential reagents if the goal is to identify a particular gene against a background of innumerable DNA sequences. Usually, the probes that are used to screen libraries are nucleotide sequences that are complementary to some part of the target gene. To make useful probes requires some information about the gene’s nucleotide sequence. Sometimes such information is available. Alternatively, if the amino acid sequence of the protein encoded by the gene is known, it is possible to work backward through the genetic code to the DNA sequence (Figure 13.12). Because the genetic code is *degenerate* (that is, several codons may specify the same amino acid; see



Known amino acid sequence:						
Phe	Met	Glu	Trp	His	Lys	Asn
Possible mRNA sequence:						
UUU	AUG	GAA	UGG	CAU	AGG	AAU
UUC		GAG		CAC	AAA	AAC

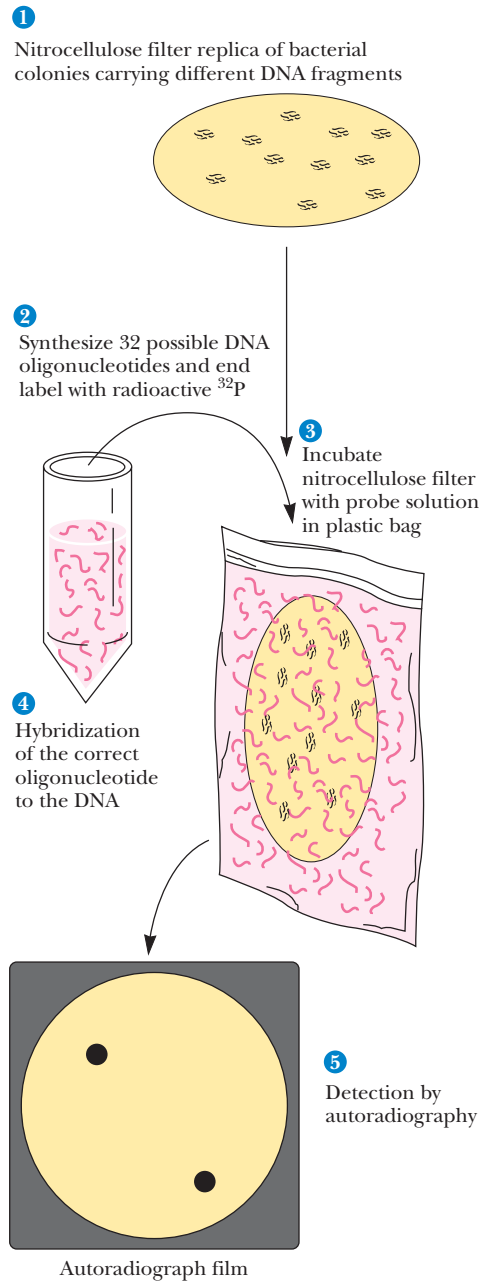


FIGURE 13.12 • Cloning genes using oligonucleotide probes designed from a known amino acid sequence. A radioactively labeled set of DNA (degenerate) oligonucleotides representing all possible mRNA coding sequences is synthesized. (In this case, there are 2⁵, or 32.) The complete mixture is used to probe the genomic library by colony hybridization (see Figure 13.11). (Adapted from Figure 19-18 in Watson, J. D., et al., 1987. Molecular Biology of the Gene. Menlo Park, CA: Benjamin-Cummings.)

Chapter 32), probes designed by this approach are usually **degenerate oligonucleotides** about 17 to 50 residues long (such oligonucleotides are so-called 17- to 50-mers). The oligonucleotides are synthesized so that different bases are incorporated at sites where degeneracies occur in the codons. The final preparation thus consists of a mixture of equal-length oligonucleotides whose sequences vary to accommodate the degeneracies. Presumably, one oligonucleotide sequence in the mixture will hybridize with the target gene. These oligonucleotide probes are at least 17-mers because shorter degenerate oligonucleotides might hybridize with sequences unrelated to the target sequence.

A piece of DNA from the corresponding gene in a related organism can also be used as a probe in screening a library for a particular gene. Such probes are termed **heterologous probes** because they are not derived from the homologous (same) organism.

Problems arise if a complete eukaryotic gene is the cloning target; eukaryotic genes can be tens or even hundreds of kilobase pairs in size. Genes this size are fragmented in most cloning procedures. Thus, the DNA identified by the probe may represent a clone that carries only part of the desired gene. However, most cloning strategies are based on a partial digestion of the genomic DNA, a technique that generates an overlapping set of genomic fragments. This being so, DNA segments from the ends of the identified clone can now be used to probe the library for clones carrying DNA sequences that flanked the original isolate in the genome. Repeating this process ultimately yields the complete gene among a subset of overlapping clones.

cDNA Libraries

cDNAs are DNA molecules copied from mRNA templates. cDNA libraries are constructed by synthesizing cDNA from purified cellular mRNA. These libraries present an alternative strategy for gene isolation, especially eukaryotic genes. Because most eukaryotic mRNAs carry 3'-poly(A) tails, mRNA can be selectively isolated from preparations of total cellular RNA by oligo(dT)-cellulose chromatography (Figure 13.13). DNA copies of the purified mRNAs are synthesized by first annealing short oligo(dT) chains to the poly(A) tails. These oligo(dT) chains serve as primers for reverse transcriptase-driven synthesis of DNA (Figure 13.14). (Random oligonucleotides can also be used as primers, with the advantages being less dependency on poly(A) tracts and increased likelihood of creating clones representing the 5'-ends of mRNAs.) **Reverse transcriptase** is an enzyme that synthesizes a DNA strand, copying RNA as the template. DNA polymerase is then used to copy the DNA strand and form a double-stranded (duplex DNA) molecule. Linkers are then added to the DNA duplexes rendered from the mRNA templates, and the cDNA is cloned into a suitable vector. Once a cDNA derived from a particular gene has been identified, the cDNA becomes an effective probe for screening genomic libraries for isolation of the gene itself.

Because different cell types in eukaryotic organisms express selected subsets of genes, RNA preparations from cells or tissues in which genes of interest are selectively transcribed are enriched for the desired mRNAs. cDNA

(Text continues on page 412.)

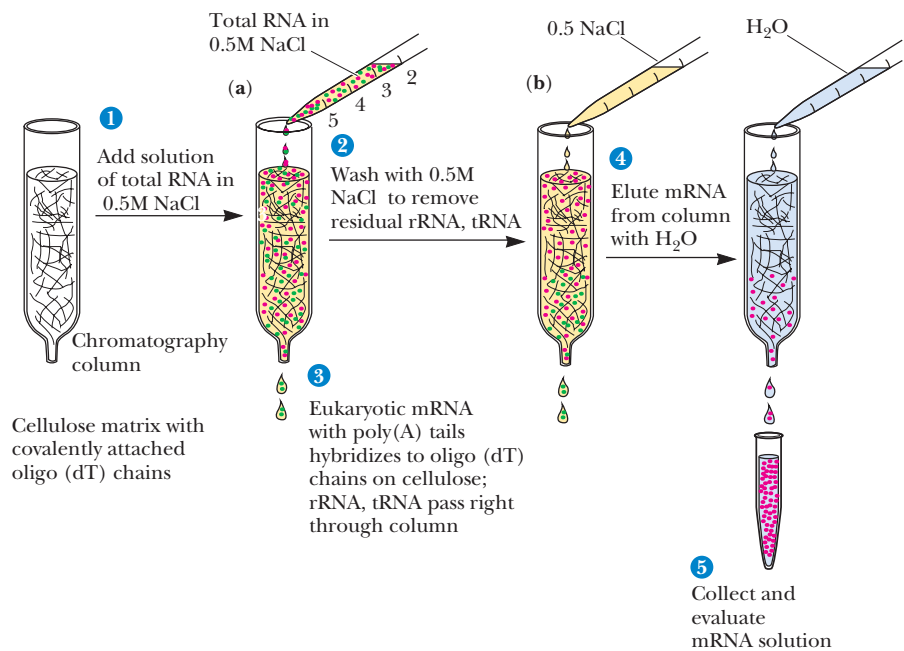


FIGURE 13.13 • Isolation of eukaryotic mRNA via oligo(dT)-cellulose chromatography. (a) In the presence of 0.5 M NaCl, the poly(A) tails of eukaryotic mRNA anneal with short oligo(dT) chains covalently attached to an insoluble chromatographic matrix such as cellulose. Other RNAs, such as rRNA (green), pass right through the chromatography column. (b) The column is washed with more 0.5 M NaCl to remove residual contaminants. (c) Then the poly(A) mRNA is recovered by washing the column with water because the base pairs formed between the poly(A) tails of the mRNA and the oligo(dT) chains are unstable in solutions of low ionic strength.

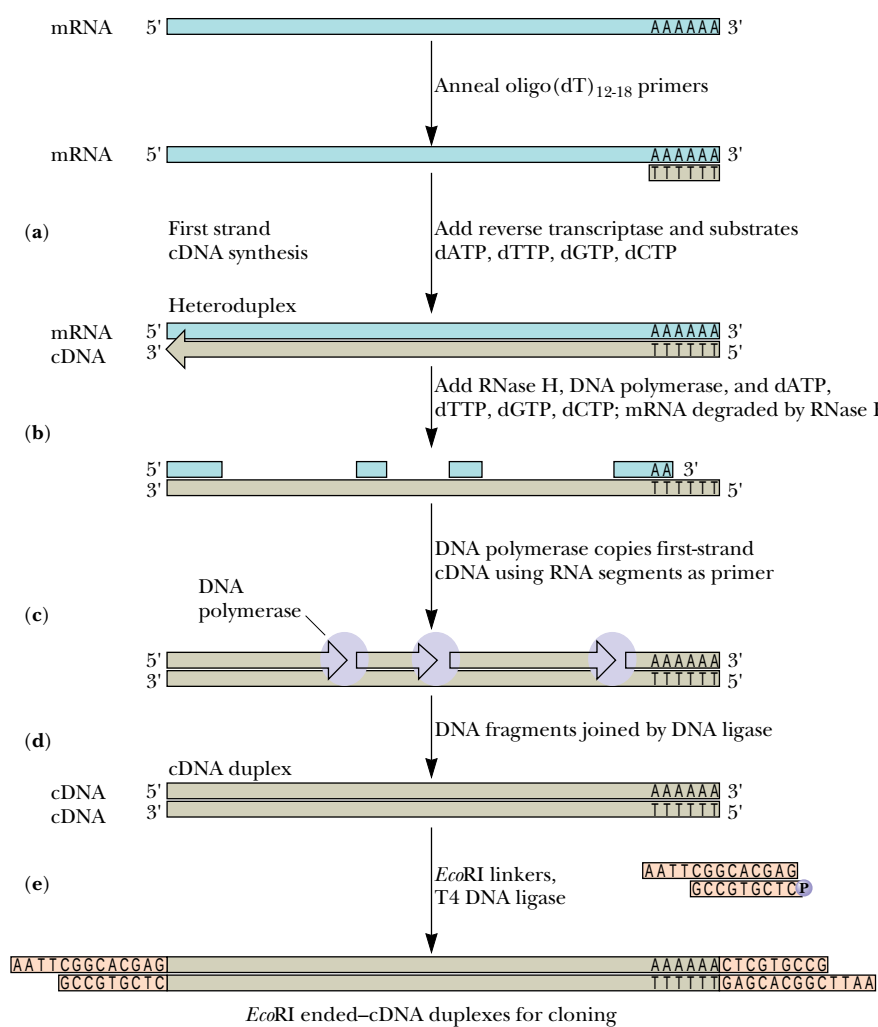


FIGURE 13.14 • Reverse transcriptase-driven synthesis of cDNA from oligo(dT) primers annealed to the poly(A) tails of purified eukaryotic mRNA. (a) Oligo(dT) chains serve as primers for synthesis of a DNA copy of the mRNA by reverse transcriptase. Following completion of first-strand cDNA synthesis by reverse transcriptase, RNase H and DNA polymerase are added (b). RNase H specifically digests RNA strands in DNA:RNA hybrid duplexes. DNA polymerase copies the first-strand cDNA, using as primers the residual RNA segments after RNase H has created nicks and gaps (c). DNA polymerase has a 5'→3' exonuclease activity that removes the residual RNA as it fills in with DNA. The nicks remaining in the second-strand DNA are sealed by DNA ligase (d), yielding duplex cDNA. EcoRI adapters with 5'-overhangs are then ligated onto the cDNA duplexes (e) using phage T4 DNA ligase to create EcoRI-ended cDNA for insertion into a cloning vector.

CRITICAL DEVELOPMENTS IN BIOCHEMISTRY

Identifying Specific DNA Sequences by Southern Blotting
(Southern Hybridization)

Any given DNA fragment is unique solely by virtue of its specific nucleotide sequence. The only practical way to find one particular DNA segment among a vast population of different DNA fragments (such as you might find in genomic DNA preparations) is to exploit its sequence specificity to identify it. In 1975, E. M. Southern invented a technique capable of doing just that.

Electrophoresis

Southern first fractionated a population of DNA fragments according to size by gel electrophoresis (see step 2 in figure). The electrophoretic mobility of a nucleic acid is inversely proportional to its molecular mass. Polyacrylamide gels are suitable for separation of nucleic acids of 25 to 2000 bp. Agarose gels are better if the DNA fragments range up to 10 times this size. Most preparations of genomic DNA show a broad spectrum of sizes, from less than 1 kbp to more than 20 kbp. Typically, no discrete-size fragments are evident following electrophoresis, just a “smear” of DNA throughout the gel.

Blotting

Once the fragments have been separated by electrophoresis (step 3), the gel is soaked in a solution of NaOH. Alkali denatures duplex DNA, converting it to single-stranded DNA. After the pH of the gel is adjusted to neutrality with buffer, a sheet of nitrocellulose soaked in a concentrated salt solution is then placed over the gel (c), and salt solution is drawn through the gel in a direction perpendicular to the direction of electrophoresis (step 4). The salt solution is pulled through the gel in one of three ways: capillary action (*blotting*), suction (*vacuum blotting*), or electrophoresis (*electroblotting*). The movement of salt solution through the gel carries the DNA to the nitrocellulose sheet. Nitrocellulose binds single-stranded DNA molecules very tightly, effectively immobilizing them in place on the sheet.* Note that the distribution pattern of the electrophoretically separated DNA

is maintained when the single-stranded DNA molecules bind to the nitrocellulose sheet (step 5 in figure). Next, the nitrocellulose is dried by baking in a vacuum oven;† baking tightly fixes the single-stranded DNAs to the nitrocellulose. Next, in the *prehybridization step*, the nitrocellulose sheet is incubated with a solution containing protein (serum albumin, for example) and/or a detergent such as sodium dodecyl sulfate. The protein and detergent molecules saturate any remaining binding sites for DNA on the nitrocellulose. Thus, no more DNA can bind nonspecifically to the nitrocellulose sheet.

Hybridization

To detect a particular DNA within the electrophoretic smear of countless DNA fragments, the prehybridized nitrocellulose sheet is incubated in a sealed plastic bag with a solution of specific probe molecules (step 6 in figure). A **probe** is usually a single-stranded DNA of defined sequence that is distinctively labeled, either with a radioactive isotope (such as ³²P) or some other easily detectable tag. The nucleotide sequence of the probe is designed to be complementary to the sought-for or *target* DNA fragment. The single-stranded probe DNA **anneals** with the single-stranded target DNA bound to the nitrocellulose through specific base pairing to form a DNA duplex. This annealing, or **hybridization** as it is usually called, labels the target DNA, revealing its position on the nitrocellulose. For example, if the probe is ³²P-labeled, its location can be detected by autoradiographic exposure of a piece of X-ray film laid over the nitrocellulose sheet (step 7 in figure).

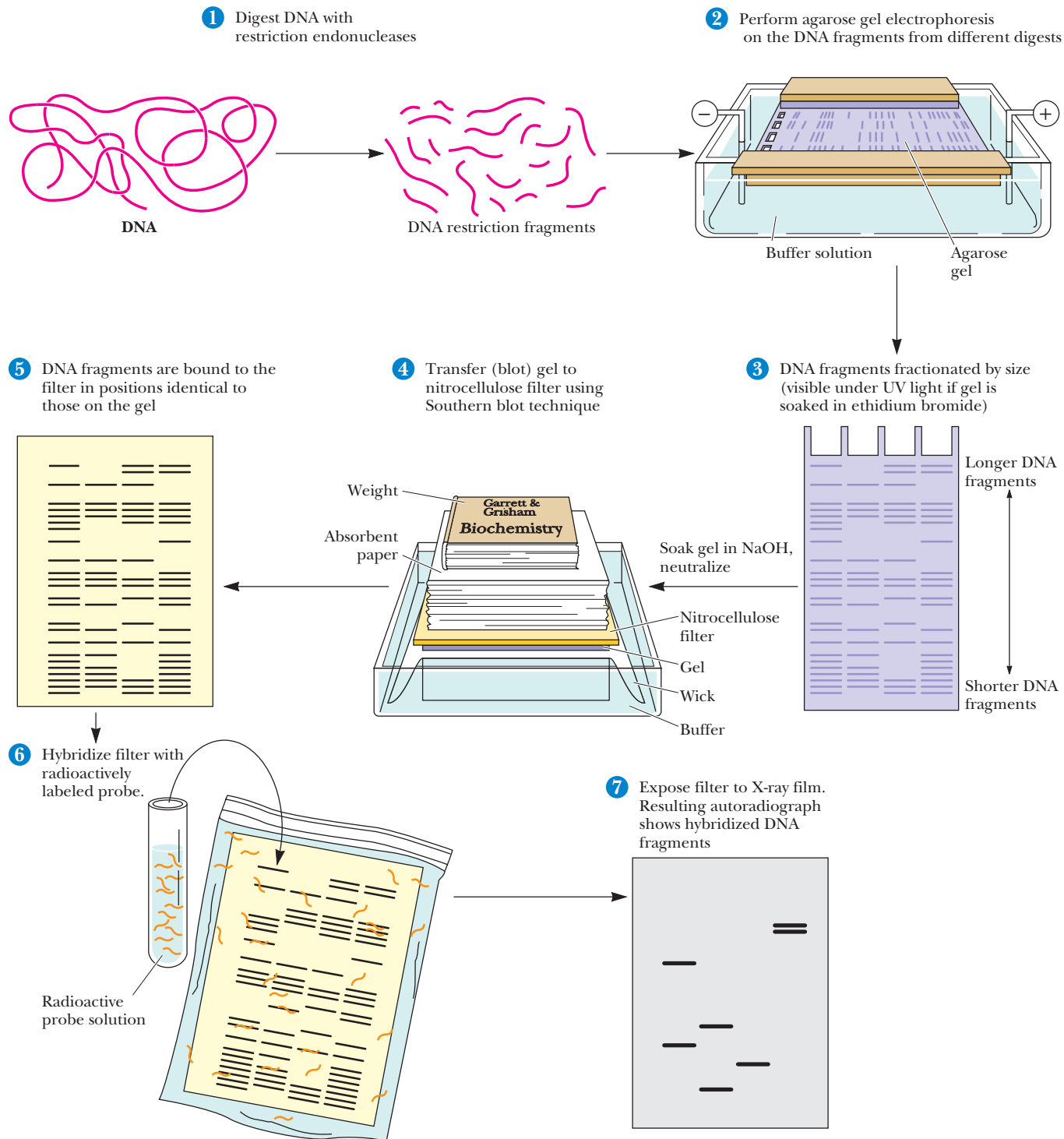
Southern’s procedure has been extended to the identification of specific RNA and protein molecules. In a play on Southern’s name, the identification of particular RNAs following separation by gel electrophoresis, blotting, and probe hybridization is called **Northern blotting**. The analogous technique for identifying protein molecules is termed **Western blotting**. In Western blotting, the probe of choice is usually an antibody specific for the target protein.

The Southern blotting technique involves the transfer of electrophoretically separated DNA fragments to a nitrocellulose sheet and subsequent detection of specific DNA sequences. A preparation of DNA fragments [typically a restriction digest, (1)] is separated according to size by gel electrophoresis (2). The separation pattern can be visualized by soaking the gel in ethidium bromide to stain the DNA and then illuminating the gel with UV light (3). Ethidium bromide molecules intercalated between the hydrophobic bases of DNA are fluorescent under UV light. The gel is soaked in strong alkali to denature the DNA and then neu-

tralized in buffer. Next, the gel is placed on a sheet of nitrocellulose (or DNA-binding nylon membrane), and concentrated salt solution is passed through the gel (4) to carry the DNA fragments out of the gel where they are bound tightly to the nitrocellulose (5). Incubation of the nitrocellulose sheet with a solution of labeled, single-stranded probe DNA (6) allows the probe to hybridize with target DNA sequences complementary to it. The location of these target sequences is then revealed by an appropriate means of detection, such as autoradiography (7). ►

*The underlying cause of DNA binding to nitrocellulose is not clear, but probably involves a combination of hydrogen bonding, hydrophobic interactions, and salt bridges.

†Vacuum drying is essential because nitrocellulose reacts violently with O₂ if heated. For this reason, nylon-based membranes are preferable to nitrocellulose membranes.



HUMAN BIOCHEMISTRY

The Human Genome Project

The Human Genome Project is a collaborative international, government- and private-sponsored effort to map and sequence the entire human genome, some 3 billion base pairs distributed among the two sex chromosomes (**X** and **Y**) and 22 **autosomes** (chromosomes that are not sex chromosomes). Initial work identified and mapped at least 3000 genetic **markers** (genes or other recognizable loci on the DNA), evenly distributed throughout the chromosomes at roughly 100-kb intervals. At the same time, determination of the entire nucleotide sequence of the human genome began. The target date for completion is 2005. An ancillary part of the project is sequencing the genomes of other species (such as yeast, *Drosophila melanogaster* [the fruit fly], mice, and *Arabidopsis thaliana* [a plant]) to reveal comparative aspects of genetic and sequence organization (Table 13.1). Information about whole genome sequences of organisms has created a new branch of science called **functional genomics**. Functional genomics addresses global issues of gene expression, such as looking at *all* the genes that are activated during major metabolic shifts (as from growth under aerobic to growth under anaerobic conditions) or during embryogenesis and development of organisms. Functional genomics also provides new insights into evolutionary relationships between organisms.

The Human Genome Project is also vital to medicine. A number of human diseases have been traced to genetic defects, whose positions within the human genome have been identified. Among these are

- cystic fibrosis* gene
- Duchenne muscular dystrophy* gene* (at 2.4 megabases, the largest known gene in any organism)
- Huntington’s disease* gene
- neurofibromatosis* gene
- neuroblastoma* gene (a form of brain cancer)
- amyotrophic lateral sclerosis* gene (Lou Gehrig’s disease)
- fragile X-linked mental retardation* gene*

as well as genes associated with the development of diabetes, breast cancer, colon cancer, and affective disorders such as *schizophrenia* and *bipolar affective disorder* (manic depression).

Table 13.1

Completed Genome Nucleotide Sequences	
Genome	Genome Size ¹ (Year Completed)
Bacteriophage ϕ X174	0.0054 (1977)
Bacteriophage λ	0.048 (1982)
<i>Marchantia</i> ² chloroplast genome	0.187 (1986)
Vaccinia virus	0.192 (1990)
Cytomegalovirus (CMV)	0.229 (1991)
<i>Marchantia</i> ² mitochondrial genome	0.187 (1992)
Variola (smallpox) virus	0.186 (1993)
<i>Hemophilus influenzae</i> ³ (Gram-negative bacterium)	1.830 (1995)
<i>Mycobacterium genatium</i> (mycobacterium)	0.58 (1995)
<i>Methanococcus jannaschii</i> (archaeobacterium)	1.67 (1996)
<i>Escherichia coli</i> (Gram-negative bacterium)	4.64 (1996)
<i>Saccharomyces cerevisiae</i> (yeast)	12.067 (1996)
<i>Bacillus subtilis</i> (Gram-positive bacterium)	4.21 (1997)
<i>Arabidopsis thaliana</i> (green plant)	100 (?)
<i>Caenorhabditis elegans</i> (simple animal: nematode worm)	100 (1998?)
<i>Drosophila melanogaster</i> (fruit fly)	165 (?)
<i>Homo sapiens</i> (human)	2900 (2005?)

¹Genome size is given as millions of base pairs (mb).
²*Marchantia* is a bryophyte (a nonvascular green plant).
³The first complete sequence for the genome of a free-living organism.

*X-chromosome linked gene. As of 1992, more than 100 disease-related genes had been mapped to this chromosome.

libraries prepared from such mRNA are representative of the pattern and extent of gene expression that uniquely define particular kinds of differentiated cells. cDNA libraries of many normal and diseased human cell types are commercially available, including cDNA libraries of many tumor cells. Comparison of normal and abnormal cDNA libraries, in conjunction with two-dimensional gel electrophoretic analysis (see Appendix to Chapter 5) of the proteins produced in normal and abnormal cells, is a promising new strategy in clinical medicine to understand disease mechanisms.

Expression Vectors

Expression vectors are engineered so that any cloned insert can be transcribed into RNA, and, in many instances, even translated into protein. cDNA expression libraries can be constructed in specially designed vectors derived from either plasmids or bacteriophage λ . Proteins encoded by the various cDNA clones within such expression libraries can be synthesized in the host cells, and if suitable assays are available to identify a particular protein, its corresponding cDNA clone can be identified and isolated. Expression vectors designed for RNA expression or protein expression, or both, are available.

RNA Expression

A vector for *in vitro* expression of DNA inserts as RNA transcripts can be constructed by putting a highly efficient promoter adjacent to a versatile cloning site. Figure 13.15 depicts such an expression vector. Linearized recombinant vector DNA is transcribed *in vitro* using SP6 RNA polymerase. Large amounts of RNA product can be obtained in this manner; if radioactive ribonucleotides are used as substrates, labeled RNA molecules useful as probes are made.

Protein Expression

Because cDNAs are DNA copies of mRNAs, cDNAs are uninterrupted copies of the exons of expressed genes. Because cDNAs lack introns, it is feasible to express these cDNA versions of eukaryotic genes in prokaryotic hosts that cannot process the complex primary transcripts of eukaryotic genes. To express a eukaryotic protein in *E. coli*, the eukaryotic cDNA must be cloned in an *expression vector* that contains regulatory signals for both transcription and translation. Accordingly, a *promoter* where RNA polymerase initiates transcription as well as a *ribosome binding site* to facilitate translation are engineered into the vector just upstream from the restriction site for inserting foreign DNA. The AUG initiation codon that specifies the first amino acid in the protein (the *translation start site*) is contributed by the insert (Figure 13.16).

Strong promoters have been constructed that drive the synthesis of foreign proteins to levels equal to 30% or more of total *E. coli* cellular protein. An example is the hybrid promoter, p_{lac} , which was created by fusing part of the promoter for the *E. coli* genes encoding the enzymes of lactose metabolism (the *lac* promoter) with part of the promoter for the genes encoding the enzymes of tryptophan biosynthesis (the *trp* promoter) (Figure 13.17). In cells carrying p_{lac} expression vectors, the p_{lac} promoter is not induced to drive transcription of the foreign insert until the cells are exposed to *inducers* that lead to its activation. Analogs of lactose (a β -galactoside) such as *isopropyl- β -thiogalactoside*, or **IPTG**, are excellent inducers of p_{lac} . Thus, expression of the foreign protein is easily controlled. (See Chapter 31 for detailed discussions of inducible gene expression.) The bacterial production of valuable eukaryotic proteins represents one of the most important uses of recombinant DNA technology. For example, human insulin for the clinical treatment of diabetes is now produced in bacteria.

Analogous systems for expression of foreign genes in eukaryotic cells include vectors carrying promoter elements derived from mammalian viruses, such as *simian virus 40* (SV40), the *Epstein-Barr virus*, and the human *cytomegalovirus* (CMV). A system for high-level expression of foreign genes uses insect cells infected with the *baculovirus* expression vector. **Baculoviruses** infect *lepidopteran* insects (butterflies and moths). In engineered baculovirus vectors, the foreign gene is cloned downstream of the promoter for **polyhedrin**, a major viral-encoded structural protein, and the recombinant vector is incorporated

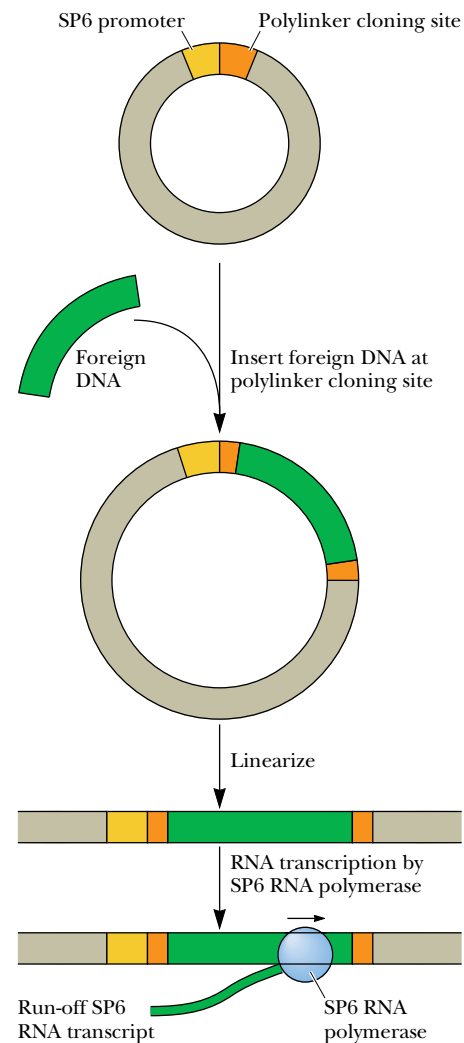


FIGURE 13.15 • Expression vectors carrying the promoter recognized by the RNA polymerase of bacteriophage SP6 are useful for making RNA transcripts *in vitro*. SP6 RNA polymerase works efficiently *in vitro* and recognizes its specific promoter with high specificity. These vectors typically have a polylinker adjacent to the SP6 promoter. Successive rounds of transcription initiated by SP6 RNA polymerase at its promoter lead to the production of multiple RNA copies of any DNA inserted at the polylinker. Before transcription is initiated, the circular expression vector is linearized by a single cleavage at or near the end of the insert so that transcription terminates at a fixed point.

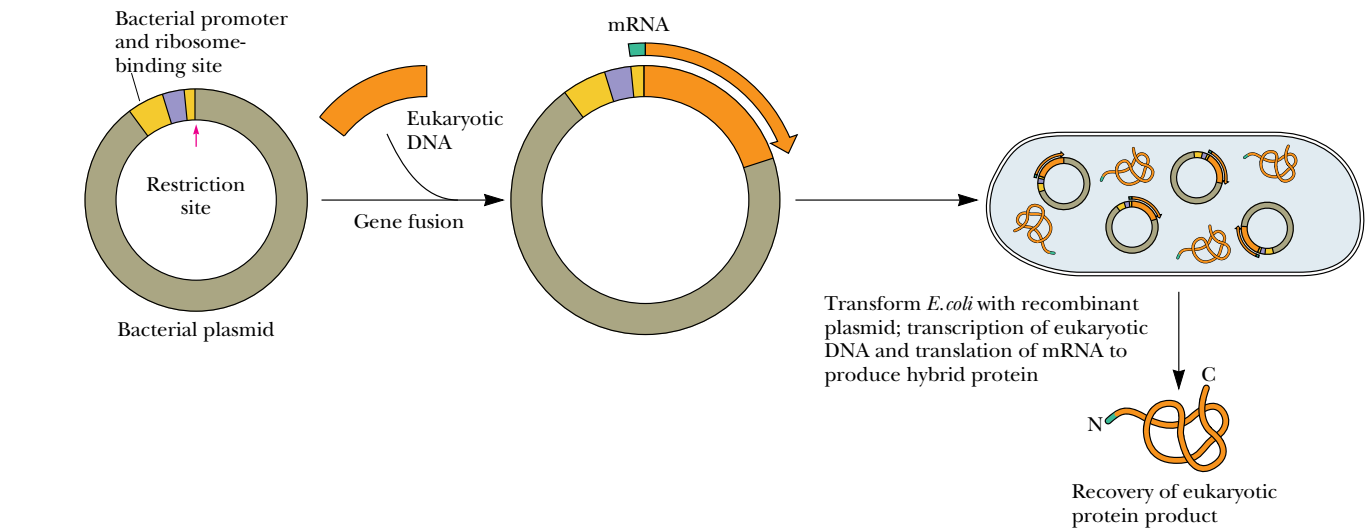


FIGURE 13.16 • A typical expression-cloning vector. Eukaryotic coding sequences are inserted at the restriction site just downstream from a promoter region where RNA polymerase binds and initiates transcription. Transcription proceeds through a region encoding a bacterial ribosome-binding site and into the cloned insert. The presence of the bacterial ribosome-binding site in the RNA transcript ensures that the RNA can be translated into protein by the ribosomes of the host bacteria. (Adapted from Figure 19-5 from Molecular Biology of the Gene, 4th edition. Copyright 1987 by James D. Watson. Reprinted by permission of Benjamin/Cummings Publishing Co., Inc.)

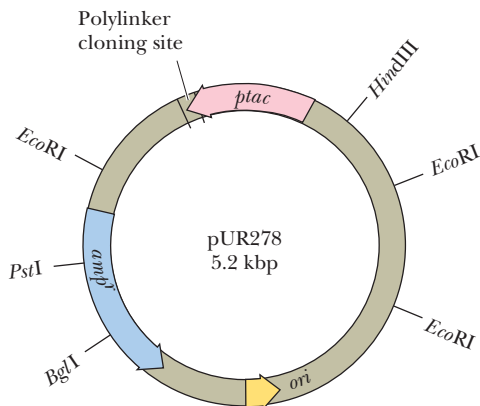


FIGURE 13.17 • A p_{lac} protein expression vector contains the hybrid promoter p_{lac} derived from fusion of the *lac* and *trp* promoters. Expression from p_{lac} is more than 10 times greater than expression from either the *lac* or *trp* promoter alone. Isopropyl- β -D-thiogalactoside, or IPTG, induces expression from p_{lac} as well as *lac*.

into insect cells grown in culture. Expression from the polyhedrin promoter can lead to accumulation of the foreign gene product to levels as high as 500 mg/L.

Screening cDNA Expression Libraries with Antibodies

Antibodies that specifically cross-react with a particular protein of interest are often available. If so, these antibodies can be used to screen a cDNA expression library to identify and isolate cDNA clones encoding the protein. The cDNA library is introduced into host bacteria, which are plated out and grown overnight, as in the colony hybridization scheme previously described. DNA-binding nylon membranes are placed on the plates to obtain a replica of the bacterial colonies. The nylon membrane is then incubated under conditions that induce protein synthesis from the cloned cDNA inserts, and the cells are treated to release the synthesized protein. The synthesized protein binds tightly to the nylon membrane, which can then be incubated with the specific antibody. Binding of the antibody to its target protein product reveals the position of any cDNA clones expressing the protein, and these clones can be recovered from the original plate. Like other libraries, expression libraries can be screened with oligonucleotide probes, too.

Fusion Protein Expression

Some expression vectors carry cDNA inserts cloned directly into the coding sequence of a vector-borne protein-coding gene (Figure 13.18). Translation of the recombinant sequence leads to synthesis of a *hybrid protein* or *fusion protein*. The N-terminal region of the fused protein represents amino acid sequences encoded in the vector, whereas the remainder of the protein is encoded by the foreign insert. Keep in mind that the triplet codon sequence within the cloned insert must be in phase with codons contributed by the vector sequences to make the right protein. The N-terminal protein sequence contributed by the vector can be chosen to suit purposes. Furthermore, adding an N-terminal sig-

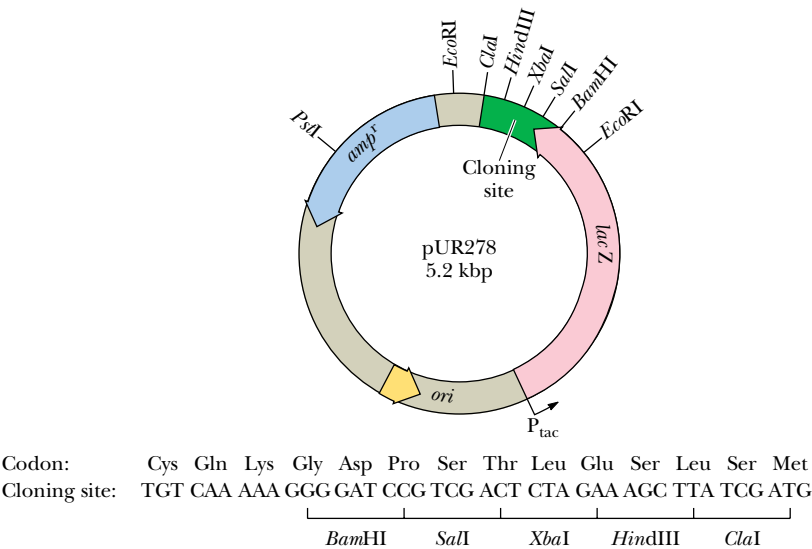


FIGURE 13.18 • A typical expression vector for the synthesis of a hybrid protein. The cloning site is located at the end of the coding region for the protein β -galactosidase. Insertion of foreign DNAs at this site fuses the foreign sequence to the β -galactosidase coding region (the *lacZ* gene). IPTG induces the transcription of the *lacZ* gene from its promoter *p_{lac}*, causing expression of the fusion protein. (Adapted from Figure 1.5.4 in Ausubel, F. M., et al., 1987. Current Protocols in Molecular Biology. New York: John Wiley & Sons.)

nal sequence that targets the hybrid protein for secretion from the cell simplifies recovery of the fusion protein. A variety of gene fusion systems have been developed to facilitate isolation of a specific protein encoded by a cloned insert. The isolation procedures are based on affinity chromatography purification of the fusion protein through exploitation of the unique ligand-binding properties of the vector-encoded protein (Table 13.2).

β -Galactosidase and Blue or White Selection

One version of these fusion protein expression vectors places the cloning site at the end of the coding region of the protein β -galactosidase, so that among other things the fusion protein is attached to β -galactosidase and can be recovered by purifying the β -galactosidase activity. Alternatively, placing the cloning site within the β -galactosidase coding region means that cloned inserts disrupt the β -galactosidase amino acid sequence, inactivating its enzymatic activity. This property has been exploited in developing a visual screening protocol that dis-

Table 13.2
Gene Fusion Systems for Isolation of Cloned Fusion Proteins

Gene Product	Origin	Molecular Mass (kD)	Secreted? ¹	Affinity Ligand
β -Galactosidase	<i>E. coli</i>	116	No	<i>p</i> -Aminophenyl- β -D-thiogalactoside (APTG)
Protein A	<i>S. aureus</i>	31	Yes	Immunoglobulin G (IgG)
Chloramphenicol acetyltransferase (CAT)	<i>E. coli</i>	24	Yes	Chloramphenicol
Streptavidin	<i>Streptomyces</i>	13	Yes	Biotin
Glutathione-S-transferase (GST)	<i>E. coli</i>	26	No	Glutathione
Maltose-binding protein (MBP)	<i>E. coli</i>	40	Yes	Starch

¹This indicates whether combined secretion–fusion gene systems have led to secretion of the protein product from the cells, which simplifies its isolation and purification.
Adapted from Uhlen, M., and Moks, T., 1990. Gene fusions for purpose of expression: An introduction. *Methods in Enzymology* **185**:129–143.

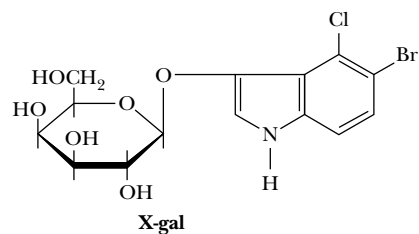


FIGURE 13.19 • The structure of 5-bromo-4-chloro-3-indolyl- β -D-galactopyranoside, or X-gal.

tinguishes those clones in the library that bear inserts from those that lack them.

Cells that have been transformed with a plasmid-based β -galactosidase expression cDNA library (or infected with a similar library constructed in a bacteriophage λ -based β -galactosidase fusion vector) are plated on media containing 5-bromo-4-chloro-3-indolyl- β -D-galactopyranoside, or **X-gal** (Figure 13.19). X-gal is a *chromogenic substrate*, a colorless substance that upon enzymatic reaction yields a colored product. Following induction with IPTG, bacterial colonies (or plaques) harboring vectors in which the β -galactosidase gene is intact (those vectors lacking inserts) express an active β -galactosidase that cleaves X-gal, liberating 5-bromo-4-chloro-indoxyl, which dimerizes to form an indigo blue product. These blue colonies (or plaques) represent clones that lack inserts. The β -galactosidase gene is inactivated in clones with inserts, so those colonies (or plaques) that remain “white” (actually, colorless) are recombinant clones.

Reporter Gene Constructs

Potential regulatory regions of genes (such as promoters) can be investigated by placing these regulatory sequences into plasmids upstream of a gene, called a **reporter gene**, whose expression is easy to measure. Such chimeric plasmids are then introduced into cells of choice (including eukaryotic cells) to assess the potential function of the nucleotide sequence in regulation because expression of the reporter gene serves as a report on the effectiveness of the regulatory element. A number of different genes have been used as reporter genes, such as the *lacZ* gene. A reporter gene with many inherent advantages is that encoding the **green fluorescent protein** (or **GFP**), described in Chapter 4. Unlike the protein expressed by other reporter gene systems, GFP does not require any substrate to measure its activity, nor is it dependent on any cofactor or prosthetic group. Detection of GFP requires only irradiation with near UV or blue light (400-nm light is optimal), and the green fluorescence (light of 500 nm) that results is easily observed with the naked eye, although it can also be measured precisely with a fluorometer. Figure 13.20 demonstrates the use of GFP as a reporter gene.

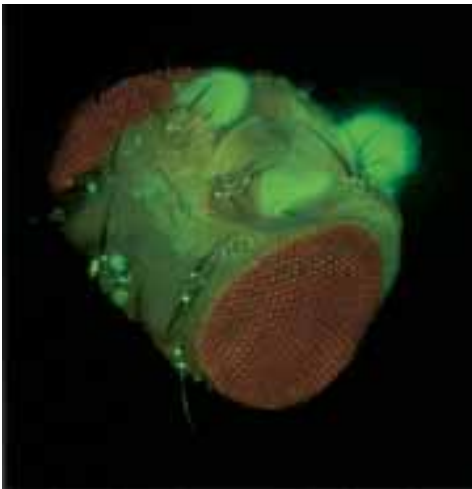
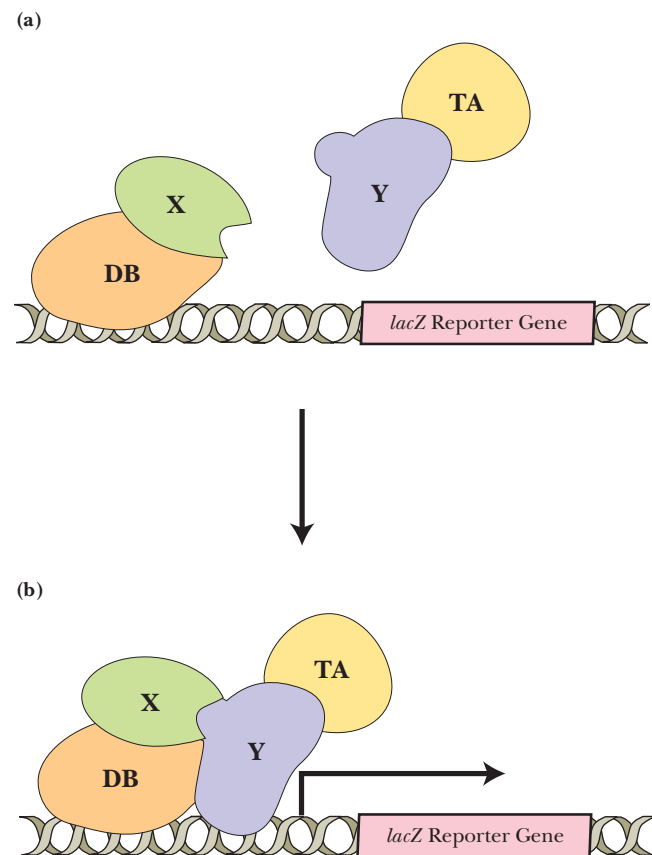


FIGURE 13.20 • Green fluorescent protein (GFP) as a reporter gene. The promoter from the *per* gene was placed upstream of the GFP gene in a plasmid and transformed into *Drosophila* (fruit flies). The *per* gene encodes a protein involved in establishing the circadian (daily) rhythmic activity of fruit flies. The fluorescence shown here in an isolated fly head follows a 24-hour rhythmic pattern and occurs to a lesser extent throughout the entire fly, indicating that *per* gene expression can occur in cells throughout the animal. Such uniformity suggests that individual cells have their own independent clocks. (Image courtesy of Jeffrey D. Plautz and Steve A. Kay, Scripps Research Institute, La Jolla, California. See also Plautz, J. D., et al., 1997. Science 278:1632–1635.)

A DEEPER LOOK

The Two-Hybrid System to Identify Proteins Involved in Specific Protein–Protein Interactions

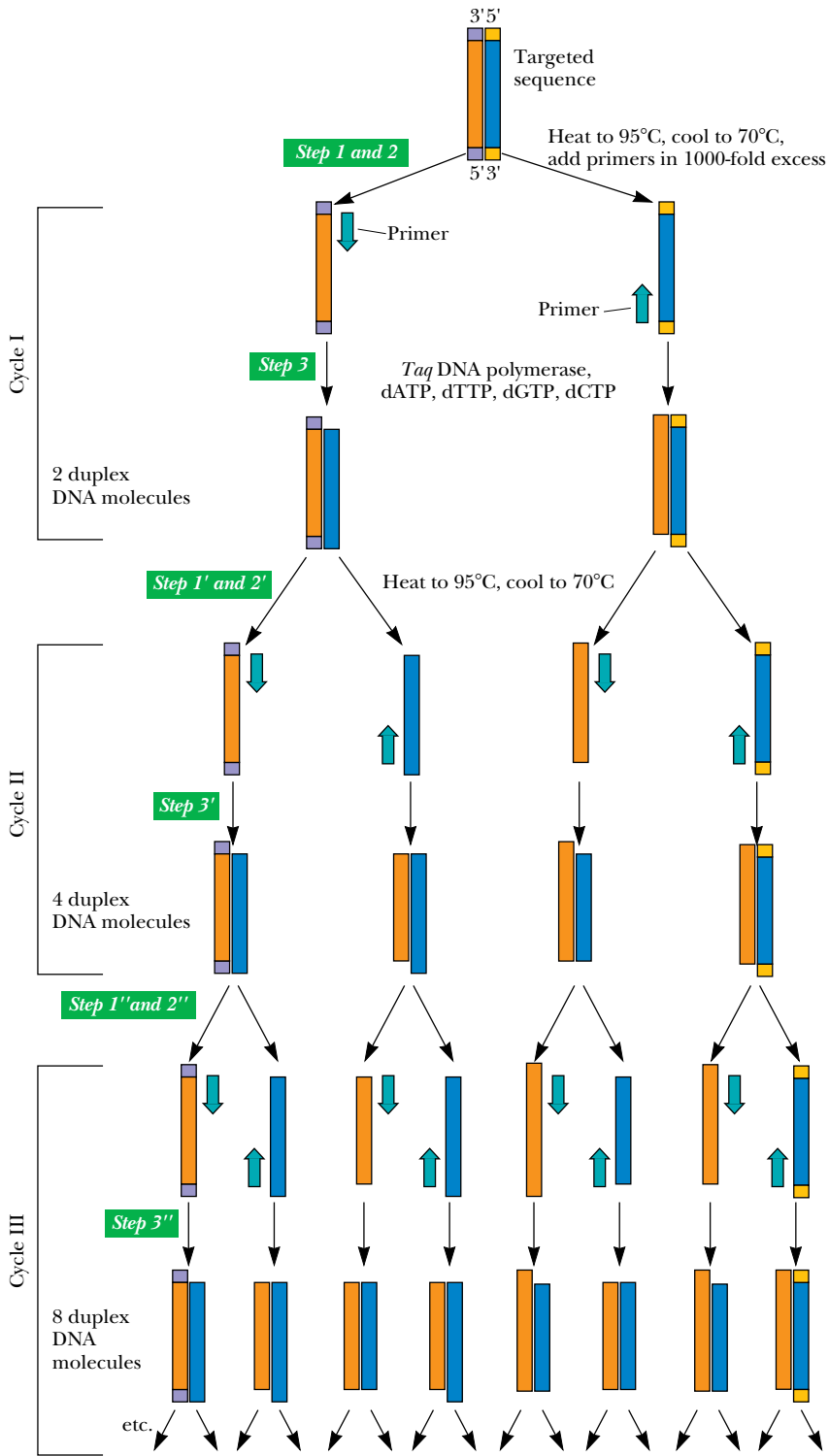
Specific interactions between proteins (so-called protein–protein interactions) lie at the heart of many essential biological processes. Stanley Fields, Cheng-Ting Chien, and their collaborators have invented a method to identify specific protein–protein interactions *in vivo* through expression of a reporter gene whose transcription is dependent on a functional transcriptional activator, the *GAL4* protein. The *GAL4* protein consists of two domains: a DNA-binding (or **DB**) domain and a transcriptional activation (or **TA**) domain. Even if expressed as separate proteins, these two domains will still work, provided they can be brought together. The method depends on two separate plasmids encoding two hybrid proteins, one consisting of the *GAL4* DB domain fused to protein X, and the other consisting of the *GAL4* TA domain fused to protein Y (figure, part a). If proteins X and Y interact in a specific protein–protein interaction, the *GAL4* DB and TA domains are brought together so that transcription of a reporter gene driven by the *GAL4* promoter can take place (figure, part b). Protein X, fused to the *GAL4* DNA binding domain (DB), serves as the “bait” to fish for the protein Y “target” and its fused *GAL4* TA domain. This method can be used to screen cells for protein “targets” that interact specifically with a particular “bait” protein. To do so, cDNAs encoding proteins from the cells of interest are inserted into the TA-containing plasmid to create fusions of the cDNA coding sequences with the *GAL4* TA domain coding sequences, so a fusion protein library is expressed. Identification of a target of the “bait” protein by this method also yields directly a cDNA version of the gene encoding the “target” protein.



13.3 • Polymerase Chain Reaction (PCR)

Polymerase chain reaction or **PCR** is a technique for dramatically amplifying the amount of a specific DNA segment. A preparation of denatured DNA containing the segment of interest serves as template for DNA polymerase, and two specific oligonucleotides serve as primers for DNA synthesis (as in Figure 13.21). These primers, designed to be complementary to the two 3′-ends of the specific DNA segment to be amplified, are added in excess amounts of 1000 times or greater (Figure 13.21). They prime the DNA polymerase–catalyzed synthesis of the two complementary strands of the desired segment, effectively doubling its concentration in the solution. Then the DNA is heated to dissociate the DNA duplexes and then cooled so that primers bind to both the newly formed and the old strands. Another cycle of DNA synthesis ensues. The protocol has been automated through the invention of **thermal cyclers** that alternately heat the reaction mixture to 95°C to dissociate the DNA, followed by cooling, annealing of primers, and another round of DNA synthesis. The isolation of heat-stable DNA polymerases from thermophilic bacteria (such as the

FIGURE 13.21 • Polymerase chain reaction (PCR). Oligonucleotides complementary to a given DNA sequence prime the synthesis of only that sequence. Heat-stable *Taq* DNA polymerase survives many cycles of heating. Theoretically, the amount of the specific primed sequence is doubled in each cycle.



Taq DNA polymerase from *Thermus aquaticus*) has made it unnecessary to add fresh enzyme for each round of synthesis. Because the amount of target DNA theoretically doubles each round, 25 rounds would increase its concentration about 33 million times. In practice, the increase is actually more like a million times, which is more than ample for gene isolation. Thus, starting with a tiny

amount of total genomic DNA, a particular sequence can be produced in quantity in a few hours.

PCR amplification is an effective cloning strategy if sequence information for the design of appropriate primers is available. Because DNA from a single cell can be used as a template, the technique has enormous potential for the clinical diagnosis of infectious diseases and genetic abnormalities. With PCR techniques, DNA from a single hair or sperm can be analyzed to identify particular individuals in criminal cases without ambiguity. **RT-PCR**, a variation on the basic PCR method, is useful when the nucleic acid to be amplified is an RNA (such as mRNA). Reverse transcriptase (RT) is used to synthesize a cDNA strand complementary to the RNA, and this cDNA serves as the template for further cycles of PCR.

In Vitro Mutagenesis

The advent of recombinant DNA technology has made it possible to clone genes, manipulate them *in vitro*, and express them in a variety of cell types under various conditions. The function of any protein is ultimately dependent on its amino acid sequence, which in turn can be traced to the nucleotide sequence of its gene. The introduction of purposeful changes in the nucleotide sequence of a cloned gene represents an ideal way to make specific structural changes in a protein. The effects of these changes on the protein’s function can then be studied. Such changes constitute *mutations* introduced *in vitro* into the gene. **In vitro mutagenesis** makes it possible to alter the nucleotide sequence of a cloned gene systematically, as opposed to the chance occurrence of mutations in natural genes.

One efficient technique for *in vitro* mutagenesis is **PCR-based mutagenesis**. Mutant primers are added to a PCR reaction in which the gene (or segment of a gene) is undergoing amplification. The *mutant primers* are primers whose sequence has been specifically altered to introduce a directed change at a particular place in the nucleotide sequence of the gene being amplified (Figure 13.22). Mutant versions of the gene can then be cloned and expressed to determine any effects of the mutation on the function of the gene product.

**13.4 • Recombinant DNA Technology:
An Exciting Scientific Frontier**

The strategies and methodologies described in this chapter are but an overview of the repertoire of experimental approaches that have been devised by molecular biologists in order to manipulate DNA and the information inherent in it. The enormous success of recombinant DNA technology means that the molecular biologist’s task in searching genomes for genes is now akin to that of a lexicographer compiling a dictionary, a dictionary in which the “letters,” i.e., the nucleotide sequences, spell out not words, but genes and what they mean. Molecular biologists have no index or alphabetic arrangement to serve as a guide through the vast volume of information in a genome; nevertheless, this information and its organization are rapidly being disclosed by the imaginative efforts and diligence of these scientists and their growing arsenal of analytical schemes.

Recombinant DNA technology now verges on the ability to engineer at will the genetic constitution of organisms for desired ends. The commercial production of therapeutic biomolecules in microbial cultures is already established (for example, the production of human insulin in quantity in *E. coli* cells). Agricultural crops with desired attributes, such as enhanced resistance to her-

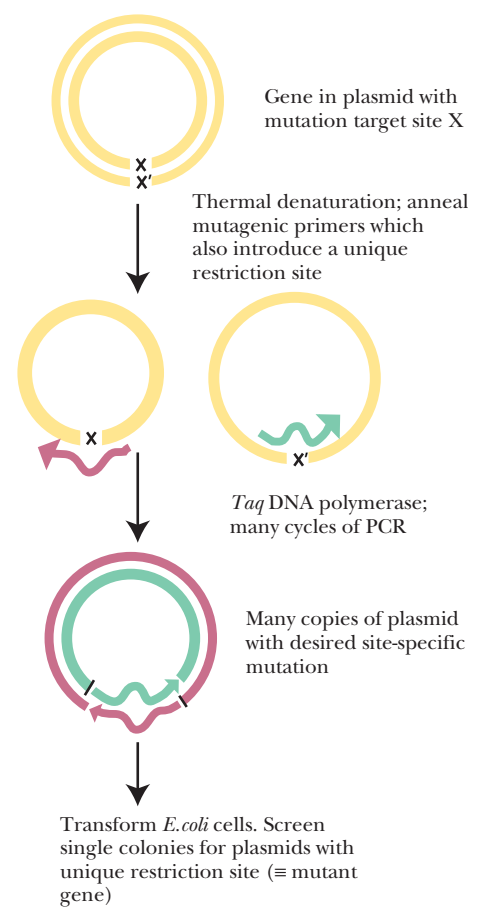


FIGURE 13.22 • One method of PCR-based site-directed mutagenesis. Template DNA strands are separated by increased temperature, and the single strands are amplified by PCR using mutagenic primers (represented as bent arrows) whose sequences introduce a single base substitution at site X (and its complementary base X’; thus the desired amino acid change in the protein encoded by the gene). Ideally, the mutagenic primers also introduce a unique restriction site into the plasmid that was not present before. Following many cycles of PCR, the DNA product can be used to transform *E. coli* cells. Single colonies of the transformed cells can be picked. The plasmid DNA within each colony can be isolated and screened for the presence of the mutation by screening for the presence of the unique restriction site by restriction endonuclease cleavage. For example, the nucleotide sequence *GGATCT* within a gene codes for amino acid residues Gly-Ser. Using mutagenic primers of nucleotide sequence *AGATCT* (and its complement *AGATCT*) changes the amino acid sequence from Gly-Ser to Arg-Ser and creates a *Bgl*III restriction site (see Table 11.5). Gene expression of the isolated mutant plasmid in *E. coli* allows recovery and analysis of the mutant protein.

HUMAN BIOCHEMISTRY

The Biochemical Defects in Cystic Fibrosis and ADA⁻ SCID

The gene defective in cystic fibrosis codes for CFTR (cystic fibrosis transmembrane conductance regulator), a membrane protein that pumps Cl⁻ out of cells. If this Cl⁻ pump is defective, Cl⁻ ions remain in cells, which then take up water from the surrounding mucus by osmosis. The mucus thickens and accumulates in various organs, including the lungs, where its presence favors infections such as pneumonia. Left untreated, children with cystic fibrosis seldom survive past the age of 5 years.

ADA⁻ SCID (adenosine deaminase–defective severe combined immunodeficiency) is a fatal genetic disorder caused by defects in the gene that encodes adenosine deaminase (ADA).

The consequence of ADA deficiency is accumulation of adenosine and 2'-deoxyadenosine, substances toxic to lymphocytes, important cells in the immune response. 2'-Deoxyadenosine is particularly toxic because its presence leads to accumulation of its nucleotide form, dATP, an essential substrate in DNA synthesis. Elevated levels of dATP actually block DNA replication and cell division by inhibiting synthesis of the other deoxynucleoside 5'-triphosphates (see Chapter 27). Accumulation of dATP also leads to selective depletion of cellular ATP, robbing cells of energy. Children with ADA⁻ SCID fail to develop normal immune responses and are susceptible to fatal infections, unless kept in protective isolation.

bicides, are in cultivation. The rat growth hormone gene has been cloned and transferred into mouse embryos, creating *transgenic mice* that at adulthood are twice normal size (see Chapter 29). Already, transgenic versions of domestic animals such as pigs, sheep, and even fish have been developed for human benefit. Perhaps most important, in a number of instances, clinical trials have been approved for **gene replacement therapy** (or, more simply, *gene therapy*) to correct particular human genetic disorders.

Human Gene Therapy

Human gene therapy seeks to repair the damage caused by a genetic deficiency through introduction of a functional version of the defective gene. To achieve this end, a cloned variant of the gene must be incorporated into the organism in such a manner that it is expressed only at the proper time *and* only in appropriate cell types. At this time, these conditions impose serious technical and clinical difficulties. Many gene therapies have received approval from the National Institutes of Health for trials in human patients, including the introduction of gene constructs into patients. Among these are constructs designed to cure ADA⁻ SCID (severe combined immunodeficiency due to adenosine deaminase [ADA] deficiency), neuroblastoma, or cystic fibrosis, or to treat cancer through expression of the E1A and p53 tumor suppressor genes.

A basic strategy in human gene therapy involves incorporation of a functional gene into target cells. The gene is typically in the form of an **expression cassette** consisting of a cDNA version of the gene downstream from a promoter that drives expression of the gene. A vector carrying such an expression cassette is introduced into target cells, either *ex vivo* via gene transfer into cultured cells in the laboratory and administration of the modified cells to the patient, or *in vivo* via direct incorporation of the gene into the cells of the patient. Because retroviruses can transfer their genetic information directly into the genome of host cells, retroviruses provide one route to permanent modification of host cells *ex vivo*. A replication-deficient version of *Maloney murine leukemia virus* can serve as a vector for expression cassettes up to 9 kb in size. Figure 13.23 describes a strategy for retrovirus vector–mediated gene delivery. In this strategy, it is hoped that the expression cassette will become stably integrated into the DNA of the patient’s own cells and expressed to pro-

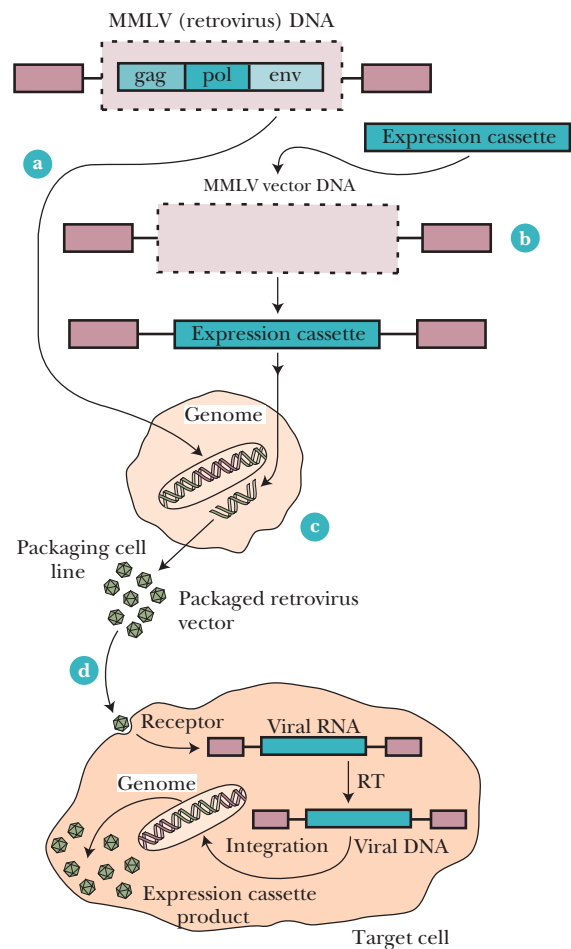


FIGURE 13.23 • Retrovirus-mediated gene delivery *ex vivo*. Retroviruses are RNA viruses that replicate their RNA genome by first making a DNA intermediate. The Maloney murine leukemia virus (MMLV) is the retrovirus used in human gene therapy. Deletion of the essential genes *gag*, *pol*, and *env* from MMLV makes it replication-deficient (so it can't reproduce) (a) and creates a space for insertion of an expression cassette (b). The modified MMLV acts as a vector for the expression cassette; although replication-defective, it is still infectious. Infection of a packaging cell line that carries intact *gag*, *pol*, and *env* genes allows the modified MMLV to reproduce (c), and the packaged retroviral viruses can be collected and used to infect a patient (d). In the cytosol of the patient's cells, a DNA copy of the viral RNA is synthesized by viral reverse transcriptase, which accompanies the viral RNA into the cells. This DNA is then randomly integrated into the host cell genome, where its expression leads to production of the expression cassette product. (Adapted from Figure 1 in Crystal, R. G., 1995. *Transfer of genes to humans: Early lessons and obstacles to success*. Science 270:404.)

duce the desired gene product. Alternatively, *adenovirus* vectors that can carry expression cassettes up to 7.5 kb are a possible *in vivo* approach to human gene therapy (Figure 13.24). Recombinant, replication-deficient adenoviruses enter target cells via specific receptors on the target cell surface; the transferred genetic information is expressed directly from the adenovirus recombinant DNA and is never incorporated into the host cell genome. Although many problems remain to be solved, human gene therapy as a clinical strategy is feasible.

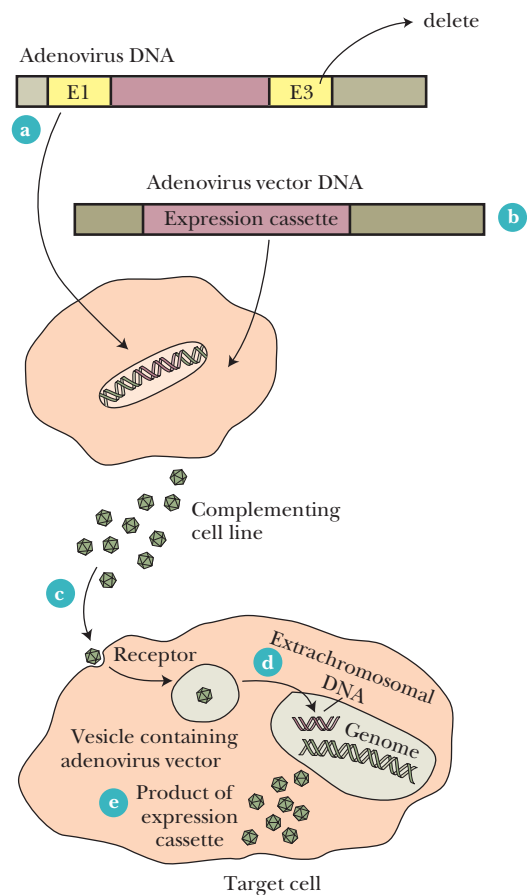


FIGURE 13.24 • Adenovirus-mediated gene delivery *in vivo*. Adenoviruses are DNA viruses. The adenovirus genome (36 kb) is divided into *early genes* (E1 through E4) and *late genes* (L1 to L5) (a). Adenovirus vectors are generated by deleting gene E1 (and sometimes E3 if more space for an expression cassette is needed) (b); deletion of E1 renders the adenovirus incapable of replication unless introduced into a complementing cell line carrying the E1 gene (c). Adenovirus progeny from the complementing cell line can be used to infect a patient. In the patient, the adenovirus vector with its expression cassette enters the cells via specific receptors (d). Its linear dsDNA ultimately gains access to the cell nucleus, where it functions extrachromosomally and expresses the product of the expression cassette (e). (Adapted from Figure 2 in Crystal, R. G., 1995. Transfer of genes to humans: Early lessons and obstacles to success. *Science* 270:404.)

PROBLEMS

1. A DNA fragment isolated from an *EcoRI* digest of genomic DNA was combined with a plasmid vector linearized by *EcoRI* digestion so sticky ends could anneal. Phage T4 DNA ligase was then added to the mixture. List all possible products of the ligation reaction.
2. The nucleotide sequence of a polylinker in a particular plasmid vector is
- GAATTCCCGGGGATCCTCTAGAGTCGACCTGCAGGCATGC—
- This polylinker contains restriction sites for *Bam*HI, *Eco*RI, *Pst*I, *Sal*I, *Sma*I, *Sph*I, and *Xba*I. Indicate the location of each restriction site in this sequence. (See Table 11.5 of restriction enzymes for their cleavage sites.)

3. A vector has a polylinker containing restriction sites in the following order: *Hind*III, *Sac*I, *Xho*I, *Bgl*II, *Xba*I, and *Cla*I.
- a. Give a possible nucleotide sequence for the polylinker.
- b. The vector is digested with *Hind*III and *Cla*I. A DNA segment contains a *Hind*III restriction site fragment 650 bases upstream from a *Cla*I site. This DNA fragment is digested with *Hind*III and *Cla*I, and the resulting *Hind*III–*Cla*I fragment is directionally cloned into the *Hind*III–*Cla*I–digested vector. Give the nucleotide sequence at each end of the vector and the insert and show that the insert can be cloned into the vector in only one orientation.
4. Yeast (*Saccharomyces cerevisiae*) has a genome size of 1.21×10^7 bp. If a genomic library of yeast DNA was constructed in a bacteriophage λ vector capable of carrying 16-kbp inserts, how many indi-

- vidual clones would have to be screened to have a 99% probability of finding a particular fragment?
5. The South American lungfish has a genome size of 1.02×10^{11} bp. If a genomic library of lungfish DNA was constructed in a cosmid vector capable of carrying inserts averaging 45 kbp in size, how many individual clones would have to be screened to have a 99% probability of finding a particular DNA fragment?
6. Given the following short DNA duplex of sequence (5'→3')

ATGCCGTAGTTCGATCATTACGATAGCATAGCACAGGGATCCA-
CATGCACACACATGACATAGGACAGATAGCAT

what oligonucleotide primers (17-mers) would be required for PCR amplification of this duplex?

7. Figure 13.5b shows a polylinker that falls within the β -galactosidase coding region of the *lacZ* gene. This polylinker serves as a cloning site in a fusion protein expression vector where the closed insert is expressed as a β -galactosidase fusion protein. Assume the vector polylinker was cleaved with *Bam*HI and then

- ligated with an insert whose sequence reads
- GATCCATTTATCCACCGGAGAGCTGGTATCCCCAAAAGACG-
GCC . . .
- What is the amino acid sequence of the fusion protein? Where is the junction between β -galactosidase and the sequence encoded by the insert? (Consult the genetic code table on the inside front cover to decipher the amino acid sequence.)
8. The amino acid sequence across a region of interest in a protein is
- Asn-Ser-Gly-Met-His-Pro-Gly-Lys-Leu-Ala-Ser-Trp-Phe-Val-Gly-Asn-Ser

The nucleotide sequence encoding this region begins and ends with an *Eco*RI site, making it easy to clone out the sequence and amplify it by the polymerase chain reaction (PCR). Give the nucleotide sequence of this region. Suppose you wished to change the middle Ser residue to a Cys to study the effects of this change on the protein's activity. What would be the sequence of the mutant oligonucleotide you would use for PCR amplification?

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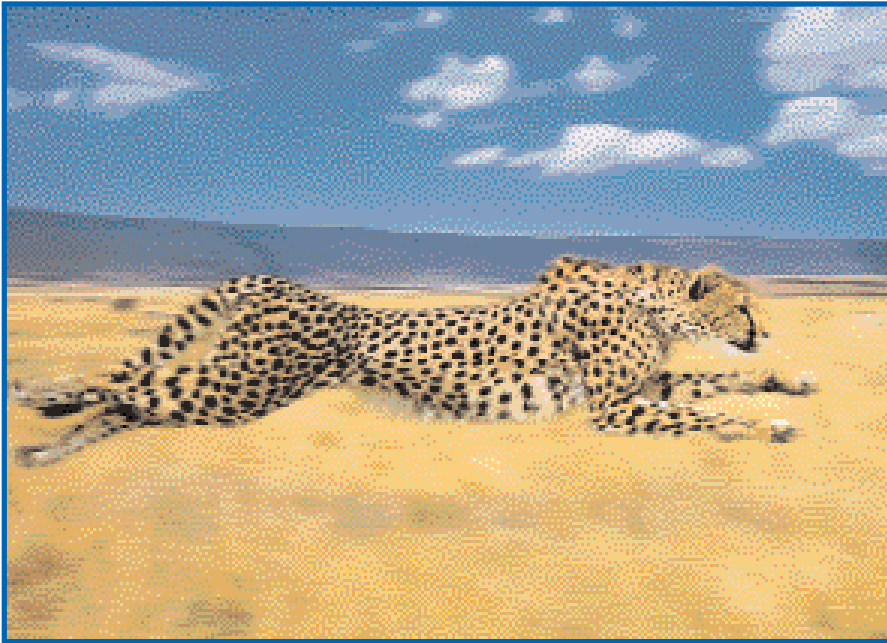
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Part II

Protein Dynamics



*Protein dynamics—the action of enzymes and molecular motors—provides the key to understanding the biochemistry of this cheetah and the grasses through which it runs.
(Frank Lane/Parfitt/Tony Stone Images)*

CHAPTER 14

Enzyme Kinetics

CHAPTER 15

Enzyme Specificity and Regulation

CHAPTER 16

Mechanisms of Enzyme Action

CHAPTER 17

Molecular Motors